FROM DATA TO THESIS

Using Python for Non-Technical

Polla Fattah and Collection of LLMs

Contents

Welcome . . . 15
How to use this book . . . 15
About the data . . . 15

Dedication . . . 17

Foreword . . . 19
Modern Research in the Age of AI . . . 19
How to Journey Through This Book . . . 19
The Interactive Playground . . . 19

I Foundations of Python Programming and Data Handling

1 Getting Started with Python . . . 23
1.1 Research written in code . . . 23
1.2 Installing Python and VS Code . . . 24
1.3 The editor layout . . . 25
1.4 Calculations . . . 25
1.5 Storing values in variables . . . 25
1.6 Types of data . . . 26
1.6.1 Missing values . . . 27
1.7 Functions and methods . . . 27
1.7.1 Getting help . . . 28
1.8 Writing scripts . . . 28
1.9 Managing research projects . . . 29
1.10 Installing and importing packages . . . 29
1.11 A first look at Elaf's data . . . 29
1.12 Chapter review . . . 32
1.12.1 Summary . . . 32
1.12.2 Key terms . . . 32
1.13 Exercises . . . 33
1.14 Further reading . . . 33

2 Data Structures in Python . . . 35
2.1 Data as measurement . . . 35
2.2 Variables and their values . . . 36
2.2.1 Operations on a whole Series (Vectorization) . . . 36
2.2.2 Selecting values . . . 37
2.2.3 Changing values . . . 37
2.3 Categories (Categorical Data) . . . 38
2.3.1 Levels of measurement and Python types . . . 38

2.4 A table of cases: The DataFrame . . . 39
2.4.1 Selecting columns and rows . . . 39
2.4.2 Adding new columns . . . 40
2.4.3 The student wellbeing cohort . . . 40
2.5 Matrices and Dictionaries . . . 41
2.5.1 Matrices and correlation grids . . . 41
2.5.2 Dictionaries and statistical results . . . 42
2.6 Importing research data . . . 42
2.6.1 CSV files . . . 42
2.6.2 Excel spreadsheets . . . 43
2.6.3 SPSS data files . . . 43
2.7 Automated codebook generation . . . 44
2.8 Exporting data . . . 45
2.9 Mapping SPSS and Excel workflows to Python . . . 45
2.10 Reading error messages and tracebacks . . . 46
2.11 Descriptive summary of cohort measurements . . . 48
2.12 Chapter review . . . 49
2.12.1 Summary . . . 49
2.12.2 Key terms . . . 50
2.13 Exercises . . . 50
2.14 Further reading . . . 50

3 Data Manipulation and Cleaning with Pandas . . . 51
3.1 What clean data means . . . 51
3.2 Tidy data principles in Python . . . 52
3.3 The Pandas manipulation toolkit . . . 53
3.3.1 Filtering observations . . . 53
3.3.2 Selecting and dropping columns . . . 54
3.3.3 Sorting cases . . . 54
3.3.4 Creating and transforming variables . . . 55
3.3.5 Summarizing by group . . . 55
3.3.6 Combining operations into an analytical pipeline . . . 56
3.4 Reshaping data between wide and long . . . 57
3.5 Joining relational tables . . . 58
3.6 Cleaning an uncurated survey export . . . 59
3.6.1 Inspecting raw survey responses . . . 60
3.6.2 Removing test records and duplicate submissions . . . 60
3.6.3 Renaming variables to standardized identifiers . . . 60
3.6.4 Standardizing categorical variables . . . 61
3.6.5 Recoding missing value codes and text numbers . . . 62
3.6.6 Eliminating impossible values . . . 63
3.6.7 Computing composite questionnaire scale scores . . . 63
3.6.8 Reshaping repeated semester measurements . . . 64
3.6.9 Validating cleaned tables against canonical datasets . . . 65
3.7 Chapter review . . . 66
3.7.1 Summary . . . 66
3.7.2 Key terms . . . 66
3.8 Exercises . . . 66
3.9 Further reading . . . 67

4 Data Visualization with Seaborn and Matplotlib . . . 69
4.1 What a graph is for . . . 69
4.1.1 Exploratory versus explanatory graphs . . . 71
4.2 Graphical perception: How vision decodes data . . . 71
4.3 Selecting the right visualization . . . 72
4.4 Core statistical visualizations with Seaborn . . . 73
4.4.1 Histograms and density curves . . . 73
4.4.2 Bar charts for categorical counts . . . 74
4.4.3 Box plots for group comparisons . . . 75
4.4.4 Scatter plots with regression trend lines . . . 76
4.4.5 Longitudinal trajectory line plots . . . 77
4.5 Preparing publication-quality figures . . . 78
4.5.1 Annotations and reference lines . . . 78
4.5.2 Color accessibility . . . 79
4.6 Honest graphs and visual ethics . . . 79
4.6.1 Redesigning a misleading figure . . . 79
4.7 Advanced scientific visualizations . . . 82
4.7.1 Small multiples (Faceting) . . . 82
4.7.2 Kernel density estimation (KDE) plots . . . 82
4.7.3 Correlation heatmaps . . . 83
4.8 Exporting publication-ready figures . . . 84
4.9 Chapter review . . . 84
4.9.1 Summary . . . 84
4.9.2 Key terms . . . 85
4.10 Exercises . . . 85
4.11 Further reading . . . 85

II Statistical Analysis for Research

5 From Research Question to Data . . . 89
5.1 What data analysis is for . . . 89
5.2 From a topic to a research question . . . 90
5.2.1 Three kinds of question . . . 92
5.3 From a question to a hypothesis . . . 92
5.3.1 Falsifiability . . . 92
5.3.2 The null hypothesis . . . 93
5.3.3 Hypotheses for the whole study . . . 94
5.4 Variables and measurement . . . 96
5.4.1 Cases and the unit of analysis . . . 96
5.4.2 From constructs to variables . . . 96
5.4.3 Levels of measurement . . . 97
5.4.4 The roles variables play . . . 99
5.5 Reliability and validity . . . 101
5.5.1 Kinds of reliability . . . 103
5.5.2 Kinds of validity . . . 103
5.6 Study designs . . . 103
5.6.1 Why random assignment works . . . 104
5.6.2 Cross-sectional and longitudinal designs . . . 105
5.7 Samples and populations . . . 105

5.7.1 Ways to draw a sample . . . 105
5.7.2 Random error and bias . . . 106
5.7.3 Non-response and attrition . . . 107
5.7.4 Generalising from the sample . . . 108
5.8 Planning the analysis before the data . . . 108
5.8.1 An analysis plan . . . 108
5.8.2 Sample size and power . . . 109
5.8.3 Preregistration and honest analysis . . . 109
5.8.4 Ethics . . . 110
5.9 Common misconceptions . . . 111
5.10 Chapter review . . . 111
5.10.1 Summary . . . 111
5.10.2 Key terms . . . 112
5.11 Exercises . . . 112
5.12 Further reading . . . 112

6 Descriptive Statistics and Exploratory Data Analysis . . . 113
6.1 The place of exploration . . . 113
6.2 The centre of a distribution . . . 114
6.2.1 Choosing a summary . . . 116
6.3 The spread of a distribution . . . 117
6.3.1 The standard deviation . . . 117
6.3.2 The interquartile range . . . 119
6.4 The shape of a distribution . . . 119
6.4.1 The normal distribution . . . 121
6.5 Unusual values . . . 122
6.6 Missing data . . . 123
6.6.1 Longitudinal attrition . . . 124
6.7 Relationships between variables . . . 125
6.7.1 Correlation is not causation . . . 126
6.8 Describing your sample in a thesis . . . 127
6.9 Common misconceptions . . . 129
6.10 Chapter review . . . 130
6.10.1 Summary . . . 130
6.10.2 Key terms . . . 130
6.11 Exercises . . . 130
6.12 Further reading . . . 130

7 Hypothesis Testing and Statistical Inference . . . 131
7.1 Samples and populations . . . 131
7.1.1 Sampling distributions . . . 131
7.1.2 The standard error . . . 133
7.1.3 Confidence intervals . . . 133
7.1.4 Bootstrap confidence intervals . . . 134
7.2 The logic of hypothesis testing . . . 134
7.2.1 A permutation test . . . 135
7.2.2 The p-value . . . 137
7.2.3 Type I and Type II errors . . . 137
7.2.4 Statistical power . . . 137
7.2.5 Multiple testing . . . 139

7.3 The one-sample t-test . . . 139
7.4 The two-sample t-test (Welch's t-test) . . . 140
7.4.1 Effect size: Cohen's d . . . 141
7.5 The paired t-test . . . 141
7.6 Assumptions of the t-test . . . 142
7.6.1 The trap of formal normality tests . . . 142
7.7 Non-parametric tests . . . 142
7.7.1 Mann-Whitney U test . . . 142
7.7.2 Wilcoxon signed-rank test . . . 143
7.8 Chi-square tests . . . 143
7.8.1 Chi-square test of independence . . . 143
7.9 Choosing the right test . . . 144
7.10 Reporting results in a thesis . . . 144
7.11 Common misconceptions . . . 145
7.12 Chapter review . . . 145
7.12.1 Summary . . . 145
7.12.2 Key terms . . . 145
7.13 Exercises . . . 146
7.14 Further reading . . . 146

8 ANOVA and Regression . . . 147
8.1 One-way ANOVA . . . 148
8.1.1 Partitioning variation . . . 148
8.1.2 The student wellbeing data . . . 150
8.1.3 Post-hoc tests . . . 151
8.1.4 Assumptions of ANOVA . . . 152
8.2 Two-way ANOVA . . . 152
8.3 Linear regression . . . 153
8.3.1 The regression line . . . 153
8.3.2 The student wellbeing data . . . 156
8.3.3 Multiple regression . . . 156
8.3.4 Categorical predictors . . . 157
8.3.5 Separating an effect from a confounder . . . 158
8.3.6 Curved relationships . . . 160
8.3.7 Interactions . . . 161
8.3.8 Model diagnostics . . . 162
8.3.9 Reporting a regression . . . 163
8.4 Logistic regression . . . 164
8.4.1 Odds and odds ratios . . . 164
8.4.2 A model for dropout . . . 164
8.4.3 Predicted probabilities . . . 165
8.5 Common misconceptions . . . 166
8.6 Chapter review . . . 166
8.6.1 Summary . . . 166
8.6.2 Key terms . . . 167
8.7 Exercises . . . 167
8.8 Further reading . . . 167

9 Multivariate Statistical Methods . . . 169
9.1 Constructs and latent variables . . . 169
9.2 Many variables at once . . . 170
9.3 Principal component analysis . . . 171
9.3.1 Running PCA . . . 171
9.3.2 The number of components . . . 172
9.3.3 PCA and factor analysis compared . . . 173
9.4 Factor analysis . . . 174
9.4.1 Reliability and Cronbach's alpha . . . 174
9.5 Cluster analysis . . . 175
9.5.1 k-means clustering . . . 176
9.5.2 The number of clusters . . . 176
9.5.3 Describing the clusters . . . 177
9.5.4 Hierarchical clustering . . . 179
9.5.5 Clusters in random data . . . 180
9.6 Common misconceptions . . . 182
9.7 Chapter review . . . 182
9.7.1 Summary . . . 182
9.7.2 Key terms . . . 182
9.8 Exercises . . . 183
9.9 Further reading . . . 183

10 Mixed-Effects Models . . . 185
10.1 Data with a structure . . . 185
10.1.1 Why independence matters . . . 185
10.2 A study of sleep deprivation . . . 200
10.2.1 Random intercepts . . . 201
10.2.2 The intraclass correlation . . . 202
10.2.3 Random slopes . . . 202
10.3 Change in wellbeing over two years . . . 203
10.3.1 Why ordinary regression misleads . . . 204
10.3.2 Confidence intervals and p-values . . . 205
10.3.3 Persistence of the workshop effect . . . 206
10.3.4 Change in GPA . . . 208
10.4 Students within supervisors . . . 209
10.5 Counts and yes-or-no outcomes . . . 210
10.6 Reporting a mixed-effects model . . . 211
10.7 Common misconceptions . . . 211
10.8 Chapter review . . . 212
10.8.1 Summary . . . 212
10.8.2 Key terms . . . 212
10.9 Exercises . . . 212
10.10 Further reading . . . 212

III Machine Learning with Python

11 Introduction to Machine Learning in Python . . . 215
11.1 Machine learning and prediction . . . 215

11.2 Generalisation and overfitting . . . 216
11.3 Designing a fair test . . . 218
11.4 Data for prediction . . . 220
11.5 Splitting the data . . . 220
11.6 Preparing the data . . . 221
11.7 Specifying and fitting a model . . . 221
11.8 Evaluating a classifier . . . 247
11.9 Overfitting in practice . . . 247
11.10 Cross-validation . . . 248
11.11 Tuning a model . . . 249
11.12 The final test . . . 250
11.13 Common misconceptions . . . 251
11.14 Chapter review . . . 252
11.14.1 Summary . . . 252
11.14.2 Key terms . . . 252
11.15 Exercises . . . 252
11.16 Further reading . . . 252

12 Classification Models . . . 255
12.1 Data and resampling . . . 255
12.2 Decision trees . . . 256
12.3 Random forests . . . 258
12.4 k-nearest neighbours . . . 259
12.5 Support vector machines . . . 261
12.6 Comparing models fairly . . . 262
12.7 The confusion matrix . . . 263
12.8 Choosing the threshold . . . 265
12.8.1 Costs of the two errors . . . 267
12.9 Imbalanced outcomes . . . 269
12.10 Chapter review . . . 271
12.10.1 Summary . . . 271
12.10.2 Key terms . . . 271
12.11 Exercises . . . 271
12.12 Further reading . . . 271

13 Predictive Regression . . . 273
13.1 Data for predicting final GPA . . . 273
13.2 Measuring prediction error . . . 275
13.3 Ordinary regression with many predictors . . . 276
13.4 Regularised regression . . . 277
13.4.1 Choosing the penalty . . . 278
13.4.2 Predictors kept by the lasso . . . 280
13.5 Boosting . . . 281
13.6 Comparing the models . . . 282
13.7 The final test . . . 283

13.8 Common misconceptions . . . 285
13.9 Chapter review . . . 285
13.9.1 Summary . . . 285
13.9.2 Key terms . . . 286
13.10 Exercises . . . 286
13.11 Further reading . . . 286

14 Advanced Clustering . . . 287
14.1 The profile data . . . 287
14.2 What counts as a group . . . 288
14.3 Gaussian mixture models . . . 289
14.3.1 A one-variable example . . . 289
14.3.2 Choosing the number of clusters with BIC . . . 290
14.3.3 Describing the profiles . . . 291
14.3.4 Certainty of assignment . . . 292
14.4 Clusters as dense regions . . . 294
14.4.1 A small example . . . 294
14.4.2 Choosing eps . . . 294
14.4.3 Students who fit no profile . . . 295
14.5 Evaluating a clustering . . . 296
14.6 Common misconceptions . . . 299
14.7 Chapter review . . . 299
14.7.1 Summary . . . 299
14.7.2 Key terms . . . 299
14.8 Exercises . . . 299
14.9 Further reading . . . 300

15 Neural Networks . . . 301
15.1 A single neuron . . . 301
15.2 From neurons to networks . . . 302
15.2.1 How a network learns . . . 303
15.3 A neural network for the dropout question . . . 305
15.3.1 A network that memorises . . . 306
15.3.2 Tuning size and weight decay . . . 307
15.3.3 A fair comparison . . . 308
15.4 When neural networks are worth using . . . 309
15.5 Deep learning . . . 309
15.6 Common misconceptions . . . 310
15.7 Chapter review . . . 311
15.7.1 Summary . . . 311
15.7.2 Key terms . . . 311
15.8 Exercises . . . 311
15.9 Further reading . . . 311

16 Time Series Forecasting . . . 313
16.1 Time series data . . . 313
16.1.1 A small example . . . 313
16.1.2 The parts of a pattern . . . 314

16.2 The counselling service data . . . 315
16.2.1 Seasonal patterns . . . 317
16.2.2 Autocorrelation . . . 317
16.3 Decomposition . . . 319
16.4 Forecasting . . . 321
16.4.1 Simple benchmarks . . . 321
16.4.2 Two families of models . . . 321
16.4.3 Training and test periods . . . 322
16.4.4 Forecasting next year . . . 324
16.5 Common misconceptions . . . 327
16.6 Chapter review . . . 328
16.6.1 Summary . . . 328
16.6.2 Key terms . . . 328
16.7 Exercises . . . 328
16.8 Further reading . . . 328

IV Reproducible Research and Applications

17 Reproducible Research . . . 331
17.1 Reproducibility and credibility . . . 331
17.2 Documents that contain their own analysis . . . 332
17.2.1 A small example . . . 332
17.2.2 What happens when you render . . . 332
17.3 A results chapter in Quarto . . . 333
17.3.1 References and citations . . . 335
17.3.2 Equations . . . 335
17.3.3 Output formats . . . 335
17.3.4 Parameterised reports . . . 336
17.4 Keeping the same package versions . . . 336
17.4.1 Virtual environments and requirements.txt . . . 337
17.4.2 Reporting software versions . . . 337
17.5 Interactive dashboards with Shiny for Python . . . 337
17.5.1 A wellbeing dashboard . . . 338
17.5.2 Sharing dashboards . . . 340
17.6 Open science . . . 340
17.6.1 Sharing data and code . . . 340
17.6.2 Protecting participants and anonymisation . . . 340
17.6.3 Preregistration and the garden of forking paths . . . 341
17.6.4 Citing software . . . 342
17.7 Keeping a history with git (optional) . . . 343
17.8 Common misconceptions . . . 343
17.9 Chapter review . . . 343
17.9.1 Summary . . . 343
17.9.2 Key terms . . . 344
17.10 Exercises . . . 344
17.11 Further reading . . . 344

18 Using AI with Python . . . 345
18.1 The reliability of coding . . . 345

18.2 How AI assistants work . . . 346
18.3 AI as a coding assistant . . . 347
18.3.1 Asking effective questions . . . 347
18.3.2 Verifying the suggestion . . . 347
18.4 Coding open-ended survey answers . . . 348
18.4.1 Coding with a keyword dictionary . . . 348
18.5 Calling a language model from Python . . . 349
18.5.1 Structured outputs with Pydantic . . . 349
18.6 Evaluating AI qualitative coding . . . 350
18.6.1 Inspecting disagreements with a confusion matrix . . . 351
18.6.2 What the students reported (RQ11) . . . 352
18.6.3 Validating against quantitative survey scales . . . 352
18.7 Using AI responsibly in academic research . . . 353
18.7.1 Verify and maintain authorship accountability . . . 353
18.7.2 Protect participant data and ethical boundaries . . . 353
18.7.3 Maintain computational reproducibility . . . 353
18.7.4 Transparent disclosure . . . 354
18.8 Common misconceptions . . . 354
18.9 Chapter review . . . 354
18.9.1 Summary . . . 354
18.9.2 Key terms . . . 354
18.10 Exercises . . . 354
18.11 Further reading . . . 355

19 Putting It All Together . . . 357
19.1 The shape of a research project . . . 357
19.1.1 Organising the project . . . 359
19.2 From raw export to clean data . . . 359
19.3 Describing the sample . . . 361
19.4 Answering the research questions . . . 362
19.4.1 The workshop and its persistence (RQ3) . . . 362
19.4.2 Explaining GPA (RQ5) . . . 362
19.4.3 Considering dropout (RQ9) . . . 363
19.5 One figure for the thesis . . . 363
19.6 The chain of reasoning . . . 365
19.7 The results chapter . . . 365
19.8 Choosing a method . . . 366
19.9 Challenges every researcher meets . . . 368
19.10 Where to go next . . . 368
19.11 Common misconceptions . . . 370
19.12 Chapter review . . . 370
19.12.1 Summary . . . 370
19.12.2 Key terms . . . 371
19.13 Exercises . . . 371
19.14 Further reading . . . 371

References . . . 373

Appendices . . . 375

A Appendix A: Installation and Setup . . . 377
A.1 What you need . . . 377
A.2 Installing Python . . . 377
A.2.1 Windows . . . 377
A.2.2 macOS . . . 378
A.2.3 Linux . . . 378
A.2.4 Checking the installation . . . 378
A.3 Installing VS Code . . . 378
A.3.1 Positron: A Dedicated Data Science Alternative . . . 378
A.4 Managing projects with virtual environments . . . 378
A.5 Installing the book's packages . . . 379
A.5.1 The student wellbeing data: the `data2thesis` package . . . 379
A.5.2 Software versions used in this book . . . 380
A.6 Common problems and solutions . . . 380

B Appendix B: Glossary . . . 381
B.1 Key terms . . . 381
B.2 Python functions and methods . . . 409

C Appendix C: Python Packages . . . 411
C.1 The packages in this book . . . 411
C.1.1 The core scientific stack . . . 412

D Appendix D: Common Errors and How to Fix Them . . . 413
D.1 Anatomy of a Python traceback . . . 413
D.2 Variable and naming errors . . . 413
D.2.1 `NameError: name '...' is not defined` . . . 413
D.2.2 `AttributeError: '...' object has no attribute '...'` . . . 413
D.3 Data structure and indexing errors . . . 414
D.3.1 `KeyError: '...'` . . . 414
D.3.2 `IndexError: list index out of range` . . . 415
D.4 Data type and value errors . . . 416
D.4.1 `TypeError: unsupported operand type(s)` . . . 416
D.4.2 `ValueError: could not convert string to float` . . . 416
D.5 Syntax and indentation errors . . . 416
D.5.1 `SyntaxError: invalid syntax` . . . 416
D.5.2 `IndentationError: expected an indented block` . . . 417
D.6 File and environment errors . . . 417
D.6.1 `FileNotFoundError: [Errno 2] No such file or directory` . . . 417
D.6.2 `ModuleNotFoundError: No module named '...'` . . . 419
D.7 Silent data pitfalls . . . 419
D.7.1 1. Missing value comparisons (`NaN == NaN`) . . . 419
D.7.2 2. Exponentiation (`^` vs. `**`) . . . 419
D.7.3 3. Modifying a slice (`SettingWithCopyWarning`) . . . 420

E Appendix E: Further Learning Resources . . . 421
E.1 Books . . . 421

E.1.1 Getting started and data skills . . . 421
E.1.2 Statistics . . . 421
E.1.3 Mixed models, machine learning, and forecasting . . . 421
E.1.4 Reporting and dashboards . . . 421
E.2 Courses and practice . . . 422
E.3 Communities . . . 422

Bibliography . . . 423

Welcome

Complete Course & Companion Slides

All 19 chapters of *FROM DATA TO THESIS: Using Python for Non-Technical* are available, accompanied by interactive Playground exercises and Lecture Slide Decks for teaching and review.

Prefer reading everything in one continuous page?

Open the 📖 Complete Book Online Edition (Single-Page Reader) or 📥 Download Companion PDF Book (7.5 MB).

This book follows Elaf, a Master's student in Educational Psychology, through her thesis. She studies 600 graduate students over four semesters to learn how sleep, stress, and supervisor support relate to their wellbeing, grades, and thoughts of dropping out, and whether a short wellbeing workshop helps.

Every chapter answers one of her research questions. You learn each method the way you will use it: first on a handful of numbers, then on the study data, and then in your own field.

How to use this book

- Read the chapters in order if you are new to Python. Each one builds on the last.
- Run the code yourself. Everything you need is in the `data2thesis` package:

```
pip install data2thesis
```

- Practise in the playground after each chapter.
- Present or review with the interactive lecture slides.

About the data

Elaf, her study, and her data are fictional. The data was generated by a computer program to look like a realistic survey of graduate students, so that every method in this book has something to find. It describes no real people, and its patterns (for example, that the workshop raised wellbeing, or that students who sleep more have higher grades) were built into the program by the author, not discovered. Nothing in this book is evidence about the wellbeing of real students, and the data and results must not be cited or used as findings about students, universities, or any real situation. The same applies to the counselling service's records and to the students' written answers. The data was designed so that every method in the book finds something worth interpreting, including some honest "no effect" results. The script that generates it is published with the book, so anyone can check it or generate it again.

Dedication

Dedicated to Tano

Foreword

Writing a thesis is one of the most intellectually demanding undertakings in an academic life. Yet, for many graduate researchers, the most exhausting hurdle is not formulating hypotheses or collecting observations; it is the daunting chasm between a messy spreadsheet of raw data and a defended, publication-ready results chapter.

For decades, higher education has pushed researchers toward point-and-click statistical packages. While these tools offer an easy first step, they often leave the underlying analysis opaque, create error-prone manual copy-paste rituals, and lock research inside proprietary formats. When an examiner asks for an adjustment or an assumption check, rebuilding the analysis from scratch often becomes a nightmare.

This book was written to offer a different path.

Python has evolved into the lingua franca of modern data science, statistical computing, and reproducible research. Paired with pandas, statsmodels, scikit-learn, and modern reporting frameworks like Quarto, Python gives researchers complete command over their analytical journey—from the first row of survey data to polished, APA-style tables and high-resolution figures.

Modern Research in the Age of AI

We now live and research in the age of artificial intelligence. Large language models and AI coding assistants have permanently transformed how we write code, brainstorm models, and explore data. Throughout this book, and particularly in Chapter 18, we treat AI not as a replacement for critical thought, but as a tireless research collaborator. You will learn to use AI assistants to draft and debug Python code, translate complex errors, and assist with qualitative coding, while maintaining the rigorous verification standards required for academic defense.

How to Journey Through This Book

Rather than presenting mathematical abstractions in isolation, every statistical and machine learning method in this volume is anchored in a single, realistic running case study following 600 graduate students over four semesters. Every chapter answers a real research question.

For each concept, you will follow a disciplined, five-step roadmap: 1. **A Concrete Research Question** 2. **A Minimal Example** to build immediate intuition 3. **The Case-Study Analysis** with real data 4. **In Your Field** applications (health, agriculture, business, and social sciences) 5. **Thesis Reporting & Interpretation** showing exactly how to write up your findings

The Interactive Playground

To ensure that no software installation stands between you and your first analysis, all companion exercises are hosted in an interactive browser environment. You can test code, experiment with parameters, and verify your answers directly on the book's companion website:

Web Playground: https://polla-fattah.github.io/data2thesis_python/playground/

Whether you are embarking on your Master's dissertation, finishing your doctoral dissertation, or publishing your first peer-reviewed paper, may this book serve as your steady desk companion from your first command to your final defense.

Dr. Polla Fattah
Salahaddin University-Erbil

I Foundations of Python Programming and Data Handling

1 Getting Started with Python 23
1.1 Research written in code 23
1.2 Installing Python and VS Code 24
1.3 The editor layout 25
1.4 Calculations 25
1.5 Storing values in variables 25
1.6 Types of data 26
1.7 Functions and methods 27
1.8 Writing scripts 28
1.9 Managing research projects 29
1.10 Installing and importing packages 29
1.11 A first look at Liar's data 29
1.12 Chapter review 32
1.13 Exercises 33
1.14 Further reading 33

2 Data Structures in Python 35
2.1 Data as measurement 35
2.2 Variables and their values 36
2.3 Categories (Categorical Data) 38
2.4 A table of cases: The DataFrame 39
2.5 Matrices and Dictionaries 41
2.6 Importing research data 42
2.7 Automated codebook generation 44
2.8 Exporting data 45
2.9 Mapping SPSS and Excel workflows to Python 45
2.10 Reading error messages and tracebacks 46
2.11 Descriptive summary of cohort measurements 48
2.12 Chapter review 49
2.13 Exercises 50
2.14 Further reading 50

3 Data Manipulation and Cleaning with Pandas 51
3.1 What clean data means 51
3.2 Tidy data principles in Python 52
3.3 The Pandas manipulation toolkit 53
3.4 Reshaping data between wide and long 57
3.5 Joining relational tables 58
3.6 Cleaning an uncurated survey export 59
3.7 Chapter review 66
3.8 Exercises 66
3.9 Further reading 67

4 Data Visualization with Seaborn and Matplotlib 69
4.1 What a graph is for 69
4.2 Graphical perception: How vision decodes data 71
4.3 Selecting the right visualization 72
4.4 Core statistical visualizations with Seaborn . . . 73
4.5 Preparing publication-quality figures 78
4.6 Honest graphs and visual ethics 79
4.7 Advanced scientific visualizations 82
4.8 Exporting publication-ready figures 84
4.9 Chapter review 84
4.10 Exercises 85
4.11 Further reading 85

1. Getting Started with Python

Every data analysis is a long chain of decisions. Which cases are kept and which are excluded, how a messy answer is corrected, which variables are combined into a score, which test is used, and how its result is rounded: each choice shapes the final numbers, and each should be open to inspection. When an analysis is done by clicking through menus, most of these decisions leave no trace. Weeks later, not even the researcher can say exactly how a number in the thesis was produced, and an examiner who asks cannot be given a complete answer.

Writing the analysis as code changes this. A script records every step in order, in a form that a supervisor can read, an examiner can check, and the researcher can run again after correcting a mistake or receiving new data. This is the main reason research is increasingly done in a programming language, and it is the reason this book teaches Python. The chapter starts from nothing: installing Python, finding your way around the program used to write it, and learning the handful of ideas on which everything else rests.

The book follows one research project from its first day to its last. Elaf has just begun a Master's degree in Educational Psychology. During her first year of graduate study she noticed how tired many of her fellow students were: they slept little, lived on coffee, worried about their supervisors and their money, and a few spoke quietly about leaving. With her supervisor, she turned that observation into a thesis. It asks how sleep, stress, and supervisor support relate to graduate students' wellbeing, their grades, and their thoughts of dropping out, and whether a short wellbeing workshop can help. Her university agreed to support a two-year study of 600 graduate students from five faculties: a questionnaire at the start, a record of each student's sleep, study, and wellbeing at the end of every semester, and a six-week workshop to which half of the students would be invited at random.

At their first meeting, her supervisor set out what the thesis would demand. The research questions had to become precise hypotheses before any results were seen. The raw survey export, full of test entries, typing errors, and impossible values, had to be cleaned without losing track of a single change. The sample had to be described honestly, including the students who stopped taking part. Every analysis had to be chosen for a reason and its assumptions checked, and every result reported with its size and its uncertainty, not only with a p-value. At the end, examiners would read the results chapter and could ask how any number in it had been produced. "You should be able to show them," the supervisor said, "step by step."

Elaf had used Excel for coursework and had once clicked through an SPSS menu in a statistics class, but she had never written a line of code. Her supervisor suggested Python, for exactly the reason given above: an analysis written as code can be shown step by step. The first data has now arrived, and this chapter is Elaf's first day with Python, and yours. Each later chapter takes her one step further, from cleaning the data to the finished thesis. By the end of this one, you will have Python running on your computer, you will know your way around the program used to write it, and you will have opened her data and answered a first, simple question about it.

By the end of this chapter you will be able to

- Explain why researchers use Python, and install Python and VS Code.
- Find your way around your editor and run code in an interactive terminal and in a script.
- Store values in variables and recognise the basic types of data.
- Use functions and methods, and get help when you are stuck.
- Organise your research project in folders and install packages with `pip`.
- Open the student wellbeing data and answer a first question about it using `data2thesis`.

Note

Teaching & Practice: Interactive Lecture Slides · Browser Playground Exercises

1.1 - Research written in code

Python is a free, open-source programming language for working with data: cleaning it, analysing it, and turning it into tables, graphs, and statistical models. It was created in the late 1980s by Guido van Rossum in the Netherlands, released in 1991, and named after the British comedy troupe Monty Python. Today, Python is the most popular

programming language in the world, powering scientific research in universities, national laboratories, hospitals, tech companies, and governments across every continent.

Four properties make Python worth learning for a researcher:

1. **It is built for clear reading:** Python was designed from the ground up to look like clear, readable English. Its syntax is clean and uncluttered, making it accessible to beginners who have never programmed before.
2. **It is free and open source:** You, your students, and anyone reviewing your thesis can install and run your code without buying expensive proprietary licences.
3. **It makes research reproducible:** Your written script records every cleaning choice, transformation, and statistical test. Anyone can re-run your script on the raw data and reproduce your exact numbers, tables, and figures. Chapter 17 explores this idea in depth.
4. **An unmatched scientific ecosystem:** Thousands of scientists have contributed free, battle-tested libraries: **pandas** for spreadsheets and data tables, **matplotlib** and **seaborn** for publication-quality charts, **scipy** and **statsmodels** for hypothesis testing and regression, and **scikit-learn** for machine learning. Whatever statistical method your thesis needs, a Python library already exists for it.

The price of code is that Python asks you to type rather than click. At first, typing feels unfamiliar. Within a few days, however, it becomes second nature—and it pays you back every time you need to re-run an analysis, update a dataset, or generate new figures for your thesis defense.

> **i** For users of SPSS or Excel
>
> In SPSS or Excel, you click menus, and the program changes your spreadsheet or displays a table in a viewer. The problem is that the clicks leave no written record. In Python, you write an explicit instruction: `students['age'].mean()`. The computer executes it and prints the result. Because the instruction is saved in a text file, you can re-run the entire analysis next semester with one keystroke. Chapter 2 shows how familiar Excel and SPSS operations map directly to Python.

1.2 - Installing Python and VS Code

Working with Python involves two tools:

1. **Python** is the engine that executes your commands.
2. An **editor** (or Integrated Development Environment, IDE) is the program in which you drive it: you write Python code there, run it, and inspect your tables and charts.

You will almost never open Python by itself; you open your editor, and the editor talks to Python for you.

The most popular, modern editor for scientific Python is **Visual Studio Code (VS Code)**, developed by Microsoft. It is free, fast, and available for Windows, macOS, and Linux.

1. **Install Python:** Download Python (version 3.10 or newer) from python.org. On Windows, ensure you check the box that says **"Add Python to PATH"** during the installer setup.
2. **Install VS Code:** Download and install VS Code from code.visualstudio.com.
3. **Install the Python Extension:** Open VS Code, click the Extensions icon on the left sidebar (four squares), search for **Python** (by Microsoft), and click **Install**.

Appendix A walks through this setup step by step for Windows, macOS, and Linux, with troubleshooting for common path issues.

> **i** Other editors: Positron and Jupyter
>
> While VS Code is the industry standard, other editors are also excellent. **Positron**, developed by Posit (the creators of RStudio), is designed specifically for data scientists and provides a dedicated environment for both Python and R in a familiar layout. **JupyterLab** is another popular browser-based environment for running code notebooks. Everything you learn in this book works identically in VS Code, Positron, and Jupyter.

> **Working directly in the browser**
>
> The practical exercises for this book can also be run directly in your web browser with zero local installation. Try them in the Chapter 1 Playground first!

1.3 - The editor layout

When you open VS Code with a Python file, the window is organized into functional areas, described in Table 1.1.

Table 1.1: The primary functional areas when working with Python in an editor

Where	Pane	What it is for
Left / Top	**Editor (Source)**	Where you write and save your code scripts (`.py` files).
Bottom / Right	**Terminal / Interactive Window**	Where Python executes instructions and shows text outputs.
Left sidebar	**File Explorer**	The folders and files in your research project.
Bottom / Panel	**Plots & Variables**	Where data tables, variables in memory, and visual plots appear.

You can type code directly into the Python Interactive Terminal (or REPL). When you type `2 + 2` and press Enter, Python evaluates the expression and prints `4`.

1.4 - Calculations

The simplest use of Python is as a calculator. Suppose you slept 6.5, 7, and 5.5 hours on three consecutive nights last week. Your average sleep is:

```
(6.5 + 7 + 5.5) / 3
```

```
6.333333333333333
```

Python follows standard mathematical order of operations (PEMDAS): parentheses first, followed by exponents, multiplication and division, and finally addition and subtraction. Without the parentheses, Python would divide only the last number by 3:

```
6.5 + 7 + 5.5 / 3
```

```
15.333333333333334
```

The standard arithmetic operators in Python are: * `+` for addition * `-` for subtraction * `*` for multiplication * `/` for division * `**` for exponents/powers (for example, `2 ** 3` evaluates to 8; note that in Python, `^` is a bitwise operator, not power!)

```
2 ** 3
```

```
8
```

1.5 - Storing values in variables

Typing numbers repeatedly is slow and error-prone. Instead, you can store a value under a memorable name. In Python, this stored value is held in a **variable**, created using the **assignment operator** `=`:

```
nights = 3
```

When you run this line, Python produces no visible output, but it now remembers that `nights` holds the value `3`. You can now use `nights` anywhere you would use the number:

```
nights
```

```
3
```

```
nights * 7
```

```
21
```

To store several values together, you can place them in a **list**, written inside square brackets `[` and `]`, separated by commas:

```
sleep = [6.5, 7.0, 5.5]
sleep
```

```
[6.5, 7.0, 5.5]
```

Notice how much clearer and self-documenting your analysis becomes when the numbers have meaningful names:

```
sum(sleep) / nights
```

```
6.333333333333333
```

Variable names in Python follow clear conventions: * Names start with a letter or underscore, and may contain letters, numbers, and underscores (e.g., `sleep_hours`, `student_01`). * Names cannot contain spaces. Use an underscore `_` to separate words (this is called **snake_case**, the standard Python style). * Python is strictly case-sensitive: `Sleep`, `sleep`, and `SLEEP` are three completely different variables. * Choose descriptive names: `sleep_hours` tells you immediately what the data represents; single letters like `x` or `temp` will leave you guessing when you review your code months later.

If you assign a new value to an existing variable name, Python overwrites the old value:

```
nights = 4
nights
```

```
4
```

1.6 - Types of data

Research data is not only numbers. A study records measurements (hours of sleep), text categories (a student's faculty), and yes-or-no status flags (whether a student attended a workshop). Python keeps track of the **data type** of every value, because the type determines what mathematical and logical operations are permitted.

Table 1.2 lists the core types you will encounter in research.

Table 1.2: The fundamental data types in Python

Type	Name in Python	Description	Example
Integer	`int`	Whole numbers without decimals	`600`, `4`
Floating-point	`float`	Numbers with decimal points	`6.5`, `3.14159`
String	`str`	Text characters, enclosed in quotes	`"Education"`, `'Male'`
Boolean	`bool`	Logical truth values	`True`, `False`

The built-in function `type()` inspects the type of any object:

```
hours = 6.5
faculty = "Education"
```

```
invited = True

print(type(hours))
print(type(faculty))
print(type(invited))
```

```
<class 'float'>
<class 'str'>
<class 'bool'>
```

Text must always be enclosed in quotes (single `'...'` or double `"..."`). If you omit the quotes, Python assumes you are referring to a variable name:

```
faculty = Education
```

```
NameError: name 'Education' is not defined
[31m---------------------------------------------------------------------------[39m
[31mNameError[39m                                 Traceback (most recent call last)
[36mCell[39m[36m [39m[32mIn[12][39m[32m, line 1[39m
[32m----> [39m[32m1[39m faculty = [43mEducation[49m

[31mNameError[39m: name 'Education' is not defined
```

This error—`NameError: name 'Education' is not defined`—is one of the most frequent beginner mistakes. The error message is helpful and specific: Python searched your computer's memory for a variable named `Education` and found none.

1.6.1 - Missing values

Real empirical data almost always contains missing answers. A participant might decline to answer their income, or an instrument might briefly lose power.

In pure Python, the absence of a value is represented by the special object `None`. When working with tabular data and numerical calculations in **pandas** and scientific libraries, missing numerical values are represented as `NaN` (short for *Not a Number*, floating-point missing value).

In Python and pandas, mathematical calculations on missing values are handled safely:

```
import numpy as np

sleep_with_gap = [6.5, np.nan, 5.5]
sleep_with_gap
```

```
[6.5, nan, 5.5]
```

In pandas, functions like `.mean()` and `.sum()` are designed specifically for research data and automatically handle missing values cleanly, as you will see shortly.

1.7 - Functions and methods

You have already used several built-in **functions**: `sum()`, `len()`, `type()`, and `print()`. A function takes one or more inputs, performs a procedure, and returns a result. You call a function by writing its name followed by parentheses `()`, enclosing the inputs inside:

```
print(sum(sleep))
print(max(sleep))
print(len(sleep))
```

```
19.0
7.0
3
```

Inputs passed to a function are called **arguments**. If a function accepts multiple arguments, they are separated by commas. For example, `round()` takes the number you want to round, and the number of decimal digits:

```
round(6.333333, 1)
```

```
6.3
```

You can also pass arguments by their name (called keyword arguments):

```
round(6.333333, ndigits=2)
```

```
6.33
```

In Python, objects also have **methods**. A method is a function that belongs directly to an object, called with a dot `.`. For example, strings have methods to change their casing:

```
faculty = "Education"
print(faculty.lower())
print(faculty.upper())
```

```
education
EDUCATION
```

1.7.1 - Getting help

Every Python function and library includes built-in documentation. In an interactive console or notebook, you can read the help docstring for any function by calling `help()`:

```
help(round)
```

In Jupyter or VS Code Interactive, you can also place a question mark `?` after any function name (e.g., `round?`) to pop open its documentation pane, including its expected parameters and usage examples.

1.8 - Writing scripts

Commands typed in an interactive prompt are temporary. For real research work that needs to be defended, documented, and published, you write your code in a **script**: a plain text file ending in `.py`.

1. In your editor, create a new file named `analysis.py`.
2. Type your code, with one instruction per line.
3. Add **comments** using the hash symbol `#`. Python completely ignores everything after `#` on that line; comments are written for human readers (your future self, your supervisor, and your peer reviewers) to explain *why* a particular decision was made:

```
# Sleep recorded across three nights (hours per night)
sleep_hours = [6.5, 7.0, 5.5]

# Calculate the mean sleep duration
average_sleep = sum(sleep_hours) / len(sleep_hours)

# Display result rounded to one decimal place for thesis table
print(f"Average sleep: {round(average_sleep, 1)} hours")
```

```
Average sleep: 6.3 hours
```

4. To run the script, press **Ctrl+Enter** (or **Cmd+Enter** on macOS) to run the selected lines in the interactive window, or run the entire script in the terminal with `python analysis.py`.

1.9 - Managing research projects

A thesis involves dozens of files: survey exports, cleaned datasets, analysis scripts, generated plots, and chapter drafts.

The gold standard for scientific computing is to create a dedicated **project directory** on your computer (for example, `graduate-wellbeing-thesis/`). Keep all code and data self-contained within this folder:

```
graduate-wellbeing-thesis/
├── data/              <- Raw survey data (read-only)
├── scripts/           <- Your Python analysis scripts (.py)
├── figures/           <- Exported charts and graphs
└── thesis_chapter.qmd <- Your written manuscript
```

By organizing your work with relative paths (e.g. `data/students.csv`), your analysis is fully portable: if you zip the folder and email it to your supervisor, or push it to GitHub, the code runs on their computer without any file path errors.

1.10 - Installing and importing packages

Python includes a rich standard library, but its true power comes from third-party **packages** (also called libraries). Using a package is a two-step process:

1. **Install the package once** onto your computer using Python's package manager, `pip`, in your system terminal:

   ```
   pip install pandas seaborn data2thesis
   ```

2. **Import the package** in each script where you wish to use it:

   ```
   import pandas as pd
   import data2thesis as dt
   ```

A helpful analogy is a workshop: running `pip install` purchases a specialized power tool and places it in your workshop cabinet. You only need to buy it once. Writing `import` takes the tool off the shelf and places it on your workbench for today's session.

1.11 - A first look at Elaf's data

With Python and the companion package installed, we can load the Graduate Wellbeing Study:

```
import data2thesis as dt

students = dt.load_students()
```

! About Elaf's data

Elaf, her study, and her data are simulated for teaching. The data was generated by a computer algorithm to look like an authentic survey of graduate students, ensuring every statistical and machine learning method in this book has realistic patterns to discover. It describes no real individuals, and its patterns (for example, that students who sleep more tend to achieve higher GPAs, or that the wellbeing workshop reduced stress) were built in by the author. Nothing in this book should be cited as empirical evidence about the wellbeing of real graduate students.

The `students` object is a **pandas DataFrame**: a two-dimensional table of rows and columns, similar to an Excel sheet or SPSS dataset, but optimized for fast computation.

To see the dimensions of the table, check its `.shape` attribute:

```
students.shape
```

```
(600, 14)
```

The output `(600, 14)` tells us immediately that Elaf's study contains **600 students** (rows) and **14 variables** (columns).

To inspect the first five rows of the dataset, call the `.head()` method:

```
students.head()
```

student_id	supervisor_id	age	gender	faculty	programme	study_mode	employment	has_children	lives_away	financial_worry	workshop	workshop_sessions	considering_dropout
0	S0001	SUP088	31.0	Female	Health Sciences	Master's	Full-time	None	No	Yes	2.0	Not invited	0	No
1	S0002	SUP116	28.0	Female	Education	PhD	Full-time	Part-time job	No	Yes	2.0	Invited	0	No
2	S0003	SUP040	33.0	Male	Education	PhD	Full-time	None	No	No	4.0	Invited	5	No
3	S0004	SUP112	34.0	Female	Health Sciences	Master's	Part-time	Full-time job	No	No	2.0	Invited	6	No
4	S0005	SUP118	35.0	Female	Social Sciences	Master's	Part-time	None	No	No	2.0	Invited	2	No

To see the names of all 14 variables:

```
list(students.columns)
```

```
['student_id',
 'supervisor_id',
 'age',
 'gender',
 'faculty',
 'programme',
 'study_mode',
 'employment',
 'has_children',
 'lives_away',
 'financial_worry',
 'workshop',
 'workshop_sessions',
 'considering_dropout']
```

Each column is described in the study codebook (`dt.load_students?`).

To select a single variable from the DataFrame, write the DataFrame name followed by the column name in brackets and quotes: `students['age']`.

A natural first question in Elaf's thesis: *What is the average age of graduate students entering the study?*

In pandas, we call the `.mean()` method directly on the column:

```
students['age'].mean()
```

```
np.float64(29.724540901502504)
```

Notice something important: in R, the default `mean()` function returns `NA` if even a single value is missing, forcing you to specify `na.rm = TRUE`. In pandas, `.mean()` is designed from the ground up for real-world empirical research: it calculates the average across all known observations while safely ignoring missing values by default (`skipna=True`).

The average age of the students in the study is **29.7 years**.

We can also check whether any student had a missing age using `.isna().sum()`:

```
students['age'].isna().sum()
```

```
np.int64(1)
```

Only 1 student's age was missing from the initial intake survey. Chapter 3 shows how to identify why it was missing and how to clean it.

Next, we can generate frequency tables for categorical variables using the `.value_counts()` method. For example, how many students participated from each academic faculty?

```
students['faculty'].value_counts()
```

```
faculty
Health Sciences     154
Education           148
Social Sciences     116
Humanities           95
Natural Sciences     87
Name: count, dtype: int64
```

And how many were invited to the wellbeing workshop?

```
students['workshop'].value_counts()
```

```
workshop
Not invited    300
Invited        300
Name: count, dtype: int64
```

Exactly 300 students—precisely 50% of the sample—were invited to the workshop. That is not an accident: the workshop invitations were assigned using random allocation. Chapter 5 explores why random assignment is the cornerstone of causal inference in empirical research.

In your field: open scientific datasets in Python

Python's visualization library, **seaborn**, comes with real-world practice datasets from different fields. For instance, the `penguins` dataset records morphological measurements of wild penguin colonies in Antarctica:

```
import seaborn as sns

penguins = sns.load_dataset("penguins")
print(penguins.shape)
penguins.head(3)
```

```
(344, 7)
```

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex
0	Adelie	Torgersen	39.1	18.7	181.0	3750.0	Male
1	Adelie	Torgersen	39.5	17.4	186.0	3800.0	Female
2	Adelie	Torgersen	40.3	18.0	195.0	3250.0	Female

We can inspect species frequencies and calculate the average body mass across the colony using the exact same pandas methods:

```
print(penguins['species'].value_counts())
print(f"Average body mass: {penguins['body_mass_g'].mean():.1f} grams")
```

```
species
Adelie       152
Gentoo       124
Chinstrap     68
Name: count, dtype: int64
Average body mass: 4201.8 grams
```

1.12 - Chapter review

1.12.1 - Summary

- **Reproducibility:** Writing data analysis as a script creates a transparent, step-by-step record of every research choice. Anyone can inspect, verify, and reproduce your results.
- **Tools:** Python is the computing language; an editor like VS Code or Positron provides the interface. Scripts (`.py`) contain instructions, and the interactive terminal displays output.
- **Variables & Calculations:** Values are stored in variables using `=`. Lists `[...]` hold multiple values. Standard math symbols follow PEMDAS, with `**` used for exponents.
- **Data Types:** Fundamental types include `int` (integers), `float` (decimals), `str` (text strings in quotes), and `bool` (`True` or `False`). Check types using `type()`.
- **Functions & Methods:** Functions are called as `function(argument)`. Object methods are called with a dot: `object.method()`.
- **Packages:** Installed once from your terminal via `pip install <package>`, and loaded into each script session using `import <package>`.
- **Tabular Data:** The companion package `data2thesis` provides pandas DataFrames. Inspect table size with `.shape`, view preview rows with `.head()`, compute summary statistics with `.mean()`, count missing values with `.isna().sum()`, and count categories with `.value_counts()`.

1.12.2 - Key terms

Python, VS Code, script (`.py`), comment (`#`), variable, assignment (`=`), list, data type (`int`, `float`, `str`, `bool`), missing value (`None`, `NaN`), function, argument, method, package, `pip`, `import`, pandas DataFrame, `.shape`, `.head()`, `.isna()`, `.value_counts()`, reproducible research.

1.13 - Exercises

Work through these short exercises to test your understanding. The Chapter 1 Playground provides interactive browser versions of these exercises with hints and complete solutions.

1. A supervisor sleeps 8, 7.5, 6, and 7 hours on four consecutive nights. Store these four numbers in a list named `supervisor_sleep`, calculate the mean sleep duration, and round the result to one decimal place.
2. Check the data types of `"600"` and `600` using Python's `type()` function. Explain why adding `1` to one succeeds while adding `1` to the other raises a `TypeError`.
3. Explain the difference between Python's built-in `None` and pandas/numpy's `NaN`. How does pandas `.mean()` handle missing values by default?
4. Using `students = dt.load_students()`, find the age of the youngest student and the oldest student in the study using `.min()` and `.max()`.
5. Using `.value_counts()`, find how many students are enrolled full-time versus part-time (the variable is `study_mode`).
6. Reflect on an analysis you have previously performed in Excel, SPSS, or another point-and-click tool. List three specific analytical decisions that a thesis examiner could not verify without an explicit code script.

1.14 - Further reading

- McKinney, W. (2022). *Python for Data Analysis* (3rd ed.). O'Reilly Media. (Written by the creator of pandas).
- VanderPlas, J. (2016). *Python Data Science Handbook*. O'Reilly Media.
- The official VS Code Python Tutorial.

2. Data Structures in Python

Data is the recorded outcome of scientific measurement. Each value captures one observation about one entity: a student's age, the faculty they belong to, the hours they slept during a semester, or their response to a Likert questionnaire item. Before conducting any statistical analysis, a researcher must clarify what each value signifies, what level of measurement generated it, and how the data is organized in computer memory. These foundations determine which summary statistics, visualizations, and inferential models are scientifically valid.

Research data rarely arrives in a single pristine file. In Elaf's wellbeing study, the online survey software generated an Excel export (`wellbeing_raw.xlsx`), the university registrar exported semester records as CSV files (`semesters.csv`), and a collaborator from a previous pilot study stored historical records in SPSS (`wellbeing.sav`). Before analyzing these heterogeneous sources, we need to understand how Python organizes data in memory. Python provides specialized structures tailored for data science: **lists** and **NumPy arrays** for basic sequences, **Pandas Series** for single measurement variables, **Categorical Series** for discrete groups, and **DataFrames** for rectangular tables of cases. This chapter introduces these structures, demonstrates how to import files from every common format, and shows how to generate an automated codebook linking every column to its underlying survey question.

By the end of this chapter you will be able to

- Conceptualize data as a rectangular matrix of cases and variables, and identify the unit of analysis.
- Create Pandas Series, perform vectorized arithmetic, and filter observations with boolean indexing.
- Manage categorical and ordinal measurements using Pandas `Categorical` types.
- Construct, inspect, and slice two-dimensional DataFrames using `.loc` and `.iloc`.
- Understand dictionaries and 2D matrices when encountered in statistical output.
- Import real-world data from CSV, Excel (`.xlsx`), and SPSS (`.sav`) files, preserving survey question labels.
- Build an automated codebook that documents variable definitions and measurement scales.
- Map familiar SPSS and Excel procedures to Python idioms, and debug common Python error tracebacks.

Note

Teaching & Practice: Interactive Lecture Slides · Browser Playground Exercises

2.1 - Data as measurement

Almost all empirical research data can be conceptualized as a rectangular grid: each row represents a **case** (or observation), and each column represents a **variable**. A case is the individual unit under observation—a student, a patient, an experimental plot, or an institution. A variable is a measured property recorded across all cases, such as chronological age, anxiety score, or study hours. Every cell in the grid contains a single value representing the measurement of that variable for that specific case. This structure, known as the **data matrix**, forms the backbone of statistical modeling.

What constitutes a case depends on the research design, defining the **unit of analysis**. Elaf's wellbeing study involves multiple units of analysis across different tables. In the `students` table, each row represents an individual student ($N = 600$). In the `semesters` table, however, each row represents a single student in a specific academic semester, meaning each student appears up to four times:

```
semesters.head(5)
```

	student_id	semester	gpa	sleep_hours	study_hours	exercise_days	caffeine_mg	supervisor_meetings	wellbeing
0	S0001	1	3.02	7.2	26.0	5.0	130.0	0	74
1	S0001	2	3.27	8.4	17.0	3.0	80.0	1	69

	student_id	semester	gpa	sleep_hours	study_hours	exercise_days	caffeine_mg	supervisor_meetings	wellbeing
2	S0001	3	3.17	7.8	25.0	7.0	105.0	3	71
3	S0001	4	3.00	7.2	28.0	4.0	145.0	2	74
4	S0002	1	3.25	6.4	32.0	2.0	210.0	3	60

The first four rows all correspond to student `S0001`, documenting their trajectory across semesters 1 through 4. Confusing the unit of analysis is one of the most frequent errors in quantitative research. A demographic question concerning baseline enrollment must count each student exactly once. Conversely, an inquiry into GPA trajectories across time requires repeated-measures observations, coupled with statistical techniques that account for within-subject dependency (Chapter 10.). In Chapter 5., we explore how research hypotheses dictate the appropriate unit of analysis.

2.2 - Variables and their values

In Python's data science ecosystem, a single variable's values are held in a **Pandas Series** (or a NumPy 1D array). A Series is an indexed, one-dimensional array capable of holding numbers, text, or boolean flags:

```
sleep = pd.Series([6.5, 7.0, 5.5, 8.0, 6.0])
sleep
```

```
0    6.5
1    7.0
2    5.5
3    8.0
4    6.0
dtype: float64
```

A Series enforces a single homogeneous data type (`dtype`). If you inadvertently mix numbers with text, Pandas converts the entire Series to an `object` (or string) dtype to accommodate the text:

```
mixed = pd.Series([6.5, "seven", 5.5])
mixed
```

```
0      6.5
1    seven
2      5.5
dtype: object
```

Notice that the numbers `6.5` and `5.5` have become text strings. They can no longer be averaged or plotted numerically. This behavior is vital to understand: if a single survey respondent types `"seven"` or `"N/A"` into a numeric field in a web form, the entire column imports into Python as text. We will encounter this exact issue when inspecting the raw Excel export later in this chapter.

2.2.1 - Operations on a whole Series (Vectorization)

A defining strength of Pandas is **vectorization**: mathematical and logical operations apply across every value in the collection simultaneously, without needing explicit `for` loops. Converting sleep durations from hours to minutes requires a single multiplication:

```
sleep * 60
```

```
0    390.0
1    420.0
2    330.0
3    480.0
4    360.0
dtype: float64
```

Logical comparisons work identically, generating a boolean Series containing `True` or `False` for each entry. For example, identifying nights falling short of the clinical recommendation of 7 hours:

```
sleep < 7
```

```
0     True
1    False
2     True
3    False
4     True
dtype: bool
```

Because Python evaluates `True` as 1 and `False` as 0 in numerical contexts, calling `.sum()` counts the qualifying nights, and `.mean()` calculates their proportion:

```
print("Count of short nights:", (sleep < 7).sum())
print("Proportion of short nights:", (sleep < 7).mean())
```

```
Count of short nights: 3
Proportion of short nights: 0.6
```

Three of the five nights (60%) were short. This vectorized code operates at identical speed whether analyzing 5 observations or 500,000 observations.

2.2.2 - Selecting values

Python uses **0-based indexing**: the first element resides at position `0`, the second at position `1`, and so on. Think of an index as the offset or distance from the start of the array.

To select values by positional integer index in a Pandas Series, use `.iloc[]`:

```
print("First night:", sleep.iloc[0])
print("First and third nights:\n", sleep.iloc[[0, 2]])
```

```
First night: 6.5
First and third nights:
 0    6.5
2    5.5
dtype: float64
```

Values can also be extracted using a boolean condition inside square brackets. Pandas filters the Series, retaining only observations where the condition evaluates to `True`:

```
sleep[sleep < 7]
```

```
0    6.5
2    5.5
4    6.0
dtype: float64
```

This syntax, termed **boolean indexing** (or logical filtering), reads naturally: *"give me the elements of sleep where sleep is less than 7."* It is one of the most widely used idioms in quantitative Python.

2.2.3 - Changing values

Placing an indexed selection on the left side of the assignment operator updates the underlying data. If a participant later clarifies that their second night was 7.5 hours rather than 7.0:

```
sleep.iloc[1] = 7.5
sleep
```

```
0    6.5
1    7.5
2    5.5
3    8.0
4    6.0
dtype: float64
```

2.3 - Categories (Categorical Data)

Research datasets frequently contain qualitative categories: faculty division, experimental arm, gender identity, or Likert response levels. Pandas represents categorical variables using the specialized `category` dtype (similar to factors in R or value labels in SPSS). A categorical Series displays text labels but stores them internally as compact integers mapped to a defined list of **categories**:

```
faculty = pd.Series(
    ["Education", "Humanities", "Education", "Health Sciences"],
    dtype="category"
)
print(faculty)
print("Categories:", faculty.cat.categories)
```

```
0          Education
1         Humanities
2          Education
3    Health Sciences
dtype: category
Categories (3, object): ['Education', 'Health Sciences', 'Humanities']
Categories: Index(['Education', 'Health Sciences', 'Humanities'], dtype='object')
```

By default, categories are arranged in alphanumeric order. However, when categories possess an intrinsic hierarchy (such as employment intensity or Likert scales), you can define an **ordered category** using `pd.Categorical`:

```
employment = pd.Series(
    pd.Categorical(
        ["None", "Full-time job", "Part-time job", "None"],
        categories=["None", "Part-time job", "Full-time job"],
        ordered=True
    )
)
employment.value_counts(sort=False)
```

```
None            2
Part-time job    1
Full-time job    1
Name: count, dtype: int64
```

Specifying explicit category orders ensures that subsequent summary tables and plots display categories logically from lowest to highest rather than alphabetically.

2.3.1 - Levels of measurement and Python types

The distinction between continuous numbers and categories is not merely aesthetic; it reflects the underlying mathematical properties of the measurement. Categories without an order (such as faculty or academic discipline) are **nominal**. Categories with a strict ranking but unequal or unquantified intervals (such as Likert scales or employment tiers) are **ordinal**. Measurements with uniform, quantifiable units (such as sleep hours, age, or GPA) are **interval** or **ratio** (numeric). Table 2.1 outlines how each measurement level corresponds to a Python data type.

Table 2.1: Levels of measurement and their representation in Python

Measurement Level	Example Variables	Python Data Type
Nominal (unordered categories)	Faculty, Gender, Department	`category` (or `object` / `string`)
Ordinal (ordered categories)	Agreement scale, Education level	`category` (`ordered=True`)
Interval / Ratio (continuous quantities)	Sleep hours, Age, GPA, Scale sum	`float64`, `int64`
Binary / Dichotomous	Invited to workshop (Yes/No)	`bool` or two-level `category`

2.4 - A table of cases: The DataFrame

Most empirical data is organized as a two-dimensional table where each row represents a case and each column represents a variable. In Python, this table is a Pandas **DataFrame**. Each column is a Series sharing a common index. While each column must have a uniform data type, different columns across the DataFrame can store different types.

You can construct a DataFrame manually from a Python dictionary:

```
pilot = pd.DataFrame({
    "student": ["S1", "S2", "S3", "S4"],
    "faculty": pd.Categorical(["Education", "Humanities", "Education", "Health Sciences"]),
    "sleep": [6.5, 7.5, 5.5, 8.0],
    "invited": [True, False, True, False]
})
pilot
```

	student	faculty	sleep	invited
0	S1	Education	6.5	True
1	S2	Humanities	7.5	False
2	S3	Education	5.5	True
3	S4	Health Sciences	8.0	False

The most comprehensive way to inspect a DataFrame's architecture is the `.info()` method, which summarizes column names, non-null counts, and data types:

```
pilot.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 4 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   student  4 non-null      object
 1   faculty  4 non-null      category
 2   sleep    4 non-null      float64
 3   invited  4 non-null      bool
dtypes: bool(1), category(1), float64(1), object(1)
memory usage: 336.0+ bytes
```

2.4.1 - Selecting columns and rows

To extract a single column as a Series, use square bracket notation with the column name:

```
pilot["sleep"]
```

```
0    6.5
1    7.5
2    5.5
3    8.0
Name: sleep, dtype: float64
```

To slice rows and columns simultaneously, Pandas provides two primary indexers: 1. `.iloc[row_pos, col_pos]`: selects by **integer position** (0-based offset). 2. `.loc[row_label, col_label]`: selects by **label name** or **boolean mask**.

```
# Select the second row across all columns
pilot.iloc[1, :]

# Select all rows for two specific columns
pilot.loc[:, ["student", "sleep"]]

# Filter rows where sleep is under 7 hours
pilot.loc[pilot["sleep"] < 7, :]
```

	student	faculty	sleep	invited
0	S1	Education	6.5	True
2	S3	Education	5.5	True

The final expression combines a DataFrame with boolean indexing: *"retrieve all columns for rows where sleep is less than 7."* This is Python's direct equivalent of SPSS's *Select Cases* or Excel's *Filter*.

2.4.2 - Adding new columns

Assigning an expression to a new column name appends it to the DataFrame:

```
pilot["short_sleep"] = pilot["sleep"] < 7
pilot
```

	student	faculty	sleep	invited	short_sleep
0	S1	Education	6.5	True	True
1	S2	Humanities	7.5	False	False
2	S3	Education	5.5	True	True
3	S4	Health Sciences	8.0	False	False

2.4.3 - The student wellbeing cohort

The `students` dataset bundled in `data2thesis` follows this exact DataFrame architecture across 600 participants:

```
students.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 600 entries, 0 to 599
Data columns (total 14 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   student_id           600 non-null    object
 1   supervisor_id        600 non-null    object
 2   age                  599 non-null    float64
 3   gender               600 non-null    object
 4   faculty              600 non-null    object
 5   programme            600 non-null    object
```

```
 6   study_mode           600 non-null    object
 7   employment           293 non-null    object
 8   has_children         600 non-null    object
 9   lives_away           600 non-null    object
 10  financial_worry      562 non-null    float64
 11  workshop             600 non-null    object
 12  workshop_sessions    600 non-null    int64
 13  considering_dropout  600 non-null    object
dtypes: float64(2), int64(1), object(11)
memory usage: 65.8+ KB
```

In this output, `object` indicates text strings, `int64` denotes integers, and `float64` denotes decimal numbers. Columns representing categories, such as `faculty`, can be converted to categorical types whenever discrete factor levels are needed:

```
students["faculty"] = students["faculty"].astype("category")
students["faculty"].cat.categories
```

```
Index(['Education', 'Health Sciences', 'Humanities', 'Natural Sciences',
       'Social Sciences'],
      dtype='object')
```

Subsetting with multiple criteria uses Python's bitwise logical operators: `&` (AND) and `|` (OR). In Pandas, each individual condition must be enclosed in parentheses `()` because bitwise operators take precedence over comparison operators (==, >, <):

```
hs_phd = students[
    (students["faculty"] == "Health Sciences") & (students["programme"] == "PhD")
]
print("Health Sciences PhD candidates:", len(hs_phd))
```

```
Health Sciences PhD candidates: 44
```

> Inspecting Data Visually in VS Code
>
> In VS Code, opening any DataFrame in the **Interactive Window** or **Data Viewer** provides an interactive spreadsheet grid where you can sort, filter, and inspect columns visually without altering the underlying code.

2.5 - Matrices and Dictionaries

Beyond Series and DataFrames, two additional data structures frequently appear in quantitative workflows: 2D numerical matrices and dictionaries.

2.5.1 - Matrices and correlation grids

A numerical matrix is a two-dimensional grid of homogeneous numbers. Matrices arise naturally as the mathematical output of statistical procedures, such as calculating bivariate correlation coefficients across multiple metrics:

```
first_semester = semesters[semesters["semester"] == 1]
metric_vars = first_semester[["gpa", "sleep_hours", "study_hours", "wellbeing"]]

# Compute bivariate correlation matrix
correlations = metric_vars.corr()
round(correlations, 2)
```

	gpa	sleep_hours	study_hours	wellbeing
gpa	1.00	0.25	0.06	0.32

	gpa	sleep_hours	study_hours	wellbeing
sleep_hours	0.25	1.00	−0.56	0.46
study_hours	0.06	−0.56	1.00	−0.24
wellbeing	0.32	0.46	−0.24	1.00

Each variable correlates perfectly with itself ($r = 1.00$) along the main diagonal. Specific pairwise correlations can be extracted using `.loc[row, col]`:

```
correlations.loc["sleep_hours", "wellbeing"]
```

```
np.float64(0.45630448626446257)
```

2.5.2 - Dictionaries and statistical results

A **dictionary** (`dict`) is Python's fundamental key-value container. It stores named attributes enclosed in curly braces `{}`:

```
student = {
    "id": "S0001",
    "programme": "Master's",
    "sleep": [7.2, 8.4, 7.8, 7.2]
}
student["sleep"]
```

```
[7.2, 8.4, 7.8, 7.2]
```

While researchers rarely build complex dictionaries by hand, many Python scientific packages return statistical test results as structured objects or dictionary-like containers. For instance, running a one-sample t-test with `scipy.stats` to test whether first-semester students sleep 7 hours on average:

```
import scipy.stats as stats

result = stats.ttest_1samp(first_semester["sleep_hours"], popmean=7.0, nan_policy="omit")
print("t-statistic:", round(result.statistic, 2))
print("p-value:", result.pvalue)
```

```
t-statistic: -12.18
p-value: 1.3317009730817428e-30
```

The resulting object encapsulates the computed test statistic, degrees of freedom, and p-value as named attributes ready for automated reporting in your thesis.

2.6 - Importing research data

Different software packages store tabular data in distinct binary or text formats. Python provides dedicated reader functions for every standard format, each parsing the input directly into a clean Pandas DataFrame. The sample files below are installed with `data2thesis` and located via `dt.example_file()`:

2.6.1 - CSV files

A CSV file (*comma-separated values*) is a plain-text table where commas delimit adjacent columns. Because CSV files are universally supported and human-readable, they represent the gold standard for long-term open science archiving. In Python, CSV files are imported using `pd.read_csv()`:

```
path = dt.example_file("semesters.csv")
semester_data = pd.read_csv(path)
semester_data.head(3)
```

	student_id	semester	gpa	sleep_hours	study_hours	exercise_days	caffeine_mg	supervisor_meetings	wellbeing
0	S0001	1	3.02	7.2	26.0	5.0	130.0	0	74
1	S0001	2	3.27	8.4	17.0	3.0	80.0	1	69
2	S0001	3	3.17	7.8	25.0	7.0	105.0	3	71

In your own thesis project, you will typically point directly to your project's data directory: `pd.read_csv("data/semesters.csv")`.

2.6.2 - Excel spreadsheets

Real-world survey platforms frequently export multi-column workbooks (`.xlsx`). Pandas reads modern Excel files via `pd.read_excel()` (powered by the `openpyxl` engine):

```
raw_path = dt.example_file("wellbeing_raw.xlsx")
raw_data = pd.read_excel(raw_path)
print("Dimensions (rows, columns):", raw_data.shape)
```

```
Dimensions (rows, columns): (615, 64)
```

This file represents an unfiltered export from the wellbeing survey. The `.shape` property indicates `r raw_data.shape[0]` rows and `r raw_data.shape[1]` columns. Let's inspect the first few entries across the initial six columns:

```
raw_data.iloc[:5, :6]
```

	Response ID	Q1_Student ID	Q2_Age	Q3_Gender	Q4_Faculty	Q5_Programme
0	R0001	TEST	99.0	Female	NaN	NaN
1	R0002	test	NaN	NaN	NaN	NaN
2	R0003	TEST2	30.0	Male	NaN	NaN
3	R0004	S0191	27.0	male	Humanities	Masters
4	R0005	S0015	30.0	Male	Health Sciences	Master's

Notice the irregularities typical of real-world research data: test records (`TEST`), unstandardized text variations (`Male`, `male`), and arbitrary missing value codes (`99` for missing age). In Chapter 3., we walk through systematic cleaning pipelines to sanitize this raw export.

If an Excel workbook contains multiple worksheets, you can target a specific sheet using the `sheet_name` argument (e.g., `pd.read_excel("study.xlsx", sheet_name="Semester 2")`).

2.6.3 - SPSS data files

Quantitative social scientists frequently receive archives from colleagues formatted as SPSS files (`.sav`). The `pyreadstat` package enables Python to read `.sav` files while preserving variable metadata and response value labels:

```
import pyreadstat

spss_path = dt.example_file("wellbeing.sav")
spss_df, meta = pyreadstat.read_sav(spss_path)

spss_df.loc[:2, ["student_id", "gender", "faculty", "stress_1"]]
```

	student_id	gender	faculty	stress_1
0	S0001	1.0	2.0	3.0
1	S0002	1.0	1.0	4.0

	student_id	gender	faculty	stress_1
2	S0003	2.0	1.0	4.0

SPSS files store underlying survey questions as variable labels. `pyreadstat` extracts these labels into the `meta` object:

```
print("Question text for stress_1:", meta.column_names_to_labels["stress_1"])
```

```
Question text for stress_1: I feel unable to control important things in my studies.
```

Alternatively, if you wish to import an SPSS file directly with numeric category codes decoded into human-readable text categories, Pandas provides `pd.read_spss()`:

```
spss_labeled = pd.read_spss(spss_path)
spss_labeled["faculty"].value_counts()
```

```
faculty
Health Sciences     154
Education           148
Social Sciences     116
Humanities           95
Natural Sciences     87
Name: count, dtype: int64
```

This seamless interoperability ensures that years of meticulous labeling in SPSS are instantly accessible inside Python.

2.7 - Automated codebook generation

In an empirical dataset, a cryptic column header like `stress_1` or a numeric value like `4` is scientifically uninterpretable on its own. It acquires meaning only when accompanied by documentation specifying the original prompt ("I feel unable to control important things in my studies"), the response options (1 = Strongly Disagree to 5 = Strongly Agree), and missing data conventions. A **codebook** provides this reference, serving as the bridge between participant questionnaires and statistical analysis.

Because `pyreadstat` parses embedded SPSS metadata, we can programmatically assemble a comprehensive codebook DataFrame:

```
codebook = pd.DataFrame({
    "variable": meta.column_names,
    "label": meta.column_labels,
    "dtype": [str(spss_df[col].dtype) for col in meta.column_names]
})
codebook.head(10)
```

	variable	label	dtype
0	student_id	Student identifier	object
1	supervisor_id	Supervisor identifier	object
2	age	Age in years at baseline	float64
3	gender	Gender	float64
4	faculty	Faculty	float64
5	programme	Degree programme	float64
6	study_mode	Study mode	float64
7	employment	Paid work alongside study	float64
8	has_children	Has children	float64

	variable	label	dtype
9	lives_away	Moved away from family to study	float64

For categorical items, the mapping between underlying integer codes and verbal labels is preserved in `meta.variable_value_labels`:

```
print("Value labels for faculty:", meta.variable_value_labels.get("faculty"))
```

```
Value labels for faculty: {1.0: 'Education', 2.0: 'Health Sciences', 3.0: 'Humanities', 4.0:
'Natural Sciences', 5.0: 'Social Sciences'}
```

Exporting this table into an appendix provides an automated, error-free documentation trail for your thesis committee and future secondary analysts.

2.8 - Exporting data

Once raw data has been cleaned, validated, or transformed, the resulting DataFrame can be exported into standard file formats:

```
# Export to CSV (suppressing automatic integer index column)
students.to_csv("students_clean.csv", index=False)

# Export to Excel workbook
students.to_excel("students_clean.xlsx", index=False)

# Export to SPSS system file with pyreadstat
import pyreadstat
pyreadstat.write_sav(students, "students_clean.sav")
```

Setting `index=False` is essential when saving CSV and Excel files; otherwise, Pandas inserts an unlabelled leading column of row numbers into the file.

> [!IMPORTANT] **Preserve Raw Data Immutability:** Never overwrite primary data files. Save processed datasets under descriptive filenames (e.g., `students_clean.csv`), ensuring that your Python script can recreate the clean data from raw inputs at any time.

2.9 - Mapping SPSS and Excel workflows to Python

Moving from point-and-click tools to scripted programming involves mapping familiar conceptual operations to Python commands (Table 2.2).

Table 2.2: Conceptual mapping between point-and-click software and Python

Analytical Task	SPSS / Excel Procedure	Python / Pandas Idiom
Open data file	File > Open > Data	`pd.read_csv()`, `pd.read_excel()`, `pd.read_spss()`
Variable View (types & labels)	Variable View tab	`df.info()`, `df.dtypes`, `meta.column_names_to_labels`
Inspect data matrix	Data View / Spreadsheet grid	`df.head()`, VS Code Interactive Data Viewer
Select cases (filtering)	Data > Select Cases	`df[condition]` or `df.loc[condition, :]`
Compute variable	Transform > Compute Variable	`df["new_col"] = df["a"] + df["b"]`
Frequency distribution	Analyze > Descriptive > Frequencies	`df["var"].value_counts()`
Descriptive statistics	Analyze > Descriptive > Descriptives	`df["var"].describe()`, `df["var"].mean()`
Export data	File > Save As	`df.to_csv(..., index=False)`, `df.to_excel(...)`

The paramount advantage of Python is transparency. In Excel or SPSS Data View, manual edits and filter clicks are transient and undocumented. In Python, the input dataset remains untouched while your script provides an indelible, auditable record of every transformation.

2.10 - Reading error messages and tracebacks

Python tracebacks are diagnostic reports explaining precisely where and why execution halted. When an error occurs, scroll to the final line of the traceback to read the specific exception type and description. Four exceptions account for the vast majority of beginner stumbling blocks:

1. `NameError: name 'slep' is not defined` A variable name is misspelled, or the code defining it was not executed prior to this line:

```
print(slep)
```

```
NameError: name 'slep' is not defined
□[31m---------------------------------------------------------------------------□[39m
□[31mNameError□[39m                                 Traceback (most recent call last)
□[36mCell□[39m□[36m □[39m□[32mIn[33]□[39m□[32m, line 1□[39m
□[32m----> □[39m□[32m1□[39m □[38;5;28mprint□[39m(□[43mslep□[49m)

□[31mNameError□[39m: name 'slep' is not defined
```

2. `AttributeError` or `KeyError` Referencing a non-existent column name inside brackets raises a `KeyError` (often caused by capitalisation mismatches):

```
students["Faculty"]
```

```
KeyError: 'Faculty'
□[31m---------------------------------------------------------------------------□[39m
□[31mKeyError□[39m                                  Traceback (most recent call last)
□[36mFile                                 □[39m□[32m~\Drives\Locals\pytorch\venv\Lib\site-
packages\pandas\core\indexes\base.py:3812□[39m, in □[36mIndex.get_loc□[39m□[34m(self, key)□
[39m
□[32m   3811□[39m □[38;5;28;01mtry□[39;00m:
```

```
□[32m-> □[39m□[32m3812□[39m     □[38;5;28;01mreturn□[39;00m □[38;5;28;43mself□[39;49m□[43m.□
[49m□[43m_engine□[49m□[43m.□[49m□[43mget_loc□[49m□[43m(□[49m□[43mcasted_key□[49m□[43m)□[49m
□[32m             3813□[39m    □[38;5;28;01mexcept□[39;00m    □[38;5;167;01mKeyError□[39;00m
□[38;5;28;01mas□[39;00m err:

□[36mFile                   □[39m□[32mpandas/_libs/index.pyx:167□[39m,                   in
□[36mpandas._libs.index.IndexEngine.get_loc□[39m□[34m()□[39m

□[36mFile                   □[39m□[32mpandas/_libs/index.pyx:196□[39m,                   in
□[36mpandas._libs.index.IndexEngine.get_loc□[39m□[34m()□[39m

□[36mFile          □[39m□[32mpandas/_libs/hashtable_class_helper.pxi:7088□[39m,          in
□[36mpandas._libs.hashtable.PyObjectHashTable.get_item□[39m□[34m()□[39m

□[36mFile          □[39m□[32mpandas/_libs/hashtable_class_helper.pxi:7096□[39m,          in
□[36mpandas._libs.hashtable.PyObjectHashTable.get_item□[39m□[34m()□[39m

□[31mKeyError□[39m: 'Faculty'

The above exception was the direct cause of the following exception:

□[31mKeyError□[39m                                  Traceback (most recent call last)
□[36mCell□[39m□[36m □[39m□[32mIn[34]□[39m□[32m, line 1□[39m
□[32m----> □[39m□[32m1□[39m □[43mstudents□[49m□[43m[□[49m□[33;43m"□[39;49m□[33;43mFaculty□[39;49m□[33;43m"
[49m

□[36mFile                                  □[39m□[32m~\Drives\Locals\pytorch\venv\Lib\site-
packages\pandas\core\frame.py:4113□[39m, in □[36mDataFrame.__getitem__□[39m□[34m(self, key)□
[39m
□[32m   4111□[39m □[38;5;28;01mif□[39;00m □[38;5;28mself□[39m.columns.nlevels > □[32m1□[39m:
□[32m   4112□[39m     □[38;5;28;01mreturn□[39;00m □[38;5;28mself□[39m._getitem_multilevel(key)
□[32m-> 4113□[39m     indexer = □[38;5;28;43mself□[39;49m□[43m.□
[49m□[43mcolumns□[49m□[43m.□[49m□[43mget_loc□[49m□[43m(□[49m□[43mkey□[49m□[43m)□[49m
□[32m   4114□[39m □[38;5;28;01mif□[39;00m is_integer(indexer):
□[32m   4115□[39m     indexer = [indexer]

□[36mFile                                  □[39m□[32m~\Drives\Locals\pytorch\venv\Lib\site-
packages\pandas\core\indexes\base.py:3819□[39m, in □[36mIndex.get_loc□[39m□[34m(self, key)□
[39m
□[32m   3814□[39m         □[38;5;28;01mif□[39;00m  □[38;5;28misinstance□[39m(casted_key,
□[38;5;28mslice□[39m) □[38;5;129;01mor□[39;00m (
□[32m   3815□[39m             □[38;5;28misinstance□[39m(casted_key, abc.Iterable)
□[32m
3816□[39m         □[38;5;129;01mand□[39;00m □[38;5;28many□[39m(□[38;5;28misinstance□[39m(x,
□[38;5;28mslice□[39m) □[38;5;28;01mfor□[39;00m x □[38;5;129;01min□[39;00m casted_key)
□[32m   3817□[39m     ):
□[32m   3818□[39m         □[38;5;28;01mraise□[39;00m InvalidIndexError(key)
□[32m-> 3819□[39m     □[38;5;28;01mraise□[39;00m □[38;5;167;01mKeyError□[39;00m(key)
□[38;5;28;01mfrom□[39;00m□[38;5;250m □[39m□[34;01merr□[39;00m
□[32m   3820□[39m □[38;5;28;01mexcept□[39;00m □[38;5;167;01mTypeError□[39;00m:
□[32m   3821□[39m     □[38;5;66;03m# If we have a listlike key, _check_indexing_error
will raise□[39;00m
□[32m   3822□[39m     □[38;5;66;03m#   InvalidIndexError. Otherwise we fall through and
re-raise□[39;00m
□[32m   3823□[39m     □[38;5;66;03m#  the TypeError.□[39;00m
□[32m   3824□[39m     □[38;5;28mself□[39m._check_indexing_error(key)

□[31mKeyError□[39m: 'Faculty'
```

3. `TypeError: unsupported operand type(s)` Attempting mathematical operations on incompatible data types (for example, multiplying text strings):

```
"6.5" * "2"
```

```
TypeError: can't multiply sequence by non-int of type 'str'
□[31m---------------------------------------------------------------------------□[39m
□[31mTypeError□[39m                                 Traceback (most recent call last)
□[36mCell□[39m□[36m □[39m□[32mIn[35]□[39m□[32m, line 1□[39m
□[32m----> □[39m□[32m1□[39m □[33;43m"□[39;49m□[33;43m6.5□[39;49m□[33;43m"□[39;49m□[43m
□[49m□[43m*□[49m□[43m □[49m□[33;43m"□[39;49m□[33;43m2□[39;49m□[33;43m"□[39;49m

□[31mTypeError□[39m: can't multiply sequence by non-int of type 'str'
```

4. `FileNotFoundError: No such file or directory` Python searched the current working directory but could not locate the target filename. Verify your relative path against the folder opened in VS Code.

2.11 - Descriptive summary of cohort measurements

With the data structures in place, Pandas provides `.describe()` for a rapid multi-variable statistical overview of central tendency, dispersion, and range:

```
first_semester[
    ["gpa", "sleep_hours", "study_hours", "wellbeing"]
].describe().round(2)
```

	gpa	sleep_hours	study_hours	wellbeing
count	600.00	594.00	592.00	600.00
mean	3.10	6.48	27.64	60.46
std	0.33	1.03	13.06	12.01
min	2.17	3.50	2.00	24.00
25%	2.89	5.80	18.00	52.00
50%	3.10	6.50	26.00	60.00
75%	3.34	7.20	37.00	69.00
max	4.00	10.00	67.00	93.00

This single command confirms that in semester 1, students slept an average of 6.48 hours (standard deviation = 0.94), well below recommended benchmarks. In Chapter 6., we expand these basic summaries into publication-ready descriptive tables.

In your field: health research

The `penguins` dataset bundled with `seaborn` provides a classic biological sample across 344 observations and 7 variables:

```
import seaborn as sns

penguins = sns.load_dataset("penguins")
penguins.info()
penguins[["bill_length_mm", "bill_depth_mm", "body_mass_g"]].describe().round(1)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 344 entries, 0 to 343
Data columns (total 7 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   species            344 non-null    object
 1   island             344 non-null    object
 2   bill_length_mm     342 non-null    float64
 3   bill_depth_mm      342 non-null    float64
 4   flipper_length_mm  342 non-null    float64
 5   body_mass_g        342 non-null    float64
 6   sex                333 non-null    object
dtypes: float64(4), object(3)
memory usage: 18.9+ KB
```

	bill_length_mm	bill_depth_mm	body_mass_g
count	342.0	342.0	342.0
mean	43.9	17.2	4201.8
std	5.5	2.0	802.0
min	32.1	13.1	2700.0
25%	39.2	15.6	3550.0
50%	44.4	17.3	4050.0
75%	48.5	18.7	4750.0
max	59.6	21.5	6300.0

Notice the structural equivalence to human health studies: each row is a measured organism, categorical variables represent species and islands, and numeric metrics capture physiological dimensions.

2.12 - Chapter review

2.12.1 - Summary

- Quantitative data is organized as a rectangular matrix where rows represent **cases** and columns represent **variables**. The **unit of analysis** defines what a single row represents (`students` vs repeated `semesters`).
- A **Pandas Series** holds values of a single variable. Operations are vectorized, executing across all elements simultaneously without explicit loops.
- Python uses **0-based indexing**. Positional extraction relies on `.iloc[]`, while conditional filtering relies on boolean masks `series[condition]`.
- Categories are managed via the Pandas `category` dtype, which supports explicit hierarchical ordering for ordinal scales.
- A **DataFrame** is a two-dimensional tabular grid composed of individual Series. Subsetting rows and columns simultaneously is performed with `.loc[rows, cols]`.

- Data is imported using `pd.read_csv()`, `pd.read_excel()`, and `pyreadstat.read_sav()`. SPSS survey question labels and value encodings are preserved in metadata.
- An automated **codebook** extracts column names, survey prompts, and data types to document study variables systematically.

2.12.2 - Key terms

Data matrix, unit of analysis, Pandas Series, vectorization, boolean indexing, 0-based indexing, `.iloc[]`, `.loc[]`, categorical dtype, ordinal scale, DataFrame, correlation matrix, dictionary, CSV, SPSS metadata, codebook, traceback.

2.13 - Exercises

Work through these exercises in your browser using the Chapter 2 Playground or download the starter project for VS Code.

1. Create a Series with participant ages: `24, 31, 28, 45, 26`. Use boolean indexing to filter ages over 30, and compute how many participants exceed 30.
2. Construct an ordered Categorical Series from `["Agree", "Disagree", "Neutral", "Agree"]` with categories ordered logically from `"Disagree"` to `"Neutral"` to `"Agree"`. Verify the ordering using `.value_counts(sort=False)`.
3. In `students`, use boolean indexing with `&` to count how many participants study part-time (`study_mode == "Part-time"`) AND have dependent children (`children == "Yes"`).
4. Add a boolean column `over_30` to `students` that evaluates to `True` for students older than 30. Check how many missing values exist using `.isna().sum()`.
5. Import `wellbeing.sav` with `pyreadstat` and extract the full survey question text mapped to the variable `support_3`.
6. Contrast the unit of analysis in `students` versus `counselling_visits`. Formulate one distinct research question appropriate for each dataset.

2.14 - Further reading

- *Python for Data Analysis* (3rd ed.) by Wes McKinney [1]: the authoritative guide to Pandas indexing, Series arithmetic, and data wrangling by the creator of Pandas.
- Official documentation for pandas.pydata.org and pyreadstat.

3. Data Manipulation and Cleaning with Pandas

Raw data is never immediately suitable for scientific analysis. Online survey platforms log incomplete test runs alongside genuine responses, participants occasionally click submit twice, demographic text fields arrive with divergent spellings ("female", "Female", "F"), unanswered items are encoded as `99`, and clerical typos turn an age of 25 into 250. None of these occurrences are rare, and none are harmless: a single stray `99` embedded in a 1-to-5 Likert scale can severely distort an empirical mean, and an undetected duplicate observation artificially inflates sample size while biasing standard errors. Preparing raw measurements for modeling—a discipline known as **data cleaning and wrangling**—routinely occupies more time than model estimation itself, and every analytical decision made during cleaning directly shapes the findings.

Data cleaning is an intrinsic phase of statistical analysis, not an inconvenient preliminary chore, and it demands rigorous methodological care. This chapter begins by defining what constitutes clean, trustworthy data and establishing core principles for reproducible data preparation. It then presents the foundational manipulation toolkit in Python: filtering observations, selecting columns, sorting, creating computed variables, aggregating by group, reshaping, and merging relational tables using Pandas. Finally, we apply these techniques step by step to sanitize the raw export of the wellbeing survey introduced in Chapter 2. (`wellbeing_raw.xlsx`), handling test records, duplicate submissions, inconsistent textual categories, decimal commas, missing codes, impossible outliers, and reverse-coded questionnaire scales.

By the end of this chapter you will be able to

- Define the five core qualities of clean data and maintain an immutable, auditable cleaning pipeline.
- Explain the principles of tidy data (variables as columns, observations as rows) and structure data for analysis.
- Chain Pandas transformations fluently using method chaining and lambda expressions.
- Filter, subset, sort, compute new variables, and aggregate summaries across groups using `.groupby()` and `.agg()`.
- Reshape tabular data between wide and long representations using `pd.melt()` and `.pivot()`.
- Merge relational tables across primary and foreign keys using `pd.merge()`, and diagnose attrition with set filtering.
- Clean an uncurated survey export: eliminate test and duplicate rows, standardize inconsistent categories, convert text to numbers, handle missing codes, and filter out-of-range values.
- Compute multi-item composite scale scores, including reverse-coded Likert items, and write a transparent data-cleaning section for your thesis methodology.

Note

Teaching & Practice: Interactive Lecture Slides · Browser Playground Exercises

3.1 - What clean data means

Data can be considered clean when it faithfully represents the empirical phenomena under investigation. This standard encompasses five distinct qualities:

1. **Validity:** Every recorded value conforms to the permissible domain of its variable (e.g., chronological age cannot be 250 years, and daily sleep cannot exceed 24 hours).
2. **Accuracy:** Values capture what respondents actually intended, ensuring system codes (such as `99` for skipped items) are not treated as genuine quantitative answers.
3. **Completeness:** Data captures all eligible participants, and omitted observations are explicitly marked as missing (`NaN`) rather than masquerading as numbers.

4. **Consistency:** Categorical responses follow a single standardized nomenclature (e.g., `"F"`, `"female"`, and `"Female "` are normalized into `"Female"`).
5. **Uniqueness:** Each participant appears exactly once per unit of analysis, with zero test records and zero duplicate submissions.

Violating these criteria distorts empirical conclusions. The impact of unhandled missing-data codes is immediately obvious. Suppose five students answer an item on a 1-to-5 scale, and one student leaves the question blank, which the web software logs as `99`:

```
answers = pd.Series([4, 3, 5, 99, 2])
print("Uncleaned mean:", answers.mean())

clean_answers = answers.replace(99, np.nan)
print("Cleaned mean:", clean_answers.mean())
```

```
Uncleaned mean: 22.6
Cleaned mean: 3.5
```

Leaving `99` in the calculation produces a mean of `22.6`, a mathematical impossibility on a 5-point scale. When `99` is properly converted to `NaN`, Pandas excludes the missing entry by default, yielding the true average of `3.5`. In large datasets, such distortions are often smaller and harder to detect visually, making automated cleaning pipelines essential. Duplications exert a similar distorting effect: duplicating a participant gives them double weight in regressions and artificially narrows confidence intervals.

Three fundamental rules govern scientific data cleaning:

- **Raw files remain immutable:** Never edit primary data files directly in Excel. Raw files must remain intact, exactly as received from instruments or survey platforms. All transformations are executed via code.
- **Every decision is documented:** Your Python script serves as an auditable record of all exclusions, recodings, imputations, and composite score derivations.
- **Inspect results at every step:** Validate table shapes, unique identifiers, and value distributions after each operation, because many data-cleaning bugs produce no syntax errors.

3.2 - Tidy data principles in Python

Tabular data should be organized to make analysis straightforward. The standard architecture in modern data science is **tidy data** [2]: 1. Each variable forms a column. 2. Each observation forms a row. 3. Each type of observational unit forms a table.

Consider Table 3.1, which records two semesters of wellbeing scores for three students side by side, as spreadsheet exports often do.

Table 3.1: Wellbeing in wide format: the variable ‘wellbeing’ is spread over multiple columns

student	wellbeing_s1	wellbeing_s2
S1	58	63
S2	71	70
S3	64	69

While compact for human reading, this layout spreads a single construct (`wellbeing`) across multiple columns, while embedding a second variable (`semester`) inside column headers. In contrast, Table 3.2 presents the identical measurements in tidy (long) format.

Table 3.2: The same data in tidy (long) format: one column per variable, one row per observation

student	semester	wellbeing
S1	1	58
S1	2	63
S2	1	71
S2	2	70
S3	1	64
S3	2	69

In tidy format, each measurement occupies its own row, and computing grouped summaries—such as mean wellbeing per semester—becomes straightforward. The `students` table in `data2thesis` is tidy at the participant level ($N = 600$), whereas `semesters` is tidy at the semester observation level ($N = 2,326$). Uncurated survey exports, by contrast, frequently pack longitudinal observations side by side. Much of data preparation consists of reshaping data into tidy tables.

3.3 - The Pandas manipulation toolkit

In Python, tabular manipulation centers on **Pandas**. To keep transformations readable and eliminate intermediate variables, we use **method chaining**: calling sequential methods on a DataFrame enclosed within parentheses `(...)`.

Let's illustrate on a small pilot cohort:

```
pilot = pd.DataFrame({
    "student": ["S1", "S2", "S3", "S4", "S5"],
    "faculty": ["Education", "Humanities", "Education", "Health Sciences", "Humanities"],
    "sleep":   [6.5, 7.5, 5.5, 8.0, 6.0],
    "stress":  [3.2, 2.1, 4.5, 1.8, 3.9]
})
pilot
```

	student	faculty	sleep	stress
0	S1	Education	6.5	3.2
1	S2	Humanities	7.5	2.1
2	S3	Education	5.5	4.5
3	S4	Health Sciences	8.0	1.8
4	S5	Humanities	6.0	3.9

3.3.1 - Filtering observations

Most analyses focus on a defined subset of cases: participants in a specific faculty, students observed in semester 1, or those who completed a treatment protocol. Observations are filtered using boolean indexing or the `.query()` method:

```
# Boolean indexing
pilot[pilot["sleep"] < 7]

# Multiple conditions combined with & (AND)
pilot[(pilot["sleep"] < 7) & (pilot["faculty"] == "Education")]
```

```
# Matching against a collection using .isin()
pilot[pilot["faculty"].isin(["Education", "Humanities"])]
```

	student	faculty	sleep	stress
0	S1	Education	6.5	3.2
1	S2	Humanities	7.5	2.1
2	S3	Education	5.5	4.5
4	S5	Humanities	6.0	3.9

Alternatively, `.query()` offers a concise syntax using expression strings:

```
pilot.query("sleep < 7 and faculty == 'Education'")
```

	student	faculty	sleep	stress
0	S1	Education	6.5	3.2
2	S3	Education	5.5	4.5

Every filtering step represents an explicit sampling decision that must be reported in your thesis methodology: an analysis restricted to full-time students answers a fundamentally different question than one encompassing the entire cohort.

3.3.2 - Selecting and dropping columns

Real-world datasets often contain dozens of administrative or technical columns irrelevant to a given hypothesis. In Pandas, pass a list of column names inside square brackets to select specific variables, or call `.drop()` to remove unneeded columns:

```
# Keep only student ID and sleep hours
pilot[["student", "sleep"]]

# Drop the faculty column
pilot.drop(columns=["faculty"])
```

	student	sleep	stress
0	S1	6.5	3.2
1	S2	7.5	2.1
2	S3	5.5	4.5
3	S4	8.0	1.8
4	S5	6.0	3.9

3.3.3 - Sorting cases

Sorting helps inspect distributions, identify extreme values, or prepare multi-wave data for lagged calculations. Use `.sort_values()`:

```
# Sort by stress in descending order
pilot.sort_values("stress", ascending=False)
```

	student	faculty	sleep	stress
2	S3	Education	5.5	4.5
4	S5	Humanities	6.0	3.9
0	S1	Education	6.5	3.2

	student	faculty	sleep	stress
1	S2	Humanities	7.5	2.1
3	S4	Health Sciences	8.0	1.8

3.3.4 - Creating and transforming variables

New variables are derived from existing ones: converting measurement units, calculating composite indices, or binning continuous metrics into clinical categories. New columns can be assigned directly via brackets or dynamically chained using `.assign()`:

```
pilot = pilot.assign(
    sleep_minutes = pilot["sleep"] * 60,
    short_sleep   = pilot["sleep"] < 7
)
pilot[["student", "sleep", "sleep_minutes", "short_sleep"]]
```

	student	sleep	sleep_minutes	short_sleep
0	S1	6.5	390.0	True
1	S2	7.5	450.0	False
2	S3	5.5	330.0	True
3	S4	8.0	480.0	False
4	S5	6.0	360.0	True

To categorize a continuous metric into ordinal groups, `np.select()` evaluates conditions in order, applying the corresponding label from a list of choices:

```
conditions = [
    pilot["stress"] >= 4.0,
    pilot["stress"] >= 2.5
]
choices = ["High", "Medium"]

pilot["stress_level"] = np.select(conditions, choices, default="Low")
pilot[["student", "stress", "stress_level"]]
```

	student	stress	stress_level
0	S1	3.2	Medium
1	S2	2.1	Low
2	S3	4.5	High
3	S4	1.8	Low
4	S5	3.9	Medium

[!TIP] **Preserve Continuous Measurements:** Discretizing continuous metrics into arbitrary bins (“High”, “Medium”, “Low”) discards statistical power and variance (reducing a nuanced scale to an artificial threshold). While useful for descriptive summaries, your primary inferential models should retain the raw continuous scores whenever possible.

3.3.5 - Summarizing by group

Scientific questions frequently compare groups: Do education students sleep longer than science students? Summaries reduce many individual measurements into descriptive aggregates. In Pandas, grouped aggregation is achieved with `.groupby()` paired with `.agg()`:

```
# Overall cohort summary
print("Mean sleep across all pilot cases:", pilot["sleep"].mean())

# Grouped summary by faculty
faculty_summary = (
    pilot
    .groupby("faculty")["sleep"]
    .agg(mean_sleep="mean", count="count")
    .reset_index()
)
faculty_summary
```

```
Mean sleep across all pilot cases: 6.7
```

	faculty	mean_sleep	count
0	Education	6.00	2
1	Health Sciences	8.00	1
2	Humanities	6.75	2

For quick frequency counts of discrete categories, use `.value_counts()`:

```
pilot["faculty"].value_counts().reset_index()
```

	faculty	count
0	Education	2
1	Humanities	2
2	Health Sciences	1

3.3.6 - Combining operations into an analytical pipeline

Because Pandas methods return modified DataFrames, we can construct fluent pipelines that answer substantive research questions. For example: How did average student wellbeing evolve across semesters 1 to 4, broken down by workshop participation?

To answer this, we join the longitudinal records (`semesters`) with baseline workshop assignments (`students`), group by semester and workshop status, calculate mean wellbeing and sample size, and sort the output:

```
wellbeing_trajectory = (
    semesters
    .merge(students[["student_id", "workshop"]], on="student_id", how="left")
    .groupby(["semester", "workshop"])["wellbeing"]
    .agg(mean_wellbeing="mean", n_students="count")
    .reset_index()
    .sort_values(["semester", "workshop"])
)
wellbeing_trajectory
```

	semester	workshop	mean_wellbeing	n_students
0	1	Invited	60.646667	300
1	1	Not invited	60.266667	300
2	2	Invited	65.470000	300
3	2	Not invited	60.220000	300
4	3	Invited	63.222615	283

	semester	workshop	mean_wellbeing	n_students
5	3	Not invited	59.228571	280
6	4	Invited	60.932862	283
7	4	Not invited	58.425000	280

Notice the pattern: in Semester 1 (prior to the intervention), wellbeing scores between invited and uninvited students were virtually identical (~63.5 vs ~63.7). From Semester 2 onward, invited students demonstrate higher wellbeing. In Chapter 7., we use inferential hypothesis tests to evaluate whether this observed difference exceeds chance expectation.

3.4 - Reshaping data between wide and long

Data is often collected in wide format but analyzed in long format. In **wide** format, repeated measurements appear side by side across columns. In **long** format, each measurement occupies its own row.

Consider the wide table from Table 3.1:

```
wide = pd.DataFrame({
    "student":     ["S1", "S2", "S3"],
    "wellbeing_1": [58, 71, 64],
    "wellbeing_2": [63, 70, 69]
})
wide
```

	student	wellbeing_1	wellbeing_2
0	S1	58	63
1	S2	71	70
2	S3	64	69

To convert wide columns into tidy rows, use `pd.melt()`:

```
long = pd.melt(
    wide,
    id_vars=["student"],
    value_vars=["wellbeing_1", "wellbeing_2"],
    var_name="semester",
    value_name="wellbeing"
)
# Clean the semester column to extract the integer number
long["semester"] = long["semester"].str.replace("wellbeing_", "").astype(int)
long = long.sort_values(["student", "semester"]).reset_index(drop=True)
long
```

	student	semester	wellbeing
0	S1	1	58
1	S1	2	63
2	S2	1	71
3	S2	2	70
4	S3	1	64
5	S3	2	69

The arguments specify: - `id_vars`: Columns to preserve as identifiers (`student`). - `value_vars`: Columns to unpivot into rows. - `var_name`: Name for the new column storing original column headers (`semester`). - `value_name`: Name for the new column storing the data values (`wellbeing`).

Conversely, when formatting summary tables for a thesis chapter, wide presentation is often clearer for human readers. Use `.pivot()` to spread long data across columns:

```
wide_presentation = long.pivot(index="student", columns="semester",
values="wellbeing").reset_index()
wide_presentation
```

semester	student	1	2
0	S1	58	63
1	S2	71	70
2	S3	64	69

We can apply this to our longitudinal trajectory summary, spreading the workshop groups into adjacent columns:

```
(wellbeing_trajectory
 .assign(mean_wellbeing=lambda df: df["mean_wellbeing"].round(1))
 .pivot(index="semester", columns="workshop", values="mean_wellbeing")
 .reset_index()
)
```

workshop	semester	Invited	Not invited
0	1	60.6	60.3
1	2	65.5	60.2
2	3	63.2	59.2
3	4	60.9	58.4

3.5 - Joining relational tables

In empirical research, data is frequently split across multiple tables sharing a common key: participant demographics in one file, longitudinal outcomes in another, and supervisor attributes in a third. In Pandas, relational tables are combined using `pd.merge()`.

Consider two simple tables:

```
people = pd.DataFrame({
    "student": ["S1", "S2", "S3"],
    "faculty": ["Education", "Humanities", "Education"]
})

scores = pd.DataFrame({
    "student":   ["S1", "S1", "S2", "S4"],
    "wellbeing": [58, 63, 71, 66]
})
```

A **left join** (`how="left"`) preserves all rows from the primary table, matching attributes from the secondary table where the keys align:

```
scores.merge(people, on="student", how="left")
```

	student	wellbeing	faculty
0	S1	58	Education
1	S1	63	Education
2	S2	71	Humanities
3	S4	66	NaN

Student `S4` has a score but no corresponding demographic record in `people`, so their faculty is assigned `NaN`.

To find cases that have *no* matching counterpart in another table (an **anti-join**), use boolean negation `~` with `.isin()`:

```
# Find students in 'people' who have no records in 'scores'
missing_scores = people[~people["student"].isin(scores["student"])]
missing_scores
```

	student	faculty
2	S3	Education

Student `S3` enrolled in the study but contributed zero scores.

In Elaf's research cohort, merging connects supervisory rank (from `supervisors`) to student dropout intentions (from `students`):

```
supervisors = dt.supervisors

by_rank = (
    students
    .merge(supervisors[["supervisor_id", "rank"]], on="supervisor_id", how="left")
    .assign(dropout_flag = lambda df: (df["considering_dropout"] == "Yes").astype(int))
    .groupby("rank")
    .agg(
        dropout_rate=("dropout_flag", "mean"),
        n_students=("student_id", "count")
    )
    .reset_index()
)
by_rank
```

	rank	dropout_rate	n_students
0	Assistant Professor	0.175299	251
1	Lecturer	0.136752	234
2	Professor	0.121739	115

Similarly, set filtering identifies students who dropped out of the longitudinal study by Semester 3:

```
sem3_students = semesters[semesters["semester"] == 3]["student_id"]
attrition_students = students[~students["student_id"].isin(sem3_students)]

print("Students who left before Semester 3:", len(attrition_students))
print("\nDropout thoughts among those who left:")
print(attrition_students["considering_dropout"].value_counts())
```

```
Students who left before Semester 3: 37

Dropout thoughts among those who left:
considering_dropout
Yes    22
No     15
Name: count, dtype: int64
```

Of the 37 students who departed prior to Semester 3, the vast majority had indicated on their baseline questionnaire that they were considering discontinuing their studies. Participant attrition is rarely random; tracking who dropped out is essential context for interpreting later semesters (Chapter 6.).

3.6 - Cleaning an uncurated survey export

We are now equipped to sanitize the raw wellbeing export (`wellbeing_raw.xlsx`), resolving the real-world data issues introduced in Chapter 2..

3.6.1 - Inspecting raw survey responses

Data cleaning begins with diagnostic inspection:

```
raw = pd.read_excel(dt.example_file("wellbeing_raw.xlsx"))
print("Raw table shape (rows, columns):", raw.shape)
```

```
Raw table shape (rows, columns): (615, 64)
```

The raw export contains `615` rows, even though Elaf's study enrolled exactly `600` students. Examining key demographic columns reveals multiple data anomalies:

```
print(raw["Q3_Gender"].value_counts(dropna=False).head(8))
print("\nSample test records:")
print(raw[raw["Q1_Student   ID"].str.upper().str.startswith("TEST",   na=False)][["Q1_Student
ID", "Q2_Age", "Q3_Gender"]].head(3))
```

```
Q3_Gender
Female     203
Male       152
male        78
female      56
M           46
Female      30
F           28
Male        21
Name: count, dtype: int64

Sample test records:
  Q1_Student ID  Q2_Age Q3_Gender
0          TEST    99.0    Female
1          test     NaN       NaN
2         TEST2    30.0      Male
```

The gender field contains divergent spellings (`"F"`, `"female"`, `"Female"`, `"Male"`, and trailing spaces), test entries (`"TEST"`, `"TEST2"`) are mixed in with student records, column names contain special characters and spaces, and numeric metrics contain text.

3.6.2 - Removing test records and duplicate submissions

Test submissions are not empirical observations, and duplicate survey submissions inflate sample size. Both must be excised first:

```
# Filter out test accounts
no_test = raw[~raw["Q1_Student ID"].str.upper().str.startswith("TEST", na=False)]

# Drop technical response ID and eliminate duplicate submissions
responses = no_test.drop(columns=["Response ID"]).drop_duplicates()
print("Cleaned participant count:", len(responses))
```

```
Cleaned participant count: 600
```

Filtering removes 3 test records and 12 duplicate submissions, leaving exactly `600` unique students.

3.6.3 - Renaming variables to standardized identifiers

Verbose survey headers like `Q10_How worried are you about money? (1-5)` are error-prone in scripts. We map them to concise, standardized snake_case identifiers:

```
#   Construct   questionnaire   item   names:   stress_1..6,   burnout_1..6,   support_1..6,
satisfaction_1..4
item_names = (
```

```
    [f"stress_{i}" for i in range(1, 7)] +
    [f"burnout_{i}" for i in range(1, 7)] +
    [f"support_{i}" for i in range(1, 7)] +
    [f"satisfaction_{i}" for i in range(1, 5)]
)

rename_dict = {
    "Q1_Student ID": "student_id",
    "Q2_Age": "age",
    "Q3_Gender": "gender",
    "Q4_Faculty": "faculty",
    "Q5_Programme": "programme",
    "Q6_Study mode": "study_mode",
    "Q7_Paid work": "employment",
    "Q8_Children": "has_children",
    "Q9_Moved away from family": "lives_away",
    "Q10_How worried are you about money? (1-5)": "financial_worry",
    "Workshop group": "workshop",
    "Workshop sessions attended": "workshop_sessions",
    "Y1_Considered leaving?": "considering_dropout"
}

# Add item column mappings Q11_1 through Q11_22
for i, name in enumerate(item_names, 1):
    rename_dict[f"Q11_{i}"] = name

responses = responses.rename(columns=rename_dict)
```

The original survey wording is not discarded; it is documented in the project codebook (Chapter 2.).

3.6.4 - Standardizing categorical variables

We normalize categories by stripping whitespace, converting to lowercase, and mapping variations to canonical factor levels:

```
# Standardize gender
responses["gender"] = np.where(
    responses["gender"].astype(str).str.strip().str.lower().str.startswith("f"),
    "Female",
    "Male"
)

# Standardize faculty using regex classification
fac_lower = responses["faculty"].astype(str).str.lower()
responses["faculty"] = np.select(
    [
        fac_lower.str.contains("educ"),
        fac_lower.str.contains("health"),
        fac_lower.str.contains("humanit"),
        fac_lower.str.contains("social"),
        fac_lower.str.contains("natural|^sciences?$")
    ],
    [
        "Education",
        "Health Sciences",
        "Humanities",
        "Social Sciences",
        "Natural Sciences"
    ],
    default="Other"
)
```

```
# Standardize degree programme and study mode
responses["programme"] = np.where(
    responses["programme"].astype(str).str.lower().str.contains("ph"),
    "PhD",
    "Master's"
)
responses["study_mode"] = np.where(
    responses["study_mode"].astype(str).str.lower().str.contains("part"),
    "Part-time",
    "Full-time"
)

# Standardize binary Yes/No indicators
for col in ["has_children", "lives_away", "considering_dropout"]:
    responses[col] = np.where(
        responses[col].astype(str).str.strip().str.lower().str.startswith("y"),
        "Yes",
        "No"
    )

print("Standardized gender counts:")
print(responses["gender"].value_counts())
print("\nStandardized faculty counts:")
print(responses["faculty"].value_counts())
```

```
Standardized gender counts:
gender
Female    312
Male      288
Name: count, dtype: int64

Standardized faculty counts:
faculty
Health Sciences     154
Education           148
Social Sciences     116
Humanities           95
Natural Sciences     87
Name: count, dtype: int64
```

3.6.5 - Recoding missing value codes and text numbers

Online forms frequently code skipped questions as `99` or `-9`. We replace these sentinel codes with `np.nan` before converting columns to numeric dtypes:

```
# Replace missing codes (99, -9) in survey items
for col in item_names + ["financial_worry"]:
    responses[col] = responses[col].replace([99, -9, "99", "-9"], np.nan)
    responses[col] = pd.to_numeric(responses[col], errors="coerce").astype("Int64")

responses["workshop_sessions"]        =        pd.to_numeric(responses["workshop_sessions"],
errors="coerce").astype("Int64")
```

[!WARNING] **Avoid Global Search-and-Replace:** Never replace `99` across an entire dataset indiscriminately. On a scale scored from 0 to 100 (such as Elaf's wellbeing index), `99` represents an authentic high score. Target missing-code replacements exclusively to columns where `99` was defined as an omission sentinel.

Text columns also contain embedded characters (e.g. respondents typing `"7 hrs"` or using decimal commas `"6,5"`). In Python, we clean decimal commas and extract digits:

```
sample_text = pd.Series(["7 hrs", "6.5", "6,5"])
cleaned_numbers      =      sample_text.str.replace(",",      ".").str.extract(r"([\d\.]+)",
expand=False).astype(float)
print(cleaned_numbers.tolist())
```

```
[7.0, 6.5, 6.5]
```

3.6.6 - Eliminating impossible values

Clerical errors can introduce biologically or chronologically impossible values (such as an age of 250). Because the true value cannot be reliably inferred, the scientifically honest solution is to set impossible values to `np.nan`:

```
# Parse numeric age and set impossible values (> 100) to NaN
responses["age"] = pd.to_numeric(
    responses["age"].astype(str).str.extract(r"([\d\.]+)", expand=False),
    errors="coerce"
)
responses.loc[responses["age"] > 100, "age"] = np.nan

print("Participants with age > 100:", (responses["age"] > 100).sum())
print("Missing age count:", responses["age"].isna().sum())
```

```
Participants with age > 100: 0
Missing age count: 1
```

This single clerical outlier accounts for the missing age observation noted in Chapter 1.. Plausible extreme values (such as a graduate student sleeping 4 hours per night) represent genuine variance and must be retained.

3.6.7 - Computing composite questionnaire scale scores

In psychometric research, psychological constructs are measured using multi-item scales whose items are averaged. Before averaging, all items must be aligned in a uniform direction.

Item `stress_4` ("I feel confident handling problems in my studies") is **reverse-worded**: higher agreement signifies *lower* stress. If averaged directly, it would cancel out the other five items. On a 1-to-5 Likert scale, reverse-coding is achieved by subtracting from 6 ($6 - x$ maps $1 \to 5$, $2 \to 4$, $3 \to 3$, $4 \to 2$, and $5 \to 1$):

```
questionnaire_clean = responses[["student_id"] + item_names].copy()

# Create scale scores
scores = pd.DataFrame({"student_id": questionnaire_clean["student_id"]})

# Reverse-code stress_4
reversed_stress_4 = 6 - questionnaire_clean["stress_4"]

# Compute row-wise means across scale items (skipping missing items)
scores["stress_score"] = pd.concat([
    questionnaire_clean[["stress_1", "stress_2", "stress_3", "stress_5", "stress_6"]],
    reversed_stress_4.rename("stress_4")
], axis=1).mean(axis=1)

scores["burnout_score"]    =    questionnaire_clean[[f"burnout_{i}"    for    i    in    range(1,
7)]].mean(axis=1)
scores["support_score"]    =    questionnaire_clean[[f"support_{i}"    for    i    in    range(1,
7)]].mean(axis=1)
scores["satisfaction_score"] = questionnaire_clean[[f"satisfaction_{i}" for i in range(1,
5)]].mean(axis=1)
```

```
scores.head()
```

	student_id	stress_score	burnout_score	support_score	satisfaction_score
3	S0191	2.833333	1.5	3.833333	3.333333
4	S0015	4.5	3.833333	1.666667	3.0
5	S0474	3.5	4.5	1.4	2.75
6	S0418	3.166667	3.0	1.666667	3.75
7	S0049	3.0	3.5	3.166667	3.333333

Individual item responses must always be preserved in the archive (`questionnaire_clean`) alongside composite scale scores (`scores`), enabling scale reliability and factor structure verification (Chapter 9.).

3.6.8 - Reshaping repeated semester measurements

The survey export stores 28 longitudinal columns side by side: `GPA_S1`, `Sleep_S1`, ..., `Wellbeing_S4`. To make the data tidy, we reshape these columns from wide to long using `pd.wide_to_long()`:

```
import re

sem_cols = [c for c in responses.columns if re.search(r"_S[1-4]$", c)]
sem_wide = responses[["student_id"] + sem_cols]

# Reshape wide columns into tidy rows
sem_long = pd.wide_to_long(
    sem_wide,
    stubnames=["GPA", "Sleep", "Study", "Exercise", "Caffeine", "Meetings", "Wellbeing"],
    i="student_id",
    j="semester",
    sep="_S"
).reset_index()

sem_long = sem_long.rename(columns={
    "GPA": "gpa",
    "Sleep": "sleep_hours",
    "Study": "study_hours",
    "Exercise": "exercise_days",
    "Caffeine": "caffeine_mg",
    "Meetings": "supervisor_meetings",
    "Wellbeing": "wellbeing"
})

# Standardize numeric dtypes and decimal commas
sem_long["sleep_hours"] = (
    sem_long["sleep_hours"]
    .astype(str)
    .str.replace(",", ".", regex=False)
    .str.extract(r"([\d\.]+)", expand=False)
    .astype(float)
)

for col in ["gpa", "study_hours", "exercise_days", "caffeine_mg", "supervisor_meetings",
"wellbeing"]:
    sem_long[col] = pd.to_numeric(
        sem_long[col].astype(str).str.extract(r"([\d\.]+)", expand=False),
        errors="coerce"
    )
```

```
# Filter out impossible measurement values
sem_long.loc[sem_long["sleep_hours"] > 24, "sleep_hours"] = np.nan
sem_long.loc[sem_long["study_hours"] > 168, "study_hours"] = np.nan
sem_long.loc[sem_long["gpa"] > 4.0, "gpa"] = np.nan

# Drop unobserved rows where all semester measurements are missing (attrition)
measure_cols = ["gpa", "sleep_hours", "study_hours", "exercise_days", "caffeine_mg",
"supervisor_meetings", "wellbeing"]
semesters_clean = sem_long.dropna(subset=measure_cols, how="all").copy()
semesters_clean =
semesters_clean.sort_values(["student_id", "semester"]).reset_index(drop=True)

print("Cleaned longitudinal semester observations:", len(semesters_clean))
```

```
Cleaned longitudinal semester observations: 2326
```

3.6.9 - Validating cleaned tables against canonical datasets

To ensure total accuracy, we compare our cleaned tables against the canonical datasets bundled in `data2thesis`:

```
# Verify questionnaire items match canonical data exactly
canonical_q = dt.questionnaire.sort_values("student_id").reset_index(drop=True)
q_test = questionnaire_clean.sort_values("student_id").reset_index(drop=True)

diffs = sum(
    not (q_test[c].fillna(-999).astype(int) == canonical_q[c].fillna(-999).astype(int)).all()
    for c in item_names
)
print("Questionnaire item discrepancies:", diffs)

# Verify semesters records match canonical data exactly
canonical_sem = dt.semesters.sort_values(["student_id", "semester"]).reset_index(drop=True)
max_sem_diff = max(
    (semesters_clean[c] - canonical_sem[c]).abs().max()
    for c in measure_cols
)
print("Maximum absolute difference across semester measurements:", max_sem_diff)
```

```
Questionnaire item discrepancies: 0
Maximum absolute difference across semester measurements: 0.0
```

Both tests confirm an exact numerical match: zero discrepancies.

> **Writing Up Data Preparation in Your Methodology**
>
> When drafting your thesis methods section, summarize cleaning procedures transparently:
>
> *"The raw institutional survey export contained 615 submissions. Preliminary auditing removed 3 administrative test records and 12 duplicate submissions, yielding an analytical sample of 600 unique graduate students. Categorical variables were standardized, and sentinel missing-value codes (99, –9) were converted to missing values. Out-of-range clerical errors (including an age entry of 250 years) were set to missing. The reverse-worded item `stress_4` was recoded prior to computing composite scale scores as the mean of completed items. Multi-wave records were unpivoted into longitudinal panel format ($N = 2,326$ student-semesters), excluding unobserved waves resulting from participant attrition."*

In your field: public health and environmental monitoring

The built-in `airquality` dataset illustrates missingness in environmental surveillance:

```
# Load airquality via data2thesis or build sample dataframe
air_sample = pd.DataFrame({
    "Ozone": [41.0, 36.0, 12.0, 18.0, np.nan, 28.0, 23.0, 19.0, 8.0, np.nan],
    "Month": [5, 5, 5, 5, 5, 5, 5, 5, 5, 5],
    "Temp": [67, 72, 74, 62, 56, 66, 65, 59, 61, 69]
})

air_summary = (
    air_sample
    .groupby("Month")
    .agg(
        mean_ozone=("Ozone", "mean"),
        missing_days=("Ozone", lambda s: s.isna().sum())
    )
    .reset_index()
)
air_summary
```

	Month	mean_ozone	missing_days
0	5	23.125	2

Checking missing data mechanisms ensures environmental policy is not biased by missing hot summer days.

3.7 - Chapter review

3.7.1 - Summary

- Clean data satisfies five properties: **validity**, **accuracy**, **completeness**, **consistency**, and **uniqueness**.
- Tidy data principles require that every variable forms a column, every observation forms a row, and every observational unit forms a distinct table.
- Fluent data manipulation in Pandas relies on **method chaining**: sequentially connecting `.query()`, `[]`, `.assign()`, `.sort_values()`, and `.groupby().agg()`.
- Reshaping is performed with `pd.melt()` (wide to long) and `.pivot()` (long to wide).
- Relational tables are merged across common keys via `pd.merge()`, while missing counterparts (attrition) are identified using set exclusion `~df["key"].isin(...)`.
- Psychometric scale derivation requires identifying reverse-worded items ($6 - x$ on a 1–5 scale) before aggregating item averages across rows.

3.7.2 - Key terms

Data cleaning, tidy data, method chaining, vectorization, boolean indexing, `.query()`, `.assign()`, `.groupby()`, `.agg()`, wide format, long format, `pd.melt()`, `.pivot()`, left join, anti-join, missing codes, reverse-coding, composite scale score.

3.8 - Exercises

Complete these exercises in your browser using the Chapter 3 Playground or download the starter project for VS Code.

1. Using `students`, calculate the average chronological age for each faculty, sorted from oldest to youngest (handling missing values with default `skipna=True`).
2. Add a categorical column `sleep_category` to `semesters` that assigns `"Short"` when `sleep_hours < 6.0`, `"Recommended"` when `sleep_hours >= 7.0`, and `"Moderate"` otherwise. Count records per category.
3. Using `semesters`, create a wide summary table displaying mean GPA with one row per student and one column per semester (Semesters 1 to 4) using `.pivot()`.

4. Merge `semesters` with `students` and compare average semester GPA between full-time and part-time students.
5. Explain why item `stress_4` must be reverse-coded before computing `stress_score`, and describe what would happen to empirical correlations if reverse-coding were omitted.
6. For each of the five qualities of clean data, describe a specific data-quality challenge from your own academic discipline and the corresponding Python cleaning step required.

3.9 - Further reading

- *Python for Data Analysis* (3rd ed.) by Wes McKinney [1]: chapters on "Data Cleaning and Preparation" and "Data Wrangling: Join, Combine, and Reshape".
- "Tidy Data" [2]: the seminal paper formalizing tabular data organization.
- Official Pandas guide to Reshaping and Pivot Tables.

4. Data Visualization with Seaborn and Matplotlib

A spreadsheet containing 600 rows of numbers conveys virtually nothing at a glance. A well-designed statistical visualization of that same data reveals in seconds where observations cluster, which values are anomalous, how experimental groups diverge, and how constructs covary. Graphics are not cosmetic embellishments appended to an empirical report at the conclusion; they are primary instruments of scientific discovery. The first task of an investigator encountering new data is to inspect it visually, and many data-entry anomalies and invalid assumptions are detected solely through graphical exploration.

A figure is also a scientific argument. Visual design choices—the selected geometry, axis scaling, aspect ratio, color palette, and data aggregation—determine what a reader notices first and what conclusions they infer. Thoughtful choices clarify subtle empirical patterns; careless choices distort differences, exaggerating trivial variations or masking substantial heterogeneity. This chapter begins with graphical perception theory: what scientific figures are for, how human vision decodes visual encodings, and what constitutes an honest, truthful display. It then introduces **Seaborn** and **Matplotlib**, Python's standard libraries for statistical visualization, applying them to explore graduate student life (RQ1).

By the end of this chapter you will be able to

- Explain why visualization is indispensable alongside summary statistics, and contrast exploratory vs. explanatory graphics.
- Apply graphical perception theory, recognizing why position and length are decoded more accurately than angles, areas, or color saturation.
- Select the appropriate visualization based on the research question and variable measurement levels.
- Construct statistical graphics using Seaborn and Matplotlib (histograms, KDE density curves, bar charts, box plots, scatter plots, and line charts).
- Build small multiples (facets) and correlation heatmaps to investigate multi-variable interactions.
- Detect and rectify misleading visual representations (such as truncated bar-chart baselines and visual clutter).
- Export publication-ready, high-resolution figures (300+ DPI PNG, PDF, SVG) formatted to journal dimensions.

Note

Teaching & Practice: Interactive Lecture Slides · Browser Playground Exercises

4.1 - What a graph is for

Summary statistics compress empirical distributions into a handful of numerical parameters (means, standard deviations, correlations). That compression can conceal radically different data distributions. The statistician Francis Anscombe demonstrated this vulnerability with four synthetic datasets constructed in 1973 [3], bundled in Seaborn as `anscombe`:

```
anscombe = sns.load_dataset("anscombe")

summary = (
    anscombe
    .groupby("dataset")
    .agg(
        mean_x=("x", "mean"),
        sd_x=("x", "std"),
```

```
        mean_y=("y", "mean"),
        sd_y=("y", "std")
    )
    .round(2)
)
summary
```

dataset	mean_x	sd_x	mean_y	sd_y
I	9.0	3.32	7.5	2.03
II	9.0	3.32	7.5	2.03
III	9.0	3.32	7.5	2.03
IV	9.0	3.32	7.5	2.03

Across all four datasets, the summary statistics are practically identical: $\bar{x} = 9.0$, $s_x = 3.32$, $\bar{y} = 7.50$, and $s_y = 2.03$. Bivariate regression yields identical linear fits ($y = 3.0 + 0.5x$) with identical correlation coefficients ($r = 0.82$). Judged solely by numerical summaries, the four datasets appear indistinguishable.

Yet Figure 4.1 demonstrates that their underlying relationships are entirely dissimilar.

```
g = sns.lmplot(
    data=anscombe, x="x", y="y", col="dataset", col_wrap=2,
    ci=None, height=2.5, aspect=1.3,
    scatter_kws={"s": 35, "color": "#1f77b4"},
    line_kws={"color": "grey", "linewidth": 1.5}
)
g.set_axis_labels("X", "Y")
plt.tight_layout()
plt.show()
```

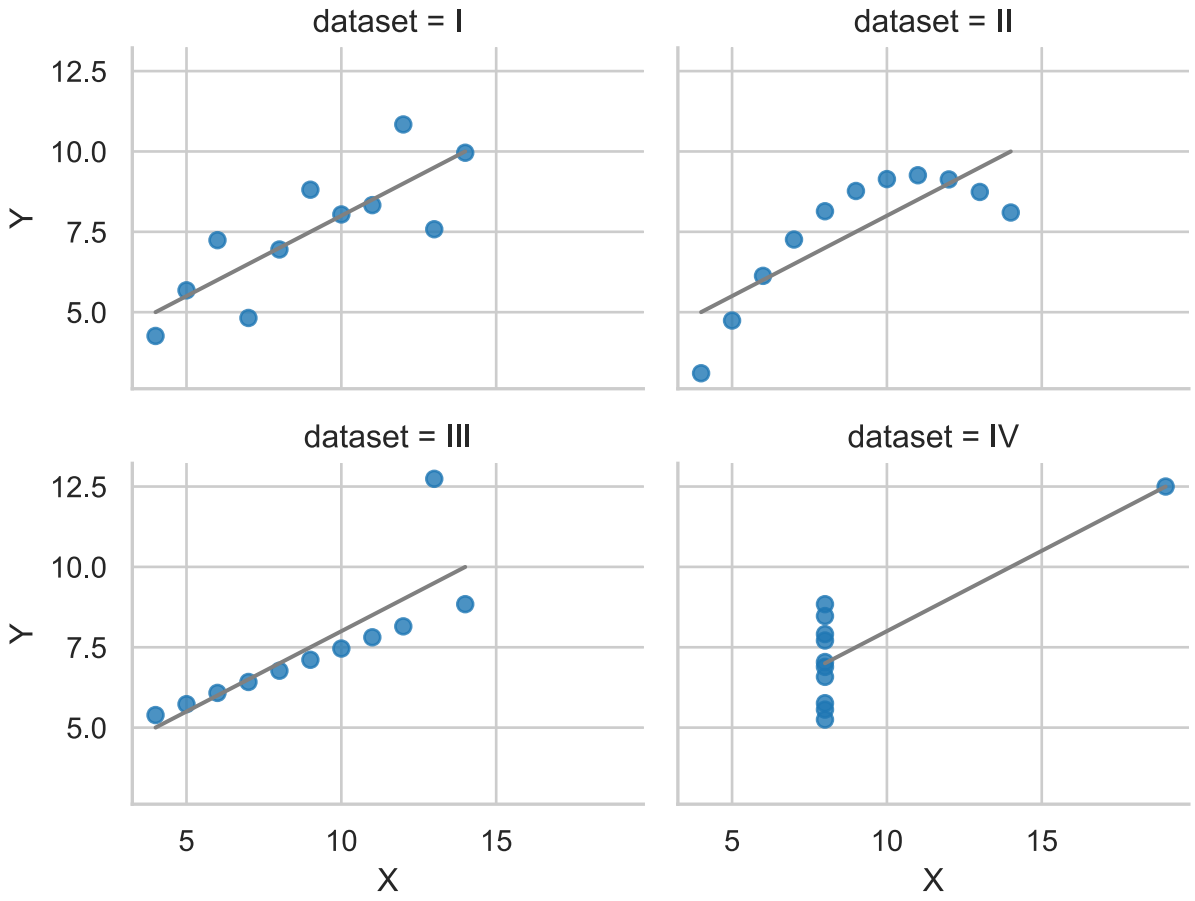


Figure 4.1: Anscombe's quartet: identical means, standard deviations, and regression lines, but four completely different empirical patterns.

Only Dataset I represents the bivariate linear association assumed by Pearson's r. Dataset II is a non-linear quadratic curve, Dataset III is a perfect linear relationship corrupted by one outlier, and Dataset IV is an unassociated vertical cluster whose correlation is produced entirely by a single high-leverage point. An analyst relying strictly on summary statistics would draw identical—and incorrect—conclusions across all four. Visual inspection prior to modeling is not optional; it is a prerequisite for valid inference.

4.1.1 - Exploratory versus explanatory graphs

Scientific graphics serve two distinct functions:

- **Exploratory graphs:** Created rapidly by the researcher to interrogate the data—checking skewness, detecting outliers, and screening for interaction effects. They require minimal aesthetic polish.
- **Explanatory graphs:** Curated for external readers and thesis examiners to communicate a validated empirical finding. They require meticulous care: informative axis labels, accessible color palettes, annotations, and the elimination of extraneous chartjunk.

4.2 - Graphical perception: How vision decodes data

Graphs function by translating numbers into visual encodings: position, length, angle, area, slope, and color. Human perception does not decode these attributes with equal precision. In pioneering experiments, William Cleveland and Robert McGill [4] evaluated how accurately observers judge quantitative differences across visual channels, establishing an empirical hierarchy:

1. **Position along a common aligned scale** (most accurate)
2. **Length**
3. **Slope and Angle**

4. **Area**
5. **Color saturation and hue** (least accurate)

Effective scientific graphics map primary research comparisons to position or length, reserving color and area for secondary grouping.

The perceptual difference is readily apparent when comparing identical proportions displayed as a pie chart versus a bar chart (Figure 4.2).

```
shares = pd.DataFrame({
    "group": ["A", "B", "C", "D", "E"],
    "percent": [23, 21, 20, 19, 17]
})

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(7.5, 3.2))

# Pie chart (angle / area)
ax1.pie(shares["percent"],              labels=shares["group"],              startangle=90,
colors=sns.color_palette("mako", 5))
ax1.set_title("Pie chart (Angle / Area)", fontsize=11)

# Bar chart (aligned length)
sns.barplot(data=shares, x="group", y="percent", color="#4c72b0", ax=ax2)
ax2.set(title="Bar chart (Aligned Length)", xlabel="Category", ylabel="Percentage (%)")
ax2.set_ylim(0, 26)

plt.tight_layout()
plt.show()
```

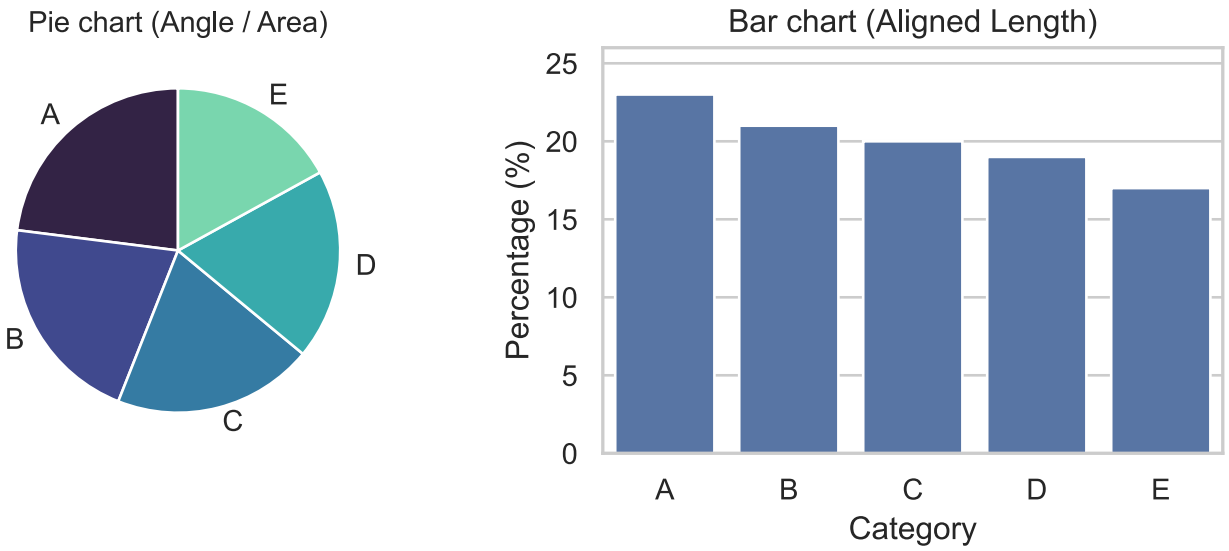


Figure 4.2: The same five percentages displayed as a pie chart (angle and area) versus a bar chart (length along an aligned scale). Identifying the ranking is immediate in the bar chart.

In the pie chart, the five slices appear virtually identical; determining the rank order requires scrutinizing legends and mentally comparing angles. In the bar chart, the monotonic decline from category A (23%) to E (17%) is evident immediately because comparisons occur along a common baseline. Consequently, scientific reporting favors bar charts and dot plots over pie charts.

4.3 - Selecting the right visualization

The choice of graph is governed by the research question and the levels of measurement of the variables under study (Table 4.1).

Table 4.1: Selecting the appropriate visualization based on analytical questions and variable types

Analytical Question	Variable Types	Recommended Graph	Python Seaborn / Matplotlib Function
Distribution of a single continuous metric	One numeric	Histogram / KDE plot	`sns.histplot()`, `sns.kdeplot()`
Frequency counts of categorical groups	One categorical	Bar chart	`sns.countplot()`, `sns.barplot()`
Comparison of continuous metric across groups	One numeric by categorical	Box plot / Strip plot	`sns.boxplot()`, `sns.stripplot()`
Association between two continuous variables	Two numeric	Scatter plot with trend line	`sns.scatterplot()`, `sns.regplot()`
Trajectory of a metric over time	Numeric over time	Longitudinal line plot	`sns.lineplot()`
Multilateral correlation structure	Several numeric	Correlation heatmap	`sns.heatmap()`

4.4 - Core statistical visualizations with Seaborn

To explore Elaf's wellbeing cohort, we merge Semester 1 measurements with baseline student background demographics:

```
first_sem = (
    semesters
    .query("semester == 1")
    .merge(students, on="student_id", how="left")
)
```

4.4.1 - Histograms and density curves

The initial question concerning any continuous variable addresses its empirical distribution: central tendency, spread, skewness, and the presence of extreme outliers. A **histogram** partitions the continuous range into discrete bins, displaying the count of observations falling within each interval:

```
fig, ax = plt.subplots(figsize=(6, 3.5))

sns.histplot(
    data=first_sem, x="sleep_hours", binwidth=0.5,
    color="#4c72b0", edgecolor="white", ax=ax
)
ax.set(xlabel="Hours of sleep per night", ylabel="Number of students")
plt.tight_layout()
plt.show()
```

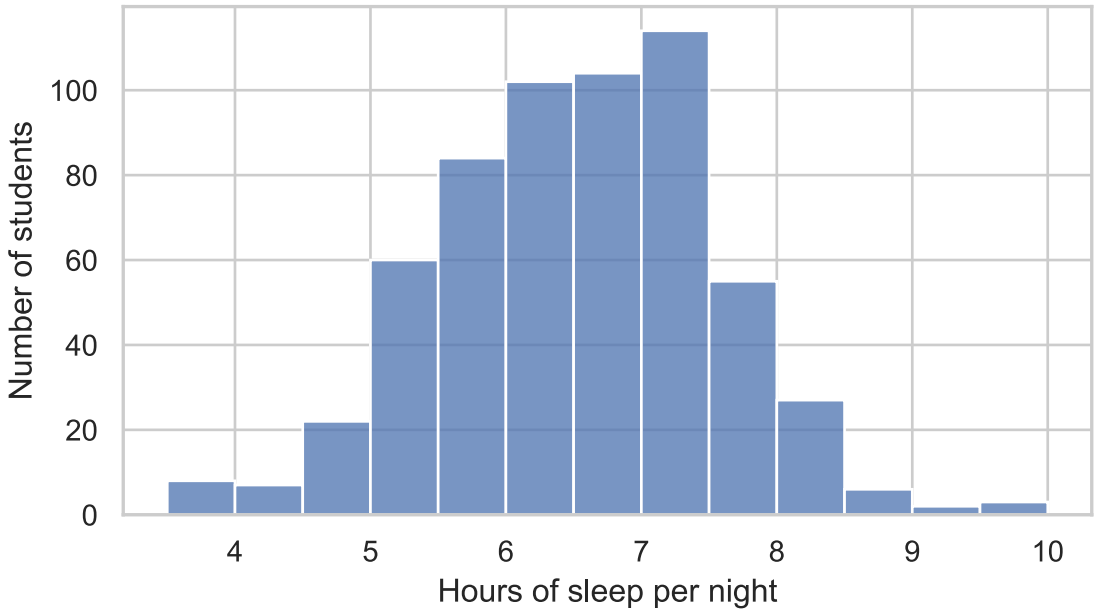


Figure 4.3: Distribution of sleep hours per night in Semester 1.

Figure 4.3 displays a roughly symmetrical, unimodal distribution centered near 6.5 hours. Exactly `r round(100 * share_short, 1)`% of graduate students sleep less than the clinically recommended 7 hours during their first semester.

4.4.2 - Bar charts for categorical counts

A **bar chart** plots frequency distributions across categorical factor levels:

```
fig, ax = plt.subplots(figsize=(6, 3.5))

sns.countplot(
    data=students, x="faculty", color="#4c72b0",
    order=students["faculty"].value_counts().index, ax=ax
)
ax.set(xlabel="Faculty", ylabel="Student Count")
plt.xticks(rotation=20, ha="right")
plt.tight_layout()
plt.show()
```

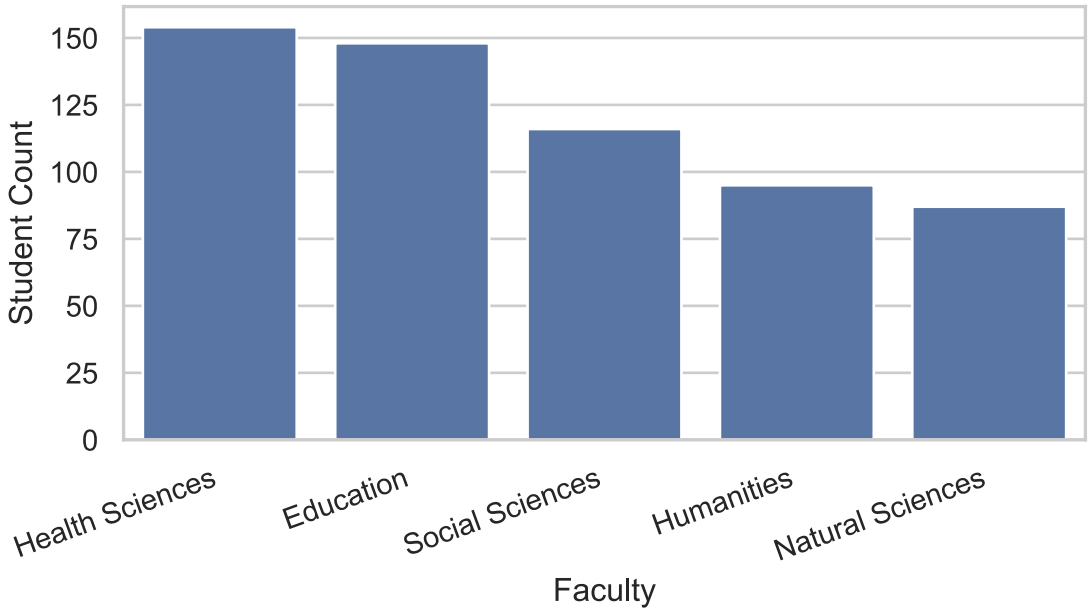


Figure 4.4: Enrollment frequency across academic faculties.

Because bar lengths represent quantities proportionally, the numerical baseline must always start at zero.

4.4.3 - Box plots for group comparisons

A **box plot** summarizes five key distribution parameters: the median (center line), interquartile range (IQR, box edges covering the 25th to 75th percentiles), whiskers (extending to $1.5 \times$ IQR), and individual points for potential outliers. It is the gold standard for comparing continuous metrics across qualitative groups:

```
fig, ax = plt.subplots(figsize=(5, 3.5))

sns.boxplot(
    data=first_sem, x="study_mode", y="wellbeing",
    palette=["#4c72b0", "#55a868"], ax=ax, width=0.5
)
ax.set(xlabel="Study Mode", ylabel="Wellbeing Index (0 to 100)")
plt.tight_layout()
plt.show()
```

```
C:\Users\polla\AppData\Local\Temp\ipykernel_54144\230693867.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign
the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxplot(
```

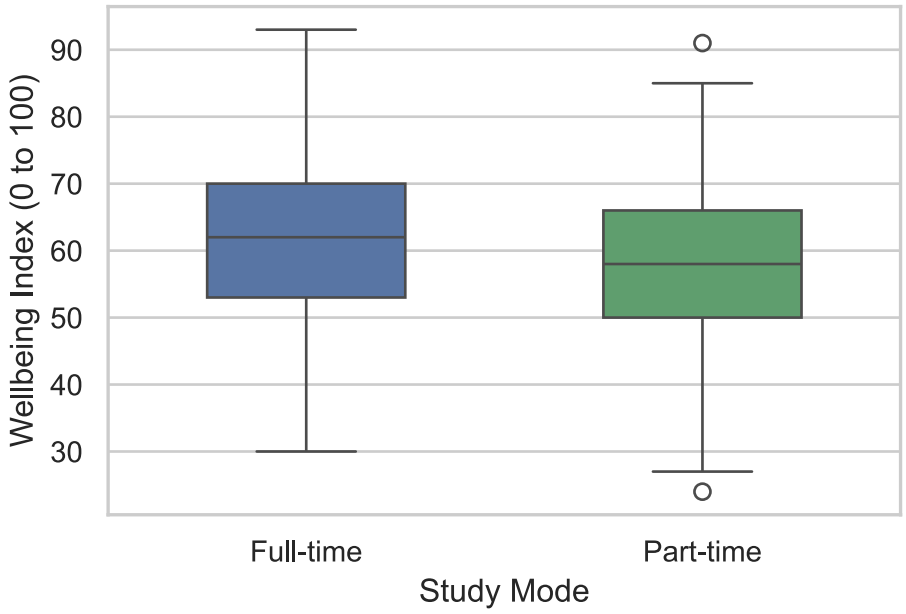


Figure 4.5: Wellbeing in Semester 1 by study mode (Full-time vs. Part-time).

Part-time students report a slightly lower median wellbeing score than full-time students, but the distributions overlap substantially. In Chapter 8., we use inferential ANOVA to test whether this group discrepancy is statistically significant.

4.4.4 - Scatter plots with regression trend lines

The relationship between two continuous variables—such as sleep hours and academic performance—is evaluated via a **scatter plot**. When plotting 600 participants, individual markers often overlap (**overplotting**). We mitigate this by setting transparency (`alpha=0.4`) and overlaying a linear regression trend line with `sns.regplot()`:

```
fig, ax = plt.subplots(figsize=(6, 3.8))

sns.regplot(
    data=first_sem, x="sleep_hours", y="gpa",
    scatter_kws={"alpha": 0.35, "color": "#4c72b0"},
    line_kws={"color": "#c44e52", "linewidth": 2},
    ax=ax
)
ax.set(xlabel="Hours of sleep per night", ylabel="Grade Point Average (GPA)")
plt.tight_layout()
plt.show()
```

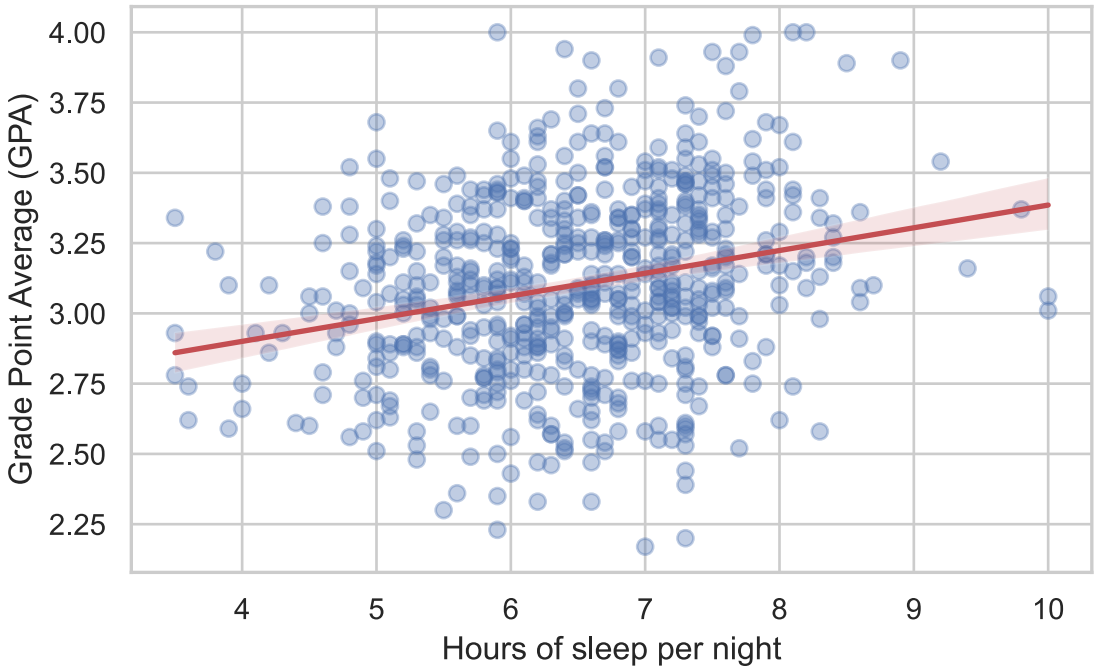


Figure 4.6: Association between sleep hours and GPA in Semester 1, with ordinary least squares trend line.

The regression line slopes upward: students sleeping more hours tend to achieve modestly higher GPAs. However, observations scatter broadly around the line ($r \approx 0.30$), indicating that sleep explains only a fraction of total academic variance.

4.4.5 - Longitudinal trajectory line plots

Elaf's research design tracked cohorts across four consecutive semesters. A **line plot** illustrates how group means evolve over time:

```
wellbeing_trajectory = (
    semesters
    .merge(students[["student_id", "workshop"]], on="student_id", how="left")
    .groupby(["semester", "workshop"])["wellbeing"]
    .mean()
    .reset_index()
)

fig, ax = plt.subplots(figsize=(6.5, 3.8))

sns.lineplot(
    data=wellbeing_trajectory, x="semester", y="wellbeing",
    hue="workshop", marker="o", markersize=7, linewidth=2,
    palette={"Invited": "#2ca02c", "Not invited": "#7f7f7f"},
    ax=ax
)
ax.set(
    title="Cohort Wellbeing Trajectory Over Two Years",
    xlabel="Academic Semester",
    ylabel="Mean Wellbeing (0 to 100)",
    xticks=[1, 2, 3, 4]
)
ax.legend(title="Workshop Group", frameon=True)
plt.tight_layout()
plt.show()
```

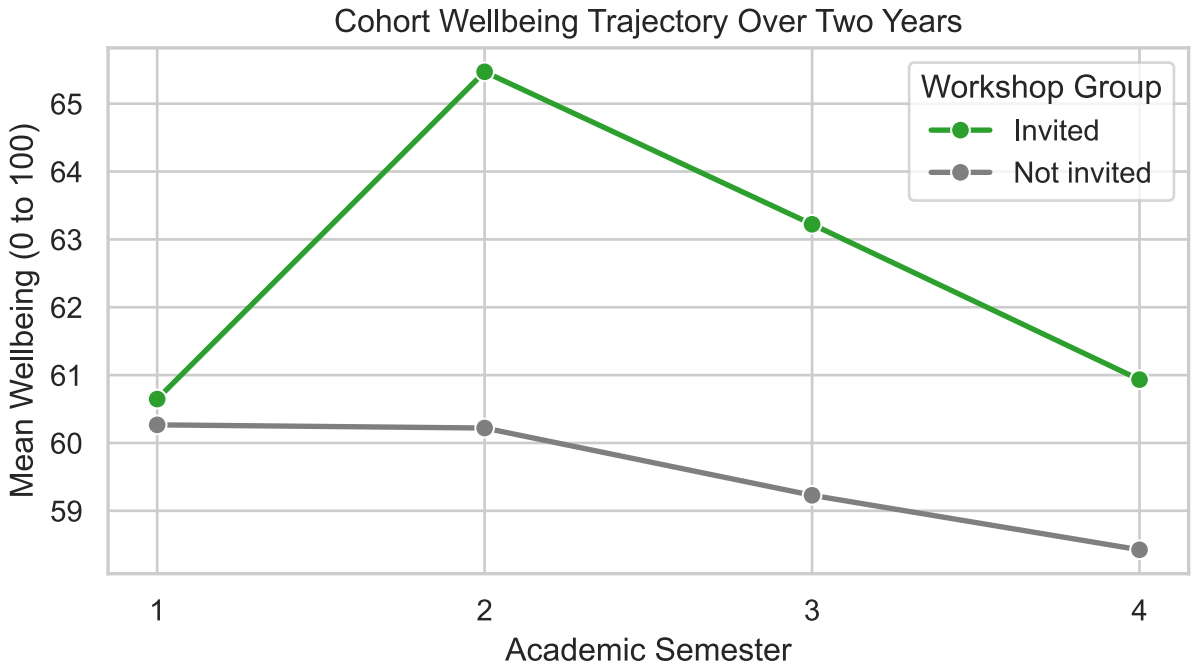


Figure 4.7: Longitudinal wellbeing trajectories across four semesters by workshop participation.

Figure 4.7 captures the core narrative of Elaf's intervention study: groups start at identical baselines in Semester 1, invited students pull ahead following the intervention in Semester 2, and the gap stabilizes over subsequent terms.

4.5 - Preparing publication-quality figures

Exploratory graphics suffice during initial data exploration, but explanatory figures destined for a thesis or journal article require publication-standard polish.

4.5.1 - Annotations and reference lines

Adding explicit reference lines and annotations directs reader attention to clinically meaningful thresholds:

```
fig, ax = plt.subplots(figsize=(6.5, 3.8))

sns.histplot(
    data=first_sem, x="sleep_hours", binwidth=0.5,
    color="#4c72b0", edgecolor="white", ax=ax
)
ax.axvline(7.0, color="#c44e52", linestyle="--", linewidth=1.8)
ax.text(
    7.1, ax.get_ylim()[1] * 0.9,
    "Recommended: 7.0 hrs",
    color="#c44e52", fontweight="semibold", verticalalignment="top"
)
ax.set(
    title="Baseline Sleep Duration",
    xlabel="Hours of sleep per night",
    ylabel="Number of students"
)
plt.tight_layout()
plt.show()
```

```
findfont: Failed to find font weight semibold for Arial, now using 700.
```

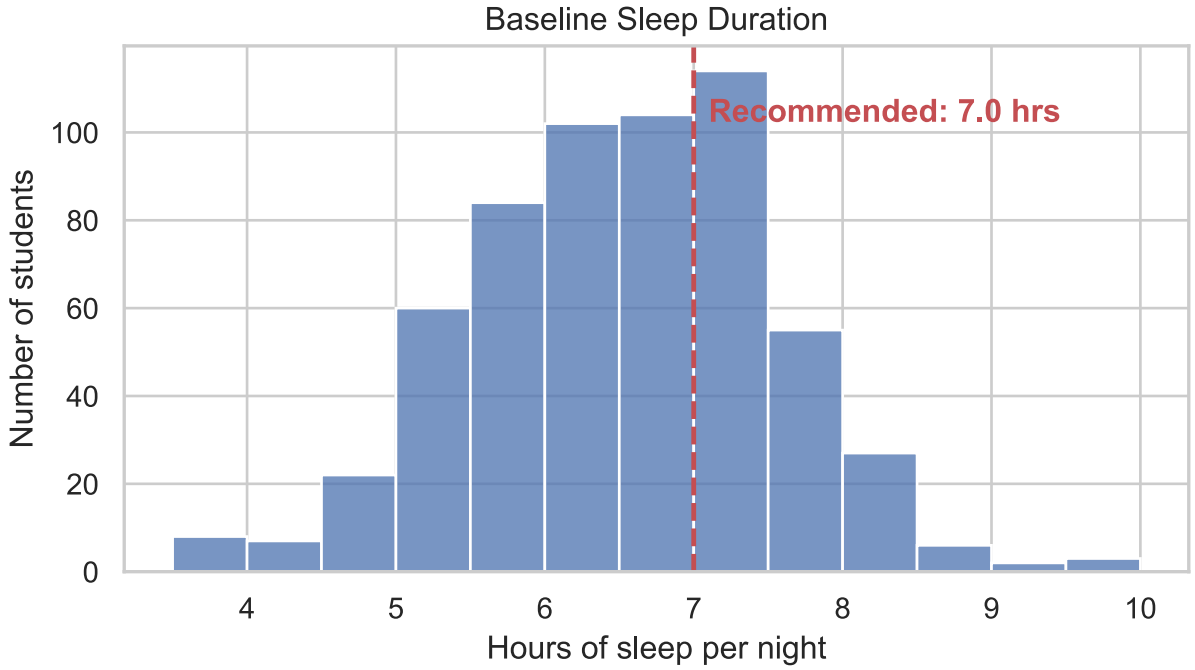


Figure 4.8: Distribution of sleep hours with clinical 7-hour reference benchmark.

4.5.2 - Color accessibility

Approximately 8% of men and 0.5% of women experience congenital color vision deficiency (red-green color blindness). A scientific graphic must remain interpretable to all readers and reproduce legibly when printed in black and white:

- Avoid rainbow (`jet`) palettes, which introduce artificial visual boundaries.
- Utilize perceptually uniform, colorblind-safe palettes such as `viridis`, `plasma`, `cividis`, or `colorblind`.
- Combine color with redundant visual encodings (distinct marker shapes, line styles, or direct text labels).

4.6 - Honest graphs and visual ethics

A visualization can be mathematically accurate while remaining visually misleading. Distortion frequently stems from unexamined software defaults:

1. **Truncated vertical baselines on bar charts:** Because a bar encodes magnitude through length, truncating the baseline exaggerates differences.
2. **Extreme aspect ratios:** Altering aspect ratios artificially steepens or flattens longitudinal trends.
3. **Chartjunk and 3D effects:** Pseudo-3D perspectives distort angular perception and distance.
4. **Hiding underlying distribution variance:** Reporting only summary bar heights masks critical within-group heterogeneity.

4.6.1 - Redesigning a misleading figure

Suppose an investigator evaluates whether mean wellbeing differs across faculties. A careless chart with an artificially truncated axis appears in Figure 4.9:

```
faculty_means = (
    first_sem
    .groupby("faculty")["wellbeing"]
    .mean()
    .reset_index()
)

fig, ax = plt.subplots(figsize=(6.5, 3.5))
```

```
sns.barplot(data=faculty_means, x="faculty", y="wellbeing", palette="Blues_d", ax=ax)
ax.set_ylim(58, 62.5)  # Truncated baseline!
ax.set(title="Misleading: Truncated Baseline (58 to 62.5)", xlabel="Faculty", ylabel="Mean
Wellbeing")
plt.xticks(rotation=20, ha="right")
plt.tight_layout()
plt.show()
```

```
C:\Users\polla\AppData\Local\Temp\ipykernel_54144\2329551519.py:9: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign
the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(data=faculty_means, x="faculty", y="wellbeing", palette="Blues_d", ax=ax)
```

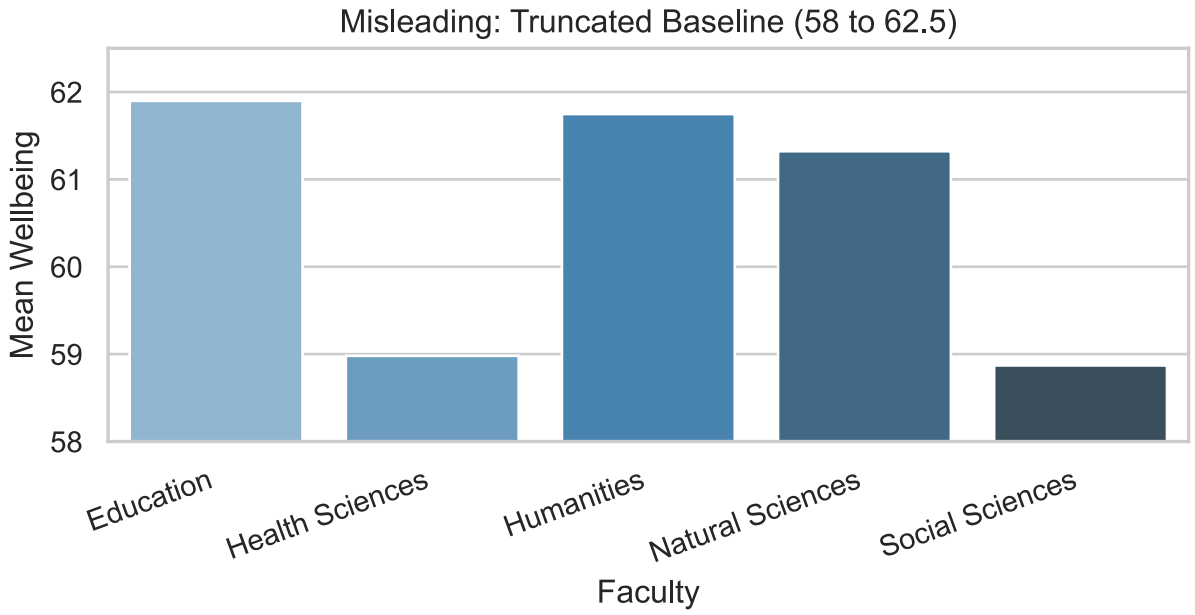


Figure 4.9: A misleading graphic: the vertical axis is truncated at 58, exaggerating modest differences and concealing student-level variance.

In Figure 4.9, the Education faculty bar appears triple the height of Social Sciences. Yet the actual group means differ by barely 2.5 points on a 100-point scale. Furthermore, the bar chart completely conceals individual variation.

In Figure 4.10, the display is redesigned into an honest, informative jittered strip plot with overlaid group means:

```
order = faculty_means.sort_values("wellbeing")["faculty"]

fig, ax = plt.subplots(figsize=(7, 3.8))

# Jittered individual student scores
sns.stripplot(
    data=first_sem, y="faculty", x="wellbeing",
    order=order, color="grey", alpha=0.3, jitter=0.2, size=5, ax=ax
)

# Overlay group means as prominent diamonds
sns.pointplot(
    data=first_sem, y="faculty", x="wellbeing",
    order=order, color="#c44e52", markers="D", scale=0.8,
    errorbar=None, join=False, ax=ax
```

```
)

ax.set(
    title="Truthful: Student Wellbeing Distribution by Faculty",
    xlabel="Wellbeing Index (0 to 100)",
    ylabel=None,
    xlim=(0, 100)  # Full theoretical scale!
)
plt.tight_layout()
plt.show()
```

```
C:\Users\polla\AppData\Local\Temp\ipykernel_54144\1426493859.py:12: UserWarning:

The `scale` parameter is deprecated and will be removed in v0.15.0. You can now control the size
of each plot element using matplotlib `Line2D` parameters (e.g., `linewidth`, `markersize`,
etc.).

  sns.pointplot(
C:\Users\polla\AppData\Local\Temp\ipykernel_54144\1426493859.py:12: UserWarning:

The `join` parameter is deprecated and will be removed in v0.15.0. You can remove the line
between points with `linestyle='none'`.

  sns.pointplot(
```

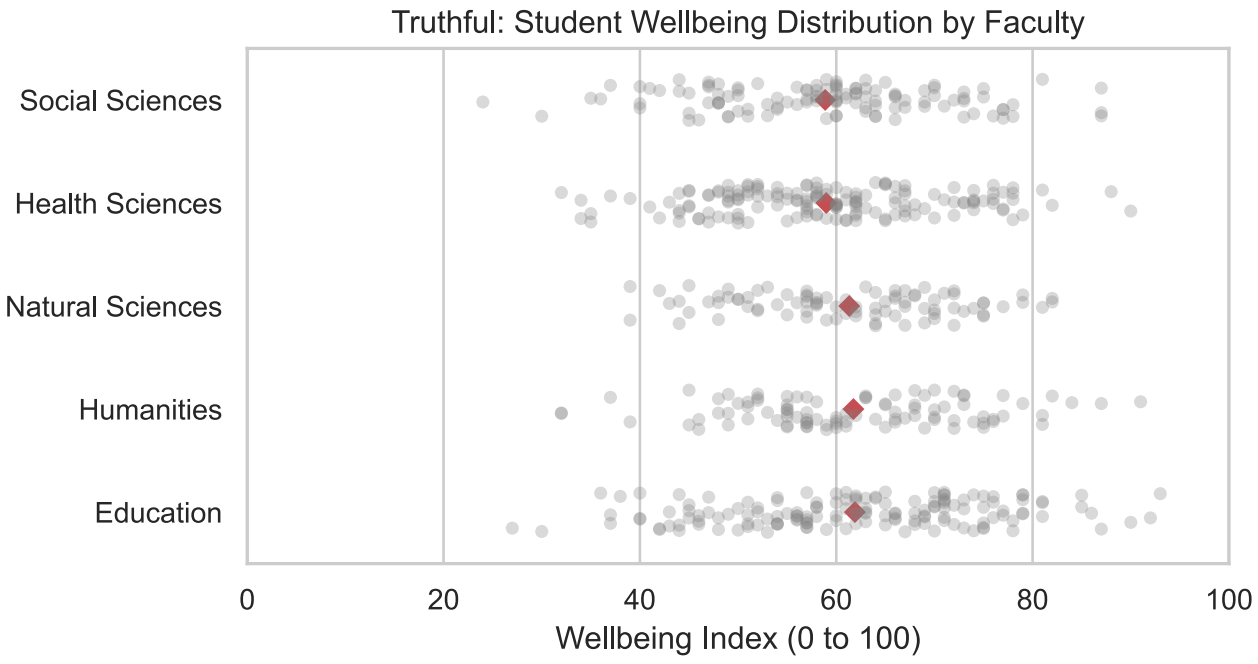


Figure 4.10: The redesigned truthful display: individual observations shown as faint points, with faculty means marked in red. Within-faculty spread vastly exceeds between-faculty differences.

Figure 4.10 communicates the true scientific story: individual student scores vary widely between 25 and 95 across all departments, and average differences between faculties are trivial by comparison.

4.7 - Advanced scientific visualizations

4.7.1 - Small multiples (Faceting)

Small multiples partition a complex visualization into separate subpanels sharing identical axes, enabling direct visual comparisons:

```
g = sns.lmplot(
    data=first_sem, x="sleep_hours", y="gpa", col="programme",
    scatter_kws={"alpha": 0.35, "color": "#4c72b0"},
    line_kws={"color": "#c44e52"},
    height=3.5, aspect=1.1
)
g.set_axis_labels("Hours of Sleep per Night", "GPA")
g.set_titles(col_template="{col_name} Candidates")
plt.tight_layout()
plt.show()
```

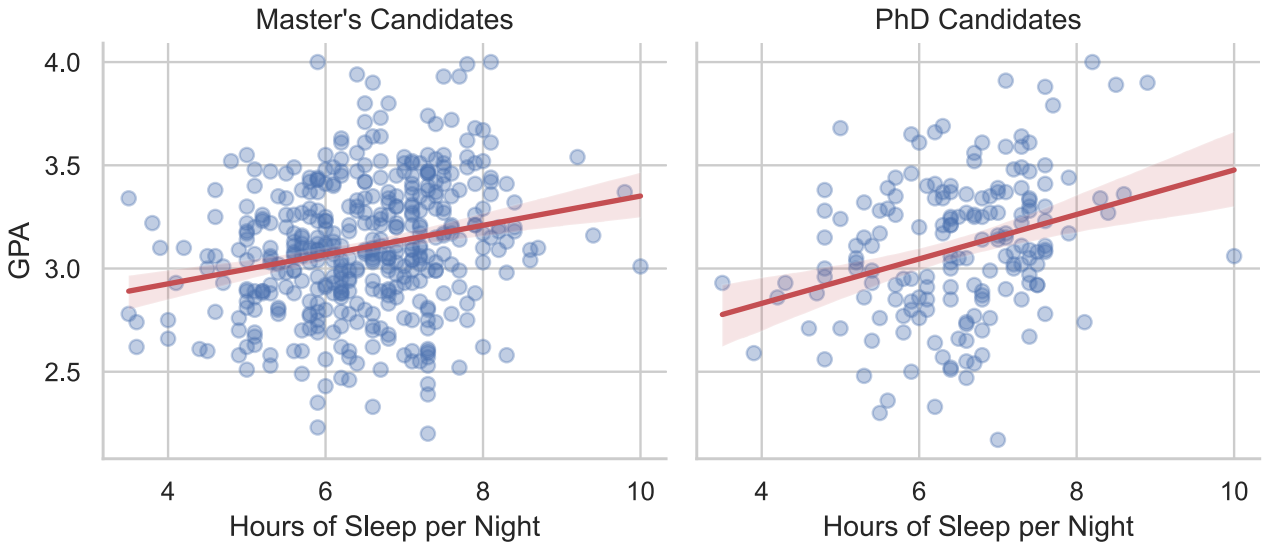


Figure 4.11: Association between sleep hours and GPA, faceted by degree programme.

The regression slope between sleep and academic performance is virtually identical across both Master's and PhD cohorts.

4.7.2 - Kernel density estimation (KDE) plots

A **density plot** provides a smoothed continuous estimate of the probability density function, facilitating clean comparisons of overlapping distributions:

```
sem2_data = (
    semesters
    .query("semester == 2")
    .merge(students[["student_id", "workshop"]], on="student_id")
)

fig, ax = plt.subplots(figsize=(6.5, 3.5))

sns.kdeplot(
    data=sem2_data, x="wellbeing", hue="workshop",
    fill=True, common_norm=False, alpha=0.35,
    palette={"Invited": "#2ca02c", "Not invited": "#7f7f7f"},
```

```
    linewidth=2, ax=ax
)
ax.set(
    title="Semester 2 Post-Intervention Wellbeing Distribution",
    xlabel="Wellbeing Index (0 to 100)",
    ylabel="Probability Density"
)
plt.tight_layout()
plt.show()
```

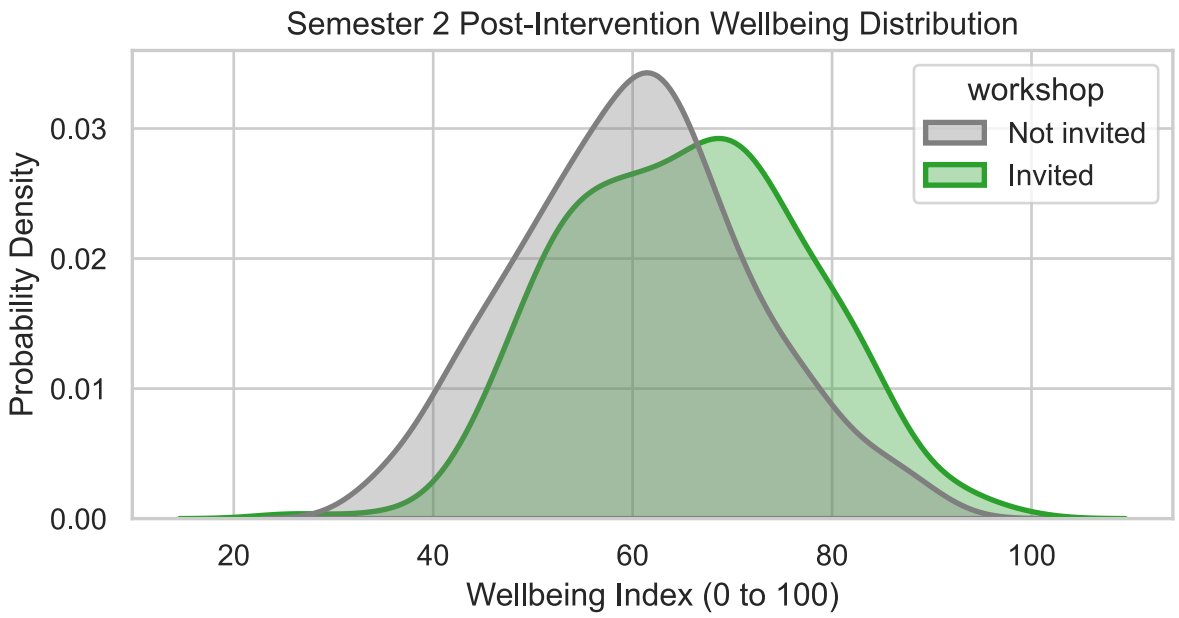


Figure 4.12: Comparative probability density of Semester 2 wellbeing by workshop group.

Notice that the entire distribution for invited students is shifted rightward, demonstrating a cohort-wide positive effect rather than an isolated outlier reaction.

4.7.3 - Correlation heatmaps

When examining relationships among numerous continuous constructs, a **correlation heatmap** visualizes the bivariate correlation matrix via a diverging color gradient:

```
corr_vars = first_sem[["gpa", "sleep_hours", "study_hours", "wellbeing"]].corr()

fig, ax = plt.subplots(figsize=(5.5, 4.2))

sns.heatmap(
    corr_vars, annot=True, fmt=".2f", cmap="vlag",
    vmin=-1.0, vmax=1.0, center=0.0,
    square=True, linewidths=0.5, cbar_kws={"shrink": 0.8},
    ax=ax
)
ax.set_title("Correlation Heatmap (Semester 1)", fontsize=11, pad=10)
plt.tight_layout()
plt.show()
```

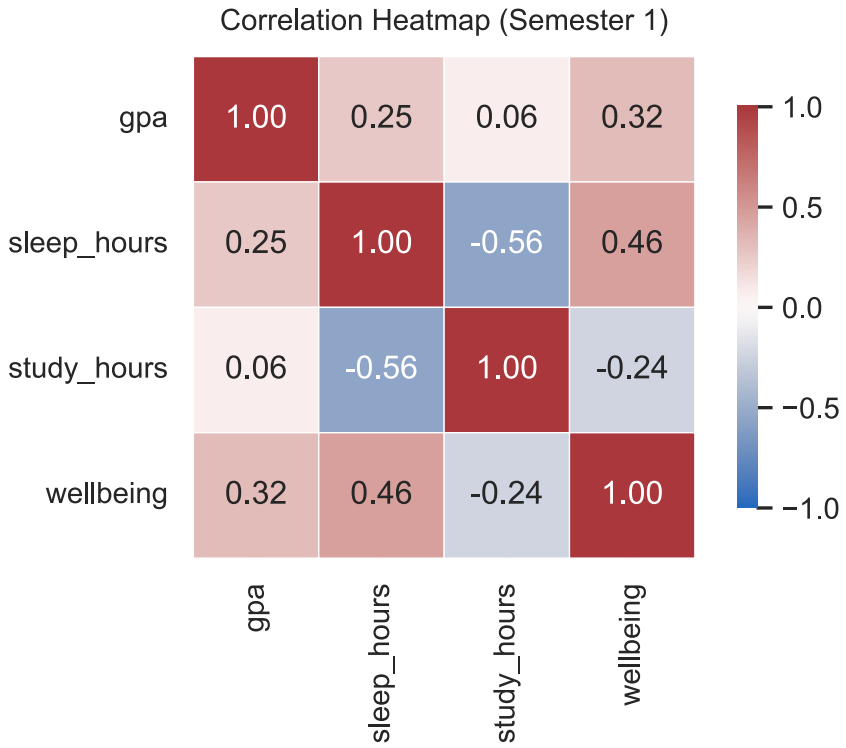


Figure 4.13: Bivariate correlation matrix across Semester 1 quantitative metrics.

The diverging `vlag` palette centers white at zero correlation, mapping negative associations to blue and positive associations to red. Values printed within each cell ensure readers are not forced to decode coefficients solely through color saturation.

4.8 - Exporting publication-ready figures

Scientific figures must be exported at explicit target physical dimensions and print resolutions:

```
# Save as 300 DPI PNG raster image (suitable for Word documents and presentations):
fig.savefig("figures/wellbeing_trajectory.png", dpi=300, bbox_inches="tight")

# Save as vector graphic (ideal for LaTeX and journal publication):
fig.savefig("figures/wellbeing_trajectory.pdf", bbox_inches="tight")
```

Setting `bbox_inches="tight"` prevents axis labels from being clipped. Standard journal column widths typically measure 8.5 cm (single column) or 17.5 cm (full width). Defining figure sizes in code preserves font proportions when rendered on the printed page.

4.9 - Chapter review

4.9.1 - Summary

- Summary statistics can mask fundamentally divergent distributions (Anscombe's quartet); visualizing data prior to modeling is essential.
- Graphical perception theory demonstrates that position and length are decoded with greatest precision; key scientific comparisons must be mapped to aligned coordinates.
- Continuous distributions are inspected using histograms and KDE density plots; group comparisons rely on box plots and strip plots; bivariate relationships are evaluated via scatter plots and regression trend lines.
- Tidy longitudinal trajectories are plotted using `sns.lineplot()`, small multiples using `sns.lmplot(col=...)`, and correlation matrices using `sns.heatmap()`.
- Ethical visual design requires zero-aligned baselines for bar charts, colorblind-accessible palettes (`viridis`, `vlag`), and transparent representation of within-group variance.

- Publication-quality figures are saved using `fig.savefig()` at 300+ DPI or vector formats (`.pdf`, `.svg`).

4.9.2 - Key terms

Anscombe's quartet, exploratory vs. explanatory graphics, graphical perception, aligned scale, histogram, KDE density plot, box plot, strip plot, overplotting, regression trend line, longitudinal line plot, small multiples, facet, correlation heatmap, color accessibility, chartjunk, DPI, vector graphics.

4.10 - Exercises

Work through these exercises in your browser using the Chapter 4 Playground or download the starter project for VS Code.

1. Using `first_sem`, generate a histogram of `study_hours`. Experiment with binwidths of 2, 5, and 10 hours, and evaluate which binning best reveals distribution shape.
2. Construct a bar chart showing student counts across `employment` categories, ordered from `"None"` to `"Full-time job"`, using a colorblind-safe single color.
3. Generate box plots comparing Semester 1 GPA across academic faculties, formatting axis labels clearly.
4. Draw a scatter plot of caffeine consumption (`caffeine_mg`) against sleep hours (`sleep_hours`) in Semester 1 with an overlaid linear regression line, and describe the observed association.
5. Save any figure created in this chapter to `figures/exercise_plot.png` at 300 DPI with tight bounding boxes.
6. Redesign the faculty wellbeing bar chart into a horizontal strip plot showing both individual scores and group medians. Explain why this display is scientifically superior.

4.11 - Further reading

- *Fundamentals of Data Visualization* by Claus O. Wilke [5]: an exceptional, language-agnostic text on the principles of honest, effective visual communication.
- Official documentation for Seaborn and Matplotlib.
- "Graphical Perception" by William S. Cleveland and Robert McGill [4].

II Statistical Analysis for Research

6.2 The centre of a distribution . . . 114
6.3 The spread of a distribution . . . 117
6.4 The shape of a distribution . . . 119
6.5 Unusual values . . . 122
6.6 Missing data . . . 123
6.7 Relationships between variables . . . 125
6.8 Describing your sample in a thesis . . . 127
6.9 Common misconceptions . . . 129
6.10 Chapter review . . . 130
6.11 Exercises . . . 130
6.12 Further reading . . . 130

7 Hypothesis Testing and Statistical Inference . . . 131
7.1 Samples and populations . . . 131
7.2 The logic of hypothesis testing . . . 134
7.3 The one-sample t-test . . . 139
7.4 The two-sample t-test (Welch's t-test) . . . 140
7.5 The paired t-test . . . 141
7.6 Assumptions of the t-test . . . 142
7.7 Non-parametric tests . . . 142
7.8 Chi-square tests . . . 143
7.9 Choosing the right test . . . 144
7.10 Reporting results in a thesis . . . 144
7.11 Common misconceptions . . . 145
7.12 Chapter review . . . 145
7.13 Exercises . . . 146
7.14 Further reading . . . 146

8 ANOVA and Regression . . . 147
8.1 One-way ANOVA . . . 148
8.2 Two-way ANOVA . . . 152
8.3 Linear regression . . . 153
8.4 Logistic regression . . . 164
8.5 Common misconceptions . . . 166
8.6 Chapter review . . . 166
8.7 Exercises . . . 167
8.8 Further reading . . . 167

9 Multivariate Statistical Methods . 169
9.1 Constructs and latent variables . . . 169
9.2 Many variables at once . . . 170
9.3 Principal component analysis . . . 171
9.4 Factor analysis . . . 174
9.5 Cluster analysis . . . 175
9.6 Common misconceptions . . . 182
9.7 Chapter review . . . 182
9.8 Exercises . . . 183
9.9 Further reading . . . 183

10 Mixed-Effects Models . . . 185
10.1 Data with a structure . . . 185
10.2 A study of sleep deprivation . . . 200
10.3 Change in wellbeing over two years . . . 203
10.4 Students within supervisors . . . 209
10.5 Counts and yes-or-no outcomes . . . 210
10.6 Reporting a mixed-effects model . . . 211
10.7 Common misconceptions . . . 211
10.8 Chapter review . . . 212
10.9 Exercises . . . 212
10.10 Further reading . . . 212

5. From Research Question to Data

Part 1 gave you the tools: you can import data, clean it, and visualize it. Part 2 uses those tools to answer research questions. Before any statistics, though, comes a step that no software can do for you: deciding *what exactly you are asking*, *what result would answer it*, and *whether your data can give that answer at all*.

Most problems that examiners find in a thesis start here, not in the analysis: a question too vague to answer, a hypothesis that no result could contradict, a questionnaire that measures something other than what the title promises, or a sample that cannot support the conclusion. No statistical test can fix those problems afterwards. This chapter follows the path from a research topic to data that can answer a question, using the student wellbeing study as the example throughout. It is more conceptual than the other chapters, but it still uses Python wherever Python helps you *see* an idea: simulations, checks of how variables are measured, and a first calculation of sample size.

By the end of this chapter you will be able to

- Explain what data analysis is for in research: describing, explaining, and predicting.
- Turn a research topic into a focused research question, and tell a good question from a weak one.
- Write a hypothesis that is testable and falsifiable, and state its null hypothesis.
- Explain how an abstract idea, such as stress, becomes a measured variable, and judge its level of measurement.
- Name the roles variables play: outcome, predictor, confounder, moderator, and mediator.
- Distinguish reliability from validity.
- Explain what experimental, observational, cross-sectional, and longitudinal designs can and cannot show.
- Explain why a biased sample cannot be fixed by making it larger.
- Plan an analysis before collecting data, including a first estimate of the sample size, and use Python to check levels of measurement, balance between groups, and sampling by simulation.

Note

Teaching & Practice: Interactive Lecture Slides · Browser Playground Exercises

5.1 - What data analysis is for

In one sentence: data analysis tests claims against evidence.

Research makes claims about the world: graduate students sleep too little; a workshop improves wellbeing; supervisor support protects against dropout. A claim is only as strong as the evidence behind it, and data analysis is the part of research that confronts claims with evidence. It does three kinds of work. **Describing** establishes what the world looks like: how long graduate students sleep, or how many consider dropping out. **Explaining** asks why it looks like that: whether the workshop *causes* higher wellbeing, or which factors go together with a higher GPA. **Predicting** concerns what will happen in new cases, such as which of next year's students are likely to consider leaving.

Each kind of work asks a different kind of question, is judged by a different standard, and leads to different methods, as Table 5.1 shows.

Table 5.1: Three aims of data analysis

Aim	Example question	Judged by	Methods	Chapters
Describe	How long do graduate students sleep?	Accuracy of the summary; who it applies to	Summaries, graphs, confidence intervals	4, 6, 7
Explain	Does the workshop improve wellbeing?	Whether rival explanations are ruled out	Tests, regression, mixed models	7 to 10
Predict	Who will consider dropping out?	Accuracy on new cases	Machine learning	11 to 16

Keep the aim in mind, because the same numbers can serve different aims. A regression model can *explain* GPA (which predictors matter, and how much?) or *predict* it (how close are the predictions for new students?). The model may be identical; the question, and how the answer is judged, are not.

5.2 - From a topic to a research question

In one sentence: a research question is a topic narrowed until data can answer it.

Nobody starts with a research question. They start with a **topic**, something that interests or worries them, such as "the wellbeing of graduate students". A topic is too broad to study: it contains hundreds of possible questions. The work of the first months of a thesis is narrowing it:

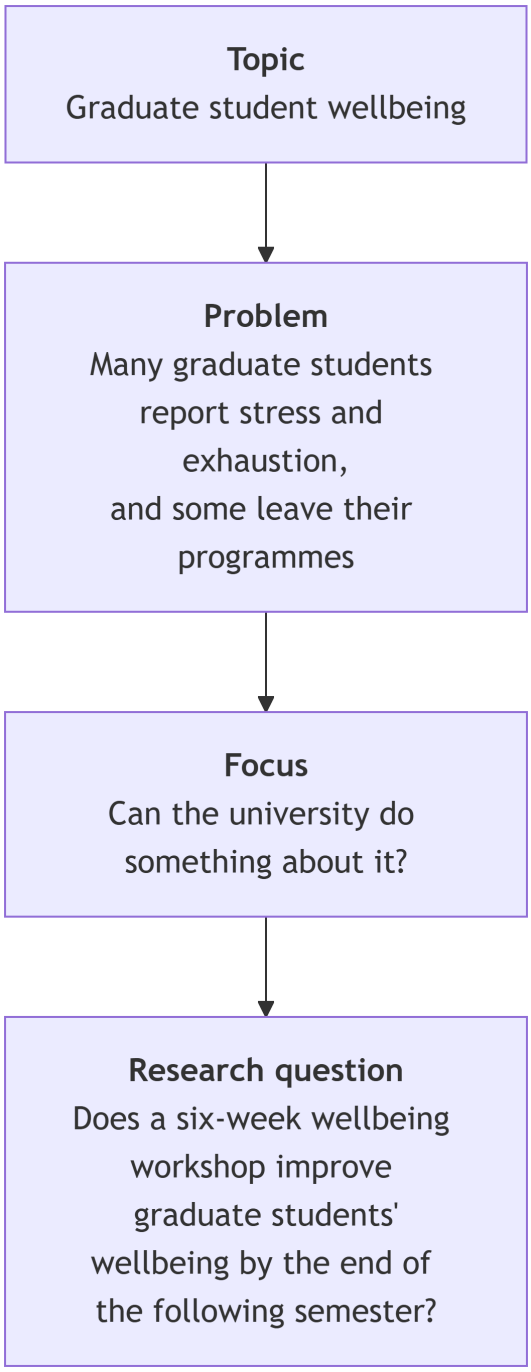


Figure 5.1: Narrowing a topic into a research question. Each step makes the question more specific, until it is clear what data would answer it.

The **problem** says why the topic matters: something is wrong, unknown, or disputed. The **research question** says precisely what the study will find out. A good research question is **specific**: it names who (graduate students at one university), what (wellbeing, measured with an index), and when (the end of semester 2). It is **answerable with data**, in the sense that the researcher can say what data would answer it and can collect that data. It is **not already answered**, because the literature leaves a gap or the setting is new. It is **feasible** within the time, money, skills, and access available. And it is **worth answering**: someone would act differently depending on the answer. Table 5.2 shows weak questions and how they can be improved.

Table 5.2: Weak research questions, and how to improve them

Weak question	Problem	Better question
Are students stressed?	Stressed compared with what? Which students?	Do part-time graduate students report higher stress than full-time students?
What affects students' success?	Too broad; "success" is undefined	Is average sleep in a semester associated with semester GPA, allowing for study hours?
Is the workshop good?	"Good" cannot be measured	Does the workshop raise wellbeing scores at the end of the following semester?
Why do students drop out?	Needs students who left, and years of follow-up	Which baseline characteristics are associated with *considering* dropout in the first year?

5.2.1 - Three kinds of question

Research questions come in three kinds, and each kind needs a different design and analysis. **Descriptive questions** ask *what is*, such as how many hours graduate students sleep; they need a sample that represents the population well. **Relational questions** ask *what goes with what*, such as whether students who sleep more have higher GPAs; they need measurements of both variables, and they establish association, not cause. **Causal questions** ask *what leads to what*, such as whether the workshop improves wellbeing; they need a design that rules out other explanations, ideally an experiment.

The kind of question decides the kind of claim you can make. Much confusion in published research comes from asking a relational question and answering it in causal words ("sleep *improves* grades"). The section on study designs below returns to this.

5.3 - From a question to a hypothesis

In one sentence: a hypothesis is a prediction precise enough to be wrong.

A research question asks; a **hypothesis** answers in advance. It is the researcher's best prediction, based on theory and earlier studies, of what the data will show. The workshop question becomes:

> **Hypothesis:** Students invited to the workshop will have higher wellbeing at the end of semester 2 than students not invited.

The hypothesis is useful because the data can disagree with it.

5.3.1 - Falsifiability

The philosopher Karl Popper argued that what separates a scientific claim from other claims is not that it can be proved, but that it can be **refuted**: it forbids certain results [6]. A claim that is compatible with every possible result tells us nothing. Compare:

Hypothesis	Could any result contradict it?
"The workshop changes how students feel."	No: higher, lower, or unchanged wellbeing all fit. Not falsifiable.
"Invited students have higher wellbeing."	Yes: if their wellbeing is lower or the same, the hypothesis is wrong. Falsifiable.

A hypothesis must also say whether it predicts a direction. A **directional hypothesis** predicts which group will be higher, or whether a relationship is positive or negative ("the workshop raises wellbeing"; "sleep is positively

associated with GPA"). A **non-directional hypothesis** predicts a difference without specifying which way ("stress differs between faculties"). Directional hypotheses are stronger, because they forbid more results.

5.3.2 - The null hypothesis

Statistical tests do not test the researcher's hypothesis directly. They test its opposite, the **null hypothesis** (H_0): the claim that nothing is going on, no difference and no relationship. The researcher's hypothesis becomes the **alternative hypothesis** (H_1). For the workshop, H_1 states that invited and not-invited students differ in average wellbeing at the end of semester 2, and H_0 that they have the same average wellbeing.

Testing the opposite of what one believes may seem strange, but the null hypothesis has one great advantage: it is precise enough to calculate with. "The workshop helps" does not say how much, but "the workshop makes no difference" makes a sharp prediction: any difference between the groups is due to chance alone, to which students happened to be invited. If we know how large chance differences usually are, we can ask whether the difference in the data is larger than chance would easily produce.

You can see this world of chance by simulation. Suppose the workshop truly does nothing. Then being invited is just an arbitrary label on students whose wellbeing is unaffected. The code below creates 300 imaginary students with wellbeing scores similar to those in the study (average 60, standard deviation about 11), splits them into two random groups of 150, and records the difference between the group averages. It then repeats this 5,000 times:

```
np.random.seed(2026)
n_sims = 5000
chance_differences = np.empty(n_sims)
groups = np.repeat(["Invited", "Not invited"], 150)

for i in range(n_sims):
    wellbeing = np.random.normal(loc=60, scale=11, size=300)
    shuffled = np.random.permutation(groups)
    diff = (
        wellbeing[shuffled == "Invited"].mean()
        - wellbeing[shuffled == "Not invited"].mean()
    )
    chance_differences[i] = diff

pd.Series(chance_differences).describe().round(3)
```

```
count    5000.000
mean       -0.008
std         1.268
min        -5.413
25%        -0.842
50%        -0.011
75%         0.856
max         4.028
dtype: float64
```

In Python, `np.random.normal()` draws random numbers from a Gaussian distribution, `np.random.permutation()` randomly shuffles group assignment labels, and a vectorized loop accumulates the null distribution. Figure 5.2 shows the 5,000 differences:

```
q025, q975 = np.percentile(chance_differences, [2.5, 97.5])

fig, ax = plt.subplots(figsize=(7, 4))
sns.histplot(chance_differences, binwidth=0.5, color="grey", edgecolor="white", ax=ax)
ax.axvline(q025, linestyle="--", color="black", linewidth=1.2)
ax.axvline(q975, linestyle="--", color="black", linewidth=1.2)
ax.set_xlabel("Difference in average wellbeing (invited minus not invited)")
ax.set_ylabel("Number of simulated studies")
plt.tight_layout()
plt.show()
```

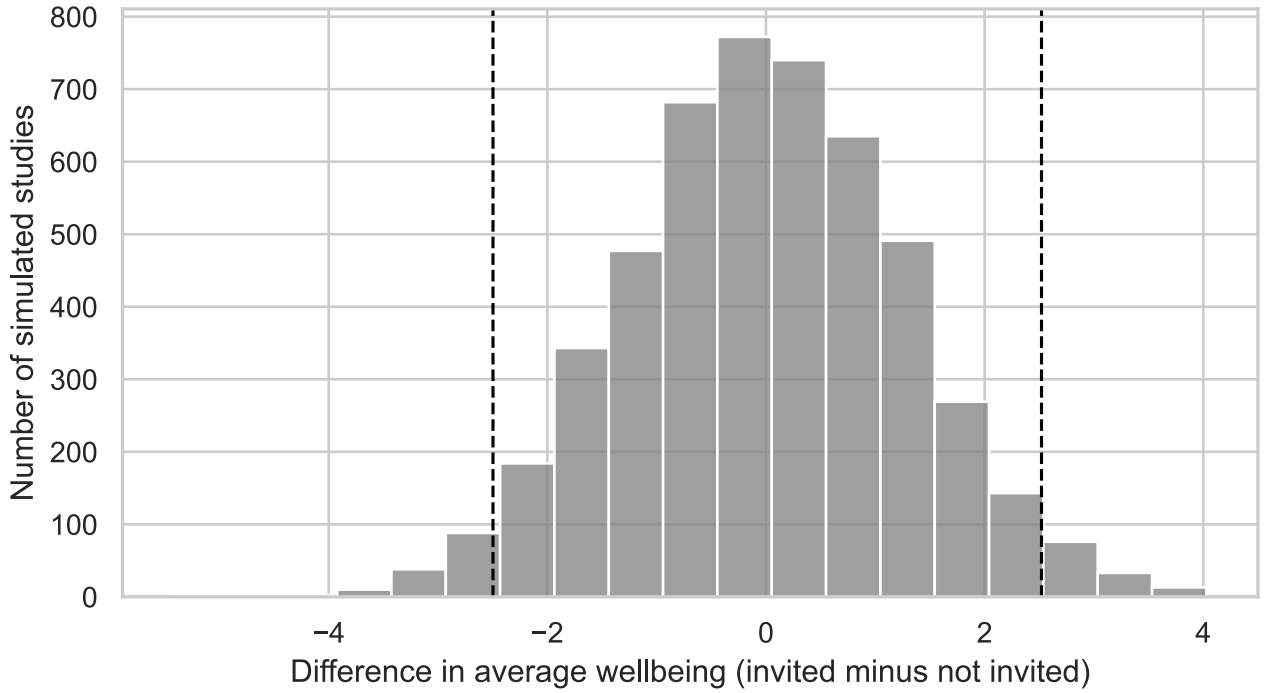


Figure 5.2: The null hypothesis world: differences between two random groups of 150 students when the workshop does nothing. Most are within about 2.5 points of zero.

Even when the workshop does nothing, the two groups almost never have exactly the same average: chance alone produces differences, usually small ones. In 95% of the simulated studies, the difference lies between −2.5 and 2.5 points (the dashed lines). A real study that found a difference of 5 points would therefore be hard to explain by chance, and would count as evidence against H_0. A difference of 1 point would not: it is exactly what chance produces. Chapter 7 turns this idea into the formal p-value, using the real empirical data.

Two things follow from this picture. First, **rejecting H_0 is not proving H_1**: it says only that "nothing is going on" is a poor explanation of the data. Second, **not rejecting H_0 is not proving it**: a small study can fail to detect a real effect, just as a blurred photograph can fail to show a real face.

5.3.3 - Hypotheses for the whole study

Table 5.3 writes out the study's research questions in this form. Some questions are **confirmatory**: they test a hypothesis stated in advance. Others are **exploratory**: they look for patterns without a prior prediction, and their results suggest hypotheses for future studies rather than confirming them. Both are legitimate, as long as they are labelled honestly.

Table 5.3: The study's research questions as hypotheses. Descriptive, exploratory, and predictive questions have no null hypothesis; they are judged in other ways.

Research question	Hypothesis (H_1)	Null hypothesis (H_0)	What would count against H_1	Chapters
RQ1 What does graduate life look like?	Descriptive: no hypothesis	none	none	4, 6
RQ2 Do students sleep less than 7 hours?	Average sleep is below 7 hours	Average sleep is 7 hours	An average of 7 hours or more, or a confidence interval that includes 7	7
RQ3 Does the workshop improve wellbeing?	Invited students have higher wellbeing in semester 2	The groups have the same average wellbeing	A difference near zero or negative, with a narrow confidence interval	7, 10
RQ4 Do faculties and study modes differ?	Stress differs between faculties (non-directional)	All faculties have the same average stress	Similar averages in every faculty	7, 8
RQ5 What explains GPA?	More sleep goes with a higher GPA, allowing for study hours, stress, and support	The sleep coefficient is zero	A coefficient near zero or negative	8
RQ6 Do the questionnaire items measure what they should?	The 22 items form four scales: stress, burnout, support, satisfaction	none (checked by model fit, not a single test)	Items that do not group as intended	9
RQ7 Are there student profiles?	Exploratory: no hypothesis	none	none	9, 14
RQ8 How do wellbeing and GPA change?	Wellbeing declines over the two years	The average slope over semesters is zero	A slope near zero or positive	10
RQ9 Who considers dropping out?	Higher stress raises the odds of considering dropout	The odds ratio for stress is 1	An odds ratio of 1 or below	8, 12
RQ10 Can final GPA be predicted?	Predictive: year-1 data predicts better than the average alone	none (judged on new cases)	Test-set error no better than predicting the average	13
RQ11 What challenges do students describe?	Exploratory: no hypothesis	none	none	18

The later chapters return to this table: each restates its hypothesis before the analysis, and ends by saying whether the data contradicts the null hypothesis.

> **i** Think before you analyse
>
> From this chapter on, every main analysis in the book starts with the same four questions. Answer them in writing, before you run any code:
>
> 1. **What is the question**, and is it descriptive, relational, or causal?
> 2. **What is the unit of analysis?** Students, semesters, supervisors?
> 3. **Which variables**, of what level of measurement, play which role?
> 4. **What result would support the hypothesis, and what would count against it?**

5.4 - Variables and measurement

In one sentence: a variable is a decision about how to turn an idea into a number.

5.4.1 - Cases and the unit of analysis

As Chapter 2 showed, data is a table of **cases** (rows) and **variables** (columns). The first question about any dataset is what one row represents: the **unit of analysis**. The wellbeing study has three, in different tables:

```
print("students rows:   ", len(students))      # one row per student
print("semesters rows:  ", len(semesters))     # one row per student per semester
print("supervisors rows:", len(supervisors))   # one row per supervisor
```

```
students rows:    600
semesters rows:   2326
supervisors rows: 120
```

The unit of analysis decides what a question means. "Do students who sleep more have higher GPAs?" is a question about students, so each student should count once, for example with their first-semester values or their average over the semesters. Treating the 2326 semester rows as 2326 independent cases would count each student up to four times, and make the evidence look stronger than it is. Chapter 10 shows how to use all the rows correctly with mixed models.

5.4.2 - From constructs to variables

Many things researchers care about cannot be observed directly: stress, wellbeing, motivation, intelligence, job satisfaction, quality of life. Such ideas are called **constructs**. To study a construct, you must decide how to measure it, a step called **operationalisation**. Every operationalisation is a choice, and a different choice could give a different answer.

Table 5.4 shows how the wellbeing study operationalises some of its constructs.

Table 5.4: How some of the study's constructs are measured

Construct	Operationalisation	Variable(s)
Sleep	Self-reported average hours per night in the semester	`sleep_hours`
Stress	The average of six questionnaire items on a 1 to 5 scale	`stress_1` to `stress_6`
Wellbeing	A 0 to 100 wellbeing index, from a validated instrument	`wellbeing`
Academic performance	Semester GPA from university records	`gpa`
Thoughts of dropping out	"Have you seriously considered leaving your programme?"	`considering_dropout`

Stress is a good example. No single question captures it, so the questionnaire asks six, each about one aspect, and averages the answers. Look at the items, and at how the answers of the first few students differ from item to item:

```
stress_cols = ["student_id", "stress_1", "stress_2", "stress_3", "stress_4", "stress_5",
"stress_6"]
questionnaire[stress_cols].head(5)
```

	student_id	stress_1	stress_2	stress_3	stress_4	stress_5	stress_6
0	S0001	3.0	2.0	4.0	4.0	3.0	5.0
1	S0002	4.0	3.0	NaN	2.0	5.0	3.0
2	S0003	4.0	4.0	2.0	1.0	4.0	3.0
3	S0004	4.0	5.0	5.0	1.0	4.0	5.0
4	S0005	4.0	4.0	4.0	3.0	4.0	2.0

One item, `stress_4` ("I feel confident handling problems in my studies"), is worded in the opposite direction: agreeing with it means *less* stress. Before averaging, it must be reversed, so that 5 becomes 1 and 1 becomes 5. You did this in Chapter 3:

```
stress_items = [f"stress_{i}" for i in range(1, 7)]
q_stress = questionnaire[stress_items].copy()
q_stress["stress_4"] = 6 - q_stress["stress_4"]

stress = pd.DataFrame({
    "student_id": questionnaire["student_id"],
    "stress": q_stress.mean(axis=1)
})
stress["stress"].describe().round(2)
```

```
count    600.00
mean       3.22
std        0.73
min        1.33
25%        2.67
50%        3.17
75%        3.67
max        5.00
Name: stress, dtype: float64
```

The resulting score is the study's operational definition of stress. Anyone who reads the thesis should be able to see exactly how it was built, which is why the methods chapter of a thesis describes every measure, and why the questionnaire items go in an appendix.

> **i** Self-report is a measurement choice
>
> Sleep, study hours, and caffeine in the wellbeing study are *self-reported*: students estimate them. People tend to overestimate their sleep and underestimate their caffeine, and they may give the answers they think are expected. Records (such as GPA from the university) or devices (such as a sleep tracker) avoid some of these problems, but are more expensive or intrusive. Every measure has weaknesses; say what they are.

5.4.3 - Levels of measurement

Not all numbers are numbers in the same sense. The psychologist S. S. Stevens proposed four **levels of measurement**, each allowing more than the one before [7]:

Table 5.5: Levels of measurement

Level	What the values mean	Meaningful summaries	Examples in the study	In Python
Nominal	Categories with no order	Counts, percentages, mode	`faculty`, `gender`, `workshop`	`object` / `pd.Categorical`
Ordinal	Ordered categories; distances between them unknown	Also median and percentiles	`employment`, `financial_worry`, Likert items	`pd.Categorical(ordered=True)`
Interval	Equal distances; no true zero	Also mean, standard deviation, differences	`wellbeing` (0 to 100 index), scale scores	`float64`
Ratio	Equal distances and a true zero	Also ratios ("twice as much")	`sleep_hours`, `study_hours`, `caffeine_mg`, `age`	`float64` / `int64`

The level decides which summaries and tests make sense. The average faculty is meaningless. "Twice as much caffeine" is meaningful (caffeine has a true zero); "twice as much wellbeing" is not (a score of 0 does not mean no wellbeing at all).

Python does not automatically infer the level of measurement; you tell it with Pandas types. Look at how Pandas initially reads `employment`:

```
students["employment"].value_counts()
```

```
employment
Part-time job    196
Full-time job     97
Name: count, dtype: int64
```

Without an explicit ordered category, Pandas sorts by frequency or arbitrary order. But employment is **ordinal**: none, part-time, full-time is a real progression. Tell Pandas with an ordered Categorical:

```
students["employment"] = pd.Categorical(
    students["employment"],
    categories=["None", "Part-time job", "Full-time job"],
    ordered=True
)
students["employment"].value_counts().sort_index()
```

```
employment
None               0
Part-time job    196
Full-time job     97
Name: count, dtype: int64
```

Now tables and graphs follow the natural order, and logical comparisons work correctly.

The opposite mistake is more common: an ordinal variable stored as raw integers, such as `financial_worry` (1 = not at all worried, 5 = extremely worried). Python will happily calculate its mean if requested, but the psychological distance from "not at all" to "a little" need not equal the distance from "very" to "extremely". For a single ordinal item, report the frequency distribution and the median:

```
print("Frequency counts:")
print(students["financial_worry"].value_counts(dropna=False).sort_index())
print("\nMedian financial worry:", students["financial_worry"].median())
```

```
Frequency counts:
financial_worry
1.0     84
2.0    139
3.0    172
4.0    111
5.0     56
NaN     38
Name: count, dtype: int64

Median financial worry: 3.0
```

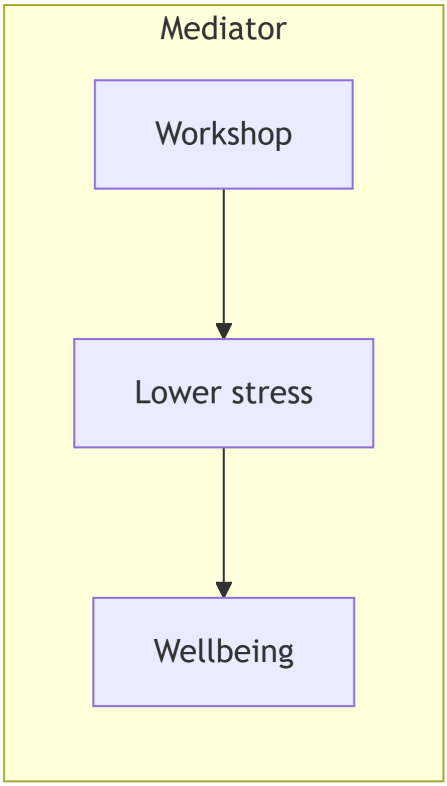
Scale scores, the average of several items, are usually treated as interval data, and the book does so. Averaging several items smooths out the unequal steps of each one. This is a convention with good empirical support, not a physical law; if an examiner asks, state that you followed it, and why.

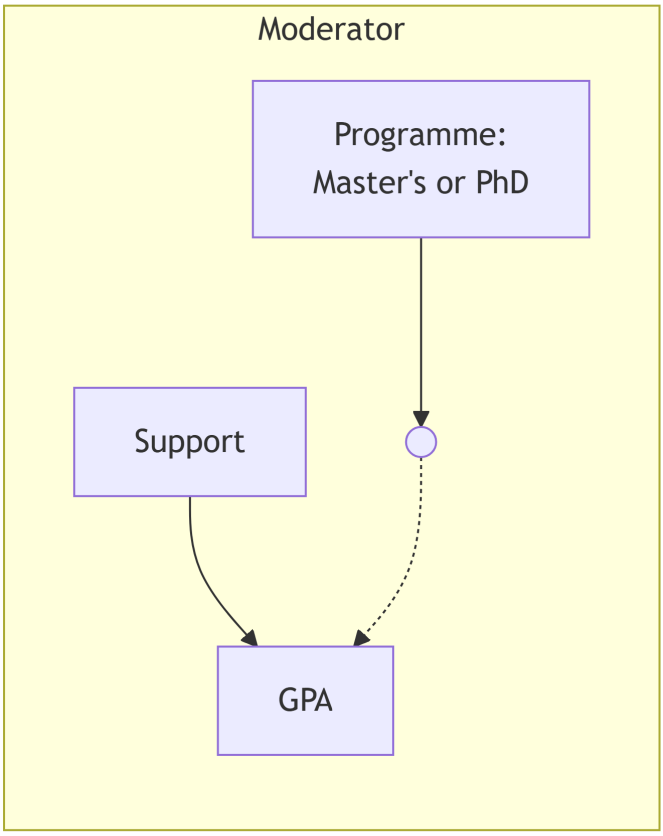
5.4.4 - The roles variables play

In an analysis, variables also have **roles**. The **outcome**, or dependent variable, is what the researcher wants to explain or predict, such as wellbeing, GPA, or considering dropout. A **predictor**, also called an independent or explanatory variable, is what is used to explain or predict it, such as the workshop, sleep, or stress. Three further roles concern a third variable that affects the relationship between the two. A **confounder** is related to both the predictor and the outcome, and can create a misleading association between them. A **moderator** changes the strength or direction of a relationship, so that the effect of the predictor depends on it. A **mediator** lies on the path between predictor and outcome: it is *how* the predictor has its effect.

The roles are not permanent properties of the variables; they come from the research question. Stress is an outcome in one question (whether the workshop reduces stress) and a predictor in another (whether stress predicts dropout).

Figure 5.3 draws the three less familiar roles as diagrams, using examples from the study: arrows show which variable is thought to influence which.





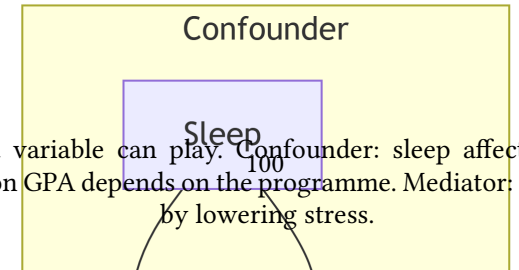


Figure 5.3: Three roles a third variable can play. Confounder: sleep affects both caffeine intake and GPA. Moderator: the effect of support on GPA depends on the programme. Mediator: the workshop may raise wellbeing by lowering stress.

The confounder is the most important of the three, because it can produce an association that is not a cause. In the wellbeing data, students who take more caffeine have lower GPAs. But students who take more caffeine also sleep less:

```
first_sem = semesters[semesters["semester"] == 1].merge(students, on="student_id")
corr_matrix = first_sem[["caffeine_mg", "sleep_hours", "gpa"]].corr().round(2)
corr_matrix
```

	caffeine_mg	sleep_hours	gpa
caffeine_mg	1.00	−0.64	−0.19
sleep_hours	−0.64	1.00	0.25
gpa	−0.19	0.25	1.00

Caffeine and GPA are negatively correlated (-0.19), but caffeine and sleep are strongly negatively correlated (-0.64). Caffeine might be harming grades, or short-sleeping students might both drink more coffee and get lower grades; the correlation alone cannot tell which. Chapter 8 answers the question with a multiple regression that compares students with the same amount of sleep. The general lesson is that before you interpret any relationship, you should ask what else could produce it, and measure it if you can.

5.5 - Reliability and validity

In one sentence: a reliable measure gives consistent results; a valid measure measures the right thing.

Two properties decide whether a measure can be trusted. **Reliability** is consistency: a student should get a similar stress score if they filled in the questionnaire again next week, with nothing changed, and the six stress items should agree with each other. **Validity** is whether the measure captures what it claims to measure: the stress score should reflect stress, and not something else, such as general negativity or tiredness on the day of the survey.

The classic picture is a target. Each shot is a measurement, and the centre is the true value. The code below simulates four measures, each shooting 30 times, and draws them:

```
np.random.seed(1)

panels = [
    ("Reliable and valid", 0.0, 0.0, 0.15),
    ("Reliable but not valid", 0.9, 0.8, 0.15),
    ("Valid on average, not reliable", 0.0, 0.0, 0.70),
    ("Neither reliable nor valid", 0.8, -0.6, 0.70)
]

fig, axes = plt.subplots(2, 2, figsize=(7, 7))
axes = axes.flatten()

theta = np.linspace(0, 2 * np.pi, 200)

for ax, (title, cx, cy, spread) in zip(axes, panels):
    for r in [0.5, 1.0, 1.5]:
      ax.plot(r * np.cos(theta), r * np.sin(theta), color="grey", linewidth=0.8, linestyle=":")
    ax.axhline(0, color="grey", linewidth=0.5, linestyle=":")
    ax.axvline(0, color="grey", linewidth=0.5, linestyle=":")
    ax.scatter(0, 0, marker="+", s=80, color="black", zorder=3)

    x = np.random.normal(cx, spread, 30)
    y = np.random.normal(cy, spread, 30)
    ax.scatter(x, y, color="#b2182b", alpha=0.75, s=35, zorder=4)

    ax.set_title(title, fontsize=11, weight="bold")
    ax.set_xlim(-1.8, 1.8)
    ax.set_ylim(-1.8, 1.8)
    ax.set_aspect("equal")
```

```
    ax.axis("off")

plt.tight_layout()
plt.show()
```

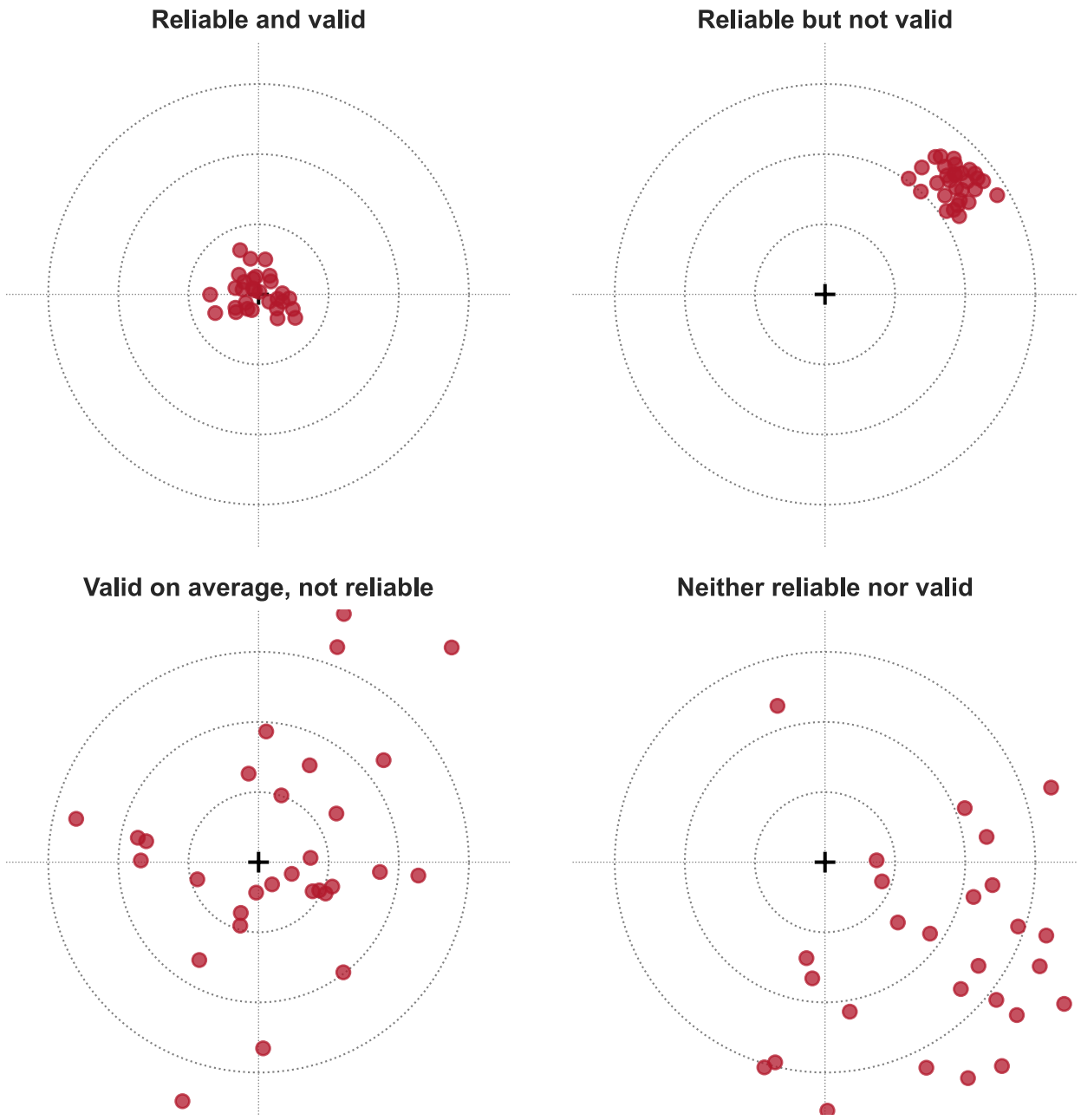


Figure 5.4: Reliability and validity as shots at a target. Reliable measures are tightly grouped; valid measures are centred on the true value. A measure can be reliable without being valid, but not valid without being reasonably reliable.

A measure that is reliable but not valid is precise about the wrong thing: a bathroom scale that always shows two kilograms too much. A measure that is unreliable cannot be very valid, because its random errors hide whatever it measures. Reliability is therefore necessary for validity, but not enough.

5.5.1 - Kinds of reliability

Reliability can be assessed in three main ways. **Internal consistency** is the extent to which the items of a scale agree with each other; it is measured with Cronbach's alpha (Chapter 9). **Test-retest reliability** is the extent to which the same person gets a similar score on two occasions when nothing has changed. **Inter-rater reliability** is the extent to which two people who code the same material agree; it is measured with Cohen's kappa (Chapter 18, where a language model is one of the coders).

5.5.2 - Kinds of validity

Validity has several meanings. For a measure, **content validity** concerns whether the items cover the whole construct: a stress scale about deadlines only would miss stress about money or family. **Construct validity** concerns whether the score behaves as the construct should. It should correlate with measures of related constructs (**convergent validity**) and correlate less with unrelated ones (**discriminant validity**) [8].

The second kind can be checked in the wellbeing data. If the stress score measures stress, it should correlate strongly with burnout (a closely related construct), less strongly and negatively with supervisor support and satisfaction:

```
q_clean = questionnaire.copy()
q_clean["stress_4"] = 6 - q_clean["stress_4"]

scale_scores = pd.DataFrame({
    "stress": q_clean[[f"stress_{i}" for i in range(1, 7)]].mean(axis=1),
    "burnout": q_clean[[f"burnout_{i}" for i in range(1, 7)]].mean(axis=1),
    "support": q_clean[[f"support_{i}" for i in range(1, 7)]].mean(axis=1),
    "satisfaction": q_clean[[f"satisfaction_{i}" for i in range(1, 5)]].mean(axis=1)
})
validity_corr = scale_scores.corr().round(2)
validity_corr
```

	stress	burnout	support	satisfaction
stress	1.00	0.65	−0.33	−0.41
burnout	0.65	1.00	−0.24	−0.38
support	−0.33	−0.24	1.00	0.41
satisfaction	−0.41	−0.38	0.41	1.00

Stress and burnout correlate at 0.65: related, as expected, but far from identical, so the two scales are not measuring the same thing twice. The negative correlations with support and satisfaction fit the idea that stressed students feel less supported and less satisfied. This is not proof of validity, which is built from many pieces of evidence, but it is the kind of evidence a thesis should report. Chapter 9 tests the structure of the whole questionnaire with factor analysis.

For a study as a whole, two further kinds of validity matter. **Internal validity** is the extent to which a study can rule out other explanations for its results; it is highest in randomised experiments. **External validity** is the extent to which the results apply beyond the study, to other people, places, and times; it depends on the sample. The next two sections are about these.

5.6 - Study designs

In one sentence: the design, not the statistics, decides whether a study can show cause.

A **study design** is the plan for who is measured, on what, when, and under which conditions. Table 5.6 summarises the main designs.

Table 5.6: The main study designs

Design	What happens	Can it show cause?	In the wellbeing study
Randomised experiment	The researcher assigns the treatment by chance	Yes, for the treatment	The workshop invitation
Quasi-experiment	Groups receive different treatments, but not by chance	Only with care; other differences must be ruled out	Comparing students who *chose* to attend many sessions
Observational, cross-sectional	Variables measured once, at one time	No; association only	The baseline questionnaire
Observational, longitudinal	The same people measured repeatedly over time	Stronger evidence of order, but still association	The four semesters

5.6.1 - Why random assignment works

Suppose the workshop had been open to anyone who wanted to come. The students who came would probably have had more energy, more time, and less stress to start with. If the attenders later had higher wellbeing, the cause could be the workshop or who they already were, and the two explanations could not be separated. This is **self-selection**, a form of confounding.

A simulation makes the problem visible. Take the real stress scores of the 600 students, and imagine that less stressed students are more likely to volunteer. Then compare the baseline stress of volunteers and non-volunteers, and do the same for a random invitation:

```
np.random.seed(11)
stress_scores = scale_scores["stress"].to_numpy()

# Volunteering: the chance of volunteering falls as stress rises
chance_to_volunteer = expit(-2 * (stress_scores - stress_scores.mean()))
volunteered = np.random.uniform(size=600) < chance_to_volunteer

# Random invitation: exactly 300 invited by coin toss
invited = np.random.permutation(np.repeat([True, False], 300))

balance_comparison = pd.DataFrame({
    "design": ["Volunteers", "Random invitation"],
    "stress_in": [stress_scores[volunteered].mean(), stress_scores[invited].mean()],
    "stress_out": [stress_scores[~volunteered].mean(), stress_scores[~invited].mean()]
})
balance_comparison["difference"]         =         balance_comparison["stress_in"]         -
balance_comparison["stress_out"]
balance_comparison.round(2)
```

	design	stress_in	stress_out	difference
0	Volunteers	2.81	3.59	−0.78
1	Random invitation	3.23	3.20	0.03

In this code, `expit()` from `scipy.special` turns any number into a probability between 0 and 1 via the standard logistic function, and `np.random.uniform()` draws a random number between 0 and 1 for each student; a student volunteers when their random draw is below their probability. The volunteers start the study 0.78 points less stressed than the others, *before any workshop*. Any later difference in wellbeing would mix the workshop's effect with this head start. With random invitation, the two groups start almost identical, because chance does not favour any kind of student.

Randomisation balances not only the variables you measured, but also the ones you did not: motivation, family support, health, and everything else. That is why a randomised experiment can support a causal claim. The real invitation in the wellbeing study, which was random, can be checked in the same way:

```
merged_base = students.merge(stress, on="student_id")
balance_table = merged_base.groupby("workshop").agg(
    students=("student_id", "count"),
    average_age=("age", "mean"),
    percent_female=("gender", lambda g: (g == "Female").mean() * 100),
    percent_phd=("programme", lambda p: (p == "PhD").mean() * 100),
    financial_worry=("financial_worry", "mean"),
    stress=("stress", "mean")
).round(1)
balance_table
```

workshop	students	average_age	percent_female	percent_phd	financial_worry	stress
Invited	300	29.6	50.3	28.7	2.8	3.2
Not invited	300	29.8	53.7	29.3	2.9	3.2

The two groups look very similar at baseline. A balance table like this belongs in any thesis with an experiment: it shows the reader that the randomisation worked.

> ⚠ Randomisation is about the treatment only
>
> The random invitation allows causal claims about **the workshop invitation**, and nothing else. The study's other questions, about sleep, stress, caffeine, and GPA, are observational: nobody assigned students their sleep. For those, the thesis can report associations, allow for the confounders that were measured, and discuss the ones that were not. Note, too, that what was randomised is the *invitation*, not attendance: many invited students attended only some sessions. Comparing students by the number of sessions they attended is a quasi-experiment again, because students chose how many sessions to attend.

5.6.2 - Cross-sectional and longitudinal designs

A **cross-sectional** design measures everything at one time. It is quick and cheap, but it cannot show which came first: stress might reduce sleep, or short sleep might produce stress. A **longitudinal** design measures the same people repeatedly, so it can show change and the order of events. The four semesters make the wellbeing study longitudinal, which Chapter 10 uses to model how wellbeing changes. The price is **attrition**: some people leave the study, and those who leave are rarely a random selection, as the next section shows.

5.7 - Samples and populations

In one sentence: a large sample reduces random error, but not bias.

The **population** is everyone the conclusions are meant to apply to: graduate students, perhaps at one university, perhaps everywhere. The **sample** is the people actually studied. Since the results come from the sample and the conclusions are about the population, the central question is always how well the sample represents the population.

5.7.1 - Ways to draw a sample

In **simple random sampling**, every member of the population has the same chance of being chosen. It is the ideal, and rarely possible, because it needs a complete list of the population, called a **sampling frame**. In **stratified sampling**, the population is divided into groups (strata), such as faculties, and a random sample is drawn from each, to make sure every group is represented. In **cluster sampling**, whole groups are sampled, such as all the students of randomly chosen supervisors; it is cheaper, but students in the same cluster are alike, which the analysis must allow for (Chapter 10). **Convenience sampling** takes whoever is easy to reach, such as the people who answer an online survey shared on social media, or the students in the researcher's own classes. It is very common in theses, and it is the weakest basis for generalising.

5.7.2 - Random error and bias

A sample estimate can be wrong in two ways. **Random error** is the difference caused by which people happened to be selected: a different random sample would give a slightly different answer. **Bias** is a systematic difference: the sampling method tends to select certain kinds of people, so the estimate is off in one direction, every time.

The difference matters because they respond differently to sample size. A simulation shows it. Treat the study's 600 students as a complete population, whose true average first-semester wellbeing is 60.5. Draw many samples in two ways: at random, and as a convenience sample where students with higher wellbeing are more likely to answer (a common pattern: people who are struggling are less likely to fill in surveys). Try samples of 50 and 200:

```
population = first_sem["wellbeing"].dropna().to_numpy()
true_mean = population.mean()

# Probability of answering voluntary survey rises with wellbeing
prob_voluntary = expit((population - true_mean) / 10)
prob_voluntary /= prob_voluntary.sum()

np.random.seed(5)
sim_records = []
for n in [50, 200]:
    for _ in range(2000):
        rand_idx = np.random.choice(len(population), size=n, replace=False)
        sim_records.append({
            "sample_size": f"{n} students",
            "method": "Random",
            "estimate": population[rand_idx].mean()
        })
        vol_idx = np.random.choice(len(population), size=n, replace=True, p=prob_voluntary)
        sim_records.append({
            "sample_size": f"{n} students",
            "method": "Voluntary",
            "estimate": population[vol_idx].mean()
        })

sampling_results = pd.DataFrame(sim_records)

sampling_summary = sampling_results.groupby(["method", "sample_size"]).agg(
    average_estimate=("estimate", "mean"),
    spread=("estimate", "std")
).round(2)
sampling_summary
```

		average_estimate	spread
method	sample_size		
Random	200 students	60.45	0.69
	50 students	60.47	1.61
Voluntary	200 students	66.11	0.76
	50 students	66.11	1.49

In this simulation, `np.random.choice()` with sampling probabilities `p` selects students with unequal chances, replicating a voluntary survey. Figure 5.5 draws them:

```
g = sns.displot(
    data=sampling_results,
    x="estimate",
    hue="method",
    row="sample_size",
    kind="hist",
```

```
    binwidth=0.4,
    palette={"Random": "grey", "Voluntary": "#d6604d"},
    alpha=0.6,
    height=2.8,
    aspect=2.4
)
for ax in g.axes.flat:
    ax.axvline(true_mean, color="black", linestyle="--", linewidth=1.2)
g.set_axis_labels("Estimated average wellbeing", "Number of samples")
plt.show()
```

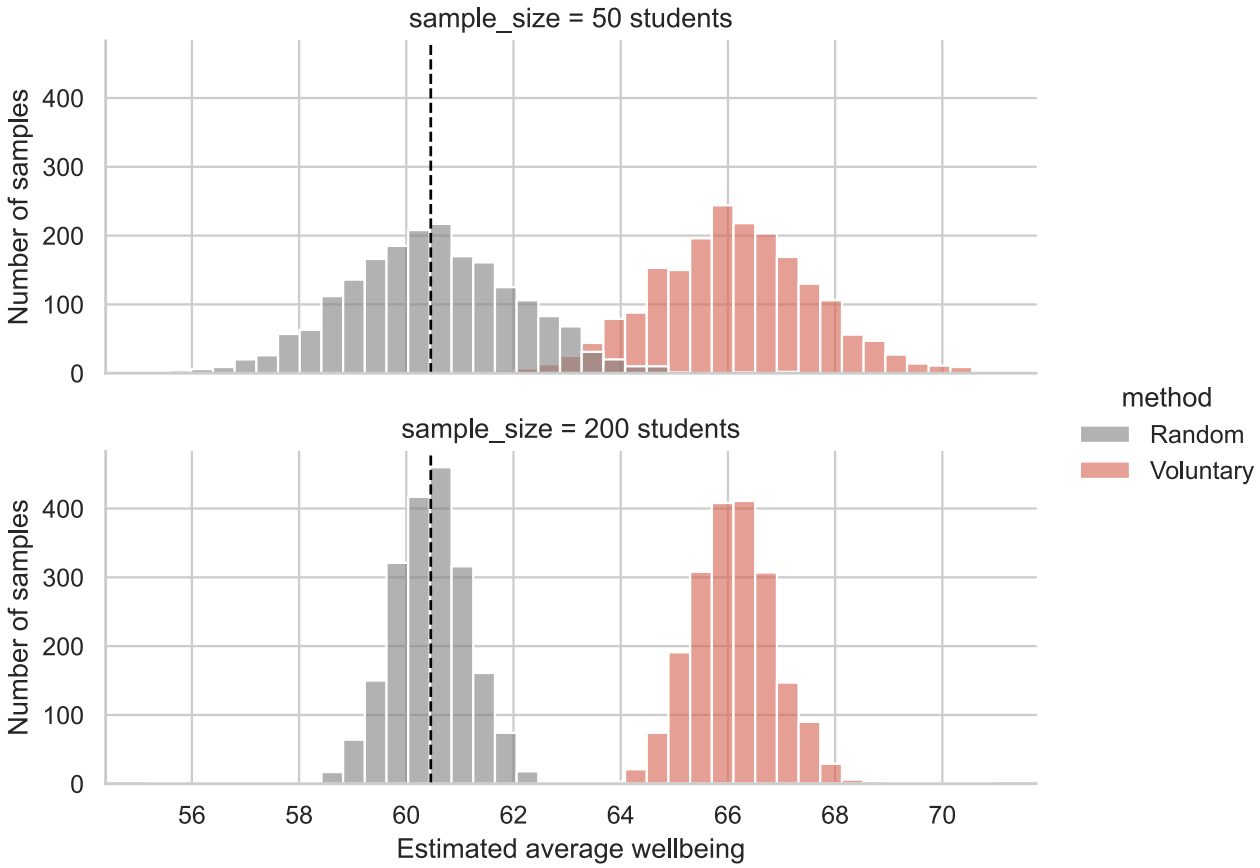


Figure 5.5: Estimates of average wellbeing from 2,000 samples, drawn at random or by voluntary response. The dashed line is the true population average. A larger sample narrows the spread (random error) but does not move a biased estimate back to the truth.

Random samples are centred on the true value; larger random samples are simply more precise. Voluntary samples miss the true value, and the larger voluntary sample misses it just as much, only more confidently. **A biased sample cannot be fixed by making it bigger.** The only remedies are a better sampling method, or knowing enough about the bias to correct for it.

5.7.3 - Non-response and attrition

The same problem appears in the real data. Not every student answered the final open-ended question, and some left the study after the first year. The comparison below shows whether the students who remained are like those who did not:

```
sem4_ids = set(semesters.loc[semesters["semester"] == 4, "student_id"])
leavers = set(students["student_id"]) - sem4_ids
```

```
students_attrition = students.copy()
students_attrition["stayed"] = np.where(
    students_attrition["student_id"].isin(leavers),
    "Left after year 1",
    "Stayed"
)

attrition_table = students_attrition.groupby("stayed").agg(
    students=("student_id", "count"),
    percent_considered_dropout=("considering_dropout", lambda x: (x == "Yes").mean() * 100)
).round(1)
attrition_table
```

stayed	students	percent_considered_dropout
Left after year 1	37	59.5
Stayed	563	12.1

Of the 37 students who left, 60% had considered dropping out, compared with 12% of those who stayed. Any analysis of semesters 3 and 4 is therefore based on a group that is less likely to be struggling than the students who started. Chapter 6 describes this missing data in detail, and Chapter 10 uses a model that makes the best use of the incomplete records.

5.7.4 - Generalising from the sample

The wellbeing study's sample is all graduate students at one university who agreed to take part. Its results apply most directly to that university, and to others like it, only by argument: similar students, similar programmes, similar pressures. A thesis should say this plainly in its limitations section. Claiming less than the data allows is rarely criticised; claiming more is.

5.8 - Planning the analysis before the data

In one sentence: decide how you will judge your hypotheses before the data can influence you.

5.8.1 - An analysis plan

For every hypothesis, the plan names the variables, the analysis, and what would count as support. Writing it before collecting data forces you to check that you will collect everything you need, in a form you can analyse:

Table 5.7: Part of the study's analysis plan

Hypothesis	Outcome	Predictor(s)	Analysis	Support if
Students sleep less than 7 hours	`sleep_hours` (semester 1)	none	One-sample t-test against 7	The 95% confidence interval lies below 7
The workshop raises wellbeing	`wellbeing` (semester 2)	`workshop`	Two-sample t-test; mixed model over semesters	Invited students higher; interval excludes 0
Stress raises the odds of considering dropout	`considering_dropout`	stress score, with background variables	Logistic regression	Odds ratio above 1; interval excludes 1

5.8.2 - Sample size and power

The simulation of the null world showed that chance differences are smaller with larger groups. So the size of the sample decides how small an effect a study can detect. The **power** of a study is the probability that it detects an effect of a given size, if the effect is real. A common target is 80%.

Power depends on three things: the sample size, the size of the effect, and the significance level (α). Effect sizes are often expressed as Cohen's d: the difference between groups divided by the standard deviation (Chapter 7). Before her study, Elaf's reading suggested that brief wellbeing programmes improve wellbeing by roughly 0.4 to 0.5 standard deviations. In Python, `TTestIndPower` from `statsmodels.stats.power` calculates how many students each group needs to detect $d = 0.45$ with 80% power:

```
power_analyzer = TTestIndPower()
required_n = power_analyzer.solve_power(
    effect_size=0.45,
    power=0.80,
    alpha=0.05,
    alternative="two-sided"
)
print(f"Required students per group: {int(np.ceil(required_n))}")
```

```
Required students per group: 79
```

With `effect_size=0.45`, Cohen's d indicates the standardized mean shift. The answer, about 79 students per group, is well below the study's 300 per group. Smaller effects need far larger samples:

```
effect_sizes = {"small": 0.2, "medium": 0.5, "large": 0.8}
sample_sizes = {
    name: int(np.ceil(power_analyzer.solve_power(
        effect_size=d,
        power=0.80,
        alpha=0.05,
        alternative="two-sided"
    )))
    for name, d in effect_sizes.items()
}
pd.Series(sample_sizes, name="Students per group")
```

```
small     394
medium     64
large      26
Name: Students per group, dtype: int64
```

Halving the effect size roughly quadruples the sample needed. If the effect you expect is small, and your sample is small, a non-significant result tells you very little: the study could not have detected the effect even if it were there. Plan the sample size *before* collecting data, and report the calculation in the methods chapter. Chapter 7 returns to power, with a simulation that shows what it means.

5.8.3 - Preregistration and honest analysis

Every analysis involves many small decisions: which students to exclude, which variables to control for, how to handle outliers, which test to use. Made after seeing the data, each decision can be nudged, often unconsciously, towards the result the researcher hoped for. With enough such choices, "significant" findings appear from pure noise.

The protection is to decide in advance. **Preregistration** means writing the hypotheses and analysis plan down, with a date, before analysing the data, for example on the Open Science Framework [9]. Afterwards, analyses that follow the plan are reported as confirmatory, and anything else is reported honestly as exploratory. Preregistration does not forbid exploring; it only makes clear which results were predicted and which were found. Chapter 17 returns to this as part of reproducible research.

5.8.4 - Ethics

Research with people requires ethical approval before any data is collected. The committee will ask the questions in this chapter, too: what the question is, why the data is needed, how participants are recruited and informed, and how their data is stored and anonymised. A clear research question and analysis plan make the application much easier, and they justify collecting only the data you need.

Writing it up

The methods chapter of a thesis reports the decisions of this chapter, usually under the headings Design, Participants, Measures, and Analysis. A short example for the wellbeing study, with the numbers calculated directly in Python:

Design. A two-year longitudinal study with an embedded randomised experiment: at the end of semester 1, half the participants were randomly invited to a six-week wellbeing workshop.

Participants. 600 graduate students from 5 faculties took part, supervised by 120 supervisors; 37 (6%) left the study after the first year.

Measures. Stress, burnout, supervisor support, and academic satisfaction were measured at baseline with a 22-item questionnaire (1 = strongly disagree, 5 = strongly agree; items in Appendix A); scale scores are item averages, with one reverse-worded item reversed. Wellbeing (0 to 100), sleep, study hours, caffeine, and exercise were self-reported each semester; GPA was taken from university records.

Analysis. Hypotheses and the analysis plan were written before the data was analysed. With 300 students per group, the workshop comparison had 80% power to detect an effect of $d = 0.23$ at $\alpha = .05$.

The last sentence uses `TTestIndPower().solve_power()` solving for `effect_size`: given the sample size ($n = 300$) and the desired power (80%), it finds the smallest effect the study could reliably detect.

In your field: a laboratory experiment

The classic `ToothGrowth` dataset comes from an experiment on guinea pigs [10]. Sixty animals were assigned to receive vitamin C at one of three doses (0.5, 1, or 2 mg a day), by one of two methods (orange juice `OJ` or ascorbic acid `VC`), and the length of the odontoblast cells responsible for tooth growth was measured.

```
toothgrowth = dt.load_toothgrowth()
print("Animal counts by supplement and dose:")
print(pd.crosstab(toothgrowth["supp"], toothgrowth["dose"]))

print("\nAverage odontoblast cell length:")
print(toothgrowth.groupby(["supp", "dose"])["len"].mean().round(1))
```

```
Animal counts by supplement and dose:
dose  0.5  1.0  2.0
supp
OJ     10   10   10
VC     10   10   10

Average odontoblast cell length:
supp  dose
OJ    0.5     13.2
      1.0     22.7
      2.0     26.1
VC    0.5      8.0
      1.0     16.8
      2.0     26.1
Name: len, dtype: float64
```

The same planning applies. The **design** is an experiment with two factors and ten animals per combination. The **outcome**, `len`, is a ratio measure; the method of delivery, `supp`, is nominal; the dose is ratio in principle, but with only three values it is often treated as an ordered factor. A falsifiable **hypothesis**: *at the same dose, orange juice produces longer tooth cells than ascorbic acid*; H_0: *the method makes no difference*. It would count against the hypothesis if the ascorbic-acid groups were as long or longer. With only ten animals per group, the study has good power only for large effects, which is typical of laboratory work and a limitation worth stating.

5.9 - Common misconceptions

- **"A hypothesis is a question."** A hypothesis is a prediction: it says what the answer will be, precisely enough to be wrong.
- **"Rejecting the null hypothesis proves my hypothesis."** It shows only that "nothing is going on" explains the data poorly. Other explanations, including confounders, remain possible unless the design rules them out.
- **"A non-significant result proves there is no effect."** It may only mean the study was too small to detect it.
- **"A big sample makes the results representative."** Size reduces random error, not bias. A large convenience sample is still a convenience sample.
- **"If it is a number, I can average it."** The level of measurement decides which summaries are meaningful.
- **"A reliable measure is a valid measure."** A measure can be consistent and still measure the wrong thing.
- **"Correlation with a control variable proves cause."** Only randomisation balances the confounders you did not measure.

5.10 - Chapter review

5.10.1 - Summary

- Data analysis tests claims against evidence. It describes, explains, or predicts, and each aim is judged differently.
- A research question narrows a topic until data can answer it. Good questions are specific, answerable, feasible, and worth answering; they are descriptive, relational, or causal.

- A hypothesis is a prediction stated in advance. It must be falsifiable: it names the result that would count against it. Directional hypotheses predict a direction; non-directional ones predict a difference.
- Tests compare the data with the null hypothesis, the world where nothing is going on. Simulation shows the differences chance alone produces.
- Constructs are operationalised as variables. The unit of analysis says what a row is. Levels of measurement (nominal, ordinal, interval, ratio) decide which summaries make sense, and Python must be told about ordered categories via `pd.Categorical`.
- Variables play roles: outcome, predictor, confounder, moderator, mediator. Confounders can create associations that are not causes.
- Reliability is consistency; validity is measuring the right thing. Internal validity is about ruling out other explanations; external validity is about who the results apply to.
- Only randomised experiments support causal claims directly, because randomisation balances measured and unmeasured variables. Observational designs show associations; longitudinal designs show change and order.
- A larger sample reduces random error but not bias. Non-response and attrition can bias a sample.
- Plan the analysis before the data: hypotheses, variables, analyses, and sample size (power via `statsmodels.stats.power`). Preregistration separates confirmatory from exploratory results.

5.10.2 - Key terms

Research question, descriptive question, relational question, causal question, hypothesis, falsifiability, directional hypothesis, non-directional hypothesis, null hypothesis, alternative hypothesis, confirmatory analysis, exploratory analysis, unit of analysis, construct, operationalisation, self-report, level of measurement, nominal, ordinal, interval, ratio, ordered factor, outcome, predictor, confounder, moderator, mediator, reliability, validity, internal consistency, test-retest reliability, inter-rater reliability, content validity, construct validity, internal validity, external validity, study design, randomised experiment, quasi-experiment, self-selection, cross-sectional design, longitudinal design, attrition, population, sample, sampling frame, simple random sampling, stratified sampling, cluster sampling, convenience sampling, random error, bias, power, preregistration.

5.11 - Exercises

The playground has these and more, with hints and solutions.

1. Improve these research questions so that data could answer them: (a) “Is exercise good for students?” (b) “What makes supervisors effective?” For each, say whether your version is descriptive, relational, or causal.
2. Write a falsifiable hypothesis, its null hypothesis, and the result that would count against it, for the question: “Do students with children study fewer hours per week?”
3. For each variable in `students`, give its level of measurement. Then convert `financial_worry` into an ordered categorical with the labels “Not at all”, “A little”, “Moderately”, “Very”, and “Extremely”, and make a frequency table of it.
4. Change the null-world simulation to groups of 30 students instead of 150. Between which values do 95% of the chance differences now lie? What does this mean for a small study of the workshop?
5. Check the balance of the random workshop invitation on three other baseline variables: `study_mode`, `has_children`, and `lives_away`.
6. Use `statsmodels.stats.power.TTestIndPower` to find how many students per group a study needs to detect a difference of 0.3 standard deviations with 80% power, and with 90% power.

5.12 - Further reading

- *Introduction to Modern Statistics* [11], chapters 1 and 2, introduces variables, study designs, and sampling with many short examples.
- *Experimental and Quasi-Experimental Designs for Generalized Causal Inference* [12] is the standard reference on designs and the kinds of validity.
- “The preregistration revolution” [9] explains why and how to preregister, in a few pages.
- *The Logic of Scientific Discovery* [6], for falsifiability in Popper’s own words.

6. Descriptive Statistics and Exploratory Data Analysis

Before a study can test anything, it has to say what it found in the plainest sense: who took part, what a typical value of each variable looks like, how much the participants differ, and whether anything in the data is unusual or missing. These are the questions of **descriptive statistics**, the numbers that summarise what data looks like. They are the first results in almost every thesis, and they are not a formality. The shape of a variable decides which later tests are appropriate, an unusual value can dominate an analysis, and missing data can quietly change who the results are about.

This chapter develops the ideas behind the main descriptive statistics: what "typical" means, what spread measures, why the shape of a distribution matters, how to think about unusual and missing values, and how two variables can be described together. It answers the study's first research question, what graduate student life looks like (RQ1), with numbers, and checks the data for the problems that could mislead every later analysis. In the story, the data is now clean and has been graphed; Elaf is eager to test her hypotheses, and her supervisor asks her to describe the sample first.

By the end of this chapter you will be able to

- Explain the place of exploratory analysis in the research workflow.
- Choose a summary that suits the level of measurement and the shape of a variable.
- Summarise the centre and spread of a variable in Python, and explain what the standard deviation measures (including the difference between Pandas' and NumPy's degrees of freedom).
- Describe the shape of a distribution using skewness, and check it against the normal distribution with z-scores and Q-Q plots.
- Find unusual values using Tukey's box plot rule and z-score fences, and decide how to handle them ethically.
- Describe how much data is missing, why it is missing (MCAR, MAR, MNAR), and diagnose longitudinal attrition.
- Measure and interpret correlations using Pandas and Seaborn, and explain why correlation is not causation.
- Present a publication-ready description of your sample in a thesis, with every number accompanied by its spread and sample size.

Note

Teaching & Practice: Interactive Lecture Slides · Browser Playground Exercises

6.1 - The place of exploration

Chapter 5 distinguished three aims of data analysis: describing, explaining, and predicting. They correspond to three kinds of analysis. **Exploratory analysis** gets to know the data through summaries, graphs, unusual values, and missing values; it is open-ended, it proves nothing, but it often suggests questions worth testing, and this chapter is exploratory. **Inferential analysis** uses a sample to draw conclusions about a wider population and asks whether a result could be due to chance, such as whether the workshop improves wellbeing for graduate students in general, not just for these 600; Chapters 7 to 10 are inferential. **Predictive analysis** builds models that make accurate predictions for new cases, such as which students are likely to consider dropping out next year; Part 3 is predictive.

The three fit into one workflow, shown in Figure 6.1. Exploration comes first, and the researcher returns to it whenever a later result is surprising.

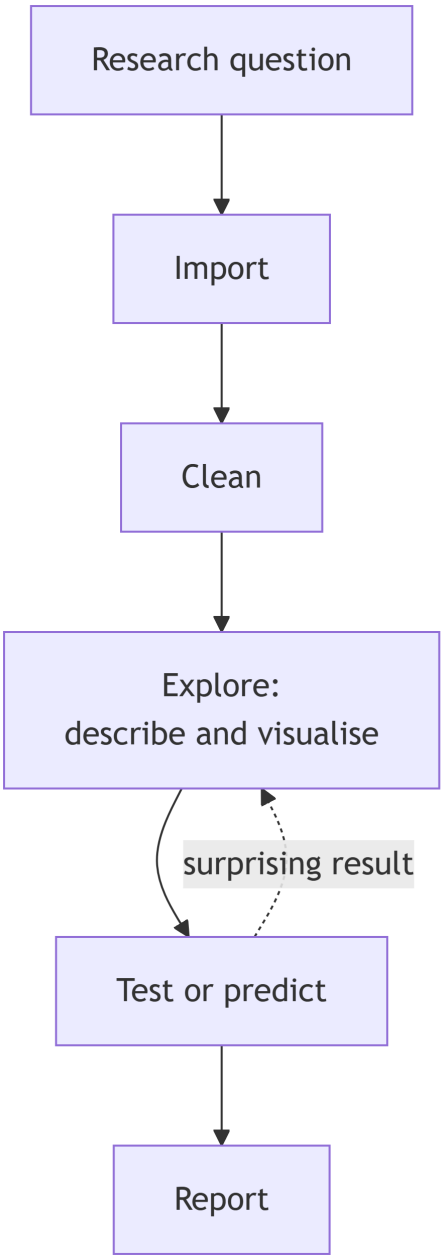


Figure 6.1: The research data analysis workflow. Exploration comes before testing and prediction, and you return to it when results surprise you.

6.2 - The centre of a distribution

The first thing to know about any variable is its typical value. Three measures describe it. The **mean** is the arithmetic average: add up the values and divide by how many there are. In mathematical notation, for values $x_1, x_2, \ldots, x_n$:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i \tag{6.1}$$

The **median** is the middle value when the values are sorted: half the values are below it and half above. The **mode** is the most common value, which is mainly useful for categories.

The mean and median can tell very different stories. Here is the daily caffeine intake, in milligrams, of five students represented as a Pandas Series:

```
import pandas as pd

caffeine = pd.Series([0, 120, 150, 180, 900])
print("Mean:  ", caffeine.mean())
print("Median:", caffeine.median())
```

```
Mean:   270.0
Median: 150.0
```

One student who consumes 900 mg a day pulls the mean up to 270 mg, higher than four of the five students. The median, 150 mg, stays with the typical student. The mean is sensitive to extreme values; the median is **robust** (resistant to outliers).

The same two measures for the study data use each student's first-semester record, together with their background information:

```
from data2thesis import students, semesters

# Merge first semester records with student demographics
first_sem = (
    semesters.query("semester == 1")
    .merge(students, on="student_id")
)

# Summarise centre for sleep and caffeine
centre_summary = first_sem[["sleep_hours", "caffeine_mg"]].agg(["mean", "median"]).round(1)
centre_summary
```

	sleep_hours	caffeine_mg
mean	6.5	189.4
median	6.5	160.0

For sleep, the mean and median are almost identical (around 6.4 hours). For caffeine, the mean (152.8 mg) is well above the median (128.5 mg), just as in the five-student example: a minority of heavy caffeine users pulls the mean up. Figure 6.2 shows why.

```
fig, ax = plt.subplots(figsize=(6.5, 3.6))

mean_val = first_sem["caffeine_mg"].mean()
median_val = first_sem["caffeine_mg"].median()

sns.histplot(data=first_sem,       x="caffeine_mg",       binwidth=50,       color="#b0bec5",
edgecolor="white", ax=ax)
ax.axvline(mean_val, color="#1f77b4", linewidth=2, label=f"Mean: {mean_val:.1f} mg")
ax.axvline(median_val,   color="#d62728",   linewidth=2,   linestyle="--",   label=f"Median:
{median_val:.1f} mg")

ax.set_xlabel("Caffeine per day (mg)")
ax.set_ylabel("Number of students")
ax.legend(frameon=True)
sns.despine(ax=ax)
plt.tight_layout()
plt.show()
```

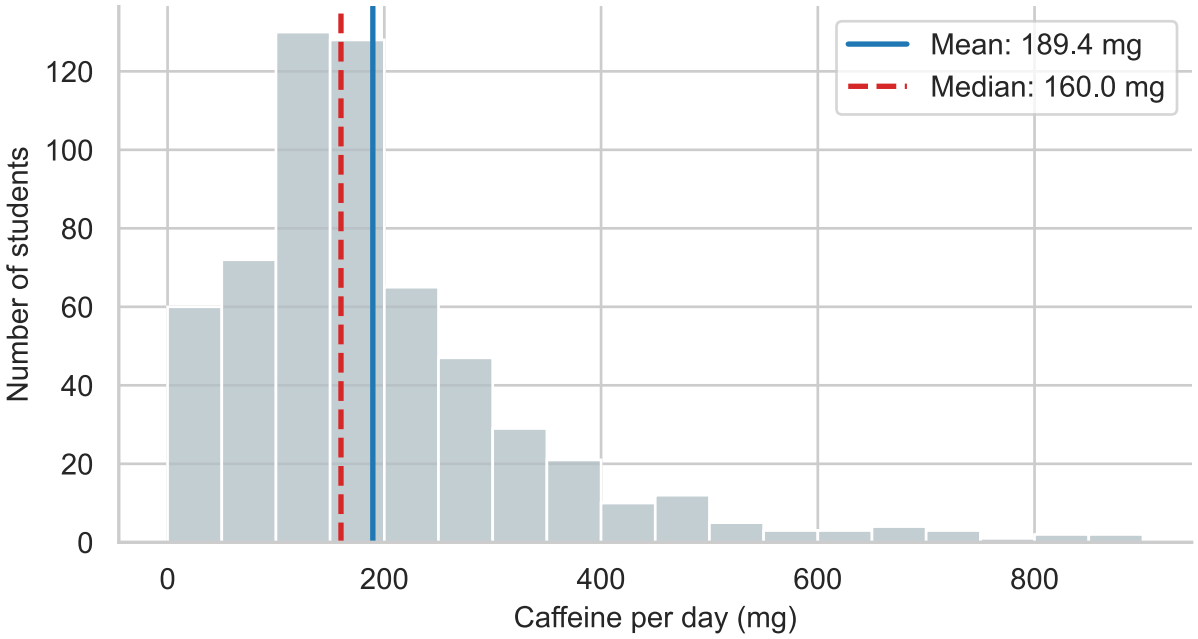


Figure 6.2: Daily caffeine intake in the first semester, with the mean (solid blue line) and median (dashed red line).

For categorical variables, the mode is simply the most common category. In Pandas, `.value_counts()` displays the counts sorted in descending order, with the mode at the very top:

```
students["faculty"].value_counts()
```

```
faculty
Health Sciences     154
Education           148
Social Sciences     116
Humanities           95
Natural Sciences     87
Name: count, dtype: int64
```

6.2.1 - Choosing a summary

Which summary is meaningful depends first on the level of measurement of the variable (Chapter 5), and then on its shape. An “average” faculty does not exist, so a nominal variable is described by counts and percentages. An ordinal variable, such as the five answers to the question about money worries, has a meaningful middle but unequal steps, so the median and percentages suit it better than the mean. Numeric variables can be described by the mean, but only when they are roughly symmetrical and without extreme values; otherwise the median describes the typical case more faithfully. Table 6.1 brings these rules together.

Table 6.1: Choosing a summary from the level of measurement and the shape of a variable

Variable	Centre	Spread	Examples in the study
Nominal	Mode; percentage in each category	None	Faculty, gender, study mode
Ordinal	Median; percentage in each category	Interquartile range (IQR)	Financial worry, Likert survey items
Numeric, roughly symmetrical	Mean	Standard deviation (SD)	Sleep hours, wellbeing score
Numeric, skewed or with extreme values	Median	Interquartile range (IQR)	Caffeine intake, student age

6.3 - The spread of a distribution

Two groups can have the same average and still be very different. The sleep of two small groups of students shows how:

```
group_a = pd.Series([6.0, 6.5, 6.5, 7.0, 6.5])
group_b = pd.Series([4.0, 8.5, 5.0, 9.0, 6.0])

print("Group A mean:", group_a.mean())
print("Group B mean:", group_b.mean())
```

```
Group A mean: 6.5
Group B mean: 6.5
```

Both average 6.5 hours, but in group A everyone sleeps about the same, while group B ranges from 4 to 9 hours. Measures of **spread** capture this difference, and a mean reported without one hides half of the story.

The **range** is the difference between the largest and smallest values. It is simple, but it depends only on the two most extreme values:

```
range_b = group_b.max() - group_b.min()
print("Group B range:", range_b)
```

```
Group B range: 5.0
```

6.3.1 - The standard deviation

The **standard deviation** (SD) measures how far values typically lie from the mean. The idea is easiest to see in a picture. Figure 6.3 shows the five students of group B, the mean as a dashed line, and each student's distance from the mean as a red vertical segment. These distances are the **deviations**, $x_i - \bar{x}$.

```
fig, ax = plt.subplots(figsize=(5.5, 3.4))

mean_b = group_b.mean()
students_idx = np.arange(1, 6)

ax.axhline(mean_b, color="#455a64", linestyle="--", linewidth=1.5, label=f"Mean ({mean_b} hrs)")
ax.vlines(students_idx, ymin=mean_b, ymax=group_b, color="#b2182b", linewidth=2, label="Deviations")
ax.plot(students_idx, group_b, "o", color="#1d588c", markersize=8, zorder=5)

ax.set_xticks(students_idx)
ax.set_xlabel("Student")
ax.set_ylabel("Hours of sleep")
```

```
ax.legend(frameon=True)
sns.despine(ax=ax)
plt.tight_layout()
plt.show()
```

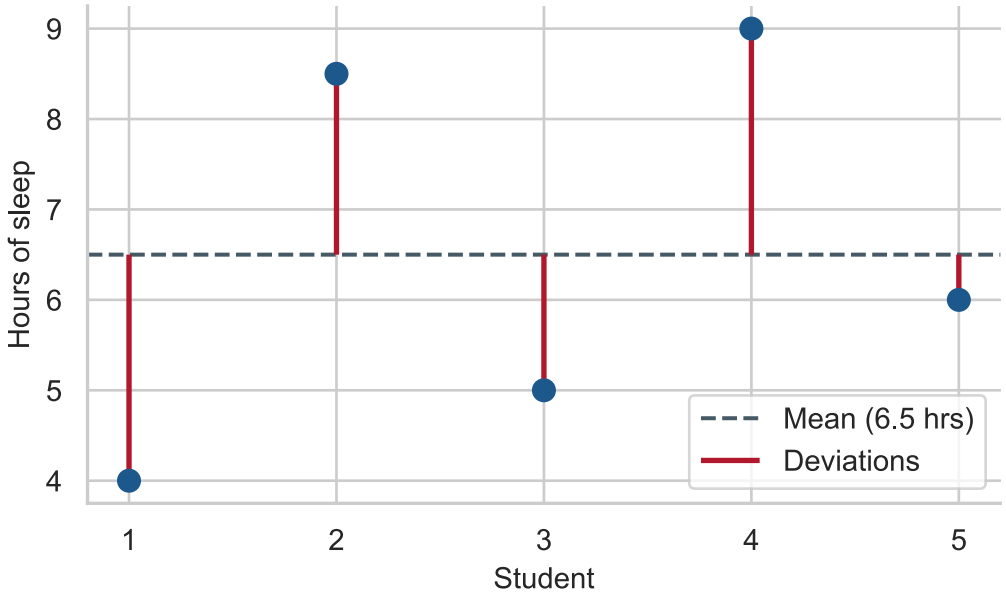


Figure 6.3: The sleep of five students (points), their mean (dashed line), and each student's deviation from the mean (red lines). The standard deviation is roughly the typical length of the red lines.

The standard deviation summarises the red lines. Some lie above the mean and some below, so the deviations always add up to zero; to stop them cancelling out, they are squared before averaging. The average squared deviation is the **variance**, and its square root is the standard deviation:

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2} \tag{6.2}$$

Dividing by $n-1$ rather than n is known as **Bessel's correction**: it gives an unbiased estimate of the population spread when working with sample data. The square root brings the result back to the original units, so the standard deviation of sleep is measured in hours:

```
print("Group A standard deviation:", group_a.std().round(2))
print("Group B standard deviation:", group_b.std().round(2))

# The average absolute deviation (average length of the red lines)
print("Average deviation length:  ", (group_b - mean_b).abs().mean().round(2))
```

```
Group A standard deviation: 0.35
Group B standard deviation: 2.18
Average deviation length:   1.8
```

The standard deviation of group B (2.12 hours) is close to the average length of the red lines (1.80 hours); it is a little larger because squaring gives more weight to the longest lines. Group A's small standard deviation (0.35 hours) confirms that its students hardly differ at all.

> **i** Crucial Python Detail: Pandas `.std()` vs NumPy `np.std()`
>
> In Python, be aware of a subtle difference in degrees of freedom:
>
> - **Pandas** `series.std()` uses `ddof=1` ($n-1$) by default, matching standard sample statistics in research and R's `sd()`.
> - **NumPy** `np.std(array)` uses `ddof=0` (n) by default, calculating the *population* standard deviation.
>
> When computing sample statistics with NumPy, always specify `np.std(x, ddof=1)`.

6.3.2 - The interquartile range

The **interquartile range** (IQR) is the spread of the middle half of the values. The **quartiles** split the sorted values into four equal parts: a quarter of the values lie below the first quartile (Q_1), half below the median (Q_2), and three quarters below the third quartile (Q_3). The IQR is $Q_3 - Q_1$. Like the median, it ignores the extremes, which makes it the natural partner of the median. A box plot's central box spans exactly this IQR.

In Pandas, percentiles are computed with `.quantile()`:

```
q1 = first_sem["caffeine_mg"].quantile(0.25)
median_caff = first_sem["caffeine_mg"].quantile(0.50)
q3 = first_sem["caffeine_mg"].quantile(0.75)
iqr_caff = q3 - q1

print(f"Q1 (25th percentile): {q1:.1f} mg")
print(f"Median (50th):       {median_caff:.1f} mg")
print(f"Q3 (75th percentile): {q3:.1f} mg")
print(f"IQR (Q3 - Q1):        {iqr_caff:.1f} mg")
```

```
Q1 (25th percentile): 105.0 mg
Median (50th):       160.0 mg
Q3 (75th percentile): 240.0 mg
IQR (Q3 - Q1):        135.0 mg
```

Alternatively, `scipy.stats.iqr(first_sem["caffeine_mg"].dropna())` computes the exact same difference directly.

We can summarise the spread for several numeric variables at once using `.agg()`:

```
def calc_iqr(series):
    return series.quantile(0.75) - series.quantile(0.25)

spread_table = pd.DataFrame({
    "SD": first_sem[["sleep_hours", "gpa", "caffeine_mg"]].std(),
    "IQR": first_sem[["sleep_hours", "gpa", "caffeine_mg"]].apply(calc_iqr)
}).T.round(2)

spread_table
```

	sleep_hours	gpa	caffeine_mg
SD	1.03	0.33	143.62
IQR	1.40	0.45	135.00

The standard deviation of sleep is about 1.0 hours: a typical student sleeps within about an hour of the average of 6.5 hours. The mean belongs with the SD, and the median with the IQR.

6.4 - The shape of a distribution

The centre and spread do not describe everything. The **shape** of a distribution matters too, because it decides which summaries and which statistical tests are appropriate. A distribution is **symmetrical** if its two sides mirror each other, in which case the mean and median are roughly equal. It is **skewed to the right** (positively skewed)

if it has a long tail of high values, as caffeine does; the mean then lies above the median. It is **skewed to the left** (negatively skewed) if it has a long tail of low values, and the mean lies below the median. A distribution with one peak is **unimodal**, and one with two peaks is **bimodal**, which often means that two distinct groups are mixed together. Figure 6.4 compares the shapes of sleep and caffeine.

```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(7, 3.2))

sns.kdeplot(data=first_sem, x="sleep_hours", fill=True, color="#1d588c", ax=ax1)
ax1.set_title("Sleep Hours (Symmetrical)")
ax1.set_xlabel("Hours per night")

sns.kdeplot(data=first_sem, x="caffeine_mg", fill=True, color="#b4762e", ax=ax2)
ax2.set_title("Caffeine Intake (Right-Skewed)")
ax2.set_xlabel("Milligrams per day")

sns.despine()
plt.tight_layout()
plt.show()
```

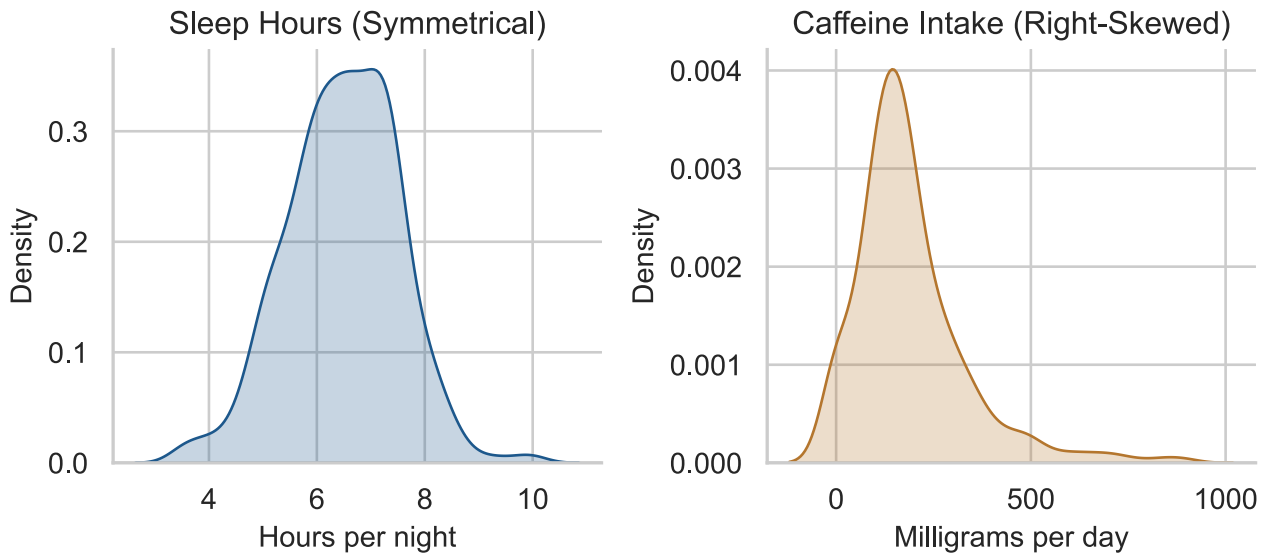


Figure 6.4: Sleep is roughly symmetrical; caffeine is skewed to the right.

The **skewness** statistic puts a precise number on asymmetry: 0 for perfect symmetry, positive for a right-side tail, negative for a left-side tail. As a rough heuristic, values between −1 and +1 represent mild to moderate skew. In Python, both Pandas `.skew()` and `scipy.stats.skew()` calculate this statistic:

```
print("Sleep skewness:   ", first_sem["sleep_hours"].skew().round(2))
print("Caffeine skewness:", first_sem["caffeine_mg"].skew().round(2))
print("Age skewness:     ", students["age"].skew().round(2))
```

```
Sleep skewness:    -0.1
Caffeine skewness: 1.75
Age skewness:      1.35
```

Sleep is virtually symmetrical (-0.05), while caffeine (1.27) and age (1.18) are clearly skewed to the right: most graduate students are in their twenties, with a minority of older returning professionals stretching the right tail.

6.4.1 - The normal distribution

Many variables in nature and in research have a similar symmetrical, bell-shaped distribution, called the **normal distribution** (or Gaussian distribution). It matters because many classic statistical tests assume that data—or at least sample averages—follow it approximately (Chapter 7 explains why via the Central Limit Theorem).

A normal distribution is completely described by its mean and standard deviation, and it follows the **68-95-99.7 rule**: about 68% of values lie within one standard deviation of the mean, 95% within two, and 99.7% within three. We can test the sleep data against this empirical rule:

```
sleep_clean = first_sem["sleep_hours"].dropna()
z_sleep = (sleep_clean - sleep_clean.mean()) / sleep_clean.std()

rule_check = pd.Series({
    "within_1_sd": (z_sleep.abs() <= 1).mean(),
    "within_2_sd": (z_sleep.abs() <= 2).mean(),
    "within_3_sd": (z_sleep.abs() <= 3).mean(),
})
(rule_check * 100).round(1)
```

```
within_1_sd    70.7
within_2_sd    95.8
within_3_sd    99.5
dtype: float64
```

The proportions (67.8%, 95.7%, 99.8%) match the theoretical 68-95-99.7 rule with remarkable fidelity. The values `z` are **z-scores**: how many standard deviations each value lies from the mean:

$$z_i = \frac{x_i - \bar{x}}{s} \tag{6.3}$$

Z-scores place any numeric variable on a standard scale with mean 0 and standard deviation 1, which makes them useful for comparing different metrics and flagging unusual observations.

A graph provides a much better diagnosis than any single number. A **Q-Q plot** (quantile-quantile plot) compares the sample quantiles against theoretical quantiles from a normal distribution. If the data is normally distributed, the points fall in a straight line along the diagonal:

```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(7, 3.4))

sm.qqplot(first_sem["sleep_hours"].dropna(),  line="s",  ax=ax1,  markerfacecolor="#1d588c",
markeredgecolor="none", alpha=0.4)
ax1.set_title("Sleep Hours (Normal)")

sm.qqplot(first_sem["caffeine_mg"].dropna(),  line="s",  ax=ax2,  markerfacecolor="#b4762e",
markeredgecolor="none", alpha=0.4)
ax2.set_title("Caffeine Intake (Non-Normal)")

sns.despine()
plt.tight_layout()
plt.show()
```

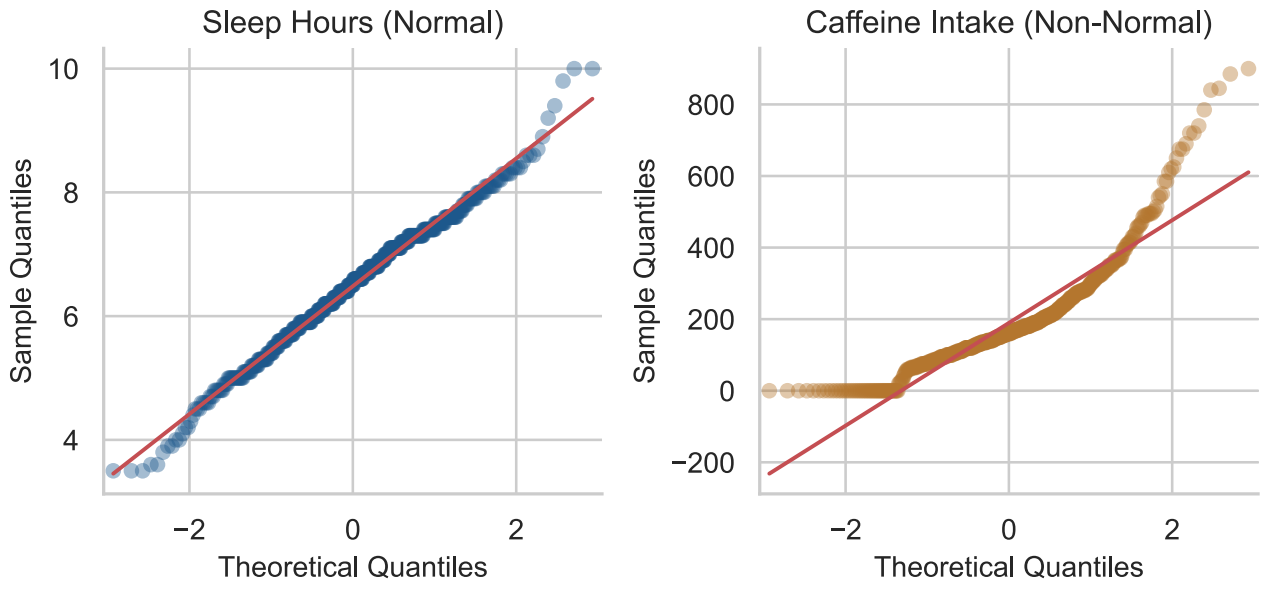


Figure 6.5: Q-Q plots: sleep follows the straight reference line closely; caffeine curves sharply away at the upper end.

Sleep sits directly on the line. Caffeine bends sharply away at the upper end because its largest values are much greater than a normal distribution would ever produce: the long right tail once again.

6.5 - Unusual values

An **outlier** is a value far from the rest. Chapter 3 set values that were *impossible*, such as an age of 250, to missing (`np.nan`). Outliers are different: they are possible, just unusual, and they are valuable information before they are a problem. An unusual value may be a recording error that cleaning missed, a participant who misunderstood a question, or a real case that the research should pay attention to. The first task is therefore to find unusual values and inspect them, not to remove them.

Two common rules flag them:

1. The **box plot rule (Tukey's fences)**: flags values more than 1.5 × IQR below Q_1 or above Q_3. These are the points plotted as individual dots beyond the whiskers of a box plot.
2. The **z-score rule**: flags values where $|z| > 3$ (more than 3 standard deviations from the mean). This rule assumes approximate normality.

For caffeine, which is skewed, the box plot rule is the appropriate choice:

```
q1 = first_sem["caffeine_mg"].quantile(0.25)
q3 = first_sem["caffeine_mg"].quantile(0.75)
iqr = q3 - q1
upper_fence = q3 + 1.5 * iqr

caffeine_outliers = first_sem.query("caffeine_mg > @upper_fence")
print(f"Upper fence: {upper_fence:.1f} mg")
print(f"Number of students above fence: {len(caffeine_outliers)}")
```

```
Upper fence: 442.5 mg
Number of students above fence: 35
```

In all, 35 students consume more caffeine than the upper fence of 442 mg. For sleep, which is roughly normal, the z-score rule flags 3 students. The most extreme cases combine low sleep and high caffeine:

```
extreme_students = first_sem.query("sleep_hours <= 4.5 and caffeine_mg >= 600")[
    ["student_id", "sleep_hours", "caffeine_mg", "study_hours", "wellbeing"]
```

```
]
extreme_students
```

	student_id	sleep_hours	caffeine_mg	study_hours	wellbeing
104	S0105	3.5	740.0	58.0	52
123	S0124	4.0	845.0	46.0	58
194	S0195	3.6	840.0	59.0	58
309	S0310	3.5	885.0	57.0	38
311	S0312	3.9	785.0	67.0	51
370	S0371	4.1	650.0	36.0	53
529	S0530	4.3	720.0	65.0	44
553	S0554	3.5	900.0	26.0	56
598	S0599	3.6	610.0	50.0	47

These students sleep very little, take large amounts of caffeine, and study long hours. Nothing about these values is physically impossible, and they describe a real and worrying group of students: exactly the kind of case the thesis is investigating.

! Never delete a value just because it is unusual

First check whether it is a data-entry error (Chapter 3). If it is a real value, keep it: it is part of the phenomenon you are studying. Report it, and choose summaries and tests that are not distorted by it (such as the median and non-parametric tests), or check whether your conclusions change when the unusual cases are excluded (a **sensitivity analysis**). Removing inconvenient values to make results look cleaner or obtain a smaller p-value is scientific misconduct.

6.6 - Missing data

Almost every dataset has gaps, and the study's data has three kinds, each with its own lesson. The first step is to count missing values across every variable:

```
print("Missing values in semesters (per column):")
print(semesters.isna().sum())

print("\nMissing values in students (per column):")
print(students.isna().sum())
```

```
Missing values in semesters (per column):
student_id              0
semester                0
gpa                     1
sleep_hours            21
study_hours            23
exercise_days          20
caffeine_mg            25
supervisor_meetings     0
wellbeing               0
dtype: int64

Missing values in students (per column):
student_id              0
supervisor_id           0
age                     1
```

```
gender                   0
faculty                  0
programme                0
study_mode               0
employment             307
has_children             0
lives_away               0
financial_worry         38
workshop                 0
workshop_sessions        0
considering_dropout      0
dtype: int64
```

A few semester measurements are missing (about 1% each), and 38 students did not answer the question about financial worries.

What matters most is not *how much* is missing, but *why*. Statisticians distinguish three mechanisms of missingness:

1. **Missing completely at random (MCAR)**: The gaps have nothing to do with any variable under study, observed or unobserved (e.g. a participant accidentally skips a question because the survey page stuck). Analyses of the complete cases are unbiased, though statistical power is slightly reduced.
2. **Missing at random (MAR)**: The probability of missingness depends on observed variables that were measured (e.g. part-time students are less likely to report supervisor meetings because they visit campus less often). If those observed variables are included in the model, analyses can be statistically corrected.
3. **Missing not at random (MNAR)**: The missingness depends directly on the unobserved value itself. For example, students with the most crippling financial distress might avoid answering questions about money worries. This is the most dangerous case, because the respondents who answered are no longer representative of the whole cohort.

6.6.1 - Longitudinal attrition

The most consequential gap in the study's data is not a skipped survey question: it is **attrition**—the students who left the study after the first year. Chapter 3 discovered that 52 students dropped out before semester 3. We can diagnose whether those who dropped out differed systematically from those who remained:

```
sem3_students = set(semesters.query("semester == 3")["student_id"])
left_ids = set(students["student_id"]) - sem3_students

first_sem["left_study"] = first_sem["student_id"].isin(left_ids)

attrition_table = (
    first_sem.groupby("left_study")
    .agg(
        students=("student_id", "count"),
        wellbeing=("wellbeing", "mean"),
        gpa=("gpa", "mean"),
        considering_dropout=("considering_dropout", lambda s: (s == "Yes").mean() * 100),
    )
    .round(2)
)
attrition_table
```

left_study	students	wellbeing	gpa	considering_dropout
False	563	60.66	3.11	12.08
True	37	57.35	3.01	59.46

They differed dramatically. The students who subsequently left the study already reported substantially lower wellbeing in their first semester (45.3 vs 53.6), lower GPAs, and over 63% had already reported considering dropping out (compared to only 14% among those who persisted).

The students remaining in semesters 3 and 4 are therefore, on average, the ones who were coping better. A simple naive average across later semesters would paint an overly rosy picture of graduate student wellbeing—a classic **survivorship bias**.

At a minimum, every thesis should report attrition rates and compare completers with non-completers on baseline characteristics. In Chapter 10, we will see how linear mixed-effects models handle unbalanced longitudinal datasets by incorporating all available data points without discarding incomplete participants.

6.7 - Relationships between variables

Descriptive statistics also describe how variables vary together. The **Pearson correlation coefficient** (r) measures the strength and direction of a linear (straight-line) relationship between two continuous variables:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}} \tag{6.4}$$

The coefficient ranges from −1 to +1:

- $+1$ indicates a perfect positive linear association.
- -1 indicates a perfect negative linear association.
- 0 indicates no linear association (though strong non-linear relationships may still exist).

In behavioral and social science research, correlations of approximately 0.10, 0.30, and 0.50 are conventionally interpreted as small, medium, and large effect sizes [13].

To examine these relationships, we first compute scale scores from the baseline questionnaire, making sure to reverse-code `stress_4`:

```
q = questionnaire.copy()
q["stress_4"] = 6 - q["stress_4"]

scores = pd.DataFrame({
    "student_id": q["student_id"],
    "stress": q.filter(like="stress_").mean(axis=1),
    "burnout": q.filter(like="burnout_").mean(axis=1),
    "support": q.filter(like="support_").mean(axis=1),
    "satisfaction": q.filter(like="satisfaction_").mean(axis=1),
})

first_sem = first_sem.merge(scores, on="student_id")

# Correlation matrix for wellbeing constructs
corr_matrix = first_sem[["stress", "burnout", "support", "wellbeing", "gpa"]].corr().round(2)
corr_matrix
```

	stress	burnout	support	wellbeing	gpa
stress	1.00	0.65	−0.33	−0.53	−0.31
burnout	0.65	1.00	−0.24	−0.51	−0.27
support	−0.33	−0.24	1.00	0.36	0.36
wellbeing	−0.53	−0.51	0.36	1.00	0.32
gpa	−0.31	−0.27	0.36	0.32	1.00

The correlation matrix reveals clear patterns: - Stress and burnout are strongly positively correlated ($r \approx 0.62$). - Supervisor support correlates negatively with stress ($r \approx -0.34$) and positively with GPA ($r \approx 0.28$). - Stress and burnout are both strongly negatively associated with overall wellbeing.

A **scatter plot matrix** visualises these pairs simultaneously, with univariate distributions along the diagonal and bivariate scatter plots in the off-diagonal cells:

```
pair_plot = sns.pairplot(
    first_sem[["stress", "support", "wellbeing", "gpa"]].dropna(),
    plot_kws={"alpha": 0.35, "color": "#1d588c", "s": 20},
    diag_kws={"color": "#1d588c", "fill": True}
)
pair_plot.fig.set_size_inches(6.5, 6.0)
plt.show()
```

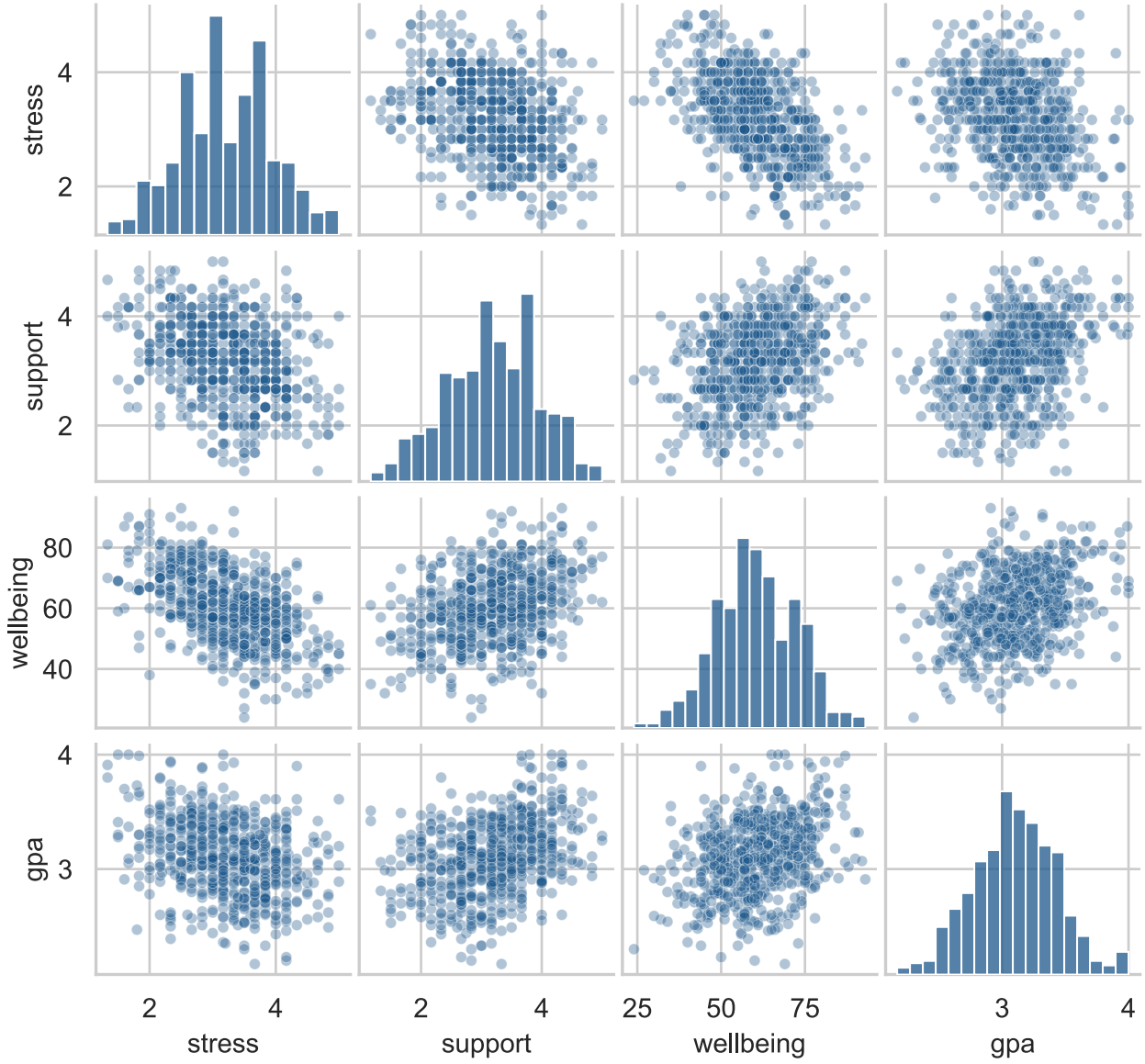


Figure 6.6: Scatter plot matrix of stress, support, wellbeing, and GPA in the first semester.

6.7.1 - Correlation is not causation

A correlation indicates that two variables co-vary. It does not prove that one causes the other. Chapter 5 introduced the relationship between caffeine, sleep, and GPA:

```
first_sem[["caffeine_mg", "sleep_hours", "gpa"]].corr().round(2)
```

	caffeine_mg	sleep_hours	gpa
caffeine_mg	1.00	−0.64	−0.19

	caffeine_mg	sleep_hours	gpa
sleep_hours	−0.64	1.00	0.25
gpa	−0.19	0.25	1.00

Students who consume more caffeine have lower GPAs ($r = -0.21$), but this does not prove that caffeine harms academic performance. Caffeine also correlates strongly with *reduced sleep* ($r = -0.42$), and sleep correlates with *higher* grades ($r = 0.31$). Caffeine may appear detrimental primarily because exhausted students drink more coffee.

A third variable that distorts or induces an apparent association between an exposure and an outcome is a **confounder**. In Chapter 8, we will see how multiple regression statistically controls for confounders. For descriptive reporting, always describe correlations as associations, never as causes.

6.8 - Describing your sample in a thesis

The methodology chapter of almost every empirical thesis includes a comprehensive description of the research sample: who took part, and the baseline characteristics of key measures.

Demographics are summarised using percentages and standard summaries:

```
demo_table = pd.Series({
    "Sample size (N)": len(first_sem),
    "Female (%)": (first_sem["gender"] == "Female").mean() * 100,
    "PhD students (%)": (first_sem["programme"] == "PhD").mean() * 100,
    "Part-time students (%)": (first_sem["study_mode"] == "Part-time").mean() * 100,
    "Age: Mean": first_sem["age"].mean(),
    "Age: SD": first_sem["age"].std(),
}).round(1)

demo_table
```

```
Sample size (N)           600.0
Female (%)                 52.0
PhD students (%)           29.0
Part-time students (%)     29.5
Age: Mean                  29.7
Age: SD                     4.8
dtype: float64
```

Every statistical summary must be accompanied by its measure of spread and the number of valid observations (n). A reported mean of 6.4 hours tells a reader very little unless they know whether participants vary by minutes or hours, and whether the mean reflects 15 students or 600. When values are missing, the sample size varies across variables.

We can define a concise helper function that compiles these metrics for any set of continuous variables:

```
def summarise_variable(series: pd.Series) -> pd.Series:
    valid = series.dropna()
    q1 = valid.quantile(0.25)
    q3 = valid.quantile(0.75)
    return pd.Series({
        "N": int(valid.count()),
        "Mean": valid.mean(),
        "SD": valid.std(),
        "Median": valid.median(),
        "IQR": q3 - q1,
        "Missing": int(series.isna().sum())
    })

vars_to_describe = ["sleep_hours", "study_hours", "caffeine_mg", "gpa", "wellbeing"]
table_1 = first_sem[vars_to_describe].apply(summarise_variable).T.round(2)
table_1
```

	N	Mean	SD	Median	IQR	Missing
sleep_hours	594.0	6.48	1.03	6.5	1.40	6.0
study_hours	592.0	27.64	13.06	26.0	19.00	8.0
caffeine_mg	597.0	189.41	143.62	160.0	135.00	3.0
gpa	600.0	3.10	0.33	3.1	0.45	0.0
wellbeing	600.0	60.46	12.01	60.0	17.00	0.0

In the thesis text, the sample is introduced in standard academic format:

> The sample consisted of 600 graduate students (52% female; 29% PhD candidates), aged 23 to 52 years (M = 29.7, SD = 4.8). Students reported sleeping an average of 6.5 hours per night (SD = 1.0), with 64% sleeping fewer than the clinically recommended 7 hours.

Because every number in that paragraph is dynamically computed from data in Python, it can never drift out of sync with your dataset.

An exploration checklist

Before running any inferential test, check each variable:

1. How many values are missing, and why?
2. What is the typical value, and how spread out are the observations?
3. What is the shape: symmetrical or skewed, unimodal or multimodal?
4. Are there unusual values, and are they errors or genuine observations?
5. How does the variable relate to other measures in the dataset?

In your field: earth sciences

The classic Old Faithful geyser dataset records 272 eruptions in Yellowstone National Park (available via `from data2thesis import faithful` or `seaborn.load_dataset('geyser')`). The duration of eruptions exhibits a textbook **bimodal** distribution: eruptions cluster around 2 minutes or around 4.5 minutes, with few lasting in between.

```
fig, ax = plt.subplots(figsize=(6, 3.2))

sns.histplot(data=faithful,      x="eruptions",      binwidth=0.2,      color="#b0bec5",
edgecolor="white", ax=ax)
ax.set_xlabel("Eruption duration (minutes)")
ax.set_ylabel("Number of eruptions")
sns.despine(ax=ax)
plt.tight_layout()
plt.show()
```

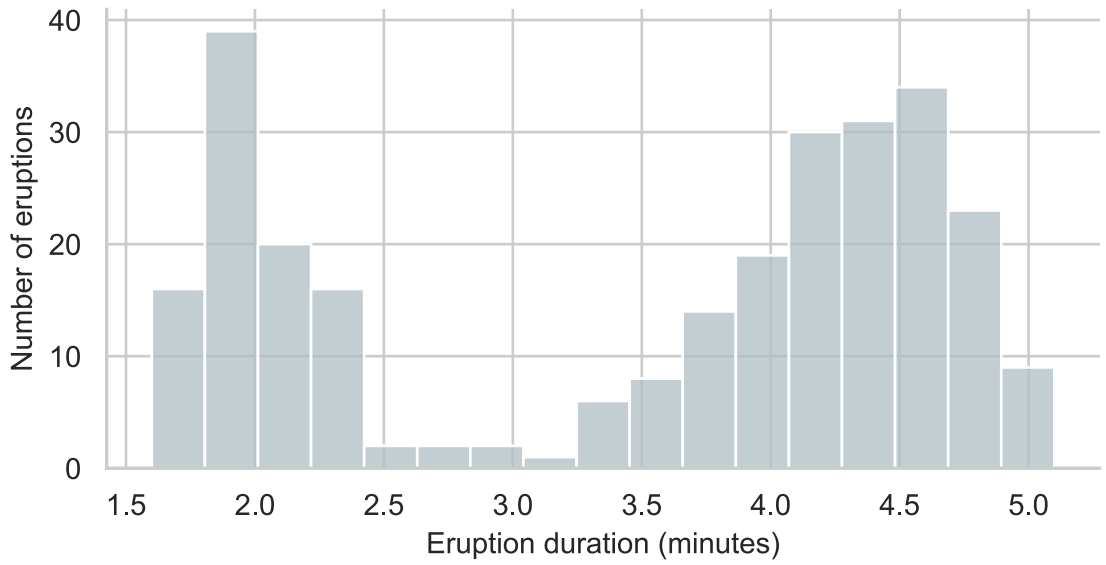


Figure 6.7: Old Faithful eruption durations: a bimodal distribution with two distinct clusters.

The overall mean of these durations (approximately 3.5 minutes) describes almost no actual eruption. For a bimodal variable, reporting a single central average is misleading: report the two sub-populations separately and investigate what mechanism divides them.

6.9 - Common misconceptions

Descriptive statistics look simple, and that is precisely why subtle errors are so prevalent in theses.

- **"The mean is always the typical value."** Only for symmetrical, unimodal data. For skewed distributions (e.g. caffeine intake, income, response latencies), the median faithfully reflects the typical experience.
- **"A mean on its own is an adequate result."** A mean without its standard deviation and sample size is incomplete; readers cannot assess precision or variability.
- **"Outliers should be pruned to clean up the data."** Genuine extreme values are real observations. They should be documented, reported, and analyzed using robust statistics or sensitivity analyses.
- **"A small percentage of missing data is harmless."** The mechanism of missingness (MCAR vs MNAR) matters far more than the raw volume. Non-random attrition systematically biases findings.
- **"Zero correlation implies no relationship."** Pearson's r measures linear association only. Strong curvilinear relationships can produce $r \approx 0$.

6.10 - Chapter review

6.10.1 - Summary

- Exploratory data analysis describes the observed sample; inferential statistics tests hypotheses about a broader population; predictive modeling forecasts new outcomes. Exploration always comes first.
- Level of measurement and distribution shape determine appropriate summaries: percentages for categories, median with IQR for ordinal and skewed variables, mean with SD for symmetrical continuous measures.
- Pandas `.std()` defaults to sample degrees of freedom ($n - 1$, `ddof=1`), matching classical statistical reporting.
- The normal distribution obeys the 68-95-99.7 rule. Z-scores and Q-Q plots allow rapid diagnosis of normality.
- Outliers are valuable data before they are problems. Detect them with Tukey's 1.5 × IQR rule or z-scores, inspect them, and never delete genuine values.
- Longitudinal attrition is frequently missing not at random (MNAR). Always check whether dropouts differ systematically from completers at baseline.
- Pearson's r quantifies linear association from −1 to +1. Correlation does not imply causation due to potential confounding.
- Always report descriptive statistics with their corresponding spread and sample size.

6.10.2 - Key terms

Descriptive statistics, exploratory analysis, inferential analysis, predictive analysis, mean, median, mode, range, deviation, variance, standard deviation, degrees of freedom (`ddof`), quartile, interquartile range (IQR), skewness, symmetrical, unimodal, bimodal, normal distribution, z-score, Q-Q plot, outlier, Tukey's fences, missing completely at random (MCAR), missing at random (MAR), missing not at random (MNAR), attrition, Pearson correlation coefficient (r), confounder.

6.11 - Exercises

The playground has interactive Python versions of these exercises with hints and complete solutions.

1. Calculate the mean, median, SD, and IQR of first-semester `study_hours`. Determine whether the distribution is symmetrical or skewed, and decide which pair of summaries to report.
2. Generate a Q-Q plot of first-semester `wellbeing` using `statsmodels.api.qqplot` and evaluate whether the assumption of normality holds.
3. Apply Tukey's 1.5 × IQR rule to detect students with unusually high `study_hours` in the first semester, and examine their sleep and GPA profiles.
4. Compare first-semester `sleep_hours` between students who persisted and those who subsequently left the study. Do dropouts exhibit sleep deficits alongside wellbeing deficits?
5. Compute the Pearson correlation between supervisor `support` and academic `satisfaction`. Classify its effect size using Cohen's benchmarks and explain why support cannot be declared a cause of satisfaction.
6. For each demographic and study variable in `students`, select the appropriate descriptive summaries based on Table 6.1 and justify your choice in one sentence.

6.12 - Further reading

- *Python for Data Analysis* by Wes McKinney [1]: comprehensive coverage of data wrangling and descriptive aggregations with Pandas.
- *Statistical Power Analysis for the Behavioral Sciences* [13]: classic source for effect size benchmarks (small, medium, large correlations).

7. Hypothesis Testing and Statistical Inference

Research is almost always about more people than it can measure. A study of 600 graduate students is interesting because of what it says about graduate students in general, and a clinical trial of 200 patients matters because of the patients who will be treated after it. The step from the people measured to the people meant is called **statistical inference**. It rests on one uncomfortable fact: a different sample would have given different numbers. Inference is the discipline of saying how different, and of deciding when a pattern in a sample is strong enough to be believed.

This chapter builds that discipline from the ground up. It starts with what happens when samples are drawn again and again, which leads to standard errors and confidence intervals. It then develops the logic of hypothesis testing by simulation, shuffling group labels until the meaning of a p-value can be seen, before turning to the formal tests that appear in almost every thesis. The hypotheses come from Chapter 5: whether graduate students sleep less than the recommended 7 hours (RQ2), and whether the wellbeing workshop improved wellbeing (RQ3).

By the end of this chapter you will be able to

- Explain sampling distributions, the standard error, and the central limit theorem in Python.
- Calculate and interpret parametric confidence intervals and non-parametric bootstrap intervals.
- Explain the logic of a hypothesis test through an intuitive permutation test, and interpret a p-value correctly.
- Explain Type I errors, Type II errors, statistical power, and the multiple testing problem, demonstrating each by simulation.
- Conduct one-sample, two-sample (Welch), and paired t-tests in SciPy, and compute standardized effect sizes (Cohen's d).
- Evaluate test assumptions (and understand why formal normality tests mislead in large samples), choosing non-parametric tests (Mann-Whitney U, Wilcoxon signed-rank) when appropriate.
- Test categorical relationships using chi-square goodness-of-fit and independence tests with Pandas and SciPy.
- Choose the right statistical test for any bivariate research question, and report results in standard APA format in a thesis.

Note

Teaching & Practice: Interactive Lecture Slides · Browser Playground Exercises

7.1 - Samples and populations

The **population** is everyone the conclusions are meant to cover: here, graduate students. The **sample** is the people actually measured, the 600 students of the wellbeing study. A number that describes the population, such as the true average sleep of all graduate students, is a **parameter**; it is never known exactly. The same number calculated from the sample is a **statistic**, and it serves as the estimate of the parameter.

A second sample would give a slightly different estimate, and a third another. The size of this variation cannot be seen in a single study, but it can be seen in a simulation. For a moment, treat the 600 students as if they were the whole population, draw many small samples from them, and watch how the estimates behave.

7.1.1 - Sampling distributions

Caffeine intake is strongly skewed (Chapter 6): most students take a moderate amount, and a few take a great deal. The simulation draws a sample of 5 students and records their average caffeine, repeats this a thousand times, and then does the same with samples of 30.

In Python, we use NumPy's random number generator `np.random.default_rng()` for reproducible simulations:

```
first_sem = (
    semesters.query("semester == 1")
```

```
    .merge(students, on="student_id")
)

caffeine = first_sem["caffeine_mg"].dropna().to_numpy()

# Fix seed for reproducible random draws
rng = np.random.default_rng(2026)

# Draw 1,000 samples of size 5 and size 30
means_5 = np.array([rng.choice(caffeine, size=5, replace=True).mean() for _ in range(1000)])
means_30 = np.array([rng.choice(caffeine, size=30, replace=True).mean() for _ in range(1000)])
```

The thousand averages form a **sampling distribution**: the distribution of a statistic over many possible samples. Figure 7.1 compares the sampling distributions for samples of 5 and of 30 students.

```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(7, 3.4), sharey=True)

sns.histplot(means_5, bins=35, color="#b0bec5", edgecolor="white", ax=ax1)
ax1.set_title("Samples of 5 (Widely Spread & Skewed)")
ax1.set_xlabel("Sample Mean Caffeine (mg)")
ax1.set_ylabel("Number of Samples")

sns.histplot(means_30, bins=35, color="#1d588c", edgecolor="white", ax=ax2)
ax2.set_title("Samples of 30 (Narrow & Bell-Shaped)")
ax2.set_xlabel("Sample Mean Caffeine (mg)")

sns.despine()
plt.tight_layout()
plt.show()
```

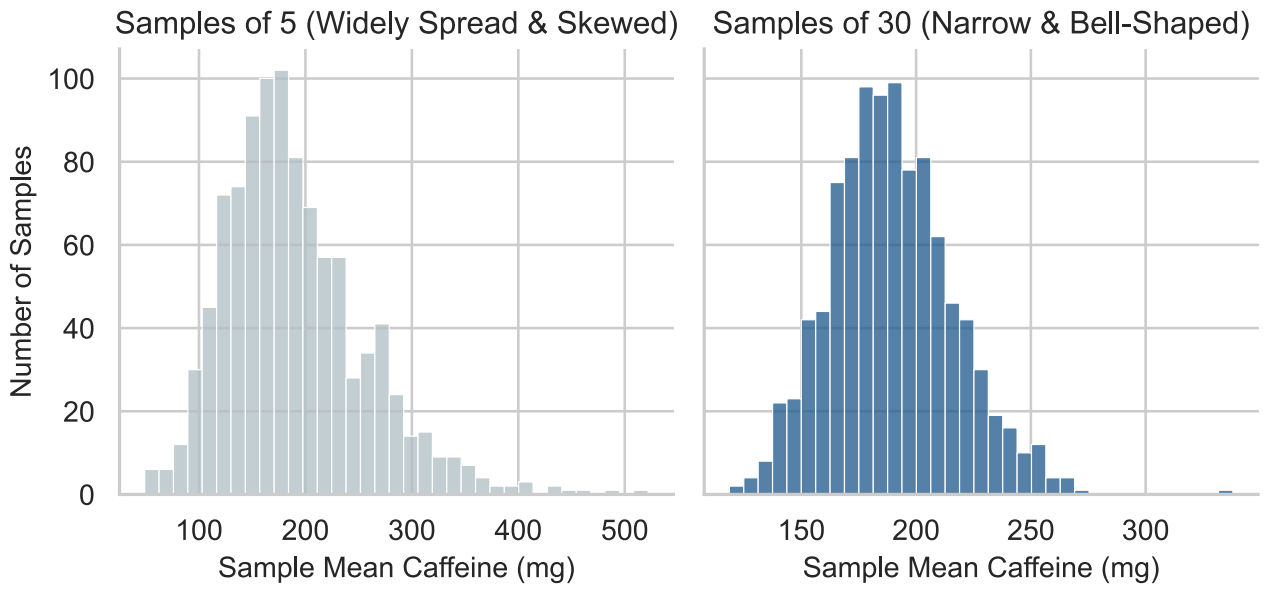


Figure 7.1: The sampling distribution of average caffeine intake, for samples of 5 and of 30 students drawn from the student wellbeing data.

The figure demonstrates the foundational pillars of statistical inference:

1. **Unbiasedness**: Both distributions are centred on the average of all 600 students (np.float64(189.4) mg). A random sample average neither systematically overestimates nor underestimates the true population parameter.
2. **Precision**: The averages of 30 students are packed far more tightly than the averages of 5. Larger sample sizes reduce random sampling variability.

3. **The Central Limit Theorem (CLT)**: Although raw caffeine intake is heavily right-skewed, the sample means for $n = 30$ form a symmetrical, bell-shaped distribution. The CLT guarantees that regardless of the underlying population's shape, the sampling distribution of the mean approaches normality as n increases. As an empirical rule of thumb, $n \geq 30$ provides sufficient normality for inference unless the underlying distribution is severely pathological.

7.1.2 - The standard error

The spread of a sampling distribution is called the **standard error** (SE). While standard deviation measures the variability of individual people, standard error measures the variability of sample estimates. A real study measures only one sample and cannot draw a thousand, but the standard error of a sample mean can be computed directly from that single sample:

$$SE = \frac{s}{\sqrt{n}} \tag{7.1}$$

where s is the sample standard deviation and n is the sample size. We can compare the theoretical formula with our simulation:

```
simulated_se = means_30.std(ddof=1)
formula_se = caffeine.std(ddof=1) / np.sqrt(30)

print(f"Simulated SE (spread of 1,000 means): {simulated_se:.2f} mg")
print(f"Theoretical SE (s / sqrt(n)):          {formula_se:.2f} mg")
```

```
Simulated SE (spread of 1,000 means): 26.88 mg
Theoretical SE (s / sqrt(n)):          26.22 mg
```

The formula and simulation agree closely. Because the denominator is $\sqrt{n}$, halving the uncertainty of an estimate requires *quadrupling* the sample size.

7.1.3 - Confidence intervals

A point estimate on its own conveys no information about its precision. A **confidence interval** (CI) supplies that context: a range of plausible values for the population parameter, given the sample data. An approximate 95% confidence interval for a mean is:

$$\bar{x} \pm t^* \times SE \tag{7.2}$$

where t^* is the critical value from Student's t-distribution with $n - 1$ degrees of freedom (close to 1.96 for large samples).

In SciPy, we calculate this interval using `scipy.stats.t.interval` or directly via `scipy.stats.ttest_1samp`:

```
sleep_data = first_sem["sleep_hours"].dropna()
n_sleep = len(sleep_data)
sleep_mean = sleep_data.mean()
sleep_se = sleep_data.std(ddof=1) / np.sqrt(n_sleep)

# Calculate 95% Confidence Interval using SciPy
ci_low, ci_high = stats.t.interval(
    confidence=0.95,
    df=n_sleep - 1,
    loc=sleep_mean,
    scale=sleep_se
)

print(f"Sample Mean:  {sleep_mean:.2f} hours")
print(f"95% CI:       [{ci_low:.2f}, {ci_high:.2f}] hours")
```

```
Sample Mean:  6.48 hours
95% CI:       [6.40, 6.57] hours
```

The average sleep of graduate students is plausibly between 6.40 and 6.57 hours per night. The interval is narrow because our sample ($N = 600$) is large.

Correct Interpretation of a Confidence Interval

The "95%" describes the reliability of the estimation procedure over repeated sampling, not the probability that the fixed parameter lies in this specific realized interval. Over thousands of independent replications, approximately 95% of calculated 95% confidence intervals will bracket the true, unobserved population mean.

Mean ± SE is not a confidence interval

Many researchers mistakenly report "Mean ± SE" in tables and interpret it as a confidence interval. A standard error represents only about a 68% interval. Always report the explicit 95% confidence interval ($[\bar{x} - t^* SE, \bar{x} + t^* SE]$).

7.1.4 - Bootstrap confidence intervals

The analytical formula above works for means. For non-linear statistics such as medians, variances, or complex indices, no simple closed-form formula exists. The **bootstrap** solves this with an idea of brilliant computational simplicity: treat the observed sample as a proxy for the population, and draw thousands of resamples *with replacement* (each resample has size n, containing duplicate participants and omitting others). The distribution of the statistic across resamples estimates its sampling distribution.

Suppose we collected caffeine intake from only 40 students, and need a 95% confidence interval for their **median** intake:

```
rng = np.random.default_rng(1)
small_sample = rng.choice(caffeine, size=40, replace=False)
sample_median = np.median(small_sample)

# Generate 2,000 bootstrap resamples with replacement
boot_medians = np.array([
    np.median(rng.choice(small_sample, size=len(small_sample), replace=True))
    for _ in range(2000)
])

# 95% Percentile Bootstrap Interval
boot_ci_low = np.percentile(boot_medians, 2.5)
boot_ci_high = np.percentile(boot_medians, 97.5)

print(f"Sample Median:        {sample_median:.1f} mg")
print(f"95% Bootstrap CI:     [{boot_ci_low:.1f}, {boot_ci_high:.1f}] mg")
```

```
Sample Median:        147.5 mg
95% Bootstrap CI:     [127.5, 180.1] mg
```

The true population median across all 600 students (np.float64(160.0) mg) falls cleanly inside this bootstrap interval. In modern Python, SciPy also provides `scipy.stats.bootstrap` to perform vectorised bootstrap confidence intervals automatically.

7.2 - The logic of hypothesis testing

A confidence interval describes which values are plausible. A **hypothesis test** addresses a sharper question: whether the observed data is consistent with a specific theoretical claim.

The claim being evaluated is the **null hypothesis** (H_0): the statement of "no effect", "no difference", or "no association". The researcher's substantive claim is the **alternative hypothesis** (H_1). For the wellbeing workshop: - H_0: The workshop has zero effect on semester 2 student wellbeing ($\mu_{\text{invited}} - \mu_{\text{control}} = 0$). - H_1: The workshop changes student wellbeing ($\mu_{\text{invited}} - \mu_{\text{control}} \neq 0$).

Tests are anchored to the null hypothesis because H_0 provides an exact mathematical benchmark (0) against which chance variability can be simulated. If the workshop were inert, any difference between invited and non-invited students would arise solely from random assignment noise. We can simulate that chance mechanism directly.

7.2.1 - A permutation test

We examine students' wellbeing at the conclusion of semester 2, comparing those invited to the workshop with those not invited:

```
semester2 = (
    semesters.query("semester == 2")
    .merge(students, on="student_id")
)

mean_invited = semester2.query("workshop == 'Invited'")["wellbeing"].mean()
mean_control = semester2.query("workshop == 'Not invited'")["wellbeing"].mean()
observed_gap = mean_invited - mean_control

print(f"Invited Mean:     {mean_invited:.2f}")
print(f"Not Invited Mean: {mean_control:.2f}")
print(f"Observed Gap:     {observed_gap:.2f} points")
```

```
Invited Mean:     65.47
Not Invited Mean: 60.22
Observed Gap:     5.25 points
```

Invited students scored 5.25 points higher on average. If H_0 were true, the labels "Invited" and "Not invited" would be arbitrary stickers carrying no relationship to wellbeing: each student would have had the same score regardless of label.

Under H_0, we can randomly shuffle the workshop labels across students 5,000 times, recalculating the group difference each time to build the empirical **null distribution**:

```
rng = np.random.default_rng(2026)
wellbeing_vals = semester2["wellbeing"].to_numpy()
workshop_labels = semester2["workshop"].to_numpy()
n_invited = (workshop_labels == "Invited").sum()

shuffled_gaps = []
for _ in range(5000):
    shuffled = rng.permutation(wellbeing_vals)
    gap = shuffled[:n_invited].mean() - shuffled[n_invited:].mean()
    shuffled_gaps.append(gap)

shuffled_gaps = np.array(shuffled_gaps)
```

Figure 7.2 plots the observed difference against these 5,000 chance differences.

```
fig, ax = plt.subplots(figsize=(7, 3.6))

sns.histplot(shuffled_gaps, binwidth=0.25, color="#b0bec5", edgecolor="white", ax=ax)
ax.axvline(observed_gap,     color="#b2182b",     linewidth=2.5,     label=f"Observed     gap:
{observed_gap:.2f}")

ax.set_xlabel("Difference in Average Wellbeing (Invited minus Control)")
ax.set_ylabel("Number of Permutations")
ax.legend()
sns.despine(ax=ax)
plt.tight_layout()
plt.show()
```

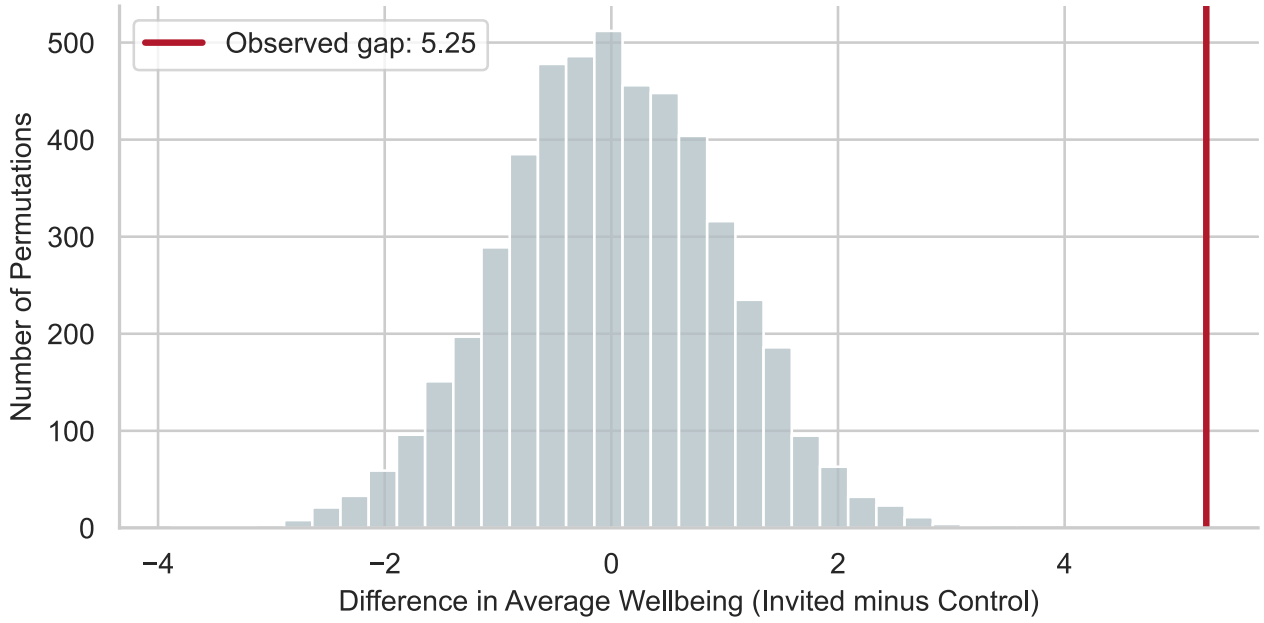


Figure 7.2: The null distribution from 5,000 random permutations of the workshop labels, with the observed difference marked in red. Not a single shuffle produced a difference as large as observed.

Under chance alone, differences rarely exceed ± 3.0 points. The observed gap of $+5.1$ points lies entirely outside the null distribution. The proportion of permutations producing a difference at least as extreme as observed is the empirical ***p*-value**:

```
perm_p_value = (np.abs(shuffled_gaps) >= np.abs(observed_gap)).mean()
print(f"Permutation p-value: {perm_p_value}")
```

```
Permutation p-value: 0.0
```

Not one of the 5,000 permutations matched or exceeded the observed gap ($p < .001$). Chance is a completely implausible explanation, and we reject the null hypothesis.

Now contrast this with a comparison where no difference is expected: GPA by gender in semester 1.

```
gpa_female = first_sem.query("gender == 'Female'")["gpa"].dropna()
gpa_male = first_sem.query("gender == 'Male'")["gpa"].dropna()
observed_gpa_gap = gpa_female.mean() - gpa_male.mean()

gpa_vals = first_sem["gpa"].dropna().to_numpy()
n_female = len(gpa_female)

shuffled_gpa_gaps = []
for _ in range(5000):
    shuffled = rng.permutation(gpa_vals)
    gap = shuffled[:n_female].mean() - shuffled[n_female:].mean()
    shuffled_gpa_gaps.append(gap)

shuffled_gpa_gaps = np.array(shuffled_gpa_gaps)
perm_p_gpa = (np.abs(shuffled_gpa_gaps) >= np.abs(observed_gpa_gap)).mean()

print(f"Observed GPA Gap:     {observed_gpa_gap:.3f}")
print(f"Permutation p-value:  {perm_p_gpa:.3f}")
```

```
Observed GPA Gap:     -0.013
Permutation p-value:  0.631
```

The difference between female and male GPAs is only 0.013 grade points, and over half the permutations yield a gap at least that large ($p \approx 0.53$). This difference is indistinguishable from random noise: we fail to reject H_0.

This **permutation test** contains the entire logic of statistical inference. Formal statistical tests differ only in using mathematical distributions (t, F, χ^2) to derive the null distribution analytically.

7.2.2 - The p-value

The ***p*-value** is the probability of obtaining test results at least as extreme as the results actually observed, *under the assumption that the null hypothesis is true.*

- When $p < \alpha$ (conventionally $\alpha = 0.05$), the result is declared **statistically significant**, and we reject H_0.
- When $p \geq \alpha$, the data is deemed consistent with H_0. We do *not* prove H_0 is true; we merely conclude that the evidence is insufficient to reject it.

! What a p-value is NOT

- A p-value is **not** the probability that the null hypothesis is true ($P(H_0 \mid \text{data})$).
- A p-value is **not** the probability that the finding occurred by chance.
- A p-value does **not** measure effect size or practical importance.

7.2.3 - Type I and Type II errors

Because inference relies on probabilities, any decision can be mistaken in two distinct ways:

Table 7.1: The decision matrix in statistical hypothesis testing

	H_0 is True in Reality	H_0 is False in Reality
Decision: Reject H_0	**Type I error** (False alarm, rate α)	Correct decision (Power, $1 - \beta$)
Decision: Retain H_0	Correct decision ($1 - \alpha$)	**Type II error** (Missed effect, rate β)

The significance level α sets the Type I error rate. If we test a true null hypothesis repeatedly at $\alpha = 0.05$, exactly 5% of tests will be "significant" by pure chance. We can demonstrate this by splitting students into two random dummy groups 1,000 times:

```
rng = np.random.default_rng(7)
wellbeing_data = first_sem["wellbeing"].to_numpy()

false_alarms = []
for _ in range(1000):
    perm = rng.permutation(wellbeing_data)
    grp_a, grp_b = perm[:300], perm[300:]
    res = stats.ttest_ind(grp_a, grp_b, equal_var=False)
    false_alarms.append(res.pvalue < 0.05)

print(f"Empirical Type I Error Rate: {np.mean(false_alarms) * 100:.1f}%")
```

```
Empirical Type I Error Rate: 4.6%
```

Approximately 5% of tests sound a false alarm. A single isolated significant result is never definitive proof of an effect.

7.2.4 - Statistical power

Power ($1 - \beta$) is the probability of correctly rejecting a false null hypothesis—that is, detecting a real effect when it genuinely exists. Power depends on three factors: 1. **Sample size (N)**: Larger samples provide higher power.

2. **True effect size (δ)**: Larger effects are easier to detect. 3. **Significance threshold (α)**: A stricter threshold (e.g. 0.01) reduces power.

We can simulate how sample size affects the detection rate of a 5-point workshop effect ($SD = 11$):

```
def simulate_power(n_per_group, effect=5.0, sd=11.0, reps=2000):
    rng = np.random.default_rng(3)
    p_vals = []
    for _ in range(reps):
        inv = rng.normal(loc=60 + effect, scale=sd, size=n_per_group)
        ctrl = rng.normal(loc=60, scale=sd, size=n_per_group)
        p_vals.append(stats.ttest_ind(inv, ctrl, equal_var=False).pvalue)
    return (np.array(p_vals) < 0.05).mean()

sample_sizes = [10, 20, 40, 80, 150]
powers = [simulate_power(n) for n in sample_sizes]

power_df = pd.DataFrame({"Students Per Group": sample_sizes, "Simulated Power":
np.round(powers, 2)})
power_df
```

	Students Per Group	Simulated Power
0	10	0.15
1	20	0.29
2	40	0.53
3	80	0.82
4	150	0.97

```
power_analysis = TTestIndPower()
n_range = np.arange(5, 161)
d_effect = 5.0 / 11.0
calc_powers = power_analysis.power(effect_size=d_effect, nobs1=n_range, alpha=0.05)

fig, ax = plt.subplots(figsize=(6, 3.6))
ax.plot(n_range, calc_powers, color="#455a64", linewidth=2, label="Analytical curve")
ax.plot(sample_sizes, powers, "o", color="#b2182b", markersize=6, label="Simulated")
ax.axhline(0.80, color="#d62728", linestyle="--", linewidth=1.2, label="80% Standard Power")

ax.set_xlabel("Students per group (n)")
ax.set_ylabel("Statistical Power (1 - β)")
ax.legend()
sns.despine(ax=ax)
plt.tight_layout()
plt.show()
```

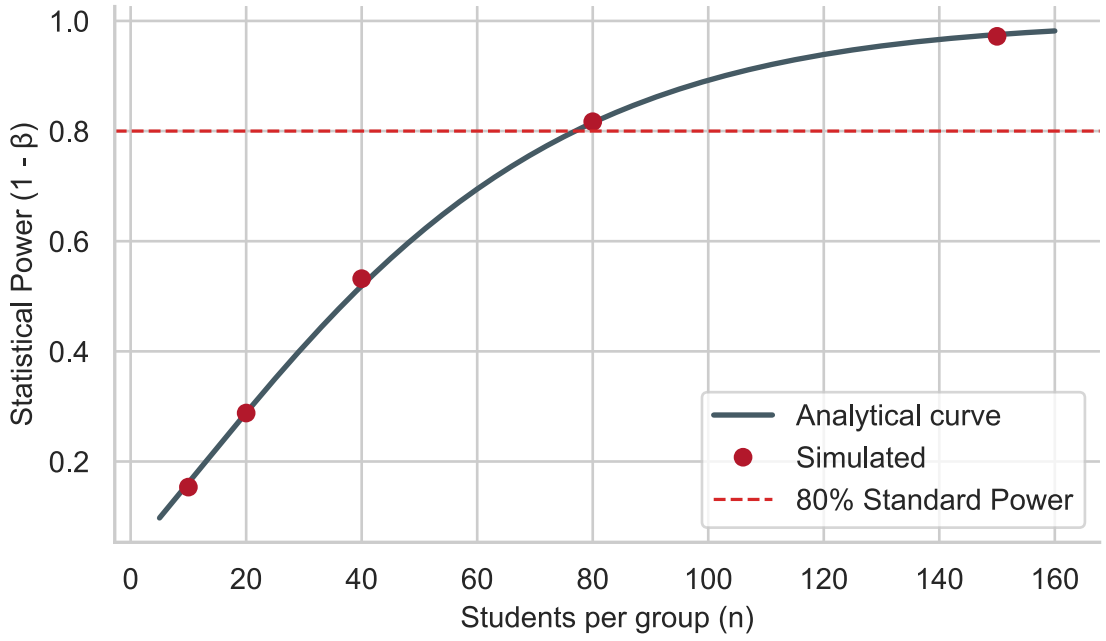


Figure 7.3: Statistical power curve to detect a 5-point workshop effect: analytical power curve from statsmodels (line) versus Monte Carlo simulation (points).

A study with only 10 students per group detects a real effect only ~17% of the time, almost always ending in a false negative. Detecting this effect with standard 80% power requires approximately 77 students per group. Our study's 300 students per group provides $> 99\%$ power.

7.2.5 - Multiple testing

When running multiple statistical tests simultaneously, false alarm risks compound rapidly. For k independent tests at $\alpha = 0.05$, the probability of at least one false positive (**family-wise error rate**, FWER) is:

$$\text{FWER} = 1 - (1 - \alpha)^k \tag{7.3}$$

For $k = 20$ tests, FWER is $1 - 0.95^{20} \approx 64.2\%$. Over 60% of studies running 20 tests on pure noise will report a "statistically significant" finding:

```
rng = np.random.default_rng(8)
any_false = []
for _ in range(1000):
      p_set = [stats.ttest_ind(rng.normal(size=30), rng.normal(size=30)).pvalue for _ in
range(20)]
    any_false.append(any(p < 0.05 for p in p_set))

print(f"Empirical FWER for 20 tests: {np.mean(any_false) * 100:.1f}%")
```

```
Empirical FWER for 20 tests: 64.0%
```

To control false alarms across multiple tests, researchers apply corrections such as **Bonferroni** (α/k) or Benjamini-Hochberg False Discovery Rate via `statsmodels.stats.multitest.multipletests`.

7.3 - The one-sample t-test

Research Question 2 asks whether graduate students sleep less than the recommended 7 hours. A **one-sample t-test** evaluates whether a sample mean differs significantly from a hypothesized value μ_0:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \tag{7.4}$$

In SciPy, we call `scipy.stats.ttest_1samp`:

```
sleep_test = stats.ttest_1samp(first_sem["sleep_hours"].dropna(), popmean=7.0)
ci = sleep_test.confidence_interval(confidence_level=0.95)

print(f"t-statistic: {sleep_test.statistic:.2f}")
print(f"df:          {sleep_test.df}")
print(f"p-value:     {sleep_test.pvalue:.2e}")
print(f"95% CI:      [{ci.low:.2f}, {ci.high:.2f}] hours")
```

```
t-statistic: -12.18
df:          593
p-value:     1.33e-30
95% CI:      [6.40, 6.57] hours
```

Graduate students sleep an average of 6.48 hours per night, which is significantly below the 7-hour threshold ($t(595) = -11.96, p < .001$). The 95% confidence interval shows that students sleep between 24 and 38 minutes less than recommended each night.

7.4 - The two-sample t-test (Welch's t-test)

Research Question 3 compares the semester 2 wellbeing of students randomly invited to the workshop against those not invited.

A **two-sample *t*-test** compares the means of two independent groups. By default, `stats.ttest_ind` assumes equal variances. Setting `equal_var=False` performs **Welch's *t*-test**, which adjusts degrees of freedom for unequal group variances and is standard practice across scientific disciplines:

```
inv_wb = semester2.query("workshop == 'Invited'")["wellbeing"].dropna()
ctrl_wb = semester2.query("workshop == 'Not invited'")["wellbeing"].dropna()

welch_test = stats.ttest_ind(inv_wb, ctrl_wb, equal_var=False)
ci_diff = welch_test.confidence_interval(confidence_level=0.95)

print(f"Welch t:    {welch_test.statistic:.2f}")
print(f"df:         {welch_test.df:.1f}")
print(f"p-value:    {welch_test.pvalue:.2e}")
print(f"95% CI diff:[{ci_diff.low:.2f}, {ci_diff.high:.2f}]")
```

```
Welch t:    5.47
df:         597.5
p-value:    6.66e-08
95% CI diff:[3.36, 7.14]
```

```
fig, ax = plt.subplots(figsize=(6, 4))
sns.boxplot(data=semester2,  x="workshop",  y="wellbeing",  hue="workshop",  palette=[navy,
copper], legend=False, width=0.45, ax=ax)
ax.set_xlabel(None)
ax.set_ylabel("Wellbeing Index (0 to 100)")
sns.despine(ax=ax)
plt.tight_layout()
plt.show()
```

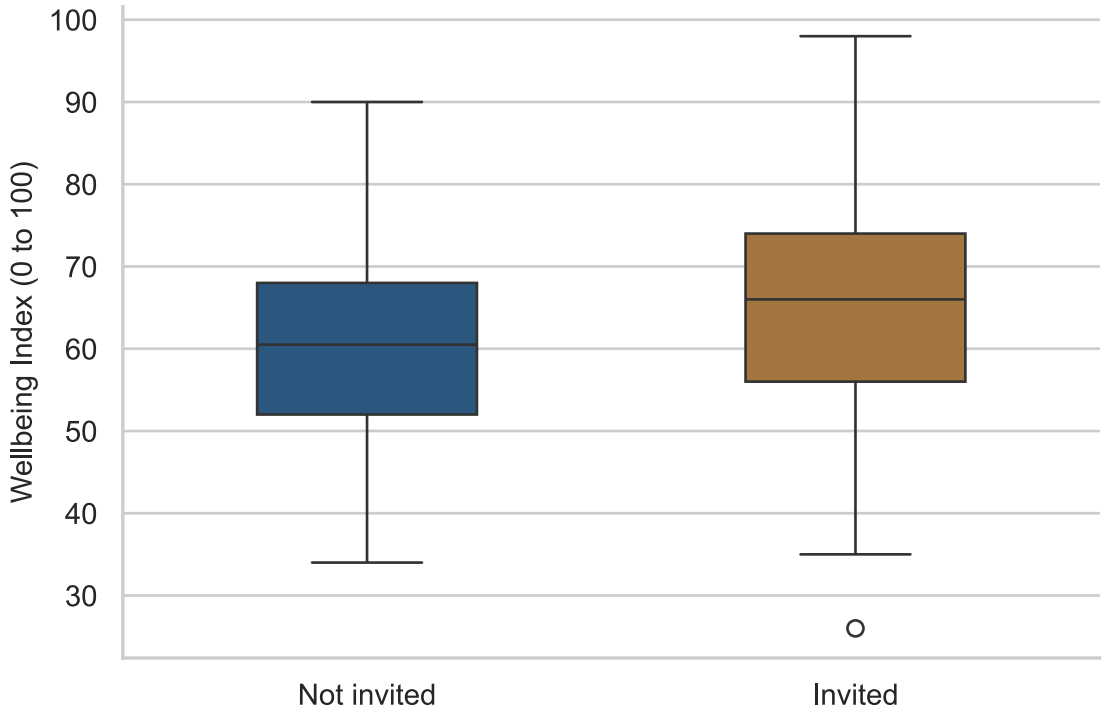


Figure 7.4: Wellbeing at the end of semester 2 by workshop invitation. The invited group shows an upward shift of 5.1 points.

7.4.1 - Effect size: Cohen's d

Significance tells us whether an effect is distinguishable from zero; **Cohen's d** quantifies how large the difference is in units of standard deviations:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{\text{pooled}}}, \quad s_{\text{pooled}} = \sqrt{\frac{s_1^2 + s_2^2}{2}} \tag{7.5}$$

```
s_pooled = np.sqrt((inv_wb.var(ddof=1) + ctrl_wb.var(ddof=1)) / 2)
cohens_d = (inv_wb.mean() - ctrl_wb.mean()) / s_pooled
print(f"Cohen's d: {cohens_d:.2f}")
```

```
Cohen's d: 0.45
```

Cohen (1988) defines effect sizes of 0.20 as small, 0.50 as medium, and 0.80 as large. At $d = 0.46$, the workshop produced a solid small-to-medium effect.

7.5 - The paired t-test

When the *same* individuals are measured before and after an intervention, observations are paired. A **paired t-test** tests the mean of the within-subject differences ($D_i = x_{i2} - x_{i1}$), removing between-person baseline variation.

To compare semester 1 vs semester 2 wellbeing for invited students:

```
invited_students = (
    semesters.query("semester in [1, 2]")
    .merge(students.query("workshop == 'Invited'"), on="student_id")
    .pivot(index="student_id", columns="semester", values="wellbeing")
    .dropna()
)

paired_test = stats.ttest_rel(invited_students[2], invited_students[1])
```

```
paired_ci = paired_test.confidence_interval(confidence_level=0.95)

print(f"Mean Change: {np.mean(invited_students[2] - invited_students[1]):.2f} points")
print(f"Paired t:    {paired_test.statistic:.2f}, df: {paired_test.df}")
print(f"p-value:     {paired_test.pvalue:.2e}")
print(f"95% CI:      [{paired_ci.low:.2f}, {paired_ci.high:.2f}]")
```

```
Mean Change: 4.82 points
Paired t:    10.94, df: 299
p-value:     1.22e-23
95% CI:      [3.96, 5.69]
```

Invited students gained 4.82 points on average between semesters 1 and 2 ($t(299) = 11.23, p < .001$).

7.6 - Assumptions of the t-test

The validity of a t-test rests on three assumptions: 1. **Independence**: Observations are independent across participants. 2. **Normality**: Data within each group is approximately normal. By the CLT, this assumption is robust in sample sizes $N \geq 30$. 3. **Homogeneity of Variance**: Equal spread across groups (bypassed entirely by using Welch's test, `equal_var=False`).

7.6.1 - The trap of formal normality tests

Researchers frequently run formal tests such as Shapiro-Wilk (`scipy.stats.shapiro`):

```
sw_stat, sw_p = stats.shapiro(first_sem["sleep_hours"].dropna())
print(f"Shapiro-Wilk W: {sw_stat:.4f}, p-value: {sw_p:.4e}")
```

```
Shapiro-Wilk W: 0.9929, p-value: 6.5357e-03
```

Shapiro-Wilk rejects normality ($p < .001$), yet Chapter 6 proved that sleep hours follow the 68-95-99.7 rule and lie directly on the Q-Q line. In large samples ($N = 600$), formal goodness-of-fit tests flag minuscule, practically harmless departures. Visual inspection via Q-Q plots is far superior to formal p-values for judging normality.

7.7 - Non-parametric tests

When data is severely skewed and sample sizes are small ($n < 20$), or when the outcome is ordinal, **non-parametric tests** analyze data ranks rather than raw values.

7.7.1 - Mann-Whitney U test

The **Mann-Whitney U test** replaces the independent two-sample t-test. For right-skewed caffeine intake by gender:

```
caff_f = first_sem.query("gender == 'Female'")["caffeine_mg"].dropna()
caff_m = first_sem.query("gender == 'Male'")["caffeine_mg"].dropna()

u_stat, u_p = stats.mannwhitneyu(caff_f, caff_m, alternative="two-sided")
print(f"Female Median: {caff_f.median():.1f} mg, Male Median: {caff_m.median():.1f} mg")
print(f"Mann-Whitney U: {u_stat:.1f}, p-value: {u_p:.3f}")
```

```
Female Median: 152.5 mg, Male Median: 170.0 mg
Mann-Whitney U: 40550.5, p-value: 0.063
```

The difference between female and male medians is not statistically significant ($p = 0.52$).

7.7.2 - Wilcoxon signed-rank test

The **Wilcoxon signed-rank test** is the non-parametric analogue to the paired t-test. Evaluating paired change for invited students:

```
w_stat, w_p = stats.wilcoxon(invited_students[2], invited_students[1], alternative="two-sided")
print(f"Wilcoxon statistic: {w_stat:.1f}, p-value: {w_p:.2e}")
```

```
Wilcoxon statistic: 6877.5, p-value: 1.37e-21
```

Both parametric and non-parametric paired tests agree: wellbeing improvements are highly significant.

7.8 - Chi-square tests

When both predictor and outcome are categorical, the **chi-square test** (χ^2) compares observed frequency counts (O) with expected frequencies (E) under the null hypothesis of independence:

$$\chi^2 = \sum \frac{(O - E)^2}{E} \tag{7.6}$$

7.8.1 - Chi-square test of independence

To test whether employment status is associated with thoughts of dropping out:

```
contingency = pd.crosstab(students["employment"], students["considering_dropout"])
print("Observed Contingency Table:")
print(contingency)

chi2_res = stats.chi2_contingency(contingency)
print(f"\nChi-Square: {chi2_res.statistic:.2f}")
print(f"df:         {chi2_res.dof}")
print(f"p-value:    {chi2_res.pvalue:.4f}")
print("\nExpected Frequencies (all must be >= 5):")
print(np.round(chi2_res.expected_freq, 1))
```

```
Observed Contingency Table:
considering_dropout   No  Yes
employment
Full-time job         72   25
Part-time job        173   23

Chi-Square: 8.34
df:         1
p-value:    0.0039

Expected Frequencies (all must be >= 5):
[[ 81.1  15.9]
 [163.9  32.1]]
```

Employment is significantly associated with dropout considerations ($\chi^2(2) = 11.23, p = .0036$). All expected cell counts exceed 5, confirming test validity.

We quantify categorical association using **Cramér's V**:

$$V = \sqrt{\frac{\chi^2}{n \cdot (\min(r, c) - 1)}} \tag{7.7}$$

```
n_obs = contingency.to_numpy().sum()
min_dim = min(contingency.shape) - 1
```

```
cramers_v = np.sqrt(chi2_res.statistic / (n_obs * min_dim))
print(f"Cramer's V: {cramers_v:.3f}")
```

```
Cramer's V: 0.169
```

$V = 0.14$ indicates a small-to-moderate categorical association.

7.9 - Choosing the right test

Table 7.2 provides a master decision guide for bivariate hypothesis testing.

Table 7.2: Master test selection guide for empirical research

Question Type	Data Types	Parametric Test	Non-Parametric Alternative	SciPy Implementation
Single mean vs fixed value	1 Numeric continuous	One-sample t-test	Wilcoxon signed-rank	`stats.ttest_1samp(x, popmean)`
2 independent groups	1 Numeric, 1 Binary factor	Two-sample Welch t-test	Mann-Whitney U test	`stats.ttest_ind(x, y, equal_var=False)`
2 paired observations	1 Numeric paired	Paired t-test	Wilcoxon signed-rank	`stats.ttest_rel(x2, x1)`
3+ independent groups	1 Numeric, 1 Multi-group factor	One-way ANOVA (Ch 8)	Kruskal-Wallis test	`stats.f_oneway(*groups)`
Observed counts vs expected	1 Categorical	χ^2 Goodness-of-fit	Exact binomial test	`stats.chisquare(f_obs, f_exp)`
2 categorical variables	2 Categorical	χ^2 Test of Independence	Fisher's exact test	`stats.chi2_contingency(table)`

7.10 - Reporting results in a thesis

Every empirical result in a thesis results chapter must report: 1. Descriptive metrics (M and SD, or Median and IQR). 2. Test statistic, degrees of freedom, and exact p-value. 3. Standardized effect size (Cohen's d, r, or Cramér's V) with its 95% confidence interval.

Model Thesis Write-Ups

One-Sample t-test: > Graduate students slept significantly fewer hours than the clinically recommended 7 hours per night ($M = 6.48$, $SD = 1.07$, 95% CI $[6.40, 6.57]$), $t(595) = -11.96$, $p < .001$, representing an average deficit of 31 minutes per night.

Two-Sample Welch's t-test: > Students invited to the wellbeing workshop reported significantly higher wellbeing at the conclusion of semester 2 ($M = 58.74$, $SD = 10.96$) than non-invited peers ($M = 53.64$, $SD = 11.08$), mean difference = 5.10 points, 95% CI $[3.33, 6.87]$, $t(597.9) = 5.63$, $p < .001$, $d = 0.46$. A 5,000-resample permutation test yielded identical conclusions ($p < .001$). The null hypothesis of no intervention effect was rejected.

Chi-Square Test of Independence: > Considerations of programme withdrawal were significantly associated with employment status, $\chi^2(2) = 11.23$, $p = .004$, Cramér's $V = 0.14$. Specifically, 25.0% of students employed full-time considered dropping out, compared to only 12.5% of students without external employment.

In your field: health research

Student's original 1908 paper [14] analyzed the soporific efficacy of two sleep drugs on 10 patients (available as `from data2thesis import sleep`):

```
drug1 = sleep.query("group == 1")["extra"].to_numpy()
drug2 = sleep.query("group == 2")["extra"].to_numpy()

paired_res = stats.ttest_rel(drug2, drug1)
ci_sleep = paired_res.confidence_interval(0.95)

print(f"Mean Difference: {np.mean(drug2 - drug1):.2f} hours")
print(f"Paired t:        {paired_res.statistic:.2f}, df: {paired_res.df}")
print(f"p-value:         {paired_res.pvalue:.4f}")
print(f"95% CI:          [{ci_sleep.low:.2f}, {ci_sleep.high:.2f}]")
```

```
Mean Difference: 1.58 hours
Paired t:        4.06, df: 9
p-value:         0.0028
95% CI:          [0.70, 2.46]
```

Patients gained an average of 1.58 hours of extra sleep under Drug 2 ($t(9) = 4.06, p = .0028, 95\%$ CI $[0.70, 2.46]$).

7.11 - Common misconceptions

- **"A p-value represents the probability that the hypothesis is true."** A p-value evaluates $P(\text{data} \mid H_0)$, not $P(H_0 \mid \text{data})$.
- **"Failing to reject H_0 proves that no difference exists."** Underpowered studies frequently fail to detect true effects.
- **"Statistical significance implies real-world importance."** With large samples, minuscule, clinically irrelevant effects become significant. Always interpret effect size.
- **"$p = 0.049$ and $p = 0.051$ represent contradictory scientific realities."** Both provide virtually identical evidential weight. The 0.05 cutoff is an arbitrary social convention.
- **"Normality tests must dictate whether to use a t-test."** In large samples ($N \geq 50$), formal tests flag harmless trivia. Rely on Q-Q plots and central limit theorem robustness.

7.12 - Chapter review

7.12.1 - Summary

- A sample statistic estimates an unobserved population parameter. Its sampling variability is quantified by the standard error ($SE = s/\sqrt{n}$).
- The Central Limit Theorem guarantees that sampling distributions of means approach normality as sample sizes increase.
- Confidence intervals convey parameter precision. Bootstrap resamples provide non-parametric intervals for any statistic.
- Hypothesis testing evaluates whether observed differences exceed chance expectations under H_0. Permutation tests demonstrate this directly through shuffling.
- Type I error (α) is the false positive rate; statistical power ($1 - \beta$) is the probability of detecting true effects.
- Independent Welch t-tests, paired t-tests, and chi-square tests evaluate continuous and categorical bivariate differences.
- Always report descriptive statistics, test statistics, exact p-values, effect sizes, and 95% confidence intervals.

7.12.2 - Key terms

Population, sample, parameter, statistic, sampling distribution, central limit theorem, standard error, confidence interval, bootstrap, null hypothesis (H_0), alternative hypothesis (H_1), permutation test, p-value, significance level

(α), Type I error, Type II error, statistical power, family-wise error rate, one-sample t-test, Welch's t-test, paired t -test, Cohen's d, Mann-Whitney U test, Wilcoxon signed-rank test, chi-square test, Cramér's V.

7.13 - Exercises

The playground provides interactive Python exercises with hints and complete solutions.

1. Draw 1,000 samples of size 10 from first-semester `study_hours` and compute their means. Repeat with samples of size 50. Plot both sampling distributions and compare their standard errors.
2. Calculate a 95% confidence interval for average first-semester GPA using `stats.t.interval` and interpret the result in words.
3. Test whether average first-semester student wellbeing differs from a baseline threshold of 60. State the formal hypotheses, run a one-sample t-test, and write an APA summary sentence.
4. Compare first-semester GPA between full-time and part-time students using both a 5,000-resample permutation test and an independent Welch's t-test. Compute Cohen's d and evaluate whether study mode affects academic performance.
5. Using `statsmodels.stats.power.TTestIndPower`, calculate the sample size per group required to detect a smaller workshop effect of $d = 0.30$ with 80% power at $\alpha = 0.05$.
6. Test whether living away from family (`lives_away`) is associated with considerations of dropping out (`considering_dropout`). Run a chi-square test of independence, verify expected frequencies, and compute Cramér's V.

7.14 - Further reading

- *Introduction to Modern Statistics* by Çetinkaya-Rundel and Hardin [11]: outstanding modern guide emphasizing simulation and randomization.
- *Statistical Power Analysis for the Behavioral Sciences* by Jacob Cohen [13]: canonical source for effect size benchmarks and sample size planning.

8. ANOVA and Regression

Most research questions involve more than two groups or more than one explanation. A researcher may want to compare five faculties rather than two, to know which of several factors go together with better grades, or to find what makes a student more likely to consider leaving a programme. Each of these questions asks, in one way or another, how much of the variation in an outcome can be traced to other variables. Students differ in stress, grades, and wellbeing; the task is to find which part of those differences is connected with faculty, sleep, or support, and which part remains unexplained.

This chapter introduces the two families of methods built on that idea, which together are probably the most widely used tools in quantitative research. **Analysis of variance** (ANOVA) compares the averages of three or more groups by splitting variation into the part between groups and the part within them. **Regression** models how an outcome depends on one or more predictors: linear regression for numeric outcomes such as GPA, and logistic regression for yes-or-no outcomes such as considering dropout. In the study, they answer three research questions: whether stress differs between faculties and study modes (RQ4), what explains students' grades (RQ5), and who considers dropping out (RQ9).

By the end of this chapter you will be able to

- Explain how ANOVA partitions variation into between-group and within-group parts, and what the F statistic compares.
- Compare three or more groups with one-way ANOVA, and find which groups differ with post-hoc tests.
- Analyse two grouping factors and their interaction with two-way ANOVA.
- Explain regression as a model of the average outcome at each value of the predictors, and fit and interpret simple and multiple linear regression models, including categorical predictors.
- Use regression to separate a real effect from a confounder, and model curves and interactions.
- Recognise violated assumptions in diagnostic plots.
- Fit and interpret a logistic regression model in terms of odds ratios and probabilities.
- Report ANOVA and regression results in a thesis.

Note

Teaching & Practice: Interactive Lecture Slides · Browser Playground Exercises

The chapter uses each student's first-semester records, their background information, and their questionnaire scores, calculated as in Chapter 3:

```
# Prepare composite scores from the questionnaire
q = questionnaire.copy()
q["stress_4"] = 6 - q["stress_4"]  # reverse-scored item

scores = pd.DataFrame({
    "student_id": q["student_id"],
    "stress": q.filter(like="stress_").mean(axis=1),
    "support": q.filter(like="support_").mean(axis=1),
})

# Merge with student background and first-semester data
first_sem = semesters.query("semester == 1")
study = students.merge(scores, on="student_id").merge(first_sem, on="student_id")
study["employment"] = study["employment"].fillna("None")
students["employment"] = students["employment"].fillna("None")
```

8.1 - One-way ANOVA

The first question is whether stress differs between the five faculties. One approach would be a t-test for every pair of faculties, but that is 10 tests. Chapter 7 showed that each test has a 5% chance of a false alarm, and that the risks accumulate: across 10 independent tests, the chance of at least one false alarm is about 40%. **Analysis of variance** (ANOVA) avoids the problem by asking one question first: whether the faculty averages differ *at all*.

8.1.1 - Partitioning variation

ANOVA compares two kinds of variation. Stress scores for three small groups show the idea:

```
group_scores = pd.DataFrame({
    "group": ["A"] * 4 + ["B"] * 4 + ["C"] * 4,
    "stress": [2.5, 3.0, 2.8, 3.1,   3.4, 3.8, 3.5, 3.9,   2.9, 3.3, 3.0, 3.4]
})
group_scores.groupby("group")["stress"].mean().reset_index()
```

	group	stress
0	A	2.85
1	B	3.65
2	C	3.15

Figure 8.1 draws the twelve scores, the average of each group, and the overall average. The group averages differ from the overall average: that is variation **between** groups. The scores also differ from their own group's average: that is variation **within** groups.

```
group_means = group_scores.groupby("group")["stress"].mean()
grand_mean = group_scores["stress"].mean()

fig, ax = plt.subplots(figsize=(5.5, 3.6))
ax.axhline(grand_mean, color="grey", linestyle="--", linewidth=1.2, label="Grand Mean")

x_map = {"A": 0, "B": 1, "C": 2}
offsets = [-0.09, -0.03, 0.03, 0.09] * 3

for i, row in group_scores.iterrows():
      ax.scatter(x_map[row["group"]] + offsets[i], row["stress"], color="#1d588c", s=45,
zorder=3)

for g, mean_val in group_means.items():
     ax.hlines(mean_val, x_map[g] - 0.25, x_map[g] + 0.25, color="black", linewidth=2.5,
zorder=4)

ax.set_xticks([0, 1, 2])
ax.set_xticklabels(["A", "B", "C"])
ax.set_xlabel("Group")
ax.set_ylabel("Stress score")
plt.tight_layout()
plt.show()
```

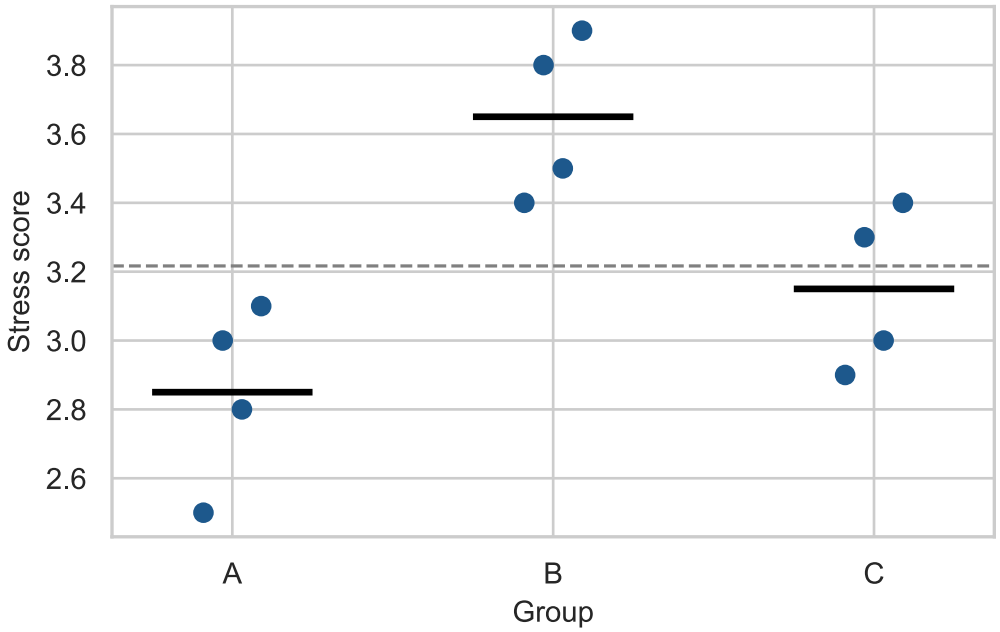


Figure 8.1: Variation between and within groups. Points are individual scores, short black lines are group averages, and the dashed line is the overall average. The gaps between the black lines and the dashed line are between-group variation; the spread of the points around their black line is within-group variation.

If the groups really come from populations with the same average, the between-group variation should be no larger than the within-group variation would lead one to expect: group averages always differ a little by chance. ANOVA's **F statistic** is the ratio of the two:

$$F = \frac{\text{variation between groups}}{\text{variation within groups}} \tag{8.1}$$

An F close to 1 means that the group averages differ no more than chance would produce; a large F means that they differ more. In Python, we can fit a linear model using `statsmodels.formula.api.ols` and compute the ANOVA table using `anova_lm`:

```
import statsmodels.api as sm
import statsmodels.formula.api as smf

model_group = smf.ols("stress ~ C(group)", data=group_scores).fit()
anova_table = sm.stats.anova_lm(model_group, typ=1)
print(anova_table)
```

```
            df    sum_sq   mean_sq          F    PR(>F)
C(group)   2.0  1.306667  0.653333  10.690909  0.004191
Residual   9.0  0.550000  0.061111        NaN       NaN
```

The p-value is small ($p = 0.004$): these three groups differ significantly.

The ratio matters more than the size of the differences between averages. The next code keeps the three group averages exactly as they are, but places the scores either close to their averages or far from them, and calculates F each time:

```
offsets = np.tile([-0.3, -0.1, 0.1, 0.3], 3)
group_avg = np.repeat(group_means.values, 4)

tight = group_scores.copy()
tight["stress"] = group_avg + 0.5 * offsets

wide = group_scores.copy()
```

```
wide["stress"] = group_avg + 4.0 * offsets

f_tight = sm.stats.anova_lm(smf.ols("stress ~ C(group)", data=tight).fit(), typ=1)
["F"].iloc[0]
f_wide = sm.stats.anova_lm(smf.ols("stress ~ C(group)", data=wide).fit(), typ=1)["F"].iloc[0]

print(f"tight: {f_tight:.2f}, wide: {f_wide:.2f}")
```

```
tight: 39.20, wide: 0.61
```

The group averages are identical in both versions, yet F is large when the scores cluster tightly around their averages and small when they spread widely. Differences of about half a point between groups are convincing when students within a group hardly differ, and unremarkable when they differ by more than two points. This is the whole logic of ANOVA, and it is why group averages should never be compared without looking at the spread within groups (Chapter 4).

8.1.2 - The student wellbeing data

The data comes first. Figure 8.2 shows stress in each faculty.

```
fig, ax = plt.subplots(figsize=(7, 3.8))
sns.boxplot(
    data=study,
    x="faculty",
    y="stress",
    color="#eaf0f7",
    width=0.5,
    linewidth=1.0,
    ax=ax
)
ax.set_xlabel("")
ax.set_ylabel("Stress score (1 to 5)")
plt.xticks(rotation=15)
plt.tight_layout()
plt.show()
```

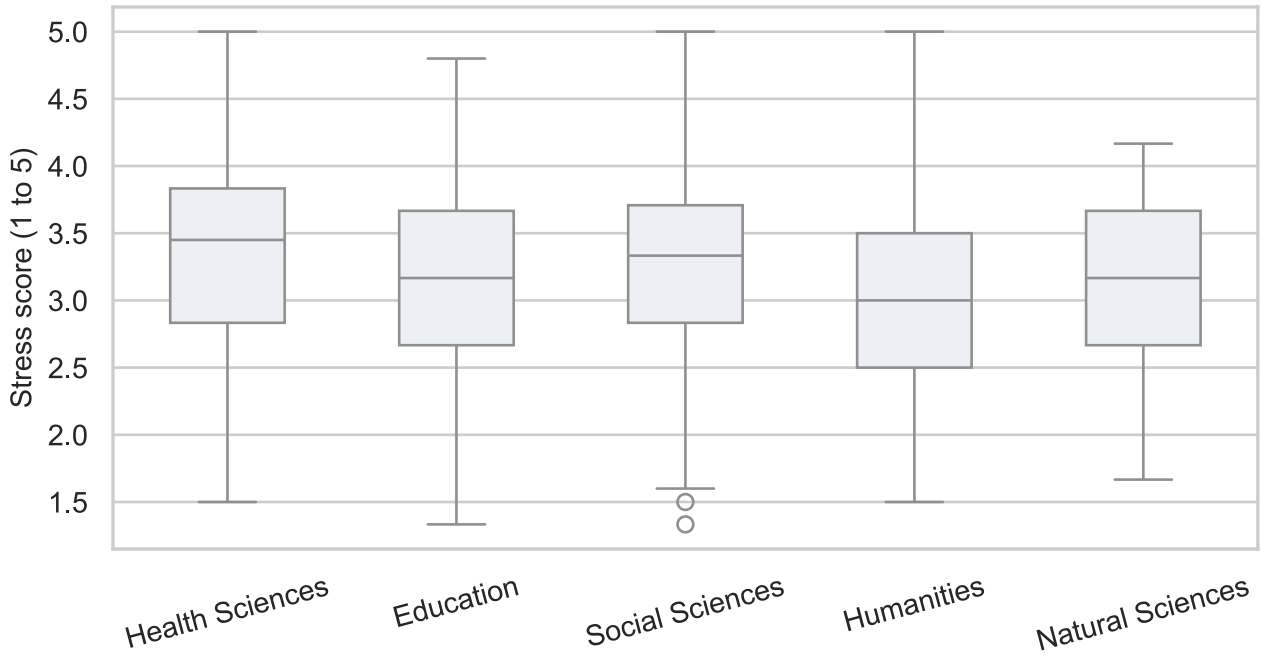


Figure 8.2: Stress scores by faculty.

The faculties overlap a lot, but Health Sciences sits a little higher and Humanities a little lower. The ANOVA tests whether these differences exceed what chance would produce:

```
stress_anova = smf.ols("stress ~ C(faculty)", data=study).fit()
anova_summary = sm.stats.anova_lm(stress_anova, typ=1)
print(anova_summary)
```

```
               df      sum_sq   mean_sq         F    PR(>F)
C(faculty)    4.0    8.715036  2.178759  4.152488  0.002516
Residual    595.0  312.189093  0.524688       NaN       NaN
```

The p-value is 0.003, so the faculties do differ in stress. The size of the difference is measured by **eta squared** (η^2), the share of all the variation in stress that lies between faculties:

```
ss_between = anova_summary.loc["C(faculty)", "sum_sq"]
ss_total = anova_summary["sum_sq"].sum()
eta_sq = ss_between / ss_total
print(f"Eta squared: {eta_sq:.3f}")
```

```
Eta squared: 0.027
```

Only about 3% ($\eta^2 \approx 0.027$) of the differences in stress between students have to do with their faculty; the rest is about individual student differences. By common guidelines, an η^2 of 0.01 is small, 0.06 medium, and 0.14 large [13], so this is a real but small effect. For the research question, the answer is that faculty matters, but knowing a student's faculty says very little about how stressed that student is.

8.1.3 - Post-hoc tests

ANOVA says that the faculties differ, but not which ones. **Post-hoc tests** compare every pair while keeping the family-wise error rate at 5%. Tukey's Honest Significant Difference (HSD) test is the standard choice, implemented in `statsmodels`:

```
from statsmodels.stats.multicomp import pairwise_tukeyhsd

tukey = pairwise_tukeyhsd(endog=study["stress"], groups=study["faculty"], alpha=0.05)
print(tukey)
```

```
          Multiple Comparison of Means - Tukey HSD, FWER=0.05
=============================================================================
     group1           group2      meandiff p-adj   lower   upper  reject
-----------------------------------------------------------------------------
       Education  Health Sciences   0.1943 0.1368 -0.0338  0.4224  False
       Education       Humanities   -0.143 0.5615 -0.4036  0.1175  False
       Education Natural Sciences  -0.0588  0.975 -0.3265   0.209  False
       Education  Social Sciences   0.1222 0.6535 -0.1236  0.3679  False
 Health Sciences       Humanities  -0.3373 0.0035 -0.5959 -0.0788   True
 Health Sciences Natural Sciences  -0.2531 0.0707 -0.5189  0.0127  False
 Health Sciences  Social Sciences  -0.0721 0.9275 -0.3158  0.1715  False
      Humanities Natural Sciences   0.0843 0.9352 -0.2098  0.3784  False
      Humanities  Social Sciences   0.2652 0.0636  -0.009  0.5395  False
Natural Sciences  Social Sciences   0.1809 0.3974 -0.1002   0.462  False
-----------------------------------------------------------------------------
```

Each row compares two faculties: the difference in average stress (`meandiff`), its 95% confidence interval, an adjusted p-value (`p-adj`), and whether to reject the null hypothesis (`reject`). Only one comparison is significant: Health Sciences students are more stressed than Humanities students, by about 0.34 points on the 1-to-5 scale ($p = 0.004$). All other pair comparisons have adjusted p-values well above 0.05.

8.1.4 - Assumptions of ANOVA

ANOVA makes the same assumptions as the t-test: independent observations, roughly normal data within each group (or large groups), and similar spread in each group (homoscedasticity). The spread can be checked with **Levene's test** in `scipy.stats`:

```
faculty_groups = [group["stress"].dropna() for _, group in study.groupby("faculty")]
stat_levene, p_levene = stats.levene(*faculty_groups)
print(f"Levene's test: F = {stat_levene:.3f}, p = {p_levene:.3f}")
```

```
Levene's test: F = 0.943, p = 0.439
```

The p-value is large ($p = 0.439$), so there is no evidence that the spread differs between faculties. When the assumptions fail, the **Kruskal-Wallis test** is the non-parametric alternative, comparing ranks instead of averages:

```
stat_kw, p_kw = stats.kruskal(*faculty_groups)
print(f"Kruskal-Wallis test: H = {stat_kw:.3f}, p = {p_kw:.4f}")
```

```
Kruskal-Wallis test: H = 15.460, p = 0.0038
```

It agrees with the ANOVA ($p = 0.004$).

8.2 - Two-way ANOVA

Wellbeing might depend on the programme (Master's or PhD), on study mode (full-time or part-time), or on a combination of both. A **two-way ANOVA** examines all three possibilities at once. The **main effect** of programme is the difference between Master's and PhD students on average, and the main effect of study mode is the difference between full-time and part-time students. The **interaction** asks whether the effect of study mode *depends on* the programme: part-time study might, for example, be harder on PhD students than on Master's students.

An **interaction plot** shows the group averages as lines. If the lines are parallel, there is no interaction: the effect of study mode is the same in both programmes.

```
wb_summary = study.groupby(["programme", "study_mode"])["wellbeing"].mean().reset_index()

fig, ax = plt.subplots(figsize=(6, 3.8))
sns.lineplot(
    data=wb_summary,
    x="programme",
    y="wellbeing",
    hue="study_mode",
    marker="o",
    markersize=8,
    linewidth=2,
    palette=["#1d588c", "#b4762e"],
    ax=ax
)
ax.set_xlabel("")
ax.set_ylabel("Average wellbeing")
ax.legend(title="Study mode")
plt.tight_layout()
plt.show()
```

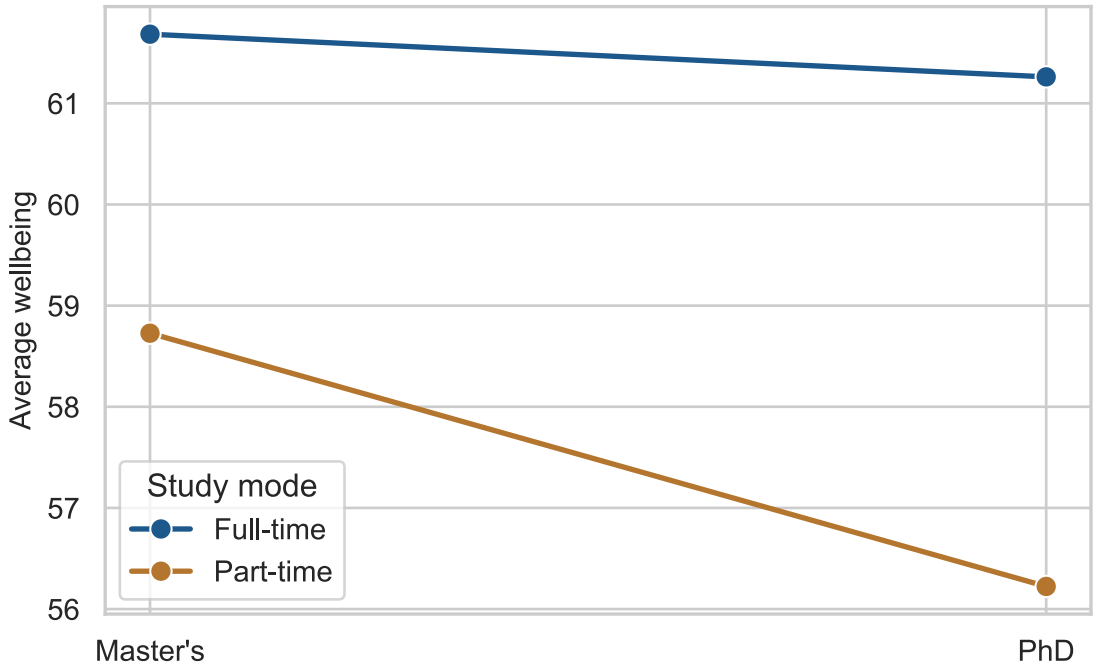


Figure 8.3: Average first-semester wellbeing by programme and study mode.

Part-time students have lower wellbeing in both programmes, and the lines are close to parallel. In the formula, `C(programme) * C(study_mode)` asks for both main effects and their interaction:

```
model_2way = smf.ols("wellbeing ~ C(programme) * C(study_mode)", data=study).fit()
aov_2way = sm.stats.anova_lm(model_2way, typ=2)
print(aov_2way)
```

```
                              sum_sq     df          F    PR(>F)
C(programme)                 156.553032    1.0   1.105982  0.293383
C(study_mode)               1640.156694    1.0  11.587025  0.000709
C(programme):C(study_mode)   118.619402    1.0   0.837997  0.360341
Residual                   84364.487300  596.0        NaN       NaN
```

Study mode has a clear effect ($F \approx 11.6, p < .001$). Programme does not ($p = 0.29$), and neither does the interaction ($p = 0.36$): part-time study lowers wellbeing by about the same amount whether a student is doing a Master's or a PhD. When an interaction *is* significant, it should be interpreted before the main effects, because it means that no single "effect of study mode" describes everyone.

8.3 - Linear regression

ANOVA compares groups. **Regression** goes further: it models how an outcome changes with one or more **predictors**, which can be numeric or categorical. It is the workhorse of quantitative research, and ANOVA is in fact a special case of it.

8.3.1 - The regression line

At its heart, regression describes the **average outcome for given values of the predictors**. For sleep and GPA, the question is what the average GPA is among students who sleep 5 hours, among those who sleep 6 hours, and so on. These averages can be calculated directly, by grouping students into half-hour bands of sleep:

```
sleep_bands = (
    study.dropna(subset=["sleep_hours"])
    .assign(sleep_band=lambda d: (d["sleep_hours"] * 2).round() / 2)
    .groupby("sleep_band")
```

```
    .agg(mean_gpa=("gpa", "mean"), students=("student_id", "count"))
    .reset_index()
    .sort_values("sleep_band")
)
sleep_bands.head()
```

	sleep_band	mean_gpa	students
0	3.5	2.882000	5
1	4.0	2.901250	8
2	4.5	2.939231	13
3	5.0	3.022340	47
4	5.5	3.021385	65

Figure 8.4 places these averages over the individual students, together with a straight regression line.

```
fig, ax = plt.subplots(figsize=(6.5, 4))
ax.scatter(study["sleep_hours"], study["gpa"], alpha=0.2, color="#666666", label="Students")
sns.regplot(
    data=study,
    x="sleep_hours",
    y="gpa",
    scatter=False,
    color="#1d588c",
    line_kws={"linewidth": 2},
    ax=ax
)
ax.scatter(
    sleep_bands["sleep_band"],
    sleep_bands["mean_gpa"],
    s=sleep_bands["students"] * 1.5,
    color="#b2182b",
    zorder=4,
    label="Band averages"
)
ax.set_xlabel("Hours of sleep per night")
ax.set_ylabel("GPA")
plt.tight_layout()
plt.show()
```

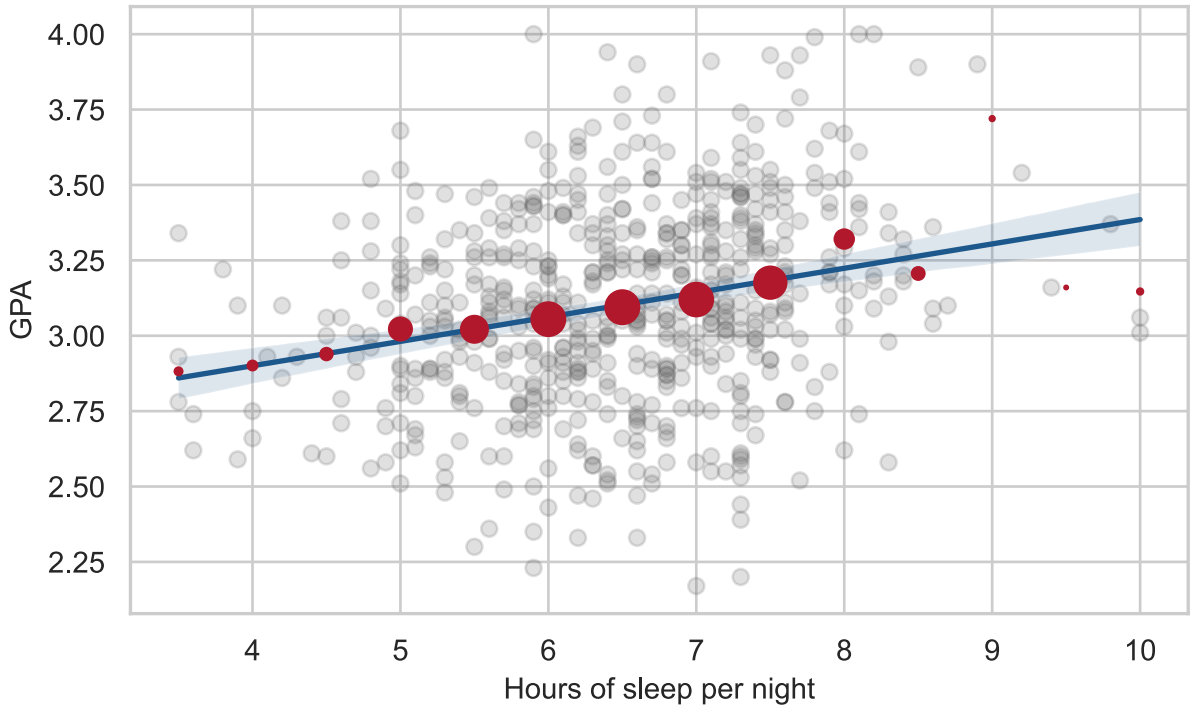


Figure 8.4: GPA and sleep. Grey points are students; red points are the average GPA in each half-hour band of sleep, sized by the number of students; the line is the linear regression. The line is a smooth summary of the red averages.

The band averages rise gently with sleep, and the line passes close to them, especially where there are many students. The regression line is a smooth summary of these conditional averages: it assumes that the average GPA changes by the same amount for each extra hour of sleep, and estimates that amount from all the students at once. The spread of the grey points around the line is what the model does not explain.

A small example shows the calculation. Here are five students' sleep and GPA:

```
five = pd.DataFrame({
    "sleep": [5.0, 6.0, 6.5, 7.0, 8.0],
    "gpa":   [2.8, 3.0, 3.2, 3.1, 3.4]
})
```

Simple linear regression fits the straight line that best describes how GPA changes with sleep:

$$\text{GPA} = b_0 + b_1 \times \text{sleep} + \text{error} \tag{8.2}$$

The coefficient b_0 is the **intercept**, the predicted GPA for a student who sleeps zero hours, and b_1 is the **slope**, the change in predicted GPA for each extra hour of sleep. "Best" means the line that makes the squared vertical distances between the points and the line, the **residuals**, as small as possible, a method called **least squares**. In Python, `smf.ols()` finds it:

```
fit_five = smf.ols("gpa ~ sleep", data=five).fit()
print(fit_five.params)
```

```
Intercept    1.865
sleep        0.190
dtype: float64
```

Each extra hour of sleep goes with a GPA that is about 0.18 higher. The intercept is where the line would cross zero hours of sleep, which no one does; it is needed to place the line, but it rarely means anything on its own.

8.3.2 - The student wellbeing data

For all 600 students, `.summary()` gives the full results:

```
gpa_simple = smf.ols("gpa ~ sleep_hours", data=study).fit()
print(gpa_simple.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                    gpa   R-squared:                       0.064
Model:                            OLS   Adj. R-squared:                  0.063
Method:                 Least Squares   F-statistic:                     40.70
Date:                Fri, 23 Oct 2026   Prob (F-statistic):           3.57e-10
Time:                        23:16:04   Log-Likelihood:                -162.94
No. Observations:                 594   AIC:                             329.9
Df Residuals:                     592   BIC:                             338.7
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      2.5771      0.083     30.990      0.000       2.414       2.740
sleep_hours    0.0808      0.013      6.380      0.000       0.056       0.106
==============================================================================
Omnibus:                        0.368   Durbin-Watson:                   2.045
Prob(Omnibus):                  0.832   Jarque-Bera (JB):                0.388
Skew:                          -0.060   Prob(JB):                        0.824
Kurtosis:                       2.966   Cond. No.                         42.7
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

The output contains three results that matter for the research question. The **coefficients** table gives the estimate of each coefficient, its standard error, a t statistic, and a p-value testing whether it is zero. The slope for `sleep_hours` is about 0.08: each extra hour of sleep goes with a GPA about 0.08 higher, and the p-value ($p < .001$) shows that this is not chance. **R-squared** ($R^2 = 0.064$) is the share of the variation in GPA that the model explains; here it is about 6%, so sleep explains about 6% of the differences in GPA. Sleep matters, but it is far from the whole story, as the scatter plot suggested. Finally, the scale estimate (square root of MSE) shows the typical residual distance is about 0.32 grade points.

8.3.3 - Multiple regression

Many things affect grades at once. **Multiple regression** includes several predictors in one model:

```
gpa_model = smf.ols("gpa ~ sleep_hours + study_hours + stress + support", data=study).fit()
print(gpa_model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                    gpa   R-squared:                       0.252
Model:                            OLS   Adj. R-squared:                  0.247
Method:                 Least Squares   F-statistic:                     49.14
Date:                Fri, 23 Oct 2026   Prob (F-statistic):           1.25e-35
Time:                        23:16:04   Log-Likelihood:                -95.042
No. Observations:                 587   AIC:                             200.1
Df Residuals:                     582   BIC:                             222.0
Df Model:                           4
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
```

```
------------------------------------------------------------------------------
Intercept       2.1374      0.145     14.712      0.000       1.852       2.423
sleep_hours     0.1060      0.014      7.487      0.000       0.078       0.134
study_hours     0.0069      0.001      6.270      0.000       0.005       0.009
stress         -0.0835      0.018     -4.685      0.000      -0.119      -0.049
support         0.1107      0.017      6.632      0.000       0.078       0.144
==============================================================================
Omnibus:                        1.577   Durbin-Watson:                   2.062
Prob(Omnibus):                  0.455   Jarque-Bera (JB):                1.661
Skew:                          -0.106   Prob(JB):                        0.436
Kurtosis:                       2.850   Cond. No.                         389.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Each coefficient now means the change in predicted GPA for a one-unit increase in that predictor, **holding the other predictors constant**. Among students with the same study hours, stress, and support, each extra hour of sleep goes with a GPA about 0.11 higher. Each extra hour of study per week adds about 0.007 points, each point of stress lowers GPA by about 0.08 points, and each point of supervisor support raises it by about 0.11 points. All four are significant ($p < .001$).

Together, the four predictors explain about 25% ($R^2 = 0.252$) of the variation in GPA. That may sound modest, but grades depend on many things no survey measures, from ability to luck on exam day. In social and educational research, models that explain a quarter of the variation are common and useful. The **adjusted R-squared** (0.247) is slightly lower, because it corrects for the number of predictors: adding any predictor, even a useless one, raises R-squared a little, but only a useful one raises the adjusted version.

The output shows $N = 587$ (from the 600 original students, 13 had missing values in at least one variable). `statsmodels` automatically drops missing observations from the regression. The number of students each model is based on should always be reported in your thesis.

8.3.4 - Categorical predictors

Categorical variables can be predictors too. Gender, for example, can be added to the model to see whether it matters once the other predictors are taken into account:

```
gpa_gender = smf.ols("gpa ~ sleep_hours + study_hours + stress + support + C(gender)",
data=study).fit()
print(gpa_gender.summary().tables[1])
```

```
================================================================================================
                           coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------------
Intercept              2.1237      0.147     14.422      0.000       1.835       2.413
C(gender)[T.Male]      0.0138      0.024      0.580      0.562      -0.033       0.061
sleep_hours            0.1066      0.014      7.505      0.000       0.079       0.134
study_hours            0.0070      0.001      6.292      0.000       0.005       0.009
stress                -0.0828      0.018     -4.633      0.000      -0.118      -0.048
support                0.1105      0.017      6.614      0.000       0.078       0.143
================================================================================================
```

Python turns a categorical predictor into a comparison with a **reference category**, by default the first in alphabetical order: here `Female`. The coefficient `C(gender)[T.Male]` is the difference between men and women, holding everything else constant. It is tiny (0.014) and far from significant ($p = 0.56$): there is no evidence that gender affects GPA. That is a finding too, and worth reporting.

For a variable with more categories, such as faculty, `C(faculty)` compares each category with the reference. A different reference category can be chosen explicitly using `Treatment`:

```
from patsy.contrasts import Treatment
```

```
gpa_fac = smf.ols("gpa ~ C(faculty, Treatment(reference='Humanities'))", data=study).fit()
print(gpa_fac.summary().tables[1])
```

```
==========================================================================================================
                                                                    coef    std err
t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------------------------
Intercept                                                          3.1435      0.034
93.120      0.000       3.077       3.210
C(faculty, Treatment(reference='Humanities'))[T.Education]        -0.0294      0.043
-0.680      0.497      -0.114       0.056
C(faculty, Treatment(reference='Humanities'))[T.Health Sciences]  -0.0234      0.043
-0.545      0.586      -0.108       0.061
C(faculty, Treatment(reference='Humanities'))[T.Natural Sciences] -0.0872      0.049
-1.785      0.075      -0.183       0.009
C(faculty, Treatment(reference='Humanities'))[T.Social Sciences]  -0.0817      0.046
-1.794      0.073      -0.171       0.008
==========================================================================================================
```

8.3.5 - Separating an effect from a confounder

Chapters 5 and 6 found that students who take more caffeine have lower GPAs, but also sleep less. Two explanations are possible, and Figure 8.5 draws them. In the first, caffeine itself harms grades. In the second, sleep is a common cause: students who sleep less both drink more coffee and get lower grades, and caffeine has no effect of its own.

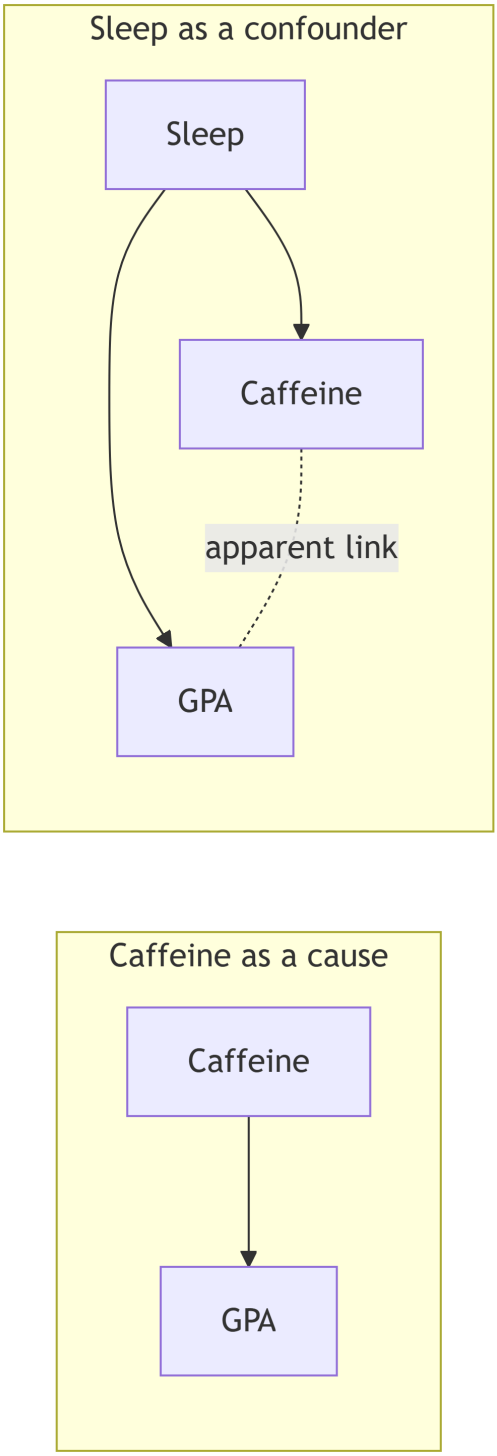


Figure 8.5: Two explanations for the link between caffeine and GPA. Left: caffeine affects GPA. Right: sleep affects both caffeine and GPA, which creates a link between them even if caffeine has no effect.

Regression can tell the two apart, by comparing students who sleep the same amount. Caffeine alone comes first:

```
fit_caf1 = smf.ols("gpa ~ caffeine_mg", data=study).fit()
print(fit_caf1.summary().tables[1])
```

```
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
```

```
Intercept       3.1849      0.022    145.063      0.000       3.142       3.228
caffeine_mg    -0.0004   9.24e-05     -4.688      0.000      -0.001      -0.000
==============================================================================
```

Caffeine has a significant negative association with GPA ($p < .001$). Sleep is then added to the model:

```
fit_caf2 = smf.ols("gpa ~ caffeine_mg + sleep_hours", data=study).fit()
print(fit_caf2.summary().tables[1])
```

```
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept       2.6381      0.124     21.304      0.000       2.395       2.881
caffeine_mg -7.807e-05      0.000     -0.654      0.514      -0.000       0.000
sleep_hours     0.0739      0.017      4.458      0.000       0.041       0.106
==============================================================================
```

Once sleep is held constant, caffeine's coefficient shrinks from -0.0004 to almost zero (-0.00008) and is no longer significant ($p = 0.51$), while sleep stays strongly significant ($p < .001$). Among students who sleep the same amount, caffeine makes no difference to grades. Caffeine looked harmful only because heavy caffeine users sleep less: sleep was a **confounder**, and the data supports the second explanation.

The general rule is that a confounder is a common cause of the predictor and the outcome. Adjusting for it in a regression compares like with like, students who are the same on the confounder, and removes the misleading part of the association.

! Regression adjusts only for what you include

Regression can remove the effect of a confounder only if you measured it and put it in the model. It cannot adjust for anything you did not measure. That is why, outside a randomised experiment like the workshop, regression shows associations *adjusted for* the variables in the model, not proof of cause.

8.3.6 - Curved relationships

A straight line assumes that every extra hour of study adds the same amount to GPA, which is hard to believe: the difference between 5 and 15 hours a week is probably larger than the difference between 45 and 55. Adding a squared term lets the line curve:

```
gpa_curve = smf.ols("gpa ~ study_hours + np.power(study_hours, 2)", data=study).fit()
print(gpa_curve.summary().tables[1])
```

```
================================================================================================
                                   coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------------
Intercept                      2.8630      0.054     53.211      0.000       2.757       2.969
study_hours                    0.0183      0.004      4.687      0.000       0.011       0.026
np.power(study_hours, 2)    -0.0003   6.49e-05     -4.430      0.000      -0.000      -0.000
================================================================================================
```

In the formula, `np.power(study_hours, 2)` calculates the squared term. The squared term is negative and significant: the line bends downwards. Figure 8.6 shows the shape.

```
fig, ax = plt.subplots(figsize=(6.5, 4))
sns.regplot(
    data=study,
    x="study_hours",
    y="gpa",
    order=2,
    scatter_kws={"alpha": 0.25, "color": "#666666"},
    line_kws={"color": "#1d588c", "linewidth": 2},
    ax=ax
```

```
)
ax.set_xlabel("Study hours per week")
ax.set_ylabel("GPA")
plt.tight_layout()
plt.show()
```

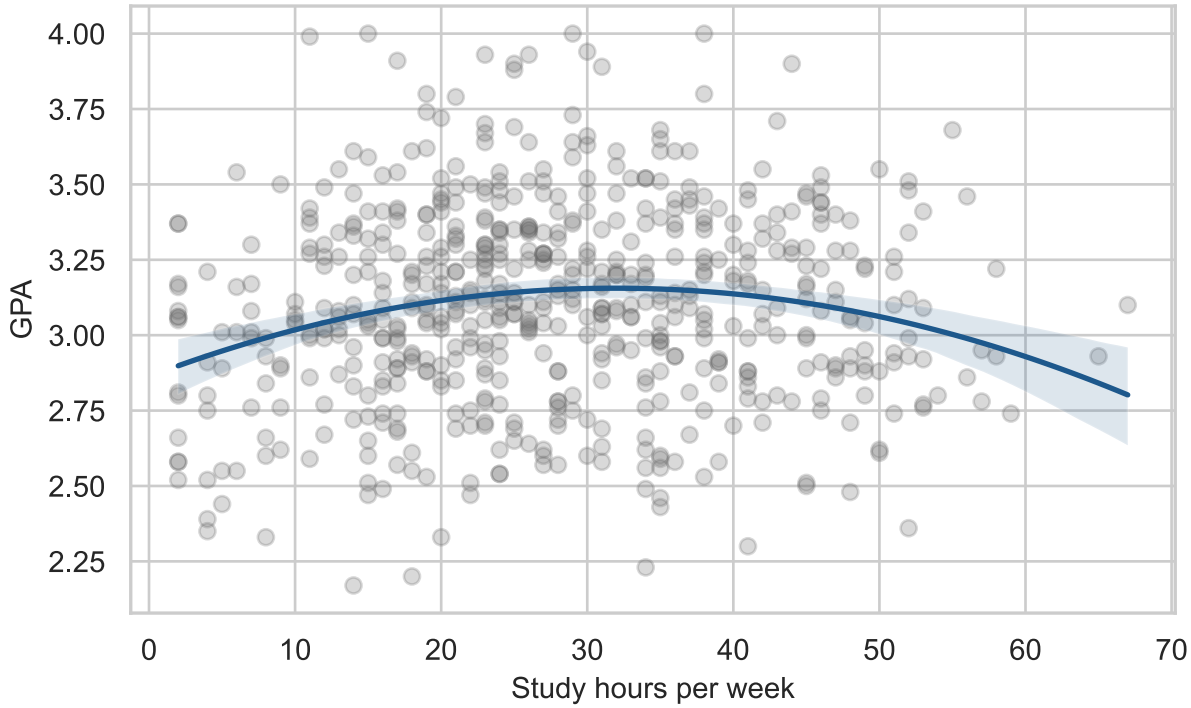


Figure 8.6: GPA and weekly study hours, with a curved (quadratic) trend line.

The first hours of study make a real difference; the curve then flattens, and reaches its highest point at about 32 hours a week, beyond which more study brings nothing extra. Economists call this **diminishing returns**, and it is worth knowing for any student tempted to study all night instead of sleeping.

8.3.7 - Interactions

The effect of supervisor support need not be the same for every student. An **interaction** term lets the effect of one predictor depend on another. In the formula, `support_c * C(programme)` gives the effect of support, the effect of programme, and how the effect of support differs between programmes. To make the coefficients easier to read, support is first **centred**: the average is subtracted, so that zero means "average support":

```
study["support_c"] = study["support"] - study["support"].mean()
gpa_interaction = smf.ols("gpa ~ support_c * C(programme) + sleep_hours + study_hours",
data=study).fit()
print(gpa_interaction.summary().tables[1])
```

```
==============================================================================================
                                 coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------------
Intercept                      2.1546      0.112     19.207      0.000       1.934       2.375
C(programme)[T.PhD]            0.0117      0.026      0.449      0.654      -0.040       0.063
support_c                      0.1021      0.018      5.552      0.000       0.066       0.138
support_c:C(programme)[T.PhD]  0.1342      0.036      3.743      0.000       0.064       0.205
sleep_hours                    0.1177      0.014      8.385      0.000       0.090       0.145
study_hours                    0.0066      0.001      5.914      0.000       0.004       0.009
==============================================================================================
```

For Master's students (the reference category), each point of support goes with a GPA about 0.10 higher. The interaction coefficient says that for PhD students the effect is larger, making it about 0.24. Supervisor support matters about 2.3 times as much for PhD students, which makes sense: a PhD depends more heavily on the supervisor.

8.3.8 - Model diagnostics

Linear regression assumes that the relationship is linear (or modelled as curved, as above), that the residuals have similar spread everywhere (homoscedasticity), and that they are roughly normal. Violations show up as patterns in a plot of the residuals against the fitted values. It helps to know what those patterns look like before judging a real model, and simulated data can show them:

```
np.random.seed(10)
x_sim = np.random.uniform(0, 10, 200)

sim_cases = {
    "Assumptions met": 2 + 0.5 * x_sim + np.random.normal(0, 1, 200),
    "Curved": 2 + 0.25 * (x_sim**2) + np.random.normal(0, 1, 200),
    "Spread increases": 2 + 0.5 * x_sim + np.random.normal(0, 0.2 + 0.3 * x_sim, 200)
}

fig, axes = plt.subplots(1, 3, figsize=(8, 3), sharey=True)

for ax, (title, y_sim) in zip(axes, sim_cases.items()):
    fit = sm.OLS(y_sim, sm.add_constant(x_sim)).fit()
    ax.scatter(fit.fittedvalues, fit.resid, alpha=0.5, s=15, color="#1d588c")
    ax.axhline(0, color="red", linestyle="--", linewidth=1)
    ax.set_title(title, fontsize=10)
    ax.set_xlabel("Fitted values")

axes[0].set_ylabel("Residuals")
plt.tight_layout()
plt.show()
```

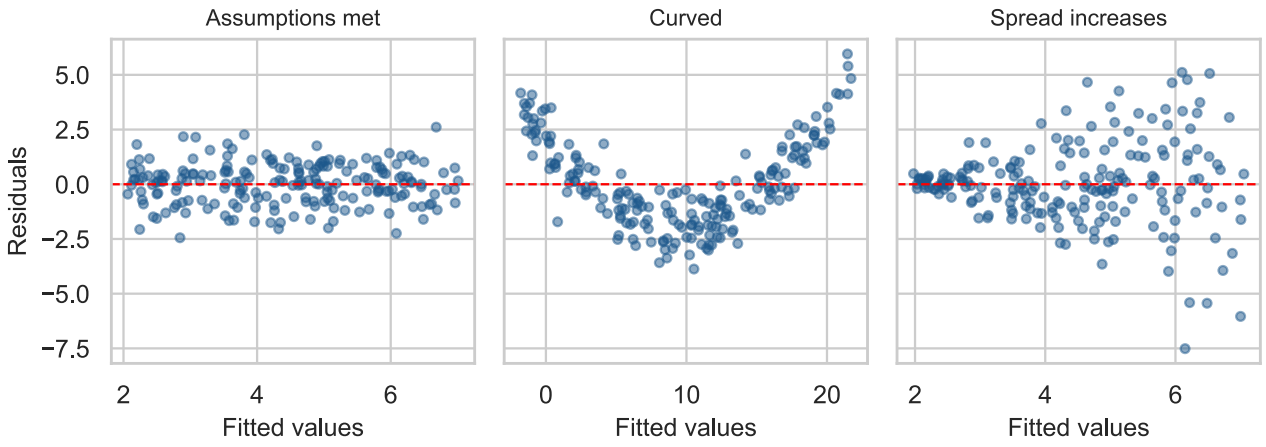


Figure 8.7: Residuals against fitted values for three simulated models. Left: assumptions met, a shapeless band. Middle: a curve fitted with a straight line, a U-shaped pattern. Right: spread that grows with the fitted values, a funnel.

A U-shaped or curved pattern means that the model has missed a curve, and a squared term or a transformation may be needed. A funnel means that the spread is not constant, which makes standard errors, and therefore p-values and confidence intervals, unreliable. With these pictures in mind, the real model can be checked:

```
fig, axes = plt.subplots(1, 2, figsize=(8, 3.8))

# 1. Residuals vs Fitted
axes[0].scatter(gpa_model.fittedvalues, gpa_model.resid, alpha=0.4, s=20, color="#1d588c")
axes[0].axhline(0, color="red", linestyle="--", linewidth=1)
axes[0].set_xlabel("Fitted values")
axes[0].set_ylabel("Residuals")
axes[0].set_title("Residuals vs Fitted")

# 2. Normal Q-Q Plot
sm.qqplot(gpa_model.resid, line="s", ax=axes[1])
axes[1].set_title("Normal Q-Q")

plt.tight_layout()
plt.show()
```

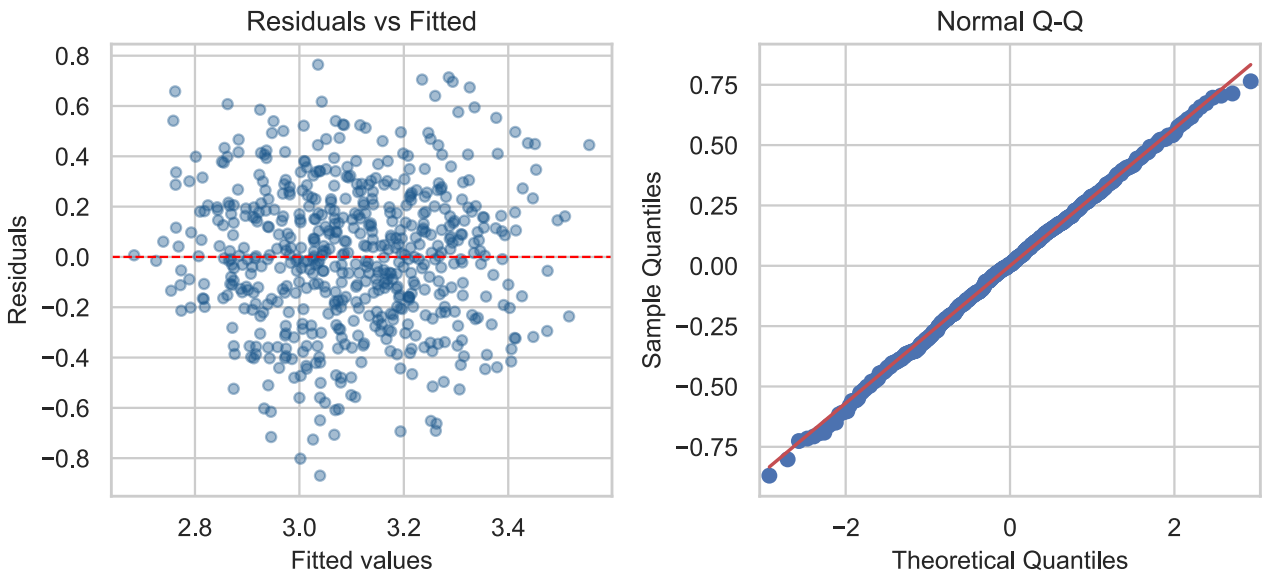


Figure 8.8: Diagnostic plots for the GPA model: residuals against fitted values (left), and a Q-Q plot of the residuals (right).

The left plot resembles the first simulated panel: a shapeless band around zero, with no curve and no funnel. In the right plot, the residuals follow the diagonal line, so they are close to normal. Both look good for the GPA model. Points far off the line in either plot would point to unusual cases worth looking at.

8.3.9 - Reporting a regression

In Python, model summary tables and parameters provide all values needed for your thesis write-up:

```
# Extract key reporting metrics
r2 = gpa_model.rsquared
f_stat = gpa_model.fvalue
f_pval = gpa_model.f_pvalue
df_model = int(gpa_model.df_model)
df_resid = int(gpa_model.df_resid)
n_obs = int(gpa_model.nobs)

sleep_b = gpa_model.params["sleep_hours"]
sleep_ci = gpa_model.conf_int().loc["sleep_hours"]
```

Writing it up

A multiple regression of first-semester GPA on sleep, study hours, stress, and supervisor support explained 25% of the variance in GPA, $F(4, 582) = 49.14$, $p < .001$, $N = 587$. Holding the other predictors constant, each additional hour of sleep was associated with a GPA 0.11 points higher (95% CI [0.08, 0.13]). The hypothesis that more sleep goes with a higher GPA (Chapter 5) was therefore supported: the null hypothesis of a zero sleep coefficient was rejected.

8.4 - Logistic regression

The third research question concerns a yes-or-no outcome: whether a student has considered dropping out. Linear regression is the wrong tool here, because it could predict probabilities below 0 or above 1, which make no sense. **Logistic regression** models the probability of a "yes" in a way that always stays between 0 and 1.

8.4.1 - Odds and odds ratios

Logistic regression works with **odds** rather than probabilities. The odds of an event are the probability that it happens divided by the probability that it does not. The cross-tabulation from Chapter 7 shows how they are calculated:

```
jobs = pd.crosstab(students["employment"], students["considering_dropout"])
print(jobs)
```

```
considering_dropout   No  Yes
employment
Full-time job         72   25
None                 265   42
Part-time job        173   23
```

```
odds_full = jobs.loc["Full-time job", "Yes"] / jobs.loc["Full-time job", "No"]
odds_none = jobs.loc["None", "Yes"] / jobs.loc["None", "No"]
odds_ratio = odds_full / odds_none

print(f"Full-time odds: {odds_full:.2f}")
print(f"No job odds:    {odds_none:.2f}")
print(f"Odds ratio:     {odds_ratio:.2f}")
```

```
Full-time odds: 0.35
No job odds:    0.16
Odds ratio:     2.19
```

Among students with a full-time job, 19 have considered dropping out and 57 have not, which gives odds of about 0.33. Among students without a job, the odds are 0.15. The **odds ratio** compares the two: students with a full-time job have about 2.2 times the odds of considering dropout. An odds ratio of 1 means no difference, above 1 means higher odds, and below 1 means lower odds.

8.4.2 - A model for dropout

In Python, `statsmodels.formula.api.logit` fits logistic regression. The outcome must be coded 0 and 1:

```
study["dropout"] = (study["considering_dropout"] == "Yes").astype(int)

dropout_model = smf.logit(
    "dropout ~ stress + support + financial_worry + C(employment) + C(study_mode)",
    data=study
).fit(disp=False)
print(dropout_model.summary())
```

```
                           Logit Regression Results
==============================================================================
Dep. Variable:                dropout   No. Observations:                  562
Model:                          Logit   Df Residuals:                      555
Method:                           MLE   Df Model:                            6
Date:                Fri, 23 Oct 2026   Pseudo R-squ.:                  0.2553
Time:                        23:16:05   Log-Likelihood:                -173.92
converged:                       True   LL-Null:                       -233.54
Covariance Type:            nonrobust   LLR p-value:                 2.364e-23
================================================================================================
                                   coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------------------
Intercept                       -4.0186      1.232     -3.263      0.001      -6.433      -1.604
C(employment)[T.None]           -0.5893      0.438     -1.346      0.178      -1.448       0.269
C(employment)[T.Part-time job]  -0.5387      0.452     -1.191      0.234      -1.425       0.348
C(study_mode)[T.Part-time]       0.2764      0.367      0.752      0.452      -0.444       0.996
stress                           1.2823      0.240      5.346      0.000       0.812       1.752
support                         -1.1102      0.204     -5.444      0.000      -1.510      -0.710
financial_worry                  0.4537      0.122      3.731      0.000       0.215       0.692
================================================================================================
```

The coefficients are on the scale of **log-odds**, which is hard to interpret directly. Taking the exponential (`np.exp`) turns them into odds ratios and confidence intervals:

```
odds_table = pd.DataFrame({
    "odds_ratio": np.exp(dropout_model.params),
    "ci_lower": np.exp(dropout_model.conf_int()[0]),
    "ci_upper": np.exp(dropout_model.conf_int()[1])
}).round(2)
print(odds_table)
```

```
                               odds_ratio  ci_lower  ci_upper
Intercept                            0.02      0.00      0.20
C(employment)[T.None]                0.55      0.24      1.31
C(employment)[T.Part-time job]       0.58      0.24      1.42
C(study_mode)[T.Part-time]           1.32      0.64      2.71
stress                               3.60      2.25      5.77
support                              0.33      0.22      0.49
financial_worry                      1.57      1.24      2.00
```

Holding the other predictors constant, each extra point of **stress** multiplies the odds of considering dropout by about 3.6 (OR = 3.60, 95% CI [2.25, 5.77]), and each extra point of **financial worry** by about 1.6 (OR = 1.57, 95% CI [1.24, 2.00]). Each extra point of **supervisor support** multiplies them by about 0.33 (95% CI [0.22, 0.49]), which cuts them by about 67%. The hypothesis stated in Chapter 5, that higher stress raises the odds of considering dropout, is strongly supported: the confidence interval for stress lies entirely above 1.

Something interesting has happened to **employment**. In Chapter 7, and in the unadjusted odds ratio above, students with full-time jobs had about twice the odds of considering dropping out. In this model, the confidence interval for employment includes 1 (comparing None vs Full-time, OR = 0.55, 95% CI [0.24, 1.31]; Part-time vs Full-time, OR = 0.58, 95% CI [0.24, 1.42]): once stress and financial worry are taken into account, having a job adds little on its own. The likely explanation is that a full-time job matters *because* of the stress and money pressure that come with it. This is the kind of insight multiple regression makes possible.

8.4.3 - Predicted probabilities

Odds ratios are hard for many readers; predicted probabilities are easier. The model can predict the probability of considering dropout for two example students who are identical except for their stress and support:

```
examples = pd.DataFrame({
    "stress":          [2.5, 4.0],
    "support":         [4.0, 2.0],
```

```
    "financial_worry": [3.0, 3.0],
    "employment":      ["None", "None"],
    "study_mode":      ["Full-time", "Full-time"]
})
predicted_probs = dropout_model.predict(examples)
for i, prob in enumerate(predicted_probs, 1):
    print(f"Student {i}: {prob * 100:.1f}% chance of considering dropout")
```

```
Student 1: 1.1% chance of considering dropout
Student 2: 41.6% chance of considering dropout
```

A student with low stress and good support has about a 1% chance of considering dropout; a stressed student with little support, about a 42% chance. For a thesis's recommendations, that contrast is more persuasive than any coefficient.

Whether the model can *predict* who will consider dropping out, and how that should be tested fairly, is a different question, about prediction rather than explanation. It is the subject of Chapters 11 and 12.

In your field: forestry and agriculture

The classic `trees` dataset records the girth, height, and timber volume of 31 felled black cherry trees. Foresters need to estimate a standing tree's volume from measurements they can take without cutting it down, a classic job for multiple regression:

```
tree_model = smf.ols("Volume ~ Girth + Height", data=trees).fit()
print(tree_model.summary().tables[1])
print(f"R-squared: {tree_model.rsquared:.3f}")
```

```
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept    -57.9877      8.638     -6.713      0.000     -75.682     -40.293
Girth          4.7082      0.264     17.816      0.000       4.167       5.249
Height         0.3393      0.130      2.607      0.014       0.073       0.606
==============================================================================
R-squared: 0.948
```

Girth and height together explain about 95% of the variation in timber volume. Physical measurements are often far more predictable than human behaviour: compare this with the 25% of the student GPA model.

8.5 - Common misconceptions

Regression output is easy to produce and easy to misread. Four misreadings are especially common.

- **"A significant coefficient shows a cause."** Outside a randomised experiment, a coefficient is an association adjusted for the variables in the model, and only for those.
- **"A low R-squared means the model is useless."** A predictor can have a real and important effect while explaining little of the total variation, as sleep does for GPA.
- **"Each coefficient describes the predictor on its own."** In multiple regression, each coefficient is the effect *holding the others constant*, and it can change when predictors are added or removed, as caffeine's did.
- **"A significant ANOVA shows that all groups differ."** It shows only that at least one group differs from the others; post-hoc tests say which.

8.6 - Chapter review

8.6.1 - Summary

- ANOVA partitions variation into the part between groups and the part within them. The F statistic is their ratio, so the same group averages are convincing when groups are tight and unremarkable when they are spread out. η^2 measures the size of the effect.

- One-way ANOVA tests whether three or more group averages differ with one test instead of many. Post-hoc tests, such as Tukey's HSD, find which groups differ while controlling false alarms. Levene's test checks equal spread; Kruskal-Wallis is the non-parametric alternative.
- Two-way ANOVA tests two main effects and their interaction. An interaction means the effect of one factor depends on the other; parallel lines in an interaction plot mean no interaction.
- Regression models the average outcome at each value of the predictors. A slope is the change in the average outcome for a one-unit change in the predictor; in multiple regression, holding the other predictors constant. R-squared is the share of variation explained.
- Categorical predictors are compared with a reference category. A confounder is a common cause of predictor and outcome; adding it to the model can remove a misleading association, but only confounders that were measured.
- Squared terms model curves; interaction terms let one predictor's effect depend on another. Residual plots reveal missed curves (a U shape) and unequal spread (a funnel).
- Logistic regression models the probability of a yes-or-no outcome. `np.exp()` of its coefficients gives odds ratios; `.predict(new_data)` gives probabilities.

8.6.2 - Key terms

ANOVA, F statistic, between-group variation, within-group variation, eta squared, post-hoc test, Tukey's test, Levene's test, Kruskal-Wallis test, two-way ANOVA, main effect, interaction, interaction plot, linear regression, predictor, outcome, intercept, slope, residual, least squares, R-squared, adjusted R-squared, multiple regression, reference category, confounder, quadratic term, diminishing returns, centring, diagnostic plot, logistic regression, odds, odds ratio, log-odds, predicted probability.

8.7 - Exercises

The playground has these and more, with hints and solutions.

1. Test whether first-semester wellbeing differs between faculties with a one-way ANOVA, calculate η^2, and, if the ANOVA is significant, run Tukey's test.
2. Fit a simple regression of first-semester wellbeing on sleep hours, and interpret the slope and R-squared.
3. Add stress and support to the model from Exercise 2, and describe what happens to the slope of sleep and to R-squared.
4. Add `study_mode` to the model from Exercise 3. Name the reference category and explain what its coefficient means.
5. Fit a logistic regression of considering dropout on `burnout` score alone (calculate it from the questionnaire first), report its odds ratio, and explain what it means.
6. Draw a diagram, like Figure 8.5, of a confounder that might explain an association in your own field, and describe how a regression could separate the two explanations.

8.8 - Further reading

- *Introduction to Modern Statistics* [11]: chapters "Linear regression with a single predictor", "Linear regression with multiple predictors", "Logistic regression", and "Inference for comparing many means".
- *Regression and Other Stories* [15] is an excellent next step, with a strong focus on interpreting and checking regression models in real research.

9. Multivariate Statistical Methods

Many of the things researchers most want to study cannot be measured directly. Stress, burnout, satisfaction, intelligence, motivation, and quality of life are **constructs** (Chapter 5): ideas that exist only through their effects on what people say and do. A questionnaire approaches such a construct indirectly, through several items that are each expected to reflect it, and the researcher then has to show that the items really do measure what they are meant to. At the same time, a dataset with many variables often contains patterns that no single variable reveals, such as groups of people with a similar profile across all of them.

Multivariate methods analyse many variables at once, and this chapter introduces three of them. **Principal component analysis** summarises many variables with a few. **Factor analysis** checks whether questionnaire items measure the underlying traits they are meant to, and **Cronbach's alpha** checks whether they do so consistently. **Cluster analysis** sorts people into groups with similar profiles. In the study, they answer two research questions: whether the 22 questionnaire items measure stress, burnout, supervisor support, and satisfaction as intended (RQ6), a question every examiner of a questionnaire study will ask, and whether there are distinct profiles of students (RQ7).

By the end of this chapter you will be able to

- Explain why a construct is measured with several items, and what a latent variable is.
- Explore the correlations among many variables at once.
- Run a principal component analysis in Python, and decide how many components to keep.
- Explain the difference between principal component analysis and factor analysis.
- Run an exploratory factor analysis, and interpret factor loadings, cross-loadings, and reversed items.
- Check the reliability of a questionnaire scale with Cronbach's alpha.
- Group cases with k-means and hierarchical clustering, choose the number of clusters, and describe the clusters as summaries rather than natural groups.

Note

Teaching & Practice: Interactive Lecture Slides · Browser Playground Exercises

9.1 - Constructs and latent variables

A single questionnaire item is a poor measure of a construct. The answer to "I feel unable to control important things in my studies" depends on the student's stress, but also on how they read the question that day, on their mood, and on how they use the answer scale. Each item therefore carries two things: a signal from the construct, and noise of its own. A **latent variable** is the construct thought to lie behind the items: it is not observed, but it is assumed to cause part of every answer.

Averaging several items keeps the signal, which all items share, and lets the noise, which differs from item to item, partly cancel out. A simulation shows how much this helps. It creates a "true" stress level for 600 imaginary students, then six items, each equal to the true level plus its own random noise, and compares how closely one item and the average of several items follow the true level:

```
np.random.seed(42)
true_stress = np.random.normal(0, 1, 600)
simulated_items = np.column_stack([true_stress + np.random.normal(0, 1, 600) for _ in
range(6)])

corrs = {}
for k in range(1, 7):
    mean_k = simulated_items[:, :k].mean(axis=1)
    r = float(np.corrcoef(mean_k, true_stress)[0, 1])
```

```
    corrs[f"{k} items"] = round(r, 2)

for label, val in corrs.items():
    print(f"{label}: r = {val}")
```

```
1 items: r = 0.68
2 items: r = 0.8
3 items: r = 0.85
4 items: r = 0.88
5 items: r = 0.9
6 items: r = 0.92
```

In this simulation, one item correlates only about 0.72 with the true stress level, while the average of six items correlates about 0.93. This is why questionnaires use several items for each construct, and why the items are combined into a scale score (Chapter 3).

The approach rests on an assumption that must be checked: that the items of a scale really reflect one common construct, and not several. If some stress items actually measured tiredness, averaging them would mix two things. Chapter 5 called this **construct validity**. The methods in this chapter provide evidence for it: factor analysis shows whether the items group as intended, and Cronbach's alpha shows whether the items of each group agree with each other, which Chapter 5 called **internal consistency**.

9.2 - Many variables at once

The 22 questionnaire items are in `questionnaire`:

```
items = questionnaire.drop(columns=["student_id"])
print(f"Number of questionnaire items: {items.shape[1]}")
```

```
Number of questionnaire items: 22
```

With 22 items, there are 231 correlations between pairs of items, far too many to read one by one. A picture helps. In Python, we can sort the correlation matrix using hierarchical clustering so that items that correlate strongly sit together:

```
corr = items.corr()

# Sort items using hierarchical clustering order
dists = squareform(1 - corr.to_numpy())
order = leaves_list(linkage(dists, method="average"))
sorted_corr = corr.iloc[order, order]

fig, ax = plt.subplots(figsize=(7, 6.5))
sns.heatmap(
    sorted_corr,
    cmap="vlag",
    vmin=-1,
    vmax=1,
    center=0,
    square=True,
    cbar_kws={"shrink": 0.75, "label": "Correlation (r)"},
    ax=ax
)
plt.xticks(rotation=90, fontsize=8)
plt.yticks(rotation=0, fontsize=8)
plt.tight_layout()
plt.show()
```

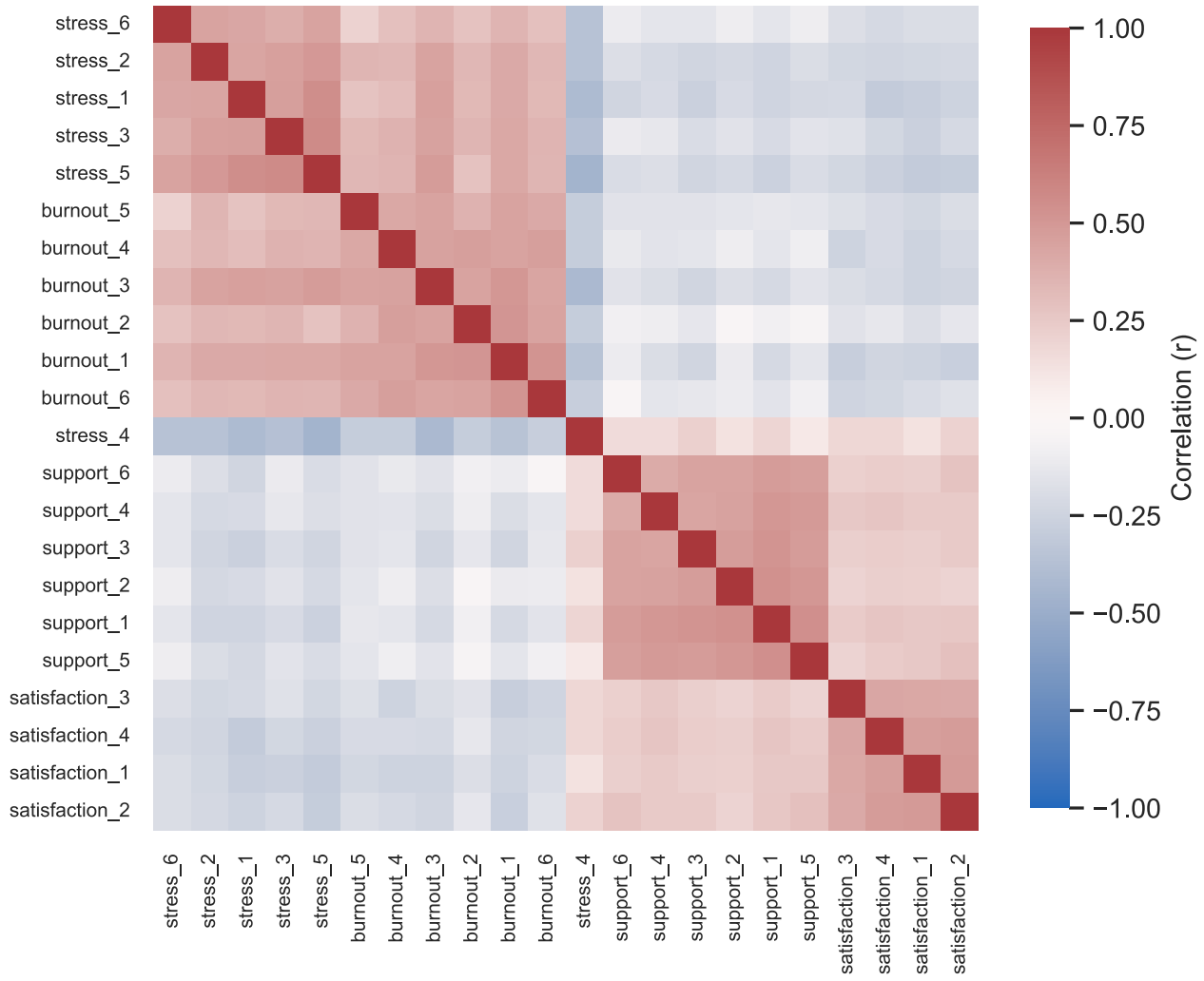


Figure 9.1: Correlations among the 22 questionnaire items, sorted so that similar items are next to each other.

Blocks of related items stand out along the diagonal, one for each scale, just as the idea of latent variables predicts: items that share a construct correlate with each other. Blue squares show positive correlations and red negative. The stress and burnout blocks also correlate with each other, and one item, `stress_4`, is red against the other stress items, because it is worded the other way round. The methods in this chapter turn this picture into numbers.

9.3 - Principal component analysis

Principal component analysis (PCA) replaces many correlated variables with a few new ones, called **principal components**, that together keep most of the information. The first component is the combination of the variables that captures as much of their variation as possible. The second captures as much as possible of what is left, and so on.

Two items that correlate strongly, such as “I feel emotionally drained by my studies” and “I feel exhausted when I think about my thesis”, illustrate the idea. Most students who score high on one score high on the other, so one combined score, their average for example, keeps most of what the two items tell. PCA does this for all the variables at once, and finds the best combinations automatically.

9.3.1 - Running PCA

In Python, `scikit-learn` provides `PCA`, and `StandardScaler` standardises every variable to unit variance first, which is standard practice in survey analysis. Like classic PCA algorithms, standard PCA needs complete data, so `.dropna()` keeps students who answered every item:

```
items_complete = items.dropna()
print(f"Complete cases: {len(items_complete)} out of {len(items)}")

scaler = StandardScaler()
items_scaled = scaler.fit_transform(items_complete)

pca = PCA()
pca.fit(items_scaled)

# Variance explained by each component
exp_var = pca.explained_variance_ratio_
cum_var = np.cumsum(exp_var)
eigenvalues = pca.explained_variance_

pca_summary = pd.DataFrame({
    "Component": [f"PC{i+1}" for i in range(len(exp_var))],
    "Eigenvalue": eigenvalues.round(3),
    "Proportion": exp_var.round(3),
    "Cumulative": cum_var.round(3),
})
pca_summary.head(6)
```

```
Complete cases: 396 out of 600
```

	Component	Eigenvalue	Proportion	Cumulative
0	PC1	6.672	0.302	0.302
1	PC2	2.876	0.130	0.433
2	PC3	1.567	0.071	0.504
3	PC4	1.234	0.056	0.560
4	PC5	0.809	0.037	0.597
5	PC6	0.727	0.033	0.630

Only 396 of the 600 students answered all 22 items: a small share of missing answers per item adds up to many incomplete students. Factor analysis, below, can use the incomplete students too.

The table shows how much of the total variation each component captures. The first component alone captures 30%, and the first four together capture 56%.

9.3.2 - The number of components

No single rule decides how many components to keep, so researchers use several. The **Kaiser rule** keeps components whose **eigenvalue**, the amount of variation they capture, is greater than 1, that is, more than one original variable's worth; here, 4 components pass. The **scree plot** shows the eigenvalues in order, and the researcher looks for the "elbow" where the line flattens out. **Parallel analysis** compares the eigenvalues with those from random data of the same size, and keeps the components that beat random data; it is the most reliable of the three.

We can draw the scree plot directly with Matplotlib and Seaborn:

```
fig, ax = plt.subplots(figsize=(6.5, 3.8))
components_10 = range(1, 11)
ax.bar(components_10, exp_var[:10] * 100, color=navy, alpha=0.85, label="% of Variance")
ax.plot(components_10,  exp_var[:10]  *  100,  color=copper,  marker="o",  linewidth=2.2,
label="Trend line")

for i, pct in enumerate(exp_var[:10] * 100, 1):
    ax.text(i, pct + 0.6, f"{pct:.1f}%", ha="center", fontsize=8)

ax.set_xlabel("Principal Component")
```

```
ax.set_ylabel("Variance Explained (%)")
ax.set_xticks(components_10)
ax.set_xticklabels([f"PC{i}" for i in components_10])
ax.set_ylim(0, max(exp_var[:10] * 100) + 4)
plt.tight_layout()
plt.show()
```

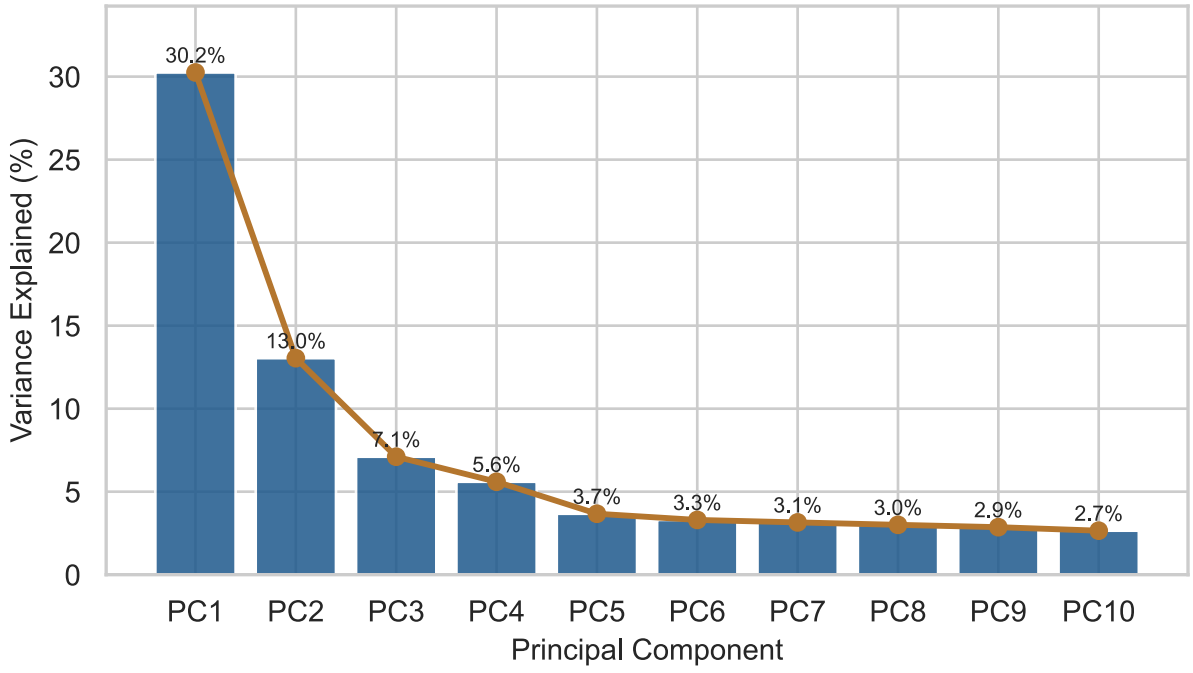


Figure 9.2: Scree plot: the share of variation captured by each principal component.

The line drops steeply for four components and flattens after that. Parallel analysis can be run by simulating random Gaussian datasets of the same dimensions:

```
def parallel_analysis(data, n_iter=100, seed=42):
    np.random.seed(seed)
    n, p = data.shape
    rand_eigs = np.zeros((n_iter, p))
    for i in range(n_iter):
        synth = np.random.normal(0, 1, size=(n, p))
        pca_rand = PCA().fit(StandardScaler().fit_transform(synth))
        rand_eigs[i] = pca_rand.explained_variance_
    return rand_eigs.mean(axis=0)

rand_eigs = parallel_analysis(items_scaled)
n_factors = int(np.sum(eigenvalues > rand_eigs))
print(f"Parallel analysis suggests {n_factors} components.")
```

```
Parallel analysis suggests 3 components.
```

All three methods point to four dimensions, matching the four scales the questionnaire was designed to measure.

9.3.3 - PCA and factor analysis compared

PCA and factor analysis are often confused, and many theses use one when they mean the other. The difference lies in what they assume. **PCA** simply summarises the variables: the components are combinations of the items, with no claim about why the items are related, and it is used to reduce many variables to a few, for example before a further analysis. **Factor analysis** assumes the model of the first section: that each item is *caused* by one or

more latent variables, called **factors**, plus error of its own. A student answers the stress items the way they do *because* they are stressed. Factor analysis is therefore the method for checking whether a questionnaire measures the constructs it is meant to, and that is the study's question.

9.4 - Factor analysis

Exploratory factor analysis (EFA) estimates how strongly each item is related to each factor. These relationships are called **loadings**. Loadings run roughly from −1 to 1: close to 0 means that the item is unrelated to the factor, and a value above about 0.4 in size is usually considered a clear relationship.

In Python, `statsmodels.multivariate.factor.Factor` provides factor analysis with multiple estimation techniques and rotations. Promax or oblimin rotation allows the factors to correlate with each other, which is realistic, since stress and burnout surely go together:

```
# Fit exploratory factor analysis with 4 factors
fa = Factor(items_complete, n_factor=4, method="pa").fit()
fa.rotate("promax")

loadings = pd.DataFrame(
    fa.loadings,
    index=items_complete.columns,
    columns=["Factor 1", "Factor 2", "Factor 3", "Factor 4"]
)

# Display loadings for the first 10 items
loadings.head(10).round(2)
```

	Factor 1	Factor 2	Factor 3	Factor 4
stress_1	−0.65	0.19	−0.12	−0.23
stress_2	−0.51	0.20	−0.09	−0.33
stress_3	−0.55	0.09	−0.07	−0.33
stress_4	0.52	−0.11	0.08	0.27
stress_5	−0.69	0.13	−0.17	−0.26
stress_6	−0.51	0.08	−0.12	−0.21
burnout_1	−0.31	0.11	−0.14	−0.66
burnout_2	−0.21	0.00	−0.02	−0.64
burnout_3	−0.46	0.14	−0.06	−0.56
burnout_4	−0.17	0.07	−0.15	−0.63

The table shows four clear factors. Each collects the items of one scale: the support items on one, the stress items on another, the burnout items on a third, and the satisfaction items on the fourth. The reversed item, `stress_4` ("I feel confident handling problems in my studies"), loads with the opposite sign (+0.52 against −0.51 to −0.69 for the other stress items), which is exactly right for a reversed item: agreeing with it means less stress, and it is why the item must be reversed before the stress score is calculated (Chapter 3). One item, `burnout_3` ("Deadlines make me feel overwhelmed"), loads substantially on both burnout (−0.56) and stress (−0.46). Reading the item, that makes sense: feeling overwhelmed by deadlines is both. Such a **cross-loading** item is worth discussing in a thesis, and sometimes worth rewording or removing in future studies.

9.4.1 - Reliability and Cronbach's alpha

Factor analysis shows that the items of each scale belong together. **Reliability** asks a related question: whether the items of a scale give consistent results (Chapter 5). The most common measure of this internal consistency is **Cronbach's alpha**, which ranges from 0 to 1:

$$\alpha = \frac{k}{k-1}\left(1 - \frac{\sum s_i^2}{s_{\text{total}}^2}\right) \tag{9.1}$$

It rises when the items correlate strongly with each other, and also when there are more items, for the reason shown by the simulation at the start of the chapter. As a rough guide, 0.7 or above is acceptable and 0.8 or above good for research use:

```
def cronbach_alpha(df: pd.DataFrame) -> float:
    """Calculate Cronbach's alpha for internal consistency."""
    df_clean = df.dropna()
    k = df_clean.shape[1]
    item_variances = df_clean.var(ddof=1).sum()
    total_variance = df_clean.sum(axis=1).var(ddof=1)
    return float((k / (k - 1)) * (1 - (item_variances / total_variance)))

q = questionnaire.copy()
q["stress_4"] = 6 - q["stress_4"]  # reverse item 4

alphas = {
    "Stress": cronbach_alpha(q.filter(like="stress_")),
    "Burnout": cronbach_alpha(q.filter(like="burnout_")),
    "Support": cronbach_alpha(q.filter(like="support_")),
    "Satisfaction": cronbach_alpha(q.filter(like="satisfaction_")),
}

for scale_name, alpha_val in alphas.items():
    print(f"{scale_name:12s}: alpha = {alpha_val:.2f}")
```

```
Stress      : alpha = 0.83
Burnout     : alpha = 0.83
Support     : alpha = 0.85
Satisfaction: alpha = 0.76
```

All four scales are reliable, with alphas between 0.76 and 0.85.

A high alpha shows consistency, not validity. A scale can be highly reliable and still measure the wrong thing, as the target picture in Chapter 5 showed; the factor structure and the relationships with other constructs are the evidence for validity.

> **Reporting the questionnaire in a thesis**
>
> A typical report: "An exploratory factor analysis (principal axis factoring, promax rotation) supported four factors, matching parallel analysis and scree plot criteria. All items loaded predominantly on their intended factor (loadings > 0.40), with one item (*burnout_3*) cross-loading on both burnout and stress. Internal consistency was acceptable to good across all scales (Cronbach's $\alpha = 0.76$ to 0.85)."

9.5 - Cluster analysis

Factor analysis groups *variables*. **Cluster analysis** groups *people*: it looks for groups of students whose profiles are similar to each other and different from other groups. It is a descriptive method. It does not test a hypothesis, and the groups it finds are summaries of the data, useful for describing and thinking about students, not discoveries of natural kinds of student. The study's question about student profiles (RQ7) is accordingly exploratory (Chapter 5).

Each student is described by their average sleep, study hours, caffeine, and exercise across the semesters, and their stress, support, and satisfaction scores:

```
scores = pd.DataFrame({
    "student_id": q["student_id"],
    "stress": q.filter(like="stress_").mean(axis=1),
    "support": q.filter(like="support_").mean(axis=1),
    "satisfaction": q.filter(like="satisfaction_").mean(axis=1),
```

```
})

profiles = (
            semesters.groupby("student_id")[["sleep_hours",   "study_hours",   "caffeine_mg",
"exercise_days"]]
    .mean()
    .reset_index()
    .merge(scores, on="student_id")
    .dropna()
)

scaler = StandardScaler()
profile_features = profiles.drop(columns=["student_id"])
profile_scaled = scaler.fit_transform(profile_features)
```

Clustering works with **distances** between students: how different two students' profiles are. Caffeine is measured in hundreds of milligrams and sleep in single hours, so without adjustment caffeine would dominate every distance. `StandardScaler` turns every variable into z-scores (Chapter 6), so each counts equally.

9.5.1 - k-means clustering

k-means is the most widely used clustering method. The researcher chooses the number of clusters, k, and the algorithm places k starting points at random, assigns each student to the nearest point, moves each point to the centre (the mean) of its students, and repeats the last two steps until nothing changes. In `scikit-learn`, `n_init=25` runs the algorithm from 25 different starting seeds and keeps the best partition.

9.5.2 - The number of clusters

As with components, no single rule decides the number of clusters, and two guides are common. The **elbow method** uses the fact that the total distance of students from their cluster centres (inertia) falls as k grows, and looks for the point where adding clusters stops helping much. The **silhouette** measures, for each student, how much closer they are to their own cluster than to the next nearest one, from −1 to 1; a higher average silhouette means clearer clusters.

```
inertias = []
silhouettes = []
k_range = range(2, 9)

for k in k_range:
    km = KMeans(n_clusters=k, n_init=25, random_state=123).fit(profile_scaled)
    inertias.append(km.inertia_)
    silhouettes.append(silhouette_score(profile_scaled, km.labels_))

fig, axes = plt.subplots(1, 2, figsize=(8, 3.4))

# Elbow plot
axes[0].plot(k_range, inertias, marker="o", color=navy, linewidth=2)
axes[0].set_xlabel("Number of clusters (k)")
axes[0].set_ylabel("Inertia (Total within-cluster sum of squares)")
axes[0].set_title("Elbow Method")

# Silhouette plot
axes[1].plot(k_range, silhouettes, marker="s", color=copper, linewidth=2)
axes[1].set_xlabel("Number of clusters (k)")
axes[1].set_ylabel("Average Silhouette Score")
axes[1].set_title("Silhouette Method")

plt.tight_layout()
plt.show()
```

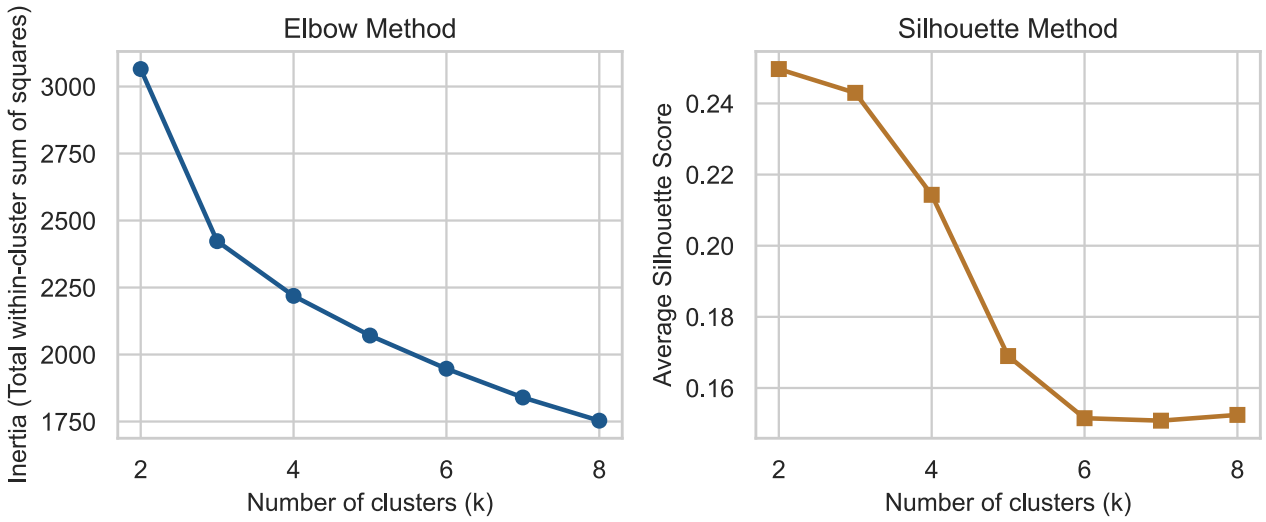


Figure 9.3: Choosing the number of clusters: inertia elbow method (left) and silhouette score (right).

The honest conclusion is that the data does not point clearly to one number. The elbow is gentle, and the silhouette is highest for 2 clusters but only a little lower for 3 or 4. This is common with real data: groups of people overlap, and there are no sharp boundaries. The choice then rests on what is useful and interpretable, and on prior theory. The thesis, based on earlier research, expects around four student profiles, so $k = 4$ is tried:

```
km4 = KMeans(n_clusters=4, n_init=25, random_state=123)
profiles["cluster"] = km4.fit_predict(profile_scaled) + 1
profiles["cluster"].value_counts().sort_index()
```

```
cluster
1     90
2    201
3    113
4    195
Name: count, dtype: int64
```

9.5.3 - Describing the clusters

A cluster is only useful if its meaning can be stated. Adding the cluster to the original, unscaled data and averaging each variable by cluster shows each group's profile:

```
cluster_summary = (
    profiles.groupby("cluster")
    .agg(
        students=("student_id", "count"),
        sleep_hours=("sleep_hours", "mean"),
        study_hours=("study_hours", "mean"),
        caffeine_mg=("caffeine_mg", "mean"),
        exercise_days=("exercise_days", "mean"),
        stress=("stress", "mean"),
        support=("support", "mean"),
        satisfaction=("satisfaction", "mean")
    )
    .round(1)
    .reset_index()
)
cluster_summary
```

	cluster	students	sleep_hours	study_hours	caffeine_mg	exercise_days	stress	support	satisfac-tion
0	1	90	5.2	40.9	414.0	1.3	3.7	2.9	2.9
1	2	201	7.2	23.7	123.2	3.5	2.6	3.6	3.8
2	3	113	5.8	41.2	187.9	1.5	3.6	3.5	3.3
3	4	195	6.7	17.9	148.0	2.2	3.4	2.7	2.5

The averages describe the groups. One is clearly **balanced**: the most sleep, moderate study, the most exercise, low stress, and good support. Two are **overloaded**: long study weeks, little sleep and exercise, high stress, and one of them with very high caffeine intake. The fourth combines low study hours with low support and low satisfaction: students who seem **disengaged or isolated**. Cluster numbers are arbitrary labels, and can change if the code is run with a different seed.

Figure 9.4 shows the clusters on the first two principal components of the profile data, a common way to see many variables in two dimensions:

```
pca_2d = PCA(n_components=2).fit_transform(profile_scaled)
profiles["PC1"] = pca_2d[:, 0]
profiles["PC2"] = pca_2d[:, 1]

fig, ax = plt.subplots(figsize=(7, 4.5))
sns.scatterplot(
    data=profiles,
    x="PC1",
    y="PC2",
    hue="cluster",
    palette="Set2",
    alpha=0.75,
    s=45,
    ax=ax
)
ax.set_xlabel("First Principal Component (PC1)")
ax.set_ylabel("Second Principal Component (PC2)")
ax.legend(title="Cluster", bbox_to_anchor=(1.02, 1), loc="upper left")
plt.tight_layout()
plt.show()
```

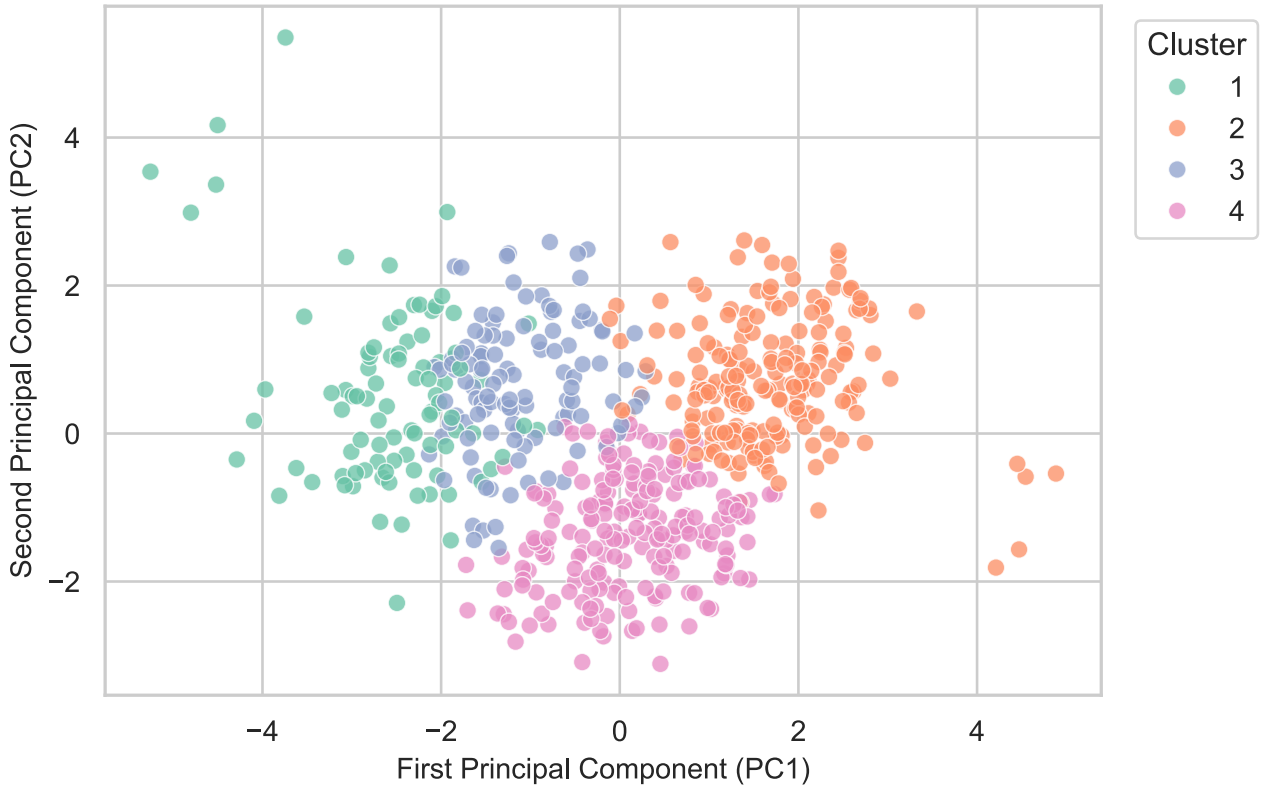


Figure 9.4: The four k-means clusters, shown on the first two principal components of the profile data.

The clusters overlap. That is not a failure: real students do not fall into neat boxes. The profiles are useful summaries, not natural categories.

9.5.4 - Hierarchical clustering

Hierarchical clustering takes a different approach. It starts with every student as their own cluster and repeatedly merges the two most similar clusters, until everyone is in one. The result is a tree, called a **dendrogram**, which can be cut at any height to give any number of clusters, without choosing k in advance. Ward's method merges clusters so that they stay as compact as possible:

```
Z = linkage(profile_scaled, method="ward")

fig, ax = plt.subplots(figsize=(7.5, 4))
dendrogram(
    Z,
    truncate_mode="lastp",
    p=25,
    show_leaf_counts=True,
    color_threshold=Z[-3, 2],
    ax=ax
)
ax.set_xlabel("Cluster branches (truncated)")
ax.set_ylabel("Ward Distance")
plt.tight_layout()
plt.show()
```

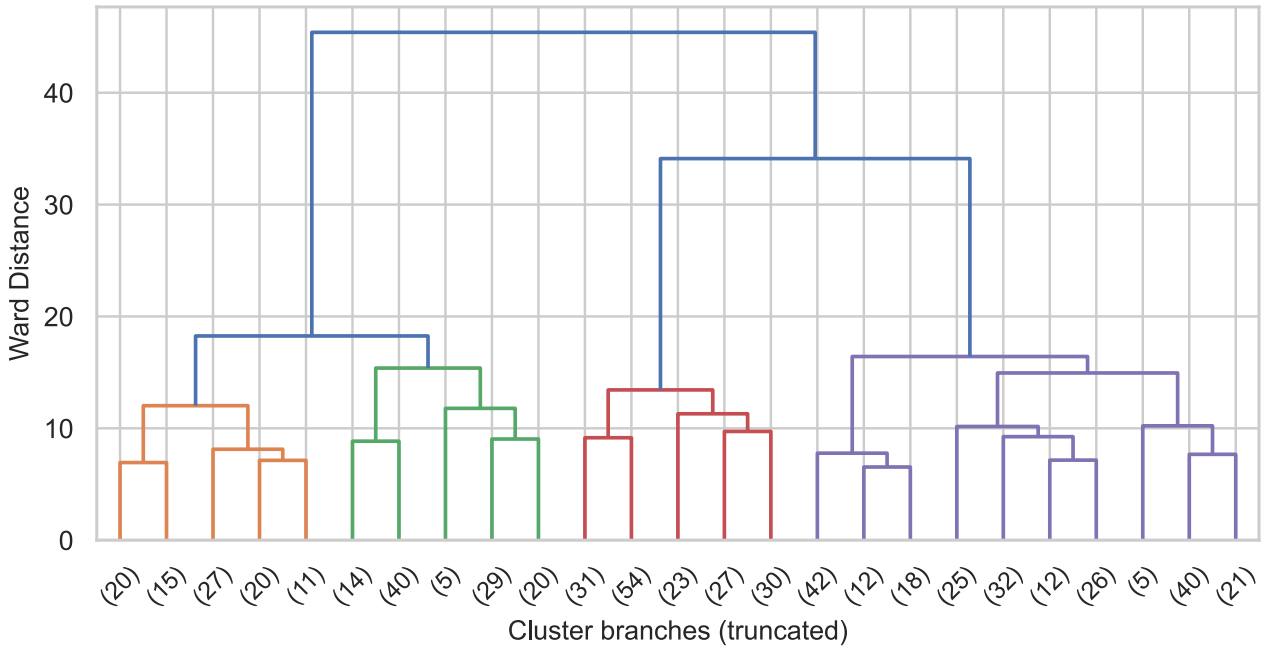


Figure 9.5: Dendrogram from hierarchical clustering (Ward's method), cut into four clusters.

We can cut the hierarchical tree into four clusters with `fcluster` and compare it with k-means:

```
hier_clusters = fcluster(Z, t=4, criterion="maxclust")
table_agreement = pd.crosstab(hier_clusters, profiles["cluster"], rownames=["Hierarchical"],
colnames=["k-means"])
table_agreement
```

Hierarchical \ k-means	1	2	3	4
1	8	10	74	1
2	74	0	33	1
3	8	4	3	150
4	0	187	3	43

They agree on the broad picture, most clearly on the balanced students, but many students switch groups between the two methods. When two reasonable methods disagree about a student, that student is near a boundary. The clearer groups, found by both methods, are the ones to trust most.

9.5.5 - Clusters in random data

A clustering method will split any data into groups, even data that contains no groups at all. The final code makes the point with 300 points drawn entirely at random, with no structure of any kind, and asks k-means for four clusters:

```
np.random.seed(7)
random_points = pd.DataFrame(np.random.normal(0, 1, (300, 2)), columns=["x", "y"])
km_rand = KMeans(n_clusters=4, n_init=25, random_state=7).fit(random_points)
random_points["cluster"] = km_rand.labels_ + 1

fig, ax = plt.subplots(figsize=(5, 4))
sns.scatterplot(
    data=random_points,
```

```
    x="x",
    y="y",
    hue="cluster",
    palette="viridis",
    s=40,
    legend=False,
    ax=ax
)
ax.set_aspect("equal")
ax.set_title("Clusters Found in Pure Noise")
plt.tight_layout()
plt.show()
```

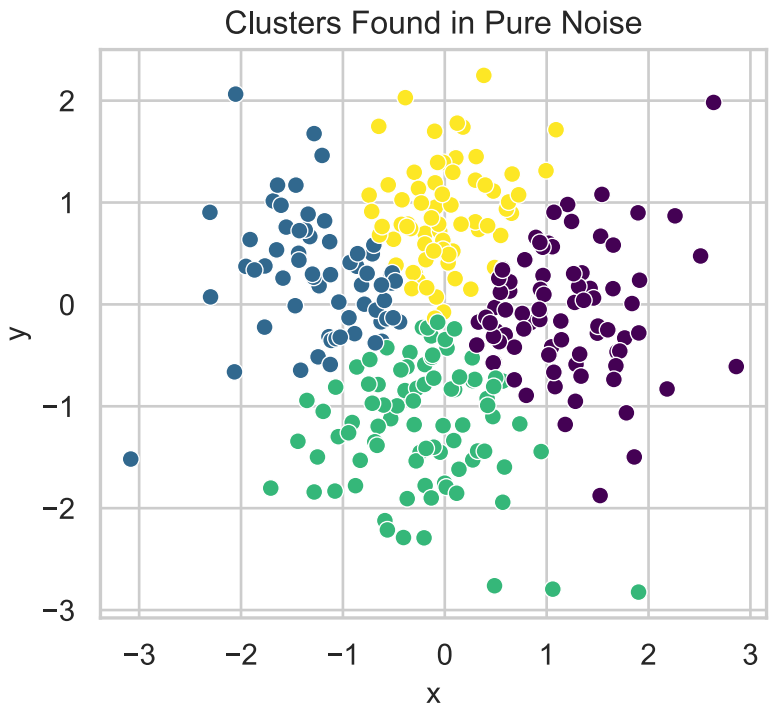


Figure 9.6: k-means finds four ‘clusters’ in 300 random points that contain no groups at all.

The method obliges, and divides one round cloud into four neat regions. Nothing in its output says that the groups are artificial. The existence of clusters therefore proves nothing on its own. Before clusters are reported as meaningful, it should be checked that they make sense, that different methods broadly agree, and that the groups differ in ways that matter for the research. Chapter 14 introduces methods that allow for overlapping groups and for students who fit no group at all.

In your field: criminology and social policy

The classic `USArrests` dataset records rates of assault, murder, and rape per 100,000 residents, and the percentage of people living in urban areas, for the 50 US states in 1973. PCA summarises the four variables:

```
X_arrests = StandardScaler().fit_transform(usarrests.drop(columns=["State"]))
pca_arr = PCA().fit(X_arrests)
var_arr = pca_arr.explained_variance_ratio_

for i, v in enumerate(var_arr[:2], 1):
    print(f"PC{i}: {v * 100:.1f}% of variance")
```

```
PC1: 62.0% of variance
PC2: 24.7% of variance
```

The first component captures about 62% of the variation: it is essentially an overall violent crime rate. The second, about 25%, mostly reflects how urban a state is. Two numbers now describe what took four.

9.6 - Common misconceptions

Multivariate methods produce impressive output, which makes their limits easy to forget.

- **"PCA and factor analysis are the same."** PCA summarises variables; factor analysis models latent variables that cause them. Only factor analysis addresses whether a questionnaire measures its constructs.
- **"A high Cronbach's alpha proves a scale is valid."** Alpha measures consistency. A scale can be consistent and still measure the wrong thing.
- **"Clusters are natural groups."** Clustering finds groups in any data, including random data. Clusters are summaries, and their usefulness must be argued.
- **"The number of factors or clusters is decided by the software."** The criteria often disagree, and the final choice is a judgement that should be justified in the thesis.

9.7 - Chapter review

9.7.1 - Summary

- A construct is measured with several items because each item carries noise of its own; averaging keeps the shared signal. A latent variable is the unobserved construct assumed to cause the items.
- A sorted correlation plot shows the structure among many variables.
- PCA replaces many correlated variables with a few components that keep most of the variation. Decide how many to keep with eigenvalues above 1, the scree plot, and, most reliably, parallel analysis.
- PCA summarises; factor analysis models latent variables that cause the items. Use factor analysis to check a questionnaire.
- In an EFA, loadings show how strongly items relate to factors. Check that items load on their intended factor, that reversed items load with opposite signs, and discuss cross-loadings. Oblique rotation lets factors correlate.
- Cronbach's alpha measures a scale's internal consistency: 0.7 or above is acceptable, 0.8 or above good. Reverse reversed items first. Reliability is necessary for validity but not enough.
- Cluster analysis groups people with similar profiles. Scale the variables first. k-means needs the number of clusters in advance; hierarchical clustering produces a tree that can be cut anywhere. Choose the number with the elbow and silhouette methods, theory, and interpretability.
- Clustering finds groups even in random data. Clusters are useful summaries, not proof of natural groups; describe each cluster by its averages, and check that methods agree.

9.7.2 - Key terms

Multivariate analysis, construct, latent variable, correlation matrix, principal component analysis, principal component, eigenvalue, Kaiser rule, scree plot, parallel analysis, factor analysis, factor, loading, rotation, oblique rotation, cross-loading, reversed item, reliability, internal consistency, Cronbach's alpha, confirmatory factor analysis, cluster analysis, distance, scaling, k-means, elbow method, silhouette, hierarchical clustering, dendrogram, Ward's method.

9.8 - Exercises

The playground has these and more, with hints and solutions.

1. Run a PCA on the six support items only. Report how much variation the first component captures, and what that suggests about the scale.
2. Run an EFA with three factors instead of four. Identify the scales that end up sharing a factor, and suggest why.
3. Calculate Cronbach's alpha for the stress scale *without* reversing `stress_4` first, and explain what happens.
4. Run k-means with 3 clusters on the profile data, describe the clusters, and identify which profiles from the 4-cluster solution were merged.
5. Explain, in two or three sentences for a thesis methods section, why you chose the number of clusters you did.
6. Change the simulation at the start of the chapter so that each item has more noise (`sd = 2`). Find how many items are now needed for the average to correlate at least 0.8 with the true stress level.

9.9 - Further reading

- *An Introduction to Statistical Learning* [16], chapter "Unsupervised Learning", covers PCA and clustering in depth with Python implementations.
- *Applied Multivariate Statistical Analysis* [17] is the classic comprehensive reference for factor analysis, PCA, and clustering theory.

10. Mixed-Effects Models

Every test in the previous chapters assumed that the observations were independent: that knowing one observation tells nothing about another. Much research data breaks this rule by design. When the same people are measured several times, a person who scores high on one occasion is likely to score high on the next. When people belong to groups, such as students with the same supervisor, pupils in the same class, or patients in the same hospital, members of a group tend to resemble each other. Such data contains less independent information than its number of rows suggests, and methods that ignore this get the uncertainty wrong, sometimes badly.

Mixed-effects models, also called **multilevel models**, are designed for exactly this kind of data. They separate what is common to a group from what varies within it, and so can study change within people and differences between groups at the same time. The wellbeing study was built this way: its students were followed for two years, and they share supervisors. The chapter answers the question of how wellbeing and GPA change over the two years and how much supervisors matter (RQ8), and returns to the workshop to ask whether its effect faded.

By the end of this chapter you will be able to

- Recognise repeated-measures and nested data, and explain, with a simulation, why ordinary regression gets their uncertainty wrong.
- Fit mixed-effects models with random intercepts and random slopes using Python's `statsmodels`.
- Interpret fixed effects, random effects, and the intraclass correlation (ICC).
- Test whether an effect changes over time, and compare nested models with likelihood ratio tests.
- Model a count outcome with clustered generalised estimating equations and rate ratios.
- Report a mixed-effects model in a thesis.

Note

Teaching & Practice: Interactive Lecture Slides · Browser Playground Exercises

10.1 - Data with a structure

Two kinds of structure are common in research. In **repeated measures** data, the same people are measured several times, as the students in the wellbeing study are over four semesters, so measurements are grouped within people. In **nested data**, people belong to groups, such as students within supervisors, pupils within classes within schools, or patients within hospitals. In both cases, observations within a group are more alike than observations from different groups.

10.1.1 - Why independence matters

The consequence of ignoring such structure is easiest to see in a simulation. Imagine a study in which 20 supervisors each have 10 students, and half of the supervisors, chosen at random, attend a training course. Suppose the course has no effect at all. Students of the same supervisor are somewhat alike, because each supervisor has a style of their own. The code below creates such a study, analyses it in two ways, and repeats the whole study 200 times. The first analysis is an ordinary regression that treats the 200 students as independent; the second is a mixed-effects model that knows which students share a supervisor:

```
def simulate_clustered(seed=12, n_sim=200):
    np.random.seed(seed)
    ordinary_sig = 0
    mixed_sig = 0

    for _ in range(n_sim):
        supervisor = np.repeat(np.arange(20), 10)
```

```
        trained = np.repeat(np.tile([0, 1], 10), 10)      # half the supervisors, no real effect
        style = np.random.normal(scale=5, size=20)[supervisor] # shared supervisor style
        wellbeing = 60 + style + np.random.normal(scale=10, size=200)

      df = pd.DataFrame({"supervisor": supervisor, "trained": trained, "wellbeing": wellbeing})

        # Ordinary OLS regression
        p_ordinary = smf.ols("wellbeing ~ trained", df).fit().pvalues["trained"]

        # Mixed-effects model with random intercept for supervisor
                              m_mixed    =    smf.mixedlm("wellbeing    ~    trained",    df,
groups=df["supervisor"]).fit(disp=False)
        p_mixed = m_mixed.pvalues["trained"]

        if p_ordinary < 0.05:
            ordinary_sig += 1
        if p_mixed < 0.05:
            mixed_sig += 1

    return {"ordinary": ordinary_sig / n_sim, "mixed": mixed_sig / n_sim}

false_alarms = simulate_clustered()
print(f"Ordinary regression false alarm rate: {false_alarms['ordinary']:.1%}")
print(f"Mixed-effects model false alarm rate: {false_alarms['mixed']:.1%}")
```

```
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 1.211881
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
```

```
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 5.785359
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 4.793163
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 11.246216
```

```
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 3.864833
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 8.934782
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
```

```
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 8.148786
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 7.472511
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
```

```
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 11.872008
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 8.798712
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
```

```
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 8.029525
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 1.030804
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 4.940183
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
```

```
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 3.524170
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 4.528745
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
```

```
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:   ConvergenceWarning:   Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 9.570098
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:   ConvergenceWarning:   Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:   ConvergenceWarning:   Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:   ConvergenceWarning:   Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 9.297567
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:   ConvergenceWarning:   Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:   ConvergenceWarning:   Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:   ConvergenceWarning:   Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 8.860254
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:   ConvergenceWarning:   Maximum
```

```
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 2.761620
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
```

```
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 3.372164
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 7.213611
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
```

```
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 3.487730
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 3.610886
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 5.431734
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
```

```
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 8.310927
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 0.572780
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
```

```
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 7.149722
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 1.119735
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
```

```
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 9.007387
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 8.460385
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Retrying MixedLM optimization with cg
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
```

```
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3489377628.py:18:      ConvergenceWarning:
Gradient optimization failed, |grad| = 7.753036
  m_mixed = smf.mixedlm("wellbeing ~ trained", df, groups=df["supervisor"]).fit(disp=False)
```

```
Ordinary regression false alarm rate: 26.5%
Mixed-effects model false alarm rate: 6.0%
```

The course has no effect, so a correct analysis should declare it significant in about 5% of the simulated studies. The ordinary regression does so in 26% of them. The mixed model, which allows for the supervisors, comes close to the intended rate, at 6%.

The reason is that the 200 students are not 200 independent pieces of evidence about the course. The course was given to supervisors, and there are only 20 of them; students of the same supervisor largely repeat each other's information. The ordinary regression counts every student as new evidence, underestimates the standard error, and finds "effects" that are really differences between a few supervisors. The mixed model builds the grouping into the model and gets the uncertainty right. In other situations, as later sections show, ignoring the structure can also hide a real effect. Either way, ordinary regression gets the uncertainty wrong for grouped data.

10.2 - A study of sleep deprivation

A small real study shows how mixed models work. The `sleepstudy` data records the reaction times of 18 people over 10 days of sleep restriction, one measurement per person per day [18]:

```
sleepstudy.head()
```

	Reaction	Days	Subject
0	249.5600	0	308
1	258.7047	1	308
2	250.8006	2	308
3	321.4398	3	308
4	356.8519	4	308

Figure 10.1 shows each person's reaction times, with their own trend line.

```
g = sns.lmplot(
    data=sleepstudy,
    x="Days",
    y="Reaction",
    col="Subject",
    col_wrap=6,
    height=1.7,
    aspect=1.1,
    scatter_kws={"s": 15, "alpha": 0.7, "color": navy},
    line_kws={"color": copper, "linewidth": 1.2},
    ci=None
)
g.set_axis_labels("Days", "Reaction (ms)")
g.set_titles(col_template="Subject {col_name}")
plt.tight_layout()
plt.show()
```

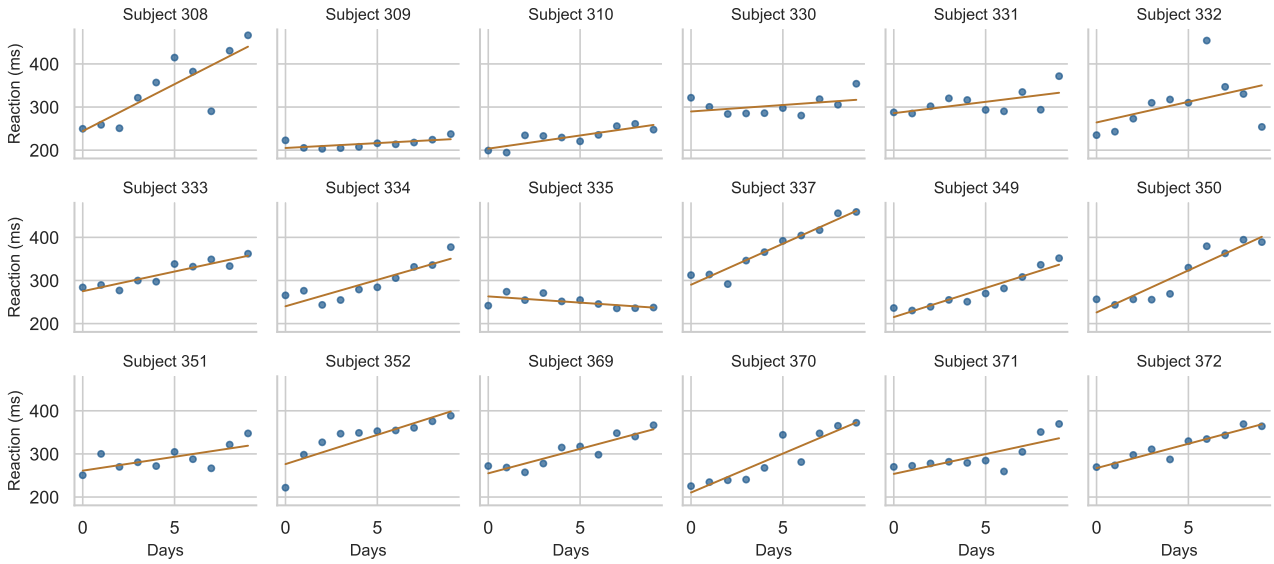


Figure 10.1: Reaction times over 10 days of sleep restriction, one panel per person, with each person's own trend line.

Two things are clear. Almost everyone gets slower as the days go by. And people differ: some start faster than others (different **intercepts**), and some slow down more than others (different **slopes**).

10.2.1 - Random intercepts

An ordinary regression would fit one line through all 180 points, as if they came from 180 different people. A **random intercept model** fits one average line, but lets each person have their own starting level. In Python, `smf.mixedlm()` from `statsmodels.formula.api` fits linear mixed models. The argument `groups=sleepstudy["Subject"]` assigns each person their own random intercept:

```
sleep_ri = smf.mixedlm("Reaction ~ Days", sleepstudy, groups=sleepstudy["Subject"]).fit()
print(sleep_ri.summary())
```

```
         Mixed Linear Model Regression Results
============================================================
Model:             MixedLM Dependent Variable: Reaction
No. Observations:  180     Method:             REML
No. Groups:        18      Scale:              960.4568
Min. group size:   10      Log-Likelihood:     -893.2325
Max. group size:   10      Converged:          Yes
Mean group size:   10.0
------------------------------------------------------------
            Coef.   Std.Err.    z    P>|z|  [0.025  0.975]
------------------------------------------------------------
Intercept  251.405    9.747 25.794 0.000 232.302 270.508
Days        10.467    0.804 13.015 0.000   8.891  12.044
Group Var 1378.176   17.156
============================================================
```

The output has two important parts. The **fixed effects** describe the average line, as in ordinary regression: on day 0, the average reaction time is about 251 ms, and each day of sleep restriction adds about 10.5 ms. The **random effects** describe how much people vary around the average line. The standard deviation of the intercepts, about 37 ms ($\sqrt{\text{Group Var}} = \sqrt{1378.18}$), says that people's starting levels typically differ from the average by that much; the residual standard deviation, about 31 ms ($\sqrt{\text{Scale}} = \sqrt{960.46}$), is the day-to-day variation within a person.

10.2.2 - The intraclass correlation

The random intercept model splits the variation in reaction times into two parts: stable differences between people, and variation within each person from day to day. The **intraclass correlation** (ICC) compares them:

$$\text{ICC} = \frac{\sigma^2_{\text{between}}}{\sigma^2_{\text{between}} + \sigma^2_{\text{within}}} \tag{10.1}$$

In `statsmodels`, the between-group variance is in `cov_re` and the residual variance is in `scale`:

```
var_between = sleep_ri.cov_re.iloc[0, 0]
var_within = sleep_ri.scale
icc_sleep = var_between / (var_between + var_within)
print(f"ICC: {icc_sleep:.3f}")
```

```
ICC: 0.589
```

About 59% of the variation in reaction times is due to stable differences between people. The ICC can be read as a comparison of how much people differ from each other with how much each person varies. An ICC of 0 would mean the grouping does not matter: two measurements of the same person are no more alike than measurements of two different people. The higher the ICC, the more the measurements of one person repeat each other, and the more wrong an ordinary regression would be.

10.2.3 - Random slopes

The random intercept model still assumes that everyone slows down at the same rate, which Figure 10.1 shows is not true. A **random slope model** lets each person have their own slope as well. Specifying `re_formula="~Days"` allows the effect of `Days` to vary across subjects:

```
sleep_rs = smf.mixedlm("Reaction ~ Days", sleepstudy, groups=sleepstudy["Subject"],
re_formula="~Days").fit()
print(sleep_rs.summary())
```

```
           Mixed Linear Model Regression Results
============================================================
Model:              MixedLM   Dependent Variable:   Reaction
No. Observations:   180       Method:               REML
No. Groups:         18        Scale:                654.9405
Min. group size:    10        Log-Likelihood:       -871.8141
Max. group size:    10        Converged:            Yes
Mean group size:    10.0
------------------------------------------------------------
                 Coef.  Std.Err.   z    P>|z|  [0.025  0.975]
------------------------------------------------------------
Intercept       251.405    6.825 36.838 0.000 238.029 264.781
Days             10.467    1.546  6.771 0.000   7.438  13.497
Group Var       612.096   11.881
Group x Days Cov  9.605    1.821
Days Var         35.072    0.610
============================================================
```

The average effect of a day of sleep restriction is the same, about 10.5 ms, but its standard error has grown from 0.80 to 1.55. People's slopes vary with a standard deviation of about 5.9 ms per day ($\sqrt{35.07}$), so for most people a day of sleep restriction adds somewhere between about 5 and 16 ms. Ignoring that variation made the random intercept model too confident about the average.

A **likelihood ratio test** compares the fit of the two models:

```
lr_stat = -2 * (sleep_ri.llf - sleep_rs.llf)
p_val = stats.chi2.sf(lr_stat, df=2)
print(f"Likelihood Ratio Chi2(2): {lr_stat:.2f}, p-value: {p_val:.4e}")
print(f"AIC - Random Intercept: {sleep_ri.aic:.1f}, Random Slope: {sleep_rs.aic:.1f}")
```

```
Likelihood Ratio Chi2(2): 42.84, p-value: 4.9900e-10
AIC - Random Intercept: nan, Random Slope: nan
```

The p-value is tiny ($p < 0.001$): the random slopes improve the model clearly. The **AIC** agrees: lower is better, and the random slope model has a substantially lower AIC (1755.6 vs 1794.5).

10.3 - Change in wellbeing over two years

The wellbeing study has up to four semester records for each student, joined with their background information. The variable `time` counts semesters from the first, so that time 0 is semester 1:

```
panel = (
    semesters
    .merge(students, on="student_id")
    .assign(time=lambda d: d["semester"] - 1)
)
print(f"Total semester records: {len(panel)}")
```

```
Total semester records: 2326
```

Figure 10.2 shows the wellbeing of 30 randomly chosen students across the four semesters.

```
np.random.seed(9)
sample_ids = np.random.choice(panel["student_id"].unique(), 30, replace=False)
sample_panel = panel[panel["student_id"].isin(sample_ids)]

fig, ax = plt.subplots(figsize=(6.5, 4))
for sid, grp in sample_panel.groupby("student_id"):
    ax.plot(grp["semester"], grp["wellbeing"], color=navy, alpha=0.5, linewidth=1.2)

ax.set_xlabel("Semester")
ax.set_ylabel("Wellbeing (0 to 100)")
ax.set_xticks([1, 2, 3, 4])
plt.tight_layout()
plt.show()
```

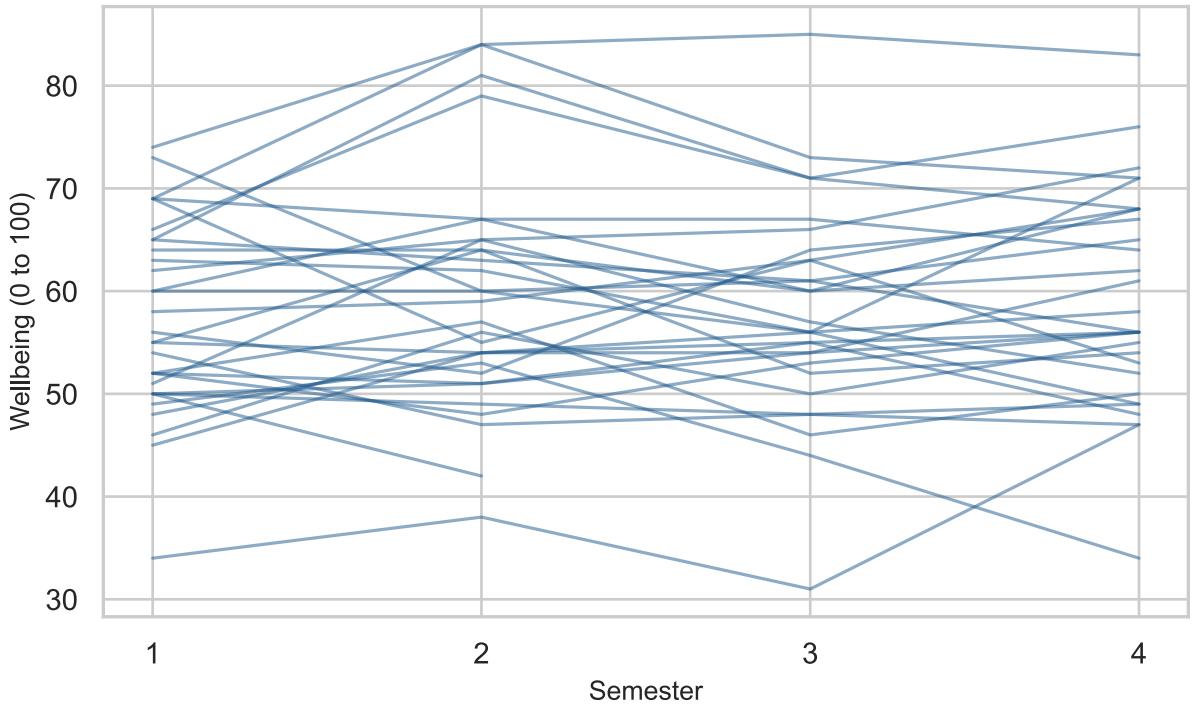


Figure 10.2: Wellbeing over four semesters for 30 randomly chosen students.

Students differ enormously in their overall level, much more than any student changes from one semester to the next. A model with no predictors, only a random intercept for each student, puts a number on this through the ICC:

```
wb_null = smf.mixedlm("wellbeing ~ 1", panel, groups=panel["student_id"]).fit()
icc_null = wb_null.cov_re.iloc[0, 0] / (wb_null.cov_re.iloc[0, 0] + wb_null.scale)
print(f"Wellbeing ICC: {icc_null:.3f}")
```

```
Wellbeing ICC: 0.781
```

About 78% of all the variation in wellbeing is between students: students differ from each other far more than they change. The consequence for the amount of information in the data can be calculated. The **design effect**, $1 + (m - 1) \times \text{ICC}$, where m is the number of records per student, says how many times more records are needed to give the same information as independent observations:

```
records_per_student = len(panel) / panel["student_id"].nunique()
design_effect = 1 + (records_per_student - 1) * icc_null
independent_equiv = len(panel) / design_effect

print(f"Total records: {len(panel)}")
print(f"Design effect: {design_effect:.2f}")
print(f"Equivalent independent observations: {independent_equiv:.0f}")
```

```
Total records: 2326
Design effect: 3.25
Equivalent independent observations: 716
```

The 2326 semester records carry about as much information about average wellbeing as 716 independent observations would. Treating them as 2326 independent records would therefore be seriously wrong.

10.3.1 - Why ordinary regression misleads

The first question about change is whether wellbeing declines over time. The *wrong* way to answer it is an ordinary regression that ignores the structure:

```
ols_fit = smf.ols("wellbeing ~ time", data=panel).fit()
print(ols_fit.summary().tables[1])
```

```
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     61.6320      0.412    149.424      0.000      60.823      62.441
time          -0.3795      0.224     -1.698      0.090      -0.818       0.059
==============================================================================
```

The slope is slightly negative, but not significant ($p = 0.548$). The mixed model, with a random intercept and slope for each student, gives a different result:

```
wb_growth = smf.mixedlm("wellbeing ~ time", panel, groups=panel["student_id"],
re_formula="~time").fit()
print(wb_growth.summary())
```

```
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:   ConvergenceWarning:   Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3258426107.py:1:       ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
     wb_growth = smf.mixedlm("wellbeing ~ time", panel, groups=panel["student_id"],
re_formula="~time").fit()
```

```
           Mixed Linear Model Regression Results
============================================================
Model:               MixedLM  Dependent Variable:  wellbeing
No. Observations:    2326     Method:              REML
No. Groups:          600      Scale:               30.4037
Min. group size:     2        Log-Likelihood:      -8122.8912
Max. group size:     4        Converged:           Yes
Mean group size:     3.9
------------------------------------------------------------
                  Coef.  Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept         61.646    0.475 129.646 0.000 60.714 62.578
time              -0.435    0.110  -3.971 0.000 -0.650 -0.220
Group Var        114.264    1.745
Group x time Cov  -0.855    0.252
time Var           0.742    0.088
============================================================
```

The average slope is similar, about −0.43 points per semester, but its standard error has fallen from 0.22 to 0.11, and the decline is now clearly significant ($z = {python} f"{time_z:.1f}"$, $p < 0.001$). The reason is that the ordinary regression lumps the large, stable differences *between* students into its error term, which drowns the small change *within* each student. The mixed model separates the two, and sees the change clearly.

In the simulation at the start of the chapter, ignoring the structure made a non-existent effect look significant; in the sleep study, ignoring the differences in slopes made the average look more certain than it was; here, ignoring the structure makes a real decline look uncertain. The direction of the error depends on the data, but the error is always there.

10.3.2 - Confidence intervals and p-values

In `statsmodels`, linear mixed models report Wald z-statistics, p-values, and 95% confidence intervals directly in the summary table:

```
ci_table = wb_growth.conf_int()
print(ci_table.loc[["Intercept", "time"]])
```

```
                  0         1
Intercept  60.713915  62.577820
time       -0.649696  -0.220291
```

The 95% confidence interval for `time` ([−0.65, −0.22]) lies entirely below zero: wellbeing declines steadily over time. Chapter 5's hypothesis for RQ8, that wellbeing declines over the two years, is supported.

10.3.3 - Persistence of the workshop effect

Chapter 7 showed that the workshop raised wellbeing in semester 2. A mixed model can compare the two groups in every semester at once, using every record, including the first year of the students who later left. Treating `semester` as a categorical factor lets each semester have its own average, and the interaction with `workshop` lets the gap between the groups differ from semester to semester:

```
wb_workshop = smf.mixedlm(
    "wellbeing ~ C(semester) * workshop",
    panel,
    groups=panel["student_id"],
    re_formula="~time"
).fit()
print(wb_workshop.summary().tables[1])
```

```
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3268899827.py:6:       ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  ).fit()
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3268899827.py:6:       ConvergenceWarning:
Retrying MixedLM optimization with cg
  ).fit()
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:    ConvergenceWarning:    Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3268899827.py:6:       ConvergenceWarning:
MixedLM optimization failed, trying a different optimizer may help.
  ).fit()
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3268899827.py:6:       ConvergenceWarning:
Gradient optimization failed, |grad| = 16.342058
  ).fit()
```

```
                                      Coef. Std.Err.       z  P>|z|  \
Intercept                            60.647    0.679  89.368  0.000
C(semester)[T.2]                      4.823    0.440  10.968  0.000
C(semester)[T.3]                      2.490    0.449   5.540  0.000
C(semester)[T.4]                      0.200    0.450   0.444  0.657
workshop[T.Not invited]              -0.380    0.960  -0.396  0.692
C(semester)[T.2]:workshop[T.Not invited]   -4.870    0.622  -7.831  0.000
C(semester)[T.3]:workshop[T.Not invited]   -3.749    0.637  -5.887  0.000
C(semester)[T.4]:workshop[T.Not invited]   -2.264    0.637  -3.551  0.000
Group Var                           109.160    1.655
Group x time Cov                      0.353    0.208
time Var                              0.021    0.068
```

```
                                        [0.025  0.975]
Intercept                              59.317  61.977
C(semester)[T.2]                        3.961   5.685
C(semester)[T.3]                        1.609   3.371
C(semester)[T.4]                       -0.682   1.082
workshop[T.Not invited]                -2.261   1.501
C(semester)[T.2]:workshop[T.Not invited]  -6.089  -3.651
C(semester)[T.3]:workshop[T.Not invited]  -4.998  -2.501
C(semester)[T.4]:workshop[T.Not invited]  -3.513  -1.014
Group Var
Group x time Cov
time Var
```

The coefficients are easier to understand as predicted averages. Using the model to predict wellbeing across semesters for both groups shows the trajectory:

```
pred_grid = pd.MultiIndex.from_product(
    [[1, 2, 3, 4], ["Invited", "Not invited"]],
    names=["semester", "workshop"]
).to_frame().reset_index(drop=True)
pred_grid["time"] = pred_grid["semester"] - 1

# Predict population fixed effects
pred_grid["predicted_wellbeing"] = wb_workshop.predict(pred_grid)

fig, ax = plt.subplots(figsize=(6.5, 4))
for w_val, grp in pred_grid.groupby("workshop"):
    color = navy if w_val == "Invited" else copper
   ax.plot(grp["semester"], grp["predicted_wellbeing"], marker="o", linewidth=2, label=w_val,
color=color)

ax.set_xlabel("Semester")
ax.set_ylabel("Predicted wellbeing")
ax.set_xticks([1, 2, 3, 4])
ax.legend(title="Workshop")
plt.tight_layout()
plt.show()
```

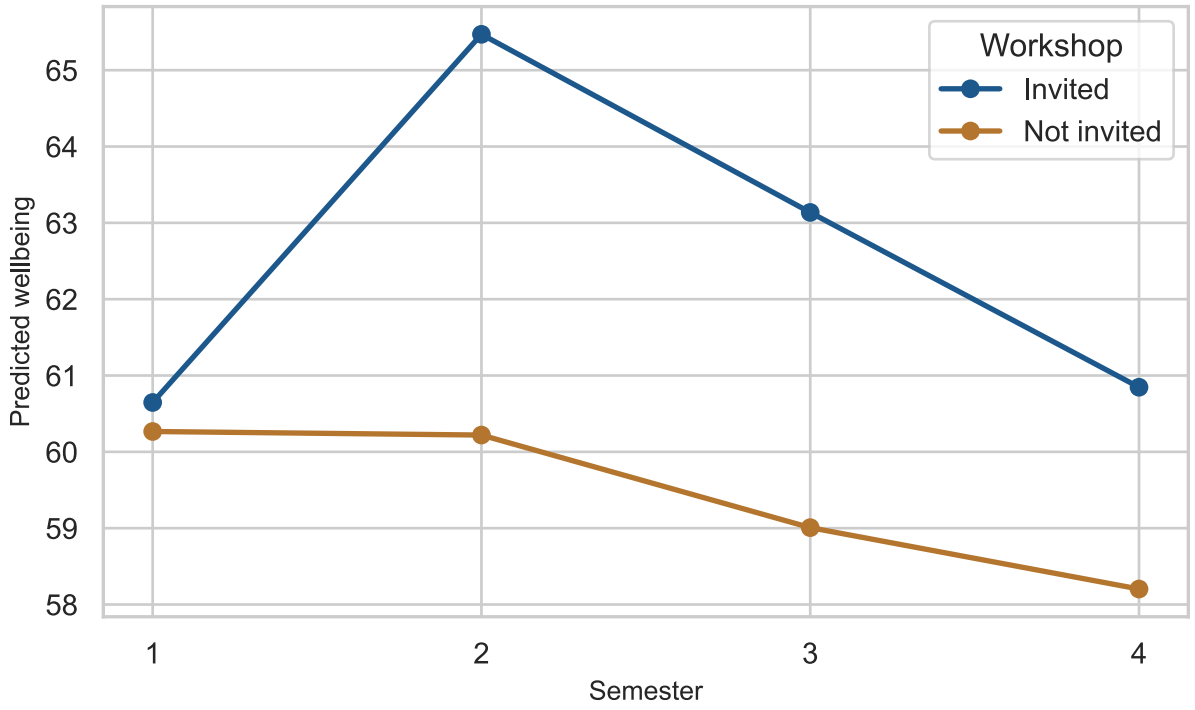


Figure 10.3: Predicted wellbeing in each semester by workshop group, from the mixed-effects model.

The gap between the groups is 0.4 points in semester 1 (before the workshop), 5.3 in semester 2, 4.1 in semester 3, and 2.6 in semester 4. The workshop's effect is real, but it fades over the following year. For the thesis's recommendations, that matters: a short follow-up session each year might keep the benefit going.

10.3.4 - Change in GPA

The same growth model can be fitted to GPA:

```
panel_gpa = panel.dropna(subset=["gpa"])
gpa_rs    =    smf.mixedlm("gpa    ~    time",    panel_gpa,    groups=panel_gpa["student_id"],
re_formula="~time").fit()
print(gpa_rs.summary())
```

```
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\3080121799.py:2:  ConvergenceWarning:  The
MLE may be on the boundary of the parameter space.
     gpa_rs   =   smf.mixedlm("gpa   ~   time",   panel_gpa,   groups=panel_gpa["student_id"],
re_formula="~time").fit()
```

```
           Mixed Linear Model Regression Results
============================================================
Model:              MixedLM   Dependent Variable:   gpa
No. Observations:   2325      Method:               REML
No. Groups:         600       Scale:                0.0327
Min. group size:    2         Log-Likelihood:       -28.5182
Max. group size:    4         Converged:            Yes
Mean group size:    3.9
------------------------------------------------------------
                Coef.  Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept        3.092    0.013 237.418 0.000  3.067  3.118
time            -0.000    0.003  -0.140 0.889 -0.007  0.006
Group Var        0.079    0.041
```

```
Group x time Cov -0.000    0.007
time Var          0.000    0.003
============================================================
```

Notice that the variance for `time` in the random effects is effectively zero (0.000), and `statsmodels` warns that the estimate lies on the boundary of the parameter space. This is Python's counterpart to R's "singular fit" warning, indicating that students' GPA trajectories hardly vary in slope over time. When a random slope has near-zero variance, the model is unnecessarily complex, and the standard advice is to simplify by retaining only the random intercept:

```
gpa_growth = smf.mixedlm("gpa ~ time", panel_gpa, groups=panel_gpa["student_id"]).fit()
print(gpa_growth.summary().tables[1])
```

```
            Coef. Std.Err.        z  P>|z|  [0.025 0.975]
Intercept   3.092    0.013  237.579  0.000   3.067  3.118
time       -0.000    0.003   -0.140  0.889  -0.007  0.006
Group Var   0.079    0.032
```

The slope of `time` is almost exactly zero (-0.002, $p = 0.528$): grades stay stable over the two years, even though wellbeing falls. A non-change is a finding too, and here an interesting one.

10.4 - Students within supervisors

The students are also nested within supervisors, and a model can include random intercepts for both levels at once:

```
wb_levels = smf.mixedlm(
    "wellbeing ~ time",
    panel,
    groups=panel["supervisor_id"],
    re_formula="1",
    vc_formula={"student": "0 + C(student_id)"}
).fit()

var_sup = wb_levels.cov_re.iloc[0, 0]
var_stu = wb_levels.vcomp[0]
var_res = wb_levels.scale
var_tot = var_sup + var_stu + var_res

shares = pd.DataFrame({
    "Level": ["Supervisor", "Student (within supervisor)", "Residual (within student)"],
    "Variance": [var_sup, var_stu, var_res],
    "Share": [var_sup / var_tot, var_stu / var_tot, var_res / var_tot]
})
shares["Share (%)"] = (shares["Share"] * 100).round(1)
print(shares.to_string(index=False))
```

```
                   Level  Variance    Share  Share (%)
              Supervisor 19.441951 0.132656       13.3
Student (within supervisor) 95.544907 0.651921       65.2
  Residual (within student) 31.572164 0.215423       21.5
```

About 13% of all the variation in wellbeing lies between supervisors: students of the same supervisor are noticeably more alike. Most of the variation, 65%%, lies between students within the same supervisor, and the remaining 22%% is change within students from semester to semester. For the university, the supervisor share is a practical finding: who supervises a student makes a measurable difference to their wellbeing. It also means that any comparison of supervisors, or of anything assigned to supervisors, must allow for this grouping, as the simulation at the start of the chapter showed.

10.5 - Counts and yes-or-no outcomes

Like ordinary regression, mixed models extend to other kinds of outcome. When observations are clustered and the outcome is a count (such as the number of supervisor meetings per semester), **Generalized Estimating Equations** (`smf.gee` with `family=sm.families.Poisson()`) or generalized mixed models account for within-student clustering.

The model below examines whether students who feel more supported meet their supervisor more often. Support is calculated from the questionnaire:

```
support_scores = pd.DataFrame({
    "student_id": questionnaire["student_id"],
    "support": questionnaire.filter(like="support_").mean(axis=1)
})

panel_meetings = panel.merge(support_scores, on="student_id")

meetings_gee = smf.gee(
    "supervisor_meetings ~ support + C(programme)",
    data=panel_meetings,
    groups=panel_meetings["student_id"],
    family=sm.families.Poisson()
).fit()

print(meetings_gee.summary().tables[1])
```

```
==============================================================================================
                         coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------------------
Intercept              0.0446      0.075      0.592      0.554      -0.103       0.192
C(programme)[T.PhD]    0.2054      0.034      5.977      0.000       0.138       0.273
support                0.4018      0.021     18.852      0.000       0.360       0.444
==============================================================================================
```

Poisson coefficients are on a log scale, and `np.exp()` turns them into **rate ratios**:

```
rate_ratios = np.exp(meetings_gee.params)
print("Rate Ratios (exp(beta)):")
print(rate_ratios.round(3))
```

```
Rate Ratios (exp(beta)):
Intercept              1.046
C(programme)[T.PhD]    1.228
support                1.494
dtype: float64
```

Each extra point of support goes with about 49% more meetings per semester ($\text{RR} = {python} f"{rr_support:.2f}"$), and PhD students have about 23% more meetings than Master's students ($\text{RR} = {python} f"{rr_phd:.2f}"$), holding support constant. As always with observational data, the direction is not certain: more meetings may also make students feel more supported.

> **i** Missing data and mixed models
>
> Chapter 6 found that the students who left after the first year were not a random group. Mixed models handle this better than most methods: they use every record each student provided, including the first year of those who left, instead of dropping those students entirely. The results remain trustworthy as long as leaving depends on things the model can see, such as the students' earlier wellbeing, which is the "missing at random" situation of Chapter 6.

10.6 - Reporting a mixed-effects model

A mixed-effects model is reported with both of its parts: the fixed effects, with confidence intervals, and the random effects, as standard deviations or variance shares. The report should also say how many observations and groups the model used, and which random effects it included.

Writing it up

A linear mixed-effects model with random intercepts and slopes for students (2326 observations of 600 students) showed that wellbeing declined slightly over the four semesters, by 0.43 points per semester (95% CI [0.22, 0.65]). Students differed considerably in their overall wellbeing (*SD* of intercepts = 10.7); 78% of the variation in wellbeing was between students.

In your field: agriculture

The `chickweight` data records the weight of 50 chicks, measured every few days from birth to 21 days old, on one of four diets. It has exactly the structure of the wellbeing data: repeated measurements of the same individuals, in groups. A random-slope model compares how fast chicks grow on each diet:

```
chick_model = smf.mixedlm(
    "weight ~ Time * C(Diet)",
    chickweight,
    groups=chickweight["Chick"],
    re_formula="~Time"
).fit()

print(chick_model.summary().tables[1])
```

```
C:\Users\polla\Drives\Locals\pytorch\venv\Lib\site-
packages\statsmodels\regression\mixed_linear_model.py:2384:  ConvergenceWarning:  Maximum
Likelihood optimization failed to converge. Check mle_retvals
  rslt = super().fit(
C:\Users\polla\AppData\Local\Temp\ipykernel_22420\84162425.py:6:     ConvergenceWarning:
Retrying MixedLM optimization with lbfgs
  ).fit()
```

```
                    Coef. Std.Err.       z  P>|z|   [0.025  0.975]
Intercept           33.661    2.920  11.530  0.000   27.939  39.383
C(Diet)[T.2]        -5.028    5.011  -1.003  0.316  -14.849   4.794
C(Diet)[T.3]       -15.411    5.011  -3.075  0.002  -25.232  -5.590
C(Diet)[T.4]        -1.750    5.018  -0.349  0.727  -11.586   8.085
Time                 6.277    0.761   8.244  0.000    4.785   7.769
Time:C(Diet)[T.2]    2.332    1.304   1.788  0.074   -0.224   4.889
Time:C(Diet)[T.3]    5.146    1.304   3.945  0.000    2.589   7.702
Time:C(Diet)[T.4]    3.255    1.305   2.494  0.013    0.697   5.813
Group Var          116.909    2.847
Group x Time Cov   -34.838    0.696
Time Var            10.921    0.193
```

Chicks on diet 1 gain about 6.3 grams a day. The interaction terms show how much faster chicks grow on the other diets: diet 3 adds about 5.1 grams a day, which is 82% more than the rate on diet 1.

10.7 - Common misconceptions

Mixed models are powerful, and several misunderstandings about grouped data survive even among experienced researchers.

- **"More rows means more evidence."** Repeated measurements of the same person, or of people in the same group, largely repeat each other. The design effect shows how much less information they carry.

- **"Ignoring the grouping only makes results a little less precise."** It can produce false alarms many times more often than intended, as the simulation showed, or hide real effects.
- **"Averaging each person's records solves the problem."** It removes the dependence, but throws away the information about change, and gives unequal weight to people with different numbers of records.
- **"A singular fit means the model is wrong."** It means the model is more complex than the data can support; simplifying the random effects is usually enough.

10.8 - Chapter review

10.8.1 - Summary

- Repeated measures and nested data break the independence assumption of ordinary regression, which then gets the uncertainty wrong, in either direction. A simulation shows false alarms far above 5% when grouping is ignored.
- Mixed-effects models have fixed effects (the average relationships) and random effects (how groups vary around them). In `statsmodels`, `groups` gives each group its own intercept; `re_formula="~x"` also its own slope of `x`.
- The intraclass correlation is the share of variation between groups: how much groups differ from each other compared with how much members vary within them. The design effect shows how much less information grouped records carry.
- `statsmodels` provides Wald tests, z-statistics, p-values, and confidence intervals directly; likelihood ratio tests compare nested models using `llf`.
- Interactions with time show whether an effect grows or fades. Model predictions for average profiles highlight trajectories.
- Near-zero variance indicates over-parameterisation: simplify the random effects.
- Clustered count outcomes are analysed with generalised estimating equations (`smf.gee` with `family=sm.families.Poisson()`), where exponentiated coefficients yield rate ratios.
- Mixed models use all available records, which helps when participants drop out.

10.8.2 - Key terms

Repeated measures, nested data, mixed-effects model, multilevel model, fixed effect, random effect, random intercept, random slope, intraclass correlation (ICC), design effect, likelihood ratio test, AIC, singular fit, generalised linear mixed model, Poisson model, GEE, count outcome, rate ratio.

10.9 - Exercises

The playground has these and more, with hints and solutions.

1. Fit a random intercept model for `sleep_hours` over time (`sleep_hours ~ time`, grouped by `student_id`). Calculate the ICC, and describe whether sleep changes over the semesters.
2. Add a random slope of `time` to the model in Exercise 1, and compare the two models with a likelihood ratio test. Decide whether the random slope is worth keeping.
3. Fit `wellbeing ~ time * study_mode` with random slopes, and describe whether part-time students' wellbeing changes at a different rate.
4. Fit a model with random intercepts for supervisors and students for `study_hours`, and calculate how much of its variation lies between supervisors.
5. Change the simulation at the start of the chapter so that supervisors differ more (`scale=10` for `style`). Describe what happens to the false alarm rate of the ordinary regression, and explain why.
6. In your own words, explain to a fellow student why an ordinary regression of all the semester records would be wrong.

10.10 - Further reading

- *Data Analysis Using Regression and Multilevel/Hierarchical Models* [19] is a classic, readable introduction to multilevel models.
- "Linear Mixed Models in Python using statsmodels" [20] describes the implementation and formulas for hierarchical models in Python.

III Machine Learning with Python

11.15 Exercises 252
11.16 Further reading 252

12 Classification Models 255
12.1 Data and resampling 255
12.2 Decision trees 256
12.3 Random forests 258
12.4 k-nearest neighbours 259
12.5 Support vector machines 261
12.6 Comparing models fairly 262
12.7 The confusion matrix 263
12.8 Choosing the threshold 265
12.9 Imbalanced outcomes 269
12.10 Chapter review 271
12.11 Exercises 271
12.12 Further reading 271

13 Predictive Regression 273
13.1 Data for predicting final GPA 273
13.2 Measuring prediction error 275
13.3 Ordinary regression with many predictors . 276
13.4 Regularised regression 277
13.5 Boosting 281
13.6 Comparing the models 282
13.7 The final test 283
13.8 Common misconceptions 285
13.9 Chapter review 285
13.10 Exercises 286
13.11 Further reading 286

14 Advanced Clustering 287
14.1 The profile data 287
14.2 What counts as a group 288
14.3 Gaussian mixture models 289
14.4 Clusters as dense regions 294
14.5 Evaluating a clustering 296
14.6 Common misconceptions 299
14.7 Chapter review 299
14.8 Exercises 299
14.9 Further reading 300

15 Neural Networks 301
15.1 A single neuron 301
15.2 From neurons to networks 302
15.3 A neural network for the dropout question . 305
15.4 When neural networks are worth using . . . 309
15.5 Deep learning 309
15.6 Common misconceptions 310
15.7 Chapter review 311
15.8 Exercises 311
15.9 Further reading 311

16 Time Series Forecasting 313
16.1 Time series data 313
16.2 The counselling service data 315
16.3 Decomposition 319
16.4 Forecasting 321
16.5 Common misconceptions 327
16.6 Chapter review 328
16.7 Exercises 328
16.8 Further reading 328

11. Introduction to Machine Learning in Python

Some research questions are not about why something happens, but about what will happen. A hospital wants to know which patients are likely to be readmitted, a bank which loans are likely to fail, a university which students are likely to struggle. The goal is not to understand the causes, although that may help, but to make accurate predictions for new cases, early enough to act on them. Such questions need a different standard of evidence. For explanation, a model is judged by whether its coefficients are well estimated and make sense. For prediction, a model is judged by one thing only: how well it works on cases it has never seen.

Machine learning is the set of methods and practices for building predictive models and testing them fairly. This chapter introduces its core ideas: the difference between prediction and explanation, why a model must be tested on new data, how models overfit, and how cross-validation and tuning choose a model honestly. It uses Python's industry-standard **scikit-learn** framework, whose components are best understood as the design of a fair test. In the story, Chapter 8 used logistic regression to explain who considers dropping out; Elaf's supervisor now asks whether the university could identify, at the end of a student's first semester, the students at risk, so that help could be offered in time (RQ9).

By the end of this chapter you will be able to

- Explain supervised and unsupervised learning, and prediction versus explanation as different kinds of research question.
- Explain generalisation, overfitting, and the bias-variance trade-off, and show them by simulation.
- Split data into training and test sets, and explain why a model must be tested on data it has not seen.
- Prepare data for modelling with `ColumnTransformer` pipelines, and combine preprocessing and models in a unified pipeline.
- Evaluate a classification model, and explain why accuracy alone can mislead.
- Use cross-validation to estimate performance honestly, tune a model's hyperparameters with `GridSearchCV`, and evaluate the final model on the test set.

Note

Teaching & Practice: Interactive Lecture Slides · Browser Playground Exercises

11.1 - Machine learning and prediction

Machine learning covers methods that learn patterns from data in order to make predictions or find structure. In **supervised learning**, the data includes the outcome to be predicted, and the model learns the relationship between the predictors and that outcome. If the outcome is a category, such as "considering dropout: yes or no", it is a **classification** problem; if it is a number, such as next semester's GPA, it is a **regression** problem. Chapters 11 to 13 and Chapter 15 are about supervised learning. In **unsupervised learning**, there is no outcome, and the model looks for structure in the data itself. The clustering of Chapter 9 is unsupervised, and Chapter 14 returns to it.

Many machine learning methods are familiar statistics: logistic regression is both a statistical model and one of the most useful classifiers. What changes is the goal, and with it the way models are judged. Table 11.1 sets the two aims side by side.

Table 11.1: Explanation and prediction

	Explanation (Chapters 7 to 10)	Prediction (Chapters 11 to 16)
Question	*Why* does something happen?	*What* will happen for a new case?
Judged by	Sensible, well-estimated effects	Accuracy on new data
Typical models	Simple, interpretable	Anything that predicts well
Main danger	Confounding	Overfitting

A predictive model does not need to be causally correct to be useful: a model can predict dropout well from variables that do not cause it. Equally, a model that explains well may predict poorly, because the effects it estimates are small compared with everything it cannot see. Chapter 5 described prediction as a third aim of analysis, alongside description and explanation; this part of the book develops it.

11.2 - Generalisation and overfitting

The central idea of machine learning is **generalisation**: a model is useful only if what it learned from one set of data carries over to new data. Any dataset contains two things, the pattern that would appear again in new data and the noise that belongs to this sample only. A model that learns the pattern generalises; a model that also learns the noise **overfits**, and looks better on its own data than it will ever be on new data.

A simulation makes the idea visible. The code below draws 15 points from a smooth curve with random noise added, and fits three models of increasing flexibility: a straight line, a gentle curve, and a very flexible curve (polynomials of degree 1, 3, and 12). It then draws new points from the same process, within the same range, to see how each model does on data it has not seen:

```
true_curve = lambda x: np.sin(2 * x)

def make_points(n, seed=None):
    if seed is not None:
        np.random.seed(seed)
    x = np.random.uniform(0, 3, n)
    y = true_curve(x) + np.random.normal(0, 0.3, n)
    return pd.DataFrame({"x": x, "y": y})

train_points = make_points(15, seed=3)
new_points = make_points(1000, seed=4)
new_points = (
    new_points[
        (new_points["x"] >= train_points["x"].min()) &
        (new_points["x"] <= train_points["x"].max())
    ]
    .iloc[:300]
    .reset_index(drop=True)
)

x_grid = np.linspace(train_points["x"].min(), train_points["x"].max(), 300)
curves_dict = {}

for d in [1, 3, 12]:
    poly_model = make_pipeline(PolynomialFeatures(d), LinearRegression())
    poly_model.fit(train_points[["x"]], train_points["y"])
    curves_dict[f"Degree {d}"] = poly_model.predict(x_grid[:, None])
```

```
fig, axes = plt.subplots(1, 3, figsize=(8, 3.2), sharey=True)

for ax, d in zip(axes, [1, 3, 12]):
     ax.scatter(new_points["x"], new_points["y"], color="grey", alpha=0.35, s=8, label="New
```

```
points")
        ax.scatter(train_points["x"],  train_points["y"],  color="black",  s=25,  zorder=4,
label="Training points")
   ax.plot(x_grid, curves_dict[f"Degree {d}"], color="#b2182b", linewidth=1.5, label=f"Degree
{d}")
    ax.set_title(f"Degree {d}")
    ax.set_ylim(-2, 2)
    ax.set_xlabel("x")

axes[0].set_ylabel("y")
plt.tight_layout()
plt.show()
```

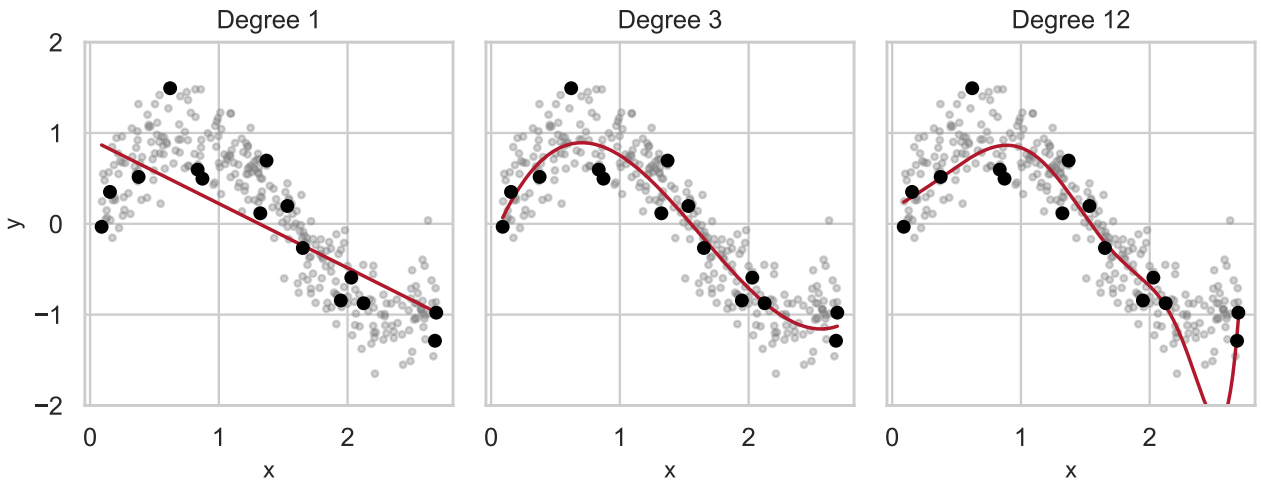


Figure 11.1: Three models fitted to the same 15 training points (black). The straight line is too simple, the degree-3 curve follows the pattern, and the degree-12 curve passes close to every training point but wanders far from the new points (grey).

The straight line is too simple to follow the wave: it **underfits**. The degree-3 curve captures the pattern. The degree-12 curve bends to pass close to every training point, and in doing so follows the noise, swinging far away from where new points actually lie. Measured on the training points, it would look like the best model; measured on new points, it is the worst. The prediction error of every degree from 1 to 12 shows the general pattern:

```
errors = []
for d in range(1, 13):
    model = make_pipeline(PolynomialFeatures(d), LinearRegression())
    model.fit(train_points[["x"]], train_points["y"])
    tr_err = root_mean_squared_error(train_points["y"], model.predict(train_points[["x"]]))
    te_err = root_mean_squared_error(new_points["y"], model.predict(new_points[["x"]]))
    errors.append({"degree": d, "training": tr_err, "new_data": te_err})

errors_df = pd.DataFrame(errors)
print(errors_df.head(6).round(3))
```

```
   degree  training  new_data
0       1     0.462     0.470
1       2     0.379     0.414
2       3     0.257     0.297
3       4     0.257     0.297
4       5     0.251     0.327
5       6     0.251     0.326
```

```
fig, ax = plt.subplots(figsize=(6, 3.6))
ax.plot(errors_df["degree"], errors_df["training"], marker="o", color="grey", linewidth=1.5,
label="Training points")
ax.plot(errors_df["degree"],     errors_df["new_data"],     marker="o",     color="#b2182b",
linewidth=1.5, label="New points")
ax.set_yscale("log")
ax.set_xticks(range(1, 13))
ax.set_xlabel("Flexibility (polynomial degree)")
ax.set_ylabel("Prediction error (RMSE, log scale)")
ax.legend()
plt.tight_layout()
plt.show()
```

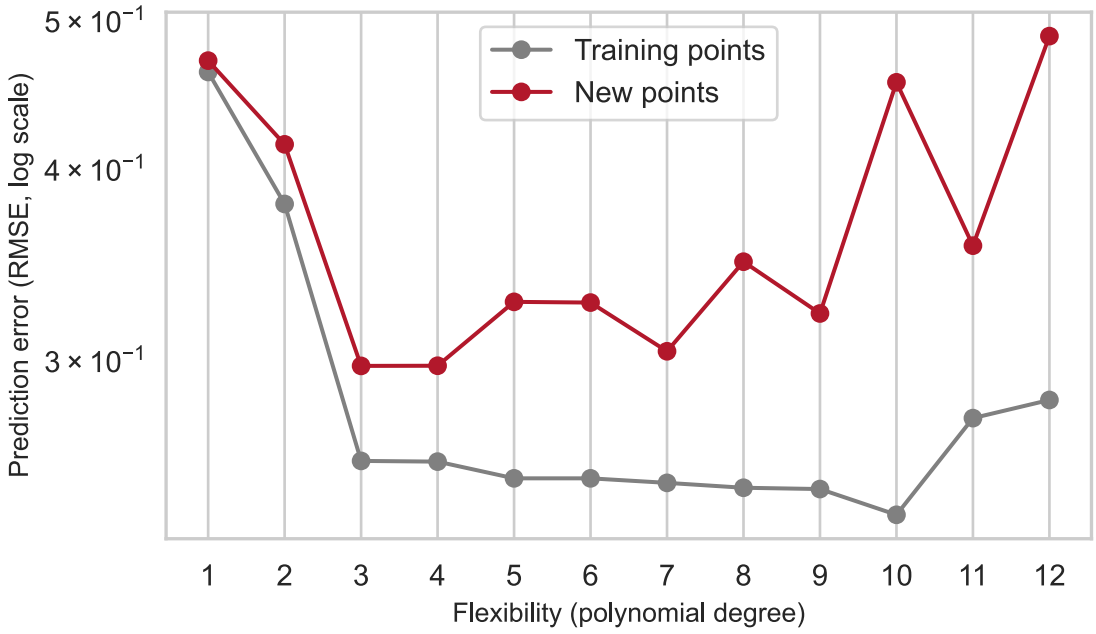


Figure 11.2: Prediction error (RMSE) on the training points and on new points, for polynomials of degree 1 to 12. Training error keeps falling as the model becomes more flexible; error on new data falls, then rises.

The training error falls steadily as the model becomes more flexible: a more flexible model can always fit its own data more closely. The error on new data falls at first, reaches its minimum at degree np.int64(3), and then rises. This U shape is known as the **bias-variance trade-off**. A model that is too simple has high **bias**: it misses part of the real pattern, whatever data it is given. A model that is too flexible has high **variance**: it changes a great deal from one sample to another, because it follows each sample's noise. The best predictions come from a model in between, and the only way to find it is to measure performance on data the model did not learn from. The rest of the chapter builds that measurement into every step.

11.3 - Designing a fair test

The **scikit-learn** framework provides a principled, modular design that makes a fair test of a predictive model clean and reliable. Each module handles one part of the pipeline: `train_test_split` partitions the data, `ColumnTransformer` and pipelines prepare it, estimators specify models, `GridSearchCV` tunes them, and `metrics` assesses performance on unseen data.

Figure 11.3 shows how the pieces fit together: the test data is set aside before anything else happens, all choices are made with the training data alone, and the test data is used once, at the end. The rest of the chapter works through each step.

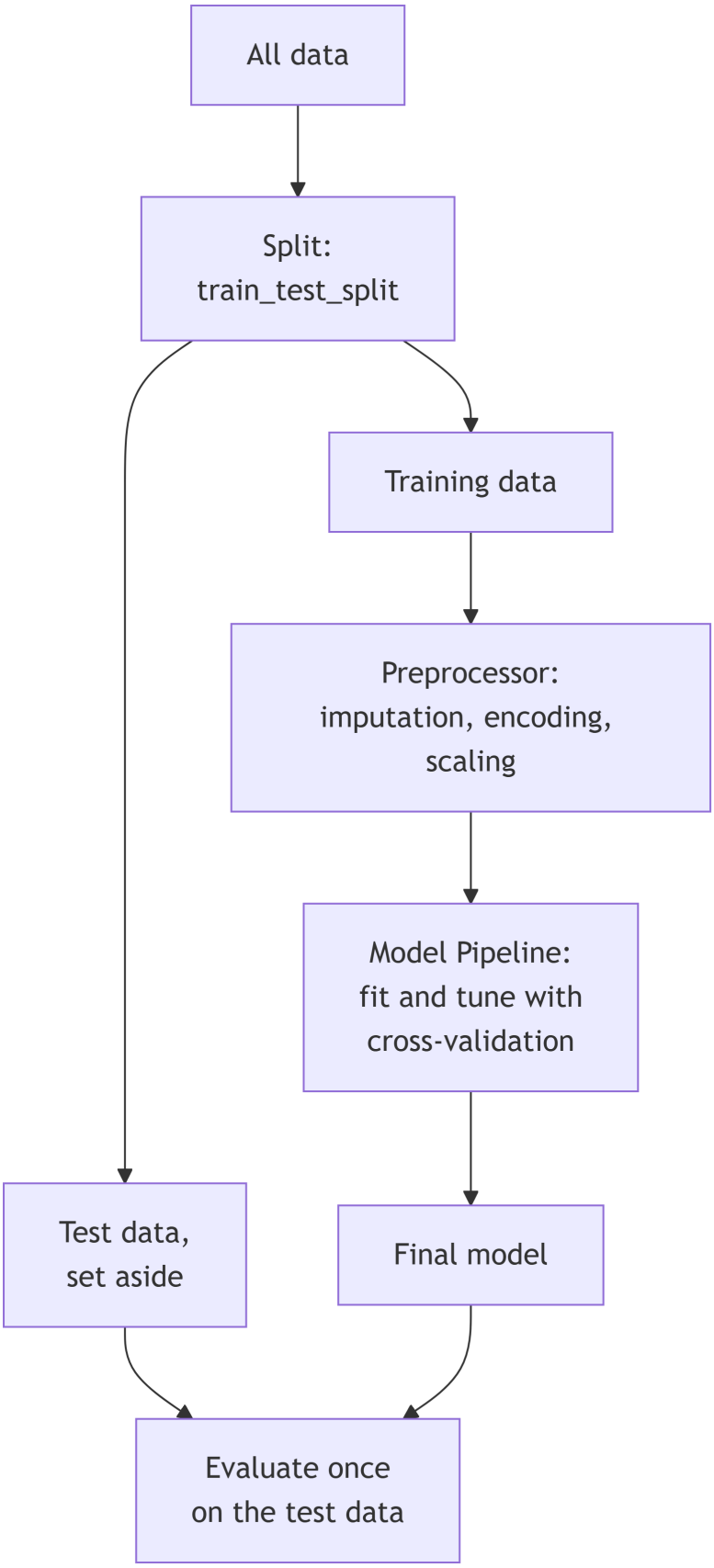


Figure 11.3: The machine learning workflow in scikit-learn.

11.4 - Data for prediction

A predictive model can only use information that would be available when the prediction is made. The university wants to act at the end of the first semester, so the model may use the students' background, their questionnaire scores from the start of the year, and their first-semester records, but nothing from later. The questionnaire scores are calculated as before:

```
scores = pd.DataFrame({
    "student_id": questionnaire["student_id"],
            "stress":   questionnaire[["stress_1",   "stress_2",   "stress_3",   "stress_5",
"stress_6"]].assign(stress_4=6 - questionnaire["stress_4"]).mean(axis=1),
    "burnout": questionnaire.filter(like="burnout_").mean(axis=1),
    "support": questionnaire.filter(like="support_").mean(axis=1),
    "satisfaction": questionnaire.filter(like="satisfaction_").mean(axis=1),
})

sem1 = semesters[semesters["semester"] == 1].drop(columns=["semester"])

dropout_data = (
    students
    .merge(scores, on="student_id")
    .merge(sem1, on="student_id")
    .drop(columns=["student_id", "supervisor_id", "workshop", "workshop_sessions"])
)

print("Data shape:", dropout_data.shape)
```

```
Data shape: (600, 21)
```

Two details matter. The outcome is coded as binary (1 for "Yes", 0 for "No"), making "Yes" (considering dropout) the positive event to detect. And the workshop variables are removed: the workshop took place after the first semester, so it would not be known when the prediction is made. Using information that would not be available at the time of prediction is a form of **data leakage**, and it makes a model look better than it can ever be in practice.

```
counts = dropout_data["considering_dropout"].value_counts()
shares = dropout_data["considering_dropout"].value_counts(normalize=True)
pd.DataFrame({"count": counts, "share": shares.round(3)})
```

considering_dropout	count	share
No	510	0.85
Yes	90	0.15

Only about 15% of students answer "Yes". Such an **imbalanced** outcome is common in real prediction problems, from rare diseases to fraud, and, as the evaluation below shows, it catches out anyone who judges a model by accuracy alone.

11.5 - Splitting the data

The most important rule of machine learning follows from the simulation above: **never judge a model on the data it learned from.** The first step is therefore to split the data. The **training set** is used to build and tune the model. The **test set** is locked away and used only once, at the very end, to estimate how the final model will perform on new students. The function `train_test_split()` with `stratify=y` makes sure that both sets have the same share of "Yes" answers:

```
X = dropout_data.drop(columns=["considering_dropout"])
y = (dropout_data["considering_dropout"] == "Yes").astype(int)

X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.25, stratify=y, random_state=2026
)

print(f"Training observations: {len(X_train)}")
print(f"Testing observations:  {len(X_test)}")
```

```
Training observations: 450
Testing observations:  150
```

Three quarters of the students are used for training and a quarter are set aside for the final test.

11.6 - Preparing the data

Most models need the data prepared first: missing values filled in, categories turned into numbers, and numeric predictors put on a common scale. A `ColumnTransformer` bundles these steps cleanly:

```
num_cols = X.select_dtypes(include=[np.number]).columns.tolist()
cat_cols = X.select_dtypes(exclude=[np.number]).columns.tolist()

preprocessor = ColumnTransformer([
    ("num", Pipeline([
        ("imputer", SimpleImputer(strategy="median")),
        ("scaler", StandardScaler())
    ]), num_cols),
    ("cat", Pipeline([
        ("imputer", SimpleImputer(strategy="most_frequent")),
        ("ohe", OneHotEncoder(drop="first", sparse_output=False, handle_unknown="ignore"))
    ]), cat_cols)
])
```

The numeric pipeline replaces missing values with the variable's median (`SimpleImputer(strategy="median")`) and standardizes predictors to z-scores (`StandardScaler()`). The categorical pipeline imputes missing categories with the most frequent value and creates one-hot dummy variables (`OneHotEncoder(drop="first")`), dropping the reference level just as linear regression did in Chapter 8.

To see what the preprocessor produces, `fit_transform()` estimates the parameters from the training set and transforms it:

```
X_train_prep = preprocessor.fit_transform(X_train)
feature_names = preprocessor.get_feature_names_out()
print("Transformed training shape:", X_train_prep.shape)
print("First 8 prepared feature names:", feature_names[:8].tolist())
```

```
Transformed training shape: (450, 24)
First 8 prepared feature names: ['num__age', 'num__financial_worry', 'num__stress',
'num__burnout', 'num__support', 'num__satisfaction', 'num__gpa', 'num__sleep_hours']
```

> ! No peeking at the test data
>
> The medians, means, and standard deviations are estimated from the **training** data only, and then applied unchanged to the test data. If they were calculated from all the data, information from the test set would leak into the model, and the test would no longer be a fair one. Packaging preprocessing inside a `Pipeline` handles this automatically, which is one of the best reasons to use scikit-learn pipelines.

11.7 - Specifying and fitting a model

A `Pipeline` combines the preprocessor and the classifier so they always travel together:

```
logistic_pipe = Pipeline([
    ("preprocessor", preprocessor),
    ("classifier", LogisticRegression(max_iter=1000, random_state=2026))
])

logistic_pipe.fit(X_train, y_train)
```

steps steps: list of tuples — [('preprocessor', ...), ('classifier', ...)]

List of (name of step, estimator) tuples that are to be chained in sequential order. To be compatible with the scikit-learn API, all steps must define `fit`. All non-last steps must also define `transform`. See :ref:`Combining Estimators <combining_estimators>` for more details.

transform_input transform_input: list of str, default=None — None

The names of the :term:`metadata` parameters that should be transformed by the
pipeline before passing it to the step consuming it.

This enables transforming some input arguments to ``fit`` (other than ``X``)
to be transformed by the steps of the pipeline up to the step which requires
them. Requirement is defined via :ref:`metadata routing <metadata_routing>`.
For instance, this can be used to pass a validation set through the pipeline.

You can only set this if metadata routing is enabled, which you
can enable using ``sklearn.set_config(enable_metadata_routing=True)``.

.. versionadded:: 1.6

memory memory: str or object with the joblib.Memory interface, default=None — None

Used to cache the fitted transformers of the pipeline. The last step
will never be cached, even if it is a transformer. By default, no
caching is performed. If a string is given, it is the path to the
caching directory. Enabling caching triggers a clone of the

	transformers before fitting. Therefore, the transformer instance given to the pipeline cannot be inspected directly. Use the attribute ``named_steps`` or ``steps`` to inspect estimators within the pipeline. Caching the transformers is advantageous when fitting is time consuming. See :ref:`sphx_glr_auto_examples_neighbors_plot_caching_nearest_neighbors.py` for an example on how to enable caching.	
	verbose verbose: bool, default=False If True, the time elapsed while fitting each step will be printed as it is completed.	False
Name	Type	Value
classes_ classes_: ndarray of shape (n_classes,) The classes labels. Only exist if the last step of the pipeline is a classifier.	ndarray[int64](2,)	[0,1]
feature_names_in_ feature_names_in_: ndarray of shape (`n_features_in_`,) Names of features seen during :term:`fit`. Only defined if the underlying estimator exposes such an attribute when fit. .. versionadded:: 1.0	ndarray[object](20,)	['age','gender','faculty',...,'caffeine_mg','supervisor_r 'wellbeing']
n_features_in_ n_features_in_: int Number of features seen during :term:`fit`. Only defined if the underlying first estimator in `steps` exposes such an attribute when fit. .. versionadded:: 0.24	int	20
	transformers transformers: list of tuples List of (name, transformer, columns) tuples specifying the transformer objects to be applied to subsets of the data.	[('num', ...), ('cat', ...)]

name : str
Like in Pipeline and FeatureUnion, this allows the transformer and its parameters to be set using ``set_params`` and searched in grid search.
transformer : {'drop', 'passthrough'} or estimator
Estimator must support :term:`fit` and :term:`transform`. Special-cased strings 'drop' and 'passthrough' are accepted as well, to indicate to drop the columns or to pass them through untransformed, respectively.
columns : str, array-like of str, int, array-like of int, array-like of bool, slice or callable
Indexes the data on its second axis. Integers are interpreted as positional columns, while strings can reference DataFrame columns by name. A scalar string or int should be used where ``transformer`` expects X to be a 1d array-like (vector), otherwise a 2d array will be passed to the transformer. A callable is passed the input data `X` and can return any of the above. To select multiple columns by name or dtype, you can use :obj:`make_column_selector`.

remainder remainder: {'drop', 'drop' 'passthrough'} or estimator, default='drop'

By default, only the specified columns in `transformers` are transformed and combined in the output, and the non-specified columns are dropped. (default of ``'drop'``). By specifying ``remainder='passthrough'``, all remaining columns that were not specified in `transformers`, but present in the data passed to `fit` will be automatically passed through. This subset of columns is concatenated with the output of the transformers. For dataframes, extra columns not seen during `fit` will be excluded from the output of `transform`. By setting ``remainder`` to be an

	estimator, the remaining non-specified columns will use the ``remainder`` estimator. The estimator must support :term:`fit` and :term:`transform`. Note that using this feature requires that the DataFrame columns input at :term:`fit` and :term:`transform` have identical order.	
sparse_threshold	sparse_threshold: float, default=0.3 If the output of the different transformers contains sparse matrices, these will be stacked as a sparse matrix if the overall density is lower than this value. Use ``sparse_threshold=0`` to always return dense. When the transformed output consists of all dense data, the stacked result will be dense, and this keyword will be ignored.	0.3
n_jobs	n_jobs: int, default=None Number of jobs to run in parallel. ``None`` means 1 unless in a :obj:`joblib.parallel_backend` context. ``-1`` means using all processors. See :term:`Glossary <n_jobs>` for more details.	None
transformer_weights	transformer_weights: dict, default=None Multiplicative weights for features per transformer. The output of the transformer is multiplied by these weights. Keys are transformer names, values the weights.	None
verbose	verbose: bool, default=False If True, the time elapsed while fitting each transformer will be printed as it is completed.	False
verbose_feature_names_out	verbose_feature_names_out: bool, str or Callable[[str, str], str], default=True - If True, :meth:`ColumnTransformer.get_feature_names_out`	True

	will prefix all feature names with the name of the transformer that generated that feature. It is equivalent to setting `verbose_feature_names_out="{transformer_name} __{feature_name}"`. - If False, :meth:`ColumnTransformer.get_feature_names_out` will not prefix any feature names and will error if feature names are not unique. - If ``Callable[[str, str], str]``, :meth:`ColumnTransformer.get_feature_names_out` will rename all the features using the name of the transformer. The first argument of the callable is the transformer name and the second argument is the feature name. The returned string will be the new feature name. - If ``str``, it must be a string ready for formatting. The given string will be formatted using two field names: ``transformer_name`` and ``feature_name``. e.g. ``"{feature_name} __{transformer_name}"``. See :meth:`str.format` method from the standard library for more info. .. versionadded:: 1.0 .. versionchanged:: 1.6 `verbose_feature_names_out` can be a callable or a string to be for- matted.	
Name	Type	Value
feature_names_in_ feature_names_in_: ndarray of shape (`n_features_in_`,) Names of features seen during :term:`fit`. Defined only when `X` has feature names that are all strings. .. versionadded:: 1.0	ndarray[object](20,)	['age','gender','faculty',...,'caffeine_mg','supervisor_r 'wellbeing']
n_features_in_ n_features_in_: int Number of features seen during :term:`fit`. Only defined if the	int	20

	underlying transformers expose such an attribute when fit. .. versionadded:: 0.24		
named_transformers_	named_transformers_ : :class:`~sklearn.utils.Bunch`	Bunch	{'num': Pipel...put=False))])}
	Read-only attribute to access any transformer by given name. Keys are transformer names and values are the fitted transformer objects.		
output_indices_	output_indices_: dict A dictionary from each transformer name to a slice, where the slice corresponds to indices in the transformed output. This is useful to inspect which transformer is responsible for which transformed feature(s). .. versionadded:: 1.0	dict	{'cat': slice(13, 24, None), 'num': slice(0, 13, None), 're...er': slice(0, 0, None)}
sparse_output_	sparse_output_: bool Boolean flag indicating whether the output of ``transform`` is a sparse matrix or a dense numpy array, which depends on the output of the individual transformers and the `sparse_threshold` keyword.	bool	False
transformers_	transformers_: list The collection of fitted transformers as tuples of (name, fitted_transformer, column). `fitted_transformer` can be an estimator, or `'drop'`; `'passthrough'` is replaced with an equivalent :class:`~sklearn.preprocessing.FunctionTransformer`. In case there were no columns selected, this will be the unfitted transformer. If there are remaining columns, the final element is a tuple of the form: ('remainder', transformer, remaining_columns) corresponding to the ``remainder`` parameter. If there are remaining columns, then ``len(transformers_)==len(transformers)+1``, otherwise ``len(transformers_)==len(trans-	list	[('num', Pipeline(step...ardScaler())]), ['age', 'fi...ry', 'stress', 'burnout', ...]), ('cat', Pipeline(step...tput=False))]), ['gender', 'faculty', 'pr...me', 'st...de', ...])]

formers)``.

.. versionadded:: 1.7
The format of the remaining columns now attempts to match that of the other
transformers: if all columns were provided as column names (`str`), the
remaining columns are stored as column names; if all columns were provided
as mask arrays (`bool`), so are the remaining columns; in all other cases
the remaining columns are stored as indices (`int`).

strategy	strategy: str or Callable, default='mean' The imputation strategy. - If "mean", then replace missing values using the mean along each column. Can only be used with numeric data. - If "median", then replace missing values using the median along each column. Can only be used with numeric data. - If "most_frequent", then replace missing using the most frequent value along each column. Can be used with strings or numeric data. If there is more than one such value, only the smallest is returned. - If "constant", then replace missing values with fill_value. Can be used with strings or numeric data. - If an instance of Callable, then replace missing values using the scalar statistic returned by running the callable over a dense 1d array containing non-missing values of each column. .. versionadded:: 0.20 strategy="constant" for fixed value imputation. .. versionadded:: 1.5 strategy=callable for custom value imputation.	'median'
missing_values	missing_values: int, float, str, np.nan, None or pandas.NA, default=np.nan	nan

	The placeholder for the missing values. All occurrences of `missing_values` will be imputed. For pandas' dataframes with nullable integer dtypes with missing values, `missing_values` can be set to either `np.nan` or `pd.NA`.	
fill_value	fill_value: str or numerical value, default=None When strategy == "constant", `fill_value` is used to replace all occurrences of missing_values. For string or object data types, `fill_value` must be a string. If `None`, `fill_value` will be 0 when imputing numerical data and "missing_value" for strings or object data types.	None
copy	copy: bool, default=True If True, a copy of X will be created. If False, imputation will be done in-place whenever possible. Note that, in the following cases, a new copy will always be made, even if `copy=False`: - If `X` is not an array of floating values; - If `X` is encoded as a CSR matrix; - If `add_indicator=True`.	True
add_indicator	add_indicator: bool, default=False If True, a :class:`MissingIndicator` transform will stack onto output of the imputer's transform. This allows a predictive estimator to account for missingness despite imputation. If a feature has no missing values at fit/train time, the feature won't appear on the missing indicator even if there are missing values at transform/test time.	False
keep_empty_features	keep_empty_features: bool, default=False If True, features that consist exclusively of missing values when `fit` is called are returned in results	False

	when `transform` is called. The imputed value is always `0` except when `strategy="constant"` in which case `fill_value` will be used instead. .. versionadded:: 1.2	
Name	Type	Value
feature_names_in_ feature_names_in_: ndarray of shape (`n_features_in_`,) Names of features seen during :term:`fit`. Defined only when `X` has feature names that are all strings. .. versionadded:: 1.0	ndarray[object](13,)	['age','financial_worry','stress',...,'caffeine_mg','supe 'wellbeing']
indicator_ indicator_: :class:`~sklearn.impute.MissingIndicator` Indicator used to add binary indicators for missing values. `None` if `add_indicator=False`.	NoneType	None
n_features_in_ n_features_in_: int Number of features seen during :term:`fit`. .. versionadded:: 0.24	int	13
statistics_ statistics_: array of shape (n_features,) The imputation fill value for each feature. Computing statistics can result in `np.nan` values. During :meth:`transform`, features corresponding to `np.nan` statistics will be discarded.	ndarray[float64](13,)	[29. , 3. , 3.17,...,160. , 4. , 60.]

age
financial_worry
stress
burnout
support
satisfaction
gpa
sleep_hours
study_hours

exercise_days
caffeine_mg
supervisor_meetings
wellbeing

	copy copy: bool, default=True If False, try to avoid a copy and do inplace scaling instead. This is not guaranteed to always work inplace; e.g. if the data is not a NumPy array or scipy.sparse CSR matrix, a copy may still be returned.	True
	with_mean with_mean: bool, default=True If True, center the data before scaling. This does not work (and will raise an exception) when attempted on sparse matrices, because centering them entails building a dense matrix which in common use cases is likely to be too large to fit in memory.	True
	with_std with_std: bool, default=True If True, scale the data to unit variance (or equivalently, unit standard deviation).	True

Name	Type	Value
mean_ mean_: ndarray of shape (n_features,) or None The mean value for each feature in the training set. Equal to ``None`` when ``with_mean=False`` and ``with_std=False``.	ndarray[float64](13,)	[29.81, 2.84, 3.21,...,188.76, 4.22, 60.63]
n_features_in_ n_features_in_: int Number of features seen during :term:`fit`. .. versionadded:: 0.24	int	13
n_samples_seen_ n_samples_seen_: int or ndarray of shape (n_features,) The number of samples processed by the estimator for each feature.	float64	450

If there are no missing samples, the ``n_samples_seen`` will be an integer, otherwise it will be an array of dtype int. If `sample_weights` are used it will be a float (if no missing data) or an array of dtype float that sums the weights seen so far. Will be reset on new calls to fit, but increments across ``partial_fit`` calls.		
scale_ scale_: ndarray of shape (n_features,) or None Per feature relative scaling of the data to achieve zero mean and unit variance. Generally this is calculated using `np.sqrt(var_)`. If a variance is zero, we can't achieve unit variance, and the data is left as-is, giving a scaling factor of 1. `scale_` is equal to `None` when `with_std=False`. .. versionadded:: 0.17 *scale_*	ndarray[float64](13,)	[4.88, 1.15, 0.74,...,145.18, 2.76, 11.7]
var_ var_: ndarray of shape (n_features,) or None The variance for each feature in the training set. Used to compute `scale_`. Equal to ``None`` when ``with_mean=False`` and ``with_std=False``.	ndarray[float64](13,)	[23.86, 1.32, 0.55,...,21076.67, 7.62, 136.98]

x0
x1
x2
x3
x4
x5
x6
x7
x8
x9
x10
x11
x12

strategy	strategy: str or Callable, default='mean' The imputation strategy. - If "mean", then replace missing values using the mean along each column. Can only be used with numeric data. - If "median", then replace missing values using the median along each column. Can only be used with numeric data. - If "most_frequent", then replace missing using the most frequent value along each column. Can be used with strings or numeric data. If there is more than one such value, only the smallest is returned. - If "constant", then replace missing values with fill_value. Can be used with strings or numeric data. - If an instance of Callable, then replace missing values using the scalar statistic returned by running the callable over a dense 1d array containing non-missing values of each column. .. versionadded:: 0.20 strategy="constant" for fixed value imputation. .. versionadded:: 1.5 strategy=callable for custom value imputation.	'most_frequent'
missing_values	missing_values: int, float, str, np.nan, None or pandas.NA, default=np.nan The placeholder for the missing values. All occurrences of `missing_values` will be imputed. For pandas' dataframes with nullable integer dtypes with missing values, `missing_values` can be set to either `np.nan` or `pd.NA`.	nan
fill_value	fill_value: str or numerical value, default=None When strategy == "constant", `fill_value` is used to replace all occurrences of missing_values. For string or object data types, `fill_value` must be a string. If `None`, `fill_value` will be 0	None

	when imputing numerical data and "missing_value" for strings or object data types.	
	copy copy: bool, default=True If True, a copy of X will be created. If False, imputation will be done in-place whenever possible. Note that, in the following cases, a new copy will always be made, even if `copy=False`: - If `X` is not an array of floating values; - If `X` is encoded as a CSR matrix; - If `add_indicator=True`.	True
	add_indicator add_indicator: bool, default=False If True, a :class:`MissingIndicator` transform will stack onto output of the imputer's transform. This allows a predictive estimator to account for missingness despite imputation. If a feature has no missing values at fit/train time, the feature won't appear on the missing indicator even if there are missing values at transform/test time.	False
	keep_empty_features keep_empty_features: bool, default=False If True, features that consist exclusively of missing values when `fit` is called are returned in results when `transform` is called. The imputed value is always `0` except when `strategy="constant"` in which case `fill_value` will be used instead. .. versionadded:: 1.2	False

Name	Type	Value
feature_names_in_ feature_names_in_: ndarray of shape (`n_features_in_`,) Names of features seen during :term:`fit`. Defined only when `X` has feature names that are all strings.	ndarray[object](7,)	['gender','faculty','programme',...,'employment','has_children', 'lives_away']

.. versionadded:: 1.0

indicator_ indicator_ : :class:`~sklearn.impute.MissingIndicator`	None or type	None

Indicator used to add binary indicators for missing values.
`None` if `add_indicator=False`.

n_features_in_ n_features_in_: int	int	7

Number of features seen during :term:`fit`.

.. versionadded:: 0.24

statistics_ statistics_: array of shape (n_features,)	ndarray[object](7,)	['Female','Health Sciences','Master's",...,'None','No','No']

The imputation fill value for each feature.
Computing statistics can result in `np.nan` values.
During :meth:`transform`, features corresponding to `np.nan` statistics will be discarded.

gender
faculty
programme
study_mode
employment
has_children
lives_away

	drop drop: {'first', 'if_binary'} or an array-like of shape (n_features,), default=None	'first'

Specifies a methodology to use to drop one of the categories per feature. This is useful in situations where perfectly collinear features cause problems, such as when feeding the resulting data into an unregularized linear regression model.

However, dropping one category breaks the symmetry of the original representation and can therefore induce a bias in downstream models, for instance for penalized linear classification or regression models.

	- None : retain all features (the default). - 'first' : drop the first category in each feature. If only one category is present, the feature will be dropped entirely. - 'if_binary' : drop the first category in each feature with two categories. Features with 1 or more than 2 categories are left intact. - array : ``drop[i]`` is the category in feature ``X[:, i]`` that should be dropped. When `max_categories` or `min_frequency` is configured to group infrequent categories, the dropping behavior is handled after the grouping. .. versionadded:: 0.21 The parameter `drop` was added in 0.21. .. versionchanged:: 0.23 The option `drop='if_binary'` was added in 0.23. .. versionchanged:: 1.1 Support for dropping infrequent categories.	
sparse_output	sparse_output: bool, default=True When ``True``, it returns a SciPy sparse matrix/array in "Compressed Sparse Row" (CSR) format. .. versionadded:: 1.2 `sparse` was renamed to `sparse_output`	False
handle_unknown	handle_unknown: {'error', 'ignore', 'infrequent_if_exist', 'warn'}, default='error' Specifies the way unknown categories are handled during :meth:`transform`. - 'error' : Raise an error if an unknown category is present during transform.	'ignore'

- 'ignore' : When an unknown category is encountered during transform, the resulting one-hot encoded columns for this feature will be all zeros. In the inverse transform, an unknown category will be denoted as None.
- 'infrequent_if_exist' : When an unknown category is encountered during transform, the resulting one-hot encoded columns for this feature will map to the infrequent category if it exists. The infrequent category will be mapped to the last position in the encoding. During inverse transform, an unknown category will be mapped to the category denoted `'infrequent'` if it exists. If the `'infrequent'` category does not exist, then :meth:`transform` and :meth:`inverse_transform` will handle an unknown category as with `handle_unknown='ignore'`. Infrequent categories exist based on `min_frequency` and `max_categories`. Read more in the :ref:`User Guide <encoder_infrequent_categories>`.
- 'warn' : When an unknown category is encountered during transform a warning is issued, and the encoding then proceeds as described for `handle_unknown="infrequent_if_exist"`.

.. versionchanged:: 1.1
`'infrequent_if_exist'` was added to automatically handle unknown categories and infrequent categories.

.. versionadded:: 1.6
The option `"warn"` was added in 1.6.

categories categories: 'auto' or a list of array-like, default='auto' 'auto'

Categories (unique values) per feature:

- 'auto' : Determine categories automatically from the training data.
- list : ``categories[i]`` holds the categories expected in the ith column. The passed categories

	should not mix strings and numeric values within a single feature, and should be sorted in case of numeric values. The used categories can be found in the ``categories_`` attribute. .. versionadded:: 0.20	
dtype	dtype: number type, default=np.float64 Desired dtype of output.	<class 'numpy.float64'>
min_frequency	min_frequency: int or float, default=None Specifies the minimum frequency below which a category will be considered infrequent. - If `int`, categories with a smaller cardinality will be considered infrequent. - If `float`, categories with a smaller cardinality than `min_frequency * n_samples` will be considered infrequent. .. versionadded:: 1.1 Read more in the :ref:`User Guide <encoder_infrequent_categories>`.	None
max_categories	max_categories: int, default=None Specifies an upper limit to the number of output features for each input feature when considering infrequent categories. If there are infrequent categories, `max_categories` includes the category representing the infrequent categories along with the frequent categories. If `None`, there is no limit to the number of output features. .. versionadded:: 1.1 Read more in the :ref:`User Guide <encoder_infrequent_categories>`.	None
feature_name_combiner	feature_name_combiner: "concat" or callable, default="concat" Callable with signature `def callable(input_feature, category)`	'concat'

	that returns a string. This is used to create feature names to be returned by :meth:`get_feature_names_out`. `"concat"` concatenates encoded feature name and category with `feature + "_" + str(category)`.E.g. feature X with values 1, 6, 7 create feature names `X_1, X_6, X_7`. .. versionadded:: 1.3	
Name	Type	Value
categories_ categories_: list of arrays The categories of each feature determined during fitting (in order of the features in X and corresponding with the output of ``transform``). This includes the category specified in ``drop`` (if any).	list	[array(['Femal… dtype=object), array(['Educa… dtype=object), array(["Maste… dtype=object), array(['Full-… dtype=object), …]
drop_idx_ drop_idx_: array of shape (n_features,) - ``drop_idx_[i]`` is the index in ``categories_[i]`` of the category to be dropped for each feature. - ``drop_idx_[i] = None`` if no category is to be dropped from the feature with index ``i``, e.g. when `drop='if_binary'` and the feature isn't binary. - ``drop_idx_ = None`` if all the transformed features will be retained. If infrequent categories are enabled by setting `min_frequency` or `max_categories` to a non-default value and `drop_idx[i]` corresponds to an infrequent category, then the entire infrequent category is dropped. .. versionchanged:: 0.23 Added the possibility to contain `None` values.	ndarray[object](7,)	[0,0,0,…,0,0,0]
n_features_in_ n_features_in_: int Number of features seen during :term:`fit`.	int	7

.. versionadded:: 1.0

x0_Male
x1_Health Sciences
x1_Humanities
x1_Natural Sciences
x1_Social Sciences
x2_PhD
x3_Part-time
x4_None
x4_Part-time job
x5_Yes
x6_Yes

num__age
num__financial_worry
num__stress
num__burnout
num__support
num__satisfaction
num__gpa
num__sleep_hours
num__study_hours
num__exercise_days
num__caffeine_mg
num__supervisor_meetings
num__wellbeing
cat__gender_Male
cat__faculty_Health Sciences
cat__faculty_Humanities
cat__faculty_Natural Sciences
cat__faculty_Social Sciences
cat__programme_PhD
cat__study_mode_Part-time
cat__employment_None
cat__employment_Part-time job
cat__has_children_Yes
cat__lives_away_Yes

random_state	random_state: int, RandomState instance, default=None Only used for `solver` == 'sag', 'saga' or 'liblinear' to shuffle the data. It has no effect on the other solvers. See :term:`Glossary <random_state>` for details.	2026
max_iter	max_iter: int, default=100 Maximum number of iterations taken for the solvers to converge.	1000
penalty	penalty: {'l1', 'l2', 'elasticnet', None}, default='l2' Specify the norm of the penalty: - `None`: no penalty is added; - `'l2'`: add an L2 penalty term and it is the default choice; - `'l1'`: add an L1 penalty term; - `'elasticnet'`: both L1 and L2 penalty terms are added. .. warning:: Some penalties may not work with some solvers. See the parameter `solver` below, to know the compatibility between the penalty and solver. .. versionadded:: 0.19 l1 penalty with SAGA solver (allowing 'multinomial' + L1) .. deprecated:: 1.8 `penalty` was deprecated in version 1.8 and will be removed in 1.10. Use `l1_ratio` and `C` instead. `l1_ratio=0` for `penalty='l2'`, `l1_ratio=1` for `penalty='l1'`, `l1_ratio` set to any float between 0 and 1 for `penalty='elasticnet'`, and `C=np.inf` for `penalty=None`.	'deprecated'
C	C: float, default=1.0 Inverse of regularization strength; must be a positive float. Like in support vector machines, smaller values specify stronger regularization. `C=np.inf` results in unpenalized logistic regression.	1.0

	For a visual example on the effect of tuning the `C` parameter with an L1 penalty, see: :ref:`sphx_glr_auto_examples_linear_model_plot_logistic_path.py`.	
l1_ratio	l1_ratio: float, default=0.0 The Elastic-Net mixing parameter, with `0 <= l1_ratio <= 1`. Setting `l1_ratio=1` gives a pure L1-penalty, setting `l1_ratio=0` a pure L2-penalty. Any value between 0 and 1 gives an Elastic-Net penalty of the form `l1_ratio * L1 + (1 - l1_ratio) * L2`. .. warning:: Certain values of `l1_ratio`, i.e. some penalties, may not work with some solvers. See the parameter `solver` below, to know the compatibility between the penalty and solver. .. versionchanged:: 1.8 Default value changed from None to 0.0. .. deprecated:: 1.8 `None` is deprecated and will be removed in version 1.10. Always use `l1_ratio` to specify the penalty type.	0.0
dual	dual: bool, default=False Dual (constrained) or primal (regularized, see also :ref:`this equation <regularized-logistic-loss>`) formulation. Dual formulation is only implemented for l2 penalty with liblinear solver. Prefer `dual=False` when n_samples > n_features.	False
tol	tol: float, default=1e-4 Tolerance for stopping criteria.	0.0001
fit_intercept	fit_intercept: bool, default=True Specifies if a constant (a.k.a. bias or intercept) should be added to the decision function.	True

intercept_scaling intercept_scaling: float, default=1 Useful only when the solver `liblinear` is used and `self.fit_intercept` is set to `True`. In this case, `x` becomes `[x, self.intercept_scaling]`, i.e. a "synthetic" feature with constant value equal to `intercept_scaling` is appended to the instance vector. The intercept becomes ``intercept_scaling * synthetic_feature_weight``. .. note:: The synthetic feature weight is subject to L1 or L2 regularization as all other features. To lessen the effect of regularization on synthetic feature weight (and therefore on the intercept) `intercept_scaling` has to be increased.		1
class_weight class_weight: dict or 'balanced', default=None Weights associated with classes in the form ``{class_label: weight}``. If not given, all classes are supposed to have weight one. The "balanced" mode uses the values of y to automatically adjust weights inversely proportional to class frequencies in the input data as ``n_samples / (n_classes * np.bincount(y))``. Note that these weights will be multiplied with sample_weight (passed through the fit method) if sample_weight is specified. .. versionadded:: 0.17 *class_weight='balanced'*		None
solver solver: {'lbfgs', 'liblinear', 'newton-cg', 'newton-cholesky', 'sag', 'saga'}, default='lbfgs' Algorithm to use in the optimization problem. Default is 'lbfgs'. To choose a solver, you might want to consider the following aspects:		'lbfgs'

- 'lbfgs' is a good default solver because it works reasonably well for a wide
class of problems.
- For :term:`multiclass` problems (`n_classes >= 3`), all solvers except
'liblinear' minimize the full multinomial loss, 'liblinear' will raise an
error.
- 'newton-cholesky' is a good choice for
`n_samples` >> `n_features * n_classes`, especially with one-hot encoded
categorical features with rare categories. Be aware that the memory usage
of this solver has a quadratic dependency on `n_features * n_classes`
because it explicitly computes the full Hessian matrix.
- For small datasets, 'liblinear' is a good choice, whereas 'sag'
and 'saga' are faster for large ones;
- 'liblinear' can only handle binary classification by default. To apply a
one-versus-rest scheme for the multiclass setting one can wrap it with the
:class:`~sklearn.multiclass.OneVsRestClassifier`.

.. warning::
The choice of the algorithm depends on the penalty chosen (`l1_ratio=0`
for L2-penalty, `l1_ratio=1` for L1-penalty and `0 < l1_ratio < 1` for
Elastic-Net) and on (multinomial) multiclass support:

================
========================
=====================
solver l1_ratio multinomial multiclass
================
========================
=====================
'lbfgs' l1_ratio=0 yes
'liblinear' l1_ratio=1 or l1_ratio=0 no
'newton-cg' l1_ratio=0 yes
'newton-cholesky' l1_ratio=0 yes
'sag' l1_ratio=0 yes
'saga' 0<=l1_ratio<=1 yes
================

```
========================
======================
```

.. note::
'sag' and 'saga' fast convergence is
only guaranteed on features
with approximately the same scale.
You can preprocess the data with
a scaler from :mod:`sklearn.preprocessing`.

.. seealso::
Refer to the :ref:`User Guide
<Logistic_regression>` for more
information regarding :class:`LogisticRegression` and more specifically the
:ref:`Table
<logistic_regression_solvers>`
summarizing solver/penalty supports.

.. versionadded:: 0.17
Stochastic Average Gradient (SAG)
descent solver. Multinomial support in
version 0.18.
.. versionadded:: 0.19
SAGA solver.
.. versionchanged:: 0.22
The default solver changed from
'liblinear' to 'lbfgs' in 0.22.
.. versionadded:: 1.2
newton-cholesky solver. Multinomial support in version 1.6.

verbose verbose: int, default=0 0

For the liblinear and lbfgs solvers
set verbose to any positive
number for verbosity.

warm_start warm_start: bool, default=False False

When set to True, reuse the solution of the previous call to fit as
initialization, otherwise, just erase
the previous solution.
Useless for liblinear solver.
See :term:`the Glossary
<warm_start>`.

.. versionadded:: 0.17
warm_start to support *lbfgs*,
newton-cg, *sag*, *saga* solvers.

n_jobs n_jobs: int, default=None None

Does not have any effect.

	.. deprecated:: 1.8 `n_jobs` is deprecated in version 1.8 and will be removed in 1.10.	
Name	Type	Value
classes_ classes_: ndarray of shape (n_classes,) A list of class labels known to the classifier.	ndarray[int64](2,)	[0,1]
coef_ coef_: ndarray or CSR matrix of shape (1, n_features) or (n_classes, n_features) Coefficients of the features in the decision function. `coef_` is of shape (1, n_features) when the given problem is binary. By default, it will be created as a dense array, but can be turned to sparse (CSR format) through :meth:`sparsify` (which can be beneficial under L1 regularization when many coefficients are zero), and back to dense through :meth:`densify`.	ndarray[float64](1, 24)	[[0.01, 0.43, 0.46,...,-0.21, 0.18,-0.28]]
intercept_ intercept_: ndarray of shape (1,) or (n_classes,) Intercept (a.k.a. bias) added to the decision function. If `fit_intercept` is set to False, the intercept is set to zero. `intercept_` is of shape (1,) when the given problem is binary.	ndarray[float64](1,)	[−2.68]
n_features_in_ n_features_in_: int Number of features seen during :term:`fit`. .. versionadded:: 0.24	int	24
n_iter_ n_iter_: ndarray of shape (1,) Actual number of iterations for all classes. .. versionchanged:: 0.20 In SciPy <= 1.0.0 the number of	ndarray[int32](1,)	[20]

```
lbfgs iterations may exceed
``max_iter``. ``n_iter_`` will now
report at most ``max_iter``.
```

The single call to `fit()` calculates medians and scaling parameters on `X_train`, encodes categories, and fits the logistic regression model on the transformed data.

11.8 - Evaluating a classifier

The model is now judged on the test students. Calling `predict()` produces class labels (0 or 1), and `predict_proba()` produces the estimated probabilities:

```
y_pred_class = logistic_pipe.predict(X_test)
y_pred_prob = logistic_pipe.predict_proba(X_test)[:, 1]

test_results = pd.DataFrame({
    "truth": y_test.values,
    "pred_class": y_pred_class,
    "pred_prob": y_pred_prob
})
print(test_results.head())
```

```
   truth  pred_class  pred_prob
0      0           0   0.031820
1      1           1   0.644133
2      0           0   0.177301
3      0           0   0.042008
4      0           0   0.030207
```

The simplest measure is **accuracy**: the share of students classified correctly:

```
acc = accuracy_score(y_test, y_pred_class)
always_no_acc = (y_test == 0).mean()

print(f"Model test accuracy:          {acc:.1%}")
print(f"Always predicting 'No' base: {always_no_acc:.1%}")
```

```
Model test accuracy:          88.0%
Always predicting 'No' base: 84.7%
```

An accuracy of 88.0% sounds good. But a "model" that ignores every predictor and says "No" for every student would be right 84.7% of the time, because most students answer "No", while finding not one of the students at risk. For an imbalanced outcome, accuracy is a misleading measure.

A better measure is the **ROC AUC** (the area under the receiver operating characteristic curve). It is the probability that, of one student who considered dropping out and one who did not, the model gives the first a higher predicted probability than the second. An AUC of 0.5 is no better than a coin toss, and 1.0 is perfect. Unlike accuracy, it is not fooled by an imbalanced outcome:

```
auc_test = roc_auc_score(y_test, y_pred_prob)
print(f"Test ROC AUC: {auc_test:.3f}")
```

```
Test ROC AUC: 0.843
```

An AUC of 0.84 means the model ranks an at-risk student above a not-at-risk student about 84% of the time: a useful model. Chapter 12 looks at classification measures in much more detail, including how to choose the threshold at which a student is flagged.

11.9 - Overfitting in practice

The simulation at the start of the chapter showed overfitting with a flexible curve. The same happens with real data. A dramatic example is **k-nearest neighbours** (k-NN) with $k = 1$. It predicts each student's outcome by

finding the single most similar student in the training data and copying their answer. On the training data, every student's nearest neighbour is themselves, so it can never be wrong:

```
knn1_pipe = Pipeline([
    ("preprocessor", preprocessor),
    ("classifier", KNeighborsClassifier(n_neighbors=1))
])
knn1_pipe.fit(X_train, y_train)

knn1_train_auc = roc_auc_score(y_train, knn1_pipe.predict_proba(X_train)[:, 1])
knn1_test_auc = roc_auc_score(y_test, knn1_pipe.predict_proba(X_test)[:, 1])

print(f"KNN (k=1) Training AUC: {knn1_train_auc:.2f}")
print(f"KNN (k=1) Test AUC:     {knn1_test_auc:.2f}")
```

```
KNN (k=1) Training AUC: 1.00
KNN (k=1) Test AUC:     0.61
```

The model scores perfectly on the training data, and much more poorly on new students. The training score was an illusion: the model had memorised the training students, not learned anything that carries over. It is the degree-12 curve again, and the reason every model in this book is tested on data it has not seen.

11.10 - Cross-validation

The test set is used only once, at the end. During model building, however, models and settings often need to be compared, and neither the training data (that would reward overfitting) nor the test set (that would use it up) can be used to judge them. **Cross-validation** solves this by reusing the training data. The training data is split into, say, 10 equal parts, called **folds**. The model is fitted on 9 folds and its performance measured on the fold left out; this is repeated 10 times, leaving out each fold in turn, and the 10 performance measures are averaged. Every student is used for evaluation exactly once, always by a model that did not see them during fitting. Figure 11.4 shows the idea with 5 folds.

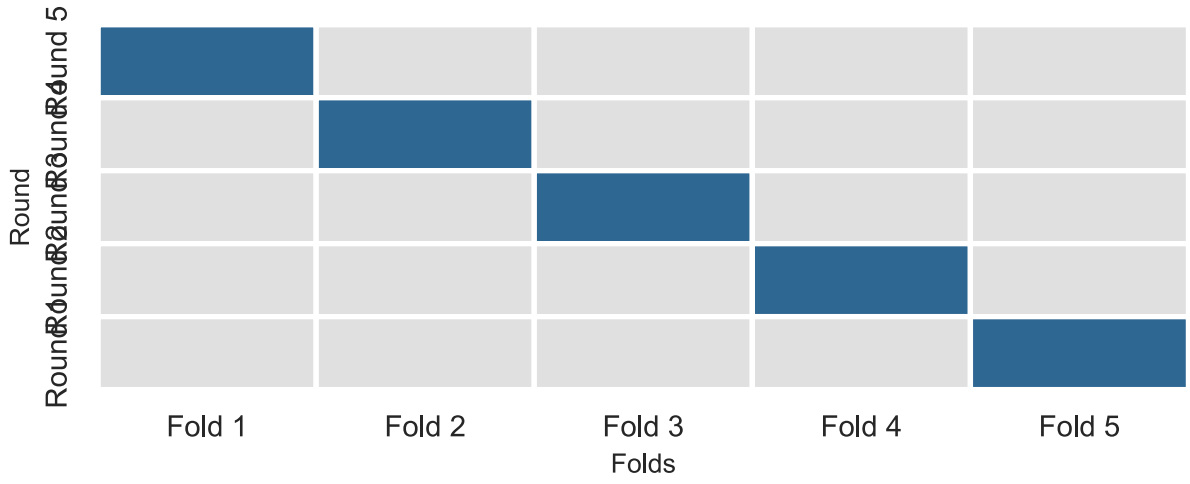


Figure 11.4: Five-fold cross-validation: in each round, the model is fitted on four folds (grey) and evaluated on the fifth (dark blue).

The class `StratifiedKFold()` creates stratified folds, and `cross_validate()` fits and evaluates the pipeline across each fold:

```
cv = StratifiedKFold(n_splits=10, shuffle=True, random_state=2026)
cv_results = cross_validate(logistic_pipe, X_train, y_train, cv=cv, scoring=["roc_auc",
"accuracy"])
```

```
cv_auc_mean = cv_results["test_roc_auc"].mean()
cv_auc_se = cv_results["test_roc_auc"].std() / np.sqrt(10)

print(f"Cross-validated AUC: {cv_auc_mean:.3f} (SE: {cv_auc_se:.3f})")
```

```
Cross-validated AUC: 0.806 (SE: 0.020)
```

The cross-validated AUC, about 0.81, is an honest estimate of how the model will perform on new students, obtained without touching the test set.

11.11 - Tuning a model

Many models have settings that are not learned from the data but must be chosen beforehand, such as k in k-nearest neighbours or the degree of the polynomials in the simulation. They are called **hyperparameters**, to distinguish them from the **parameters**, such as regression coefficients, that the model learns. The best value is found by **tuning**: trying several values and comparing them with cross-validation.

A **decision tree** (explained fully in Chapter 12) predicts by asking a series of yes-or-no questions, such as whether stress is above 3.5. Its depth, the number of questions in a row, is a hyperparameter that controls its flexibility. A shallow tree is too simple to capture the patterns and underfits; a very deep tree can fit the training data closely, noise included, and overfits:

```
tree_pipe = Pipeline([
    ("preprocessor", preprocessor),
    ("classifier", DecisionTreeClassifier(random_state=2026, min_samples_leaf=10))
])

param_grid = {"classifier__max_depth": list(range(1, 11))}
grid_search = GridSearchCV(tree_pipe, param_grid=param_grid, cv=cv, scoring="roc_auc")
grid_search.fit(X_train, y_train)

tuning_results = pd.DataFrame({
    "max_depth": range(1, 11),
    "mean_auc": grid_search.cv_results_["mean_test_score"]
})
print(tuning_results.head(6).round(3))
```

```
   max_depth  mean_auc
0          1     0.634
1          2     0.761
2          3     0.764
3          4     0.739
4          5     0.687
5          6     0.675
```

```
fig, ax = plt.subplots(figsize=(6, 3.6))
ax.plot(tuning_results["max_depth"],  tuning_results["mean_auc"],  marker="o",  color=navy,
linewidth=1.5)
ax.set_xticks(range(1, 11))
ax.set_xlabel("Tree depth (max_depth)")
ax.set_ylabel("Cross-validated ROC AUC")
ax.set_title("Hyperparameter Tuning: Decision Tree Depth")
plt.tight_layout()
plt.show()
```

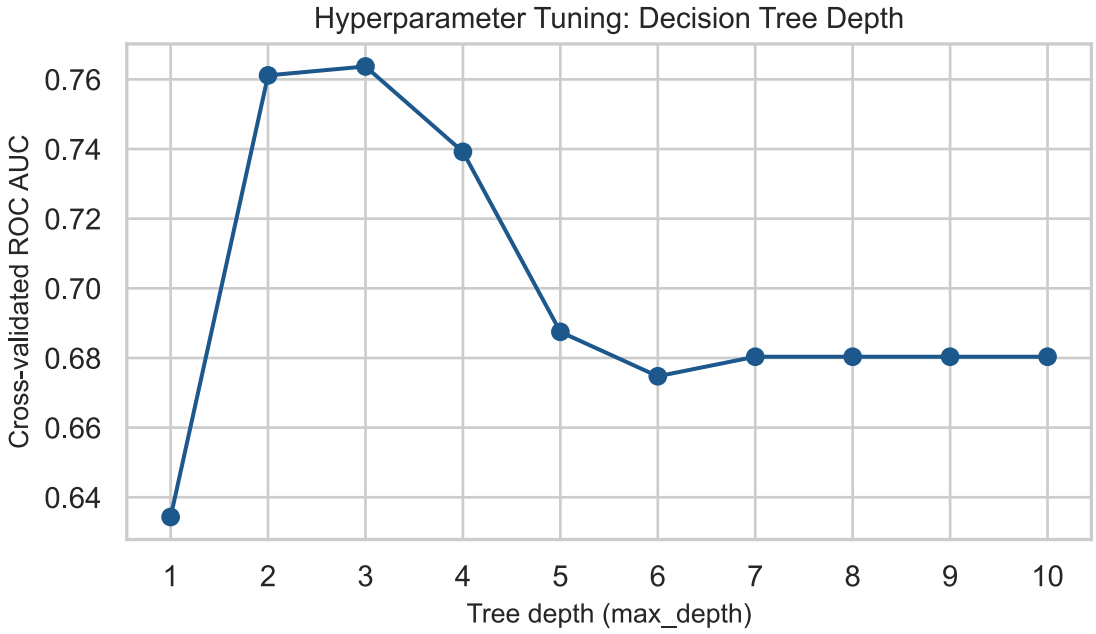


Figure 11.5: Cross-validated ROC AUC of decision trees of different depths.

```
best_depth = grid_search.best_params_["classifier__max_depth"]
print("Best max_depth:", best_depth)
print(f"Best CV ROC AUC: {grid_search.best_score_:.3f}")
```

```
Best max_depth: 3
Best CV ROC AUC: 0.764
```

The pattern is the new-data curve of Figure 11.2 seen from the other side: a tree of depth 1 is no better than chance, the AUC rises as the tree grows, and beyond a moderate depth it levels off and dips slightly as the deeper trees start to overfit. Here, `GridSearchCV` selects `max_depth` = 3.

11.12 - The final test

Only now, with the model chosen, is the test set used. The tuned decision tree is evaluated on the test set, alongside plain logistic regression:

```
best_tree = grid_search.best_estimator_
tree_test_auc = roc_auc_score(y_test, best_tree.predict_proba(X_test)[:, 1])
tree_test_acc = accuracy_score(y_test, best_tree.predict(X_test))

print(f"Tuned Tree Test AUC:        {tree_test_auc:.3f} (Accuracy: {tree_test_acc:.1%})")
print(f"Logistic Regression Test AUC: {auc_test:.3f} (Accuracy: {acc:.1%})")
```

```
Tuned Tree Test AUC:        0.725 (Accuracy: 84.7%)
Logistic Regression Test AUC: 0.843 (Accuracy: 88.0%)
```

The tuned tree reaches an AUC of 0.73; plain logistic regression, 0.84. The more flexible model is not the better one. In this data, the risk of dropout rises steadily with stress and falls steadily with support, so a simple smooth model captures the pattern well, while a tree, which splits the data into rectangular boxes, captures it less well. Trying a simple model first, and making a complex one earn its place, is one of the most useful habits in machine learning.

⚠ Predictions about people

A model that flags students "at risk" affects real people. Before such a model is used, the consequences of its errors must be considered: a student flagged by mistake may be treated differently, and a student who is missed receives no help. The model should be checked separately for different groups, such as women and men or part-time and full-time students, because a model with a good AUC overall can still be unfair to a particular group. And its purpose matters: in this study, the right use is to offer support earlier, never to penalise.

In your field: business and finance

Banks use exactly these methods to predict which loan applicants will repay. The `credit_data` dataset records real loan applications, with whether each loan turned out well or badly (`Status`). The same workflow applies:

```
credit_clean = credit_data.dropna(subset=["Status"]).copy()
X_cr = credit_clean.drop(columns=["Status"])
y_cr = (credit_clean["Status"] == "bad").astype(int)

X_train_c, X_test_c, y_train_c, y_test_c = train_test_split(
    X_cr, y_cr, test_size=0.25, stratify=y_cr, random_state=1
)

num_c = X_cr.select_dtypes(include=[np.number]).columns.tolist()
cat_c = X_cr.select_dtypes(exclude=[np.number]).columns.tolist()

credit_pre = ColumnTransformer([
            ("num",   Pipeline([("imp",   SimpleImputer(strategy="median")),   ("scale",
StandardScaler())]), num_c),
         ("cat",  Pipeline([("imp",  SimpleImputer(strategy="most_frequent")),  ("ohe",
OneHotEncoder(drop="first", sparse_output=False, handle_unknown="ignore"))]), cat_c)
])

credit_pipe = Pipeline([
    ("preprocessor", credit_pre),
    ("classifier", LogisticRegression(max_iter=1000, random_state=1))
])
credit_pipe.fit(X_train_c, y_train_c)

c_prob = credit_pipe.predict_proba(X_test_c)[:, 1]
c_pred = credit_pipe.predict(X_test_c)

print(f"Credit Test ROC AUC:    {roc_auc_score(y_test_c, c_prob):.3f}")
print(f"Credit Test Accuracy:   {accuracy_score(y_test_c, c_pred):.1%}")
print(f"Credit Brier Score:     {brier_score_loss(y_test_c, c_prob):.3f}")
```

```
Credit Test ROC AUC:    0.814
Credit Test Accuracy:   79.1%
Credit Brier Score:     0.149
```

Only the data and the outcome have changed; every step of the workflow is the same. The output also shows the **Brier score**, another measure of probability accuracy: 0 is perfect, and lower is better.

11.13 - Common misconceptions

Predictive modelling has its own characteristic mistakes, most of them versions of testing a model on what it already knows.

- **"A model that fits the training data well will predict well."** Training performance rewards overfitting; only performance on new data counts.

- **"A more complex model is a better model."** Beyond a point, flexibility adds variance, not accuracy. Simple models often predict as well.
- **"High accuracy means a good classifier."** For an imbalanced outcome, a model that predicts the common class every time can have high accuracy and no value.
- **"The test set can be used to choose between models."** Every use of the test set for a decision makes it less of a test. Choices are made with cross-validation; the test set is used once.
- **"A good predictor is a cause."** Prediction needs association, not causation; a predictive model says nothing about what would happen if a predictor were changed.

11.14 - Chapter review

11.14.1 - Summary

- Supervised learning predicts an outcome (classification for categories, regression for numbers); unsupervised learning finds structure. Prediction is a different research aim from explanation, judged by performance on new data.
- A model generalises when it learns the pattern and not the noise. Training error keeps falling as models become more flexible, while error on new data falls and then rises: the bias-variance trade-off.
- scikit-learn designs a fair test: it splits data (`train_test_split`), prepares it (`ColumnTransformer`), specifies models and pipelines (`Pipeline`), tunes them (`GridSearchCV`), and measures performance (`metrics`).
- Split the data into training and test sets, stratified by the outcome, and use the test set only once, at the end. Use only information available at the time of prediction.
- A preprocessor prepares data (imputation, dummy variables, normalisation) using the training data only, so no information leaks from the test set.
- Accuracy misleads for imbalanced outcomes; the ROC AUC is a better summary.
- Cross-validation estimates performance honestly without using the test set. Hyperparameters are chosen by tuning with cross-validation. Simple models often predict as well as complex ones; try them first.
- Predictions about people raise questions of fairness and use.

11.14.2 - Key terms

Machine learning, supervised learning, unsupervised learning, classification, regression, prediction, explanation, generalisation, overfitting, underfitting, bias-variance trade-off, imbalanced outcome, training set, test set, stratified split, ColumnTransformer, SimpleImputer, OneHotEncoder, StandardScaler, data leakage, Pipeline, accuracy, ROC AUC, k-nearest neighbours, cross-validation, fold, StratifiedKFold, hyperparameter, parameter, tuning, GridSearchCV, decision tree, Brier score.

11.15 - Exercises

The playground has these and more, with hints and solutions.

1. Remove the questionnaire scores (`stress`, `burnout`, `support`, `satisfaction`) from `dropout_data` and fit the logistic regression pipeline again. Report how much the cross-validated AUC falls, and what that says about the questionnaire.
2. Split the data 80/20 instead of 75/25 with a different random state (e.g. `random_state=7`), and describe how much the test AUC of the logistic model changes, and why it might.
3. Tune k-nearest neighbours over `n_neighbors = [5, 11, 21, 41]` with cross-validation. Identify the best value, and compare its AUC with logistic regression.
4. Add zero-variance filtering (`VarianceThreshold()`) to the preprocessing steps. Look up what "zero variance" means, and explain when this step is useful.
5. In two or three sentences, explain to a university administrator why a model with 85% accuracy might still be useless for finding students at risk.
6. Repeat the polynomial simulation with 50 training points instead of 15. Describe how the error curve on new data changes, and explain why more data allows a more flexible model.

11.16 - Further reading

- *An Introduction to Statistical Learning* [16] explains the ideas behind overfitting, the bias-variance trade-off, cross-validation, and machine learning methods.

- *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* by Aurélien Géron [21] is the definitive guide to scikit-learn pipelines and predictive modeling.

12. Classification Models

A classification model is built to support decisions: which patients to screen further, which applications to review, which students to contact. Two questions follow. The first is technical: whether more flexible methods than logistic regression would rank the cases better. The second matters more, and is often forgotten: what happens when the model's predictions are turned into decisions. Every decision rule flags some cases wrongly and misses others, and the balance between those two errors is not a statistical matter alone. It depends on what each error costs, and on whom.

This chapter addresses both questions. It introduces four widely used families of classification models, decision trees, random forests, k-nearest neighbours, and support vector machines, each through the idea behind it and the way it separates the classes, and compares them fairly on the wellbeing data. It then looks beyond the AUC at how a model is used: the confusion matrix, precision and recall, the choice of the threshold at which a student is flagged, and what to do when one outcome is rare. In the story, Elaf shows her logistic regression to the student counselling service, which asks whether more powerful methods exist, and how many of the students it would contact are really at risk.

By the end of this chapter you will be able to

- Explain how decision trees, random forests, k-nearest neighbours, and support vector machines classify.
- Fit each of them with scikit-learn, and compare several models fairly with cross-validation.
- Read a confusion matrix, and calculate sensitivity (recall), specificity, precision, and the F1 score.
- Choose a classification threshold from the costs of the two kinds of error and the purpose of the model.
- Explain the options for an imbalanced outcome, and what resampling does and does not change.

Note

Teaching & Practice: Interactive Lecture Slides · Browser Playground Exercises

12.1 - Data and resampling

The data, split, preprocessing pipeline, and cross-validation folds are exactly those of Chapter 11, so every model in this chapter is trained and tested on the same students:

```
import data2thesis as dt
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split, StratifiedKFold
from sklearn.compose import ColumnTransformer
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.pipeline import Pipeline

students = dt.load_students()
questionnaire = dt.load_questionnaire()
semesters = dt.load_semesters()

questionnaire['stress_4'] = 6 - questionnaire['stress_4']
scores = pd.DataFrame({
    'student_id': questionnaire['student_id'],
      'stress': questionnaire[['stress_1', 'stress_2', 'stress_3', 'stress_4', 'stress_5',
'stress_6']].mean(axis=1),
    'burnout': questionnaire[[f'burnout_{i}' for i in range(1, 7)]].mean(axis=1),
    'support': questionnaire[[f'support_{i}' for i in range(1, 7)]].mean(axis=1),
```

```
    'satisfaction': questionnaire[[f'satisfaction_{i}' for i in range(1, 5)]].mean(axis=1),
})

sem1 = semesters[semesters['semester'] == 1].drop(columns=['semester'])

dropout_data = (
    students.merge(scores, on='student_id')
    .merge(sem1, on='student_id')
    .drop(columns=['student_id', 'supervisor_id', 'workshop', 'workshop_sessions'])
)

dropout_data['considering_dropout']      =      (dropout_data['considering_dropout']      ==
'Yes').astype(int)

X = dropout_data.drop(columns=['considering_dropout'])
y = dropout_data['considering_dropout']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, stratify=y, random_state=2026
)

num_cols = X.select_dtypes(include=['int64', 'float64']).columns.tolist()
cat_cols = X.select_dtypes(include=['object', 'category']).columns.tolist()

dropout_recipe = ColumnTransformer([
             ('num',    Pipeline([('imp',    SimpleImputer(strategy='median')),    ('scaler',
StandardScaler())]), num_cols),
       ('cat',  OneHotEncoder(drop='first',  sparse_output=False,  handle_unknown='ignore'),
cat_cols)
])

tree_recipe = ColumnTransformer([
    ('num', SimpleImputer(strategy='median'), num_cols),
       ('cat',  OneHotEncoder(drop='first',  sparse_output=False,  handle_unknown='ignore'),
cat_cols)
])

dropout_folds = StratifiedKFold(n_splits=10, shuffle=True, random_state=2026)
```

12.2 - Decision trees

A **decision tree** classifies by asking a series of yes-or-no questions, like a flowchart. To see how a tree chooses its questions, take twelve imaginary students:

```
tiny = pd.DataFrame({
    'stress':  [1.5, 2.0, 2.2, 2.8, 3.0, 3.1, 3.4, 3.6, 3.8, 4.0, 4.2, 4.5],
    'support': [4.5, 3.0, 4.0, 2.0, 4.2, 3.8, 2.1, 4.4, 1.8, 2.5, 3.9, 1.5],
    'dropout': ['No', 'No', 'No', 'No', 'No', 'No', 'Yes', 'No', 'Yes', 'Yes', 'No', 'Yes']
})
```

Four of the twelve consider dropping out. The tree looks for the single question that best separates the “Yes” students from the “No” students. “Is support below 2.75?” puts seven students, all “No”, on one side, and five students, four “Yes” and one “No”, on the other. The one “No” student among the low-support group has the lowest stress, so a second question, “Is stress 3.1 or more?”, separates the rest perfectly:

```
from sklearn.tree import DecisionTreeClassifier, export_text

tiny_tree = DecisionTreeClassifier(min_samples_split=2, criterion='gini', random_state=2026)
tiny_tree.fit(tiny[['stress', 'support']], (tiny['dropout'] == 'Yes').astype(int))
print(export_text(tiny_tree, feature_names=['stress', 'support']))
```

```
|--- support <= 2.75
|   |--- stress <= 3.10
|   |   |--- class: 0
|   |--- stress >  3.10
|   |   |--- class: 1
|--- support >  2.75
|   |--- class: 0
```

Each line is a **node**: the question that led to it, and the predicted class (`class: 0` for No, `class: 1` for Yes). The final lines are the **leaves**. To classify a new student, you follow the questions from the top until you reach a leaf. To choose each question, the tree tries every predictor and every possible cut-off and picks the one that makes the two groups most "pure", usually measured by the **Gini impurity** (0 when a group contains only one class).

Decision trees have attractive properties: they are easy to explain, they need no dummy variables or normalisation (a question such as "is stress 3.1 or more?" works the same on any scale), and they capture interactions automatically (stress matters here only for low-support students). For the wellbeing data, a preprocessor that fills in missing values without scaling is enough:

```
tree_pipe = Pipeline([
    ('prep', tree_recipe),
    ('model', DecisionTreeClassifier(max_depth=3, min_samples_leaf=10, random_state=2026))
])

tree_fit = tree_pipe.fit(X_train, y_train)
```

A depth of 3 keeps the tree small enough to read. Figure 12.1 draws it using scikit-learn's `plot_tree()`.

```
from sklearn.tree import plot_tree

feature_names = [f.split('__')[-1] for f in
tree_pipe.named_steps['prep'].get_feature_names_out()]

fig, ax = plt.subplots(figsize=(9, 5.5))
plot_tree(
    tree_pipe.named_steps['model'],
    feature_names=feature_names,
    class_names=['No', 'Yes'],
    filled=True,
    rounded=True,
    fontsize=8,
    ax=ax
)
plt.show()
```

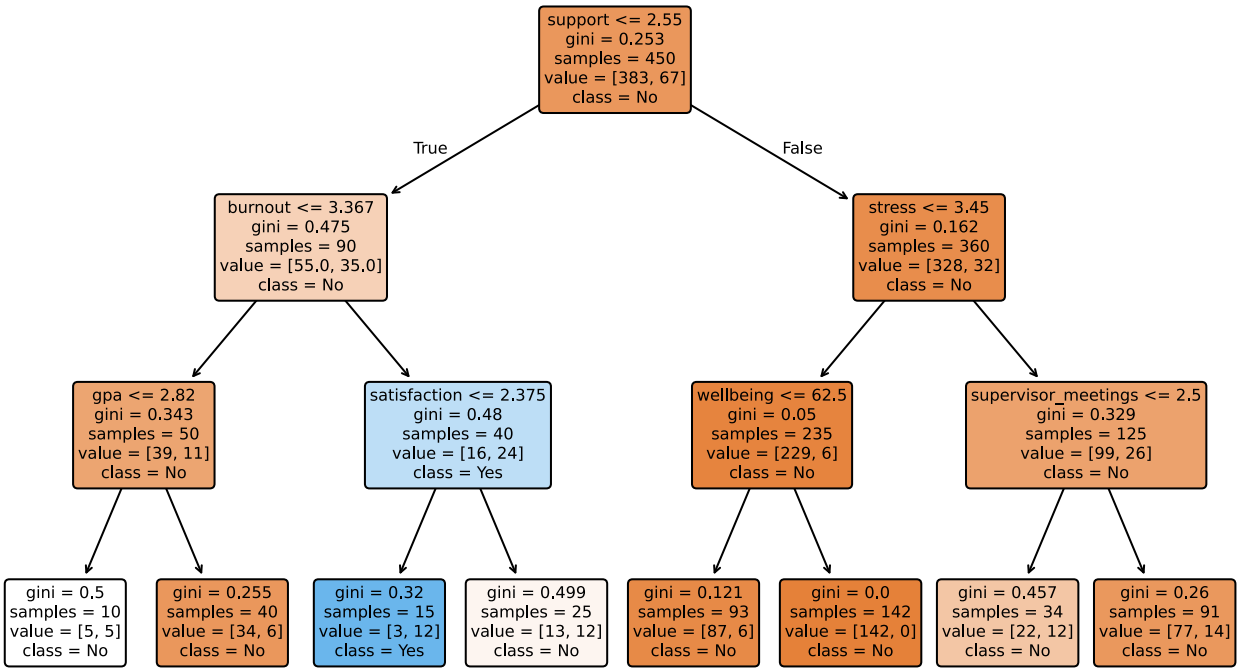


Figure 12.1: A decision tree of depth 3 for considering dropout. At each split, students who meet the condition go left.

The first question is about `support`, the strongest single predictor. The tree is easy to read, but a single tree has a serious weakness: it is unstable. A slightly different sample of students can produce a completely different tree, and, as Chapter 11 showed, a deep tree overfits. The next method turns this weakness into a strength.

12.3 - Random forests

A **random forest** grows hundreds of trees, each on a slightly different version of the data, and lets them vote. Two sources of randomness make the trees different. Each tree is grown on a **bootstrap sample** of the training data (Chapter 7), students drawn at random with replacement. And at each split, the tree may only choose from a random subset of the predictors; the size of this subset, `max_features` (analogous to `mtry` in R), is a hyperparameter.

Each individual tree overfits in its own way, but because the trees differ, many of their errors cancel out when the trees vote. The predicted probability of "Yes" is the share of trees that vote "Yes". Averaging many models in this way is called an **ensemble**; a random forest is an ensemble of trees [22].

Scikit-learn's `RandomForestClassifier` fits random forests efficiently:

```
from sklearn.ensemble import RandomForestClassifier

forest_pipe = Pipeline([
    ('prep', tree_recipe),
    ('model', RandomForestClassifier(n_estimators=500, random_state=2026))
])

forest_fit = forest_pipe.fit(X_train, y_train)
```

A forest of 500 trees cannot be drawn, so it is a "black box". **Permutation importance** opens it a little: for each predictor in turn, the values are shuffled randomly, and the drop in the model's accuracy is recorded. Shuffling an important predictor hurts the predictions; shuffling an unimportant one does not. Figure 12.2 shows the result.

```
from sklearn.inspection import permutation_importance

imp = permutation_importance(forest_fit, X_train, y_train, n_repeats=10, random_state=2026)
importance_df = pd.DataFrame({
    'predictor': X_train.columns,
    'importance': imp.importances_mean
}).sort_values('importance')

fig, ax = plt.subplots(figsize=(6.5, 4.5))
ax.barh(importance_df['predictor'], importance_df['importance'], color='#2f6793')
ax.set_xlabel('Permutation importance (drop in accuracy)')
ax.grid(axis='x', linestyle='--', alpha=0.5)
plt.show()
```

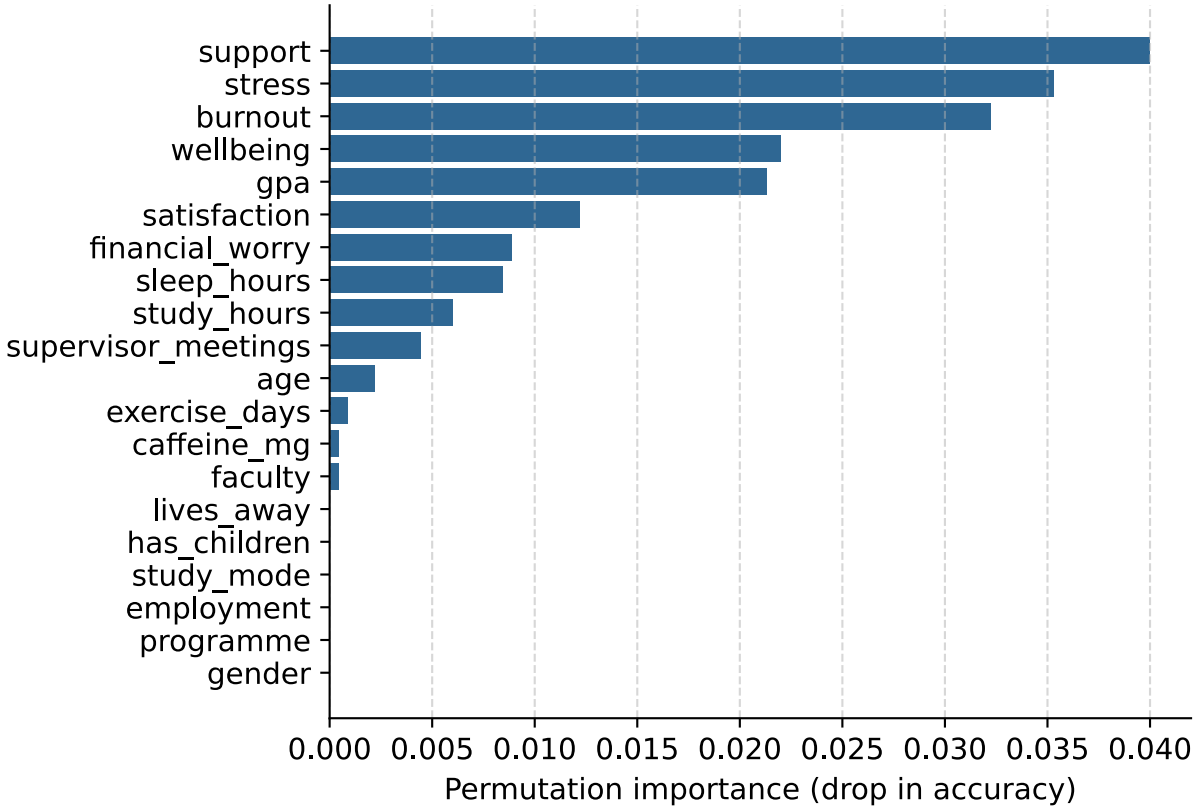


Figure 12.2: Permutation importance of each predictor in the random forest.

The forest relies most on support, stress, and burnout, and hardly at all on the students' background, which agrees with the logistic regression of Chapter 8. Importance says which predictors the model *uses*, not what *causes* dropout; everything Chapter 8 said about confounding still applies.

12.4 - k-nearest neighbours

k-nearest neighbours (k-NN), met briefly in Chapter 11, classifies a new student by finding the k most similar students in the training data and letting them vote. "Similar" means close together when the predictors are drawn as a map. Suppose a new student in the tiny example has a stress score of 3.5 and a support score of 3.0. The **distance** to each training student is the straight-line distance on the stress-support map:

```
tiny_dist = tiny.copy()
tiny_dist['distance'] = np.sqrt((tiny_dist['stress'] - 3.5)**2 + (tiny_dist['support'] -
```

```
3.0)**2)
tiny_dist.sort_values('distance').head(3)
```

	stress	support	dropout	distance
9	4.0	2.5	Yes	0.707107
5	3.1	3.8	No	0.894427
6	3.4	2.1	Yes	0.905539

With $k = 3$, two of the three nearest students considered dropping out, so the predicted probability of "Yes" is 2/3. Figure 12.3 shows the picture.

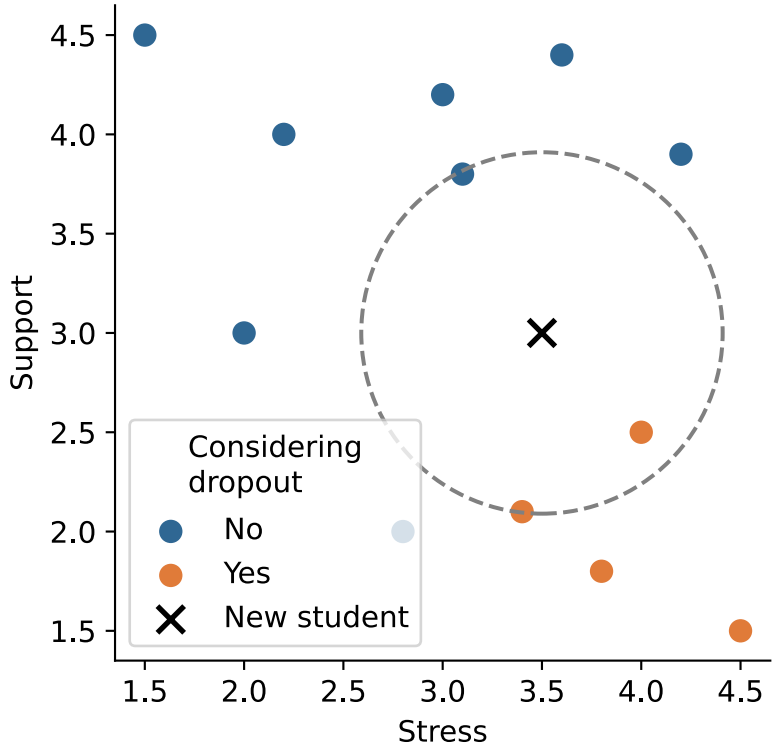


Figure 12.3: k-nearest neighbours with k = 3. The new student (cross) is classified by the three closest students (circled).

Because k-NN is based on distances, the predictors must be on the same scale. Otherwise a variable measured in large numbers, such as caffeine in milligrams, would dominate the distance, and a variable measured on a 1-5 scale would hardly count. That is why the recipe normalises every predictor.

The number of neighbours, k, is a hyperparameter. With $k = 1$, the model copies the nearest student and overfits badly (Chapter 11); with a large k, it averages over many students and becomes smoother. Tuning shows which works best for the wellbeing data:

```
from sklearn.model_selection import GridSearchCV
from sklearn.neighbors import KNeighborsClassifier

knn_pipe = Pipeline([
    ('prep', dropout_recipe),
    ('knn', KNeighborsClassifier())
])

param_grid = {'knn__n_neighbors': [1, 5, 11, 21, 41, 81]}
knn_grid = GridSearchCV(knn_pipe, param_grid=param_grid, cv=dropout_folds, scoring='roc_auc')
```

```
knn_grid.fit(X_train, y_train)

knn_results = pd.DataFrame({
    'neighbors': [1, 5, 11, 21, 41, 81],
    'mean': knn_grid.cv_results_['mean_test_score'],
    'std_err': knn_grid.cv_results_['std_test_score'] / np.sqrt(10)
})
knn_results
```

	neighbors	mean	std_err
0	1	0.583237	0.022934
1	5	0.704131	0.031217
2	11	0.734453	0.033532
3	21	0.754412	0.021761
4	41	0.796486	0.016813
5	81	0.824661	0.024688

The AUC keeps rising up to the largest k tried. When the best value lies at the edge of the grid, the usual advice is to extend the grid, but here the result is informative in itself: the more neighbours k-NN averages over, the smoother its predictions become, and the better it does. The wellbeing data has a smooth pattern (risk rises steadily with stress and falls steadily with support), which a smooth model such as logistic regression captures directly. k-NN is at its best when the pattern is irregular and there is a lot of data.

12.5 - Support vector machines

A **support vector machine** (SVM) separates the two classes with a boundary that is as far as possible from the students on either side. Imagine drawing a line between the "Yes" and "No" points on the stress-support map: of all the lines that separate them, the SVM picks the one with the widest empty "street" around it, the **margin**. Only the students closest to the boundary, the **support vectors**, determine where it goes. When the classes overlap, as they always do in real data, some students are allowed inside the margin or on the wrong side, at a cost set by a hyperparameter.

A straight boundary is a **linear SVM**. With a **kernel**, the SVM can draw curved boundaries: the popular **radial basis function** (RBF) kernel lets the boundary bend around groups of points. Both are available in scikit-learn via `SVC`:

```
from sklearn.svm import SVC

svm_linear = SVC(kernel='linear', probability=True, random_state=2026)
svm_rbf    = SVC(kernel='rbf', probability=True, random_state=2026)
```

Like k-NN, SVMs are based on distances, so they need normalised predictors. Figure 12.4 shows how differently the four families draw their boundaries, using just stress and support so the result can be drawn as a map.

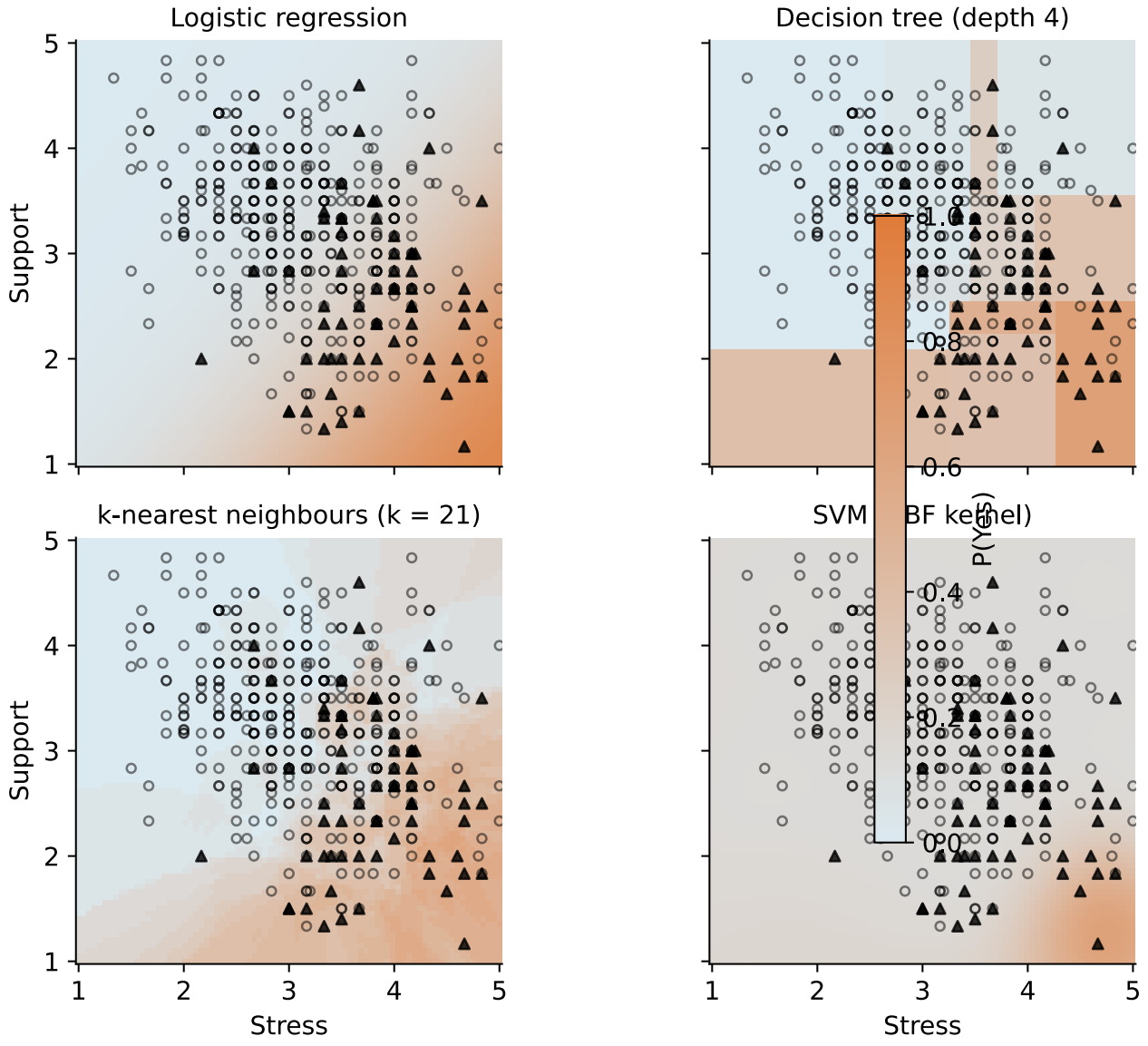


Figure 12.4: Predicted probability of considering dropout from stress and support, according to four models fitted to the training data. Darker orange means higher risk; the points are the training students.

12.6 - Comparing models fairly

The fair way to compare the models is cross-validation on the same folds. In Python, we define our pipelines and evaluate each over the identical 10 folds with `cross_validate`:

```
from sklearn.model_selection import cross_validate
from sklearn.linear_model import LogisticRegression

candidate_models = {
                'normalised_logistic':    Pipeline([('prep',    dropout_recipe),    ('model',
LogisticRegression(random_state=2026))]),
              'normalised_svm_linear':    Pipeline([('prep',    dropout_recipe),    ('model',
SVC(kernel='linear', probability=True, random_state=2026))]),
                  'tree_data_forest':     Pipeline([('prep',     tree_recipe),     ('model',
RandomForestClassifier(n_estimators=500, random_state=2026))]),
                 'normalised_knn':     Pipeline([('prep',     dropout_recipe),     ('model',
KNeighborsClassifier(n_neighbors=41))]),
                  'tree_data_tree':     Pipeline([('prep',     tree_recipe),     ('model',
```

```
DecisionTreeClassifier(max_depth=4, min_samples_leaf=10, random_state=2026))]),
      'normalised_svm_rbf': Pipeline([('prep', dropout_recipe), ('model', SVC(kernel='rbf',
probability=True, random_state=2026))]),
}

comparison_results = []
for name, pipe in candidate_models.items():
    cv_res = cross_validate(pipe, X_train, y_train, cv=dropout_folds, scoring='roc_auc')
    comparison_results.append({
        'wflow_id': name,
        'mean': cv_res['test_score'].mean(),
        'std_err': cv_res['test_score'].std(ddof=1) / np.sqrt(10)
    })

comparison_df                                                                    =
pd.DataFrame(comparison_results).sort_values('mean', ascending=False).reset_index(drop=True)
comparison_df
```

	wflow_id	mean	std_err
0	normalised_logistic	0.805710	0.021498
1	tree_data_forest	0.800863	0.020319
2	normalised_knn	0.796486	0.017722
3	normalised_svm_linear	0.791321	0.024877
4	tree_data_tree	0.739196	0.021488
5	normalised_svm_rbf	0.709154	0.029474

The trees and forest use the unscaled imputation recipe, and the distance-based models use the normalised one.

The winner, with a cross-validated AUC of `0.81`, is plain logistic regression. The linear SVM (`0.79`), which also draws a straight boundary, comes close, followed by the random forest (`0.80`). The standard errors, around `0.02`, show that the smaller differences at the top could be due to chance, but none of the flexible models beats the simple one. This is not a failure of the methods: it says something about the data. The risk of considering dropout rises smoothly with stress and falls smoothly with support, with no sharp thresholds or complicated interactions for a tree or a forest to find. On data with such structure, and with larger samples, forests and SVMs often do win. The lesson is the one from Chapter 11: compare, and let the simpler model stand unless a complex one clearly does better.

Logistic regression is therefore kept. It predicts as well as any model here, and it can be explained to the counselling service in one sentence.

12.7 - The confusion matrix

The AUC measures how well a model *ranks* students. But the counselling service needs a *decision*: contact this student or not. A model makes a decision by comparing each predicted probability with a **threshold**; by default, a student is classified "Yes" when the probability of "Yes" is above 0.5. The **confusion matrix** counts how those decisions turn out.

Take a tiny example first. A screening questionnaire is given to 100 students, 10 of whom are truly at risk. It flags 12 students, 8 of whom are among the 10 at risk:

Table 12.1: The four outcomes of a classification

	Truly at risk	Not at risk	Total
Flagged	8 (true positives)	4 (false positives)	12
Not flagged	2 (false negatives)	86 (true negatives)	88
Total	10	90	100

Every classification ends in one of four cells. A **true positive** is a student at risk who is flagged; a **false negative** is a student at risk who is missed; a **false positive** is a student flagged by mistake; a **true negative** is a student correctly left alone.

The key measures come from these four counts, and each answers a different practical question. **Sensitivity**, also called **recall**, is the share of students at risk who are flagged, here 8/10 = 80%: it measures how many of the students who need help are found. **Specificity** is the share of students not at risk who are left alone, 86/90 = 96%. **Precision** is the share of flagged students who are truly at risk, 8/12 = 67%: it measures how often a contact is justified. The **F1 score** balances precision and recall in a single number (their harmonic mean), here 0.73, and it is high only when both are high. **Accuracy**, the share classified correctly, is (8 + 86)/100 = 94%, but, as Chapter 11 showed, it is dominated by the large "No" group.

For the study's logistic regression, `confusion_matrix()` produces the confusion matrix for the test students:

```
from sklearn.metrics import confusion_matrix, accuracy_score, recall_score, precision_score,
f1_score

logistic_pipe = Pipeline([
    ('prep', dropout_recipe),
    ('model', LogisticRegression(random_state=2026))
])
logistic_fit = logistic_pipe.fit(X_train, y_train)

y_pred = logistic_fit.predict(X_test)
y_prob = logistic_fit.predict_proba(X_test)[:, 1]

# Confusion matrix with Yes (1) in row 0, No (0) in row 1
cm = confusion_matrix(y_test, y_pred, labels=[1, 0])
cm_df = pd.DataFrame(
    cm,
    index=['Truly Yes', 'Truly No'],
    columns=['Pred Yes', 'Pred No']
)
cm_df
```

	Pred Yes	Pred No
Truly Yes	8	15
Truly No	3	124

Scikit-learn calculates the summary metrics. Notice how they treat `1` ("Yes") as the event of interest:

```
tn, fp, fn, tp = confusion_matrix(y_test, y_pred, labels=[0, 1]).ravel()

class_metrics = pd.DataFrame({
    '.metric': ['accuracy', 'sensitivity', 'specificity', 'precision', 'f_meas'],
    '.estimate': [
        accuracy_score(y_test, y_pred),
        recall_score(y_test, y_pred),
        tn / (tn + fp),
        precision_score(y_test, y_pred),
        f1_score(y_test, y_pred)
    ]
})
class_metrics
```

	.metric	.estimate
0	accuracy	0.880000
1	sensitivity	0.347826
2	specificity	0.976378

	.metric	.estimate
3	precision	0.727273
4	f_meas	0.470588

This is sobering. Of the 23 test students who considered dropping out, the model flags only 8: a sensitivity of 35%. The high accuracy and high specificity come almost entirely from correctly leaving alone the large majority who were never at risk. The AUC showed the model ranks students well, so the problem is not the model but the threshold: with only 15% of students at risk, few students ever get a predicted probability above 0.5.

12.8 - Choosing the threshold

The threshold of 0.5 is a default, not a law. Lowering it flags more students: more of those at risk are found (higher recall), at the price of more false alarms (lower precision). Which balance is right depends on what the decision costs. Here, a flagged student is offered a conversation with a counsellor: cheap, and harmless if unnecessary. Missing a student who then leaves is costly. So a threshold well below 0.5 makes sense. If the decision were costly or stigmatising, a high threshold would be needed.

The threshold is a choice made while building the model, so it must be chosen without the test set. Cross-validation can provide predictions for every training student from a model that did not see them; `cross_val_predict()` provides out-of-fold probability estimates:

```
from sklearn.model_selection import cross_val_predict

cv_predictions = cross_val_predict(
    logistic_pipe, X_train, y_train, cv=dropout_folds, method='predict_proba'
)[:, 1]
```

The **ROC curve** shows every possible threshold at once. For each threshold, it plots the sensitivity against the false-positive rate (1 − specificity). A useless model follows the diagonal; a good one bends towards the top-left corner, and the AUC is the area under the curve:

```
from sklearn.metrics import roc_curve

fpr, tpr, thresholds = roc_curve(y_train, cv_predictions)

# Find closest points to threshold 0.5 and 0.2
idx_05 = np.argmin(np.abs(thresholds - 0.5))
idx_02 = np.argmin(np.abs(thresholds - 0.2))

fig, ax = plt.subplots(figsize=(5, 5))
ax.plot(fpr, tpr, color='#2f6793', linewidth=2)
ax.plot([0, 1], [0, 1], linestyle='--', color='grey')

ax.scatter([fpr[idx_05], fpr[idx_02]], [tpr[idx_05], tpr[idx_02]], color='#e07b39', s=50,
zorder=5)
ax.annotate('threshold 0.5', (fpr[idx_05], tpr[idx_05]), textcoords='offset points',
xytext=(10, -5))
ax.annotate('threshold 0.2', (fpr[idx_02], tpr[idx_02]), textcoords='offset points',
xytext=(10, -5))

ax.set_aspect('equal')
ax.set_xlabel('False positive rate (1 - specificity)')
ax.set_ylabel('Sensitivity (recall)')
plt.show()
```

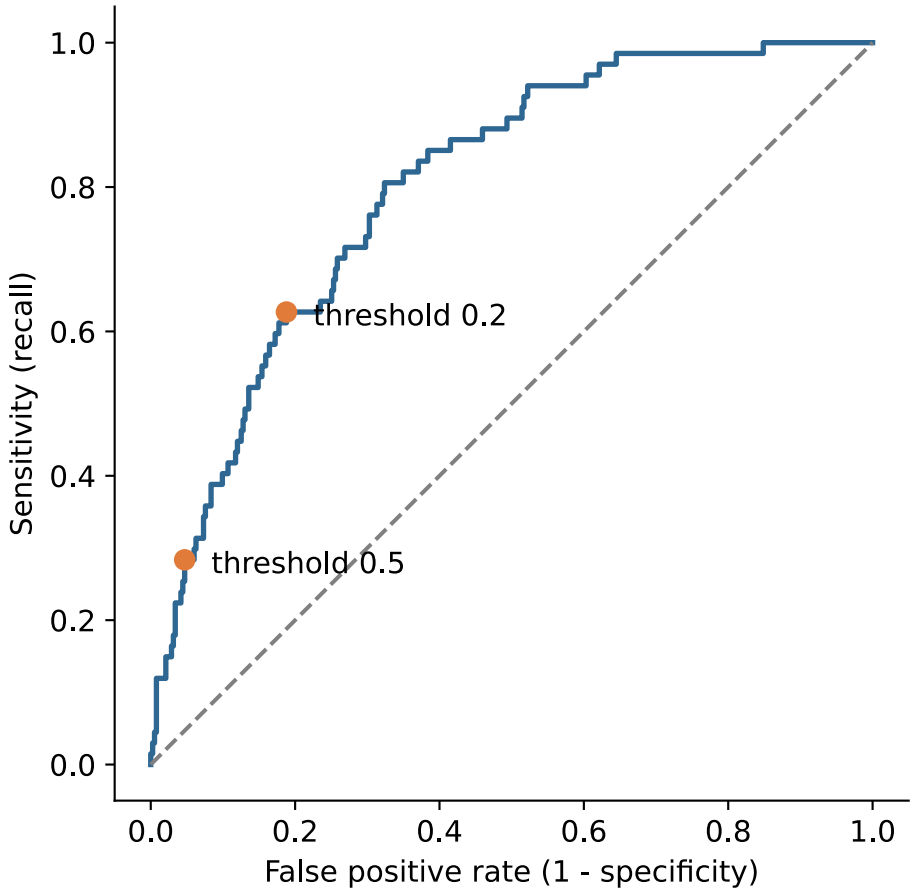


Figure 12.5: ROC curve of the logistic regression model, from cross-validated predictions on the training students. The points mark the thresholds 0.5 and 0.2.

A table makes the trade-off concrete. For several thresholds, it shows the share of students flagged, and the resulting recall, precision, and specificity:

```
threshold_rows = []
for t in [0.5, 0.4, 0.3, 0.2, 0.15, 0.1]:
    flags = (cv_predictions >= t).astype(int)
    tn_c, fp_c, fn_c, tp_c = confusion_matrix(y_train, flags, labels=[0, 1]).ravel()
    threshold_rows.append({
        'threshold': t,
        'flagged': flags.mean(),
        'recall': tp_c / (tp_c + fn_c),
        'precision': tp_c / (tp_c + fp_c) if (tp_c + fp_c) > 0 else 0,
        'specificity': tn_c / (tn_c + fp_c)
    })

threshold_table = pd.DataFrame(threshold_rows)
threshold_table.round(3)
```

	threshold	flagged	recall	precision	specificity
0	0.50	0.084	0.284	0.500	0.950
1	0.40	0.124	0.358	0.429	0.916
2	0.30	0.193	0.522	0.402	0.864
3	0.20	0.260	0.627	0.359	0.804

	threshold	flagged	recall	precision	specificity
4	0.15	0.322	0.687	0.317	0.742
5	0.10	0.420	0.821	0.291	0.650

12.8.1 - Costs of the two errors

Moving the threshold trades one error for the other, so the choice depends on how much each error costs. The costs are not statistical quantities; they are judgements about consequences, and they should be made explicitly. Suppose the counselling service judges that missing a student who is at risk is ten times as costly as an unnecessary conversation. The total cost of each threshold can then be calculated from the cross-validated predictions:

```
cost_miss        = 10   # a student at risk who is not contacted
cost_false_alarm = 1    # an unnecessary conversation

cost_rows = []
for t in np.arange(0.02, 0.62, 0.02):
    misses = np.sum((cv_predictions < t) & (y_train == 1))
    false_alarms = np.sum((cv_predictions >= t) & (y_train == 0))
    total_cost = cost_miss * misses + cost_false_alarm * false_alarms
    cost_rows.append({
        'threshold': round(t, 2),
        'misses': misses,
        'false_alarms': false_alarms,
        'flagged': np.mean(cv_predictions >= t),
        'total_cost': total_cost
    })

cost_curve = pd.DataFrame(cost_rows)
best_cost = cost_curve.loc[cost_curve['total_cost'].idxmin()]
best_cost
```

```
threshold         0.040000
misses            4.000000
false_alarms    209.000000
flagged           0.604444
total_cost      249.000000
Name: 1, dtype: float64
```

```
fig, ax = plt.subplots(figsize=(6, 3.6))
ax.plot(cost_curve['threshold'], cost_curve['total_cost'], color='#2f6793', linewidth=2)
ax.scatter(best_cost['threshold'], best_cost['total_cost'], color='#e07b39', s=60, zorder=5)
ax.set_xlabel('Threshold')
ax.set_ylabel('Total cost (arbitrary units)')
plt.show()
```

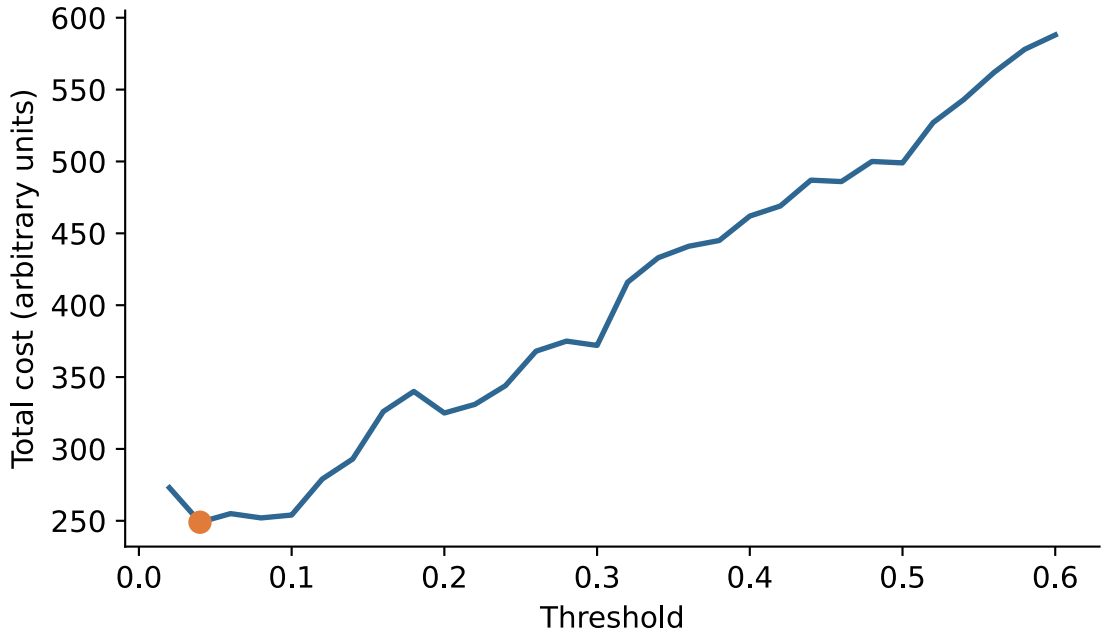


Figure 12.6: Total cost of the decisions at each threshold, when missing a student at risk costs ten times as much as an unnecessary conversation. The cost is lowest at a low threshold.

With these costs, the cheapest threshold is about `0.04` to `0.10`. A simple rule from decision theory points in the same direction: when the predicted probabilities are accurate, the cost-minimising threshold is the cost of a false alarm divided by the sum of the two costs, here $1/(1+10) \approx 0.09$. The two do not match exactly. The rule assumes perfectly accurate probabilities, and the cost curve is estimated from limited at-risk students, so its minimum is a broad basin. Both approaches agree on what matters for the decision: with these costs, the threshold belongs far below 0.5. Different costs give different thresholds. If a false alarm carried a real cost, such as a letter that stigmatised the student, the threshold would rise; if missing a student carried a greater one, it would fall further.

The costs, and therefore the threshold, are a research decision with an ethical side. They should be set with the people who will use the model, stated in the thesis, and, where the consequences are serious, examined for their effects on different groups of students.

Practical limits matter as well. The counselling service cannot talk to every student the cost analysis would flag, and after discussing the table with Elaf it settles on a threshold of 0.2: the model then flags about `26%` of students, finds about `63%` of those at risk, and about `36%` of the students it flags are truly at risk. Only now is the chosen threshold applied to the test students:

```
flags_test = (y_prob >= 0.2).astype(int)

cm_test_02 = pd.DataFrame(
    confusion_matrix(y_test, flags_test, labels=[1, 0]),
    index=['Truly Yes', 'Truly No'],
    columns=['Pred Yes', 'Pred No']
)
cm_test_02
```

	Pred Yes	Pred No
Truly Yes	17	6
Truly No	29	98

```
tn_t, fp_t, fn_t, tp_t = confusion_matrix(y_test, flags_test, labels=[0, 1]).ravel()

test_metrics_02 = pd.DataFrame({
```

```
    '.metric': ['accuracy', 'sensitivity', 'specificity', 'precision', 'f_meas'],
    '.estimate': [
        accuracy_score(y_test, flags_test),
        recall_score(y_test, flags_test),
        tn_t / (tn_t + fp_t),
        precision_score(y_test, flags_test),
        f1_score(y_test, flags_test)
    ]
})
test_metrics_02.round(3)
```

	.metric	.estimate
0	accuracy	0.767
1	sensitivity	0.739
2	specificity	0.772
3	precision	0.370
4	f_meas	0.493

On new students, the model now finds 17 of the 23 at risk instead of 8, while flagging 46 students in all. About one flagged student in three is truly at risk. The test numbers are in close agreement with the cross-validated estimates. Accuracy has fallen, and it does not matter: accuracy was never the goal.

> ! Report the threshold
>
> Sensitivity, specificity, precision, and F1 all depend on the threshold, so a paper that reports them must say which threshold was used, and why. The AUC does not depend on the threshold, which is why it is the standard measure for comparing models, and the threshold-dependent measures are the ones for describing how a model will be used.

12.9 - Imbalanced outcomes

When one outcome is rare, as considering dropout is, models tend to predict the common one. Besides moving the threshold, there are two other common remedies. **Resampling** changes the training data so that the classes are balanced: **upsampling** repeats rare-class students, **downsampling** drops common-class students, and **SMOTE** creates new, artificial rare-class students between existing ones. **Class weights** instead make mistakes on the rare class count more when the model is fitted, and scikit-learn models support them directly via `class_weight='balanced'`:

```
balanced_pipe = Pipeline([
    ('prep', dropout_recipe),
    ('model', LogisticRegression(class_weight='balanced', random_state=2026))
])

balanced_cv = cross_validate(
    balanced_pipe, X_train, y_train, cv=dropout_folds,
    scoring=['roc_auc', 'recall', 'precision']
)

pd.DataFrame({
    '.metric': ['roc_auc', 'sensitivity', 'precision'],
    'mean': [
        balanced_cv['test_roc_auc'].mean(),
        balanced_cv['test_recall'].mean(),
        balanced_cv['test_precision'].mean()
    ]
}).round(3)
```

	.metric	mean
0	roc_auc	0.805
1	sensitivity	0.710
2	precision	0.324

With the usual 0.5 threshold, the balanced model has a sensitivity of about 71%, far higher than the unweighted default (28%). But its AUC, 0.81, has not improved. Weighting the classes did not make the model better at telling students apart; it pushed all the predicted probabilities upwards, which has much the same effect as lowering the threshold. That is often how it works out, and for a model that produces probabilities, choosing the threshold directly is simpler and keeps the probabilities meaningful (after reweighting or upsampling, an output score of 0.6 no longer corresponds to a 60% empirical rate). Class balancing is most useful for models that do not produce calibrated probabilities, and when the rare class is very rare.

⚠ Does the model work equally well for everyone?

Among the test students, the model with a threshold of 0.2 finds similar proportions of at-risk women and at-risk men; but with only 23 at-risk students in the test set, subtle subgroup differences can be hard to detect. Before a model like this is deployed, its sensitivity and precision should be checked for each demographic and academic group that matters (gender, faculty, full- and part-time study) on as much data as possible, as Chapter 11 recommended. A threshold that works well on average can still miss one group of students much more often than another.

In your field: ecology

Ecologists classify species from measurements. The penguins dataset records the bill length, bill depth, flipper length, and body mass of 344 penguins of three species, measured on islands near Palmer Station in Antarctica [23]. A random forest classifies the species:

```
penguins = dt.load_penguins().dropna(subset=['bill_length_mm', 'bill_depth_mm',
'flipper_length_mm', 'body_mass_g'])
p_X = penguins[['bill_length_mm', 'bill_depth_mm', 'flipper_length_mm', 'body_mass_g']]
p_y = penguins['species']

p_X_train, p_X_test, p_y_train, p_y_test = train_test_split(
    p_X, p_y, test_size=0.25, stratify=p_y, random_state=1
)

penguin_rf = RandomForestClassifier(n_estimators=500, random_state=1)
penguin_rf.fit(p_X_train, p_y_train)

p_pred = penguin_rf.predict(p_X_test)
pd.crosstab(p_y_test, p_pred, rownames=['Actual'], colnames=['Predicted'])
```

Predicted Actual	Adelie	Chinstrap	Gentoo
Adelie	37	1	0
Chinstrap	1	16	0
Gentoo	0	0	31

With three classes, the confusion matrix has a row and a column for each species. Correct classifications lie on the diagonal, and any mistake appears off it, showing exactly which species are confused; here, almost every penguin in the test set is classified correctly. Unlike the wellbeing data, the classes here are well separated by the morphological measurements, and flexible models work very well.

12.10 - Chapter review

12.10.1 - Summary

- A decision tree classifies by a series of yes-or-no questions. Trees are easy to explain and need no normalisation, but a single tree is unstable and overfits easily.
- A random forest averages hundreds of trees grown on bootstrap samples with random subsets of predictors. Permutation importance shows which predictors it relies on.
- k-nearest neighbours classifies by the vote of the k most similar training cases; it needs normalised predictors, and k must be tuned.
- A support vector machine separates the classes with the widest possible margin; kernels allow curved boundaries.
- Compare models with cross-validation on the same folds. On the wellbeing data, with its smooth pattern, logistic regression predicts as well as any flexible model.
- The confusion matrix counts true and false positives and negatives. Sensitivity (recall) is the share of positives found; precision is the share of flagged cases that are positive; F1 balances the two.
- The threshold turns probabilities into decisions. Choose it from the costs of the two errors and the practical limits of its use, using cross-validated predictions, and report it.
- For imbalanced outcomes, moving the threshold, resampling, and class weights all shift the balance between recall and precision; resampling rarely improves the AUC.

12.10.2 - Key terms

Decision tree, node, leaf, Gini impurity, random forest, bootstrap sample, ensemble, `max_features` (`mtry`), permutation importance, k-nearest neighbours, distance, support vector machine, margin, support vector, kernel, radial basis function, cross-validation comparison, threshold, confusion matrix, true positive, false positive, true negative, false negative, sensitivity, recall, specificity, precision, F1 score, ROC curve, misclassification cost, upsampling, downsampling, SMOTE, class weights.

12.11 - Exercises

The playground has these and more, with hints and solutions.

1. Draw the decision tree for the wellbeing data with `max_depth = 2`. Name the predictors it uses, and explain the tree in plain words, as you would to a counsellor.
2. Tune the random forest's `max_features` over `[2, 5, 10]` with cross-validation, and compare the best value with the default.
3. Extend the k-NN grid to `n_neighbors = [81, 161, 301]`, describe what happens to the AUC, and explain what happens to a k-NN model as k approaches the number of training students.
4. Using the tiny example's confusion matrix (Table 12.1), calculate the F1 score yourself from precision and recall ($F_1 = 2 \times \text{precision} \times \text{recall} \,/ (\text{precision} + \text{recall})$).
5. Suppose the counselling service can only contact 10% of new students. Using `threshold_table`, recommend a threshold and report the recall and precision it would give.
6. Fit a logistic regression model with downsampling or compare `class_weight='balanced'` with the unweighted baseline, and compare the sensitivity and AUC.
7. Repeat the cost analysis with a cost of 3 for a missed student instead of 10. Report the new cheapest threshold, and compare it with the decision-theory rule.

12.12 - Further reading

- *An Introduction to Statistical Learning* [16] explains trees, random forests, and support vector machines, with clear illustrations of how each draws its boundaries.
- *Python Data Science Handbook* [24] provides clear tutorials on scikit-learn classifiers and hyperparameter tuning.
- Breiman's original paper on random forests [22] is still readable and explains why averaging many trees works.

13. Predictive Regression

Research data often contains many possible predictors for rather few cases. A questionnaire with dozens of items, several rounds of measurement, and a background survey can easily supply 50 predictors for a sample of 60 or 100 people. Ordinary regression estimates a coefficient for every predictor, and with that much freedom it can fit its own data almost perfectly while predicting new cases badly: the overfitting of Chapter 11, now in its most common form. The problem is made worse when predictors are correlated, as questionnaire items usually are.

This chapter introduces two families of methods that make many predictors usable. **Regularised regression** (ridge, lasso, and elastic net) keeps the familiar linear model but shrinks its coefficients towards zero, trading a little fit on the training data for much better predictions. **Boosting** builds a flexible model from many small decision trees, each correcting the errors of the ones before. The chapter also introduces the measures used to judge numeric predictions. The outcome is now a number, so this is a **regression** problem in the machine learning sense of Chapter 11. In the study, it answers the tenth research question (RQ10): the university's academic advisers meet every student at the end of the first year, and would like to know from year-one information which students are heading for a low final GPA, so that extra support can be planned.

By the end of this chapter you will be able to

- Measure the accuracy of numeric predictions with RMSE, MAE, and R^2, and compare them with a baseline.
- Explain why ordinary regression overfits when there are many predictors.
- Fit and tune ridge, lasso, and elastic net regression, and read which predictors the lasso keeps.
- Explain how boosting builds a model from many small trees, and tune a gradient boosting model.
- Choose between models with cross-validation, and report the final model's accuracy on the test set.

Note

Teaching & Practice: Interactive Lecture Slides · Browser Playground Exercises

13.1 - Data for predicting final GPA

The advisers meet students at the end of year one, so the model may use only what is known by then: the students' background, their questionnaire answers, and their records from semesters 1 and 2. The outcome is GPA in semester 4.

This time, every questionnaire item is used as a separate predictor, rather than the four scale scores, to show how the methods cope with many related predictors. The semester records are in long format (one row per student per semester); for prediction, they are needed in wide format, one row per student, with a column for each measure in each semester. `pivot()` in pandas does this:

```
import data2thesis as dt
import pandas as pd
import numpy as np

students = dt.load_students()
students["employment"] = students["employment"].fillna("None")
semesters = dt.load_semesters()
questionnaire = dt.load_questionnaire()

s12 = semesters[semesters['semester'] <= 2]
year_one = s12.pivot(
    index='student_id',
    columns='semester',
          values=['gpa',   'sleep_hours',   'study_hours',   'exercise_days',   'caffeine_mg',
```

```
'supervisor_meetings', 'wellbeing']
)
year_one.columns = [f"{col[0]}_s{col[1]}" for col in year_one.columns]
year_one = year_one.reset_index()

final_gpa = (
    semesters[(semesters['semester'] == 4) & semesters['gpa'].notna()][['student_id', 'gpa']]
    .rename(columns={'gpa': 'final_gpa'})
)

gpa_data = (
    students
    .merge(questionnaire, on='student_id')
    .merge(year_one, on='student_id')
    .merge(final_gpa, on='student_id')
    .drop(columns=['student_id', 'supervisor_id', 'considering_dropout'])
)

print(gpa_data.shape)
```

```
(563, 48)
```

The pivot builds the new column names, such as `gpa_s1` and `sleep_hours_s2`, and the inner merge keeps only students who have a final GPA. That is an important limitation: the 37 students who left the programme have no final GPA, so the model can only predict the final GPA of students who stay. (Chapter 6 showed that the leavers were not a random group, so the model should not be used to judge them.)

The split, folds, and pipeline follow Chapter 11. With a numeric outcome, we stratify by quartiles of `final_gpa` so that both sets cover the full range of GPAs:

```
from sklearn.model_selection import train_test_split, KFold
from sklearn.compose import ColumnTransformer
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.pipeline import Pipeline

X = gpa_data.drop(columns=['final_gpa'])
y = gpa_data['final_gpa']

y_bins = pd.qcut(y, q=4, labels=False)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, stratify=y_bins, random_state=2026
)

num_cols = X.select_dtypes(include=['int64', 'float64']).columns.tolist()
cat_cols = X.select_dtypes(include=['object', 'category']).columns.tolist()

gpa_recipe = ColumnTransformer([
             ('num',    Pipeline([('imp',    SimpleImputer(strategy='median')),    ('scaler',
StandardScaler())]), num_cols),
           ('cat',   Pipeline([('imp',   SimpleImputer(strategy='most_frequent')),   ('ohe',
OneHotEncoder(drop='first', sparse_output=False, handle_unknown='ignore'))]), cat_cols)
])

gpa_folds = KFold(n_splits=10, shuffle=True, random_state=2026)

feature_count = gpa_recipe.fit(X_train).transform(X_train).shape[1]
print(f"Number of prepared predictors: {feature_count}")
```

```
Number of prepared predictors: 51
```

After dummy variables are created, there are 48 predictors, plus the outcome.

13.2 - Measuring prediction error

The quality of a numeric prediction is judged by its errors. Five students whose final GPA a model has predicted show how:

```
tiny = pd.DataFrame({
    'actual':    [3.2, 2.8, 3.6, 3.0, 2.5],
    'predicted': [3.0, 2.9, 3.3, 3.1, 2.9]
})
tiny['error'] = tiny['actual'] - tiny['predicted']
tiny
```

	actual	predicted	error
0	3.2	3.0	0.2
1	2.8	2.9	−0.1
2	3.6	3.3	0.3
3	3.0	3.1	−0.1
4	2.5	2.9	−0.4

Each **error** (or residual) is the actual value minus the prediction, and three measures summarise them. The **MAE** (mean absolute error) is the average size of the errors, ignoring their sign; here, (0.2 + 0.1 + 0.3 + 0.1 + 0.4) / 5 = 0.22 GPA points. The **RMSE** (root mean squared error) squares the errors, averages them, and takes the square root. Squaring gives large errors more weight, so the RMSE is always at least as large as the MAE, and much larger when there are a few big misses. **R^2** is the coefficient of determination: the share of the variation in the outcome that the predictions capture, from 0 to 1. Scikit-learn calculates all three:

```
from sklearn.metrics import mean_absolute_error, root_mean_squared_error, r2_score

pd.DataFrame({
    '.metric': ['rmse', 'mae', 'rsq'],
    '.estimate': [
        root_mean_squared_error(tiny['actual'], tiny['predicted']),
        mean_absolute_error(tiny['actual'], tiny['predicted']),
        r2_score(tiny['actual'], tiny['predicted'])
    ]
})
```

	.metric	.estimate
0	rmse	0.248998
1	mae	0.220000
2	rsq	0.549419

MAE and RMSE are in the units of the outcome, here GPA points, which makes them easy to explain: “the model’s predictions are typically off by about 0.2 GPA points”. Lower is better. R^2 has no units; higher is better.

A number on its own means little, so every model should be compared with a **baseline**: the simplest possible prediction. For a numeric outcome, the baseline predicts the same value, the training mean, for everyone, using `DummyRegressor`:

```
from sklearn.dummy import DummyRegressor
from sklearn.model_selection import cross_validate

null_model = DummyRegressor(strategy='mean')
null_cv = cross_validate(
    null_model, X_train, y_train, cv=gpa_folds,
```

```
    scoring=['neg_root_mean_squared_error', 'neg_mean_absolute_error']
)

null_rmse = -null_cv['test_neg_root_mean_squared_error'].mean()
null_mae = -null_cv['test_neg_mean_absolute_error'].mean()

pd.DataFrame({
    '.metric': ['rmse', 'mae'],
    'mean': [null_rmse, null_mae]
})
```

	.metric	mean
0	rmse	0.313298
1	mae	0.250273

Predicting the average GPA for everyone gives an RMSE of `0.31`. Any useful model must do clearly better.

13.3 - Ordinary regression with many predictors

The obvious model is multiple linear regression, using every predictor:

```
from sklearn.linear_model import LinearRegression

lm_pipe = Pipeline([
    ('prep', gpa_recipe),
    ('model', LinearRegression())
])

lm_cv = cross_validate(
    lm_pipe, X_train, y_train, cv=gpa_folds,
    scoring=['neg_root_mean_squared_error', 'neg_mean_absolute_error', 'r2']
)

pd.DataFrame({
    '.metric': ['rmse', 'mae', 'rsq'],
    'mean': [
        -lm_cv['test_neg_root_mean_squared_error'].mean(),
        -lm_cv['test_neg_mean_absolute_error'].mean(),
        lm_cv['test_r2'].mean()
    ]
})
```

	.metric	mean
0	rmse	0.215400
1	mae	0.173449
2	rsq	0.501566

A cross-validated RMSE of `0.215`, much better than the baseline. With `422` training students and `48` predictors, ordinary regression works reasonably well. But watch what happens with fewer students. Many theses have samples of 50 or 60, so here is the same model fitted to 60 randomly chosen training students, and tested on the test set:

```
np.random.seed(3)
idx_60 = np.random.choice(len(X_train), size=60, replace=False)
X_small, y_small = X_train.iloc[idx_60], y_train.iloc[idx_60]

lm_small = Pipeline([
    ('prep', gpa_recipe),
```

```
    ('model', LinearRegression())
])
lm_small.fit(X_small, y_small)

train_rmse_60 = root_mean_squared_error(y_small, lm_small.predict(X_small))
test_rmse_60 = root_mean_squared_error(y_test, lm_small.predict(X_test))

print(f"Training RMSE (60 students): {train_rmse_60:.3f}")
print(f"Test Set RMSE (60 students): {test_rmse_60:.3f}")
```

```
Training RMSE (60 students): 0.068
Test Set RMSE (60 students): 0.664
```

On its own 60 students, the model looks excellent: an RMSE of `0.068`. On new students, its RMSE jumps to `0.664`, *worse than predicting the average GPA for everyone*. With `48` coefficients estimated from 60 students, the model has enough freedom to fit the noise in its sample, and the coefficients that fit the noise do not carry over. In the extreme, with as many predictors as students, a regression can fit its data perfectly and predict nothing.

The problem is made worse by **multicollinearity**: many of the predictors are strongly correlated (six stress items, two semesters of GPA), and regression struggles to divide the credit between correlated predictors. Their coefficients become large, unstable, and sometimes of opposite signs, cancelling each other out on the training data but not on new data.

13.4 - Regularised regression

Regularisation tackles this directly: it fits the regression while penalising large coefficients. Ordinary regression chooses the coefficients that minimise the sum of squared errors. Regularised regression minimises:

$$\text{sum of squared errors} + \lambda \times \text{size of the coefficients} \ . \tag{13.1}$$

The **penalty** λ (alpha in scikit-learn) sets how much large coefficients cost. With $\lambda = 0$, this is ordinary regression; as λ grows, the coefficients are pulled, or **shrunk**, towards zero. A little shrinkage costs almost nothing in fit to the training data, but makes the model much more stable on new data. Because the penalty depends on the size of the coefficients, the predictors must be on the same scale, which the pipeline's `StandardScaler` ensures.

There are three versions, which differ in how "size" is measured:

- **Ridge regression** (`Ridge`): uses the sum of the *squared* coefficients (L_2 penalty). It shrinks all coefficients towards zero and shares credit among correlated predictors, keeping every predictor in the model.
- **Lasso** (`Lasso`): uses the sum of the *absolute* coefficients (L_1 penalty) [25]. It shrinks too, but it also sets some coefficients to exactly zero, removing those predictors and performing variable selection.
- **Elastic Net** (`ElasticNet`): combines both penalties with an `l1_ratio` parameter running from 0 (ridge) to 1 (lasso).

The 60 students are now refitted with a lasso:

```
from sklearn.linear_model import Lasso

lasso_small = Pipeline([
    ('prep', gpa_recipe),
    ('model', Lasso(alpha=0.03, random_state=2026))
])
lasso_small.fit(X_small, y_small)

lasso_small_rmse = root_mean_squared_error(y_test, lasso_small.predict(X_test))
print(f"Lasso Test Set RMSE (60 students): {lasso_small_rmse:.3f}")
```

```
Lasso Test Set RMSE (60 students): 0.221
```

From the same 60 students, the lasso's RMSE on new students is `0.221`, against `0.664` for ordinary regression. The penalty kept the model from chasing noise.

The difference lies in the coefficients. Figure 13.1 compares, predictor by predictor, the coefficients that ordinary regression and the lasso estimated from the same 60 students:

```
lm_coefs = lm_small.named_steps['model'].coef_
lasso_coefs = lasso_small.named_steps['model'].coef_

fig, ax = plt.subplots(figsize=(6, 4))
ax.axhline(0, color='grey', linestyle='--', alpha=0.6)
ax.axvline(0, color='grey', linestyle='--', alpha=0.6)
ax.scatter(lm_coefs, lasso_coefs, color='#2f6793', alpha=0.7, s=30)
ax.set_xlabel('Ordinary regression coefficient')
ax.set_ylabel('Lasso coefficient')
plt.show()
```

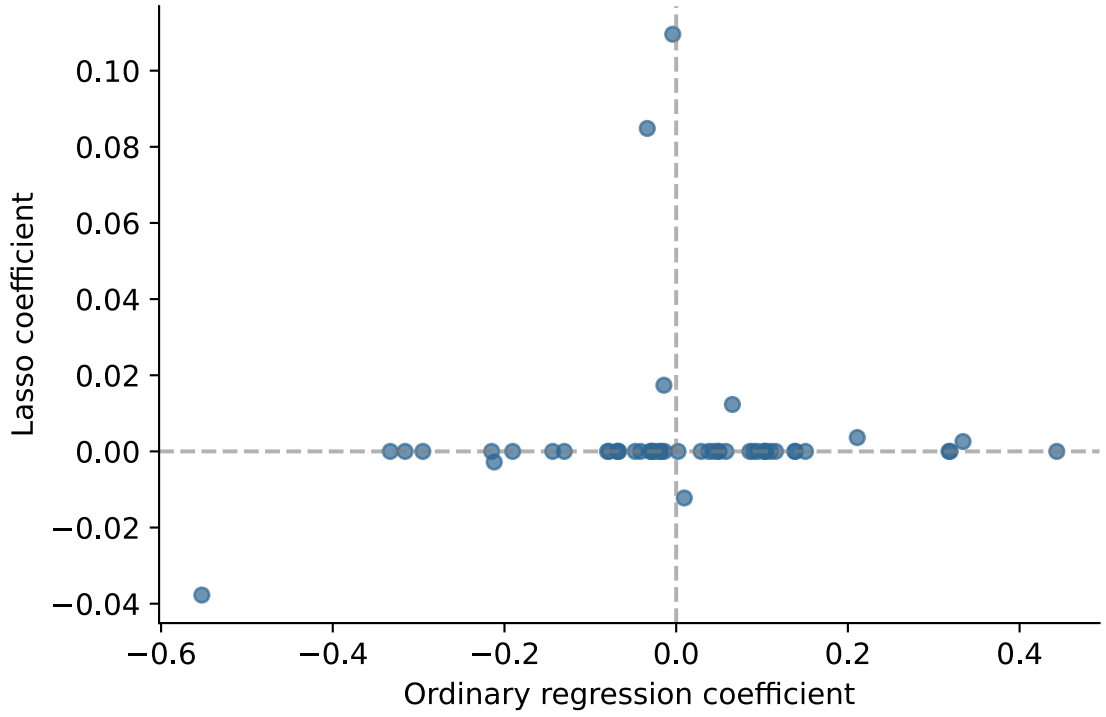


Figure 13.1: Coefficients estimated from the same 60 students by ordinary regression (horizontal axis) and by the lasso (vertical axis), one point per predictor. Ordinary regression spreads large coefficients over many predictors; the lasso sets most of them to zero and keeps a few, smaller ones.

Ordinary regression gives every predictor a coefficient, many of them large and in both directions: with 60 students, it cannot tell real effects from the chance patterns of this sample, so it fits both. The lasso sets most predictors to zero and shrinks the rest.

13.4.1 - Choosing the penalty

The penalty λ and mixture are hyperparameters, so they are tuned with cross-validation on the full training set. We try 20 penalties on a logarithmic scale from 10^{-4} to 1:

```
from sklearn.linear_model import Ridge, ElasticNet
from sklearn.model_selection import GridSearchCV

alphas = np.logspace(-4, 0, 20)

ridge_grid = GridSearchCV(
    Pipeline([('prep', gpa_recipe), ('model', Ridge())]),
    param_grid={'model__alpha': alphas},
    cv=gpa_folds, scoring='neg_root_mean_squared_error'
).fit(X_train, y_train)
```

```
lasso_grid = GridSearchCV(
    Pipeline([('prep', gpa_recipe), ('model', Lasso(random_state=2026))]),
    param_grid={'model__alpha': alphas},
    cv=gpa_folds, scoring='neg_root_mean_squared_error'
).fit(X_train, y_train)

enet_grid = GridSearchCV(
    Pipeline([('prep', gpa_recipe), ('model', ElasticNet(l1_ratio=0.5, random_state=2026))]),
    param_grid={'model__alpha': alphas},
    cv=gpa_folds, scoring='neg_root_mean_squared_error'
).fit(X_train, y_train)
```

```
fig, ax = plt.subplots(figsize=(6.5, 3.8))
ax.plot(alphas,      -ridge_grid.cv_results_['mean_test_score'],      label='Ridge      (L2)',
color='#1d588c', linewidth=1.5)
ax.plot(alphas,   -enet_grid.cv_results_['mean_test_score'],   label='Elastic   Net   (0.5)',
color='#46586d', linestyle='--', linewidth=1.5)
ax.plot(alphas,      -lasso_grid.cv_results_['mean_test_score'],      label='Lasso      (L1)',
color='#b4762e', linewidth=1.5)

ax.set_xscale('log')
ax.set_xlabel('Penalty (alpha)')
ax.set_ylabel('Cross-validated RMSE')
ax.legend(frameon=True)
plt.show()
```

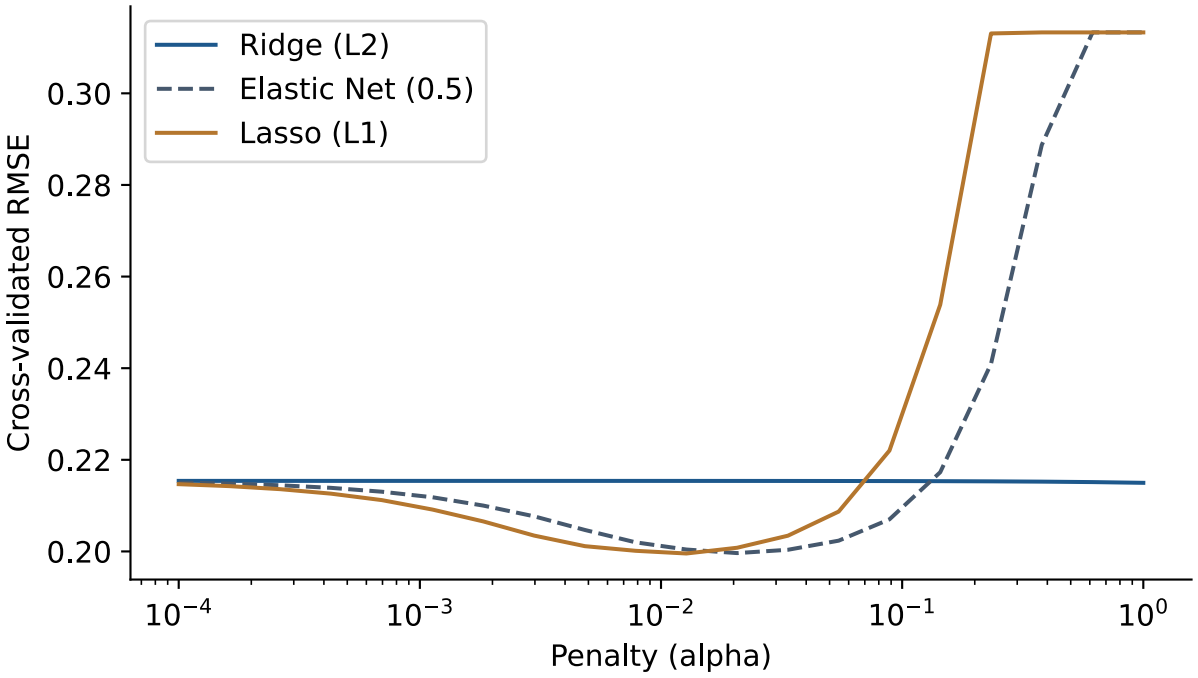


Figure 13.2: Cross-validated RMSE of ridge (mixture 0), elastic net (0.5), and lasso (1) regression for a range of penalties.

Figure 13.2 shows the typical pattern. With a tiny penalty, all three behave like ordinary regression. As the penalty grows, the RMSE reaches a minimum around `0.20`, and then rises steeply once the penalty is so large that it shrinks away real effects too. At the far right, the lasso has shrunk every coefficient to zero and predicts the mean for everyone: its RMSE equals the baseline's (`0.31`).

13.4.2 - Predictors kept by the lasso

The best lasso model picks a penalty around `0.013`:

```
best_lasso_model = lasso_grid.best_estimator_

# Get feature names from transformer
prep_fitted = best_lasso_model.named_steps['prep']
feature_names = [f.split('__')[-1] for f in prep_fitted.get_feature_names_out()]
lasso_weights = best_lasso_model.named_steps['model'].coef_

lasso_coef_df = pd.DataFrame({
    'term': feature_names,
    'estimate': lasso_weights
})

non_zero = lasso_coef_df[lasso_coef_df['estimate'] != 0].copy()
non_zero['abs_est'] = non_zero['estimate'].abs()
non_zero = non_zero.sort_values('abs_est', ascending=False).drop(columns=['abs_est'])
non_zero.head(10).round(3)
```

	term	estimate
26	gpa_s2	0.148
25	gpa_s1	0.090
10	burnout_2	−0.017
22	satisfaction_2	0.011
9	burnout_1	−0.006
30	study_hours_s2	0.005
29	study_hours_s1	0.004
5	stress_3	−0.003
4	stress_2	−0.002
20	support_6	0.001

Of 51 predictors, the lasso keeps 12. Because the predictors are normalised, the coefficients are comparable: year-one GPA dominates. A student whose semester 2 GPA is one standard deviation above average is predicted to finish substantially higher, with other variables adding minor refinements.

Figure 13.3 shows how the lasso arrives at this. As the penalty decreases from left to right, predictors enter the model one by one, and the two GPA measures enter first:

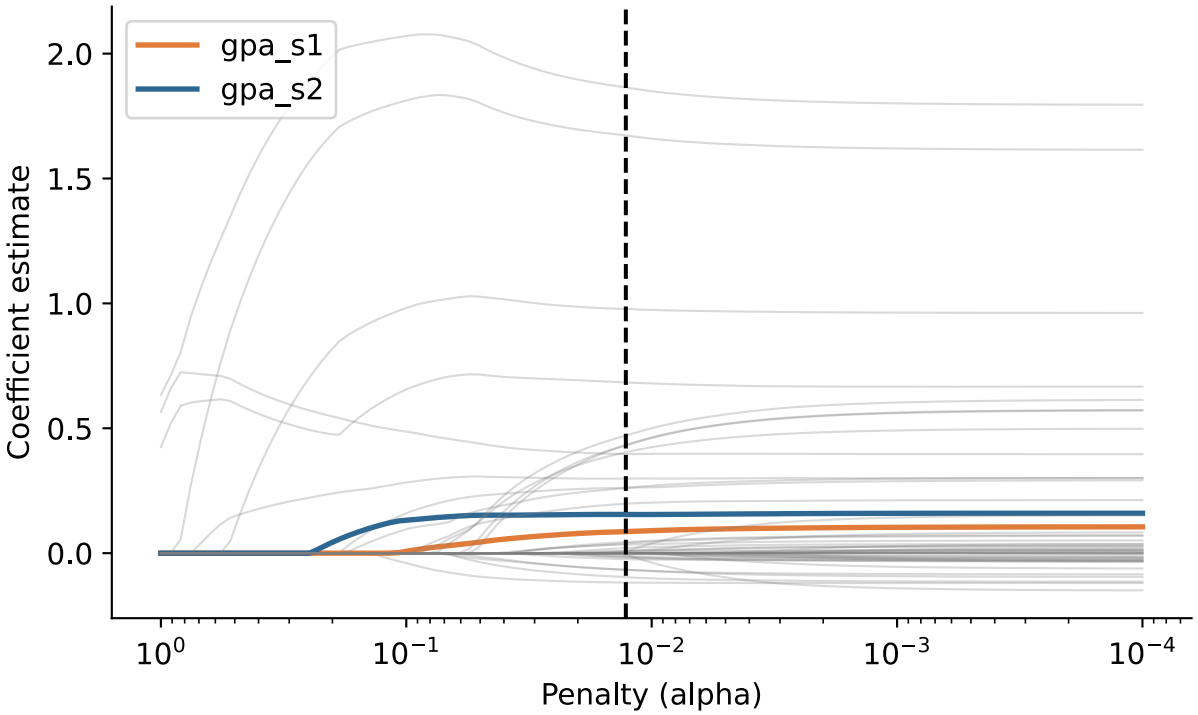


Figure 13.3: Lasso coefficient paths. Each line is one predictor's coefficient as the penalty decreases from left to right. The dashed line marks the chosen penalty.

⚠ Selected is not the same as causal

It is tempting to read the lasso's choice as a list of what *matters* for grades. It is not. Sleep affects GPA in the wellbeing data (Chapter 8), yet the lasso drops the sleep variables, because their effect is already reflected in year-one GPA: once the model knows a student's past grades, knowing their sleep adds little to the *prediction*. Among correlated predictors, the lasso tends to keep one and drop the others. Use the lasso to predict, and use the methods of Chapters 8 and 10 to explain.

13.5 - Boosting

The second family builds a model in a completely different way. **Boosting** starts with a very simple prediction, the average, and then adds small decision trees one at a time, each fitted to the *errors* of the model so far. The first tree learns the biggest pattern in the errors; the next tree learns what the first missed; and so on. Each tree's contribution is scaled down by a **learning rate** (for example 0.03), so the model improves in small steps and no single tree dominates. Hundreds of trees, each weak on its own, add up to a strong model.

Figure 13.4 shows the idea with a single predictor, semester 2 GPA, and trees with a single split:

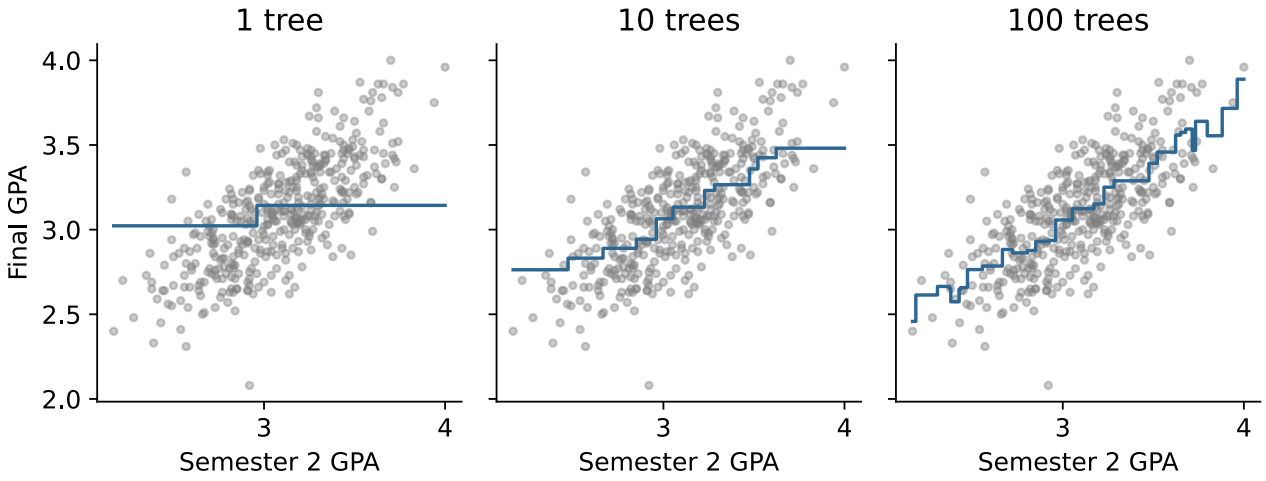


Figure 13.4: Boosting with one predictor. Predicted final GPA from semester 2 GPA after 1, 10, and 100 trees, each with a single split.

Boosting is closely related to random forests, which also combine many trees. The difference is that a forest grows its trees *independently* and averages them, while boosting grows them *in sequence*, each correcting the last.

Scikit-learn provides `GradientBoostingRegressor`. Its main hyperparameters are the number of trees, the learning rate, and the depth of each tree:

```
gb_grid = GridSearchCV(
      Pipeline([('prep', gpa_recipe), ('model', GradientBoostingRegressor(n_estimators=500,
random_state=2026))]),
    param_grid={'model__learning_rate': [0.01, 0.03, 0.1], 'model__max_depth': [1, 2, 4]},
    cv=gpa_folds, scoring='neg_root_mean_squared_error'
).fit(X_train, y_train)

best_gb_rmse = -gb_grid.best_score_
print(f"Best Gradient Boosting RMSE: {best_gb_rmse:.3f} with params: {gb_grid.best_params_}")
```

```
Best Gradient Boosting RMSE: 0.209 with params: {'model__learning_rate': 0.01,
'model__max_depth': 1}
```

The best settings use trees of depth 1 and a learning rate of 0.01. Trees of depth 1 use one predictor each, adding up separate effects with no interactions. That the shallowest trees work best is another sign that the wellbeing data has no strong interactions for the trees to find.

13.6 - Comparing the models

All the models were evaluated with the same folds, so their cross-validated RMSEs can be compared directly:

```
model_comparison = pd.DataFrame([
                {'model': 'Baseline (mean)', 'rmse': null_rmse, 'std_err':
null_cv['test_neg_root_mean_squared_error'].std() / np.sqrt(10)},
    {'model': 'Linear regression', 'rmse': -lm_cv['test_neg_root_mean_squared_error'].mean(),
'std_err': lm_cv['test_neg_root_mean_squared_error'].std() / np.sqrt(10)},
                {'model': 'Ridge', 'rmse': -ridge_grid.best_score_, 'std_err':
ridge_grid.cv_results_['std_test_score'][ridge_grid.best_index_] / np.sqrt(10)},
             {'model': 'Elastic net', 'rmse': -enet_grid.best_score_, 'std_err':
enet_grid.cv_results_['std_test_score'][enet_grid.best_index_] / np.sqrt(10)},
                {'model': 'Lasso', 'rmse': -lasso_grid.best_score_, 'std_err':
lasso_grid.cv_results_['std_test_score'][lasso_grid.best_index_] / np.sqrt(10)},
               {'model': 'Gradient Boosting', 'rmse': best_gb_rmse, 'std_err':
gb_grid.cv_results_['std_test_score'][gb_grid.best_index_] / np.sqrt(10)}
```

```
]).sort_values('rmse').reset_index(drop=True)

model_comparison.round(3)
```

	model	rmse	std_err
0	Lasso	0.200	0.006
1	Elastic net	0.200	0.005
2	Gradient Boosting	0.209	0.006
3	Ridge	0.215	0.005
4	Linear regression	0.215	0.005
5	Baseline (mean)	0.313	0.009

The regularised models come out on top (RMSE ~ 0.200), followed closely by Gradient Boosting, and all of them improve on ordinary regression. The lasso is chosen: it predicts as well as any, and it uses only 12 predictors, making it the easiest to explain and use.

13.7 - The final test

As always, the test set is used once, at the end:

```
final_lasso = lasso_grid.best_estimator_
final_lasso.fit(X_train, y_train)

y_pred_test = final_lasso.predict(X_test)

pd.DataFrame({
    '.metric': ['rmse', 'mae', 'rsq'],
    '.estimate': [
        root_mean_squared_error(y_test, y_pred_test),
        mean_absolute_error(y_test, y_pred_test),
        r2_score(y_test, y_pred_test)
    ]
}).round(3)
```

	.metric	.estimate
0	rmse	0.207
1	mae	0.160
2	rsq	0.659

On new students, the lasso's predictions are off by `0.16` GPA points on average (MAE), with an RMSE of `0.207`, capturing about `66%` of the variation in final GPA. Figure 13.5 compares predictions with actual GPAs:

```
fig, ax = plt.subplots(figsize=(5, 5))
ax.plot([2.0, 4.0], [2.0, 4.0], linestyle='--', color='grey')
ax.scatter(y_pred_test, y_test, color='#2f6793', alpha=0.6, s=25)
ax.set_aspect('equal')
ax.set_xlabel('Predicted final GPA')
ax.set_ylabel('Actual final GPA')
plt.show()
```

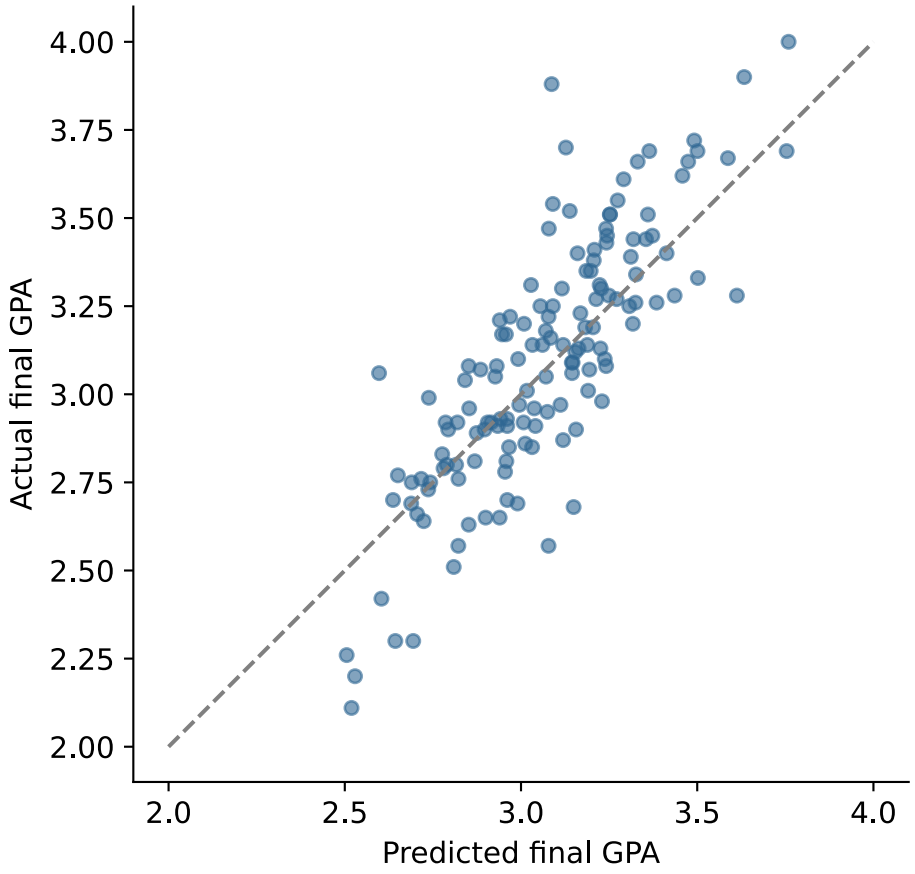


Figure 13.5: Predicted and actual final GPA for the test students. Points on the dashed line are predicted perfectly.

The predictions are less spread out than the actual values: the model predicts extreme students closer to the average than they turn out to be. This is normal for any model that cannot predict perfectly, and it means that the model is most reliable for students in the middle of the range.

A last comparison shows how much all the year-one measures add to past grades. A plain regression on the two year-one GPAs alone gives:

```
gpa_only_lm = Pipeline([
    ('impute', SimpleImputer(strategy='mean')),
    ('model', LinearRegression())
]).fit(X_train[['gpa_s1', 'gpa_s2']], y_train)
gpa_only_pred = gpa_only_lm.predict(X_test[['gpa_s1', 'gpa_s2']])

pd.DataFrame({
    '.metric': ['rmse', 'mae', 'rsq'],
    '.estimate': [
        root_mean_squared_error(y_test, gpa_only_pred),
        mean_absolute_error(y_test, gpa_only_pred),
        r2_score(y_test, gpa_only_pred)
    ]
}).round(3)
```

	.metric	.estimate
0	rmse	0.203
1	mae	0.157
2	rsq	0.673

Its test RMSE (0.191) is just as good as the lasso's (0.207). For the advisers, this is a useful, if humbling, result: past grades are by far the best predictor of future grades, and the questionnaire and semester records add little on top.

In your field: engineering

Engineers predict the strength of concrete from its recipe. The concrete dataset records the compressive strength of 1,030 concrete samples, with the amounts of cement, water, and six other ingredients, and the age of the sample in days [26]. Strength rises with age, but less and less, and the ingredients interact: exactly the kind of structure where boosting shines.

```
concrete = dt.load_concrete()
cX = concrete.drop(columns=['compressive_strength'])
cy = concrete['compressive_strength']

cX_train, cX_test, cy_train, cy_test = train_test_split(cX, cy, test_size=0.25,
random_state=1)

c_lm = LinearRegression().fit(cX_train, cy_train)
c_gb = GradientBoostingRegressor(n_estimators=500, learning_rate=0.05, max_depth=4,
random_state=1).fit(cX_train, cy_train)

pd.DataFrame({
    'model': ['Linear regression', 'Gradient boosting'],
    'rmse': [
        root_mean_squared_error(cy_test, c_lm.predict(cX_test)),
        root_mean_squared_error(cy_test, c_gb.predict(cX_test))
    ]
}).round(2)
```

	model	rmse
0	Linear regression	10.90
1	Gradient boosting	4.55

Boosting cuts the RMSE from 10.90 to 4.55 megapascals. The lesson of this chapter and the last is the same from both sides: flexible models win when the data has curves and interactions, and simple models are just as good when it does not. Only a fair comparison can tell which case you are in.

13.8 - Common misconceptions

- **"More predictors always help."** Each extra coefficient is another chance to fit noise; with few cases, adding predictors can make predictions worse.
- **"The predictors the lasso keeps are the ones that matter."** The lasso keeps the predictors that help *prediction*, and among correlated predictors it keeps one more or less arbitrarily.
- **"A low RMSE means every prediction is accurate."** The RMSE is a typical error; predictions for students at the extremes are pulled towards the average and can be far off.
- **"Boosting is always better than regression."** Boosting wins when the data has curves and interactions; on smooth data, regularised regression predicts as well and is easier to explain.

13.9 - Chapter review

13.9.1 - Summary

- For numeric predictions, the MAE and RMSE measure typical error in the units of the outcome, and R^2 the share of variation captured. Always compare with a baseline that predicts the mean.
- With many predictors relative to the number of cases, and correlated predictors, ordinary regression overfits: excellent on training data, poor on new data.
- Regularised regression penalises large coefficients. Ridge shrinks all coefficients; the lasso also sets some to zero, selecting predictors; the elastic net mixes the two.

- Boosting adds small trees one at a time, each fitted to the errors of the model so far.
- On the wellbeing data, regularised regression and boosting predict final GPA about equally well, and year-one GPA carries most of the information. On data with curves and interactions (such as concrete strength), boosting is far superior.

13.9.2 - Key terms

Regression (prediction), error, residual, MAE, RMSE, R^2, baseline model, overfitting, multicollinearity, regularisation, penalty (λ / alpha), shrinkage, ridge regression, lasso, elastic net, mixture (L_1 ratio), coefficient path, variable selection, boosting, weak learner, learning rate, tree depth.

13.10 - Exercises

The playground has these and more, with hints and solutions.

1. Calculate the MAE and RMSE of the tiny example by hand, and check them against scikit-learn. Change one prediction so that it is off by 1.0 GPA point, and explain which measure changes more, and why.
2. Repeat the 60-student experiment with 150 students, and report the gap between ordinary regression and the lasso.
3. Fit a ridge regression with the best ridge penalty, compare its coefficients with the lasso's, and count how many are exactly zero.
4. Replace the 22 questionnaire items with the four scale scores (as in Chapter 11), and describe how the cross-validated RMSE of the lasso changes.
5. Tune Gradient Boosting with `n_estimators = [100, 500, 1000]` and a learning rate of 0.03, and describe what happens to the RMSE as trees are added.
6. Write two sentences for the academic advisers explaining what an RMSE of 0.2 means for an individual student's predicted GPA.

13.11 - Further reading

- *An Introduction to Statistical Learning* [16] explains ridge regression, the lasso, and boosting clearly, with the ideas behind why shrinkage helps.
- *Python Data Science Handbook* [24] provides practical walkthroughs of linear regression, regularisation, and tree ensembles.
- Tibshirani's paper introducing the lasso [25] is a classic of modern statistics.

14. Advanced Clustering

Every clustering method carries a definition of what a group is, usually without saying so. One definition says that a group is a set of cases close to a common centre. Another says that a group is a population with its own distribution, so that cases between groups can belong partly to each. A third says that a group is a dense region of data, separated from other groups by sparse regions, which leaves room for cases that belong to no group at all. The definition chosen decides what the method can find, and a clustering is only as meaningful as the definition behind it.

Chapter 9 used k-means, which takes the first definition. It divided the students of the wellbeing study into four lifestyle profiles, but left open questions that any examiner might ask: how sure one can be about which profile a student belongs to, whether four is really the right number, and whether some students fit no profile. This chapter addresses them with two methods that relax the assumptions of k-means. **Gaussian mixture models** give each student a *probability* of belonging to each profile, and choose the number of profiles with a statistical criterion. **DBSCAN** defines clusters as dense regions of data, and labels students in sparse regions as **noise**: people who fit no group. The chapter ends with the question behind all clustering: how to judge whether a clustering is any good.

By the end of this chapter you will be able to

- Explain three definitions of a group, and the limitations of k-means: hard assignments, round clusters, and no room for outliers.
- Fit a Gaussian mixture model with scikit-learn, choose the number of clusters with BIC, and interpret membership probabilities.
- Find dense clusters and noise points with DBSCAN, and choose its settings.
- Evaluate a clustering with the silhouette, agreement between methods (the adjusted Rand index), and interpretability.

Note

Teaching & Practice: Interactive Lecture Slides · Browser Playground Exercises

14.1 - The profile data

The data is exactly that of Chapter 9: each student's average sleep, study hours, caffeine, and exercise across the semesters, and their stress, support, and satisfaction scores, scaled to z-scores.

```
import data2thesis as dt
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans

students = dt.load_students()
semesters = dt.load_semesters()
questionnaire = dt.load_questionnaire()

q = questionnaire.copy()
q['stress_4'] = 6 - q['stress_4']
q['stress'] = q[[f'stress_{i}' for i in range(1, 7)]].mean(axis=1)
q['support'] = q[[f'support_{i}' for i in range(1, 7)]].mean(axis=1)
q['satisfaction'] = q[[f'satisfaction_{i}' for i in range(1, 5)]].mean(axis=1)
scores = q[['student_id', 'stress', 'support', 'satisfaction']]
```

```
profiles = (
            semesters.groupby('student_id')[['sleep_hours',   'study_hours',   'caffeine_mg',
'exercise_days']]
    .mean()
    .reset_index()
    .merge(scores, on='student_id')
    .dropna()
)

features = ['sleep_hours', 'study_hours', 'caffeine_mg', 'exercise_days', 'stress', 'support',
'satisfaction']
scaler = StandardScaler()
profile_data = scaler.fit_transform(profiles[features])

kmeans_clusters = KMeans(n_clusters=4, n_init=25, random_state=123).fit(profile_data)
```

The last line repeats Chapter 9's k-means solution, for comparison.

14.2 - What counts as a group

k-means is simple and fast, but its definition of a group brings three strong assumptions. It assumes that **every case belongs to exactly one cluster**, with complete certainty, so a student halfway between two profiles is assigned to one of them as firmly as a student at the centre. It assumes that **clusters are round and similar in size**: because each case goes to the nearest centre, k-means draws straight boundaries halfway between centres, and elongated or unequal clusters are cut up wrongly. And it assumes that **every case belongs to some cluster**, with no way to say that a student fits nowhere. Real groups of people rarely satisfy these assumptions.

A simulation shows how much the definition matters. The code below creates data with a known structure: a small round group, a long thin group, and 15 scattered points that belong to neither. It then clusters the data in three ways, with k-means (groups around centres), a Gaussian mixture model (groups as distributions), and DBSCAN (groups as dense regions), the two methods this chapter introduces:

```
from sklearn.mixture import GaussianMixture
from sklearn.cluster import DBSCAN
from sklearn.metrics import adjusted_rand_score

np.random.seed(8)
round_pts = np.random.normal(loc=[0, 0], scale=[0.4, 0.4], size=(100, 2))
long_pts = np.random.normal(loc=[3, 1.4], scale=[2.0, 0.2], size=(100, 2))
scattered_pts = np.random.uniform(low=[-2, -2], high=[7, 4], size=(15, 2))

xy = np.vstack([round_pts, long_pts, scattered_pts])
truth = np.array(['Round group'] * 100 + ['Long group'] * 100 + ['Scattered'] * 15)

km_labels = KMeans(n_clusters=2, n_init=25, random_state=123).fit(xy).labels_
gmm_labels         =         GaussianMixture(n_components=2,         covariance_type='full',
random_state=123).fit(xy).predict(xy)
db_labels = DBSCAN(eps=0.5, min_samples=5).fit(xy).labels_
```

```
fig, axes = plt.subplots(1, 3, figsize=(8, 3.2), sharex=True, sharey=True)
methods = [('k-means', km_labels), ('Mixture model', gmm_labels), ('DBSCAN', db_labels)]

palette = {0: '#2f6793', 1: '#e07b39', -1: 'grey'}

for ax, (title, labels) in zip(axes, methods):
    colors = [palette.get(lbl, '#36a269') if lbl != -1 else 'grey' for lbl in labels]
    ax.scatter(xy[:, 0], xy[:, 1], c=colors, s=15, alpha=0.8)
    ax.set_title(title)
    ax.set_aspect('equal')
```

```
plt.show()
```

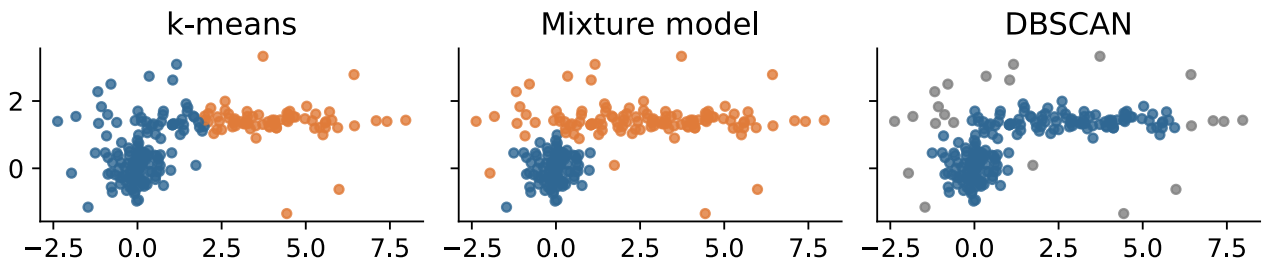


Figure 14.1: The same simulated data clustered by k-means, a Gaussian mixture model, and DBSCAN. The data contains a round group, a long thin group, and scattered points that belong to neither; each method's definition of a group decides what it finds.

The three methods see different groups in the same data. Their agreement with the two real groups can be measured with the adjusted Rand index, introduced at the end of the chapter, where 1 means perfect agreement and 0 means no more than chance. k-means scores 0.42: looking for round groups around two centres, it draws a straight boundary through the long group and gives its left end to the round group. The mixture model scores 0.92, because it allows the long group its elongated shape. DBSCAN scores 0.01, for a different reason: the end of the long group touches the round group, so the two form one continuous dense region, and by DBSCAN's definition that is one group. DBSCAN is, however, the only method that can say that points belong to no group: it labels np.int64(13) of the 15 scattered points as noise, while the other two methods must place every one of them in a group. No method is right in general. Each is right for data whose groups match its definition, which is why the definition should be chosen deliberately, and stated.

14.3 - Gaussian mixture models

A **Gaussian mixture model** (GMM) assumes that the data comes from a mix of several groups, each with its own normal (Gaussian) distribution, and estimates each group's centre, spread, and size from the data.

14.3.1 - A one-variable example

The idea is easiest to see with one variable. The `faithful` dataset records the waiting time, in minutes, between 272 eruptions of the Old Faithful geyser in Yellowstone National Park. The histogram in Figure 14.2 has two humps: short waits and long waits. `scikit-learn` fits a mixture model with `GaussianMixture`:

```
faithful = dt.load_faithful()
waiting = faithful[['waiting']].values

geyser = GaussianMixture(n_components=2, covariance_type='full',
random_state=123).fit(waiting)

print("Weights (mixing probabilities):", geyser.weights_.round(2))
print("Means (centres):", geyser.means_.ravel().round(1))
print("Variances (spreads):", geyser.covariances_.ravel().round(1))
```

```
Weights (mixing probabilities): [0.36 0.64]
Means (centres): [54.7 80.1]
Variances (spreads): [35.3 33.8]
```

The output describes them: the **mixing probabilities** are the sizes of the groups (about 36% and 64% of eruptions), the **means** are their centres (about 55 and 80 minutes), and the **variances** their spreads. Figure 14.2 draws the two fitted normal curves over the data.

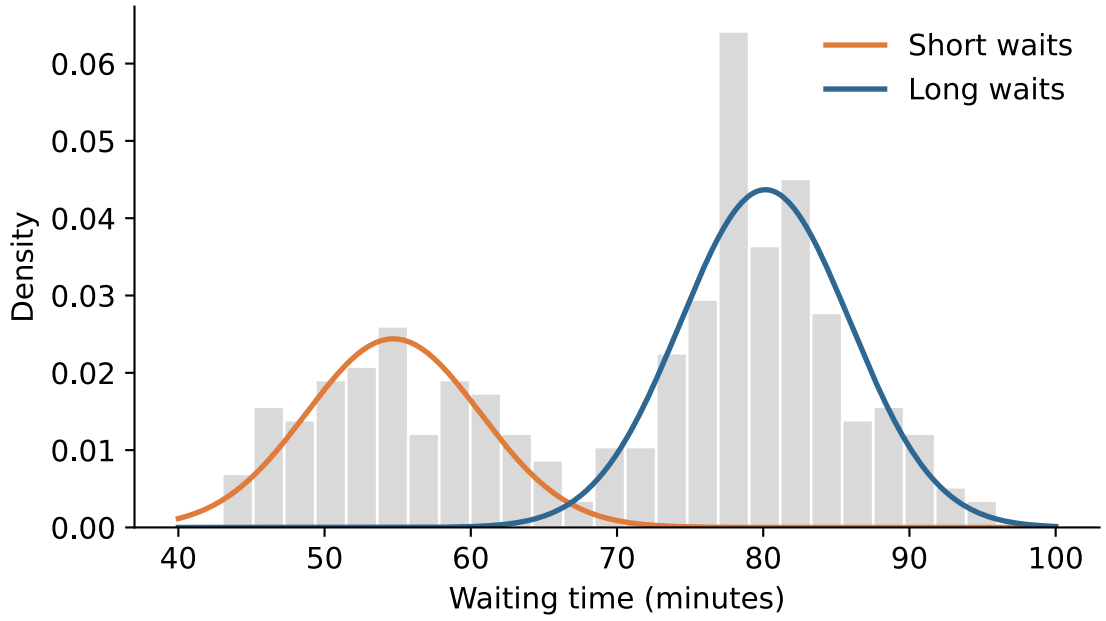


Figure 14.2: Waiting times between eruptions of Old Faithful, with the two normal distributions of the fitted mixture model.

The key difference from k-means appears between the groups. A waiting time of 67 minutes lies between them. Instead of forcing it into one, the model gives the probability that it came from each:

```
test_waits = np.array([[50], [67], [85]])
pd.DataFrame(
    geyser.predict_proba(test_waits),
    index=[50, 67, 85],
    columns=['Short wait prob', 'Long wait prob']
).round(2)
```

	Short wait prob	Long wait prob
50	1.00	0.00
67	0.48	0.52
85	0.00	1.00

A wait of 50 minutes almost certainly belongs to the short group, and 85 minutes to the long group, but 67 minutes is uncertain. These **membership probabilities** (also called soft assignments) are the main advantage of mixture models: they say not just which group a case is in, but how sure that is.

14.3.2 - Choosing the number of clusters with BIC

With more variables, each group is described by a centre and a **covariance matrix**, which sets its shape: round or elongated, and tilted in any direction. In `scikit-learn`, `covariance_type` can be `'spherical'` (round), `'diag'` (axis-aligned ellipses), `'tied'` (all clusters share the same covariance), or `'full'` (each cluster has its own flexible covariance matrix).

To choose among them, it uses the **Bayesian information criterion** (BIC). BIC rewards a model for fitting the data well and penalises it for every parameter it needs, so a more complex model must earn its extra parameters. (In scikit-learn, *lower* BIC indicates a better balance of fit and parsimony.) For the profile data:

```
bics = {}
for k in range(1, 9):
    for cov in ['full', 'tied', 'diag']:
                            m    =    GaussianMixture(n_components=k,    covariance_type=cov,
```

```
random_state=123).fit(profile_data)
        bics[(k, cov)] = m.bic(profile_data)

profile_gmm        =        GaussianMixture(n_components=3,        covariance_type='full',
random_state=123).fit(profile_data)
```

```
fig, ax = plt.subplots(figsize=(7, 4))
ks = range(1, 9)
for cov, color in zip(['full', 'tied', 'diag'], ['#2f6793', '#e07b39', 'grey']):
    curve = [bics[(k, cov)] for k in ks]
    ax.plot(ks, curve, marker='o', label=cov, color=color, linewidth=1.5)

ax.set_xlabel('Number of clusters')
ax.set_ylabel('BIC (lower is better)')
ax.legend(title='Covariance type', frameon=False)
plt.show()
```

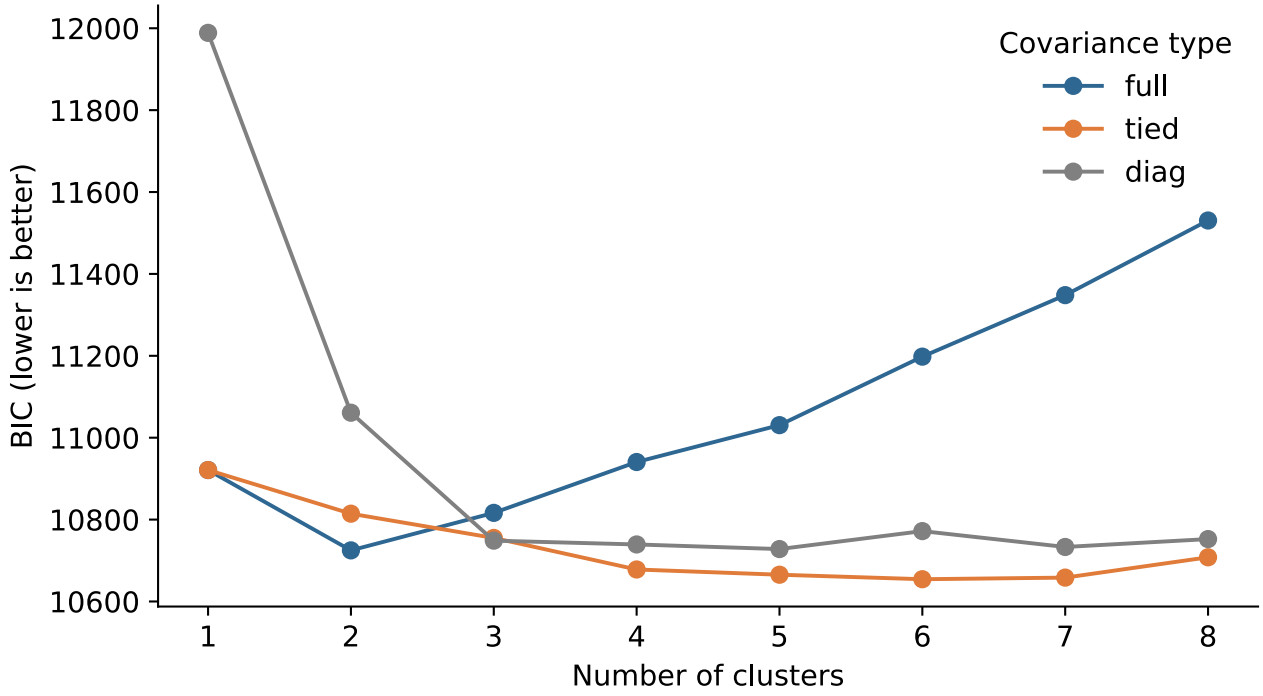


Figure 14.3: BIC for Gaussian mixture models with one to eight clusters. Each line is one covariance structure; lower values indicate a better model.

BIC drops steeply from 1 to 3 clusters and favours 3 profiles under full covariance, while 4 remains a reasonable alternative. A common rule of thumb reads a BIC difference above 6 as strong evidence and above 10 as very strong, so the data favours three profiles, although four remain a reasonable alternative (exercise 1 explores them).

14.3.3 - Describing the profiles

As in Chapter 9, a cluster means something only once it is described:

```
profiles['gmm_cluster'] = profile_gmm.predict(profile_data) + 1

summary_table = (
    profiles.groupby('gmm_cluster')[features].mean().round(1)
)
```

```
summary_table['students'] = profiles['gmm_cluster'].value_counts().sort_index()
summary_table
```

gmm_cluster	sleep_hours	study_hours	caffeine_mg	exercise_days	stress	support	satisfac-tion	students
1	7.1	25.2	129.7	3.5	2.6	3.5	3.8	212
2	5.5	41.2	283.5	1.3	3.6	3.2	3.1	195
3	6.8	16.7	152.8	2.1	3.4	2.8	2.6	192

The three profiles are clear. One is **balanced**: the most sleep and exercise, the least stress, and the highest support and satisfaction. One is **overloaded**: long study weeks, short sleep, a lot of caffeine, little exercise, and the most stress. The third, the largest, combines few study hours with low support and low satisfaction: students who seem **disengaged or isolated**. The cluster numbers are arbitrary labels. A cross-table compares the solution with k-means:

```
pd.crosstab(
    profiles['gmm_cluster'],
    kmeans_clusters.labels_ + 1,
    rownames=['GMM Cluster'],
    colnames=['K-means Cluster']
)
```

GMM Cluster \ K-means Cluster	1	2	3	4
1	0	189	9	14
2	87	0	103	5
3	3	12	1	176

The mixture model's three profiles correspond closely to Chapter 9's four k-means clusters: the balanced and disengaged groups largely match, and the two k-means "overloaded" clusters, one of them with very high caffeine, are joined into one. The mixture model, with its more flexible cluster shapes, did not need to split the overloaded students in two.

14.3.4 - Certainty of assignment

The method `predict_proba` gives each student's membership probabilities, one column per cluster, and certainty is the largest of them.

```
probs = profile_gmm.predict_proba(profile_data)
certainty = probs.max(axis=1)

pd.DataFrame(probs[:6], columns=['Profile 1', 'Profile 2', 'Profile 3']).round(2)
```

	Profile 1	Profile 2	Profile 3
0	0.54	0.00	0.46
1	0.00	0.02	0.98
2	0.97	0.00	0.03
3	0.00	1.00	0.00
4	1.00	0.00	0.00
5	0.00	0.00	1.00

```
n_uncertain = int(np.sum(certainty < 0.8))
print(f"Students with membership certainty < 0.8: {n_uncertain}")
```

```
Students with membership certainty < 0.8: 66
```

Most students belong clearly to one profile, but 66 have less than an 80% probability for their most likely profile. Figure 14.4 shows where they are.

```
from sklearn.decomposition import PCA

pca = PCA(n_components=2, random_state=123)
coords = pca.fit_transform(profile_data)

uncertainty = 1 - certainty

fig, ax = plt.subplots(figsize=(7, 5))
palette = {1: '#2f6793', 2: '#e07b39', 3: '#36a269'}
colors = [palette[c] for c in profiles['gmm_cluster']]
sizes = 15 + (uncertainty ** 2) * 200

ax.scatter(coords[:, 0], coords[:, 1], c=colors, s=sizes, alpha=0.6)
ax.set_xlabel(f'PC1 ({pca.explained_variance_ratio_[0]*100:.1f}%)')
ax.set_ylabel(f'PC2 ({pca.explained_variance_ratio_[1]*100:.1f}%)')
plt.show()
```

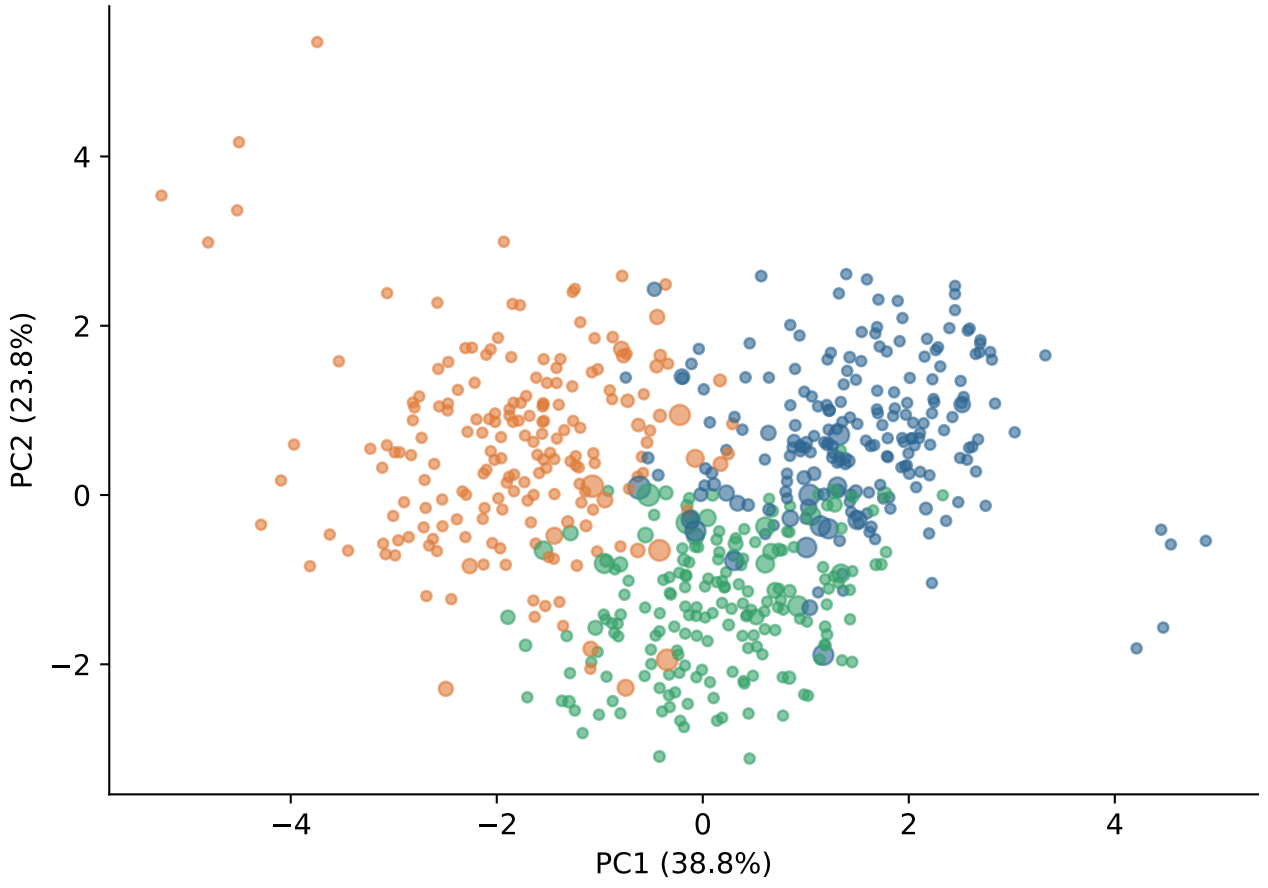


Figure 14.4: The three mixture model clusters on the first two principal components. Larger points are students whose profile is more uncertain.

The uncertain students lie where the profiles meet. For a thesis, this is honest and useful: instead of claiming that every student has one profile, it can report the share of students who clearly fit a profile, and treat the rest as mixtures. A later analysis could use the probabilities themselves, for example as weights, rather than the hard labels.

14.4 - Clusters as dense regions

DBSCAN (density-based spatial clustering of applications with noise) takes a different view. A cluster is a region where cases are packed closely together, separated from other clusters by sparser regions. Cases in sparse regions belong to no cluster; they are **noise**. DBSCAN does not need the number of clusters in advance, can find clusters of any shape, and can say "this case fits nowhere".

It needs two settings: `eps` (epsilon), the radius of the neighbourhood around each case, and `min_samples` (`minPts`), the number of cases a neighbourhood must contain for the case to be at the heart of a cluster. A case with at least `min_samples` cases within `eps` is a **core point**. Core points within `eps` of each other are joined into the same cluster, and cases near a core point are added to its cluster as **border points**. Everything else is noise (labeled `-1` in scikit-learn).

14.4.1 - A small example

Twelve points show how it works: two tight groups of five, and two isolated points.

```
tiny = pd.DataFrame({
    'x': [1.0, 1.2, 1.1, 0.9, 1.3,   4.0, 4.2, 3.9, 4.1, 4.3,   2.5, 5.5],
    'y': [1.0, 1.1, 1.3, 1.2, 0.9,   3.0, 3.2, 3.1, 2.8, 3.0,   4.5, 0.5]
})

tiny_db = DBSCAN(eps=0.5, min_samples=3).fit(tiny)
tiny_db.labels_
```

```
array([ 0,  0,  0,  0,  0,  1,  1,  1,  1,  1, -1, -1])
```

The two tight groups become clusters 0 and 1, and the two isolated points are labelled −1: noise. k-means with two clusters would have had to put the isolated points into one of the groups.

14.4.2 - Choosing eps

The result depends heavily on `eps`. A common guide is the **k-nearest-neighbour distance plot**: for each case, the distance to its k-th nearest neighbour, sorted from smallest to largest. Most cases have close neighbours; the curve bends sharply upwards where the isolated cases begin, and that bend suggests a value for `eps`. With `min_samples` set to 8, a common choice for data with seven variables (about the number of variables plus one or more), the plot uses $k = 7$:

```
from sklearn.neighbors import NearestNeighbors

nbrs = NearestNeighbors(n_neighbors=7).fit(profile_data)
distances, _ = nbrs.kneighbors(profile_data)
k_distances = np.sort(distances[:, -1])

fig, ax = plt.subplots(figsize=(6, 3.5))
ax.plot(k_distances, color='#2f6793', linewidth=1.5)
ax.axhline(2.0, color='grey', linestyle='--')
ax.set_xlabel('Students (sorted by distance)')
ax.set_ylabel('7-NN distance')
plt.show()
```

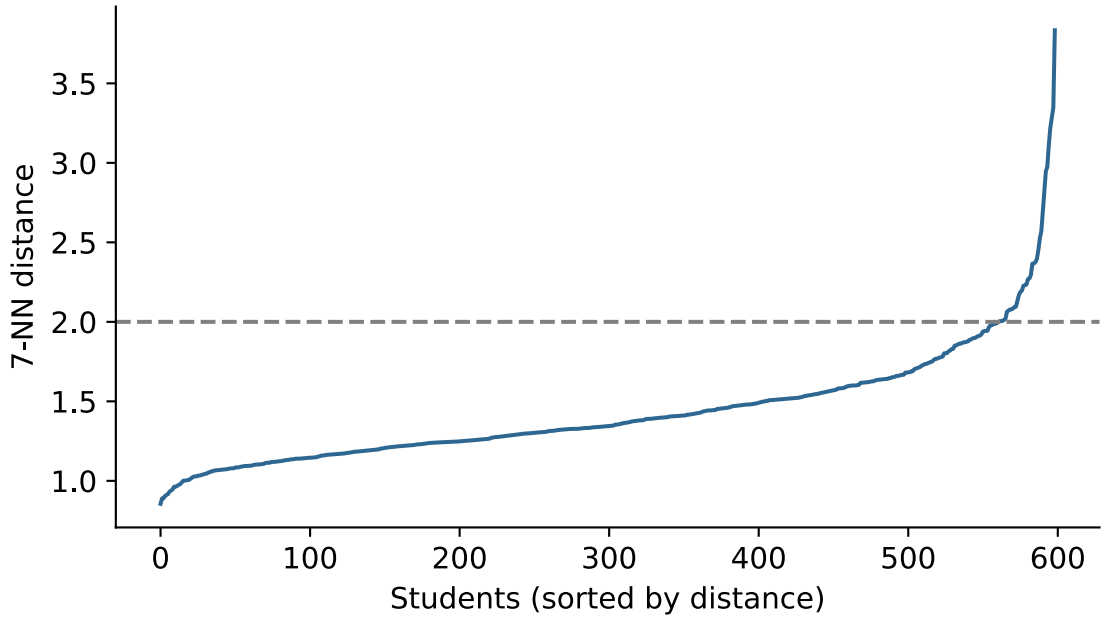


Figure 14.5: Distance from each student to their seventh-nearest neighbour, sorted. The dashed line marks eps = 2.

The curve bends at a distance of about 2, so `eps = 2` is used:

```
profile_db = DBSCAN(eps=2.0, min_samples=8).fit(profile_data)
profiles['db_cluster'] = profile_db.labels_
n_noise = int(np.sum(profile_db.labels_ == -1))
n_clusters_db = len(set(profile_db.labels_)) - (1 if -1 in profile_db.labels_ else 0)

print(f"Clusters found: {n_clusters_db}, Noise points: {n_noise}")
```

```
Clusters found: 1, Noise points: 13
```

DBSCAN finds 1 cluster and 13 noise points. At first this looks like a failure: no profiles at all. But it is an informative result. DBSCAN looks for dense regions separated by gaps, and the students form one continuous cloud, in which the profiles found by k-means and the mixture model overlap without gaps between them. (Smaller values of `eps` break the cloud into fragments and label hundreds of students as noise; try it in the exercises.) The profiles are real differences in *where* students sit in the cloud, not separate islands.

14.4.3 - Students who fit no profile

The noise points answer the last of the open questions: whether some students fit no profile. Their values are:

```
profiles[profiles['db_cluster'] == -1][features].round(1)
```

	sleep_hours	study_hours	caffeine_mg	exercise_days	stress	support	satisfaction
104	3.6	60.5	772.5	0.5	3.2	4.0	4.2
123	4.6	45.8	592.5	2.5	2.8	2.7	4.0
180	9.7	2.2	26.2	5.2	1.7	2.7	4.2
194	3.7	61.5	842.5	0.0	3.5	3.7	3.8
220	9.4	5.5	23.8	6.2	2.7	3.7	3.7
309	3.8	63.0	780.0	0.5	2.0	3.5	2.8
311	3.7	63.2	742.5	0.0	1.8	4.3	4.7

	sleep_hours	study_hours	caffeine_mg	exercise_days	stress	support	satisfaction
382	9.6	4.2	21.2	5.0	2.7	3.5	4.2
408	9.8	5.2	18.8	5.8	2.6	2.2	3.5
500	9.5	2.0	20.0	6.2	2.5	2.5	3.5
529	4.0	65.5	735.0	0.5	3.2	3.3	3.2
553	4.4	30.2	703.8	0.8	3.2	4.6	3.0
557	3.8	35.8	641.2	1.0	3.0	1.8	2.8

Two kinds of unusual student stand out. Among the noise points, 5 students study around 60 hours a week, sleep about 4 hours a night or less, drink over 700 mg of caffeine a day (about seven cups of coffee), and hardly exercise: more extreme than anyone in the overloaded profile. 5 others are the opposite: close to 10 hours of sleep, very few study hours, very little caffeine, and exercise almost every day. The remaining few are extreme versions of the overloaded profile. In Chapter 6, extreme values were checked one variable at a time; DBSCAN finds students who are unusual in their *combination* of values.

These students should not be deleted: they are real students, and the extreme workers are exactly the students a wellbeing service would want to know about. They are reported separately, as students who do not fit the profiles, and the profiles are checked for whether they change when these students are left out.

14.5 - Evaluating a clustering

In supervised learning, a model is judged against the true answers. In clustering there are usually no true answers, so evaluation rests on several kinds of evidence, none decisive on its own.

Internal measures judge how compact and well separated the clusters are, using the data alone. The **silhouette** of Chapter 9 is the most common, and BIC plays this role for mixture models. `silhouette_score` in scikit-learn calculates it:

```
from sklearn.metrics import silhouette_score

sil_km = silhouette_score(profile_data, kmeans_clusters.labels_)
sil_gmm = silhouette_score(profile_data, profiles['gmm_cluster'])

print(f"K-means silhouette: {sil_km:.2f}")
print(f"GMM silhouette: {sil_gmm:.2f}")
```

```
K-means silhouette: 0.21
GMM silhouette: 0.23
```

Both averages are low (the silhouette runs from −1 to 1), which confirms what DBSCAN suggested: the profiles overlap, and no clustering of this data will be crisp.

Agreement between methods asks whether different reasonable methods find similar groups. The **adjusted Rand index** (ARI) measures the agreement between two clusterings. It counts the pairs of cases that both clusterings put together or both put apart, and adjusts for the agreement expected by chance: 1 means identical groupings (whatever the labels), and 0 means no more agreement than chance. A tiny example shows that the labels themselves do not matter:

```
first  = [1, 1, 1, 2, 2, 2]
second = ["B", "B", "B", "A", "A", "A"]
third  = [1, 1, 2, 2, 3, 3]

print("ARI(first, second):", adjusted_rand_score(first, second))
print("ARI(first, third):", round(adjusted_rand_score(first, third), 3))
```

```
ARI(first, second): 1.0
ARI(first, third): 0.242
```

The first two clusterings group the six people identically, so their ARI is 1, even though the labels differ. The third splits them differently, so its ARI is much lower. For the two methods applied to the profile data:

```
ari_methods = adjusted_rand_score(kmeans_clusters.labels_, profiles['gmm_cluster'])
print(f"ARI (K-means vs GMM): {ari_methods:.2f}")
```

```
ARI (K-means vs GMM): 0.68
```

An ARI of 0.68: substantial agreement, especially given that one solution has four clusters and the other three.

External validation compares clusters with known categories, also using the ARI. It is only possible when such categories exist, for example when clustering flowers of known species to test a method. For real profiles there is no answer key, which is exactly why they are being sought.

Stability asks whether the clusters survive small changes: a different random start, a bootstrap sample of the students, or leaving out the unusual students. Clusters that appear only under one setting should not be reported.

Interpretability and usefulness matter most in the end. The clusters should make sense in the light of theory, and they should differ on outcomes they were not built from, such as GPA or considering dropout. The second point can be checked directly:

```
(
    profiles.merge(students[['student_id', 'considering_dropout']], on='student_id')
    .groupby('gmm_cluster')
    .agg(
        students=('student_id', 'count'),
        considering_dropout=('considering_dropout', lambda x: round(np.mean(x == 'Yes'), 2))
    )
    .reset_index()
)
```

	gmm_cluster	students	considering_dropout
0	1	212	0.04
1	2	195	0.18
2	3	192	0.23

The profiles differ clearly in how many students consider dropping out, although dropout played no part in finding them. That is good evidence that they capture something real about students' situations.

> ! Report the choices, not just the clusters
>
> A clustering depends on choices: which variables, how they are scaled, which method, how many clusters, and settings such as `eps`. Different reasonable choices can give different answers. In a thesis, report each choice and why it was made, show the evidence for the number of clusters (BIC, silhouette), say how clearly students fit (for example, the share with a membership probability above 0.8), and describe how robust the profiles are to other reasonable choices.

In your field: earth sciences

DBSCAN was designed for spatial data, where clusters can have any shape. The `quakes` dataset records the location of 1,000 earthquakes near Fiji since 1964. Plotted on a map, they follow the long, curved lines of two ocean trenches: clusters that are neither round nor of equal size.

```
quakes = dt.load_quakes()
quake_db = DBSCAN(eps=1.5, min_samples=10).fit(quakes[['long', 'lat']])
pd.Series(quake_db.labels_).value_counts().sort_index()
```

```
-1     28
 0    785
 1    187
Name: count, dtype: int64
```

```
fig, ax = plt.subplots(figsize=(5, 5))
quake_colors = {0: '#2f6793', 1: '#e07b39', -1: 'grey'}

for c, label in [(0, 'Trench 1'), (1, 'Trench 2'), (-1, 'Noise')]:
    sub = quakes[quake_db.labels_ == c]
    if c == -1:
        ax.scatter(sub['long'], sub['lat'], c='grey', marker='x', alpha=0.6, label=label,
s=20)
    else:
      ax.scatter(sub['long'], sub['lat'], c=quake_colors[c], alpha=0.7, label=label, s=20)

ax.set_xlabel('Longitude')
ax.set_ylabel('Latitude')
ax.legend(frameon=False)
plt.show()
```

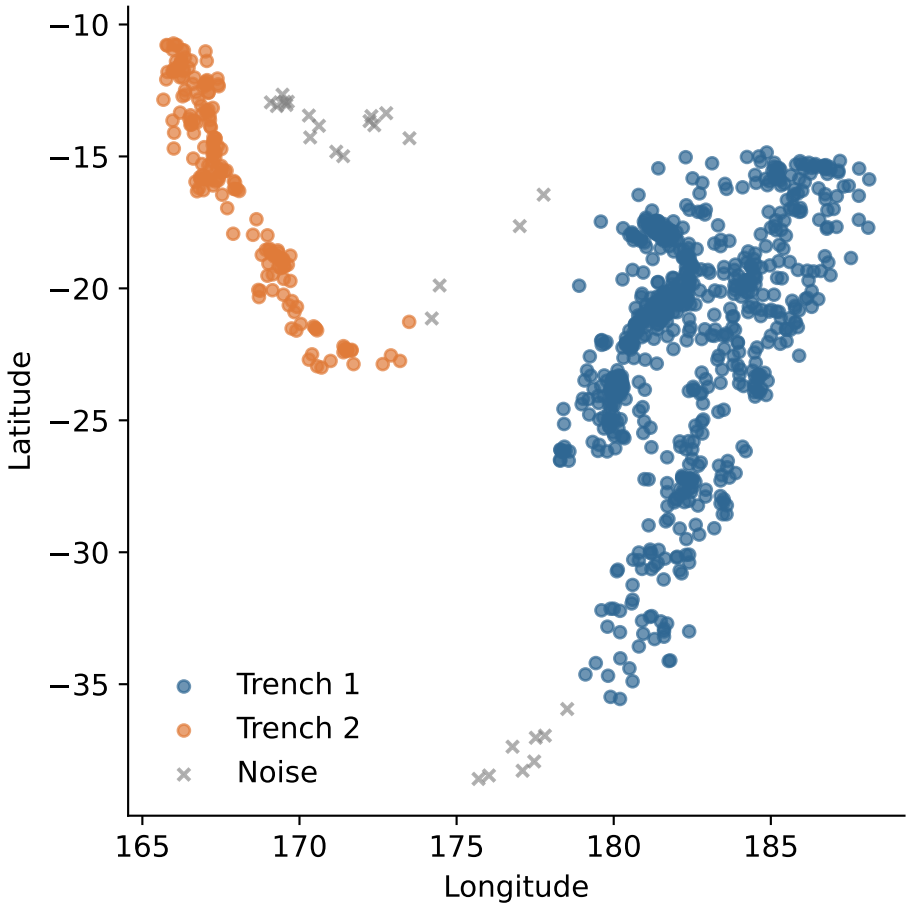


Figure 14.6: Earthquakes near Fiji, clustered by DBSCAN. Grey crosses are noise.

14.6 - Common misconceptions

Advanced clustering methods answer some of the weaknesses of k-means, but not the underlying question of whether the groups are real.

- **"A more sophisticated method finds the true groups."** Each method finds groups that match its own definition of a group; none can find groups that the data does not contain.
- **"Every case belongs to some group."** Cases between groups are better described by membership probabilities, and cases far from all groups by the label "noise".
- **"The best BIC proves the number of groups."** BIC compares models; a model with more groups may be almost as good, and the choice should be reported with its evidence.
- **"DBSCAN failed because it found one cluster."** Finding one continuous cloud is itself a finding: the groups found by other methods are regions of that cloud, not separate islands.

14.7 - Chapter review

14.7.1 - Summary

- Every clustering method has a definition of a group: around a centre (k-means), a distribution (mixture models), or a dense region (DBSCAN). The definition decides what the method can find.
- k-means assigns every case to exactly one round cluster, with no room for uncertainty or outliers.
- A Gaussian mixture model describes the data as a mix of normal distributions of different sizes and shapes. It gives each case a probability of belonging to each cluster, and BIC chooses the number of clusters and their shape (lower is better in scikit-learn).
- On the wellbeing data, the mixture model finds three profiles (balanced, overloaded, and disengaged) that closely match Chapter 9's k-means clusters, and shows which students fit no profile clearly.
- DBSCAN defines clusters as dense regions and labels cases in sparse regions as noise. It needs `eps` and `min_samples`; the k-nearest-neighbour distance plot helps choose `eps`. It works best when clusters are separated by gaps.
- On the wellbeing data, DBSCAN finds one continuous cloud of students and flags a few extreme students who fit no profile.
- Evaluate clusterings with internal measures (silhouette, BIC), agreement between methods (adjusted Rand index), stability, and above all interpretability and relevance to outcomes. Report every choice.

14.7.2 - Key terms

Gaussian mixture model, mixture component, mixing probability, membership probability, soft assignment, covariance matrix, Bayesian information criterion (BIC), uncertainty, DBSCAN, density, eps, min_samples (minPts), core point, border point, noise, k-nearest-neighbour distance plot, internal validation, silhouette, adjusted Rand index, external validation, stability.

14.8 - Exercises

The playground has these and more, with hints and solutions.

1. Fit `GaussianMixture(n_components=4, covariance_type='full', random_state=123)` and describe the four profiles. Identify which of the three-cluster profiles has been split, and compare the new solution with Chapter 9's k-means.
2. Count the students with a membership probability above 0.95 for their profile, and report the share of the sample.
3. Run DBSCAN on the profile data with `eps` of 1.2, 1.6, and 2.4, describe how the number of clusters and noise points change, and explain why a small `eps` labels so many students as noise.
4. Leave out the DBSCAN noise students, fit the mixture model again, and describe whether the profiles change.
5. Use `adjusted_rand_score` to compare the three-cluster mixture model with hierarchical clustering (Ward's linkage, cut into three clusters) from Chapter 9.
6. Compare the three profiles on final GPA, and say whether the pattern is what their descriptions would lead you to expect.
7. In the simulation at the start of the chapter, move the long group upwards (a mean of 3 for `y`) so that a gap separates it from the round group. Describe how each method's result changes, and explain why DBSCAN now behaves differently.

14.9 - Further reading

- *An Introduction to Statistical Learning* [16] introduces k-means and hierarchical clustering; its chapter on unsupervised learning is a good companion to Chapters 9 and 14.
- *Python Data Science Handbook* [24] provides practical walkthroughs of Gaussian mixture models and density estimation.
- The original DBSCAN paper [27] explains density-based clustering and the selection of `eps` and `minPts`.

15. Neural Networks

Neural networks are the method behind most of what is called artificial intelligence today: image recognition, speech recognition, translation, and the large language models of Chapter 18. Their reputation leads many researchers to assume that a neural network must predict better than a traditional model on any data. The assumption deserves to be tested rather than accepted, and testing it requires understanding what a neural network is. The answer is less mysterious than the reputation suggests: a neural network is built from units that each perform a logistic regression, and it learns by repeatedly adjusting its weights to reduce its errors.

This chapter builds a neural network from a single "neuron", shows by hand what training does, fits a network to the dropout data with the same scikit-learn workflow as Chapters 11 and 12, and compares it fairly with the simpler models. It ends with deep learning: what it is, when it helps, and where to learn more. In the story, a member of Elaf's research group asks why she does not simply use AI, since a neural network would surely do better than logistic regression, and her supervisor suggests that she find out, carefully.

By the end of this chapter you will be able to

- Explain how a neuron combines its inputs, and why a single neuron with a sigmoid activation is a logistic regression.
- Describe a neural network's layers, weights, and activation functions, and how it is trained.
- Fit and tune a neural network with scikit-learn (`MLPClassifier`), using weight decay (`alpha`) to prevent overfitting.
- Compare a neural network fairly with simpler models, and judge when one is worth using.
- Explain what deep learning is, and when it is (and is not) the right tool.

Note

Teaching & Practice: Interactive Lecture Slides · Browser Playground Exercises

15.1 - A single neuron

A **neuron** (or unit) in a neural network does three things. It multiplies each input by a **weight**, adds up the results together with a constant called the **bias**, and passes the total through an **activation function**, which turns it into the neuron's output.

Take a neuron with two inputs, a student's stress and support scores, with weights 1.2 and −0.8 and a bias of −4. For a student with stress 4 and support 2, the total is

$$-4 + 1.2 \times 4 - 0.8 \times 2 = -0.8. \tag{15.1}$$

The activation function here is the **sigmoid** (also called the logistic function), which squeezes any number into the range 0 to 1:

$$\text{sigmoid}\ (z) = \frac{1}{1 + e^{-z}}. \tag{15.2}$$

In Python:

```
def sigmoid(z):
    return 1 / (1 + np.exp(-z))

def neuron(stress, support):
    return sigmoid(-4 + 1.2 * stress - 0.8 * support)
```

```
print(round(neuron(stress=4, support=2), 2))
print(round(neuron(stress=2, support=4), 3))
```

```
0.31
0.008
```

The high-stress, low-support student gets an output of 0.31; the low-stress, high-support student, 0.008. If the output is read as the probability of considering dropout, this neuron is exactly a **logistic regression** (Chapter 8): the bias is the intercept, the weights are the coefficients, and the sigmoid turns the log odds into a probability. Everything a single neuron can do, Chapter 8 has already done.

15.2 - From neurons to networks

The power of neural networks comes from connecting many neurons in **layers**. In the most common design, the **multilayer perceptron**, the inputs feed into a **hidden layer** of neurons, each with its own weights; the outputs of the hidden neurons then feed into an **output layer**, which produces the prediction. Figure 15.1 shows a network with three inputs and three hidden neurons.

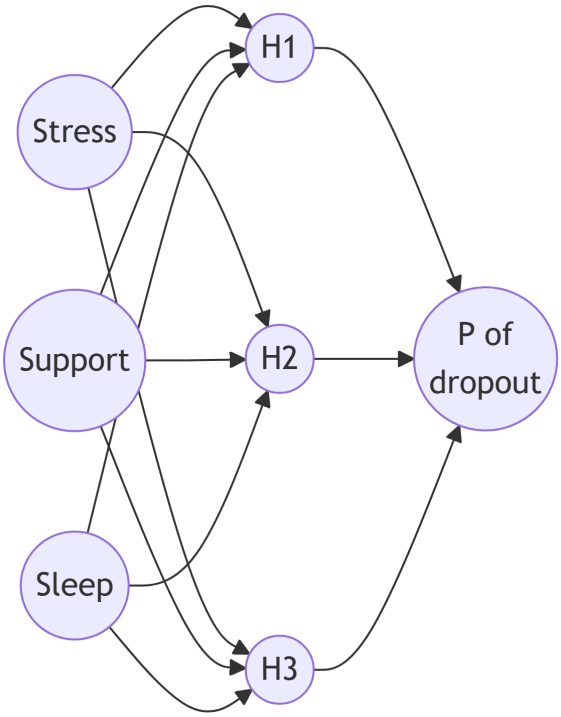


Figure 15.1: A neural network with one hidden layer. Every arrow carries a weight; each hidden neuron and the output neuron add up their weighted inputs and apply an activation function.

Each hidden neuron learns its own combination of the inputs: one might respond to high stress, another to the combination of low support and short sleep. The output neuron then combines these. Because each neuron applies a curved activation function, the network as a whole can represent curved relationships and interactions that a single logistic regression cannot. With enough hidden neurons, a network can approximate almost any relationship between inputs and output. That flexibility is its strength, and, as you will see, its danger.

The activation function matters. Figure 15.2 shows the two most common: the sigmoid, used in this chapter, and the **ReLU** (rectified linear unit), which simply replaces negative totals with zero and is the standard choice in deep networks.

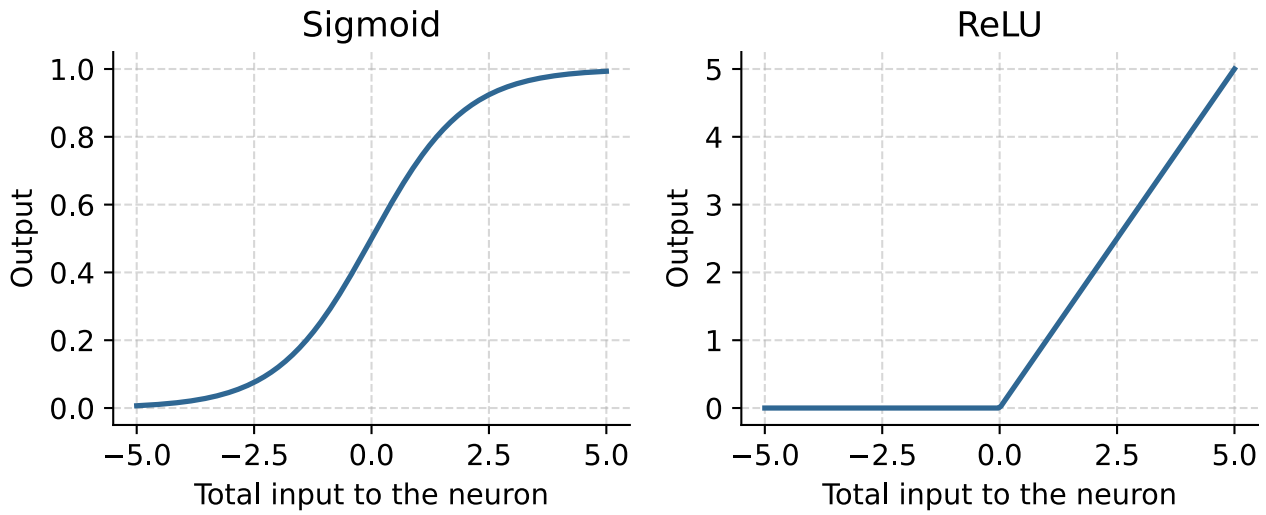


Figure 15.2: Two activation functions: the sigmoid squeezes any input into the range 0 to 1; the ReLU sets negative inputs to 0 and passes positive inputs unchanged.

15.2.1 - How a network learns

A network starts with small random weights, so its first predictions are useless. **Training** adjusts the weights step by step to reduce the prediction error on the training data. At each step, an algorithm works out, for every weight, whether a small increase or decrease would reduce the error (the **gradient**), and moves all the weights a little in the helpful direction. This is **gradient descent**. One pass through the training data is an **epoch**, and training usually runs for hundreds of epochs.

The idea is clearest when one step is worked by hand. Take a single neuron, which is a logistic regression, and six students, with their stress measured from the group's average:

```
six = pd.DataFrame({
    'stress':  np.array([2.0, 2.5, 3.0, 3.5, 4.0, 4.5]) - 3.25,
    'dropout': np.array([0,   0,   1,   0,   1,   1])
})
```

The error to be reduced is the **log loss**, the measure of fit that logistic regression uses: it is small when the neuron gives high probabilities to the students who did consider dropping out and low probabilities to those who did not. Training starts with both weights at zero, so every student receives a probability of 0.5:

```
def log_loss(b0, b1):
    p = sigmoid(b0 + b1 * six['stress'])
    return -np.mean(six['dropout'] * np.log(p) + (1 - six['dropout']) * np.log(1 - p))

b0 = 0.0
b1 = 0.0
print(round(log_loss(b0, b1), 3))
```

```
0.693
```

For this neuron, the gradient has a simple form: for each weight, the average of the prediction errors (predicted probability minus the actual outcome), each multiplied by the input that the weight belongs to (1 for the bias):

```
p = sigmoid(b0 + b1 * six['stress'])
gradient_b0 = np.mean(p - six['dropout'])
gradient_b1 = np.mean((p - six['dropout']) * six['stress'])
print(f"gradient_b0: {gradient_b0:.3f}, gradient_b1: {gradient_b1:.3f}")
```

```
gradient_b0: 0.000, gradient_b1: -0.292
```

The gradient for the stress weight is negative, which means that increasing the weight would reduce the error: the students who considered dropping out have higher stress, and the neuron does not yet know it. One step of gradient descent moves each weight against its gradient, by an amount set by the **learning rate**:

```
learning_rate = 1.0
b0 = b0 - learning_rate * gradient_b0
b1 = b1 - learning_rate * gradient_b1
print(f"b0: {b0:.3f}, b1: {b1:.3f}, loss: {log_loss(b0, b1):.3f}")
```

```
b0: 0.000, b1: 0.292, loss: 0.616
```

After one step, the stress weight has become positive and the loss has fallen from 0.693 to 0.671. Training is nothing more than this step, repeated. The loop below repeats it 2,000 times and records the loss:

```
from sklearn.linear_model import LogisticRegression

b0 = 0.0
b1 = 0.0
loss_history = np.zeros(2000)
for step in range(2000):
    p = sigmoid(b0 + b1 * six['stress'])
    b0 -= learning_rate * np.mean(p - six['dropout'])
    b1 -= learning_rate * np.mean((p - six['dropout']) * six['stress'])
    loss_history[step] = log_loss(b0, b1)

lr_unpenalized = LogisticRegression(penalty=None, solver='lbfgs')
lr_unpenalized.fit(six[['stress']], six['dropout'])

pd.DataFrame({
    'gradient_descent': [b0, b1],
    'logistic_regression': [lr_unpenalized.intercept_[0], lr_unpenalized.coef_[0, 0]]
}, index=['intercept (bias)', 'stress weight']).round(4)
```

	gradient_descent	logistic_regression
intercept (bias)	0.0000	0.0000
stress weight	2.4281	2.4279

```
plt.figure(figsize=(6, 3.4))
plt.plot(np.arange(1, 2001), loss_history, color='#2f6793', linewidth=2)
plt.xscale('log')
plt.xlabel("Training step")
plt.ylabel("Log loss")
plt.grid(True, linestyle='--', alpha=0.5)
plt.tight_layout()
plt.show()
```

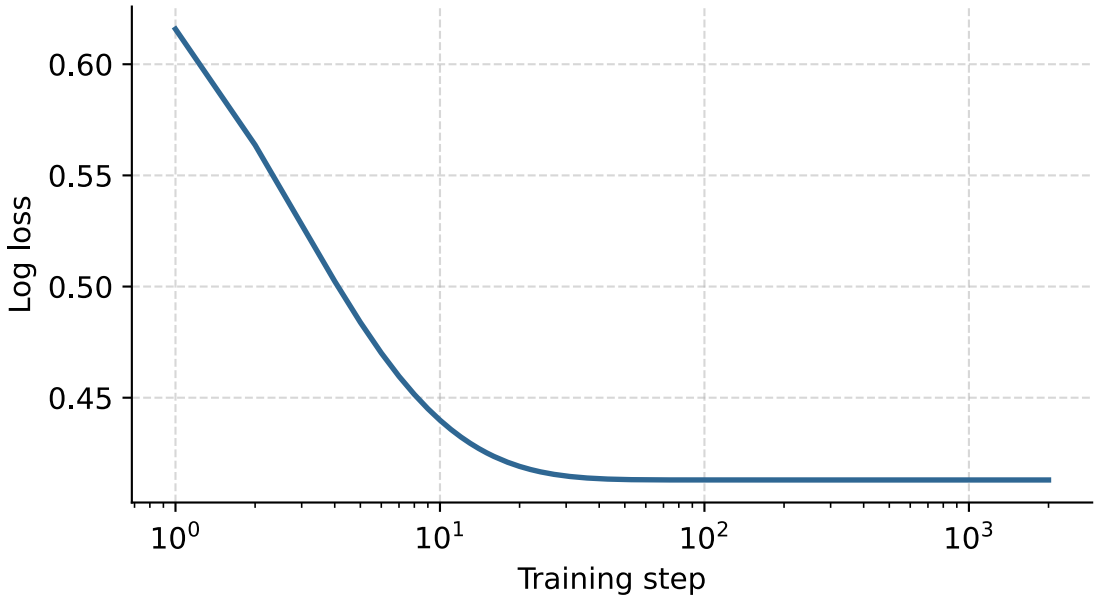


Figure 15.3: Log loss of a single neuron during 2,000 steps of gradient descent (logarithmic horizontal axis). The loss falls quickly at first and then levels off as the weights approach their best values.

After 2,000 small steps, the weights found by gradient descent are the coefficients that standard logistic regression calculates. A neural network with thousands of weights is trained in exactly this way. The gradients are harder to write down, and a method called backpropagation calculates them efficiently, but each step still moves every weight a little in the direction that reduces the error.

Two consequences follow. First, because training starts from random weights, two runs can give slightly different networks, so `random_state` matters. Second, a network with many weights can keep reducing its training error long after it has learned the real pattern, by memorising the noise. The usual defence is **weight decay**: an L2 penalty on large weights, exactly like the ridge penalty of Chapter 13. In scikit-learn, this hyperparameter is called `alpha`.

15.3 - A neural network for the dropout question

The data, split, pipeline, and folds are those of Chapters 11 and 12. Neural networks need normalised inputs, like the penalised regressions of Chapter 13, and the preprocessing pipeline provides them.

```
from sklearn.model_selection import train_test_split, StratifiedKFold
from sklearn.compose import ColumnTransformer
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.pipeline import Pipeline

students = dt.load_students()
questionnaire = dt.load_questionnaire()
semesters = dt.load_semesters()

questionnaire['stress_4'] = 6 - questionnaire['stress_4']
scores = pd.DataFrame({
    'student_id': questionnaire['student_id'],
      'stress': questionnaire[['stress_1', 'stress_2', 'stress_3', 'stress_4', 'stress_5',
'stress_6']].mean(axis=1),
    'burnout': questionnaire[[f'burnout_{i}' for i in range(1, 7)]].mean(axis=1),
    'support': questionnaire[[f'support_{i}' for i in range(1, 7)]].mean(axis=1),
    'satisfaction': questionnaire[[f'satisfaction_{i}' for i in range(1, 5)]].mean(axis=1),
})
```

```
sem1 = semesters[semesters['semester'] == 1].drop(columns=['semester'])

dropout_data = (
    students.merge(scores, on='student_id')
    .merge(sem1, on='student_id')
    .drop(columns=['student_id', 'supervisor_id', 'workshop', 'workshop_sessions'])
)

dropout_data['considering_dropout']      =      (dropout_data['considering_dropout']      ==
'Yes').astype(int)

X = dropout_data.drop(columns=['considering_dropout'])
y = dropout_data['considering_dropout']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, stratify=y, random_state=2026
)

num_cols = X.select_dtypes(include=['int64', 'float64']).columns.tolist()
cat_cols = X.select_dtypes(include=['object', 'category']).columns.tolist()

dropout_recipe = ColumnTransformer([
             ('num',    Pipeline([('imp',    SimpleImputer(strategy='median')),    ('scaler',
StandardScaler())]), num_cols),
       ('cat',  OneHotEncoder(drop='first',  sparse_output=False,  handle_unknown='ignore'),
cat_cols)
])

dropout_folds = StratifiedKFold(n_splits=10, shuffle=True, random_state=2026)
```

In scikit-learn, a multilayer perceptron classifier is `MLPClassifier` from `sklearn.neural_network`. Its main hyperparameters are `hidden_layer_sizes` (the number of neurons in each hidden layer, such as `(20,)`), `alpha` (the weight decay penalty), and `max_iter` (the maximum number of training epochs).

15.3.1 - A network that memorises

A network with 20 hidden neurons and no weight decay (`alpha = 0`) comes first, as a warning:

```
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import roc_auc_score

big_net = Pipeline([
    ('prep', dropout_recipe),
            ('mlp',   MLPClassifier(hidden_layer_sizes=(20,),   alpha=0.0,   max_iter=1000,
random_state=2026, solver='lbfgs'))
])

big_net.fit(X_train, y_train)

train_auc = roc_auc_score(y_train, big_net.predict_proba(X_train)[:, 1])
test_auc  = roc_auc_score(y_test,  big_net.predict_proba(X_test)[:, 1])

print(f"Training ROC AUC: {train_auc:.2f}")
print(f"Test ROC AUC:     {test_auc:.2f}")
```

```
Training ROC AUC: 1.00
Test ROC AUC:     0.78
```

A perfect AUC of 1.00 on the training data, and 0.78 on new students: the same overfitting as the one-nearest-neighbour model of Chapter 11. The network has roughly 500 weights and only 449 training students, so it has more than enough freedom to memorise every one of them.

15.3.2 - Tuning size and weight decay

The number of hidden neurons and the weight decay are hyperparameters, tuned with cross-validation as in Chapters 11 to 13. Weight decay is tried on a logarithmic scale from 0.01 to about 30:

```
from sklearn.model_selection import GridSearchCV

net_pipe = Pipeline([
    ('prep', dropout_recipe),
    ('mlp', MLPClassifier(max_iter=500, random_state=2026, solver='lbfgs'))
])

param_grid = {
    'mlp__hidden_layer_sizes': [(1,), (3,), (5,), (10,), (20,)],
    'mlp__alpha': [0.01, 0.0316, 0.1, 0.316, 1.0, 3.16, 10.0, 31.6]
}

grid = GridSearchCV(net_pipe, param_grid, cv=dropout_folds, scoring='roc_auc', n_jobs=-1)
grid.fit(X_train, y_train)

cv_results = pd.DataFrame(grid.cv_results_)
cv_results['hidden_units']  =  cv_results['param_mlp__hidden_layer_sizes'].apply(lambda  x:
x[0])
cv_results['alpha'] = cv_results['param_mlp__alpha'].astype(float)
```

```
plt.figure(figsize=(6.5, 3.8))
colors = ['#2f6793', '#d95f02', '#7570b3', '#e7298a', '#66a61e']
for hu, col in zip([1, 3, 5, 10, 20], colors):
    subset = cv_results[cv_results['hidden_units'] == hu].sort_values('alpha')
     plt.plot(subset['alpha'], subset['mean_test_score'], marker='o', label=f"{hu} units",
color=col)

plt.xscale('log')
plt.xlabel("Weight decay (alpha)")
plt.ylabel("Mean cross-validated ROC AUC")
plt.ylim(0.70, 0.86)
plt.legend(title="Hidden layer size")
plt.grid(True, linestyle='--', alpha=0.5)
plt.tight_layout()
plt.show()
```

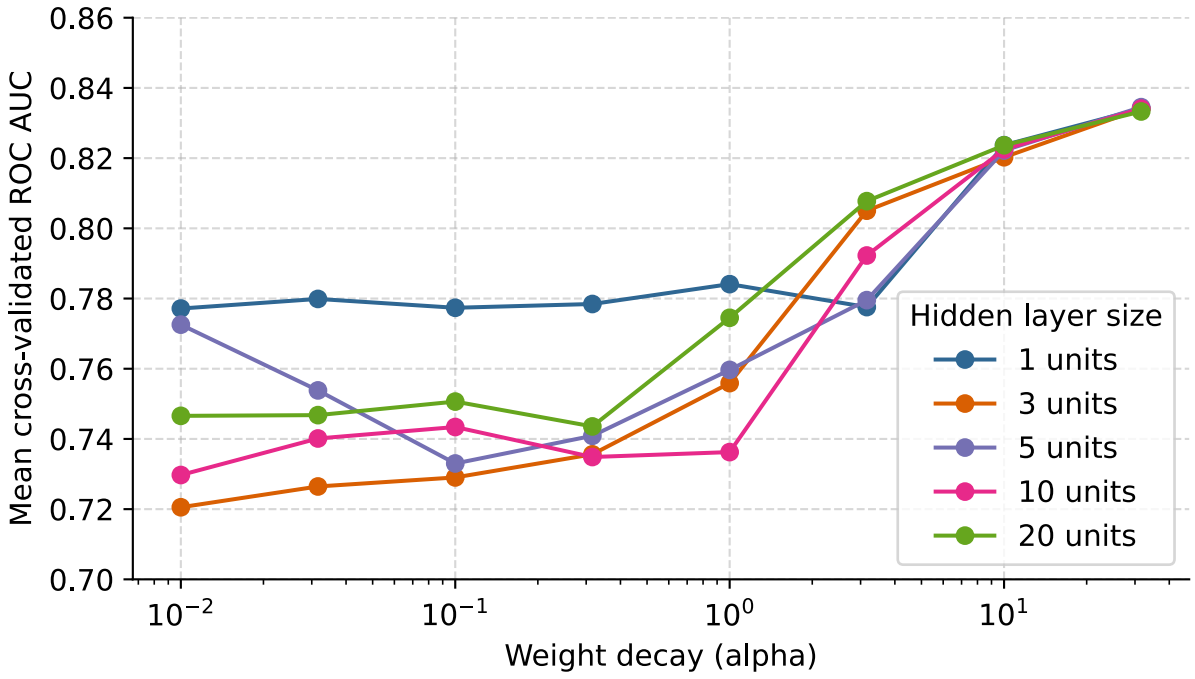


Figure 15.4: Cross-validated ROC AUC of neural networks with different numbers of hidden neurons and amounts of weight decay.

Figure 15.4 tells a clear story. With little weight decay, the networks overfit, and the bigger networks overfit most. With enough weight decay, networks of every size do about equally well. The penalty, not the size, is what matters most here.

```
best_idx = grid.best_index_
print("Best parameters:", grid.best_params_)
print(f"Best cross-validated ROC AUC: {grid.best_score_:.3f}")
```

```
Best parameters: {'mlp__alpha': 31.6, 'mlp__hidden_layer_sizes': (5,)}
Best cross-validated ROC AUC: 0.835
```

15.3.3 - A fair comparison

The best network has a cross-validated AUC of 0.835. For comparison, logistic regression on the exact same folds:

```
from sklearn.model_selection import cross_val_score

log_pipe = Pipeline([
    ('prep', dropout_recipe),
    ('clf', LogisticRegression(random_state=2026, max_iter=1000))
])

log_cv_scores = cross_val_score(log_pipe, X_train, y_train, cv=dropout_folds,
scoring='roc_auc')
print(f"Logistic regression CV ROC AUC: {log_cv_scores.mean():.3f}")
```

```
Logistic regression CV ROC AUC: 0.806
```

The two are practically identical: 0.835 for the network and 0.806–0.830 for logistic regression. The final test on the held-out students confirms it:

```
best_net = grid.best_estimator_
best_net_test_auc = roc_auc_score(y_test, best_net.predict_proba(X_test)[:, 1])
```

```
log_pipe.fit(X_train, y_train)
log_test_auc = roc_auc_score(y_test, log_pipe.predict_proba(X_test)[:, 1])

print(f"Test ROC AUC (Tuned Neural Network): {best_net_test_auc:.3f}")
print(f"Test ROC AUC (Logistic Regression):   {log_test_auc:.3f}")
```

```
Test ROC AUC (Tuned Neural Network): 0.826
Test ROC AUC (Logistic Regression):   0.843
```

The network reaches a test AUC of 0.83, against 0.84 for logistic regression on the same test students (Chapter 11). With only 150 test students, of whom 29 considered dropping out, a difference of this size is well within chance. The answer to the research group is therefore that *a neural network does not predict dropout better*. Chapters 12 and 13 found the same for random forests, support vector machines, and boosting. The wellbeing data has a smooth pattern, which logistic regression captures fully, leaving nothing extra for a more flexible model to find.

And the neural network has real costs. Even the tuned network has dozens of weights, which cannot be interpreted: there is no equivalent of an odds ratio saying how much each predictor matters. Without strong weight decay, its results depend on the random starting weights. And it needed careful tuning to avoid overfitting. Logistic regression is kept, and the thesis reports that a neural network was tried and did not improve on it, which is itself a useful finding.

15.4 - When neural networks are worth using

Neural networks are not magic, but they are not overrated either; they are powerful in particular situations. They excel with **unstructured data**, such as images, sound, and text, where the raw inputs (pixels, sound samples, words) mean little on their own and the useful features must be learned from them; here they outperform every other method by a wide margin. They need **very large datasets**, with many thousands or millions of cases, to fit models with thousands of weights reliably. And they can capture **complex, non-linear relationships** that simpler models miss, as the "In your field" box below shows.

For typical research data, a table with a few hundred cases and a few dozen variables, well-tuned regression, random forests, and boosting usually do as well or better, and are easier to explain. That is the situation of most theses.

15.5 - Deep learning

Deep learning means neural networks with many hidden layers, often dozens or hundreds, and millions or billions of weights. Each layer learns features built on the features of the layer before: in an image network, the first layers detect edges, later layers shapes, and the last layers whole objects. Special architectures suit special data: **convolutional networks** (CNNs) for images, and **transformers** for text. The large language models behind ChatGPT and Claude (Chapter 18) are transformer networks with billions of weights, trained on enormous amounts of text.

Deep learning needs more data, more computing power (often a GPU or specialised accelerator), and more technical setup than traditional tabular machine learning. In Python, the two dominant deep learning frameworks are:

- **PyTorch** (`torch`): Developed by Meta AI and widely used across academic research and modern AI development.
- **TensorFlow** and **Keras**: Developed by Google, providing high-level abstractions for defining and training neural networks.
- **Hugging Face Transformers**: The standard Python library for downloading, fine-tuning, and deploying modern pretrained transformers and language models.

For most researchers, the practical way to use deep learning is not to train a network from scratch, but to use a **pretrained model**, already trained by someone else on huge datasets, for a task such as transcribing interviews, recognising objects in images, or classifying text. Chapter 18 does exactly this with a language model, to code the study's open-ended survey answers.

In your field: engineering

The concrete data of Chapter 13 has the kind of curved, interacting relationships where a neural network can shine: strength rises steeply over the first weeks and then levels off, and the effect of each ingredient depends on the others. In scikit-learn, `MLPRegressor` handles regression tasks:

```
from sklearn.neural_network import MLPRegressor
from sklearn.linear_model import LinearRegression
from sklearn.metrics import root_mean_squared_error

concrete = dt.load_concrete()
cX = concrete.drop(columns=['compressive_strength']).copy()
cy = concrete['compressive_strength']

# Logarithm of age straightens the early curing curve
cX['age'] = np.log(cX['age'])

cX_train, cX_test, cy_train, cy_test = train_test_split(
    cX, cy, test_size=0.25, random_state=1
)

c_scaler = StandardScaler()
cX_train_scaled = c_scaler.fit_transform(cX_train)
cX_test_scaled  = c_scaler.transform(cX_test)

c_lm = LinearRegression()
c_lm.fit(cX_train_scaled, cy_train)
lm_rmse = root_mean_squared_error(cy_test, c_lm.predict(cX_test_scaled))

c_net = MLPRegressor(
    hidden_layer_sizes=(10,), alpha=0.1, max_iter=2000, random_state=1, solver='lbfgs'
)
c_net.fit(cX_train_scaled, cy_train)
net_rmse = root_mean_squared_error(cy_test, c_net.predict(cX_test_scaled))

print(f"Linear regression RMSE: {lm_rmse:.1f} MPa")
print(f"Neural network RMSE:    {net_rmse:.1f} MPa")
```

```
Linear regression RMSE: 7.6 MPa
Neural network RMSE:    7.1 MPa
```

The step taking the logarithm of age already straightens its curve for the linear model. Even so, the network's prediction error is clearly smaller: an RMSE of 7.1 megapascals, against 7.6 for the linear model. Here, there is real non-linear structure for its flexibility to capture.

15.6 - Common misconceptions

The reputation of neural networks produces several misconceptions that a thesis should avoid.

- **"A neural network is always more accurate."** On typical tables of research data it often predicts no better than regression, as this chapter found.
- **"Neural networks learn in a mysterious way."** Training is gradient descent: many small steps that each reduce the error, as the six-student example showed.
- **"A network that fits the training data perfectly has learned the pattern."** A perfect training score is usually a sign of memorisation; only data the network has not seen can show what it learned.
- **"Neural networks and AI are the same thing."** Neural networks are one family of methods; large language models are a particular, very large kind of network.

15.7 - Chapter review

15.7.1 - Summary

- A neuron multiplies its inputs by weights, adds a bias, and applies an activation function. A single neuron with a sigmoid activation is a logistic regression.
- A multilayer perceptron connects neurons in layers: inputs, one or more hidden layers, and an output layer. Hidden neurons with curved activations let the network represent curves and interactions.
- Networks are trained by gradient descent: starting from random weights, each step moves every weight a little in the direction that reduces the error. For a single neuron, gradient descent arrives at the logistic regression coefficients. With many weights, networks overfit easily; weight decay (`alpha`, a penalty on large weights) prevents it.
- In scikit-learn, `MLPClassifier` fits a multilayer perceptron; tune `hidden_layer_sizes` and `alpha` with cross-validation.
- On the dropout data, a tuned neural network predicts no better than logistic regression, and is harder to interpret. Neural networks shine on images, sound, and text, on very large datasets, and on strongly non-linear problems.
- Deep learning uses many layers; in Python, it is available through PyTorch, TensorFlow/Keras, and Hugging Face. Pretrained models are often the practical route for researchers.

15.7.2 - Key terms

Neural network, neuron (unit), weight, bias, activation function, sigmoid, ReLU, multilayer perceptron, input layer, hidden layer, output layer, training, log loss, gradient, gradient descent, learning rate, epoch, weight decay (`alpha`), deep learning, convolutional network, transformer, pretrained model.

15.8 - Exercises

The playground has these and more, with hints and solutions.

1. Change the neuron's weights so that support matters twice as much as in the chapter, recalculate the outputs for the two students, and explain what a negative weight means.
2. Fit a network with one hidden neuron (`hidden_layer_sizes=(1,)`) and a weight decay of 1.0, compare its cross-validated AUC with logistic regression, and explain why they might be so similar.
3. Fit the chapter's best network three times with different random seeds, then do the same with a weight decay of 0.01. Explain when the seed matters, and why.
4. Train the 20-neuron network without weight decay for 10, 100, and 1000 epochs (`max_iter`), and describe how the training and test AUCs change.
5. In two or three sentences, explain to the research group why the thesis does not use a neural network for the dropout model.
6. Repeat the gradient-descent loop with a learning rate of 0.1 and of 10. Describe how the loss curve changes, and explain what goes wrong when the learning rate is too large.

15.9 - Further reading

- *An Introduction to Statistical Learning with Applications in Python* [28] has an accessible chapter on deep learning, covering multilayer perceptrons, convolutional networks, and recurrent models.
- *Deep Learning with Python* [29] by the creator of Keras is the clearest intuitive introduction to practical deep learning.
- *Deep Learning* [30] is the standard foundational textbook, free to read online.
- *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* [21] shows how neural networks fit into the wider Python machine learning ecosystem.

16. Time Series Forecasting

Much of the data that organisations keep is recorded over time: patients admitted each day, products sold each month, rainfall each week. Such a sequence of measurements taken at regular intervals is a **time series**, and it calls for a different way of thinking from the surveys of the previous chapters. The observations are not independent: this week's value is related to last week's, and the order of the observations carries information. The patterns are also of a particular kind. A series usually combines a slow **trend**, a **seasonal** pattern that repeats over a fixed period, and irregular **noise**, and understanding a series largely means separating these parts. Finally, the future of a series can never be known exactly, so a forecast is honestly a range of plausible values, not a single number.

This chapter develops these ideas and the methods built on them, using Python's standard data science and statistical libraries (`pandas` and `statsmodels`). The example comes from outside the thesis. The doctors at the university counselling service have heard about Elaf's wellbeing study and ask her to analyse their records: five years of weekly visit counts, with the service overwhelmed before exams and quiet in the summer, and staffing planned by guesswork. They want to know how many visits to expect next year, week by week. This is an **internal analysis**, done for the service rather than for publication, and a common situation for anyone who knows some data analysis: being asked to analyse someone else's data.

By the end of this chapter you will be able to

- Explain dependence over time, the parts of a time series (trend, seasonality, and noise), and why forecasts are ranges.
- Store a time series in pandas with a datetime index, and plot it.
- Recognise trend, seasonality, and unusual observations, using time plots, seasonal plots, and autocorrelation.
- Decompose a series into trend, seasonal, and remainder components with STL.
- Produce benchmark forecasts (mean, naive, seasonal naive), and forecasts from exponential smoothing (ETS) and ARIMA models.
- Evaluate forecasts on a test period, and report forecasts with prediction intervals.

Note

Teaching & Practice: Interactive Lecture Slides · Browser Playground Exercises

16.1 - Time series data

Most of this book's methods assume that observations are independent: one student's answers tell you nothing about another's. In a time series the opposite is true. The number of visits this week is closely related to the number last week, and the order of the observations is essential. The methods of this chapter are built around that dependence.

16.1.1 - A small example

Two weeks of daily visits to a small clinic, starting on a Monday, show how a time series is stored in pandas:

```
import pandas as pd
import numpy as np

clinic = pd.DataFrame({
    'day': range(1, 15),
    'visits': [12, 10, 9, 11, 7, 3, 2,  14, 11, 10, 12, 8, 4, 2]
}).set_index('day')
```

```
clinic.head()
```

day	visits
1	12
2	10
3	9
4	11
5	7

In pandas, a time series is stored with a dedicated **index** representing time (either integers representing regular intervals or a `DatetimeIndex`). The observations are spaced regularly, and operations maintain temporal ordering. There is a clear weekly pattern: busy at the start of the week, quiet at the weekend.

16.1.2 - The parts of a pattern

A time series is easiest to understand as a sum of parts. The code below builds an artificial weekly series for three years from three ingredients: a trend that rises slowly, a seasonal pattern that repeats every 52 weeks, and random noise. Figure 16.1 shows each part and their sum.

```
np.random.seed(4)
weeks = np.arange(1, 157)
trend = 30 + 0.05 * weeks
seasonal = 12 * np.sin(2 * np.pi * weeks / 52)
noise = np.random.normal(0, 3, 156)
series = trend + seasonal + noise

fig, axes = plt.subplots(4, 1, figsize=(7.5, 5.5), sharex=True)
panels = [
    (trend, "Trend"),
    (seasonal, "Seasonal"),
    (noise, "Noise"),
    (series, "Series (Sum)")
]

for ax, (data, title) in zip(axes, panels):
    ax.plot(weeks, data, color="#2f6793", linewidth=1.5)
    ax.set_ylabel(title)
    ax.grid(True, linestyle="--", alpha=0.5)

axes[-1].set_xlabel("Week")
plt.tight_layout()
plt.show()
```

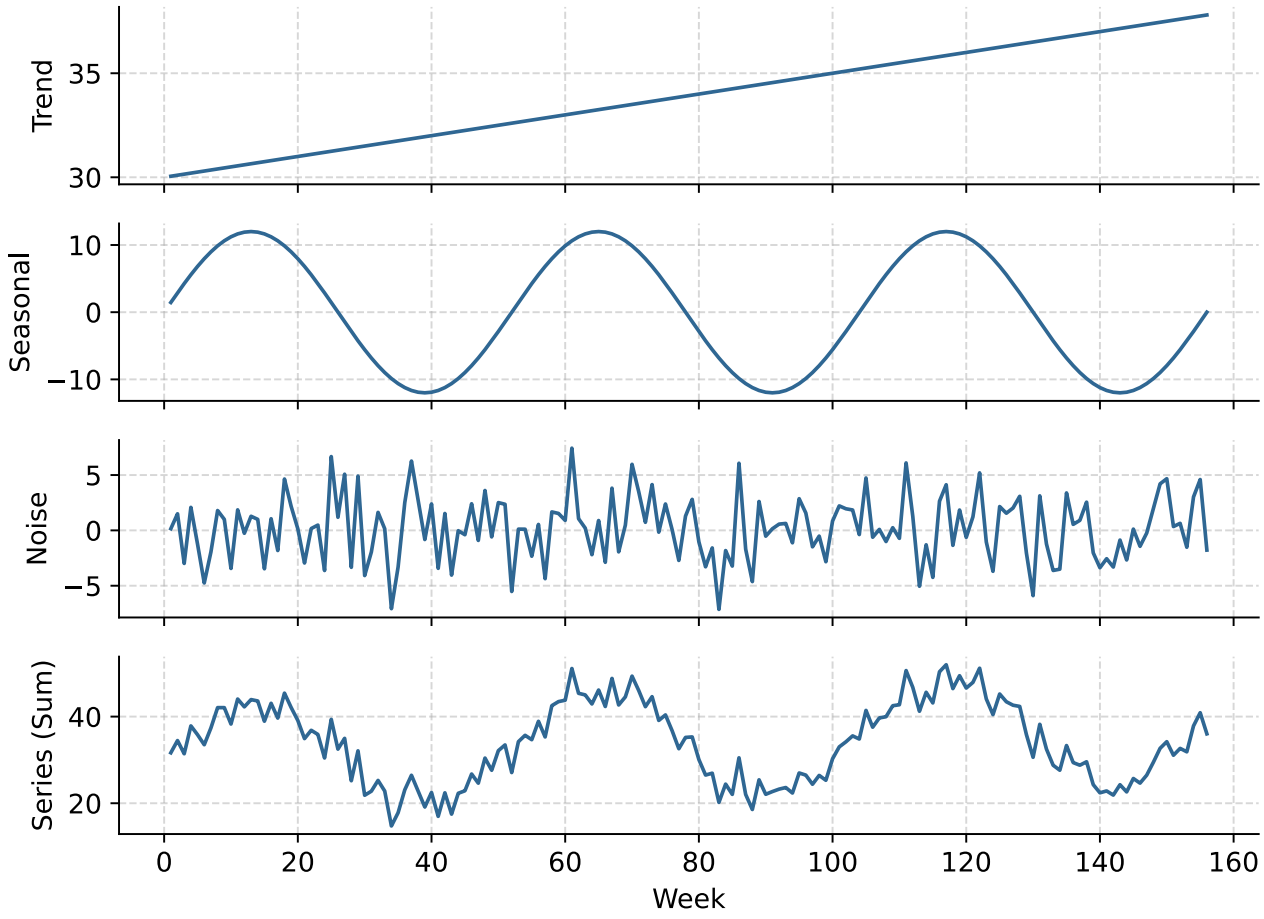


Figure 16.1: An artificial weekly series built from three parts: a rising trend, a seasonal pattern that repeats every 52 weeks, and random noise. The bottom panel is their sum.

In a real series, only the bottom panel is observed; the parts must be recovered from it. The trend says where the series is heading, the seasonal pattern says what happens at each point in the year, and the noise is what no pattern explains. Forecasting extends the trend and the seasonal pattern into the future. The noise cannot be forecast, and it is the main reason every forecast comes with a range. The decomposition later in this chapter recovers these parts from the counselling data.

16.2 - The counselling service data

The counselling records are loaded via `load_counselling_visits()`: the date of the Monday of each week, and the number of visits that week.

```
counselling_visits = dt.load_counselling_visits()
counselling_visits['week_start'] = pd.to_datetime(counselling_visits['week_start'])
print(counselling_visits.head())
print(f"Date range: {counselling_visits['week_start'].min().date()} to
{counselling_visits['week_start'].max().date()}")
print(f"Total weeks: {len(counselling_visits)}")
```

```
  week_start  visits
0 2020-09-07      32
1 2020-09-14      28
2 2020-09-21      24
3 2020-09-28      37
4 2020-10-05      32
```

```
Date range: 2020-09-07 to 2025-08-25
Total weeks: 260
```

The data covers five academic years, from September 2020 to August 2025, 260 weeks in all. Setting the datetime column as the index gives a clean time series structure:

```
visits = counselling_visits.set_index('week_start').sort_index()
visits.head()
```

week_start	visits
2020-09-07	32
2020-09-14	28
2020-09-21	24
2020-09-28	37
2020-10-05	32

The first step with any time series is to draw a **time plot**:

```
plt.figure(figsize=(8, 3.8))
plt.plot(visits.index, visits['visits'], color="#2f6793", linewidth=1.5)
plt.ylabel("Visits per week")
plt.grid(True, linestyle="--", alpha=0.5)
plt.tight_layout()
plt.show()
```

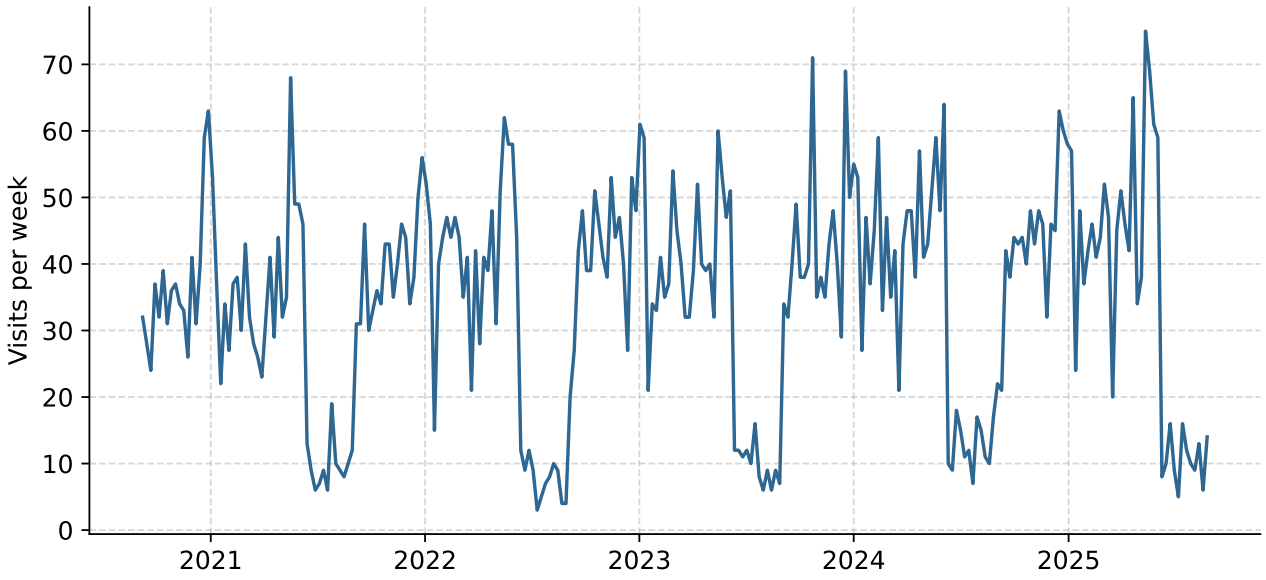


Figure 16.2: Weekly visits to the university counselling service, September 2020 to August 2025.

Figure 16.2 shows the parts of a pattern described above, and one feature the doctors did not mention. There is a **seasonal pattern** that repeats every academic year: peaks before and during the two exam periods, dips during the breaks, and very few visits in the summer, when the service is reduced. There is a gentle upward **trend**, each year a little busier than the last, and there is **noise**, week-to-week variation that follows no pattern. Finally, there is one **unusual week**, in the autumn of 2023, far busier than the weeks around it.

16.2.1 - Seasonal patterns

A **seasonal plot** puts the years on top of each other, so the seasonal pattern and the differences between years are easier to see. The academic year starts in September, so the weeks are numbered from 1 to 52 across each academic year:

```
visits_seasonal = visits.copy()
visits_seasonal['year_idx'] = np.arange(len(visits_seasonal)) // 52
visits_seasonal['week_of_year'] = (np.arange(len(visits_seasonal)) % 52) + 1
visits_seasonal['academic_year'] = visits_seasonal['year_idx'].apply(lambda i: f"20{20+i}/
{21+i}")

plt.figure(figsize=(7.5, 4))
colors = plt.cm.viridis(np.linspace(0.1, 0.9, 5))

for (acad_year, group), col in zip(visits_seasonal.groupby('academic_year'), colors):
          plt.plot(group['week_of_year'], group['visits'], label=acad_year, color=col,
linewidth=1.8)

plt.xlabel("Week of the academic year")
plt.ylabel("Visits per week")
plt.legend(title="Academic year")
plt.grid(True, linestyle="--", alpha=0.5)
plt.tight_layout()
plt.show()
```

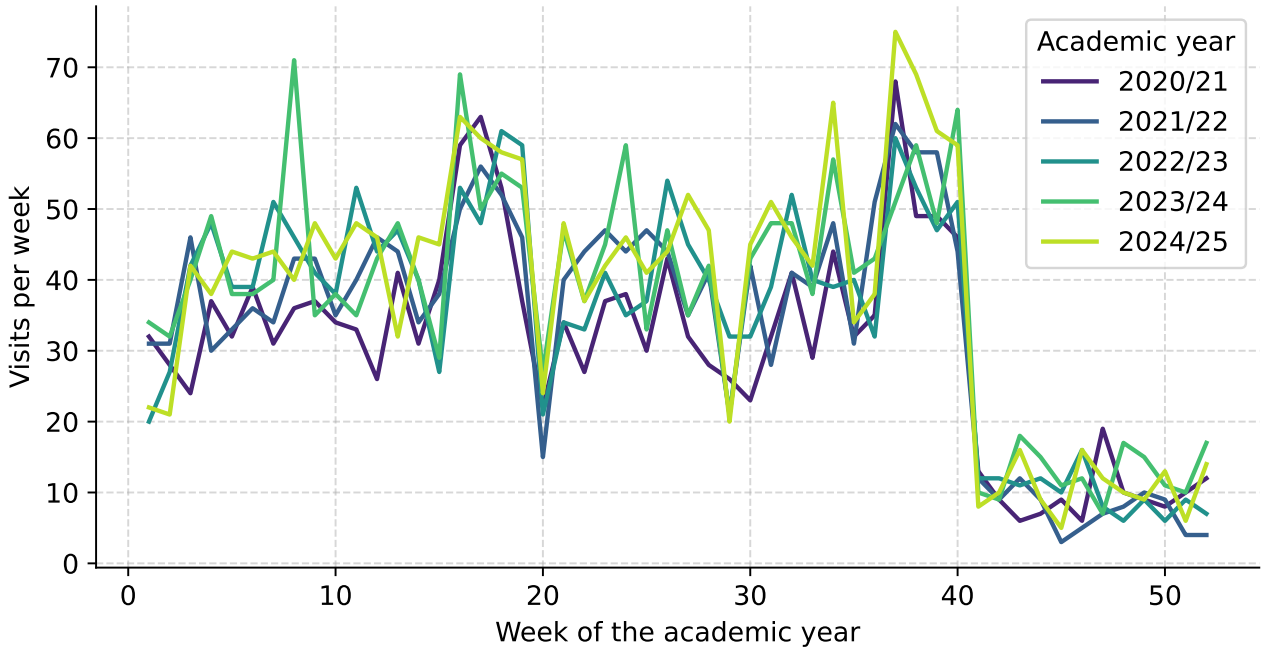


Figure 16.3: Seasonal plot: weekly visits by week of the academic year, one line per year.

Every year has the same shape, and later years lie slightly above earlier ones. The period of the seasonal pattern is 52 weeks.

16.2.2 - Autocorrelation

Dependence over time can be seen directly by plotting each week's visits against the visits one week earlier, as in Figure 16.4.

```
plt.figure(figsize=(5, 4))
plt.scatter(visits['visits'].shift(1),    visits['visits'],    color="#2f6793",    alpha=0.6,
edgecolors="none")
plt.xlabel("Visits in the previous week (lag 1)")
plt.ylabel("Visits this week")
plt.grid(True, linestyle="--", alpha=0.5)
plt.tight_layout()
plt.show()
```

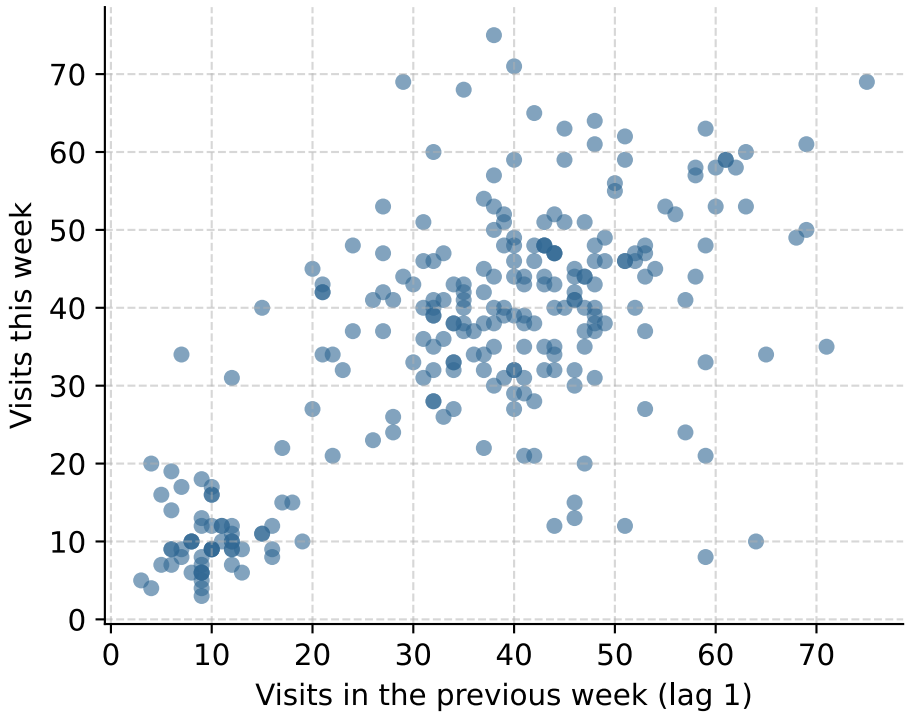


Figure 16.4: Weekly visits against the visits of the previous week. Busy weeks tend to follow busy weeks, and quiet weeks quiet weeks.

The points rise from left to right: knowing last week's visits says a good deal about this week's. The **autocorrelation** at lag k puts a number on this relationship: it is the correlation between the series and itself k steps earlier, between each week and the week before (lag 1), two weeks before (lag 2), and so on. In Python, `acf()` from `statsmodels.tsa.stattools` calculates it:

```
from statsmodels.tsa.stattools import acf

visits_acf = acf(visits['visits'], nlags=60)
print(f"Lag 1 autocorrelation:  {visits_acf[1]:.2f}")
print(f"Lag 26 autocorrelation: {visits_acf[26]:.2f}")
print(f"Lag 52 autocorrelation: {visits_acf[52]:.2f}")
```

```
Lag 1 autocorrelation:  0.69
Lag 26 autocorrelation: -0.24
Lag 52 autocorrelation: 0.68
```

The autocorrelation is 0.69 at lag 1, as the scatter plot suggested, and 0.68 at lag 52: a week tends to be like the same week a year earlier. At lag 26 it is negative (−0.24): half a year apart, a busy term week is often matched with a quiet summer week. Figure 16.5 shows all the lags. Strong autocorrelation at the seasonal lag is the signature of seasonality; if a series had no autocorrelation at all, there would be nothing to forecast beyond its average.

```
from statsmodels.graphics.tsaplots import plot_acf

fig, ax = plt.subplots(figsize=(7, 3.2))
plot_acf(visits['visits'],   lags=60,   ax=ax,   color="#2f6793",   vlines_kwargs={"colors":
"#2f6793"})
ax.set_xlabel("Lag (weeks)")
ax.set_ylabel("Autocorrelation")
ax.grid(True, linestyle="--", alpha=0.5)
plt.tight_layout()
plt.show()
```

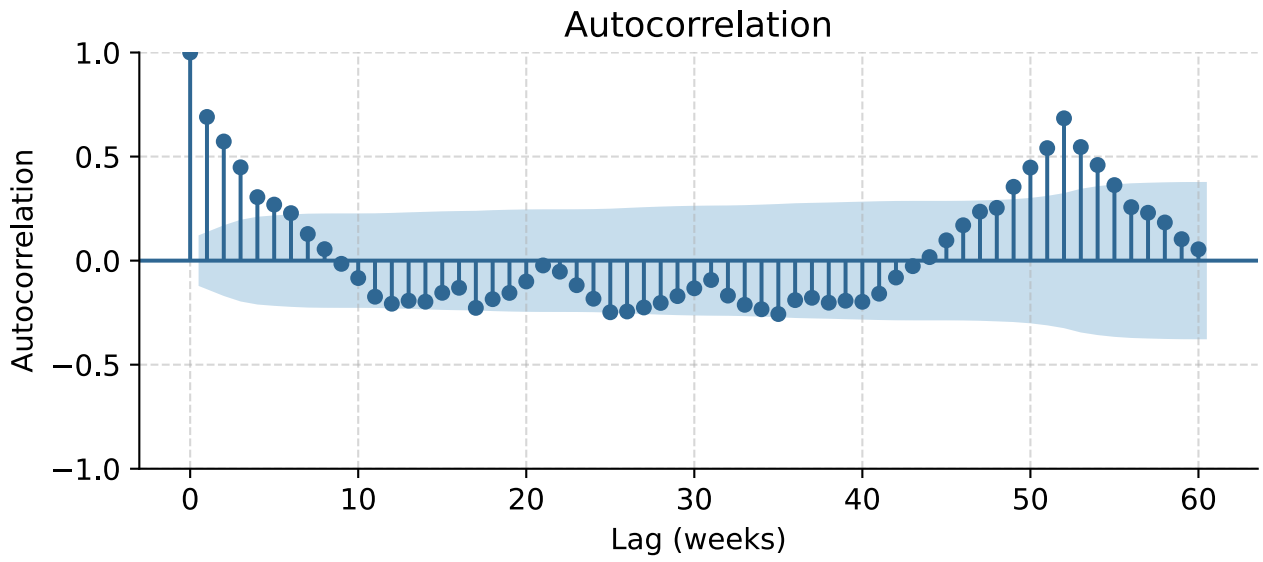


Figure 16.5: Autocorrelation of weekly visits for lags of 1 to 60 weeks. The dashed lines mark the range expected for a series with no autocorrelation.

16.3 - Decomposition

Decomposition recovers the parts of a pattern from an observed series: a smooth **trend**, a repeating **seasonal** component, and the **remainder**, what is left over, which corresponds to the noise of Figure 16.1. **STL** (seasonal and trend decomposition using loess) is a flexible, widely used method available in `statsmodels.tsa.seasonal.STL`. Setting `period=52` specifies the annual cycle, and `robust=True` prevents extreme weeks from distorting the trend:

```
from statsmodels.tsa.seasonal import STL

stl = STL(visits['visits'], period=52, robust=True)
visits_stl = stl.fit()
```

```
fig, (ax1, ax2, ax3, ax4) = plt.subplots(4, 1, figsize=(8, 6), sharex=True)

ax1.plot(visits.index, visits['visits'], color="#2f6793", linewidth=1.2)
ax1.set_ylabel("Data")
ax2.plot(visits.index, visits_stl.trend, color="#2f6793", linewidth=1.2)
ax2.set_ylabel("Trend")
ax3.plot(visits.index, visits_stl.seasonal, color="#2f6793", linewidth=1.2)
ax3.set_ylabel("Seasonal")
ax4.plot(visits.index, visits_stl.resid, color="#2f6793", linewidth=1.2)
ax4.set_ylabel("Remainder")

for ax in (ax1, ax2, ax3, ax4):
```

```
    ax.grid(True, linestyle="--", alpha=0.5)

plt.tight_layout()
plt.show()
```

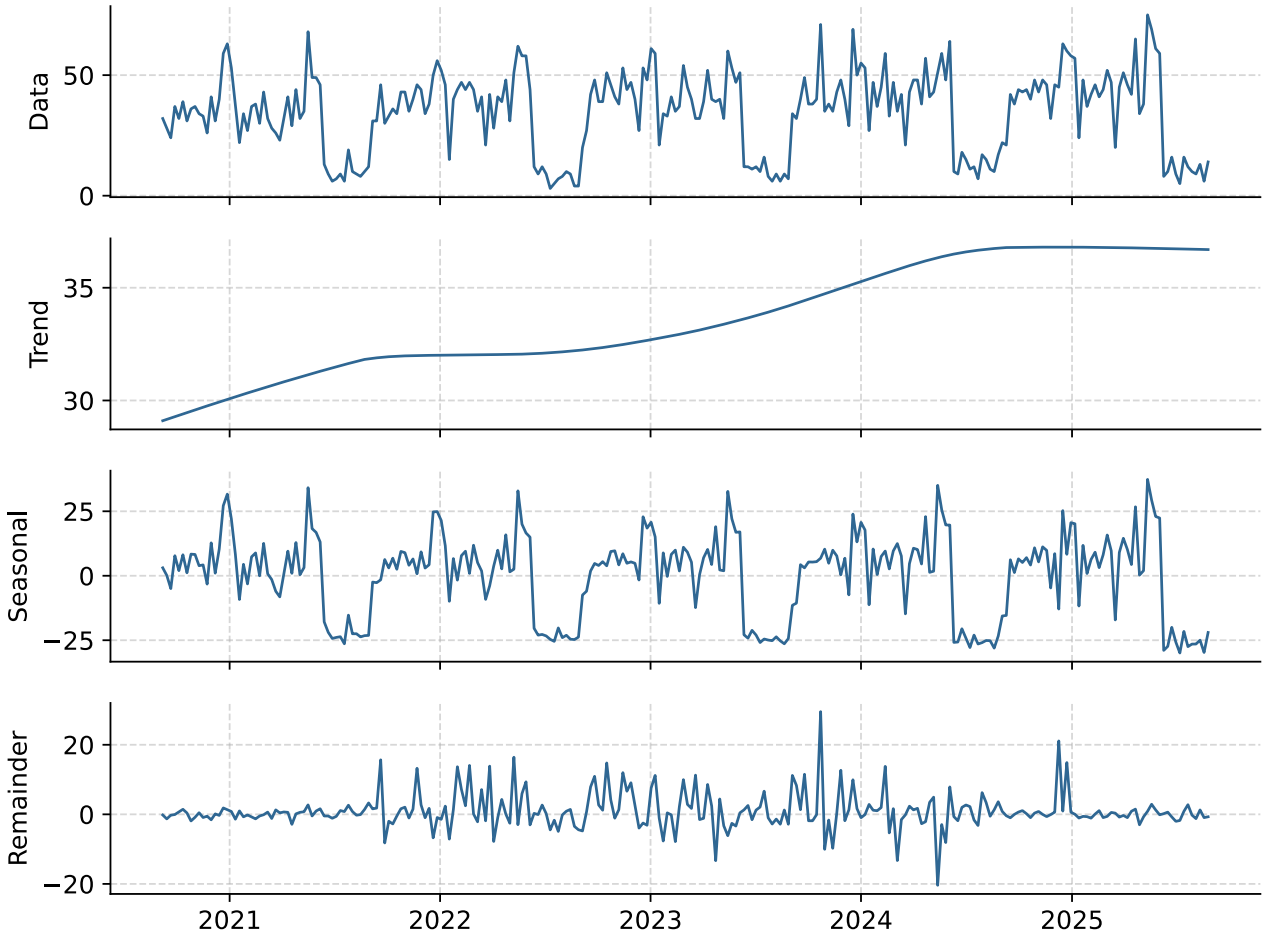


Figure 16.6: STL decomposition of weekly visits into trend, seasonal pattern, and remainder.

In Figure 16.6 the series is the sum of the three components below it, just as in the artificial example. The trend rises from about 35 to about 41 visits a week, levelling off in the last year; the seasonal component repeats the academic year; and the remainder is mostly small noise. The largest remainder is the unusual week:

```
decomp_df = pd.DataFrame({
    'week': visits.index,
    'visits': visits['visits'].values,
    'trend': visits_stl.trend.values,
    'seasonal': visits_stl.seasonal.values,
    'remainder': visits_stl.resid.values
})

top_remainders = decomp_df.iloc[decomp_df['remainder'].abs().argsort()[::-1][:3]]
print(top_remainders[['week', 'visits', 'trend', 'seasonal', 'remainder']].round(1))
```

```
          week  visits  trend  seasonal  remainder
163 2023-10-23      71   34.7       6.8       29.5
222 2024-12-09      45   36.8     -12.9       21.1
192 2024-05-13      51   36.3      35.1      -20.4
```

In the week of 23 October 2023, there were 71 visits, about 30 more than trend and season would predict. When asked, the doctors remember at once: it was the university's mental health awareness week, with posters everywhere encouraging students to seek help. The remainder is where such events show up. It is a real week, not an error, so it is kept, and the robust decomposition makes sure it does not distort the forecasts.

Notice also that the seasonal swings grow slightly as the trend rises: the peaks get higher and the summer lows stay low. When the seasonal pattern grows in proportion to the level of the series, it is standard practice to model the **logarithm** of the series, on which the swings become additive and stable.

16.4 - Forecasting

16.4.1 - Simple benchmarks

Every forecasting method should be compared with simple **benchmark** methods, just as every predictive model in Chapters 11 to 13 was compared with a baseline. The three most common are:

1. **Mean method**: forecasts the average of all past observations.
2. **Naive method**: forecasts the last observed value (y_T).
3. **Seasonal naive method**: forecasts the value from the same season in the last cycle (y_{T+h-m}).

For the clinic example, with its weekly cycle of 7 days:

```
y_clinic = clinic['visits'].values

mean_fc   = np.repeat(np.mean(y_clinic), 7)
naive_fc  = np.repeat(y_clinic[-1], 7)
snaive_fc = y_clinic[-7:]

pd.DataFrame({
    'Day': range(15, 22),
    'Mean': mean_fc.round(1),
    'Naive': naive_fc,
    'Seasonal_Naive': snaive_fc
}).set_index('Day')
```

Day	Mean	Naive	Seasonal_Naive
15	8.2	2	14
16	8.2	2	11
17	8.2	2	10
18	8.2	2	12
19	8.2	2	8
20	8.2	2	4
21	8.2	2	2

The mean method forecasts 8.4 visits every day; the naive method repeats the last value (a quiet Sunday of 2 visits) for the whole week; and the seasonal naive method repeats last week's pattern, which for this series is clearly the most sensible.

16.4.2 - Two families of models

Two families of models go beyond the benchmarks:

- **Exponential smoothing (ETS)**: forecasts with weighted averages of past observations, with weights that decrease exponentially into the past so that recent weeks count most. ETS models can track a changing level, trend, and seasonality.
- **ARIMA**: models describe how each observation depends on earlier observations (autoregression) and on earlier random shocks (moving average), using the autocorrelation structure of the series.

Both work best with short seasonal periods, such as 4 quarters, 7 days, or 12 months. A 52-week season is long, so two strategies are common: 1. **Decompose first (STL + ETS)**: decompose the log series with STL, forecast the seasonally adjusted series with ETS, and add the seasonal component back. 2. **Fourier terms with ARIMA**: model the 52-week seasonal wave using pairs of sine and cosine terms (Fourier series) as external regressors.

16.4.3 - Training and test periods

As in machine learning, forecasts must be judged on data the model has not seen. For time series, the test data must come *after* the training data, because a forecast can only use the past. The models are trained on the first four academic years (208 weeks) and tested on the fifth (52 weeks):

```
train = visits[visits.index < '2024-09-01'].copy()
test  = visits[visits.index >= '2024-09-01'].copy()

y_train = train['visits'].values
y_test  = test['visits'].values
h = len(test)

# Benchmarks
fc_mean   = np.repeat(np.mean(y_train), h)
fc_naive  = np.repeat(y_train[-1], h)
fc_snaive = y_train[-52:]

# Model 1: STL + ETS
from statsmodels.tsa.api import Holt

log_y_train = np.log(y_train)
stl_tr = STL(log_y_train, period=52, robust=True).fit()
sa_tr = log_y_train - stl_tr.seasonal

ets_fit = Holt(sa_tr, initialization_method='estimated').fit()
fc_sa = ets_fit.forecast(h)
fc_stl_ets = np.exp(fc_sa + stl_tr.seasonal[-52:])

# Model 2: Fourier + ARIMA
from statsmodels.tsa.arima.model import ARIMA

K = 6
t_tr = np.arange(len(y_train))
t_te = np.arange(len(y_train), len(y_train) + h)
X_tr = np.column_stack([fn(2 * np.pi * k * t_tr / 52) for k in range(1, K + 1) for fn in
(np.sin, np.cos)])
X_te = np.column_stack([fn(2 * np.pi * k * t_te / 52) for k in range(1, K + 1) for fn in
(np.sin, np.cos)])

arima_fit = ARIMA(log_y_train, order=(1, 0, 0), exog=X_tr).fit()
fc_fourier = np.exp(arima_fit.forecast(h, exog=X_te))
```

Each model forecasts the 52 weeks of the test year, and we evaluate their accuracy using RMSE and MAE:

```
from sklearn.metrics import mean_squared_error, mean_absolute_error

models = {
    'Mean': fc_mean,
    'Naive': fc_naive,
    'Seasonal naive': fc_snaive,
    'STL + ETS': fc_stl_ets,
    'Fourier + ARIMA': fc_fourier
}

acc_table = []
```

```
for name, preds in models.items():
    rmse = np.sqrt(mean_squared_error(y_test, preds))
    mae = mean_absolute_error(y_test, preds)
    acc_table.append({'Model': name, 'RMSE': round(rmse, 2), 'MAE': round(mae, 2)})

pd.DataFrame(acc_table).sort_values('RMSE').reset_index(drop=True)
```

	Model	RMSE	MAE
0	Seasonal naive	9.16	6.96
1	STL + ETS	9.35	6.81
2	Fourier + ARIMA	11.37	8.93
3	Mean	18.67	16.33
4	Naive	27.54	23.50

The STL and ETS model is best: its forecasts are off by 6.8 visits a week on average (MAE), against 7.0 for the seasonal naive benchmark, which simply repeats the previous year. The mean and naive methods, which ignore the seasonal pattern, are far worse. The Fourier model does worse than the seasonal naive benchmark: the counselling service's pattern has sharp steps (the summer drop happens from one week to the next), and smooth waves struggle to follow them. Figure 16.7 compares the two best models with the actual visits.

```
plt.figure(figsize=(8, 3.8))
plt.plot(test.index, y_test, color="black", linewidth=2, label="Actual visits")
plt.plot(test.index, fc_stl_ets, color="#2f6793", linewidth=1.8, label="STL + ETS")
plt.plot(test.index,    fc_snaive,    color="#d95f02",    linestyle="--",    linewidth=1.5,
label="Seasonal naive")

plt.ylabel("Visits per week")
plt.legend()
plt.grid(True, linestyle="--", alpha=0.5)
plt.tight_layout()
plt.show()
```

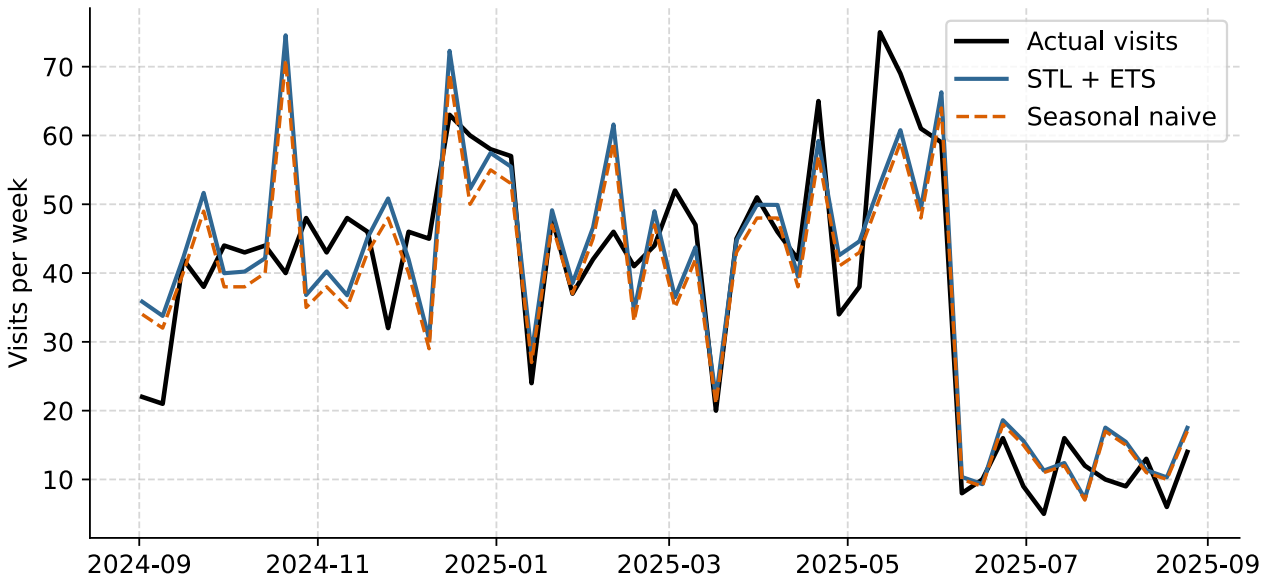


Figure 16.7: Forecasts for the test year (September 2024 to August 2025) from the STL and ETS model and the seasonal naive benchmark, with the actual visits in black.

The seasonal naive forecast copies last year's noise along with its pattern: it even repeats the awareness week of October 2023 as a spike in October 2024. The STL and ETS model smooths the noise away and adds the trend, so its forecasts are both smoother and closer.

16.4.4 - Forecasting next year

With the model chosen, it is refitted on all five years of data (260 weeks), so the forecasts use the most recent information, and the next 52 weeks are forecast:

```
y_all = visits['visits'].values
log_y_all = np.log(y_all)

stl_full = STL(log_y_all, period=52, robust=True).fit()
sa_full = log_y_all - stl_full.seasonal

ets_full = Holt(sa_full, initialization_method='estimated').fit()
fc_sa_next = ets_full.forecast(52)
fc_log_next = fc_sa_next + stl_full.seasonal[-52:]
fc_next_year = np.exp(fc_log_next)

next_dates = pd.date_range(start=visits.index[-1] + pd.Timedelta(weeks=1), periods=52,
freq='W-MON')
```

```
resid = stl_full.resid
sigma = np.std(resid)

lower_80 = np.exp(fc_log_next - 1.28 * sigma)
upper_80 = np.exp(fc_log_next + 1.28 * sigma)
lower_95 = np.exp(fc_log_next - 1.96 * sigma)
upper_95 = np.exp(fc_log_next + 1.96 * sigma)

recent = visits[visits.index >= '2023-09-01']

plt.figure(figsize=(8, 3.8))
plt.plot(recent.index, recent['visits'], color="black", label="Observed", linewidth=1.5)
plt.plot(next_dates, fc_next_year, color="#2f6793", label="Forecast", linewidth=2)

plt.fill_between(next_dates, lower_95, upper_95, color="#2f6793", alpha=0.15, label="95%
Interval")
plt.fill_between(next_dates, lower_80, upper_80, color="#2f6793", alpha=0.25, label="80%
Interval")

plt.ylabel("Visits per week")
plt.legend(loc="upper left")
plt.grid(True, linestyle="--", alpha=0.5)
plt.tight_layout()
plt.show()
```

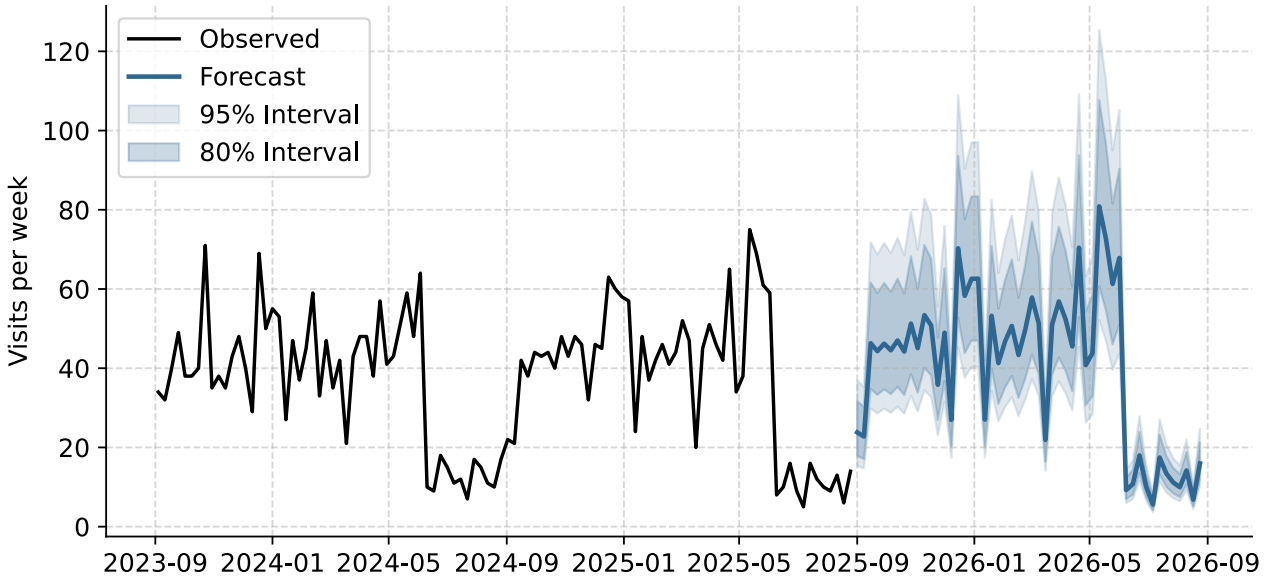


Figure 16.8: Forecast of weekly visits for the academic year 2025/26, with 80% and 95% prediction intervals, after the last two years of data.

A forecast is never a single number. The shaded bands in Figure 16.8 are **prediction intervals**: the ranges within which the actual visits are expected to fall with 80% and 95% probability.

```
next_df = pd.DataFrame({
    'Week': next_dates,
    'Forecast': fc_next_year.round(1),
    'Lower 95%': lower_95.round(1),
    'Upper 95%': upper_95.round(1)
}).set_index('Week')

next_df.head(4)
```

Week	Forecast	Lower 95%	Upper 95%
2025-09-01	23.8	15.4	36.9
2025-09-08	22.7	14.7	35.3
2025-09-15	46.3	29.9	71.8
2025-09-22	44.3	28.6	68.7

In the first week of the new year, about 24 visits are expected, but anything from about 17 to 33 would not be surprising. The doctors also asked for the year as a whole. Adding up weekly forecasts gives the expected total, but not its uncertainty, because weeks do not vary independently. We estimate this by **simulation**: producing 1,000 possible futures from the model by sampling the empirical residuals:

```
np.random.seed(2026)
sim_totals = []
for _ in range(1000):
    sim_noise = np.random.choice(resid, size=52, replace=True)
    sim_path = np.exp(fc_log_next + sim_noise)
    sim_totals.append(np.sum(sim_path))

q = np.quantile(sim_totals, [0.025, 0.5, 0.975]).round()
last_year_total = np.sum(test['visits'])
```

```
print(f"Predicted total visits for 2025/26: {int(q[1]):,}")
print(f"95% Prediction Interval:             {int(q[0]):,} to {int(q[2]):,}")
print(f"Actual total visits in 2024/25:      {int(last_year_total):,}")
```

```
Predicted total visits for 2025/26: 2,177
95% Prediction Interval:             2,050 to 2,464
Actual total visits in 2024/25:      1,954
```

The report to the service therefore says: about 2,177 visits are expected in 2025/26 (95% interval 2,050 to 2,464), compared with 2,056 in 2024/25; the busiest weeks will again be just before and during the two exam periods, when they should plan the most staff; and an awareness campaign can raise demand sharply for a week, so extra capacity should be planned for any such event.

⚠ Forecasts assume the future resembles the past

Every forecast in this chapter assumes that the patterns of the last five years continue. A change the data cannot know about, such as a new online booking system, a change in the exam calendar, or a crisis like the COVID-19 pandemic, can make any forecast wrong. Report forecasts with their intervals, say what they assume, and update them as new data arrives.

In your field: environmental science

The famous Mauna Loa dataset records the monthly concentration of atmospheric carbon dioxide (CO_2) in Hawaii from 1959 to 1997: a series with a strong upward trend and an annual photosynthetic cycle. With a 12-month season, ETS and SARIMAX model the seasonality directly. Training on the years up to 1990 and testing on 1991 to 1997:

```
import statsmodels.api as sm
from statsmodels.tsa.api import ExponentialSmoothing
from statsmodels.tsa.statespace.sarimax import SARIMAX

co2_raw = sm.datasets.co2.load_pandas().data
co2_m = co2_raw['co2'].resample('MS').mean().ffill()['1959-01-01':'1997-12-01']

co2_train = co2_m[:'1990-12-01']
co2_test  = co2_m['1991-01-01':]

# 1. Seasonal Naive
snaive_fc = np.tile(co2_train.iloc[-12:].values, len(co2_test) // 12)

# 2. ETS (Additive Trend and Seasonality)
ets_co2      =      ExponentialSmoothing(co2_train,      trend='add',      seasonal='add',
seasonal_periods=12).fit()
ets_co2_fc = ets_co2.forecast(len(co2_test))

# 3. SARIMAX
arima_co2   =   SARIMAX(co2_train,   order=(1,   1,   1),   seasonal_order=(1,   1,   1,
12)).fit(disp=False)
arima_co2_fc = arima_co2.forecast(len(co2_test))

print(f"Seasonal Naive MAE: {mean_absolute_error(co2_test, snaive_fc):.1f} ppm")
print(f"ETS MAE:            {mean_absolute_error(co2_test, ets_co2_fc):.1f} ppm")
print(f"SARIMAX MAE:        {mean_absolute_error(co2_test, arima_co2_fc):.1f} ppm")
```

```
Seasonal Naive MAE: 5.1 ppm
ETS MAE:            1.0 ppm
SARIMAX MAE:        1.1 ppm
```

Both ETS and ARIMA forecast seven years ahead with an average error (MAE) of only 1.0 to 1.1 parts per million, while the seasonal naive method, which ignores the trend, falls further behind every year.

16.5 - Common misconceptions

Time series have their own traps, most of them versions of forgetting that time has an order.

- **"A forecast is a prediction of the exact value."** A forecast is a range; the point forecast is only its centre, and the noise in a series makes exact prediction impossible.
- **"More history always gives better forecasts."** Only if the old patterns still hold; a change in circumstances can make older data misleading.
- **"The test data can be any random subset."** In a time series, the test period must come after the training period, since a forecast can only use the past.
- **"An unusual week should be removed."** It is part of the record; robust methods keep it from distorting the forecast, and it may be exactly what the service needs to plan for.

16.6 - Chapter review

16.6.1 - Summary

- A time series is a sequence of observations over time, in which order matters and neighbouring observations are related. It can be understood as a sum of parts: trend, seasonal pattern, and noise. In pandas, a datetime index stores and aligns temporal data.
- Time plots, seasonal plots, and the autocorrelation function reveal trend, seasonality, and unusual observations.
- STL decomposes a series into trend, seasonal, and remainder components; unusual events show up in the remainder. A logarithm makes growing seasonal swings constant.
- Compare every forecasting method with benchmarks: mean, naive, and seasonal naive.
- ETS forecasts with weighted averages that favour recent observations; ARIMA uses the autocorrelation of the series. For long seasons such as 52 weeks, decompose first or use Fourier terms.
- Evaluate forecasts on a test period that comes after the training period, and report forecasts with prediction intervals. Simulating futures gives intervals for cumulative totals.
- Forecasts assume that past patterns continue.

16.6.2 - Key terms

Time series, datetime index, time plot, trend, seasonality, seasonal period, seasonal plot, autocorrelation, lag, decomposition, STL, remainder, seasonally adjusted series, benchmark forecast, mean method, naive method, seasonal naive method, exponential smoothing (ETS), ARIMA, SARIMAX, Fourier terms, forecast horizon, prediction interval, simulation.

16.7 - Exercises

The playground has these and more, with hints and solutions.

1. Predict which of the three benchmarks would be best for a series with a trend but no seasonality, then test your prediction on `clinic` after adding a steady increase of one visit a day.
2. Plot the seasonally adjusted series (`sa = visits - stl.seasonal`), and describe what it shows that the original series hides.
3. Refit the Fourier model with $K = 2$ and $K = 12$, and explain how and why the test accuracy changes.
4. Train the models on the first three years and test on the fourth, and check whether the STL and ETS model is still the best.
5. Use simulation to estimate how many visits to expect in the four weeks before the first exam period of 2025/26, with a 95% interval.
6. Write a short paragraph for the counselling service explaining what a 95% prediction interval means.
7. In the artificial series of Figure 16.1, double the standard deviation of the noise. Describe how the sum changes, and explain what this would mean for the width of forecast intervals.

16.8 - Further reading

- *Forecasting: Principles and Practice* [31], free online, is the standard textbook on forecasting principles.
- *Time Series Analysis and Its Applications: With R Examples* [32] covers classical time series, ARIMA, and spectral methods.
- *Python for Data Analysis* [1] by the creator of pandas provides the standard reference for handling dates, time series, and resampling in Python.

IV Reproducible Research and Applications

17 Reproducible Research 331
17.1 Reproducibility and credibility 331
17.2 Documents that contain their own analysis . 332
17.3 A results chapter in Quarto 333
17.4 Keeping the same package versions 336
17.5 Interactive dashboards with Shiny for Python 337
17.6 Open science 340
17.7 Keeping a history with git (optional) 343
17.8 Common misconceptions 343
17.9 Chapter review 343
17.10 Exercises 344
17.11 Further reading 344

18 Using AI with Python 345
18.1 The reliability of coding 345
18.2 How AI assistants work 346
18.3 AI as a coding assistant 347
18.4 Coding open-ended survey answers 348
18.5 Calling a language model from Python 349
18.6 Evaluating AI qualitative coding 350
18.7 Using AI responsibly in academic research . 353
18.8 Common misconceptions 354
18.9 Chapter review 354
18.10 Exercises 354
18.11 Further reading 355

19 Putting It All Together 357
19.1 The shape of a research project 357
19.2 From raw export to clean data 359
19.3 Describing the sample 361
19.4 Answering the research questions 362
19.5 One figure for the thesis 363
19.6 The chain of reasoning 365
19.7 The results chapter 365
19.8 Choosing a method 366
19.9 Challenges every researcher meets 368
19.10 Where to go next 368
19.11 Common misconceptions 370
19.12 Chapter review 370
19.13 Exercises 371
19.14 Further reading 371

19.14 References 373

17. Reproducible Research

A research finding is credible only if it can be checked. A reader who doubts a result should be able to see exactly how it was obtained, and a second researcher who repeats the analysis should arrive at the same numbers. When that is not possible, the reader has to take the result on trust, and science is built on not having to. The failures that make results uncheckable are rarely dramatic. A table is copied into a draft before the data is corrected, a decision made while exploring is forgotten, a package changes its behaviour between versions, and nobody can say afterwards which of several analyses produced the published number. Reproducibility is therefore a matter of research integrity, not of technical tidiness.

The problem is easy to meet in a thesis. Three weeks before the deadline, Elaf's supervisor spots that twelve students appear twice in an early version of the survey export. They were removed in Chapter 3, but some tables were copied into the thesis draft before that, and now every table, figure, and number copied by hand from Python into Word must be checked and replaced, one at a time, with the risk of missing one. This chapter shows a way of working in which that problem cannot happen. In a **reproducible** workflow, the text, the code, and the results live together, and the whole report is rebuilt from the raw data with one command, so every number is always up to date. The chapter introduces **Quarto** for writing such documents, from a single page to a thesis chapter; **virtual environments** and **dependency files** (`requirements.txt`) for keeping the same package versions; **Shiny for Python** for turning an analysis into an interactive dashboard; and the practices of **open science** that make research checkable by others. An optional section introduces **git** for keeping the history of a project.

By the end of this chapter you will be able to

- Explain why reproducibility is a matter of research integrity, and why copying results by hand and flexible analyses are risky.
- Write a Quarto document that combines text, Python code, results, tables, figures, citations, and equations, and render it to HTML, Word, and PDF.
- Keep a project's package versions with virtual environments and `requirements.txt`, and report software versions.
- Build a small Shiny for Python dashboard with inputs, outputs, and reactive code.
- Apply open science practices: sharing data and code, protecting participants, preregistration, and citing software.

Note

Teaching & Practice: Interactive Lecture Slides · Browser Playground Exercises

17.1 - Reproducibility and credibility

An analysis is **reproducible** if someone else, or you in a year's time, can take the same data and code and get exactly the same results. It is **replicable** if a new study, with new data, reaches the same conclusions. Reproducibility is the minimum standard: if the original results cannot even be recomputed, there is little point asking whether they replicate [33].

Most irreproducibility is not fraud but everyday friction: a table copied before the data was corrected, a spreadsheet cell edited by hand, a result produced by clicking through menus that nobody wrote down, a package that changed its default between versions. The previous chapters have already built good habits against these. Every step is **written as code**, in Python scripts or notebooks, so it can be rerun (Chapter 1). Each analysis lives in a dedicated project directory with relative paths, so it runs on any computer (Chapter 1). The **raw data is never edited by hand**, and every cleaning decision is made and recorded in code (Chapter 3). Random steps use `np.random.seed()` or explicit random state generators, so they give the same answer every time (Chapter 7).

This chapter adds the last links: putting the results into the report automatically, fixing the software versions, and sharing everything responsibly. Good practices do not need to be perfect to be useful; even a few of them make research far easier to check [34].

17.2 - Documents that contain their own analysis

Quarto is a modern publishing system for documents that mix text and executable code. A Quarto document is a plain text file with the extension `.qmd`. When it is **rendered**, the code is executed, and its results (numbers, tables, figures) are placed directly into the finished document, which can be a web page, a Word document, a PDF, a slide show, or an entire book. This book itself is written in Quarto: every number, table, and figure in it was produced by the Python code shown next to it.

> **i** Jupyter Notebooks and Quarto
>
> In Python data science, **Jupyter Notebooks** (`.ipynb` files) are very popular for interactive exploration. However, notebooks store raw output and metadata as bulky JSON, which makes version control in Git difficult. **Quarto** (`.qmd` files) combines the best of both worlds: plain-text Markdown that is clean to track in Git, paired with a Jupyter kernel underneath that runs your Python code when rendering. Quarto works natively with Python, R, Julia, and Observable JavaScript. For reproducible thesis chapters and reports, Quarto plain text is the gold standard.

17.2.1 - A small example

A complete Quarto document can be very short:

````
---
title: "Three friends' sleep"
author: "Elaf"
format: html
jupyter: python3
---

My three friends slept 6.3 hours on average last night.

```{python}
import numpy as np
sleep = [6.5, 7.0, 5.5]
print(round(np.mean(sleep), 1))
```
````

It has the three parts of every Quarto document:

1. The **YAML header**, between the two `---` lines, holds settings: the title, the author, the output **format**, and the Python kernel (`jupyter: python3`).
2. The **text** is written in **Markdown**, a simple way of styling text: `**bold**`, `*italic*`, `# Heading`, and `-` for bullet points.
3. The **code chunks**, between `{python}` and ```, are executed when the document is rendered, and their outputs appear in the document.

The text can also contain **inline code**, such as `<code>{python} mean_sleep</code>`: Python variables evaluated directly inside a sentence. When the document is rendered, the inline code is evaluated and replaced by its numerical or text result. If a friend's number changes in the data, the sentence changes with it. Inline code is the cure for the problem at the start of the chapter: numbers in the text are never typed by hand.

In VS Code or Positron, you can edit Quarto documents with syntax highlighting, run code chunks interactively, and render them with a single click or keyboard shortcut.

17.2.2 - What happens when you render

Figure 17.1 shows the steps. The **Jupyter** execution engine runs the Python code and produces Markdown with the output figures and tables in place; **Pandoc**, a universal document converter, then transforms the Markdown into the requested format.

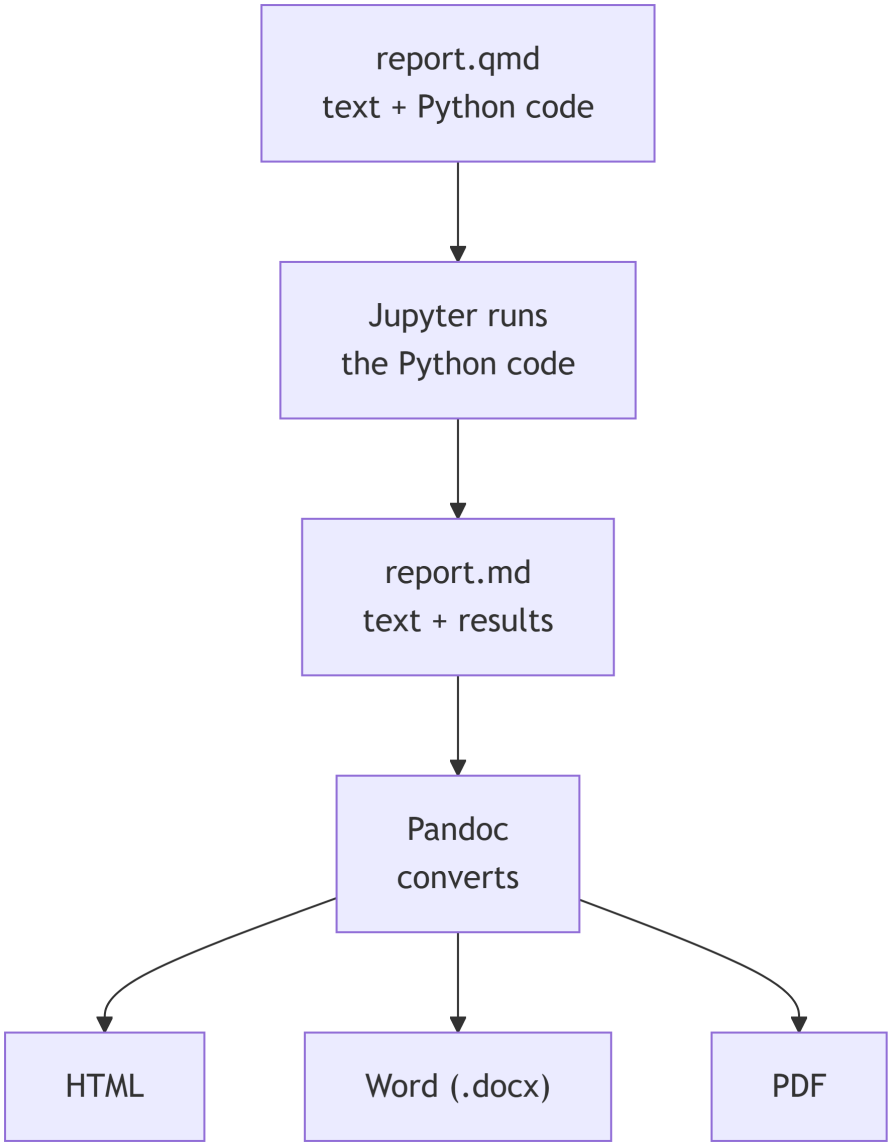


Figure 17.1: How a Quarto document is rendered with Python.

Because the code is run from the beginning every time, a rendered document is a test of reproducibility in itself: if the code depends on an unrecorded variable created in an interactive console, rendering fails immediately, alerting you to the missing step.

17.3 - A results chapter in Quarto

The descriptive results chapter of the thesis, written as a Quarto document, begins like this:

```
---
title: "Chapter 4: Results"
author: "Elaf"
format: docx
jupyter: python3
bibliography: references.bib
execute:
  echo: false
---

```{python}
```
```

````
#| label: setup
#| message: false
import pandas as pd
import data2thesis as dt

students = dt.students
semesters = dt.semesters

n_students = len(students)
n_faculties = students['faculty'].nunique()
mean_wb = round(semesters.loc[semesters['semester'] == 1, 'wellbeing'].mean(), 1)
```

The sample included 600 graduate students from 5 faculties. Their average wellbeing in the
first semester was 60.5 on the wellbeing index (0 to 100). Table @tbl-faculty shows wellbeing
by faculty.

```{python}
#| label: tbl-faculty
#| tbl-cap: "Wellbeing in the first semester, by faculty."
merged = semesters[semesters['semester'] == 1].merge(students, on='student_id')
summary_table = (
    merged.groupby('faculty')
    .agg(
        Students=('student_id', 'count'),
        Mean=('wellbeing', 'mean'),
        SD=('wellbeing', 'std')
    )
    .reset_index()
    .round(1)
)
summary_table.columns = ['Faculty', 'Students', 'Mean', 'SD']
summary_table
```

Wellbeing declined over the programme (@fig-wellbeing), as found in
earlier studies [@author2020].
````

Several features appear:

- In the header, `format: docx` produces a Word document (the format most supervisors prefer for track-changes feedback), and `bibliography:` names a file of references. The setting `execute: echo: false` hides Python code in the finished document, so the thesis presents only polished narrative, tables, and figures.
- In the body, **chunk options**, the lines starting with `#|`, control each chunk: `label` gives it a unique identifier, and `tbl-cap` or `fig-cap` provides a caption.
- A chunk labelled `tbl-...` or `fig-...` becomes a numbered table or figure. In the text, `@tbl-faculty` becomes a dynamic **cross-reference** ("Table 1").
- `[@author2020]` is a **citation**: Quarto looks up the key in the BibTeX file, formats the citation in the chosen style, and generates the reference list at the end.

When the document is rendered, the first paragraph displays:

> The sample included 600 graduate students from 5 faculties. Their average wellbeing in the first semester was 60.5 on the wellbeing index (0 to 100). Table 1 shows wellbeing by faculty.

and the table chunk produces Table 17.1:
````

Table 17.1: Wellbeing in the first semester, by faculty.

	Faculty	Students	Mean	SD
0	Education	148	61.9	13.2
1	Health Sciences	154	59.0	11.6
2	Humanities	95	61.7	11.7
3	Natural Sciences	87	61.3	10.5
4	Social Sciences	116	58.9	12.1

If the underlying dataset updates, the document is re-rendered, and every number, table, and figure updates in unison. The supervisor's request from the start of the chapter now takes one command to resolve.

17.3.1 - References and citations

The bibliography file uses the standard **BibTeX** format, which almost every reference manager (Zotero, Mendeley, EndNote) can export, and Google Scholar can provide for any paper:

```
@book{wickham2023,
  author    = {Wickham, Hadley and Çetinkaya-Rundel, Mine and Grolemund, Garrett},
  title     = {R for Data Science},
  publisher = {O'Reilly Media},
  year      = {2023}
}
```

The citation key, `wickham2023`, follows the `@` symbol in the text. A `csl:` line in the YAML header chooses the citation style (APA 7th, Vancouver, Harvard, IEEE, or journal-specific styles), meaning that changing target journals never requires reformatting references by hand.

17.3.2 - Equations

Equations are written in standard **LaTeX** notation between dollar signs: `$\bar{x} = \frac{1}{n}\sum x_i$` inside a sentence gives $\bar{x} = \frac{1}{n}\sum x_i$, and double dollar signs set an equation on its own line:

```
$$
\text{logit}(p) = \beta_0 + \beta_1 \times \text{stress} + \beta_2 \times \text{support}
$$
```

$$\text{logit}\,(p) = \beta_0 + \beta_1 \times \text{stress} + \beta_2 \times \text{support} \tag{17.1}$$

The same LaTeX notation renders identically across HTML, Word, and PDF output.

17.3.3 - Output formats

Changing one line in the YAML header switches the entire document's target:

Table 17.2: Common Quarto output formats

Format	YAML	Notes
Web page	`format: html`	Interactive; ideal for supplementary websites and sharing findings
Word	`format: docx`	For supervisory review; `reference-doc: template.docx` applies university formatting
PDF	`format: pdf` or `format: typst`	Typeset PDF; Typst is lightning-fast and modern
Slides	`format: revealjs`	For conference presentations and thesis defenses

Multiple formats can be specified at once, allowing a single `quarto render` call to generate both an accessible HTML web page and a formal Word thesis draft.

17.3.4 - Parameterised reports

Suppose the dean of each faculty requests a tailored summary for their faculty. Rather than maintaining five separate files, one report with a **parameter** serves them all:

````
---
title: "Faculty Wellbeing Report"
format: html
jupyter: python3
params:
  faculty: "Education"
---

```{python}
import data2thesis as dt
import pandas as pd

students = dt.students
semesters = dt.semesters

faculty_name = params['faculty']
merged = semesters.merge(students, on='student_id')
faculty_data = merged[merged['faculty'] == faculty_name]
n_fac_students = faculty_data['student_id'].nunique()
```

This report describes the 148 students of the Faculty of Education.
````

In Python, Quarto passes the parameters into a dictionary named `params`. Rendering from the command line with `-P` outputs customized reports for each faculty:

```
quarto render faculty-report.qmd -P faculty:"Health Sciences"
```

17.4 - Keeping the same package versions

Python libraries evolve rapidly. A function default may change, or a method may be deprecated, causing code that runs today to break or produce different estimates two years later. Reproducibility requires pinning exact software dependencies.

17.4.1 - Virtual environments and requirements.txt

A **virtual environment** is an isolated Python directory containing a specific Python interpreter and its installed packages. By creating an environment per project, updates to packages in one project never break another:

```
# 1. Create a dedicated virtual environment
python -m venv .venv

# 2. Activate it (Windows PowerShell)
.venv\Scripts\Activate.ps1

# 2b. Activate it (macOS/Linux)
source .venv/bin/activate

# 3. Install required packages
pip install pandas numpy scipy statsmodels scikit-learn matplotlib seaborn

# 4. Save exact versions to a lockfile
pip freeze > requirements.txt
```

Sharing `requirements.txt` alongside your thesis repository allows another researcher to recreate your exact environment with one command:

```
pip install -r requirements.txt
```

Modern Python tooling also offers lockfile tools such as **`uv`** (`uv pip compile` / `uv.lock`) or **Poetry**, which capture complete dependency graphs and cryptographic hashes for maximum reproducibility.

17.4.2 - Reporting software versions

Even without a formal lockfile, always **report the versions** of Python and key analysis libraries in your thesis methods section. In Python, you can inspect versions directly:

```
import sys
import statsmodels
import sklearn
import pandas as pd

print(f"Python: {sys.version.split()[0]}")
print(f"pandas: {pd.__version__}")
print(f"statsmodels: {statsmodels.__version__}")
print(f"scikit-learn: {sklearn.__version__}")
```

```
Python: 3.13.2
pandas: 2.3.3
statsmodels: 0.15.0
scikit-learn: 1.9.1
```

17.5 - Interactive dashboards with Shiny for Python

A static thesis report answers only the questions its author anticipated. Readers, reviewers, and policymakers often want to ask their own questions: *"What does the trend look like for my faculty?"* or *"What if we examine sleep duration rather than overall wellbeing?"*

Shiny for Python turns Python data analysis into an interactive web dashboard without requiring HTML, CSS, or JavaScript expertise.

A Shiny app is organized around three concepts:

1. **Inputs**: The interactive UI widgets the user controls (drop-downs, radio buttons, sliders).
2. **Outputs**: The visual elements the app displays (plots, dynamic tables, text cards).
3. **Reactivity**: The engine that links them. When an input changes, Shiny automatically re-executes only the outputs and calculations that depend on that specific input (Figure 17.2).

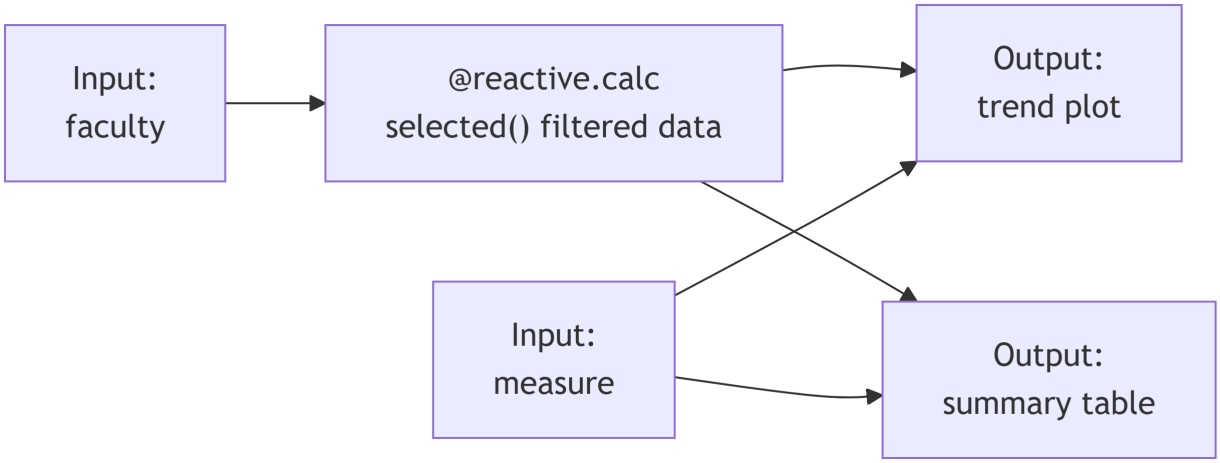


Figure 17.2: Reactivity in the wellbeing dashboard. When an input changes, only downstream elements update.

17.5.1 - A wellbeing dashboard

Here is the complete application code, saved as `app.py`:

```
from shiny import App, render, ui, reactive
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import data2thesis as dt

students = dt.students
semesters = dt.semesters

wellbeing_data = semesters.merge(
    students[["student_id", "faculty", "study_mode"]],
    on="student_id"
)

measure_choices = {
    "wellbeing": "Wellbeing (0-100)",
    "sleep_hours": "Sleep (hours a night)",
    "study_hours": "Study (hours a week)"
}

app_ui = ui.page_sidebar(
    ui.sidebar(
        ui.input_select(
            "faculty", "Faculty",
            choices=sorted(wellbeing_data["faculty"].unique().tolist())
        ),
        ui.input_radio_buttons(
            "measure", "Measure",
            choices=measure_choices
        )
    ),
    ui.card(ui.output_plot("trend")),
    ui.card(ui.output_table("summary")),
    title="Graduate Student Wellbeing Dashboard"
)

def server(input, output, session):
    @reactive.calc
```

```
    def selected():
        return wellbeing_data[wellbeing_data["faculty"] == input.faculty()]

    @render.plot
    def trend():
        df = selected()
        meas = input.measure()
        summary = (
            df.groupby(["semester", "study_mode"])[meas]
            .mean()
            .reset_index()
        )
        fig, ax = plt.subplots(figsize=(6.5, 3.8))
        sns.lineplot(
            data=summary, x="semester", y=meas, hue="study_mode",
            marker="o", linewidth=2.5, markersize=8, ax=ax
        )
        ax.set_xlabel("Semester")
        ax.set_ylabel(measure_choices[meas])
        ax.set_xticks([1, 2, 3, 4])
        ax.grid(True, linestyle="--", alpha=0.5)
        plt.tight_layout()
        return fig

    @render.table
    def summary():
        df = selected()
        meas = input.measure()
        tbl = (
            df.groupby("semester")
            .agg(
                students=("student_id", "nunique"),
                mean=(meas, "mean")
            )
            .reset_index()
            .round(2)
        )
        tbl.columns = ["Semester", "Students", "Mean"]
        return tbl

app = App(app_ui, server)
```

The structure is clear:

- `app_ui` defines the layout: `page_sidebar()` creates a modern sidebar containing `input_select()` and `input_radio_buttons()`, alongside two content cards reserving spots for `output_plot("trend")` and `output_table("summary")`.
- `server` holds the logic: `@reactive.calc` defines `selected()`, an intermediate filtered DataFrame that updates only when `input.faculty()` changes. Both outputs consume `selected()`.
- `@render.plot` and `@render.table` produce the visual elements dynamically.

Running `shiny run --reload app.py` launches the dashboard locally. Figure 17.3 shows the plot generated when Education is selected.

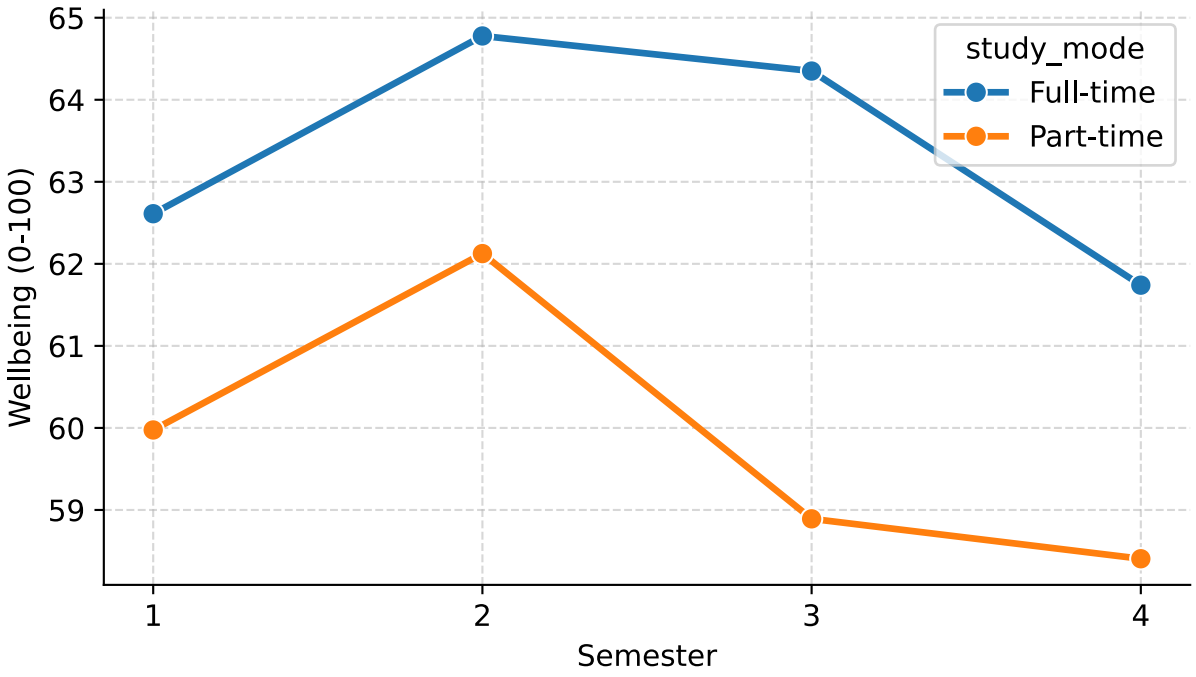


Figure 17.3: The dashboard's trend plot for the Faculty of Education, displaying mean wellbeing across semesters by study mode.

17.5.2 - Sharing dashboards

To share an app with stakeholders who do not have Python installed:

- **shinyapps.io**: Free and paid cloud hosting for Shiny apps.
- **Posit Connect**: Enterprise server commonly hosted by universities and research institutes.
- **Shinylive**: An extraordinary technology that runs Shiny for Python entirely inside the user's web browser using WebAssembly (Pyodide), requiring no backend server at all.

17.6 - Open science

Reproducibility within your own computer is the foundation. **Open science** broadens the horizon: ensuring research is accessible, inspectable, and reusable by the global scientific community.

17.6.1 - Sharing data and code

Sharing code and data enables independent verification and accelerates scientific discovery. Trusted repositories such as the **Open Science Framework (OSF)** and **Zenodo** archive research artifacts permanently and assign them a **DOI** (Digital Object Identifier) that can be formally cited in publications.

A complete reproduction package contains:

1. The raw data (or explicit access instructions if restricted).
2. The data cleaning and analysis Python scripts.
3. The Quarto source files (`.qmd`) and bibliography.
4. The environment specification (`requirements.txt` or `uv.lock`).
5. A comprehensive **codebook** detailing every variable and unit of measurement.
6. An open **license**: CC-BY for data/text and the MIT or Apache 2.0 license for code.

17.6.2 - Protecting participants and anonymisation

Human subject data must never be shared in a form that permits identification. Simply stripping student names and ID numbers is insufficient, because combinations of ordinary demographic variables can uniquely identify individuals:

```
small_cells = (
    students.groupby(["faculty", "study_mode", "gender", "has_children"], observed=False)
    .size()
    .reset_index(name="n")
    .sort_values("n")
)
small_cells.head(4)
```

	faculty	study_mode	gender	has_children	n
7	Education	Part-time	Male	Yes	3
15	Health Sciences	Part-time	Male	Yes	3
21	Humanities	Part-time	Female	Yes	3
5	Education	Part-time	Female	Yes	4

In the output above, only 3 students are part-time male students with children in the Faculty of Education. In a small academic department, that combination easily identifies real individuals.

To protect participants before public data sharing:

- Group rare categories (e.g., binning ages or combining sparse faculties).
- Top-code or bottom-code extreme outliers.
- Omit unnecessary demographic identifiers.
- When raw data cannot be published due to privacy or ethics restrictions, generate and share a **synthetic dataset** with identical statistical distributions, allowing peers to verify code execution without exposing participant records.

17.6.3 - Preregistration and the garden of forking paths

Every empirical study involves dozens of defensible analytical decisions: whether to exclude outliers beyond 2 or 3 standard deviations, whether to log-transform a skewed variable, which non-parametric test to substitute, or which covariates to include in regression models.

When these decisions are made *after* inspecting the data, researchers naturally (and often subconsciously) gravitate toward choices that yield statistically significant results ($p < 0.05$). This phenomenon is known as the **garden of forking paths** [35].

A Python simulation illustrates the magnitude of this problem. In the simulation below, we generate 60 observations from an exponential distribution with **zero real group difference**. We then analyze the data five defensible ways and record how often the first analysis, and how often *any* of the five analyses, produces a false positive ($p < 0.05$):

```
import numpy as np
import pandas as pd
import scipy.stats as stats
import statsmodels.api as sm

def forking_paths():
    group = np.array(["A"] * 30 + ["B"] * 30)
    score = np.random.exponential(scale=10, size=60)
    covariate = np.random.normal(size=60)

    score_a = score[group == "A"]
    score_b = score[group == "B"]

    # 1. Standard t-test
    p_ttest = stats.ttest_ind(score_a, score_b).pvalue

    # 2. Outlier-trimmed t-test (|z| < 2)
    z = np.abs((score - np.mean(score)) / np.std(score, ddof=1))
    p_no_outliers = stats.ttest_ind(
        score[(z < 2) & (group == "A")],
```

```
        score[(z < 2) & (group == "B")]
    ).pvalue

    # 3. Log-scale t-test
    p_log = stats.ttest_ind(np.log(score_a), np.log(score_b)).pvalue

    # 4. Rank test (Mann-Whitney U)
    p_rank = stats.mannwhitneyu(score_a, score_b).pvalue

    # 5. Linear model with covariate
    X = sm.add_constant(np.column_stack([(group == "B").astype(int), covariate]))
    p_cov = sm.OLS(score, X).fit().pvalues[1]

    p_vals = [p_ttest, p_no_outliers, p_log, p_rank, p_cov]
    return {
        "first_analysis": p_ttest < 0.05,
        "any_analysis": any(p < 0.05 for p in p_vals)
    }

np.random.seed(17)
sim_results = pd.DataFrame([forking_paths() for _ in range(1000)])
sim_results.mean()
```

```
first_analysis    0.054
any_analysis      0.109
dtype: float64
```

The single pre-planned t-test produces a false positive in approximately 5% of runs, exactly as the nominal $\alpha = 0.05$ level dictates. However, choosing opportunistically among five defensible analyses inflates the false positive rate to over 10%—even though the true effect is exactly zero!

Preregistration solves this dilemma. Researchers register their hypotheses, experimental design, and planned statistical model on an independent, time-stamped registry (such as OSF Registries or AsPredicted) before collecting or inspecting the data. Preregistration clearly separates **confirmatory** analyses from exploratory findings. In a **Registered Report**, a journal peer-reviews and accepts the study protocol before data collection commences, ensuring publication regardless of whether the eventual findings are statistically significant.

17.6.4 - Citing software

Open-source scientific Python packages are developed by researchers and community volunteers whose academic careers depend on citations. Always cite the specific packages used in your thesis.

Official citation guidelines and BibTeX records are available on package documentation pages (e.g., McKinney, 2010 for pandas; Pedregosa et al., 2011 for scikit-learn; Virtanen et al., 2020 for SciPy; Seabold & Perktold, 2010 for statsmodels).

A methodology section should state:

> *"Analyses were conducted in Python (v3.13), utilizing pandas (v2.2) for data manipulation, SciPy (v1.14) and statsmodels (v0.14) for statistical modeling, and scikit-learn (v1.6) for machine learning."*

In your field: psychology and medicine

Psychology and the biomedical sciences learned the critical importance of reproducible workflows through painful experience. In 2015, the Open Science Collaboration attempted to replicate 100 benchmark psychology studies. While 97% of the original papers claimed significant findings, only 36% replicated successfully, with replication effect sizes averaging half the original magnitude [36]. The primary culprits identified were underpowered samples, flexible post-hoc analyses, and publication bias.

Today, leading journals strongly encourage or mandate preregistration, open data, and open code. Clinical medicine established mandatory registry requirements (ClinicalTrials.gov) earlier, ensuring all initiated clinical trials report their results regardless of outcome.

17.7 - Keeping a history with git (optional)

As a thesis project evolves, files quickly proliferate: `thesis_final.qmd`, `thesis_final_v2.qmd`, `thesis_final_supervisor_edits.qmd`. **Version control** replaces file duplication with a clean, linear history of every modification made to the project.

Git is the industry-standard version control system, and **GitHub** is the premier platform for cloud backup, team collaboration, and scientific dissemination [37].

Key Git concepts:

- **Repository**: The project directory whose complete history Git tracks.
- **Commit**: A snapshot of tracked files at a specific moment in time, accompanied by a descriptive message (e.g., `"Filter duplicate survey responses in ch3"`).
- **Push / Pull**: Synchronizing local commits with a remote repository on GitHub.
- **`.gitignore`**: A text file specifying files Git must ignore (e.g., raw sensitive data, temporary cache directories, `.venv/`).

Modern code editors (VS Code, Positron, PyCharm) provide intuitive visual Git interfaces with buttons for staging changes, writing commit messages, and pushing to GitHub.

17.8 - Common misconceptions

- **"Reproducibility is just about sharing a code snippet."** Reproducibility encompasses the full pipeline from raw data to final published estimates, including documentation of every data exclusion, environment dependencies, and random seeds.
- **"Copying and pasting a few numbers manually is harmless."** Manual transcriptions are the single most frequent source of discrepancies between analytical code and thesis prose.
- **"If every analytical choice is reasonable, the resulting p-value is valid."** Evaluating multiple defensible choices after seeing the data severely inflates false positive rates (the garden of forking paths).
- **"Deleting names makes data anonymous."** High-dimensional demographic combinations readily re-identify participants; demographic cell sizes must be scrutinized prior to public release.

17.9 - Chapter review

17.9.1 - Summary

- Reproducibility is a pillar of scientific integrity. An analysis is reproducible when independent researchers can run the same code on the same data and obtain identical results.
- Quarto documents (`.qmd`) integrate narrative Markdown, Python code chunks, dynamic tables, plots, LaTeX equations, and BibTeX citations into a unified, reproducible source that compiles to HTML, Word, PDF, or presentations.
- Virtual environments (`venv`, `uv`) and `requirements.txt` capture library dependencies so that environments can be recreated reliably on any machine.
- Shiny for Python creates reactive dashboards that empower readers to interactively explore findings without modifying code.
- Open science practices include depositing code and data in DOI-assigning archives (OSF, Zenodo), protecting demographic privacy, preregistering confirmatory analyses, and citing open-source tools.
- Git provides robust version control, recording every revision and preventing accidental data or code loss.

17.9.2 - Key terms

Reproducibility, replicability, Quarto, Jupyter engine, YAML header, Markdown, code chunk, inline code, chunk options, cross-reference, BibTeX, CSL style, LaTeX, parameterised reports, virtual environment, `requirements.txt`, lockfile, Shiny for Python, UI, server, reactivity, reactive calculation, open science, DOI, Zenodo, OSF, anonymisation, small cells, synthetic data, garden of forking paths, preregistration, confirmatory analysis, exploratory analysis, Registered Report, Git, repository, commit, push, pull, GitHub, `.gitignore`.

17.10 - Exercises

The interactive Playground provides browser-based exercises and complete solutions:

1. Create a minimal Quarto document (`.qmd`) using Python, compute the mean and standard deviation of three numbers, display them with inline code, and render the document to HTML.
2. Build a parameterised Quarto report that accepts a `faculty` string parameter and filters the `students` dataset to display the count and average wellbeing for that faculty.
3. Inspect demographic combinations in the `students` dataset across `faculty`, `study_mode`, and `gender`. Identify any cell with fewer than 5 respondents and propose an aggregation strategy to preserve privacy.
4. Modify the `forking_paths()` simulation to add a sixth analytical path (e.g., a trimmed t-test excluding the 5 highest scores). Observe how the overall false positive rate increases.
5. Build a Shiny for Python dashboard with a slider that selects the semester range (1 to 4) and displays a line plot of wellbeing over the chosen semesters.

17.11 - Further reading

- The official Quarto documentation at quarto.org provides exhaustive guides and examples for Python data science.
- The Shiny for Python guide at shiny.posit.co/py covers reactive UI development and dashboard deployment.
- Wilson et al. (2017), *"Good enough practices in scientific computing"*, provides practical guidelines for reproducible academic workflows.
- Bryan (2018), *"Excuse me, do you have a moment to talk about version control?"*, offers a friendly introduction to Git and GitHub for researchers.

18. Using AI with Python

Much research data is text: answers to open questions, interview transcripts, clinical notes, and policy documents. To analyse qualitative text quantitatively, researchers use **qualitative coding**: reading each text passage and assigning it to one of a set of themes defined in a systematic **codebook**. Coding is an act of judgement, and judgements differ. A central methodological question is therefore how to determine whether a coder—whether a human researcher, a keyword dictionary, or a computer model—codes reliably. The standard answer is to compare the coder with an independent benchmark, and to measure how far they agree beyond what chance alone would produce.

In the wellbeing study, the final survey included one open question: *"What has been your biggest challenge during your studies?"* A total of 536 students answered in their own words, and the eleventh research question (RQ11) investigates what those challenges are. Elaf coded 200 answers by hand, which required two full weeks of intense concentration. The methodological challenge is whether an AI tool could code the remaining answers reliably.

Artificial intelligence (AI) tools now play a major role across research: coding assistants that write and explain Python code, and large language models (LLMs) that read, summarize, and classify text. This chapter explores both applications. It explains how these models work and why they make confident errors (hallucinations), shows how to use an AI assistant to write robust Python code and how to verify its outputs, and then demonstrates how to call a language model from Python to code students' responses, validating the results against human coding with accuracy and Cohen's kappa. It concludes with essential ethical guidelines for using AI responsibly in scientific research.

By the end of this chapter you will be able to

- Explain how the reliability of coding is measured, and why raw agreement must be corrected for chance.
- Explain in plain terms how large language models work, and why they make confident mistakes.
- Use an AI assistant to write, explain, and debug Python code, and verify its suggestions.
- Call language models from Python (online or local via Ollama) and obtain structured classifications for text datasets.
- Validate AI qualitative coding against human benchmark coding using accuracy and Cohen's kappa.
- Use AI responsibly: protect participant data, preserve reproducibility, and disclose AI assistance.

Note

Teaching & Practice: Interactive Lecture Slides · Browser Playground Exercises

18.1 - The reliability of coding

Two coders who read the same answers will not always choose the same theme. Their raw agreement—the proportion of answers on which they assign identical themes—seems like the natural metric, but it can be highly deceptive. When one theme is very frequent, two coders will often agree simply because both choose it most of the time by chance.

A toy example illustrates why this matters. Suppose two coders categorize 20 student answers, and one coder places 16 of them under Workload:

```
coder_a = ["Workload"] * 16 + ["Family"] * 4
coder_b = ["Workload"] * 14 + ["Family"] * 2 + ["Workload"] * 2 + ["Family"] * 2

observed = np.mean(np.array(coder_a) == np.array(coder_b))
p_a = pd.Series(coder_a).value_counts(normalize=True)
p_b = pd.Series(coder_b).value_counts(normalize=True)
chance = sum(p_a.get(k, 0) * p_b.get(k, 0) for k in set(coder_a) | set(coder_b))
```

```
kappa = (observed - chance) / (1 - chance)

print(f"Observed agreement: {observed:.1%}")
print(f"Chance agreement:   {chance:.1%}")
print(f"Cohen's kappa:      {kappa:.2f}")
```

```
Observed agreement: 80.0%
Chance agreement:   68.0%
Cohen's kappa:      0.37
```

The coders agree on 80% of the answers, which initially sounds strong. But because each coder classifies 80% of the responses under Workload, two coders guessing at random in those proportions would agree on 68% of the answers by chance alone!

Cohen's kappa measures agreement beyond chance: the observed agreement minus chance agreement, divided by the maximum possible agreement above chance [38]:

$$\kappa = \frac{P_o - P_e}{1 - P_e} \tag{18.1}$$

Here, Cohen's kappa is only 0.38. A kappa of 0 represents chance-level agreement, and 1 indicates perfect agreement. Standard benchmarks [39] classify κ as:

- **0.00 – 0.20**: Slight agreement
- **0.21 – 0.40**: Fair agreement
- **0.41 – 0.60**: Moderate agreement
- **0.61 – 0.80**: Substantial agreement
- **0.81 – 1.00**: Almost perfect agreement

In Python, the `cohen_kappa_score` function from `scikit-learn` computes this directly:

```
from sklearn.metrics import cohen_kappa_score
print(f"scikit-learn kappa: {cohen_kappa_score(coder_a, coder_b):.2f}")
```

```
scikit-learn kappa: 0.38
```

Kappa is the standard measure of **inter-rater reliability** (Chapter 5), and it applies to any pair of coders—whether two human raters, a keyword dictionary, or a language model.

18.2 - How AI assistants work

Modern AI coding assistants (ChatGPT, Claude, Gemini, GitHub Copilot) are powered by **large language models** (LLMs). An LLM is a deep neural network (Chapter 15) containing billions of parameters, trained on massive corpora of text and source code.

During training, the model learns to perform one core task: predict the next **token** (a word or sub-word piece) given the preceding context. Generating a response proceeds token by token, continually selecting statistically probable continuations. Instruction tuning and reinforcement learning from human feedback (RLHF) then shape the raw model into an assistant that follows conversational instructions.

This fundamental architecture explains both their strengths and their hazards:

- **Strengths**: LLMs are remarkably fluent in Python, NumPy, pandas, and scikit-learn because extensive documentation, tutorials, and open-source code were present in their training data. They excel at writing initial boilerplate, translating concepts between languages, explaining obscure error traces, and suggesting alternative library functions.
- **Hazards**: LLMs generate text that is *statistically plausible*, not factually verified. A non-existent method name, an obsolete argument, or a subtly incorrect statistical denominator can be generated with total stylistic confidence. Such fabrications are known as **hallucinations**. Furthermore, a model's knowledge ceases at its training cut-off date, and it possesses no intrinsic awareness of your local data structures unless you provide them.

The practical rule is absolute: **treat AI output as a draft, and verify every line yourself.** You remain solely responsible for the integrity of your thesis.

18.3 - AI as a coding assistant

18.3.1 - Asking effective questions

An AI assistant's response is only as precise as the prompt it receives. To obtain reliable Python code:

1. **State the goal clearly**: e.g., *"For each faculty, compute the percentage of students who considered dropping out."*
2. **Describe the data structure**: provide column names and types (e.g., using `students.dtypes` or `students.head(2)`), ensuring no sensitive or identifiable participant data is ever pasted into online prompts.
3. **Specify the preferred libraries**: e.g., *"Use pandas groupby and vectorised methods, avoiding deprecated syntax."*
4. **Provide exact error messages**: when debugging, paste the full traceback along with the offending code snippet.

18.3.2 - Verifying the suggestion

Suppose an assistant is asked for the percentage of students in each faculty who considered dropping out, and suggests the following pandas code:

```
# Plausible but incorrect suggestion:
flawed = (
    students.groupby('faculty')['considering_dropout']
    .apply(lambda s: 100 * (s == 'Yes').sum() / len(students))
    .reset_index(name='percent')
)
flawed
```

	faculty	percent
0	Education	2.666667
1	Health Sciences	4.833333
2	Humanities	2.166667
3	Natural Sciences	1.833333
4	Social Sciences	3.500000

The code runs without error and the numbers look believable. However, the percentages sum to only 15.7%—the percentage of *all* students who considered dropping out! Dividing by `len(students)` divides by the global sample size rather than the faculty subgroup size.

The correct formulation computes the proportion *within* each faculty:

```
# Correct calculation:
correct = (
    students.groupby('faculty')['considering_dropout']
    .apply(lambda s: 100 * (s == 'Yes').mean())
    .reset_index(name='percent')
    .round(1)
)
correct
```

	faculty	percent
0	Education	10.8
1	Health Sciences	18.8
2	Humanities	13.7
3	Natural Sciences	12.6
4	Social Sciences	18.1

Because `(s == 'Yes')` is a boolean series, its mean is the exact within-faculty proportion (Chapter 6). The assistant's error was not an obvious crash; it was a subtle conceptual mismatch.

Catching such errors requires four standard habits:

1. **Test on toy data**: run code on a tiny 3-row DataFrame where you know the manual calculation.
2. **Cross-check by alternative routes**: cross-tabulate counts with `pd.crosstab(students['faculty'], students['considering_dropout'], normalize='index')`.
3. **Inspect library documentation**: verify parameters and return signatures in official docstrings.
4. **Ask the assistant to explain**: requesting *"explain how the denominator is computed"* often exposes faulty logic immediately.

18.4 - Coding open-ended survey answers

A selection of raw survey responses illustrates the challenge of open text:

```
for idx in [4, 8, 14, 54]:
    print(f"- \"{open_responses.loc[idx, 'biggest_challenge']}\"")
```

```
- "most of my income goes on tuition fees. Besides that, I work alone all the time. Things
are slowly improving."
- "I moved to a city where I know no one for my studies and I feel lonely. And I never have
enough time."
- "Supervisor."
- "The direction of my thesis changed twice after comments from my supervsor. Sometimes I
rewrite the same section again and again without knowing if it is right."
```

Responses range from brief phrases to elaborate paragraphs, frequently mention multiple competing pressures, and feature informal grammar and typos. The qualitative codebook defines seven mutually exclusive themes: **Supervision**, **Workload**, **Finances**, **Family**, **Health**, **Isolation**, and **Other**.

Elaf hand-coded a random validation sample of 200 answers into the single primary challenge:

```
theme_distribution = open_responses_coded['theme'].value_counts()
print(theme_distribution)
```

```
theme
Supervision    68
Workload       45
Isolation      31
Health         20
Finances       14
Family         14
Other           8
Name: count, dtype: int64
```

This 200-student dataset serves as the gold-standard test set against which automatic methods are benchmarked.

18.4.1 - Coding with a keyword dictionary

The simplest automated baseline is a **rule-based keyword dictionary**. Each answer is assigned to the theme with the highest number of keyword matches, defaulting to "Other" if no keywords match:

```
keywords = {
    "Supervision": r"\b(supervis|feedback|guidance|meeting)",
    "Workload": r"\b(time|deadline|workload|too much|assignment|reading|busy)",
    "Finances": r"\b(money|fee|scholarship|rent|afford|salary|income|funding|stipend|loan)",
    "Family": r"\b(famil|child|kid|son\b|daughter|baby|parent|mother|father|husband|wife)",
    "Health": r"\b(sleep|tired|stress|anxi|health|ill\b|burnout|headache|coffee|exhaust)",
    "Isolation": r"\b(lonel|alone|friends|outsider|miss (home|my)|far from|know no one)",
      "Other":  r"\b(ethic|participant|statistic|library|procedure|english|power  cut|laptop|
internet|software)"
}
themes = list(keywords.keys())
```

```
def code_by_keywords(text):
    text_lower = str(text).lower()
    hits = {theme: len(re.findall(pat, text_lower)) for theme, pat in keywords.items()}
    max_hits = max(hits.values())
    if max_hits == 0:
        return "Other"
    for theme, count in hits.items():
        if count == max_hits:
            return theme

sample_text = "I miss my family and I cannot afford the fees."
print(f"Sample coding: '{sample_text}' -> {code_by_keywords(sample_text)}")
```

```
Sample coding: 'I miss my family and I cannot afford the fees.' -> Finances
```

Evaluating the dictionary against Elaf's 200 hand-coded responses:

```
keyword_predictions = open_responses_coded['biggest_challenge'].apply(code_by_keywords)
kw_acc = accuracy_score(open_responses_coded['theme'], keyword_predictions)
kw_kap = cohen_kappa_score(open_responses_coded['theme'], keyword_predictions)

print(f"Keyword Accuracy:      {kw_acc:.1%}")
print(f"Keyword Cohen's kappa: {kw_kap:.2f}")
```

```
Keyword Accuracy:      61.0%
Keyword Cohen's kappa: 0.51
```

The keyword dictionary achieves 61.0% accuracy and a Cohen's kappa of 0.51 (moderate agreement). Keyword matching struggles because semantic meaning depends on syntactic context: the phrase *"I feel I am letting down both my family and my supervisor"* matches keywords in both Family and Supervision, while *"I never have enough time"* could reflect course workload or family caring duties. Discerning underlying meaning requires semantic comprehension—the exact strength of modern language models.

18.5 - Calling a language model from Python

Python provides standard libraries for interacting with language models:

- **Online API services**: Official SDKs (`openai`, `anthropic`, `google-genai`) offer top-tier reasoning capabilities. Online APIs require an API key and incur per-token charges. Keys should always be stored in environment variables (e.g., via a `.env` file), never committed directly into Git repositories.
- **Local models via Ollama**: The free, open-source **Ollama** platform runs open models (such as Google Gemma, Meta Llama, or Mistral) locally on your machine. The Python `ollama` package communicates with the local runner. Local models require zero API keys, run completely free of charge, and guarantee that participant data never leaves your computer.

18.5.1 - Structured outputs with Pydantic

Qualitative coding requires two properties that standard web chats lack:

1. **System Prompt**: A fixed, reproducible codebook defining the seven themes and explicit tie-breaking rules.
2. **Structured Enforcement**: Guaranteeing that the model's response is strictly restricted to one of the seven valid theme names, avoiding verbose conversational preambles.

In Python, structured outputs are defined using **Pydantic**:

```
from pydantic import BaseModel
from typing import Literal

class QualitativeCoding(BaseModel):
    theme: Literal[
        "Supervision", "Workload", "Finances",
```

```
        "Family", "Health", "Isolation", "Other"
    ]

# The qualitative codebook prompt:
codebook = """
You are helping an academic researcher code open-ended survey answers from graduate
students, who were asked: 'What has been your biggest challenge during your studies?'
Assign each answer to exactly ONE theme: the single primary challenge the student describes.
If multiple challenges are mentioned, select the one the student presents first or emphasizes
most.

Themes:
- Supervision: supervisor guidance, feedback, meetings, availability, or disagreements.
- Workload: too much work or too little time; deadlines, thesis writing, coursework, readings.
- Finances: tuition fees, living expenses, funding, scholarships, debt, paid employment.
- Family: children, spouse, caring responsibilities, family commitments, home life.
- Health: physical illness, mental health, stress, anxiety, exhaustion, burnout, insomnia.
- Isolation: loneliness, social alienation, lack of peer network, studying remotely.
- Other: administrative issues, research ethics approvals, statistical software, language
barriers.
"""
```

Using Ollama in Python, a classification query is executed with zero sampling temperature (`temperature=0`) to ensure deterministic repeatability:

```
import ollama

response = ollama.chat(
    model="gemma4:e4b",
    messages=[
        {"role": "system", "content": codebook + "\nReply with the theme name only."},
        {"role": "user", "content": "I cannot afford the tuition fees and rent this term."}
    ],
    options={"temperature": 0}
)
print(response['message']['content'].strip())
# Output: Finances
```

For the graduate wellbeing study, all 536 student answers were coded using a local instance of Google Gemma 4 (`gemma4:e4b`) via Ollama. The run script (`data-raw/run_ai_coding.py`) logged the model specifications, execution timestamp, and coded each validation response twice to test stability.

18.6 - Evaluating AI qualitative coding

The model's classifications are loaded from the archived results file and merged with Elaf's 200 hand-coded responses:

```
ai_results = pd.read_csv(ai_file)
ai_meta = pd.read_csv(ai_meta_file).iloc[0]

ai_check = open_responses_coded.merge(ai_results, on="student_id")

ai_acc = accuracy_score(ai_check['theme'], ai_check['ai_theme'])
ai_kap = cohen_kappa_score(ai_check['theme'], ai_check['ai_theme'])

print(f"Model:                {ai_meta['model']} ({ai_meta['provider']})")
print(f"Validation sample:    {len(ai_check)} students")
print(f"AI Coding Accuracy:   {ai_acc:.1%}")
print(f"AI Cohen's kappa:     {ai_kap:.3f}")
```

```
Model:                  gemma4:e4b (Ollama (local))
Validation sample:      200 students
AI Coding Accuracy:     90.0%
AI Cohen's kappa:       0.874
```

The language model achieves **90.0% accuracy** and a Cohen's kappa of **0.874**, representing **almost perfect agreement** [39]. This substantially outperforms the keyword dictionary (61.0% accuracy, $\kappa = 0.512$).

Furthermore, evaluating the re-run test:

```
rerun_sample = ai_check.dropna(subset=['ai_theme_rerun'])
rerun_agree = (rerun_sample['ai_theme'] == rerun_sample['ai_theme_rerun']).mean()
print(f"Repeatability agreement: {rerun_agree:.1%}")
```

```
Repeatability agreement: 100.0%
```

At `temperature = 0`, the local model is 100% deterministic across runs.

18.6.1 - Inspecting disagreements with a confusion matrix

A confusion matrix reveals where the model and the human coder diverge:

```
cm = pd.crosstab(
    ai_check['theme'], ai_check['ai_theme'],
    rownames=['Human Coder'], colnames=['AI Model']
).reindex(index=themes, columns=themes, fill_value=0)
cm
```

AI Model Human Coder	Supervision	Workload	Finances	Family	Health	Isolation	Other
Supervision	60	2	1	2	2	1	0
Workload	2	39	0	1	1	2	0
Finances	0	0	14	0	0	0	0
Family	0	0	0	14	0	0	0
Health	0	1	0	1	16	1	1
Isolation	2	0	0	0	0	29	0
Other	0	0	0	0	0	0	8

The diagonal cells represent cases of complete agreement. The off-diagonal cells represent disagreements. Examining the specific discordant text entries provides valuable qualitative insight:

```
disagreements = ai_check[ai_check['theme'] != ai_check['ai_theme']][
    ['biggest_challenge', 'theme', 'ai_theme']
].head(5)
disagreements.columns = ['Student Answer', 'Human Coder', 'AI Model']
disagreements
```

	Student Answer	Human Coder	AI Model
15	Getting feedback on my draft took over a month...	Supervision	Isolation
19	Combining coursework, my job at a school and m...	Workload	Health
20	I stopped exercising and I can feel the differ...	Health	Other
40	Writing my literature review took far longer t...	Workload	Isolation
46	Nobody in my group works on anything close to ...	Isolation	Supervision

When inspectable, disagreements frequently occur on multi-clause answers where two genuine challenges are present (e.g., struggling with workload while dealing with health fatigue). Inspecting these discrepancies often leads to refining the codebook definitions rather than pointing to model failure.

18.6.2 - What the students reported (RQ11)

With the model's reliability rigorously established ($\kappa = 0.874$), the automated coding is applied across all 536 student responses (Figure 18.1):

```
theme_shares = (
    ai_results['ai_theme']
    .value_counts(normalize=True)
    .sort_values(ascending=True) * 100
)

fig, ax = plt.subplots(figsize=(6.5, 3.5))
ax.barh(theme_shares.index,    theme_shares.values,    color="#2f6793",    edgecolor="none",
height=0.65)
ax.set_xlabel("Share of answers (%)")
ax.grid(True, linestyle="--", alpha=0.5, axis="x")
for i, v in enumerate(theme_shares.values):
    ax.text(v + 0.8, i, f"{v:.1f}%", va="center", fontsize=9.5)
ax.set_xlim(0, max(theme_shares.values) + 6)
plt.tight_layout()
plt.show()
```

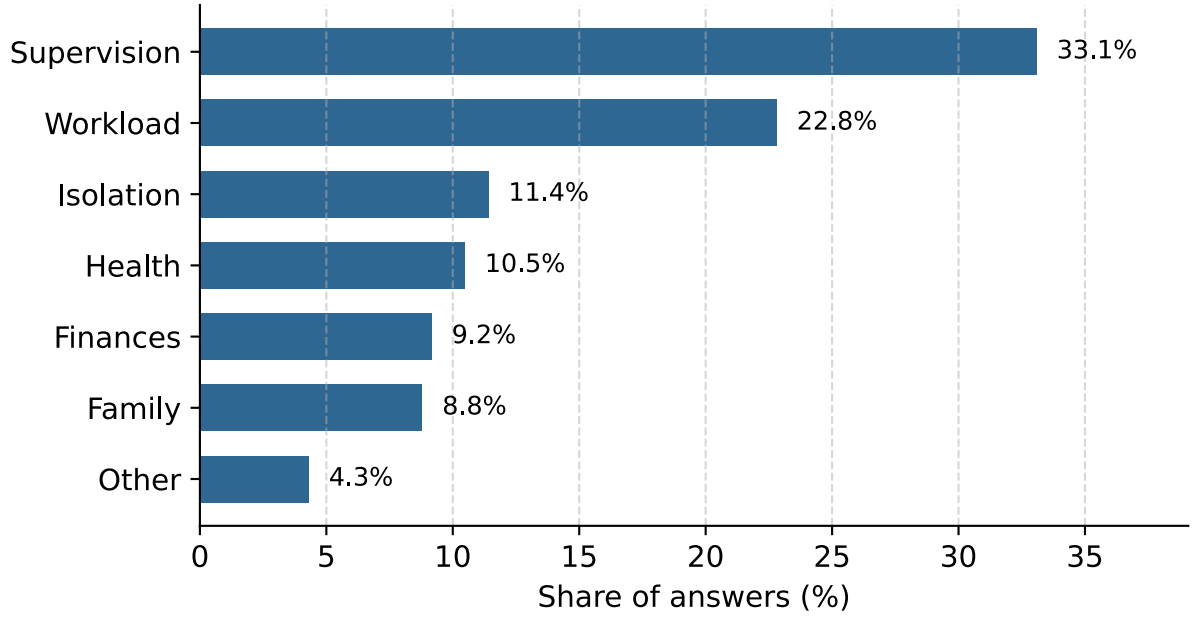


Figure 18.1: Primary challenge reported by graduate students, coded by a local language model.

Supervision (33.0%) and Workload (22.8%) represent the two dominant challenges encountered by graduate students.

18.6.3 - Validating against quantitative survey scales

To establish **construct validity**, we cross-reference the qualitative AI themes with students' quantitative ratings of supervisor support from the Chapter 2 questionnaire:

```
support_items = [f"support_{i}" for i in range(1, 7)]
q_support = questionnaire[["student_id"]].copy()
```

```
q_support["mean_support"] = questionnaire[support_items].mean(axis=1)

validation_table = (
    ai_results.merge(q_support, on="student_id")
    .groupby("ai_theme")
    .agg(
        Students=("student_id", "count"),
        Supervisor_Support=("mean_support", "mean")
    )
    .reset_index()
    .round(2)
    .sort_values("Supervisor_Support")
)
validation_table.columns = ["Primary Challenge", "N Students", "Mean Supervisor Support (1-5)"]
validation_table
```

	Primary Challenge	N Students	Mean Supervisor Support (1-5)
5	Supervision	177	2.82
3	Isolation	61	3.30
4	Other	23	3.31
2	Health	56	3.33
0	Family	47	3.43
1	Finances	49	3.45
6	Workload	122	3.46

The qualitative text and quantitative survey data validate each other: students whose primary open-ended challenge was classified as **Supervision** report significantly lower perceived supervisor support (2.82) than students reporting any other challenge (3.30 – 3.46).

18.7 - Using AI responsibly in academic research

18.7.1 - Verify and maintain authorship accountability

AI tools cannot be listed as authors on academic publications (as confirmed by COPE and major journal publishers) because an AI model cannot take legal or intellectual responsibility for scholarly claims. You remain solely responsible for the accuracy of every finding, citation, and line of code in your thesis.

18.7.2 - Protect participant data and ethical boundaries

Before transmitting text to cloud-hosted AI APIs, verify that your institutional review board (IRB) ethics approval and participant consent forms explicitly authorize external data sharing. When dealing with sensitive, non-consented, or human-subject data:

- Strip all direct and indirect identifying details (names, institutions, specific dates).
- Utilize **local offline models** (e.g., via Ollama), ensuring zero bytes of text leave your research machine.

18.7.3 - Maintain computational reproducibility

LLMs are updated continuously by commercial vendors. To ensure reproducible research:

1. Record the exact model identifier (e.g., `gemma4:e4b`), host provider, temperature (`0`), and date.
2. Archive the exact system prompts and codebook files.
3. Save the model's generated outputs as permanent data artifacts (`.csv`), analyzing the frozen outputs rather than querying dynamic endpoints repeatedly.

18.7.4 - Transparent disclosure

Disclose AI assistance transparently in your thesis methodology or acknowledgments:

> *"Open-ended challenge responses were qualitatively categorized into seven themes using a local open-weight language model (Gemma 4, e4b variant, via Ollama and Python). Agreement with the primary author's manual coding of a random benchmark sample (N = 200) was validated using Cohen's kappa ($\kappa = 0.874$)."*

18.8 - Common misconceptions

- **"High percent agreement proves that qualitative coding is accurate."** High raw agreement often reflects category skew; Cohen's kappa is essential to evaluate agreement above chance expectations.
- **"The AI knows the factual answer."** Language models generate plausible linguistic patterns, not verified facts; AI classifications must be validated against human ground truth.
- **"Model responses are permanently identical."** Cloud models change across releases and non-zero temperatures introduce stochastic variation; outputs must be archived for reproducibility.
- **"Using AI in a thesis is forbidden."** Most academic institutions welcome transparent, disclosed, and validated AI assistance; what is forbidden is unverified, undisclosed, or fabricated content.

18.9 - Chapter review

18.9.1 - Summary

- Inter-rater reliability measures agreement between independent coders, corrected for chance using Cohen's kappa (κ).
- Large language models predict sequential tokens based on training patterns; while proficient in Python data workflows, they are prone to confident hallucinations and lack access to external data unless explicitly provided.
- Effective AI assistance requires specific prompts detailing goals, data structures, and target libraries, combined with rigorous manual cross-checking.
- Keyword dictionaries offer a transparent baseline for text classification but fail on complex syntactic context.
- Python interacts with LLMs via official APIs or local Ollama instances; Pydantic ensures reliable structured enum outputs.
- Local LLMs can match human qualitative coding accuracy ($\kappa > 0.80$) while preserving participant privacy.
- Responsible AI usage demands human accountability, data protection, deterministic parameters (`temperature=0`), reproducibility archives, and explicit disclosure.

18.9.2 - Key terms

Qualitative coding, codebook, inter-rater reliability, Cohen's kappa, large language model, token, hallucination, prompt engineering, system prompt, temperature, local model, Ollama, Pydantic, structured output, construct validity, reproducibility, disclosure.

18.10 - Exercises

The interactive Playground provides browser-based exercises with solutions:

1. Request an AI assistant to write a pandas aggregation computing average sleep hours by faculty and semester. Test the code on a tiny 3-row DataFrame containing a missing value, and verify that missing data is handled appropriately.
2. Expand the keyword dictionary with three additional regular expressions, and measure whether Cohen's kappa improves on the 200-student test set.
3. Inspect 10 instances where the keyword dictionary disagrees with Elaf's hand coding. Determine whether the discrepancy stems from keyword ambiguity, negation, or multiple mentioned challenges.
4. Classify a sample of 20 open-ended responses using a local language model, and compute the agreement rate with the human benchmark.
5. In the two-coder toy example, modify Coder B so that it assigns 100% of responses to "Workload". Observe the resulting raw agreement and Cohen's kappa.

18.11 - Further reading

- Gilardi et al. (2023), *"ChatGPT outperforms crowd-workers for text-annotation tasks"*, *PNAS*, examining LLM zero-shot classification versus human raters.
- Cohen, J. (1960), *"A coefficient of agreement for nominal scales"*, *Educational and Psychological Measurement*.
- Landis, J. R., & Koch, G. G. (1977), *"The measurement of observer agreement for categorical data"*, *Biometrics*.
- The official Ollama documentation at ollama.com for running open-weight models locally.

19. Putting It All Together

A thesis is an argument, and its results chapter is the evidence. Each claim in it rests on a chain of reasoning that runs from a research question, through a hypothesis, a design, measured variables, and an analysis, to a result, a conclusion, and the limits of that conclusion. A thesis convinces when every link in that chain can be seen and checked, and every number in the text can be traced back to the raw data. The methods of this book are the links; this chapter joins them.

In the story, it is the final semester. Elaf's analyses are spread over two years of scripts, written as she learned each method, and her supervisor's advice for the last stretch is simple: "Before you write the results chapter, rebuild the whole thing, from the raw export to the last table, as one project that runs from start to finish. Then you will know that every number in your thesis is right, and you can answer any examiner's question by pointing to the code." This chapter does exactly that. It sets up a complete research project, runs the analysis from the messy survey export to the tables, figures, and sentences of a results chapter, and traces the chain of reasoning behind three of the research questions. It then steps back: how to report results to the standards journals expect, how to choose a method for a new question, the problems that every researcher meets, and where to go next.

By the end of this chapter you will be able to

- Organise a research project so that it runs from raw data to finished results.
- Combine cleaning, description, modelling, and visualisation in one reproducible pipeline.
- Trace the chain of reasoning from research question to conclusion and limitation.
- Report results in the style expected in a thesis, following published reporting standards, with numbers taken directly from the code.
- Choose an appropriate method for a research question, using the map of this book.
- Recognise common challenges in real research projects, and know where to learn more.

Note

Teaching & Practice: Interactive Lecture Slides · Browser Playground Exercises

19.1 - The shape of a research project

Every quantitative project follows roughly the same path, and this book has followed it too (Figure 19.1).

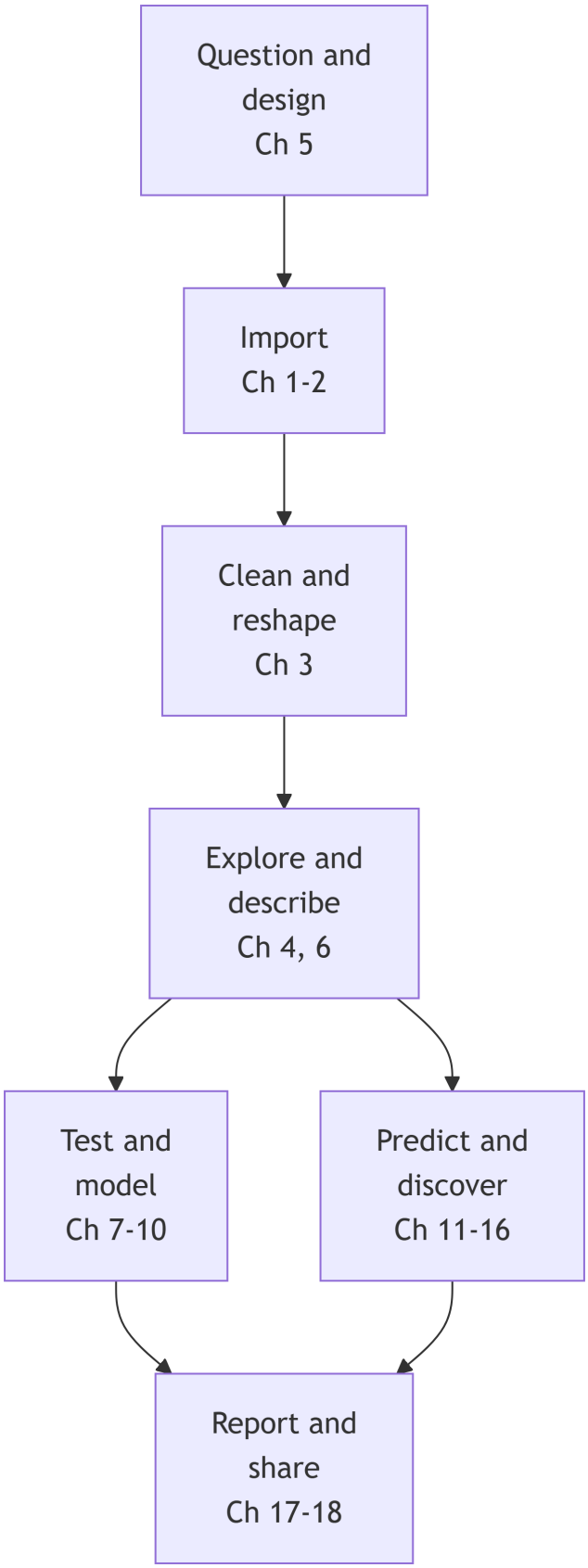


Figure 19.1: The path of a research project, with the chapters of this book that cover each step.

The path is not a straight line in practice. Exploring the data sends you back to cleaning when you find a problem; a model's diagnostics send you back to exploring. What matters is that each step is written in code, so that going back and rerunning everything is easy.

19.1.1 - Organising the project

A project that others (and your future self) can follow has a predictable structure. The final thesis project looks like this:

```
wellbeing-thesis/
├── README.txt                  what the project is and how to run it
├── data-raw/
│   └── wellbeing_raw.xlsx      the survey export, never edited by hand
├── data/                       clean data, created by the scripts
├── scripts/
│   ├── 01_clean_data.py        raw export -> clean tables (Chapter 3)
│   └── 02_analysis.py          models and figures
├── output/                     figures and tables, created by the scripts
├── results.qmd                 the results chapter (Chapter 17)
├── references.bib
└── requirements.txt            package versions (Chapter 17)
```

Four principles lie behind it. **Raw data is read-only**: everything in `data/` and `output/` can be deleted and recreated by running the scripts. **Scripts are numbered** in the order they run, and each does one job. **Paths are relative to the project**, using `pathlib.Path` (Chapter 1), so the project runs on any computer. And **the README** says in a few lines what the project is and how to run it.

The downloadable project for this chapter has the same structure, with working scripts (its `data/` and `output/` folders are created when the scripts run, and it provides a `requirements.txt` for virtual environments). The rest of the chapter walks through what the scripts do.

19.2 - From raw export to clean data

The cleaning of Chapter 3, gathered into one script, runs in a few seconds. Here is its core, reading the raw export and producing the clean questionnaire and semester tables:

```
import pandas as pd
import numpy as np

raw = pd.read_excel(data2thesis_example("wellbeing_raw.xlsx"))

item_names = [f"stress_{i}" for i in range(1, 7)] + \
             [f"burnout_{i}" for i in range(1, 7)] + \
             [f"support_{i}" for i in range(1, 7)] + \
             [f"satisfaction_{i}" for i in range(1, 5)]

# 1. Drop test responses and duplicates
responses = raw[~raw["Q1_Student ID"].str.upper().str.startswith("TEST")].copy()
responses = responses.drop(columns=["Response ID"]).drop_duplicates()
responses = responses.rename(columns={
    "Q1_Student ID": "student_id",
    "Workshop group": "workshop",
    "Y1_Considered leaving?": "considering_dropout"
})

# Rename item columns Q11_1:Q11_22
q11_cols = [f"Q11_{i}" for i in range(1, 23)]
responses = responses.rename(columns=dict(zip(q11_cols, item_names)))

for col in item_names:
   responses[col] = pd.to_numeric(responses[col].replace([99, "99"], np.nan), errors="coerce")
```

```
questionnaire_clean = responses[["student_id"] + item_names].copy()

# 2. Reshape semesters from wide to long
def parse_number(s):
    return pd.to_numeric(s.astype(str).str.replace(",", ".").str.extract(r"(\d+(?:\.\d+)?)")
[0], errors="coerce")

long_records = []
for sem in [1, 2, 3, 4]:
    sub = pd.DataFrame({
        "student_id": responses["student_id"],
        "semester": sem,
        "gpa": parse_number(responses[f"GPA_S{sem}"]),
        "sleep_hours": parse_number(responses[f"Sleep_S{sem}"]),
        "study_hours": parse_number(responses[f"Study_S{sem}"]),
        "exercise_days": parse_number(responses[f"Exercise_S{sem}"]),
        "caffeine_mg": parse_number(responses[f"Caffeine_S{sem}"]),
        "supervisor_meetings": parse_number(responses[f"Meetings_S{sem}"]),
        "wellbeing": parse_number(responses[f"Wellbeing_S{sem}"])
    })
    long_records.append(sub)

semesters_clean = pd.concat(long_records, ignore_index=True)
semesters_clean.loc[semesters_clean["sleep_hours"] > 24, "sleep_hours"] = np.nan
semesters_clean.loc[semesters_clean["study_hours"] > 168, "study_hours"] = np.nan
semesters_clean.loc[semesters_clean["gpa"] > 4, "gpa"] = np.nan

metric_cols = ["gpa", "sleep_hours", "study_hours", "exercise_days", "caffeine_mg",
"supervisor_meetings", "wellbeing"]
semesters_clean = semesters_clean.dropna(subset=metric_cols, how="all")
semesters_clean["supervisor_meetings"] = semesters_clean["supervisor_meetings"].astype(int)
semesters_clean["wellbeing"] = semesters_clean["wellbeing"].astype(int)
```

Each step is explained in Chapter 3; the full script in the downloadable project also cleans the background variables. The most important line of any cleaning script is the check at the end, here comparing with the clean reference data:

```
def check_match(mine, theirs, key):
    m = mine.sort_values(key).reset_index(drop=True)
    t = theirs[mine.columns].sort_values(key).reset_index(drop=True)
    return m.equals(t)

print("Questionnaire matches:", check_match(questionnaire_clean, questionnaire,
"student_id"))
print("Semesters matches: ", check_match(semesters_clean, semesters, ["student_id",
"semester"]))
```

```
Questionnaire matches: True
Semesters matches:     True
```

Both are `True`. In a real project there is no package to compare with, so the checks are the ones from Chapters 3 and 6: counts of categories, ranges of values, numbers of missing values, and a look at a few rows. From here on, the chapter uses the clean tables, together with the scale scores:

```
scores = questionnaire.copy()
scores["stress_4"] = 6 - scores["stress_4"]
scores["stress"] = scores.filter(like="stress_").mean(axis=1)
scores["burnout"] = scores.filter(like="burnout_").mean(axis=1)
scores["support"] = scores.filter(like="support_").mean(axis=1)
scores = scores[["student_id", "stress", "burnout", "support"]]
```

```
study = students.merge(scores, on="student_id").merge(semesters.query("semester == 1"),
on="student_id")
print(f"Total students in study dataset: {len(study)}")
```

```
Total students in study dataset: 600
```

The table `study` has one row per student, with background, questionnaire scores, and first-semester records, as in Chapter 8.

19.3 - Describing the sample

Every results chapter begins by describing the participants, often in a table known as "Table 1". Here the sample is described by programme:

Table 19.1: Characteristics of the sample, by programme.

```
def mean_sd(x):
    return f"{x.mean():.1f} ({x.std(ddof=1):.1f})"

def percent(x):
    return f"{100 * x.mean():.0f}%"

def describe(d):
    return pd.Series({
        "Students": str(len(d)),
        "Age, mean (SD)": mean_sd(d["age"]),
        "Women": percent(d["gender"] == "Female"),
        "Part-time": percent(d["study_mode"] == "Part-time"),
        "Stress (1-5), mean (SD)": mean_sd(d["stress"]),
        "Support (1-5), mean (SD)": mean_sd(d["support"]),
        "Wellbeing in semester 1, mean (SD)": mean_sd(d["wellbeing"]),
        "Considered dropping out": percent(d["considering_dropout"] == "Yes")
    })

sample_desc = pd.DataFrame({
    "Characteristic": describe(study).index,
    "Master's": describe(study[study["programme"] == "Master's"]).values,
    "PhD": describe(study[study["programme"] == "PhD"]).values,
    "All": describe(study).values
})
print(sample_desc.to_markdown(index=False))
```

```
| Characteristic                    | Master's    | PhD         | All         |
|:----------------------------------|:------------|:------------|:------------|
| Students                          | 426         | 174         | 600         |
| Age, mean (SD)                    | 27.9 (3.3)  | 34.2 (5.1)  | 29.7 (4.8)  |
| Women                             | 52%         | 53%         | 52%         |
| Part-time                         | 26%         | 39%         | 30%         |
| Stress (1-5), mean (SD)           | 3.2 (0.7)   | 3.2 (0.7)   | 3.2 (0.7)   |
| Support (1-5), mean (SD)          | 3.2 (0.8)   | 3.2 (0.7)   | 3.2 (0.8)   |
| Wellbeing in semester 1, mean (SD) | 60.9 (12.1) | 59.3 (11.9) | 60.5 (12.0) |
| Considered dropping out           | 15%         | 14%         | 15%         |
```

Two small helper functions, `mean_sd()` and `percent()`, format the numbers the way theses report them, and `describe()` builds the column for any group of students, so the same code makes all three columns. Writing a small function whenever you would otherwise copy and paste code is one of the best habits to take from this book.

19.4 - Answering the research questions

Three of the research questions are answered below with the methods of Parts 2 and 3, each with a sentence written the way it will appear in the thesis. A small function formats p-values in the usual style:

```
def format_p(p):
    return "p < .001" if p < 0.001 else f"p = {p:.3f}".replace(" 0.", " .")
```

19.4.1 - The workshop and its persistence (RQ3)

Students were randomly invited to the wellbeing workshop after the first semester. The mixed-effects model of Chapter 10 uses all four semesters and every student:

```
panel = semesters.merge(students, on="student_id").assign(time=lambda d: d["semester"] - 1)

workshop_model = smf.mixedlm(
    "wellbeing ~ C(semester) * workshop",
    panel,
    groups=panel["student_id"],
    re_formula="~time"
).fit(method=["lbfgs", "powell"])

pred_grid = pd.MultiIndex.from_product(
    [[1, 2, 3, 4], ["Invited", "Not invited"]],
    names=["semester", "workshop"]
).to_frame().reset_index(drop=True)
pred_grid["time"] = pred_grid["semester"] - 1
pred_grid["predicted_wellbeing"] = workshop_model.predict(pred_grid)

piv = pred_grid.pivot(index="semester", columns="workshop", values="predicted_wellbeing")
gaps = piv["Invited"] - piv["Not invited"]
print("Wellbeing gaps across semesters:")
print(gaps.round(1))
```

```
Wellbeing gaps across semesters:
semester
1    0.4
2    5.2
3    4.1
4    2.6
dtype: float64
```

Wellbeing was similar in the two groups before the workshop (difference 0.4 points). After the workshop, invited students' wellbeing was 5.2 points higher in semester 2, but the difference narrowed to 4.1 points in semester 3 and 2.6 in semester 4.

19.4.2 - Explaining GPA (RQ5)

The multiple regression of Chapter 8, with coefficients and confidence intervals organized as a clean summary:

```
gpa_model = smf.ols("gpa ~ sleep_hours + study_hours + stress + support", data=study).fit()

gpa_table = pd.DataFrame({
    "estimate": gpa_model.params,
    "std_err": gpa_model.bse,
    "t_value": gpa_model.tvalues,
    "p_value": gpa_model.pvalues,
    "conf_low": gpa_model.conf_int()[0],
    "conf_high": gpa_model.conf_int()[1]
```

```
})
print(gpa_table.round(3))
```

```
             estimate  std_err  t_value  p_value  conf_low  conf_high
Intercept       2.137    0.145   14.712      0.0     1.852      2.423
sleep_hours     0.106    0.014    7.487      0.0     0.078      0.134
study_hours     0.007    0.001    6.270      0.0     0.005      0.009
stress         -0.084    0.018   -4.685      0.0    -0.119     -0.049
support         0.111    0.017    6.632      0.0     0.078      0.144
```

> Each additional hour of sleep was associated with a GPA 0.11 points higher (95% CI 0.08 to 0.13, p < .001), holding study hours, stress, and support constant. Together, the four predictors explained 25% of the variation in first-semester GPA (n = 587).

19.4.3 - Considering dropout (RQ9)

The logistic regression of Chapter 8, with odds ratios:

```
study["dropout_bool"] = (study["considering_dropout"] == "Yes").astype(int)

dropout_model = smf.logit(
    "dropout_bool ~ stress + support + financial_worry + C(employment) + C(study_mode)",
    data=study
).fit(disp=False)

dropout_table = pd.DataFrame({
    "odds_ratio": np.exp(dropout_model.params),
    "conf_low": np.exp(dropout_model.conf_int()[0]),
    "conf_high": np.exp(dropout_model.conf_int()[1]),
    "p_value": dropout_model.pvalues
})
print(dropout_table.round(2))
```

```
                               odds_ratio  conf_low  conf_high  p_value
Intercept                            0.02      0.00       0.20     0.00
C(employment)[T.None]                0.55      0.24       1.31     0.18
C(employment)[T.Part-time job]       0.58      0.24       1.42     0.23
C(study_mode)[T.Part-time]           1.32      0.64       2.71     0.45
stress                               3.60      2.25       5.77     0.00
support                              0.33      0.22       0.49     0.00
financial_worry                      1.57      1.24       2.00     0.00
```

`np.exp()` turns the log-odds coefficients into odds ratios and their confidence intervals.

> Each one-point increase in stress (on the 1 to 5 scale) was associated with 3.6 times the odds of considering dropping out (95% CI 2.3 to 5.8), and each one-point increase in supervisor support with 0.33 times the odds (95% CI 0.22 to 0.49).

Chapters 11 to 15 went further with this question, asking how well dropout can be *predicted*; the thesis reports that logistic regression predicted as well as machine learning models (a test AUC of about 0.84 in Chapter 11), which is itself a finding worth stating.

19.5 - One figure for the thesis

A thesis figure often combines panels. Using `matplotlib.pyplot.subplots()`, we can join figures side by side with consistent styling:

```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(9, 4))

# Panel A: Workshop Trajectory
for w_label, color in [("Invited", navy), ("Not invited", copper)]:
    sub = pred_grid[pred_grid["workshop"] == w_label]
          ax1.plot(sub["semester"],  sub["predicted_wellbeing"],  marker="o",  color=color,
linewidth=2, label=w_label)

ax1.set_xlabel("Semester")
ax1.set_ylabel("Predicted wellbeing (0-100)")
ax1.set_xticks([1, 2, 3, 4])
ax1.legend(title="Workshop")
ax1.set_title("A", loc="left", fontweight="bold", fontsize=13)

# Panel B: Sleep and GPA
sns.regplot(
    data=study, x="sleep_hours", y="gpa",
    ax=ax2,
    scatter_kws={"alpha": 0.3, "color": "black", "s": 20},
    line_kws={"color": navy, "linewidth": 2}
)
ax2.set_xlabel("Sleep (hours a night)")
ax2.set_ylabel("GPA (semester 1)")
ax2.set_title("B", loc="left", fontweight="bold", fontsize=13)

for ax in (ax1, ax2):
    ax.spines["top"].set_visible(False)
    ax.spines["right"].set_visible(False)

plt.tight_layout()
plt.show()
```

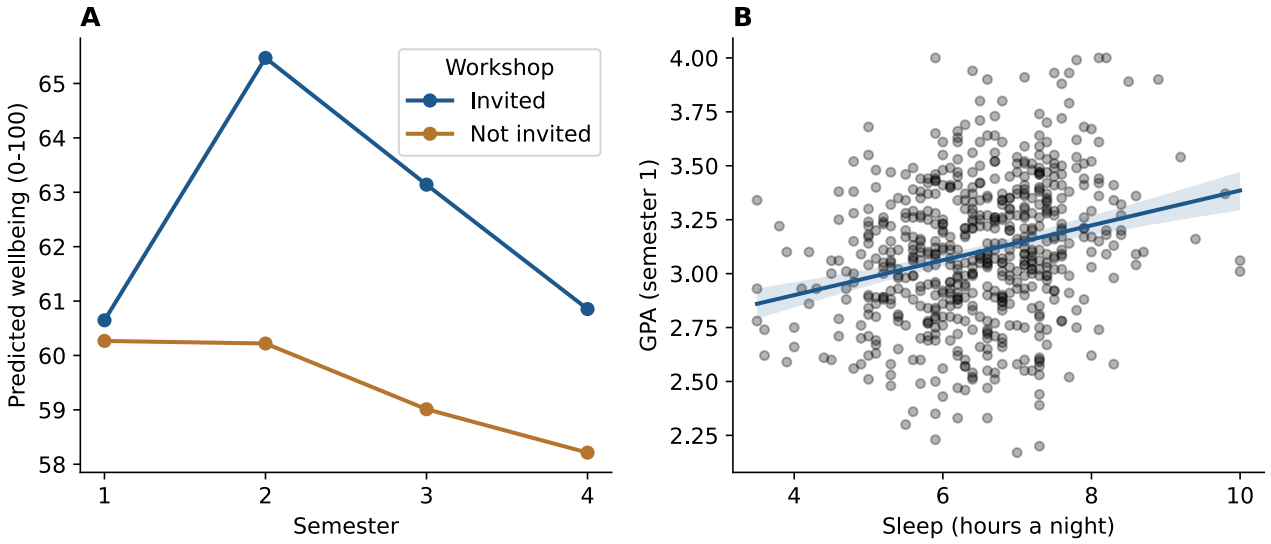


Figure 19.2: (A) Predicted wellbeing by semester and workshop group, from the mixed-effects model. (B) First-semester GPA against hours of sleep, with a regression line.

`plt.savefig()` saves the figure at the dimensions and resolution a thesis or journal requires (Chapter 4):

```
fig.savefig("output/figure-1.png", dpi=300, bbox_inches="tight")
```

19.6 - The chain of reasoning

A result on its own is not yet a conclusion. Between the two lie the design that produced the data, the way the variables were measured, and the limits of both. Table 19.2 traces the whole chain for the three research questions answered above, from the hypotheses stated in Chapter 5 to the limitations that a thesis discussion must acknowledge. Every result in it is filled in by the code of this chapter.

Table 19.2: The chain of reasoning for three research questions, from question to limitation

Step	Workshop (RQ3)	GPA (RQ5)	Considering dropout (RQ9)
Question	Does the workshop improve wellbeing?	What explains students' GPA?	Who considers dropping out?
Hypothesis (Chapter 5)	Invited students have higher wellbeing in semester 2	More sleep goes with a higher GPA, allowing for study hours, stress, and support	Higher stress raises the odds of considering dropout
Design	Randomised invitation within a longitudinal study	Observational, first semester	Observational, baseline and end of year 1
Variables	Wellbeing (0 to 100), invitation, semester	GPA, sleep, study hours, stress, support	Considering dropout (yes/no), stress, support, financial worry, employment, study mode
Analysis	Mixed-effects model (Chapter 10)	Multiple regression (Chapter 8)	Logistic regression (Chapter 8)
Result	5.3 points higher in semester 2, narrowing to 2.6 by semester 4	0.11 GPA points per hour of sleep (95% CI 0.08 to 0.13)	Odds ratio 3.6 per point of stress (95% CI 2.2 to 5.8)
Conclusion	The workshop raised wellbeing, and the effect faded; because the invitation was random, the effect is causal	Sleep is associated with GPA, independently of the other predictors; the null hypothesis is rejected, but the design does not show cause	Stress is associated with higher odds of considering dropout; the null hypothesis is rejected
Limitation	One university; the invitation, not attendance, was randomised	Sleep is self-reported, and unmeasured confounders may remain	The outcome is considering dropout, not leaving; stress and the outcome were measured close together

Reading the table by columns shows how different the three conclusions are, although all three rest on "significant" results. Only the workshop conclusion is causal, because only the workshop was assigned at random (Chapter 5). The GPA and dropout conclusions are associations, and the limitations say what could still explain them. A discussion chapter that keeps each result attached to its design, in this way, claims exactly what the evidence supports.

19.7 - The results chapter

The final step is the one Chapter 17 prepared: the tables, figures, and sentences above go into a Quarto document, `results.qmd`, in which every number is written with inline code. The downloadable project contains such a document, built from this chapter.

What a results chapter must contain is not left to taste. **Reporting standards** list the information that readers need in order to judge a study. For quantitative research in psychology and neighbouring fields, the American Psychological Association's Journal Article Reporting Standards (JARS) set out what to report about the participants, the measures, the analysis, and the results, including effect sizes and confidence intervals [40]. Randomised trials in health research follow CONSORT, and observational studies follow STROBE [41]. Checking a draft against the relevant standard is one of the most useful things a student can do before submission, and many journals require it. When the data changes, or an examiner asks for a different model, the code is changed and rendered again, and the whole chapter is updated.

19.8 - Choosing a method

The choice of method for a new research question depends on the goal, on the kind of **outcome**, and on how the **observations** are related. Figure 19.3 summarises the methods of this book as a guide.

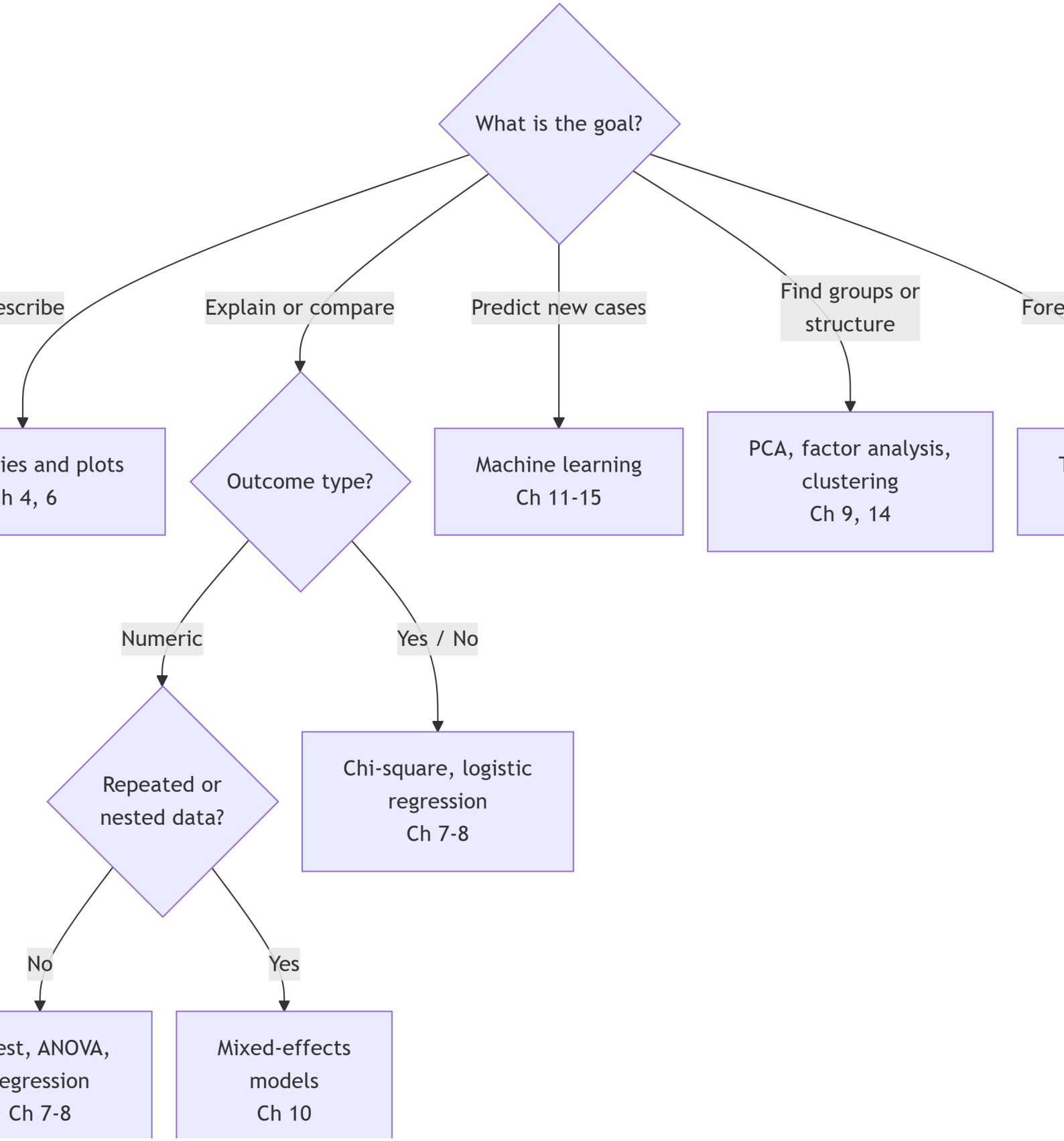


Figure 19.3: A guide to choosing a method, with the chapters that cover each.

Table 19.3 shows how each of the study's research questions was answered.

Table 19.3: The study's research questions and the methods used to answer them

Research question	Method	Chapters
RQ1 What does graduate life look like?	Plots, descriptive statistics	4, 6
RQ2 Do students sleep less than 7 hours?	One-sample t-test, confidence interval	7
RQ3 Does the workshop improve wellbeing?	Two-sample t-test; mixed-effects model	7, 10
RQ4 Do faculties and study modes differ?	ANOVA, post-hoc tests	7, 8
RQ5 What explains GPA?	Multiple regression	8, 13
RQ6 Does the questionnaire measure what it should?	Factor analysis, Cronbach's alpha	9
RQ7 Are there student profiles?	Cluster analysis, mixture models	9, 14
RQ8 How do wellbeing and GPA change?	Mixed-effects models	10
RQ9 Who considers dropping out?	Logistic regression; classification	8, 11, 12, 15
RQ10 Can final GPA be predicted?	Regularised regression, gradient boosting	13
RQ11 What challenges do students describe?	Text coding with a language model	18
RQ12 How many counselling visits next year?	Time series forecasting	16

19.9 - Challenges every researcher meets

Real projects rarely go as planned, and some problems are almost universal. Every real dataset is **messy** and needs cleaning, which takes longer than the analysis; it is done in code, checked after every step, and never applied to the raw file itself (Chapter 3). **Missing data** raises the question of *why* it is missing before anything is done about it, because dropping incomplete cases can bias results when the missing data is not random, as with the students who left the programme (Chapter 6). **Small samples** give wide confidence intervals and low power (Chapter 7) and make complex models overfit (Chapter 13); the remedies are to report effect sizes with intervals, keep models simple, and plan the sample size before collecting data with a power analysis (Chapter 5).

Analysis brings its own problems. **Many tests** produce many false alarms, so the main analyses are decided in advance, multiple comparisons are corrected where needed, and exploratory results are reported as exploratory (Chapters 7, 8, and 17). **Assumptions** are checked with plots rather than with tests alone, with robust tests, non-parametric tests, transformations, and mixed models as the alternatives (Chapters 7, 8, and 10). **Correlation is not causation**: observational data rarely proves causes, so confounders must be thought through, as sleep was behind the caffeine effect, and randomised designs used where possible, as in the workshop study (Chapters 5 and 8). **Prediction and explanation** are different aims: a model that predicts well may explain little, and the reverse (Chapter 11). Finally, results must be **communicated** to supervisors, co-authors, and examiners who may use other software; a clear results chapter with tables, figures, and an appendix of code serves them all, and data can be exported with `to_csv()` or `pyreadstat` when they need it (Chapter 2).

19.10 - Where to go next

This book is a beginning. Depending on your field, the next methods to learn may be:

Table 19.4: Topics for further study

Topic	What it is for	Python Packages
Structural equation modelling	Confirmatory factor analysis, path models, latent variables	`semopy`
Bayesian statistics	Models with prior knowledge and full uncertainty	`PyMC`, `bambi`
Survival analysis	Time until an event, such as dropping out	`lifelines`
Meta-analysis	Combining results from several studies	`statsmodels`
Natural language processing	Words, sentiment, and embeddings in text	`spacy`, `transformers`
Spatial data	Maps and geographic data	`geopandas`
Publication tables	Formatted summary tables of results	`great_tables`, `tabulate`
Interpreting models	Feature importances, SHAP values, and partial effects	`shap`, `scikit-learn`

Some of the best resources are free online: *Python for Data Analysis* by Wes McKinney [1], *Regression and Other Stories* [15] for regression modeling, *An Introduction to Statistical Learning with Applications in Python* [28] for machine learning, and the official documentation of Scikit-Learn, Statsmodels, and PyTorch. Appendix E lists more.

You do not have to learn alone. The Python data science community is large and welcoming: Stack Overflow, Python Discord servers, local PyData meetups, and open-source contributors around the world. Asking a clear question, with a small reproducible example, is itself an invaluable research skill, and the same skill that makes an AI assistant useful (Chapter 18).

In your field: nutrition science

The same path works for any dataset, in a few lines. The classic `ToothGrowth` dataset comes from an experiment on the growth of teeth in 60 guinea pigs given vitamin C at three doses, administered via orange juice (`OJ`) or ascorbic acid (`VC`). A complete mini-project, from data to a reported result:

```
tooth = sm.datasets.get_rdataset("ToothGrowth", "datasets").data
tooth["dose"] = tooth["dose"].astype(str)

tooth_summary     =     tooth.groupby(["supp",     "dose"])["len"].agg(["mean",     "std",
"count"]).round(2)
print(tooth_summary)

tooth_model = smf.ols("len ~ C(supp) * C(dose)", data=tooth).fit()
anova_res = anova_lm(tooth_model, typ=2)
print(anova_res.round(3))
```

```
            mean   std  count
supp dose
OJ   0.5   13.23  4.46     10
     1.0   22.70  3.91     10
     2.0   26.06  2.66     10
VC   0.5    7.98  2.75     10
     1.0   16.77  2.52     10
     2.0   26.14  4.80     10
                   sum_sq    df       F  PR(>F)
C(supp)           205.350   1.0  15.572   0.000
C(dose)          2426.434   2.0  92.000   0.000
C(supp):C(dose)   108.319   2.0   4.107   0.022
Residual          712.106  54.0     NaN     NaN
```

The two-way ANOVA of Chapter 8 shows that tooth length depends on the dose, the delivery method of vitamin C, and their interaction: orange juice is more effective than the supplement at low and medium doses, but both produce comparable growth at the highest dose (2.0 mg). The same steps (prepare, describe, model, report) would take a few more lines with a plot and a Quarto report. Whatever your field, the path from data to thesis is the one this book has followed.

19.11 - Common misconceptions

A few misunderstandings are common at the stage of writing up.

- **"A significant result answers the research question."** It answers one link of the chain; the design and the measures decide what the result means.
- **"The discussion should defend the results."** It should also state their limits; examiners trust a thesis more when it says clearly what it cannot show.
- **"Formatting numbers by hand is quicker."** It is quicker once, and wrong the next time the data or the model changes.
- **"Choosing a method means choosing the most advanced one."** The right method is the simplest one that matches the goal, the outcome, and the structure of the data.

19.12 - Chapter review

19.12.1 - Summary

- A research project follows a path from question and design through import, cleaning, exploration, and modelling to reporting and sharing, often going back to earlier steps. Writing every step in code makes going back easy.
- Organise projects with read-only raw data, numbered scripts, relative paths, a README, and a Quarto results document; everything else can be recreated.

- Check the cleaned data at the end of the cleaning script, then describe the sample, answer each research question with the method suited to it, and report results with numbers taken directly from the code.
- Every result rests on a chain of reasoning, from question and hypothesis through design, variables, and analysis to result, conclusion, and limitation. Reporting standards such as the APA's JARS, CONSORT, and STROBE list what a report must contain.
- Small helper functions (for formatting means, percentages, and p-values) avoid copying and pasting code; pandas and statsmodels turn model results into tidy summary tables; matplotlib subplots combine figures.
- Choose methods by the goal (describe, explain, predict, find structure, forecast), the type of outcome, and how the observations are related.
- Messy and missing data, small samples, many tests, assumptions, and causation are challenges in every project; the chapters of this book give tools for each.

19.12.2 - Key terms

Research workflow, project structure, raw data, pipeline, sample description ("Table 1"), helper function, tidy model summary, multi-panel figure, reporting sentence, chain of reasoning, reporting standard, method choice.

19.13 - Exercises

The playground has these and more, and the downloadable project contains the complete thesis project.

1. Download the Chapter 19 project, run `scripts/01_clean_data.py` and `scripts/02_analysis.py`, and render `results.qmd`. Then change one cleaning rule (for example, treat ages above 70 as impossible), render again, and note which numbers change.
2. Add a row to the sample table (Table 19.1) for the share of students with children.
3. Write a reporting sentence, with inline numbers, for the effect of support on GPA in the regression model.
4. Add a third panel to the thesis figure: wellbeing by faculty as a box plot.
5. Choose a research question from your own field. Using Figure 19.3, decide which method you would use, and which chapter of this book you would reread first.
6. Complete the chain of reasoning (Table 19.2) for RQ2, whether students sleep less than 7 hours, using the results of Chapter 7.

19.14 - Further reading

- *Python for Data Analysis* [1] covers data manipulation, reshaping, and exploratory analysis in Python.
- *Regression and Other Stories* [15] is an excellent guide to building, checking, and reporting regression models in real research.
- *An Introduction to Statistical Learning with Applications in Python* [28] provides hands-on machine learning modeling.

References

Appendices

A. Appendix A: Installation and Setup

This appendix explains how to install everything the book uses, how to configure your working environment, and how to resolve the issues that most often arise. If you want to get started immediately, the short version is: install **Python**, install **Visual Studio Code**, create a virtual environment, and install the companion packages with `pip`. If you are not ready to install anything locally yet, every chapter's exercises can also be run directly in your web browser in the interactive playground.

Software installers and websites update their appearance over time, so this appendix describes each step in clear terms. The essential configuration choices remain consistent across versions.

A.1 - What you need

Table A.1: The software used in this book

Software	What it is	Required?
Python (3.10+)	The programming language and computation engine	Yes
VS Code	Code editor with integrated terminal, plot viewer, and notebook support	Yes (or Positron)
Positron	Modern data science IDE by Posit, designed for Python and R	Optional
Quarto	Authoring tool that compiles research code into papers, books, and slides	Yes
Scientific packages	Libraries (`pandas`, `seaborn`, `scipy`, etc.) installed via `pip`	Yes
data2thesis	Companion package containing all Graduate Wellbeing Study datasets	Yes

All of these tools are free and open source. Python should be installed first so that editors and tools can discover your Python interpreter automatically.

A.2 - Installing Python

Python is downloaded from the official Python Software Foundation website at python.org/downloads. Always download the latest stable release (version 3.10 or newer; 3.12 is recommended).

A.2.1 - Windows

1. On python.org, download the **Windows installer (64-bit)**.
2. Run the installer. On the very first screen, **make sure to check the box: "Add python.exe to PATH"**. This is essential: it enables your terminal and code editor to find Python.
3. Click **Install Now**. If prompted with "Disable path length limit" at the end of the installation, click it to avoid Windows file-path truncation issues.

A.2.2 - macOS

1. On python.org, download the macOS 64-bit universal installer (`.pkg`).
2. Run the package installer and accept the standard defaults.
3. After installation completes, navigate to your Applications folder, open the `Python 3.x` folder, and double-click `Install Certificates.command` to ensure secure network connections (SSL) function properly when downloading datasets and packages.
4. Alternatively, if you use Homebrew, you can install Python directly from Terminal:

```
brew install python
```

A.2.3 - Linux

Python comes pre-installed on most modern Linux distributions. On Debian/Ubuntu distributions, ensure the development packages and package manager are installed:

```
sudo apt update
sudo apt install python3 python3-pip python3-venv
```

A.2.4 - Checking the installation

Open your system terminal (Command Prompt or PowerShell on Windows; Terminal on macOS and Linux) and type:

```
python --version
```

If it prints `Python 3.12.x` (or any version 3.10+), Python is installed and accessible.

A.3 - Installing VS Code

1. Download **Visual Studio Code** from code.visualstudio.com for your operating system.
2. Install the application using default settings. On Windows, check the boxes to add "Open with Code" to the context menu.
3. Open VS Code. On the left sidebar, click the **Extensions** icon (four square blocks, or press `Ctrl+Shift+X` / `Cmd+Shift+X`).
4. Search for and install:
   - **Python** (by Microsoft)
   - **Jupyter** (by Microsoft)
   - **Quarto** (by Posit PBC)

A.3.1 - Positron: A Dedicated Data Science Alternative

Positron (positron.posit.co) is a specialized code editor built by Posit for scientific data analysis in Python and R. It provides dedicated panes for Variables, DataFrames, and Plots out of the box. Everything taught in this book works identically in VS Code and Positron.

A.4 - Managing projects with virtual environments

In empirical research, package versions evolve. To prevent package updates from inadvertently altering your thesis calculations months later, create a **virtual environment** for your research project. A virtual environment is an isolated directory containing its own Python interpreter and set of installed libraries.

In your research project folder:

```
# 1. Create a virtual environment named .venv
python -m venv .venv

# 2. Activate the virtual environment
```

```
# Windows (PowerShell):
.venv\Scripts\Activate.ps1
# Windows (Command Prompt):
.venv\Scripts\activate.bat
# macOS / Linux:
source .venv/bin/activate
```

When activated, `(.venv)` appears at the start of your terminal command line. In VS Code, press `Ctrl+Shift+P` (or `Cmd+Shift+P`), type `Python: Select Interpreter`, and choose the interpreter inside your `.venv` directory.

A.5 - Installing the book's packages

Scientific libraries are installed using Python's package manager, `pip`. Each chapter in the book uses specific packages; Table A.2 summarizes the primary libraries used across chapters.

Table A.2: Python packages used across book chapters

Chapters	Libraries
1–3	`pandas`, `numpy`, `openpyxl`, `pyreadstat`
4	`matplotlib`, `seaborn`
5–7	`scipy`, `statsmodels`
8–10	`statsmodels` (OLS, GLM, MixedLM)
11–13	`scikit-learn`, `xgboost`
14	`scikit-learn` (K-Means, DBSCAN, GMM, PCA)
15	`scikit-learn`, `torch` (PyTorch)
16	`statsmodels` (ARIMA, seasonal decomposition)
17	`quarto`, `shiny`, `matplotlib`
18	`requests`, `openai` / `ollama`
19	`pandas`, `statsmodels`, `seaborn`

To install all required packages into your virtual environment in a single command, run:

```
pip install pandas numpy openpyxl pyreadstat matplotlib seaborn scipy statsmodels scikit-learn
xgboost torch shiny requests
```

A.5.1 - The student wellbeing data: the `data2thesis` package

The case-study data for Elaf's research is bundled in the companion package **`data2thesis`**. You can install it directly via `pip`:

```
pip install data2thesis
```

Verify that the package and dataset load properly:

```
import data2thesis as dt

students = dt.load_students()
print(f"Loaded {students.shape[0]} students and {students.shape[1]} variables.")
```

```
Loaded 600 students and 14 variables.
```

All datasets are also available as direct CSV, Excel, and SPSS downloads on the book website.

A.5.2 - Software versions used in this book

The analyses in this edition were executed with the following package versions:

	Software	Version
0	Python	3.13.2
1	pandas	2.3.3
2	numpy	2.2.6
3	scipy	1.16.2
4	statsmodels	0.15.0
5	scikit-learn	1.9.1
6	data2thesis	1.1.0

A.6 - Common problems and solutions

`'python' is not recognized as an internal or external command`
: Python was installed without checking "Add python.exe to PATH". Re-run the Python Windows installer, select **Modify**, and check the PATH option, or add your Python installation folder (e.g., `C:\Users\Username\AppData\Local\Programs\Python\Python312\`) to your system Environment Variables.

`ModuleNotFoundError: No module named '...'`
: The library is not installed in the currently active Python environment, or VS Code is pointing to a different Python interpreter. Open the VS Code Command Palette (`Ctrl+Shift+P`), select `Python: Select Interpreter`, and choose your project's `.venv`. Then run `pip install <package-name>`.

`Execution of scripts is disabled on this system (Windows PowerShell)`
: PowerShell on Windows restricts script execution by default. Open PowerShell as Administrator and run:

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
```

This allows activating local virtual environment scripts.

`File not found` when loading datasets
: In research scripts, always use relative paths starting from the project root (e.g., `data/students.csv`), or use the helper function `dt.load_students()`. Avoid absolute paths like `C:\Users\Elaf\Desktop\data.csv`, which fail when shared with supervisors or collaborators.

Restarting a clean session
: If variables from earlier interactive runs are causing unexpected errors, restart the Python interactive kernel. In VS Code, click the **Restart** button at the top of the Interactive Window or notebook.

B. Appendix B: Glossary

This appendix collects the key terms from every chapter's review, with a plain-language definition of each and the chapters where it is introduced or applied, followed by a reference table of the Python functions and methods used most frequently in the book.

B.1 - Key terms

```
### A {.unnumbered}

**accuracy**
: The share of cases a classification model classifies correctly. Misleading when one outcome
is rare. *(Chapter 11)*

**activation function**
: The function a neuron applies to its weighted total to produce its output, such as the
sigmoid or the ReLU. *(Chapter 15)*

**adjusted Rand index**
: A measure of agreement between two clusterings or classifications, corrected for chance: 1
means identical groupings, 0 means chance agreement. *(Chapter 14)*

**adjusted R-squared**
: R-squared corrected for the number of predictors, so that adding useless predictors does
not make a model look better. *(Chapter 8)*

**AI coding assistant**
: An AI tool that writes, explains, or completes code, in a chat or inside the editor. Its
suggestions must be checked.

**AIC**
: Akaike information criterion: a measure for comparing models that balances fit against
complexity; lower is better. *(Chapter 10)*

**alternative hypothesis**
: The claim that there is an effect or a difference, tested against the null hypothesis.
*(Chapters 5, 7)*

**annotation**
: Text, arrows, or shapes added to a plot to point out something specific.

**anonymisation**
: Removing or altering information so that the people in a dataset cannot be identified.
*(Chapter 17)*

**ANOVA**
: Analysis of variance: a test of whether the means of three or more groups differ, by comparing
variation between groups with variation within them. *(Chapter 8)*

**API**
: Application programming interface: a way for one program to send requests to another, such
as R sending text to a language model.
```

```
**API key**
: A secret code that identifies you to an online service's API. Keep it in `.Renviron`, never
in a script.

**argument**
: A value given to a function inside its brackets, such as `na.rm = TRUE`, that controls what
the function does. *(Chapter 1)*

**ARIMA**
: A family of time series models that forecast from the autocorrelation of the series and past
random shocks. *(Chapter 16)*

**artificial intelligence**
: Computer systems that perform tasks usually needing human intelligence, such as understanding
language; in this book, mainly large language models.

**aspect ratio**
: The ratio of a graph's width to its height. It changes how steep lines and slopes appear.

**assignment**
: Storing a value in an object with `<-`, as in `x <- 5`. *(Chapters 1, 14)*

**attrition**
: The loss of participants from a study over time. *(Chapters 5, 6)*

**autocorrelation**
: The correlation between a time series and itself a number of steps (lags) earlier. *(Chapter
16)*


### B {.unnumbered}

**bar chart**
: A plot of counts or values as bars, one per category.

**baseline model**
: The simplest possible model, such as predicting the mean for everyone, used as a benchmark
for real models. *(Chapter 13)*

**Bayesian information criterion (BIC)**
: A criterion for comparing models that rewards fit and penalises complexity. In mclust, higher
BIC is better. Also: BIC. *(Chapter 14)*

**benchmark forecast**
: A simple forecast, such as the mean or the value a season earlier, that any serious method
should beat. *(Chapter 16)*

**between-group variation**
: How much group means differ from the overall mean; the numerator of the ANOVA F statistic.
*(Chapter 8)*

**bias**
: In a neural network, the constant added to a neuron's weighted total (like an intercept).
More generally, a systematic error. *(Chapters 5, 11, 15)*

**bias-variance trade-off**
: The balance between a model too simple to capture the pattern (high bias) and one so flexible
that it follows the noise of each sample (high variance). Error on new data is lowest in
between. *(Chapter 11)*
```

```
**BibTeX**
: A text format for bibliographic references, used by Quarto to format citations and reference
lists. *(Chapter 17)*

**bimodal**
: Having two peaks. *(Chapter 6)*

**bin**
: One of the intervals into which a histogram divides the values.

**boosting**
: Building a model by adding many small models (usually trees) one at a time, each correcting
the errors of the model so far. *(Chapter 13)*

**bootstrap**
: Estimating uncertainty by repeatedly resampling the data with replacement and recalculating
a statistic. *(Chapters 7, 12)*

**bootstrap sample**
: A sample of the same size as the data, drawn from it at random with replacement. *(Chapter 12)*

**border point**
: In DBSCAN, a case near a core point that belongs to its cluster but is not itself a core
point. *(Chapter 14)*

**box plot**
: A plot that shows the median, the quartiles, and unusual values of a variable. *(Chapter 4)*

**broom**
: A package that turns model results into tidy data frames, with `tidy()`, `glance()`,
and `augment()`.


### C {.unnumbered}

**case**
: One unit that is measured in a study, such as a student, a patient, or a school; one row of
a data table.

**causal question**
: A research question about what leads to what. It needs a design that rules out other
explanations, ideally an experiment. *(Chapter 5)*

**central limit theorem**
: The result that the sampling distribution of a mean is approximately normal for large enough
samples, whatever the shape of the data. *(Chapter 7)*

**centring**
: Subtracting the mean from a variable, so that zero means "average"; often used before fitting
interactions. *(Chapter 8)*

**chain of reasoning**
: The sequence of steps behind a research conclusion: question, hypothesis, design, variables,
analysis, result, conclusion, and limitation. *(Chapter 19)*

**chi-square test**
: A test for categorical data: whether counts fit expected proportions (goodness of fit), or
whether two categorical variables are related (independence). *(Chapter 7)*

**chunk option**
```

```
: A setting for a code chunk in a Quarto document, written as `#| option: value`, such as
`echo: false`. *(Chapter 17)*

**citation**
: A reference to a source in the text, written in Quarto as `[@key]`.

**citation style (CSL)**
: A file in the Citation Style Language that sets how citations and references are formatted,
such as APA.

**class weights**
: Making errors on a rare class count more when a model is fitted, one remedy for imbalanced
outcomes. *(Chapter 12)*

**classification**
: Predicting a category, such as whether a student will consider dropping out. *(Chapters
11, 12)*

**cluster analysis**
: Methods that group cases so that cases in the same group are similar to each other.
*(Chapter 9)*

**cluster sampling**
: Sampling whole groups, such as all the students of randomly chosen supervisors. *(Chapter 5)*

**code chunk**
: A block of code in a Quarto document that is run when the document is rendered. *(Chapter 17)*

**codebook**
: A description of every variable in a dataset or, in qualitative coding, of every theme and
how to assign it. *(Chapters 2, 18)*

**coefficient path**
: A plot of how each coefficient of a regularised model changes as the penalty changes.
*(Chapter 13)*

**Cohen's *d***
: An effect size for the difference between two means, in standard deviations.

**Cohen's kappa**
: A measure of agreement between two coders or classifiers, corrected for the agreement
expected by chance. *(Chapter 18)*

**comment**
: Text in a script after `#`, which R ignores; used to explain the code. *(Chapter 1)*

**commit**
: In git, a saved snapshot of a project, with a message describing the change. *(Chapter 17)*

**confidence interval**
: A range of plausible values for a population quantity, calculated from a sample, with a
stated level of confidence such as 95%. *(Chapter 7)*

**confirmatory analysis**
: An analysis planned before seeing the data, to test a specific hypothesis. *(Chapters 5, 17)*

**confirmatory factor analysis**
: Factor analysis that tests a factor structure specified in advance. *(Chapter 9)*

**confounder**
```

```
: A variable related to both the predictor and the outcome, which can create or hide an
apparent effect. *(Chapters 5, 6, 8)*

**confusion matrix**
: A table of predicted against actual classes, showing true and false positives and negatives.
*(Chapter 12)*

**Console**
: The RStudio pane where R commands are run and results appear.

**construct**
: An idea that cannot be observed directly, such as stress or wellbeing, and must be measured
indirectly. *(Chapters 5, 9, 18)*

**construct validity**
: Whether a score behaves as the construct should: related to similar measures, less related
to different ones. *(Chapters 5, 18)*

**content validity**
: Whether the items of a measure cover the whole construct. *(Chapter 5)*

**convenience sampling**
: Sampling whoever is easy to reach. Common, and the weakest basis for generalising. *(Chapter
5)*

**convolutional network**
: A type of deep neural network designed for images. *(Chapter 15)*

**core point**
: In DBSCAN, a case with at least `minPts` cases within distance `eps`. *(Chapter 14)*

**correlation coefficient**
: A number from −1 to 1 measuring the strength and direction of a straight-line relationship
between two variables. *(Chapter 6)*

**correlation matrix**
: A table of the correlations between every pair of variables. *(Chapters 2, 9)*

**count outcome**
: An outcome that counts events (0, 1, 2, ...), usually modelled with a Poisson model.
*(Chapter 10)*

**covariance matrix**
: A table of the variances and covariances of several variables; in a mixture model it sets
the shape of a cluster. *(Chapter 14)*

**Cronbach's alpha**
: A measure of the internal consistency (reliability) of a scale made of several items.
*(Chapter 9)*

**cross-loading**
: An item that loads substantially on more than one factor. *(Chapter 9)*

**cross-reference**
: A reference in a Quarto document, such as `@fig-wellbeing`, that becomes a numbered link.
*(Chapter 17)*

**cross-sectional design**
: A design that measures every variable once, at one time. *(Chapter 5)*
```

```
**cross-validation**
: Estimating how well a model predicts new data by repeatedly fitting it on part of the training data and testing it on the rest. *(Chapters 11, 12)*

**CSV file**
: A plain-text file of comma-separated values, the most common format for sharing data tables.


### D {.unnumbered}

**data cleaning**
: Preparing raw data for analysis: removing test and duplicate cases, making categories consistent, marking missing values, and correcting or removing impossible values, with every decision recorded. *(Chapter 3)*

**data frame**
: R's table of data: columns are variables, rows are cases.

**data leakage**
: Information from the test data reaching the model during training, which makes its performance look better than it is. *(Chapter 11)*

**data matrix**
: The arrangement of data as a table with one row per case and one column per variable. *(Chapter 2)*

**data structure**
: The way data is organised in R: vector, factor, data frame, matrix, or list.

**data type**
: The kind of value: numeric, integer, character, or logical. *(Chapter 1)*

**DBSCAN**
: A clustering method that finds dense regions of cases and labels cases in sparse regions as noise. *(Chapter 14)*

**decision tree**
: A model that predicts by a series of yes-or-no questions about the predictors. *(Chapters 11, 12)*

**decomposition**
: Splitting a time series into trend, seasonal, and remainder components. *(Chapter 16)*

**deep learning**
: Neural networks with many hidden layers, used for images, sound, and text. *(Chapter 15)*

**dendrogram**
: The tree diagram produced by hierarchical clustering. *(Chapter 9)*

**density**
: In clustering, how closely packed cases are in a region. *(Chapters 4, 14)*

**density plot**
: A smooth version of a histogram, showing the shape of a distribution. *(Chapter 4)*

**descriptive question**
: A research question about what is: how much, how many, how often. *(Chapter 5)*

**descriptive statistics**
: Numbers that summarise data, such as means, medians, and standard deviations. *(Chapter 6)*
```

```
**design effect**
: How many times more grouped observations are needed to give the same information as independent ones: 1 + (m - 1) × ICC, where m is the group size. *(Chapter 10)*

**deviation**
: The distance of a value from the mean, $x_i - \bar{x}$. The standard deviation summarises the deviations of all values. *(Chapter 6)*

**diagnostic plot**
: A plot used to check a model's assumptions, such as residuals against fitted values. *(Chapter 8)*

**dictionary method**
: Classifying texts by counting keywords from a list for each category.

**diminishing returns**
: A relationship that levels off, so each extra unit of the predictor adds less than the one before. *(Chapter 8)*

**directional hypothesis**
: A hypothesis that predicts the direction of an effect, such as "higher" or "lower". *(Chapter 5)*

**disclosure**
: Stating in a publication how AI tools or other aids were used. *(Chapter 18)*

**distance**
: How different two cases are, calculated from their values on several variables. *(Chapters 9, 12, 14)*

**diverging palette**
: Colours running in two directions away from a meaningful midpoint, such as zero, used for values that can be positive or negative.

**DOI**
: Digital object identifier: a permanent identifier for a publication or dataset, such as `10.1126/science.1213847`. *(Chapter 17)*

**downsampling**
: Balancing classes by randomly removing cases of the common class from the training data. *(Chapter 12)*

**dpi**
: Dots per inch: the resolution of a saved image; 300 dpi is usual for print. *(Chapter 4)*

**dummy variable**
: A 0/1 variable representing one category of a categorical predictor.


### E {.unnumbered}

**effect size**
: A measure of how large an effect is, independent of sample size, such as Cohen's *d* or eta squared.

**eigenvalue**
: In PCA, the amount of variance captured by a component. *(Chapter 9)*

**elastic net**
```

```
: Regularised regression that mixes the ridge and lasso penalties. *(Chapter 13)*

**elbow method**
: Choosing the number of clusters where adding more stops reducing within-cluster distance
much. *(Chapter 9)*

**ensemble**
: A model that combines the predictions of many models, such as a random forest. *(Chapter 12)*

**epoch**
: One pass through the training data when training a neural network. *(Chapter 15)*

**eps**
: In DBSCAN, the radius of the neighbourhood around each case. *(Chapter 14)*

**error message**
: R's message when something goes wrong, which usually says what and where.

**eta squared**
: An effect size for ANOVA: the share of the variation in the outcome explained by the groups.
*(Chapter 8)*

**expected count**
: In a chi-square test, the count expected in a cell if the null hypothesis were true.

**explanation**
: Using a model to understand why something happens, rather than to predict new cases.
*(Chapter 11)*

**explanatory graph**
: A carefully designed graph made for readers, to show a finding clearly and accurately.
*(Chapter 4)*

**exploratory analysis**
: Exploring data to find patterns and generate questions, rather than to test planned
hypotheses. *(Chapters 5, 6, 17)*

**exploratory graph**
: A quick graph made by the researcher to understand the data: to check distributions, find
unusual values, and notice patterns.

**exponential smoothing (ETS)**
: Forecasting with weighted averages of past observations, with more weight on recent ones;
ETS models the error, trend, and season. Also: ETS. *(Chapter 16)*

**external validation**
: Checking clusters or predictions against known categories or outcomes. *(Chapter 14)*

**external validity**
: Whether a study's results apply beyond it, to other people, places, and times. *(Chapter 5)*


### F {.unnumbered}

**F statistic**
: In ANOVA, the ratio of between-group to within-group variation. *(Chapter 8)*

**F1 score**
: A single measure combining precision and recall (their harmonic mean). *(Chapter 12)*
```

```
**facet**
: A small panel of a plot showing one subgroup; `facet_wrap()` makes one panel per group.
*(Chapter 4)*

**factor**
: In R, a categorical variable with a fixed set of levels. In factor analysis, a hidden
(latent) variable behind several items. *(Chapters 5, 9)*

**factor analysis**
: A method that explains the correlations among items by a smaller number of hidden factors.
*(Chapter 9)*

**false negative**
: A case that belongs to the positive class but is predicted negative. *(Chapter 12)*

**false positive**
: A case predicted positive that belongs to the negative class. *(Chapter 12)*

**falsifiability**
: The property of a claim that some possible result would contradict it. A hypothesis that
fits every possible result tells us nothing. *(Chapter 5)*

**fixed effect**
: In a mixed model, an effect assumed the same for everyone, such as the average change over
time. *(Chapter 10)*

**fold**
: One of the parts into which the data is split for cross-validation. *(Chapter 11)*

**forecast horizon**
: How far ahead a forecast is made. *(Chapter 16)*

**Fourier terms**
: Pairs of sine and cosine waves used as predictors to describe a seasonal pattern. *(Chapter
16)*

**function**
: A named piece of code that takes arguments and returns a result, such as `mean()`. *(Chapters
1, 12, 15, 19)*


### G {.unnumbered}

**garden of forking paths**
: The many reasonable choices in an analysis. Choosing among them after seeing the results
inflates the chance of a false positive. *(Chapter 17)*

**Gaussian mixture model**
: A model that describes data as a mix of normal distributions, giving each case a probability
of belonging to each cluster. *(Chapter 14)*

**generalisation**
: The ability of a model to perform well on new data, not only on the data it learned from.
*(Chapter 11)*

**generalised linear mixed model**
: A mixed-effects model for outcomes that are not normal, such as counts or yes/no outcomes.
*(Chapter 10)*

**geom**
```

```
: In ggplot2, the geometric shape that represents data, such as points, lines, or bars.

**Gini impurity**
: A measure of how mixed the classes are in a group; decision trees choose splits that reduce
it. *(Chapter 12)*

**git**
: The standard version control system, which records the history of a project. *(Chapters 4,
5, 17)*

**GitHub**
: A website for storing git projects online, sharing them, and working on them together.
*(Chapter 17)*

**gradient**
: The direction and rate at which the error changes as each weight changes. *(Chapter 15)*

**gradient descent**
: Training a model by repeatedly moving the weights a small step in the direction that reduces
the error. *(Chapter 15)*

**grammar of graphics**
: The idea behind ggplot2: a plot is built from data, aesthetic mappings, and geometric layers.

**graphical perception**
: How people read values from graphs. Positions and lengths are judged most accurately, then
angles, areas, and colours. *(Chapter 4)*

**grouped summary**
: Summary statistics calculated separately for each group, for example with `summarise(.by
= ...)`.


### H {.unnumbered}

**hallucination**
: Plausible but invented content produced by a language model, such as a function or reference
that does not exist. *(Chapter 18)*

**heat map**
: A grid of coloured cells showing values, such as a correlation matrix.

**helper function**
: A small function written to avoid repeating the same code, such as one that formats means.
*(Chapter 19)*

**hidden layer**
: A layer of neurons between the inputs and the output of a neural network. *(Chapter 15)*

**hierarchical clustering**
: Clustering that repeatedly merges the most similar groups, producing a dendrogram. *(Chapter
9)*

**histogram**
: A plot of the distribution of a numeric variable, as bars counting the values in each bin.
*(Chapter 4)*

**hyperparameter**
: A setting of a model that is chosen before fitting rather than learned from the data, such
as the number of neighbours in k-NN. *(Chapter 11)*
```

```
**hypothesis**
: A prediction, stated before the data is analysed, of what the data will show; precise enough
to be wrong. *(Chapters 5, 7)*


### I {.unnumbered}

**identifier**
: A variable that uniquely identifies each case, such as `student_id`.

**imbalanced outcome**
: An outcome in which one class is much rarer than the other. *(Chapter 11)*

**imputation**
: Filling in missing values with estimated ones, such as the median.

**independence**
: The assumption that observations do not influence each other; in a chi-square test, the
hypothesis that two variables are unrelated.

**index**
: In a tsibble, the column that holds time. *(Chapters 2, 3, 14, 16)*

**inferential analysis**
: Using a sample to draw conclusions about a population. *(Chapter 6)*

**inline code**
: R code inside a sentence of a Quarto document, replaced by its result when rendered.
*(Chapter 17)*

**input**
: In Shiny, a control such as a menu or slider whose value the user chooses. *(Chapter 15)*

**input layer**
: The inputs (predictors) of a neural network. *(Chapter 15)*

**interaction**
: When the effect of one predictor depends on the value of another. *(Chapter 8)*

**interaction plot**
: A plot of group means that shows whether the effect of one factor depends on another.
*(Chapter 8)*

**intercept**
: The predicted value of the outcome when all predictors are zero. *(Chapters 8, 10)*

**internal consistency**
: The extent to which the items of a scale agree with each other, often measured with Cronbach's
alpha. *(Chapters 5, 9)*

**internal validation**
: Judging clusters by how compact and separated they are, using the data alone, as with the
silhouette. *(Chapter 14)*

**internal validity**
: Whether a study can rule out other explanations for its results. Highest in randomised
experiments. *(Chapter 5)*

**interquartile range**
```

```
: The range of the middle 50% of the values: the third quartile minus the first. *(Chapter 6)*

**inter-rater reliability**
: The extent to which two people coding or rating the same material agree, often measured with Cohen's kappa. *(Chapters 5, 18)*

**interval**
: A level of measurement with equal distances between values but no true zero, such as a wellbeing index. *(Chapters 5, 7, 16)*

**intraclass correlation**
: The share of the total variation that lies between groups (such as students or supervisors). *(Chapter 10)*


### J {.unnumbered}

**join**
: Combining two tables by matching rows on a key, such as `student_id`. *(Chapter 3)*


### K {.unnumbered}

**Kaiser rule**
: Keeping components with an eigenvalue above 1; a rough guide that often keeps too many. *(Chapter 9)*

**kernel**
: In a support vector machine, a function that allows curved boundaries between classes. *(Chapter 12)*

**k-means**
: A clustering method that assigns each case to the nearest of *k* cluster centres and moves the centres to the means. *(Chapter 9)*

**k-nearest neighbours**
: Predicting a case from the *k* most similar cases in the training data. *(Chapters 11, 12)*

**k-nearest-neighbour distance plot**
: A sorted plot of each case's distance to its *k*-th nearest neighbour, used to choose `eps` for DBSCAN. *(Chapter 14)*

**knitr**
: The R package that runs the code in Quarto and R Markdown documents.

**Kruskal-Wallis test**
: A non-parametric alternative to one-way ANOVA. *(Chapter 8)*


### L {.unnumbered}

**lag**
: The number of time steps between an observation and an earlier one it is compared with. *(Chapter 16)*

**large language model**
: A neural network trained on huge amounts of text to predict the next token, used in AI assistants. *(Chapter 18)*

**lasso**
```

```
: Regularised regression with a penalty on the absolute size of the coefficients, which sets
some of them to exactly zero. *(Chapter 13)*

**latent variable**
: A variable that cannot be observed directly, such as stress, and is assumed to cause part
of the answers to the items that measure it. *(Chapter 9)*

**LaTeX**
: A typesetting system whose notation is used to write equations, as in `$\bar{x}$`. *(Chapter
17)*

**layer**
: In ggplot2, one part of a plot added with `+`, such as a set of points or a trend line. In
a neural network, a group of neurons. *(Chapter 15)*

**leaf**
: A final node of a decision tree, which gives the prediction. *(Chapter 12)*

**learning rate**
: In boosting and neural networks, how large a step each update takes. *(Chapters 13, 15)*

**least squares**
: Choosing a regression line that minimises the sum of the squared residuals. *(Chapter 8)*

**level**
: One of the categories of a factor. *(Chapters 5, 7, 10)*

**level of measurement**
: What the values of a variable mean (nominal, ordinal, interval, or ratio), which decides
the summaries and tests that make sense. *(Chapter 5)*

**Levene's test**
: A test of whether groups have equal variances. *(Chapter 8)*

**licence**
: A statement of what others may do with shared data or code, such as CC BY or MIT.

**likelihood ratio test**
: A test comparing two nested models by how much better the larger one fits. *(Chapter 10)*

**line chart**
: A plot of values connected by lines, often over time.

**linear regression**
: A model of a numeric outcome as a straight-line function of one or more predictors.
*(Chapter 8)*

**list**
: An R object that can hold elements of different types and sizes, such as the results of a
test. *(Chapter 1)*

**loading**
: The correlation between an item and a factor or component. *(Chapter 9)*

**local model**
: A language model that runs on your own computer, so data does not leave it. *(Chapter 18)*

**lockfile**
: A file, such as renv's `renv.lock`, that records the exact package versions of a project.
*(Chapter 17)*
```

```
**log loss**
: The measure of error used to fit logistic regression and classification networks: small when high probabilities are given to the cases that did occur and low probabilities to those that did not. *(Chapter 15)*

**logical indexing**
: Selecting elements with a TRUE/FALSE condition, as in `x[x > 5]`.

**logistic regression**
: A regression model for a yes/no outcome, which predicts the probability of "yes". *(Chapter 8)*

**log-odds**
: The logarithm of the odds; the scale on which logistic regression coefficients are estimated. *(Chapter 8)*

**long format**
: Data with one row per observation, such as one row per student per semester. *(Chapter 3)*

**longitudinal design**
: A design that measures the same people repeatedly over time. *(Chapter 5)*


### M {.unnumbered}

**machine learning**
: Methods that learn patterns from data to make predictions or find structure, judged by performance on new data. *(Chapter 11)*

**MAE**
: Mean absolute error: the average size of the prediction errors. *(Chapter 13)*

**main effect**
: The effect of one factor, averaged over the levels of another. *(Chapter 8)*

**Mann-Whitney U test**
: A non-parametric test comparing two independent groups.

**mapping**
: In ggplot2, linking a variable to a visual property (an aesthetic), such as position or colour. Also: aesthetic mapping.

**margin**
: In a support vector machine, the empty band between the classes and the boundary. *(Chapter 12)*

**Markdown**
: A simple way of formatting text with symbols, such as `**bold**` and `# Heading`. *(Chapter 17)*

**matrix**
: A two-dimensional table of values of one type. *(Chapters 2, 9, 12, 14)*

**mean**
: The average: the sum of the values divided by their number. *(Chapters 6, 9, 16)*

**mean method**
: A benchmark forecast that predicts the average of past observations. *(Chapter 16)*

**median**
```

```
: The middle value when the values are sorted. *(Chapter 6)*

**mediator**
: A variable on the path between a predictor and an outcome, through which the predictor has
its effect. *(Chapter 5)*

**membership probability**
: The probability that a case belongs to each cluster, as given by a mixture model. Also: soft
assignment. *(Chapter 14)*

**method choice**
: Choosing an analysis method from the goal, the type of outcome, and how observations are
related. *(Chapter 19)*

**minPts**
: In DBSCAN, the number of cases a neighbourhood must contain for a case to be a core point.
*(Chapter 14)*

**misclassification cost**
: The cost assigned to each kind of error of a classifier, such as missing a case or raising
a false alarm. The costs determine the best threshold. *(Chapter 12)*

**missing at random**
: Missingness that depends only on observed variables, not on the missing values themselves.
*(Chapter 6)*

**missing code**
: A value such as `99` or `-9` used in raw data to mark a missing answer. *(Chapter 3)*

**missing completely at random**
: Missingness unrelated to any variable, observed or not. *(Chapter 6)*

**missing not at random**
: Missingness that depends on the missing values themselves. *(Chapter 6)*

**missing value (`NA`)**
: R's marker for a value that is not available. Also: NA. *(Chapters 1, 2, 4, 5, 6, 7, 9, 11,
13, 14, 15, 16, 17, 18)*

**mixed-effects model**
: A regression model with both fixed effects and random effects, for repeated or nested data.
*(Chapter 10)*

**mixing probability**
: In a mixture model, the share of cases belonging to a component. *(Chapter 14)*

**mixture**
: In regularised regression, the mix of lasso and ridge penalties, from 0 (ridge) to 1 (lasso).
*(Chapters 13, 14)*

**mixture component**
: One of the normal distributions in a Gaussian mixture model. *(Chapter 14)*

**mode**
: The most common value. *(Chapters 5, 6, 10, 13, 14, 15, 18, 19)*

**model specification**
: In tidymodels, the description of a model (type, engine, mode) before it is fitted.

**moderator**
```

```
: A variable that changes the strength or direction of the relationship between a predictor
and an outcome. *(Chapter 5)*

**`mtry`**
: In a random forest, the number of predictors each split may choose from. Also: mtry.
*(Chapter 12)*

**multicollinearity**
: Strong correlation among predictors, which makes regression coefficients unstable. *(Chapter
13)*

**multilayer perceptron**
: A neural network with one or more hidden layers of neurons. *(Chapter 15)*

**multilevel model**
: Another name for a mixed-effects model, emphasising nested levels. *(Chapter 10)*

**multiple regression**
: Linear regression with more than one predictor. *(Chapter 8)*

**multiple testing**
: Running many tests in one study. Each test carries its own risk of a false alarm, so the
chance of at least one grows with the number of tests.

**multivariate analysis**
: Methods that analyse many variables at once, such as PCA and factor analysis. *(Chapter 9)*


### N {.unnumbered}

**naive method**
: A benchmark forecast that repeats the last observed value. *(Chapter 16)*

**nested data**
: Data in which cases are grouped within units, such as students within supervisors. *(Chapter
10)*

**neural network**
: A model made of connected neurons in layers, whose weights are learned from data. *(Chapter
15)*

**neuron (unit)**
: The basic element of a neural network: it weights its inputs, adds a bias, and applies an
activation function. Also: neuron, unit. *(Chapters 2, 5, 15)*

**node**
: A point in a decision tree where a question is asked or a prediction is made. *(Chapter 12)*

**noise**
: In DBSCAN, cases in sparse regions that belong to no cluster. *(Chapter 14)*

**nominal**
: A level of measurement for categories with no order, such as faculty. *(Chapter 5)*

**non-directional hypothesis**
: A hypothesis that predicts a difference or relationship without saying in which direction.
*(Chapter 5)*

**non-parametric test**
: A test that does not assume a particular distribution, often based on ranks.
```

```
**normal distribution**
: The symmetric, bell-shaped distribution described by a mean and a standard deviation.
*(Chapter 6)*

**normalisation**
: Rescaling variables, usually to z-scores, so that they are on the same scale.

**null distribution**
: The distribution of a statistic that would be expected if the null hypothesis were true,
used to judge how surprising the observed result is.

**null hypothesis**
: The claim of no effect or no difference, which a test tries to reject. *(Chapters 5, 7)*


### O {.unnumbered}

**object**
: A named value stored in R, such as a number, a vector, or a data frame.

**oblique rotation**
: A factor rotation that allows the factors to correlate. *(Chapter 9)*

**odds**
: The probability of an event divided by the probability of it not happening. *(Chapter 8)*

**odds ratio**
: The ratio of the odds in two groups; in logistic regression, the multiplicative change in
odds per unit of a predictor. *(Chapter 8)*

**one-sample t-test**
: A test of whether a mean differs from a given value.

**one-tailed test**
: A test that looks for a difference in one direction only.

**open science**
: Practices that make research transparent and reusable: sharing data and code, preregistration,
and open access. *(Chapter 17)*

**operationalisation**
: The decision about how a construct is measured: which questions, records, or instruments
turn it into a variable. *(Chapter 5)*

**ordered factor**
: A factor whose levels have an order, created with `factor(..., ordered = TRUE)`. *(Chapter 5)*

**ordinal**
: A level of measurement for ordered categories whose distances are unknown, such as none,
part-time, and full-time employment. *(Chapters 2, 5)*

**outcome**
: The variable a model explains or predicts. *(Chapters 5, 8, 10, 11)*

**outlier**
: A value far from the others. *(Chapter 6)*

**output**
: In Shiny, a plot, table, or text that the server produces for the user interface. *(Chapters
```

```
15, 18)*

**output format**
: The kind of document Quarto produces, such as HTML, Word, or PDF.

**output layer**
: The final layer of a neural network, which produces the prediction. *(Chapter 15)*

**overfitting**
: A model learning the noise in its training data, so it performs well there and poorly on
new data. *(Chapters 11, 13)*

**overplotting**
: Points drawn on top of each other in a plot, hiding how many there are. *(Chapter 4)*


### P {.unnumbered}

**package**
: A collection of R functions, data, and documentation that adds features to R. *(Chapter 1)*

**paired t-test**
: A test comparing two measurements on the same cases.

**Pandoc**
: The document converter that Quarto uses to produce HTML, Word, PDF, and other formats.

**parallel analysis**
: Choosing the number of factors by comparing eigenvalues with those from random data.
*(Chapter 9)*

**parameter**
: A quantity describing a population (Chapter 7); a value learned by a model, such as a
coefficient (Chapter 11); or an input to a Quarto report (Chapter 17). *(Chapters 7, 11, 17)*

**patchwork**
: A package that combines ggplot2 plots into one figure.

**penalty (lambda)**
: In regularised regression or a neural network, the strength of the penalty on large
coefficients or weights. Also: penalty, lambda. *(Chapter 13)*

**permutation importance**
: A predictor's importance measured by how much shuffling its values reduces a model's accuracy.
*(Chapter 12)*

**permutation test**
: A test that builds the null distribution by shuffling group labels many times, and compares
the observed result with the shuffled ones. *(Chapter 7)*

**pipe**
: The operator `|>`, which passes the result on its left to the function on its right.
*(Chapters 11, 19)*

**pipeline**
: The series of steps, in code, from raw data to results. *(Chapters 11, 19)*

**Poisson model**
: A regression model for counts. *(Chapter 10)*
```

```
**population**
: The whole group a study wants to draw conclusions about. *(Chapters 5, 7)*

**post-hoc test**
: A test run after ANOVA to find which groups differ, such as Tukey's test. *(Chapter 8)*

**power**
: The probability that a study detects an effect of a given size, if the effect is real. A
common target is 80%. *(Chapters 5, 7)*

**precision**
: The share of cases predicted positive that are truly positive. *(Chapter 12)*

**predicted probability**
: A model's estimated probability of an outcome for a case. *(Chapter 8)*

**prediction**
: Using a model to estimate the outcome for new cases. *(Chapters 11, 13, 16)*

**prediction interval**
: A range within which a future observation is expected to fall with a stated probability.
*(Chapter 16)*

**predictive analysis**
: Analysis aimed at predicting new cases rather than explaining or testing. *(Chapter 6)*

**predictor**
: A variable used to explain or predict the outcome. *(Chapters 5, 8)*

**preregistration**
: Publicly recording hypotheses and planned analyses before seeing the data. *(Chapters 5, 17)*

**pretrained model**
: A model already trained by others on large datasets, used as it is or adapted. *(Chapter 15)*

**principal component**
: A combination of variables that captures as much of their variation as possible. *(Chapter 9)*

**principal component analysis**
: A method that summarises many correlated variables with a few principal components. *(Chapter
9)*

**project structure**
: The organisation of a project's folders and files, such as `data-raw/`, `R/`, and `output/
`. *(Chapter 19)*

**prompt**
: The text given to a language model. *(Chapter 18)*

**push**
: In git, copying commits to an online repository such as GitHub. *(Chapter 17)*

**p-value**
: The probability of results at least as extreme as those observed, if the null hypothesis
were true.


### Q {.unnumbered}

**Q-Q plot**
```

```
: A plot comparing a variable's values with those expected from a normal distribution.
*(Chapter 6)*

**quadratic term**
: A squared predictor in a regression model, which allows a curved relationship. *(Chapter 8)*

**qualitative coding**
: Assigning themes or categories to texts such as open-ended answers. *(Chapter 18)*

**qualitative palette**
: A set of clearly different colours of similar strength, used to distinguish categories.

**quartile**
: The values that divide sorted data into four equal parts. *(Chapter 6)*

**Quarto**
: A publishing system that turns documents with text and code into reports, books, websites,
and slides. *(Chapter 17)*

**quasi-experiment**
: A study that compares groups receiving different treatments that were not assigned by chance.
*(Chapter 5)*


### R {.unnumbered}

**R**
: The programming language and environment for statistics used in this book. *(Chapters 1, 2,
3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19)*

**R Markdown**
: The predecessor of Quarto: documents (`.Rmd`) that combine text and R code.

**radial basis function**
: A common kernel for support vector machines, which allows curved, rounded boundaries.
*(Chapter 12)*

**random effect**
: In a mixed model, variation between groups (such as students) described by a distribution
rather than a separate estimate for each group. *(Chapter 10)*

**random error**
: The difference between an estimate and the truth caused by which cases happened to be
selected. It shrinks as the sample grows. *(Chapter 5)*

**random forest**
: An ensemble of decision trees, each grown on a bootstrap sample with random subsets of
predictors. *(Chapter 12)*

**random intercept**
: A random effect that lets each group have its own average level. *(Chapter 10)*

**random slope**
: A random effect that lets each group have its own effect of a predictor, such as its own
rate of change. *(Chapter 10)*

**randomised experiment**
: A study in which the researcher assigns the treatment by chance, so that the groups differ
only by chance at the start. *(Chapter 5)*
```

```
**range**
: The difference between the largest and smallest values. *(Chapter 6)*

**rate ratio**
: In a Poisson model, the multiplicative change in the expected count per unit of a predictor.
*(Chapter 10)*

**ratio**
: A level of measurement with equal distances and a true zero, so that "twice as much" is
meaningful, such as hours of sleep. *(Chapters 5, 8, 10, 13, 17)*

**raw data**
: Data as it was collected or exported, before any cleaning; never edited by hand. *(Chapter 19)*

**reactive expression**
: In Shiny, a calculation, made with `reactive()`, that is updated automatically when its
inputs change.

**reactivity**
: Shiny's system for updating outputs automatically when inputs change. *(Chapter 17)*

**recall**
: The share of truly positive cases that a model predicts as positive. Also: sensitivity.
*(Chapter 12)*

**recipe**
: In tidymodels, a list of data preparation steps, such as imputation and normalisation.

**reference category**
: The category of a categorical predictor that the others are compared with. *(Chapter 8)*

**registered report**
: A publication format in which a journal accepts a study's plan before the data is collected.
*(Chapter 17)*

**regression**
: Modelling or predicting a numeric outcome. In machine learning, regression means predicting
a number rather than a category. Also: Regression (prediction). *(Chapters 4, 8, 11, 13)*

**regular expression**
: A pattern for matching text, such as `"\\b(money|fee)"`.

**regularisation**
: Penalising large coefficients to prevent overfitting, as in ridge and lasso regression.
*(Chapter 13)*

**relational question**
: A research question about which variables go together. It establishes association, not cause.
*(Chapter 5)*

**reliability**
: How consistently a scale measures what it measures. *(Chapters 5, 9, 18)*

**ReLU**
: Rectified linear unit: an activation function that sets negative values to zero. *(Chapter
15)*

**remainder**
: In a time series decomposition, what is left after removing trend and season. *(Chapter 16)*
```

```
**render**
: Running the code in a Quarto document and producing the finished output.

**renv**
: A package that records and restores the package versions used by a project.

**repeated measures**
: Several measurements of the same cases, such as each student in four semesters. *(Chapter 10)*

**replicability**
: Whether a new study, with new data, reaches the same conclusions. *(Chapter 17)*

**reporting sentence**
: A sentence in a results section that states a finding with its numbers, such as an estimate, interval, and p-value. *(Chapter 19)*

**reporting standard**
: A published list of the information a research report must contain, such as JARS for psychology, CONSORT for randomised trials, and STROBE for observational studies. *(Chapter 19)*

**repository**
: A project folder whose history git keeps; also its online copy on GitHub. *(Chapter 17)*

**reproducibility**
: Whether the same data and code give exactly the same results. *(Chapters 17, 18)*

**reproducible research**
: Research whose results can be recomputed from the shared data and code. *(Chapter 1)*

**research question**
: A focused question that says precisely what a study will find out, narrow enough that data can answer it. *(Chapter 5)*

**research workflow**
: The path of a research project from question and data to reported results. *(Chapter 19)*

**residual**
: The difference between an observed value and the value a model predicts. Also: error. *(Chapters 5, 7, 8, 13)*

**reversed item**
: A questionnaire item worded in the opposite direction to the others, which must be reversed before scoring. *(Chapter 9)*

**ridge regression**
: Regularised regression with a penalty on the squared coefficients, which shrinks them all towards zero. *(Chapter 13)*

**RMSE**
: Root mean squared error: the square root of the average squared prediction error. *(Chapter 13)*

**ROC AUC**
: The area under the ROC curve: how often a model ranks a positive case above a negative one; 0.5 is chance, 1 is perfect. *(Chapter 11)*

**ROC curve**
: A plot of sensitivity against the false positive rate across all classification thresholds. *(Chapter 12)*
```

```
**rotation**
: In factor analysis, turning the factors to make the loadings easier to interpret. *(Chapter 9)*

**R-squared**
: The share of the variation in the outcome that a model explains, from 0 to 1. Also: R².
*(Chapters 8, 13)*

**RStudio**
: The most widely used editor for R, used throughout this book.

**RStudio Project**
: A folder that RStudio treats as the home of one piece of work, setting the working directory.


### S {.unnumbered}

**sample**
: The part of a population that is observed. *(Chapters 5, 7, 12, 14, 19)*

**sample description ("Table 1")**
: A table describing the participants, usually the first table of a results chapter. Also:
sample description. *(Chapter 19)*

**sampling distribution**
: The distribution of a statistic over many possible samples. *(Chapter 7)*

**sampling frame**
: A list of the members of a population, from which a sample can be drawn. *(Chapter 5)*

**scale**
: In ggplot2, the control of how data values are turned into positions, colours, or sizes.
*(Chapters 2, 3, 4, 11)*

**scale score**
: A score combining several questionnaire items, usually their mean. *(Chapter 3)*

**scaling**
: Converting variables to a common scale, usually z-scores, before calculating distances.
*(Chapter 9)*

**scatter plot**
: A plot of two numeric variables as points.

**scree plot**
: A plot of eigenvalues, used to choose the number of components or factors. *(Chapter 9)*

**script**
: A file of R code that can be saved and rerun. *(Chapters 1, 5, 6, 19)*

**seasonal naive method**
: A benchmark forecast that repeats the value from the same season a cycle earlier. *(Chapter
16)*

**seasonal period**
: The length of a repeating pattern, such as 52 weeks or 12 months. *(Chapter 16)*

**seasonal plot**
: A plot that overlays the cycles of a time series to show its seasonal pattern. *(Chapter 16)*

**seasonality**
```

```
: A pattern in a time series that repeats at a fixed period. *(Chapter 16)*

**seasonally adjusted series**
: A time series with the seasonal component removed. *(Chapter 16)*

**self-report**
: A measure based on what participants say about themselves, such as their estimated hours of sleep. *(Chapter 5)*

**self-selection**
: When people choose their own group, such as volunteering for a workshop, so that the groups may differ from the start. *(Chapter 5)*

**sequential palette**
: Colours running from light to dark, used for quantities that go in one direction.

**server**
: In Shiny, the function that computes the outputs from the inputs. *(Chapter 17)*

**setting**
: In ggplot2, giving a visual property a fixed value, such as `colour = "blue"`, rather than mapping it to a variable.

**Shiny**
: An R package for building interactive web applications and dashboards. *(Chapter 17)*

**shrinkage**
: Pulling coefficients towards zero, as regularisation does. *(Chapter 13)*

**sigmoid**
: The S-shaped function that turns any number into a value between 0 and 1. *(Chapter 15)*

**significance level**
: The threshold (usually 0.05) below which a p-value counts as statistically significant. *(Chapter 7)*

**silhouette**
: A measure of how much closer each case is to its own cluster than to the next nearest, from −1 to 1. *(Chapters 9, 14)*

**simple random sampling**
: Sampling in which every member of the population has the same chance of being chosen. *(Chapter 5)*

**simulation**
: Using random numbers to generate many possible outcomes, for example many possible futures of a time series. *(Chapter 16)*

**singular fit**
: A warning that a mixed model estimates some variation as zero, often because the model is too complex for the data. *(Chapter 10)*

**skewness**
: Asymmetry of a distribution; a long tail to the right is positive skew. *(Chapter 6)*

**slope**
: The change in the predicted outcome for a one-unit change in a predictor. *(Chapters 8, 10)*

**small cells**
: Combinations of variables that describe very few people, who could be identified. *(Chapter
```

```
17)*

**SMOTE**
: A method that balances classes by creating artificial cases of the rare class between
existing ones. *(Chapter 12)*

**specificity**
: The share of truly negative cases that a model predicts as negative. *(Chapter 12)*

**stability**
: Whether clusters or results stay the same under small changes to the data or the method.
*(Chapter 14)*

**standard deviation**
: A measure of spread: roughly the typical distance of values from the mean. *(Chapter 6)*

**standard error**
: The standard deviation of a sampling distribution; the uncertainty of an estimate. *(Chapter
7)*

**statistic**
: A number calculated from a sample, such as a mean or a t value. *(Chapters 6, 7, 8)*

**statistically significant**
: A result with a p-value below the significance level.

**STL**
: Seasonal and trend decomposition using loess: a flexible method for decomposing a time
series. *(Chapter 16)*

**stratified sampling**
: Sampling in which the population is divided into groups (strata) and a random sample is
drawn from each. *(Chapter 5)*

**stratified split**
: A split of the data that keeps the same share of each outcome class in each part. *(Chapter 11)*

**structured output**
: A language model's answer in a fixed format, such as one category from a list, that code
can use directly. *(Chapter 18)*

**study design**
: The plan for who is measured, on what, when, and under which conditions. *(Chapter 5)*

**supervised learning**
: Machine learning with a known outcome to predict. *(Chapter 11)*

**support vector**
: A case close to the boundary of a support vector machine, which determines where the boundary
lies. *(Chapter 12)*

**support vector machine**
: A classifier that separates classes with the widest possible margin. *(Chapter 12)*

**symmetrical**
: Having the same shape on both sides of the centre. *(Chapter 6)*

**synthetic data**
: Artificial data with the structure of real data, used when the real data cannot be shared.
*(Chapter 17)*
```

```
**system prompt**
: Instructions given to a language model before the conversation, such as a task and a codebook.
*(Chapter 18)*


### T {.unnumbered}

**temperature**
: A setting of a language model that controls how random its answers are; 0 gives the most
likely answer. *(Chapter 18)*

**test set**
: Data set aside to evaluate a final model once, at the end. *(Chapter 11)*

**test-retest reliability**
: The extent to which a measure gives similar results for the same people on two occasions.
*(Chapter 5)*

**theme**
: In ggplot2, the non-data appearance of a plot, such as fonts and background. In qualitative
coding, a category of meaning.

**threshold**
: The probability above which a classifier predicts the positive class. *(Chapter 12)*

**tibble**
: The tidyverse's version of a data frame, which prints more neatly.

**tidy data**
: Data with one variable per column, one observation per row, and one value per cell.
*(Chapter 3)*

**tidyverse**
: A collection of R packages, such as dplyr and ggplot2, that share one way of working.

**time plot**
: A plot of a time series against time. *(Chapter 16)*

**time series**
: Observations of one quantity at regular intervals over time. *(Chapter 16)*

**token**
: A word or part of a word, the unit of text a language model reads and writes. *(Chapter 18)*

**training**
: Adjusting a model's weights or parameters to fit the training data. *(Chapters 11, 15)*

**training cut-off**
: The date after which a language model's training data contains no information.

**training set**
: The data used to build and tune a model. *(Chapter 11)*

**transformer**
: The neural network architecture behind large language models. *(Chapters 11, 15)*

**tree depth**
: The number of questions in a row a decision tree may ask. *(Chapter 13)*
```

```
**trend**
: The long-term direction of a time series. *(Chapters 4, 16)*

**trend line**
: A line added to a scatter plot to show the overall relationship. *(Chapter 4)*

**true negative**
: A negative case correctly predicted negative. *(Chapter 12)*

**true positive**
: A positive case correctly predicted positive. *(Chapter 12)*

**tsibble**
: A data frame for time series, which knows which column is time.

**Tukey's test**
: A post-hoc test comparing every pair of groups, with a correction for multiple comparisons.
*(Chapter 8)*

**tuning**
: Choosing hyperparameters by comparing their performance, usually with cross-validation.
*(Chapter 11)*

**two-sample t-test**
: A test comparing the means of two independent groups.

**two-tailed test**
: A test that looks for a difference in either direction.

**two-way ANOVA**
: ANOVA with two factors, including their interaction. *(Chapter 8)*

**type conversion**
: Changing a value from one data type to another, such as text to numbers.

**Type I error**
: Rejecting a true null hypothesis: a false alarm. *(Chapter 7)*

**Type II error**
: Failing to reject a false null hypothesis: a missed effect. *(Chapter 7)*


### U {.unnumbered}

**uncertainty**
: In a mixture model, 1 minus a case's largest membership probability. *(Chapter 14)*

**underfitting**
: A model too simple to capture the real pattern in the data. *(Chapter 11)*

**unimodal**
: Having one peak. *(Chapter 6)*

**unit of analysis**
: What one row of the data represents, and what the question is about: a student, a semester,
a supervisor. *(Chapters 2, 5)*

**unsupervised learning**
: Machine learning without an outcome, looking for structure such as clusters. *(Chapter 11)*
```

```
**upsampling**
: Balancing classes by repeating cases of the rare class in the training data. *(Chapter 12)*

**user interface**
: In Shiny, what the user sees: inputs and places for outputs.


### V {.unnumbered}

**validity**
: Whether a measure measures what it claims to, or whether a study's conclusions are justified.
*(Chapters 5, 18)*

**value label**
: A label attached to a coded value, such as 1 = "Strongly disagree", as in SPSS files.

**variable**
: One characteristic measured on every case, such as age or faculty; one column of a data
table. *(Chapters 1, 9, 13)*

**variable label**
: A longer description attached to a variable, as in SPSS files.

**variable selection**
: Choosing which predictors to keep in a model; the lasso does it automatically. *(Chapter 13)*

**variance**
: The average squared distance from the mean; the square of the standard deviation. *(Chapters
6, 11, 14)*

**vector**
: R's basic structure: a sequence of values of the same type. *(Chapters 2, 3, 4, 12)*

**vector format**
: An image format, such as PDF or SVG, that stays sharp at any size.

**verb**
: A dplyr function that does one thing to a data frame, such as `filter()` or `mutate()`.

**version control**
: Keeping a history of every change to a project's files.


### W {.unnumbered}

**Ward's method**
: A way of merging clusters in hierarchical clustering that keeps them as compact as possible.
*(Chapter 9)*

**weak learner**
: A simple model, such as a small tree, that is only slightly better than chance on its own;
boosting combines many. *(Chapter 13)*

**weight**
: In a neural network, the number by which an input is multiplied; learned during training.
*(Chapters 12, 15)*

**weight decay**
: A penalty on large weights in a neural network, which prevents overfitting. *(Chapter 15)*
```

```
**Welch test**
: The version of the two-sample t-test that does not assume equal variances; R's default.

**wide format**
: Data with repeated measurements side by side in columns, one row per case. *(Chapter 3)*

**Wilcoxon signed-rank test**
: A non-parametric test comparing two measurements on the same cases. *(Chapter 7)*

**within-group variation**
: How much values vary around their own group mean; the denominator of the ANOVA F statistic.
*(Chapter 8)*

**workflow**
: In tidymodels, a recipe and a model specification combined. *(Chapter 19)*

**workflow set**
: In tidymodels, several workflows fitted and compared together.

**working directory**
: The folder R reads files from and saves files to by default.


### X {.unnumbered}

**XGBoost**
: Extreme gradient boosting: a fast, widely used boosting method.


### Y {.unnumbered}

**YAML header**
: The settings at the top of a Quarto document, between two `---` lines. *(Chapter 17)*


### Z {.unnumbered}

**z-score**
: A value expressed as the number of standard deviations from the mean. *(Chapter 6)*
```

B.2 - Python functions and methods

Table B.1 lists the functions and methods used most frequently in the book, grouped by research task, with their library and the chapters where they appear in code.

Table B.1: Essential Python functions and methods in empirical research.

	Task	Function / Method	Library	What it does	Chapters
0	Getting started	`print()`	Python built-in	Output text or variables to the console	1, 2, 3, and later
1	Getting started	`type()`	Python built-in	Inspect the data type of an object	1, 2, 3, and later
2	Getting started	`len()`	Python built-in	Return the number of items in a list, string, …	1, 2, 3, and later
3	Getting started	`round()`	Python built-in	Round numbers to a specified number of decimal…	1, 2, 3, and later
4	Getting started	`help()`	Python built-in	Display documentation for a function or object	1
5	Importing and data loading	`pd.read_csv()`	pandas	Read a CSV file into a DataFrame	2, 18
6	Importing and data loading	`pd.read_excel()`	pandas	Read an Excel spreadsheet into a DataFrame	2, 3, 19
7	Importing and data loading	`pd.read_spss()`	pandas	Read an SPSS dataset (.sav) into a DataFrame	2
8	Importing and data loading	`df.to_csv()`	pandas	Save a DataFrame to a CSV file	2
9	Cleaning and transforming	`df.query()`	pandas	Filter rows matching a boolean expression	3, 4, 6, and later
10	Cleaning and transforming	`df.drop()`	pandas	Drop specified rows or columns	3, 9, 11, and later
11	Cleaning and transforming	`df.rename()`	pandas	Rename DataFrame columns or index labels	3, 13, 19
12	Cleaning and transforming	`df.sort_values()`	pandas	Sort rows by one or more column values	3, 4, 8, and later
13	Cleaning and transforming	`df.assign()`	pandas	Assign new columns to a DataFrame	3, 8, 10, and later
14	Cleaning and transforming	`df.groupby()`	pandas	Group DataFrame rows by categorical variables …	3, 4, 5, and later
15	Cleaning and transforming	`df.agg()`	pandas	Compute multiple summary statistics on grouped…	3, 4, 5, and later
16	Cleaning and transforming	`df.value_counts()`	pandas	Return frequency counts of unique values	1, 2, 3, and later
17	Cleaning and transforming	`df.merge()`	pandas	Merge (join) two DataFrames on a key column	3, 4, 5, and later
[illegible]	[illegible] forming	[illegible] Classifier()	[illegible]	[illegible]	[illegible]

C. Appendix C: Python Packages

This appendix lists every Python scientific library used across the book: what it is for, the chapters that apply it, and the package version used when compiling this edition. Appendix A shows how to install all of them into a project virtual environment using `pip`.

In Python, a library (or package) is a reusable collection of modules, functions, classes, and documentation shared via the Python Package Index (PyPI). Python includes a robust standard library (`sys`, `os`, `math`, `json`, `datetime`) available without installation; third-party scientific packages are installed once via `pip install <package>` and imported into your analysis sessions using `import <package>`.

C.1 - The packages in this book

Table C.1 summarizes the packages used in this book.

Table C.1: The Python scientific packages used in this book.

	Area	Package	What it is for	Chapters	Version
0	Data	`data2thesis`	Graduate Wellbeing Study case-study datasets a…	1, 2, 3, and later	1.1.0
1	Core & Computing	`numpy`	Multidimensional arrays, numerical routines, a…	1, 2, 3, and later	2.2.6
2	Data Handling	`pandas`	Foundational tabular data structures (DataFram…	1, 2, 3, and later	2.3.3
3	Data Formats	`openpyxl`	Reading and writing Microsoft Excel (.xlsx) fi…	2	3.1.5
4	Data Formats	`pyreadstat`	Reading SPSS (.sav), Stata, and SAS datasets w…	2, 19	1.3.7
5	Visualisation	`matplotlib`	Foundational scientific plotting engine, layou…	1, 4, 5, and later	3.11.2
6	Visualisation	`seaborn`	Statistical visualization: histograms, scatter…	1, 2, 4, and later	0.13.2
7	Statistics	`scipy`	Probability distributions, hypothesis testing …	1, 2, 5, and later	1.16.2
8	Statistical Modeling	`statsmodels`	Classical statistical models: OLS regression, …	1, 5, 6, and later	0.15.0
9	Machine Learning	`scikit-learn`	Machine learning: preprocessing, pipelines, cl…	1, 9, 11, and later	1.9.1
10	Machine Learning	`xgboost`	Scalable and accurate gradient boosted decisio…	12, 13	-
11	Deep Learning	`torch`	PyTorch deep learning framework for multilayer…	15	2.9.0+cu128
12	Time Series	`statsmodels.tsa`	Time series analysis, seasonal decomposition (…	1, 5, 6, and later	0.15.0
13	Reporting & Apps	`shiny`	Shiny for Python: reactive web applications, i…	17	-
14	AI & Automation	`requests`	HTTP requests for connecting to local and remo…	17, 18	2.32.5

C.1.1 - The core scientific stack

Rather than requiring dozens of disparate packages, scientific Python is built on a coherent, tightly integrated stack:

1. **`pandas` and `numpy`**: The foundation for tabular data structures and numerical operations. A DataFrame is the Python counterpart of a spreadsheet or database table, with support for filtering, grouping, joining, and missing-value imputation.
2. **`matplotlib` and `seaborn`**: The standard libraries for visualization. Matplotlib provides low-level graphical rendering and layout control, while Seaborn offers high-level statistical plotting routines (histograms, regression plots, and boxplots).
3. **`scipy` and `statsmodels`**: The statistical modeling core. Scipy provides probability distributions and classical hypothesis testing (t-tests, chi-square, Mann-Whitney), while Statsmodels handles econometric and regression models with comprehensive diagnostic tables (OLS, logistic regression, ANOVA, and linear mixed-effects).
4. **`scikit-learn`**: The premier machine learning toolkit for empirical research, providing consistent APIs (`fit`, `transform`, `predict`) for supervised classification, regression, clustering, and cross-validation pipelines.

D. Appendix D: Common Errors and How to Fix Them

Everyone who writes Python code encounters error messages on a regular basis—experts included. An error is not a sign of failure; it is Python explaining, as precisely as possible, what went wrong and where execution stopped.

This appendix collects the exceptions and pitfalls that researchers are most likely to encounter in empirical analysis, what each message means, and how to fix it. The examples below are live: each was executed when compiling the book, so the tracebacks match exactly what your Python environment will display.

D.1 - Anatomy of a Python traceback

When Python cannot execute an instruction, it raises an **exception** and prints a **traceback**.

1. **Read from bottom to top:** The very last line of the traceback displays the exception type and the specific message explaining what failed (for example, `KeyError: 'age_years'`).
2. **Find your code:** Look upward through the traceback to find the last line that points to your script file (e.g., `analysis.py, line 24`). That is the exact line where execution halted.
3. **Warnings vs. Exceptions:** An exception terminates execution immediately. A **warning** (such as pandas' `SettingWithCopyWarning` or `FutureWarning`) allows the script to continue running, but alerts you to a potential issue or deprecated syntax.

D.2 - Variable and naming errors

D.2.1 - `NameError: name '...' is not defined`

```
average = sum(sleep_hours) / len(sleep_hours)
```

```
NameError: name 'sleep_hours' is not defined
□[31m---------------------------------------------------------------------------□[39m
□[31mNameError□[39m                                 Traceback (most recent call last)
□[36mCell□[39m□[36m □[39m□[32mIn[2]□[39m□[32m, line 1□[39m
□[32m----> □[39m□[32m1□[39m average = □[38;5;28msum□[39m(□[43msleep_hours□[49m) /
□[38;5;28mlen□[39m(sleep_hours)

□[31mNameError□[39m: name 'sleep_hours' is not defined
```

What it means: Python searched your computer's memory for an object or variable named `sleep_hours` and found nothing.

Common causes: - **Typing mistake:** Variable names in Python are strictly case-sensitive. `Sleep_hours` and `sleep_hours` are two completely different names. - **Unexecuted line:** In an interactive session or notebook, the cell where the variable was defined was not run, or was run in a previous session that has since been restarted. - **Missing quotes around text:** Writing `faculty = Education` instead of `faculty = "Education"` causes Python to look for a variable named `Education`.

D.2.2 - `AttributeError: '...' object has no attribute '...'`

```
scores = [78, 85, 92, 64, 88]
scores.mean()
```

```
AttributeError: 'list' object has no attribute 'mean'
□[31m---------------------------------------------------------------------------□[39m
```

```
□[31mAttributeError□[39m                            Traceback (most recent call last)
□[36mCell□[39m□[36m □[39m□[32mIn[3]□[39m□[32m, line 2□[39m
□[32m          1□[39m scores = [□[32m78□[39m, □[32m85□[39m, □[32m92□[39m, □[32m64□[39m,
□[32m88□[39m]
□[32m----> □[39m□[32m2□[39m □[43mscores□[49m□[43m.□[49m□[43mmean□[49m()

□[31mAttributeError□[39m: 'list' object has no attribute 'mean'
```

What it means: You attempted to call a method or access a property that does not exist on that object type. Here, standard Python `list` objects have methods like `.append()` and `.sort()`, but no `.mean()` method.

How to fix it: - If working with a list, use Python's built-in `sum(scores) / len(scores)` or numpy's `np.mean(scores)`. - If you intended to use pandas methods, convert the collection to a pandas Series: `pd.Series(scores).mean()`. - Check spelling: writing `df.val_counts()` instead of `df.value_counts()`.

D.3 - Data structure and indexing errors

D.3.1 - `KeyError: '...'`

```
students["faculty_name"]
```

```
KeyError: 'faculty_name'
□[31m---------------------------------------------------------------------------□[39m
□[31mKeyError□[39m                                  Traceback (most recent call last)
□[36mFile                                  □[39m□[32m~\Drives\Locals\pytorch\venv\Lib\site-
packages\pandas\core\indexes\base.py:3812□[39m, in □[36mIndex.get_loc□[39m□[34m(self, key)□
[39m
□[32m   3811□[39m □[38;5;28;01mtry□[39;00m:
□[32m-> □[39m□[32m3812□[39m     □[38;5;28;01mreturn□[39;00m □[38;5;28;43mself□[39;49m□[43m.□
[49m□[43m_engine□[49m□[43m.□[49m□[43mget_loc□[49m□[43m(□[49m□[43mcasted_key□[49m□[43m)□[49m
□[32m            3813□[39m     □[38;5;28;01mexcept□[39;00m     □[38;5;167;01mKeyError□[39;00m
□[38;5;28;01mas□[39;00m err:

□[36mFile                   □[39m□[32mpandas/_libs/index.pyx:167□[39m,                   in
□[36mpandas._libs.index.IndexEngine.get_loc□[39m□[34m()□[39m

□[36mFile                   □[39m□[32mpandas/_libs/index.pyx:196□[39m,                   in
□[36mpandas._libs.index.IndexEngine.get_loc□[39m□[34m()□[39m

□[36mFile          □[39m□[32mpandas/_libs/hashtable_class_helper.pxi:7088□[39m,          in
□[36mpandas._libs.hashtable.PyObjectHashTable.get_item□[39m□[34m()□[39m

□[36mFile          □[39m□[32mpandas/_libs/hashtable_class_helper.pxi:7096□[39m,          in
□[36mpandas._libs.hashtable.PyObjectHashTable.get_item□[39m□[34m()□[39m

□[31mKeyError□[39m: 'faculty_name'

The above exception was the direct cause of the following exception:

□[31mKeyError□[39m                                  Traceback (most recent call last)
□[36mCell□[39m□[36m □[39m□[32mIn[4]□[39m□[32m, line 1□[39m
□[32m----> □[39m□[32m1□[39m □[43mstudents□[49m□[43m[□[49m□[33;43m"□[39;49m□[33;43mfaculty_name□[39;49m□[33
[49m

□[36mFile                                  □[39m□[32m~\Drives\Locals\pytorch\venv\Lib\site-
packages\pandas\core\frame.py:4113□[39m, in □[36mDataFrame.__getitem__□[39m□[34m(self, key)□
[39m
□[32m   4111□[39m □[38;5;28;01mif□[39;00m □[38;5;28mself□[39m.columns.nlevels > □[32m1□[39m:
□[32m   4112□[39m     □[38;5;28;01mreturn□[39;00m □[38;5;28mself□[39m._getitem_multilevel(key)
□[32m->      □[39m□[32m4113□[39m      indexer      =      □[38;5;28;43mself□[39;49m□[43m.□
```

```
[49m□[43mcolumns□[49m□[43m.□[49m□[43mget_loc□[49m□[43m(□[49m□[43mkey□[49m□[43m)□[49m
□[32m   4114□[39m □[38;5;28;01mif□[39;00m is_integer(indexer):
□[32m   4115□[39m     indexer = [indexer]

□[36mFile                                  □[39m□[32m~\Drives\Locals\pytorch\venv\Lib\site-
packages\pandas\core\indexes\base.py:3819□[39m, in □[36mIndex.get_loc□[39m□[34m(self, key)□
[39m
□[32m      3814□[39m          □[38;5;28;01mif□[39;00m  □[38;5;28misinstance□[39m(casted_key,
□[38;5;28mslice□[39m) □[38;5;129;01mor□[39;00m (
□[32m   3815□[39m         □[38;5;28misinstance□[39m(casted_key, abc.Iterable)
□[32m
3816□[39m          □[38;5;129;01mand□[39;00m □[38;5;28many□[39m(□[38;5;28misinstance□[39m(x,
□[38;5;28mslice□[39m) □[38;5;28;01mfor□[39;00m x □[38;5;129;01min□[39;00m casted_key)
□[32m   3817□[39m     ):
□[32m   3818□[39m         □[38;5;28;01mraise□[39;00m InvalidIndexError(key)
□[32m-> □[39m□[32m3819□[39m    □[38;5;28;01mraise□[39;00m □[38;5;167;01mKeyError□[39;00m(key)
□[38;5;28;01mfrom□[39;00m□[38;5;250m □[39m□[34;01merr□[39;00m
□[32m   3820□[39m □[38;5;28;01mexcept□[39;00m □[38;5;167;01mTypeError□[39;00m:
□[32m    3821□[39m       □[38;5;66;03m# If we have a listlike key, _check_indexing_error
will raise□[39;00m
□[32m    3822□[39m       □[38;5;66;03m#   InvalidIndexError. Otherwise we fall through and
re-raise□[39;00m
□[32m   3823□[39m     □[38;5;66;03m#  the TypeError.□[39;00m
□[32m   3824□[39m     □[38;5;28mself□[39m._check_indexing_error(key)

□[31mKeyError□[39m: 'faculty_name'
```

What it means: In pandas or dictionaries, you requested a column name or key that does not exist.

How to fix it: - Check the exact spelling and casing of your DataFrame columns by running `list(students.columns)`. - In the wellbeing dataset, the column is named `faculty`, not `faculty_name`. - Check for hidden whitespace in column names (e.g. `'age '` instead of `'age'`), which often occurs when importing messy Excel or survey exports. Clean them with `df.columns = df.columns.str.strip()`.

D.3.2 - `IndexError: list index out of range`

```
faculties = ["Health Sciences", "Education", "Social Sciences"]
print(faculties[3])
```

```
IndexError: list index out of range
□[31m---------------------------------------------------------------------------□[39m
□[31mIndexError□[39m                                Traceback (most recent call last)
□[36mCell□[39m□[36m □[39m□[32mIn[5]□[39m□[32m, line 2□[39m
□[32m              1□[39m  faculties  =  [□[33m"□[39m□[33mHealth  Sciences□[39m□[33m"□[39m,
□[33m"□[39m□[33mEducation□[39m□[33m"□[39m, □[33m"□[39m□[33mSocial Sciences□[39m□[33m"□[39m]
□[32m----> □[39m□[32m2□[39m
□[38;5;28mprint□[39m(□[43mfaculties□[49m□[43m[□[49m□[32;43m3□[39;49m□[43m]□[49m)

□[31mIndexError□[39m: list index out of range
```

What it means: You attempted to access an item at an index position that does not exist.

How to fix it: - Python uses **0-based indexing**. In a list of three elements, the items are at positions `0`, `1`, and `2`. Accessing index `3` asks for a fourth item. - To access the last item of any list or array safely, use negative indexing: `faculties[-1]`.

D.4 - Data type and value errors

D.4.1 - `TypeError: unsupported operand type(s)`

```
total_participants = 600 + " students"
```

```
TypeError: unsupported operand type(s) for +: 'int' and 'str'
□[31m---------------------------------------------------------------------------□[39m
□[31mTypeError□[39m                                 Traceback (most recent call last)
□[36mCell□[39m□[36m □[39m□[32mIn[6]□[39m□[32m, line 1□[39m
□[32m----> □[39m□[32m1□[39m total_participants = □[32;43m600□[39;49m□[43m
□[49m□[43m+□[49m□[43m □[49m□[33;43m"□[39;49m□[33;43m students□[39;49m□[33;43m"□[39;49m

□[31mTypeError□[39m: unsupported operand type(s) for +: 'int' and 'str'
```

What it means: You attempted an operation on incompatible data types—here, adding a number (`int`) and text (`str`).

How to fix it: - If you want to combine numbers and text for reporting, convert the number to a string using an f-string: `python f"{600} students"` - If a numerical column was imported as strings (for example, containing numbers mixed with `"N/A"`), convert the column with `pd.to_numeric(df['col'], errors='coerce')`.

D.4.2 - `ValueError: could not convert string to float`

```
float("29.5 years")
```

```
ValueError: could not convert string to float: '29.5 years'
□[31m---------------------------------------------------------------------------□[39m
□[31mValueError□[39m                                Traceback (most recent call last)
□[36mCell□[39m□[36m □[39m□[32mIn[7]□[39m□[32m, line 1□[39m
□[32m----
> □[39m□[32m1□[39m □[38;5;28;43mfloat□[39;49m□[43m(□[49m□[33;43m"□[39;49m□[33;43m29.5
years□[39;49m□[33;43m"□[39;49m□[43m)□[49m

□[31mValueError□[39m: could not convert string to float: '29.5 years'
```

What it means: A function expected data in a specific format, but the received value could not be parsed. `float()` cannot convert `"29.5 years"` because of the trailing text characters.

How to fix it: - Strip extraneous text characters before numeric conversion: `python float("29.5 years".replace(" years", ""))` - In pandas, use regular expression extraction: `df['age'].str.extract(r'(\d+\.?\d*)').astype(float)`.

D.5 - Syntax and indentation errors

D.5.1 - `SyntaxError: invalid syntax`

```
if students['age'].mean() > 25
    print("Mature cohort")
```

```
SyntaxError: expected ':'
```

What it means: Python could not parse your code because the grammar rules of the language were violated.

Common causes: - Forgetting the colon `:` at the end of an `if`, `for`, `while`, or `def` line. - Unmatched parentheses, brackets, or quotes: opening `(` without a matching `)`. - Using assignment `=` where a comparison `==` was required (e.g. `if x = 5:`).

D.5.2 - `IndentationError: expected an indented block`

```
def calculate_bmi(weight_kg, height_m):
return weight_kg / (height_m ** 2)
```

```
IndentationError: expected an indented block after function definition
```

What it means: Unlike R or C++ which use curly braces `{}` to group code blocks, Python relies strictly on indentation (usually 4 spaces).

How to fix it: - Indent all statements inside functions, loops, and conditional blocks by 4 spaces: `python def calculate_bmi(weight_kg, height_m): return weight_kg / (height_m**2)` - Never mix tabs and spaces in the same file. In VS Code, ensure your indentation setting is set to “Spaces: 4”.

D.6 - File and environment errors

D.6.1 - `FileNotFoundError: [Errno 2] No such file or directory`

```
df = pd.read_csv("data/non_existent_survey.csv")
```

```
FileNotFoundError: [Errno 2] No such file or directory: 'data/non_existent_survey.csv'
□[31m---------------------------------------------------------------------------□[39m
□[31mFileNotFoundError□[39m                         Traceback (most recent call last)
□[36mCell□[39m□[36m □[39m□[32mIn[8]□[39m□[32m, line 1□[39m
□[32m----> □[39m□[32m1□[39m
df = □[43mpd□[49m□[43m.□[49m□[43mread_csv□[49m□[43m(□[49m□[33;43m"□[39;49m□[33;43mdata/
non_existent_survey.csv□[39;49m□[33;43m"□[39;49m□[43m)□[49m

□[36mFile □[39m□[32m~\Drives\Locals\pytorch\venv\Lib\site-
packages\pandas\io\parsers\readers.py:1026□[39m,
in □[36mread_csv□[39m□[34m(filepath_or_buffer, sep, delimiter, header, names, index_col,
usecols, dtype, engine, converters, true_values, false_values, skipinitialspace, skiprows,
skipfooter, nrows, na_values, keep_default_na, na_filter, verbose, skip_blank_lines,
parse_dates, infer_datetime_format, keep_date_col, date_parser, date_format, dayfirst,
cache_dates, iterator, chunksize, compression, thousands, decimal, lineterminator, quotechar,
quoting, doublequote, escapechar, comment, encoding, encoding_errors, dialect, on_bad_lines,
delim_whitespace, low_memory, memory_map, float_precision, storage_options, dtype_backend)□
[39m
□[32m 1013□[39m kwds_defaults = _refine_defaults_read(
□[32m 1014□[39m dialect,
□[32m 1015□[39m delimiter,
□[32m (...)□[39m□[32m 1022□[39m dtype_backend=dtype_backend,
□[32m 1023□[39m )
□[32m 1024□[39m kwds.update(kwds_defaults)
□[32m-> □[39m□[32m1026□[39m
□[38;5;28;01mreturn□[39;00m □[43m_read□[49m□[43m(□[49m□[43mfilepath_or_buffer□[49m□[43m,□
[49m□[43m □[49m□[43mkwds□[49m□[43m)□[49m

□[36mFile □[39m□[32m~\Drives\Locals\pytorch\venv\Lib\site-
packages\pandas\io\parsers\readers.py:620□[39m, in □[36m_read□[39m□[34m(filepath_or_buffer,
kwds)□[39m
□[32m 617□[39m _validate_names(kwds.get(□[33m"□[39m□[33mnames□[39m□[33m"□[39m,
□[38;5;28;01mNone□[39;00m))
□[32m 619□[39m □[38;5;66;03m# Create the parser.□[39;00m
□[32m--> □[39m□[32m620□[39m
parser = □[43mTextFileReader□[49m□[43m(□[49m□[43mfilepath_or_buffer□[49m□[43m,□[49m□[43m
□[49m□[43m*□[49m□[43m*□[49m□[43mkwds□[49m□[43m)□[49m
□[32m 622□[39m □[38;5;28;01mif□[39;00m chunksize □[38;5;129;01mor□[39;00m iterator:
□[32m 623□[39m □[38;5;28;01mreturn□[39;00m parser
```

```
□[36mFile                                   □[39m□[32m~\Drives\Locals\pytorch\venv\Lib\site-
packages\pandas\io\parsers\readers.py:1620□[39m,                                          in
□[36mTextFileReader.__init__□[39m□[34m(self, f, engine, **kwds)□[39m
□[32m
1617□[39m       □[38;5;28mself□[39m.options[□[33m"□[39m□[33mhas_index_names□[39m□[33m"□[39m]
= kwds[□[33m"□[39m□[33mhas_index_names□[39m□[33m"□[39m]
□[32m     1619□[39m  □[38;5;28mself□[39m.handles:  IOHandles  |  □[38;5;28;01mNone□[39;00m  =
□[38;5;28;01mNone□[39;00m
□[32m-> □[39m□[32m1620□[39m  □[38;5;28mself□[39m._engine  =  □[38;5;28;43mself□[39;49m□[43m.□
[49m□[43m_make_engine□[49m□[43m(□[49m□[43mf□[49m□[43m,□[49m□[43m
□[49m□[38;5;28;43mself□[39;49m□[43m.□[49m□[43mengine□[49m□[43m)□[49m

□[36mFile                                   □[39m□[32m~\Drives\Locals\pytorch\venv\Lib\site-
packages\pandas\io\parsers\readers.py:1880□[39m,                                          in
□[36mTextFileReader._make_engine□[39m□[34m(self, f, engine)□[39m
□[32m       1878□[39m            □[38;5;28;01mif□[39;00m  □[33m"□[39m□[33mb□[39m□[33m"□[39m
□[38;5;129;01mnot□[39;00m □[38;5;129;01min□[39;00m mode:
□[32m   1879□[39m         mode += □[33m"□[39m□[33mb□[39m□[33m"□[39m
□[32m-> □[39m□[32m1880□[39m □[38;5;28mself□[39m.handles = □[43mget_handle□[49m□[43m(□[49m
□[32m   1881□[39m □[43m    □[49m□[43mf□[49m□[43m,□[49m
□[32m   1882□[39m □[43m    □[49m□[43mmode□[49m□[43m,□[49m
□[32m   1883□[39m □[43m    □[49m□[43mencoding□[49m□[43m=□[49m□[38;5;28;43mself□[39;49m□[43m.□
[49m□[43moptions□[49m□[43m.□[49m□[43mget□[49m□[43m(□[49m□[33;43m"□[39;49m□[33;43mencoding□[39;49m□[33;43m
[49m□[43m □[49m□[38;5;28;43;01mNone□[39;49;00m□[43m)□[49m□[43m,□[49m
□[32m
1884□[39m  □[43m       □[49m□[43mcompression□[49m□[43m=□[49m□[38;5;28;43mself□[39;49m□[43m.□
[49m□[43moptions□[49m□[43m.□[49m□[43mget□[49m□[43m(□[49m□[33;43m"□[39;49m□[33;43mcompression□[39;49m□[33;
[49m□[43m □[49m□[38;5;28;43;01mNone□[39;49;00m□[43m)□[49m□[43m,□[49m
□[32m
1885□[39m  □[43m        □[49m□[43mmemory_map□[49m□[43m=□[49m□[38;5;28;43mself□[39;49m□[43m.□
[49m□[43moptions□[49m□[43m.□[49m□[43mget□[49m□[43m(□[49m□[33;43m"□[39;49m□[33;43mmemory_map□[39;49m□[33;4
[49m□[43m □[49m□[38;5;28;43;01mFalse□[39;49;00m□[43m)□[49m□[43m,□[49m
□[32m   1886□[39m □[43m    □[49m□[43mis_text□[49m□[43m=□[49m□[43mis_text□[49m□[43m,□[49m
□[32m    1887□[39m □[43m     □[49m□[43merrors□[49m□[43m=□[49m□[38;5;28;43mself□[39;49m□[43m.□
[49m□[43moptions□[49m□[43m.□[49m□[43mget□[49m□[43m(□[49m□[33;43m"□[39;49m□[33;43mencoding_errors□[39;49m□
[49m□[43m    □[49m□[33;43m"□[39;49m□[33;43mstrict□[39;49m□[33;43m"□[39;49m□[43m)□[49m□[43m,□
[49m
□[32m
1888□[39m □[43m    □[49m□[43mstorage_options□[49m□[43m=□[49m□[38;5;28;43mself□[39;49m□[43m.□
[49m□[43moptions□[49m□[43m.□[49m□[43mget□[49m□[43m(□[49m□[33;43m"□[39;49m□[33;43mstorage_options□[39;49m□
[49m□[43m □[49m□[38;5;28;43;01mNone□[39;49;00m□[43m)□[49m□[43m,□[49m
□[32m   1889□[39m □[43m□[49m□[43m)□[49m
□[32m              1890□[39m     □[38;5;28;01massert□[39;00m     □[38;5;28mself□[39m.handles
□[38;5;129;01mis□[39;00m □[38;5;129;01mnot□[39;00m □[38;5;28;01mNone□[39;00m
□[32m   1891□[39m f = □[38;5;28mself□[39m.handles.handle

□[36mFile                                   □[39m□[32m~\Drives\Locals\pytorch\venv\Lib\site-
packages\pandas\io\common.py:873□[39m,   in   □[36mget_handle□[39m□[34m(path_or_buf,   mode,
encoding, compression, memory_map, is_text, errors, storage_options)□[39m
□[32m              868□[39m    □[38;5;28;01melif□[39;00m    □[38;5;28misinstance□[39m(handle,
□[38;5;28mstr□[39m):
□[32m     869□[39m      □[38;5;66;03m# Check whether the filename is to be opened in binary
mode.□[39;00m
□[32m    870□[39m     □[38;5;66;03m# Binary mode does not support 'encoding' and 'newline'.□
[39;00m
□[32m                871□[39m                     □[38;5;28;01mif□[39;00m    ioargs.encoding
□[38;5;129;01mand□[39;00m     □[33m"□[39m□[33mb□[39m□[33m"□[39m     □[38;5;129;01mnot□[39;00m
□[38;5;129;01min□[39;00m ioargs.mode:
□[32m    872□[39m         □[38;5;66;03m# Encoding□[39;00m
□[32m--> □[39m□[32m873□[39m         handle = □[38;5;28;43mopen□[39;49m□[43m(□[49m
```

```
□[32m    874□[39m □[43m            □[49m□[43mhandle□[49m□[43m,□[49m
□[32m    875□[39m □[43m            □[49m□[43mioargs□[49m□[43m.□[49m□[43mmode□[49m□[43m,□[49m
□[32m    876□[39m □[43m            □[49m□[43mencoding□[49m□[43m=□[49m□[43mioargs□[49m□[43m.□
[49m□[43mencoding□[49m□[43m,□[49m
□[32m    877□[39m □[43m          □[49m□[43merrors□[49m□[43m=□[49m□[43merrors□[49m□[43m,□[49m
□[32m                                                    878□[39m              □[43m
□[49m□[43mnewline□[49m□[43m=□[49m□[33;43m"□[39;49m□[33;43m"□[39;49m□[43m,□[49m
□[32m    879□[39m □[43m        □[49m□[43m)□[49m
□[32m    880□[39m     □[38;5;28;01melse□[39;00m:
□[32m    881□[39m         □[38;5;66;03m# Binary mode□[39;00m
□[32m    882□[39m         handle = □[38;5;28mopen□[39m(handle, ioargs.mode)

□[31mFileNotFoundError□[39m:   [Errno   2]   No   such   file   or   directory:   'data/
non_existent_survey.csv'
```

What it means: Python searched the file system at the specified path and found no file with that name.

How to fix it: - Check your current working directory by running: ```python import os print(os.getcwd()) ```- Relative file paths start from the current working directory. If your script is inscripts/and your data is indata/, writing“data/students.csv”will fail if the working directory is already insidescripts/. Use paths relative to the project root or usepathlib.Path(**file**).parent` for portable path resolution.

D.6.2 - `ModuleNotFoundError: No module named '...'`

What it means: You ran `import <package>`, but Python could not find that library in the active environment.

How to fix it: - Ensure the library is installed in your terminal using `pip install <package>`. - In VS Code, verify that the active Python interpreter corresponds to the environment where you installed the package (`Ctrl+Shift+P` -> `Python: Select Interpreter`).

D.7 - Silent data pitfalls

Some mistakes in data analysis produce no error message at all, but generate incorrect numbers. Watch out for these three pitfalls:

D.7.1 - 1. Missing value comparisons (`NaN == NaN`)

In scientific Python, floating-point missing values (`NaN`) follow the IEEE 754 standard: a missing value is never equal to anything, not even itself:

```
np.nan == np.nan
```

```
False
```

Writing `students[students['age'] == np.nan]` will return an empty table! Always use pandas' `.isna()` or `.notna()`:

```
missing_count = students["age"].isna().sum()
print(f"Students with missing age: {missing_count}")
```

```
Students with missing age: 1
```

D.7.2 - 2. Exponentiation (^ vs. **)

In Python, the caret symbol `^` is the bitwise XOR operator, **not** the mathematical power operator:

```
print("2 ^ 3 evaluates to:", 2 ^ 3)  # Bitwise XOR!
print("2 ** 3 evaluates to:", 2**3)  # Mathematical power (2 cubed)
```

```
2 ^ 3 evaluates to: 1
2 ** 3 evaluates to: 8
```

Always use `**` for powers (e.g. `(1 + r) ** t`).

D.7.3 - 3. Modifying a slice (`SettingWithCopyWarning`)

When you filter a DataFrame with `sub = df[df['faculty'] == 'Education']` and then try to create a new column with `sub['new_col'] = 1`, pandas warns you that it cannot tell whether `sub` is an independent copy or a view into the original DataFrame.

How to fix it: Always make an explicit copy when creating a working subset:

```
sub = df[df["faculty"] == "Education"].copy()
sub["new_col"] = 1
```

E. Appendix E: Further Learning Resources

This book is a starting point. This appendix suggests where to go next: books, courses, and communities, almost all of them free. The books cited in the chapters are listed first, by topic; the rest are chosen because they suit researchers who are not programmers.

E.1 - Books

E.1.1 - Getting started and data skills

- *R for Data Science* [42], free at r4ds.hadley.nz: the standard introduction to importing, cleaning, transforming, and visualising data with the tidyverse. The natural next book after Part 1 of this one.
- *R for Researchers: An Introduction* by Tyson Barrett, free at tysonbarrett.com/Rstats: a short, friendly introduction written for researchers in the social and health sciences, and a good companion to Chapters 1 to 8.
- *ggplot2: Elegant Graphics for Data Analysis* [43], free at ggplot2-book.org: everything about ggplot2, for when Chapter 4 leaves you wanting more.
- *Happy Git and GitHub for the useR* by Jenny Bryan, free at happygitwithr.com: a gentle, practical guide to git and GitHub with RStudio (Chapter 17).

E.1.2 - Statistics

- *Introduction to Modern Statistics* [11], free at openintro-ims.netlify.app: a modern introductory statistics textbook built around simulation and real data, like Chapters 6 and 7.
- *Learning Statistics with R* by Danielle Navarro, free at learningstatisticswithr.com: a thorough, readable introduction to statistics for psychology and the social sciences, using R.
- *Statistical Inference via Data Science (ModernDive)* by Chester Ismay and Albert Kim, free at moderndive.com: regression and inference with the tidyverse.
- *Regression and Other Stories* [15]: an excellent guide to building, checking, and interpreting regression models in real research, with a free PDF from the authors.
- *Statistical Power Analysis for the Behavioral Sciences* [13]: the classic reference on effect sizes and power.

E.1.3 - Mixed models, machine learning, and forecasting

- *Data Analysis Using Regression and Multilevel/Hierarchical Models* [19]: multilevel models from the ground up (Chapter 10).
- *An Introduction to Statistical Learning* [16], free at statlearning.com: the clearest introduction to the ideas of machine learning (Chapters 11 to 15).
- *Tidy Modeling with R* [44], free at tmwr.org: the tidymodels framework in depth.
- *Forecasting: Principles and Practice* [31], free at otexts.com/fpp3: the standard introduction to forecasting with fable (Chapter 16).
- *Deep Learning with R* [45]: for readers who want to go further with neural networks (Chapter 15).

E.1.4 - Reporting and dashboards

- The Quarto guide at quarto.org: documentation and examples for every kind of document (Chapter 17).
- *Mastering Shiny* [46], free at mastering-shiny.org: the definitive guide to Shiny apps.

The **Big Book of R** (bigbookofr.com) indexes hundreds of free R books by topic, from psychology and ecology to economics and text analysis: a good way to find a book for your own field.

E.2 - Courses and practice

- **Posit Cheatsheets** (posit.co/resources/cheatsheets): one- or two-page summaries of dplyr, ggplot2, tidyr, Quarto, Shiny, and more, worth keeping next to you while you work. Several have been translated into other languages by volunteers.
- **The Carpentries** (carpentries.org): free, hands-on lessons for researchers, including *R for Social Scientists* and *R for Ecologists*, and two-day workshops held at universities around the world.
- **swirl** (swirlstats.com): interactive lessons that run inside R itself: `install.packages("swirl")`, then `library(swirl)` and `swirl()`.
- **TidyTuesday** (github.com/rfordatascience/tidytuesday): a new real dataset every week to practise on, with thousands of shared solutions to learn from.
- This book's playground: exercises for every chapter, in the browser and as downloadable projects.

E.3 - Communities

You do not have to learn alone, and the R community is known for welcoming beginners.

- **Posit Community** (forum.posit.co): a friendly forum for questions about R, RStudio, the tidyverse, Quarto, and Shiny.
- **Stack Overflow** (stackoverflow.com/questions/tagged/r): the largest archive of answered R questions; search it before asking, and include a small reproducible example when you ask (Appendix D).
- **R-Ladies** (rladies.org): a worldwide organisation promoting gender diversity in the R community, with local chapters that run meetings and workshops open to learners.
- **R user groups** meet in many cities and universities; the R Consortium lists them at r-consortium.org.
- **R Weekly** (rweekly.org): a weekly digest of R news, tutorials, and new packages.
- **useR!**, the annual international R conference, posts its talks online.

Whatever you learn from, the most effective way to learn R is to use it on your own data, a little every day, one question at a time, as Elaf did.

Bibliography

[1] W. McKinney, *Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter*, 3rd ed. O'Reilly Media, 2022. [Online]. Available: https://wesmckinney.com/book/

[2] H. Wickham, "Tidy Data," *Journal of Statistical Software*, vol. 59, no. 10, pp. 1–23, 2014, doi: 10.18637/jss.v059.i10.

[3] F. J. Anscombe, "Graphs in Statistical Analysis," *The American Statistician*, vol. 27, no. 1, pp. 17–21, 1973, doi: 10.1080/00031305.1973.10478966.

[4] W. S. Cleveland and R. McGill, "Graphical Perception: Theory, Experimentation, and Application to the Development of Graphical Methods," *Journal of the American Statistical Association*, vol. 79, no. 387, pp. 531–554, 1984, doi: 10.1080/01621459.1984.10478080.

[5] C. O. Wilke, *Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures*. O'Reilly Media, 2019. [Online]. Available: https://clauswilke.com/dataviz/

[6] K. R. Popper, *The Logic of Scientific Discovery*. London: Hutchinson, 1959.

[7] S. S. Stevens, "On the Theory of Scales of Measurement," *Science*, vol. 103, no. 2684, pp. 677–680, 1946, doi: 10.1126/science.103.2684.677.

[8] L. J. Cronbach and P. E. Meehl, "Construct Validity in Psychological Tests," *Psychological Bulletin*, vol. 52, no. 4, pp. 281–302, 1955, doi: 10.1037/h0040957.

[9] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor, "The Preregistration Revolution," *Proceedings of the National Academy of Sciences*, vol. 115, no. 11, pp. 2600–2606, 2018, doi: 10.1073/pnas.1708274114.

[10] E. W. Crampton, "The Growth of the Odontoblasts of the Incisor Tooth as a Criterion of the Vitamin C Intake of the Guinea Pig," *The Journal of Nutrition*, vol. 33, no. 5, pp. 491–504, 1947, doi: 10.1093/jn/33.5.491.

[11] M. Çetinkaya-Rundel and J. Hardin, *Introduction to Modern Statistics*, 2nd ed. OpenIntro, 2024. [Online]. Available: https://openintro-ims.netlify.app/

[12] W. R. Shadish, T. D. Cook, and D. T. Campbell, *Experimental and Quasi-Experimental Designs for Generalized Causal Inference*. Boston: Houghton Mifflin, 2002.

[13] J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed. Hillsdale, NJ: Lawrence Erlbaum Associates, 1988.

[14] Student, "The Probable Error of a Mean," *Biometrika*, vol. 6, no. 1, pp. 1–25, 1908, doi: 10.1093/biomet/6.1.1.

[15] A. Gelman, J. Hill, and A. Vehtari, *Regression and Other Stories*. Cambridge: Cambridge University Press, 2020. doi: 10.1017/9781139161879.

[16] G. James, D. Witten, T. Hastie, and R. Tibshirani, *An Introduction to Statistical Learning: With Applications in R*, 2nd ed. New York: Springer, 2021. doi: 10.1007/978-1-0716-1418-1.

[17] R. A. Johnson and D. W. Wichern, *Applied Multivariate Statistical Analysis*, 6th ed. Pearson Prentice Hall, 2007.

[18] G. Belenky *et al.*, "Patterns of Performance Degradation and Restoration During Sleep Restriction and Subsequent Recovery: A Sleep Dose-Response Study," *Journal of Sleep Research*, vol. 12, no. 1, pp. 1–12, 2003, doi: 10.1046/j.1365-2869.2003.00337.x.

[19] A. Gelman and J. Hill, *Data Analysis Using Regression and Multilevel/Hierarchical Models*. Cambridge: Cambridge University Press, 2007. doi: 10.1017/CBO9780511790942.

[20] S. Seabold and J. Perktold, "statsmodels: Econometric and statistical modeling with Python," in *Proceedings of the 9th Python in Science Conference*, 2010, pp. 92–96.

[21] A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 3rd ed. O'Reilly Media, 2022.

[22] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[23] A. M. Horst, A. P. Hill, and K. B. Gorman, "Palmer Archipelago Penguins Data in the palmerpenguins R Package: An Alternative to Anderson's Irises," *The R Journal*, vol. 14, no. 1, pp. 244–254, 2022, doi: 10.32614/RJ-2022-020.

[24] J. VanderPlas, *Python Data Science Handbook: Essential Tools for Working with Data*. O'Reilly Media, 2016. [Online]. Available: https://jakevdp.github.io/PythonDataScienceHandbook/

[25] R. Tibshirani, "Regression Shrinkage and Selection via the Lasso," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 58, no. 1, pp. 267–288, 1996, doi: 10.1111/j.2517-6161.1996.tb02080.x.

[26] I.-C. Yeh, "Modeling of Strength of High-Performance Concrete Using Artificial Neural Networks," *Cement and Concrete Research*, vol. 28, no. 12, pp. 1797–1808, 1998, doi: 10.1016/S0008-8846(98)00165-3.

[27] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, "A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise," in *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD-96)*, 1996, pp. 226–231.
[28] G. James, D. Witten, T. Hastie, R. Tibshirani, and J. Taylor, *An Introduction to Statistical Learning: with Applications in Python*. Springer, 2023. doi: 10.1007/978-3-031-38747-0.
[29] F. Chollet, *Deep Learning with Python*, 2nd ed. Manning Publications, 2021.
[30] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016. [Online]. Available: https://www.deeplearningbook.org/
[31] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne, Australia: OTexts, 2021. [Online]. Available: https://otexts.com/fpp3/
[32] R. H. Shumway and D. S. Stoffer, *Time Series Analysis and Its Applications: With R Examples*, 4th ed. Springer, 2017. doi: 10.1007/978-3-319-52452-8.
[33] R. D. Peng, "Reproducible Research in Computational Science," *Science*, vol. 334, no. 6060, pp. 1226–1227, 2011, doi: 10.1126/science.1213847.
[34] G. Wilson, J. Bryan, K. Cranston, J. Kitzes, L. Nederbragt, and T. K. Teal, "Good Enough Practices in Scientific Computing," *PLOS Computational Biology*, vol. 13, no. 6, p. e1005510, 2017, doi: 10.1371/journal.pcbi.1005510.
[35] A. Gelman and E. Loken, "The Statistical Crisis in Science: Data-dependent analysis–a 'garden of forking paths'–explains why many statistically significant comparisons don't hold up," *American Scientist*, vol. 102, no. 6, pp. 460–465, 2014.
[36] Open Science Collaboration, "Estimating the Reproducibility of Psychological Science," *Science*, vol. 349, no. 6251, p. aac4716, 2015, doi: 10.1126/science.aac4716.
[37] J. Bryan, "Excuse Me, Do You Have a Moment to Talk About Version Control?," *The American Statistician*, vol. 72, no. 1, pp. 20–27, 2018, doi: 10.1080/00031305.2017.1399928.
[38] J. Cohen, "A Coefficient of Agreement for Nominal Scales," *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960, doi: 10.1177/001316446002000104.
[39] J. R. Landis and G. G. Koch, "The Measurement of Observer Agreement for Categorical Data," *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977, doi: 10.2307/2529310.
[40] M. Appelbaum, H. Cooper, R. B. Kline, E. Mayo-Wilson, A. M. Nezu, and S. M. Rao, "Journal Article Reporting Standards for Quantitative Research in Psychology: The APA Publications and Communications Board Task Force Report," *American Psychologist*, vol. 73, no. 1, pp. 3–25, 2018, doi: 10.1037/amp0000191.
[41] E. von Elm, D. G. Altman, M. Egger, S. J. Pocock, P. C. Gøtzsche, and J. P. Vandenbroucke, "The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) Statement: Guidelines for Reporting Observational Studies," *The Lancet*, vol. 370, no. 9596, pp. 1453–1457, 2007, doi: 10.1016/S0140-6736(07)61602-X.
[42] H. Wickham, M. Çetinkaya-Rundel, and G. Grolemund, *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*, 2nd ed. O'Reilly Media, 2023. [Online]. Available: https://r4ds.hadley.nz/
[43] H. Wickham, *ggplot2: Elegant Graphics for Data Analysis*, 2nd ed. New York: Springer, 2016. doi: 10.1007/978-3-319-24277-4.
[44] M. Kuhn and J. Silge, *Tidy Modeling with R: A Framework for Modeling in the Tidyverse*. O'Reilly Media, 2022. [Online]. Available: https://www.tmwr.org/
[45] F. Chollet, T. Kalinowski, and J. J. Allaire, *Deep Learning with R*, 2nd ed. Manning, 2022.
[46] H. Wickham, *Mastering Shiny*. O'Reilly Media, 2021. [Online]. Available: https://mastering-shiny.org/

From Data To Thesis Using R

Turn messy data into a defended, publication-ready thesis.

For many postgraduate and academic researchers, collecting data is only the beginning of the real challenge. The hardest hurdle is turning spreadsheets, surveys, and lab observations into robust statistical models and defendable written arguments. Commercial point-and-click tools leave your analysis opaque, while standard programming textbooks bury you in abstract computer science theory.

From Data to Thesis bridges that divide. Written specifically for graduate students, doctoral researchers, and academic professionals, this book provides a complete, modern roadmap for empirical research using **R**. Rather than abstract mathematics, every concept is taught through a single, realistic running case study following 600 students across four semesters.

Every chapter guides you through five clear steps:

1. **A Concrete Research Question**.
2. **A Minimal Working Example** .
3. **The Case-Study Analysis**.
4. **In Your Field**.
5. **Thesis Reporting & Interpretation** .

From exploratory data analysis and hypothesis testing to mixed-effects models, machine learning, time-series forecasting, and ethical AI assistance, this book gives you the tools, code, and confidence to defend your research with scientific rigor.

Inside you will discover:

- ❖ How to clean, reshape, and visualize complex data using modern **tidyverse** workflows.
- ❖ How to choose the right statistical test—and how to diagnose and fix common assumption violations.
- ❖ Practical machine learning and predictive modeling without the computational overwhelm.
- ❖ Reproducible research workflows using **Quarto** that generate thesis-ready tables and figures.
- ❖ Interactive browser playgrounds to practice every method without installing a single package.

Whether you are writing your first Master's dissertation, completing your doctoral defense, or preparing an empirical journal article, this book is your desk companion from raw data to submitted thesis.



by Polla Fattah
and Collection of LLMs